

Direct guaranteed lower eigenvalue bounds with quasi-optimal adaptive mesh-refinement

Dissertation

zur Erlangung des akademischen Grades
doctor rerum naturalium (Dr. rer. nat.)
im Fach Mathematik

eingereicht an der
Mathematisch-Naturwissenschaftlichen Fakultät
der Humboldt-Universität zu Berlin

von

M. Sc. Sophie Louise Puttkammer

Präsidentin der Humboldt-Universität zu Berlin
Prof. Dr. Julia von Blumenthal

Dekanin der Mathematisch-Naturwissenschaftlichen Fakultät
Prof. Dr. Caren Tischendorf

Gutachter*innen

1. Prof. Dr. Carsten Carstensen (Humboldt-Universität zu Berlin)
2. Prof. Dr. Neela Nataraj (Indian Institute of Technology Bombay)
3. Prof. Dr. Dietmar Gallistl (Friedrich-Schiller-Universität Jena)

Tag der mündlichen Prüfung
24. Januar 2023

Abstract

Guaranteed lower eigenvalue bounds (GLB) for elliptic eigenvalue problems of partial differential equation (PDE) are of high relevance in theory and praxis, e.g., in a safety analysis in computational mechanics, for the detection of spectral gaps, or for valid bounds of the continuity constants of Sobolev embeddings. Due to the Rayleigh-Ritz (or) min-max principle all conforming finite element methods (FEM) provide guaranteed upper eigenvalue bounds. A post-processing for nonconforming FEM of Carstensen and Gedicke (*Math. Comp.*, 83.290 (2014), pp. 2605–2629) as well as Carstensen and Gallistl (*Numer. Math.*, 126.1 (2014), pp. 33–51) computes GLB. However, the maximal mesh-size enters as a global parameter in the eigenvalue bound and may cause significant underestimation for adaptive mesh-refinement. There are numerical examples, where this post-processing on locally refined meshes fails completely.

Inspired by a recent skeletal method from Carstensen, Zhai, and Zhang (*SIAM J. Numer. Anal.*, 58.1 (2020), pp. 109–124) this thesis presents on the one hand a modified hybrid high-order (HHO) method ($m = 1$) and on the other hand a general framework for an extra-stabilized nonconforming Crouzeix-Raviart ($m = 1$) or Morley ($m = 2$) FEM. These novel methods compute direct GLB for the m -Laplace operator in that a specific smallness assumption on the maximal mesh-size guarantees that the computed k -th discrete eigenvalue is a lower bound for the k -th Dirichlet eigenvalue. This GLB property as well as a priori convergence rates are established in any space dimension. The novel ansatz allows for adaptive mesh-refinement necessary to recover optimal convergence rates for non-smooth eigenfunctions. Striking numerical evidence indicates the superiority of the new adaptive eigensolvers. For the extra-stabilized nonconforming methods (a generalization of) known abstract arguments entitled as the axioms of adaptivity from Carstensen, Feischl, Page, and Praetorius (*Comput. Math. Appl.*, 67.6 (2014), pp. 1195–1253) as well as Carstensen and Rabus (*SIAM J. Numer. Anal.*, 55.6 (2017), pp. 2644–2665) allow to prove the convergence of the GLB towards a simple eigenvalue with optimal rates.

Zusammenfassung

Garantierte untere Eigenwertschranken (GLB) für elliptische Eigenwertprobleme partieller Differentialgleichungen (PDE) sind in der Theorie sowie in praktischen Anwendungen wie Sicherheitsanalysen in der numerischen Mechanik, beim Erkennen spektraler Lücken oder für garantierte Schranken der Stetigkeitskonstanten in Sobolev-Einbettungen relevant. Auf Grund des Rayleigh-Ritz- (oder) min-max-Prinzips berechnen alle konformen Finite-Elemente-Methoden (FEM) garantierte obere Eigenwertschranken. Ein Postprocessing nichtkonformer Methoden von Carstensen und Gedicke (*Math. Comp.*, 83.290 (2014), pp. 2605–2629) sowie Carstensen und Gallistl (*Numer. Math.*, 126.1 (2014), pp. 33–51) erlaubt die Berechnung von GLB. In diesen Eigenwertschranken ist die maximale Netzweite ein globaler Parameter, dies kann bei adaptiver Netzverfeinerung zu deutlichen Unterschätzungen führen. In einigen numerischen Beispielen versagt dieses Postprocessing für lokal verfeinerte Netze komplett.

Diese Dissertation präsentiert, inspiriert von einer neuen skeletal-Methode von Carstensen, Zhai und Zhang (*SIAM J. Numer. Anal.*, 58.1 (2020), pp. 109–124), einerseits eine modifizierte hybrid-high-order (HHO) Methode ($m = 1$) und andererseits ein allgemeines Framework für extra-stabilisierte nichtkonforme Crouzeix-Raviart ($m = 1$) bzw. Morley ($m = 2$) FEM. Diese neuen Methoden berechnen direkte GLB für den m -Laplace-Operator, bei denen eine leicht überprüfbare Bedingung an die maximale Netzweite garantiert, dass der berechnete k -te diskrete Eigenwert eine untere Schranke für den k -ten Dirichlet-Eigenwert ist. Diese GLB-Eigenschaft und a priori Konvergenzraten werden für jede Raumdimension etabliert. Der neu entwickelte Ansatz erlaubt adaptive Netzverfeinerung, die für optimale Konvergenzraten auch bei nichtglatten Eigenfunktionen erforderlich ist. Die Überlegenheit der neuen adaptiven FEM wird durch eine Vielzahl repräsentativer numerischer Beispiele illustriert. Für die mit den extra-stabilisierten nichtkonformen Methoden bestimmten GLB wird Konvergenz mit optimalen Raten gegen einen einfachen Eigenwert gezeigt, indem die Axiome der Adaptivität von Carstensen, Feischl, Page und Praetorius (*Comput. Math. Appl.*, 67.6 (2014), pp. 1195–1253) sowie Carstensen und Rabus (*SIAM J. Numer. Anal.*, 55.6 (2017), pp. 2644–2665) geeignet verallgemeinert werden.

Contents

1	Introduction	1
2	Preliminaries	11
2.1	Functional analysis background	11
2.2	Triangulation and piecewise spaces	15
2.3	General eigenvalue problem	19
2.4	Inequalities	23
3	Discrete spaces	25
3.1	Hsieh-Clough-Tocher	26
3.2	Worsey-Farin	27
3.2.1	Bernstein polynomials in a simplex	28
3.2.2	Smoothness across an interface	31
3.2.3	Piecewise quadratic polynomials in a WF partition of a tetrahedron	35
3.2.4	The WF finite element	38
3.2.5	The WF on adjacent tetrahedra	46
3.2.6	The shape-regularity of the WF partition	48
3.2.7	The scaling of the WF basis functions	51
3.3	Crouzeix-Raviart	59
3.3.1	Interpolation operator	60
3.3.2	Conforming companion	62
3.4	Morley	68
3.4.1	Interpolation operator	71
3.4.2	Conforming companion	76
4	A modified HHO Method	93
4.1	Introduction to HHO methods	93
4.2	The modified HHO method	94
4.2.1	Operators of interest	96
4.2.2	Properties of the discrete bilinear forms	98
4.2.3	Matrix eigenvalue problem	101
4.3	Guaranteed lower eigenvalue bound property	101
4.4	Convergence analysis	103
4.4.1	Convergence analysis for the source problem	104
4.4.2	Convergence rates for the eigenvalue problem	109

4.5	Lowest-order case	111
4.5.1	Comparison with the Crouzeix-Raviart method	111
4.5.2	Estimate on the constant τ	116
5	The extra-stabilized method	119
5.1	Introduction to the extra-stabilized method	119
5.1.1	Properties of the interpolation operator	120
5.1.2	Properties of the conforming companion	122
5.2	Guaranteed lower eigenvalue bound property	124
5.3	Comparison and general remarks	128
5.3.1	Reduced eigenvalue problem	128
5.3.2	Comparison with the nonconforming method	130
5.3.3	Comparison with the lowest-order modified HHO method	131
5.4	A priori convergence analysis	134
5.4.1	Convergence analysis for the source problem	135
5.4.2	Convergence rates for the eigenvalue problem	140
5.4.3	Medius analysis	142
5.5	Optimality of the adaptive algorithm	154
5.5.1	Adaptive algorithm	154
5.5.2	Stability and reduction	157
5.5.3	Discrete reliability	159
5.5.4	Quasiorthogonality	163
5.5.5	Proof of optimality	166
5.5.6	Global convergence	167
6	A review and extension of the axioms of adaptivity	169
6.1	The general adaptive algorithm and axioms of adaptivity	169
6.2	The general optimality result	170
6.3	Preparation of the optimality proof	171
6.4	Proof of optimality	176
7	Numerical experiments	179
7.1	Numerical realization	179
7.1.1	Domains and refinement	179
7.1.2	Computation of reference eigenvalues	180
7.2	The Laplace eigenvalue problem	182
7.2.1	Setting the parameter	183
7.2.2	Adaptive mesh-refinement	183
7.2.3	Displayed quantities	184
7.2.4	Dumbbell-slit domain	185
7.2.5	L-shaped domain	187
7.2.6	Slit domain	190
7.2.7	Four-slit domain	192

7.2.8	Isospectral domains	194
7.2.9	Conclusions	197
7.3	The bi-Laplace eigenvalue problem	198
7.3.1	Setting the parameter	199
7.3.2	Adaptive mesh-refinement	199
7.3.3	Displayed quantities	200
7.3.4	Dumbbell-slit domain	200
7.3.5	L-shaped domain	202
7.3.6	Slit domain	203
7.3.7	Four-slit domain	204
7.3.8	Conclusions	207
8	Conclusion and outlook	209
A	List of common notations	211
A.1	Elementary notation	211
A.2	Measure theory notation and derivatives	211
A.3	Triangulation notation	212
B	Some fundamental estimates	215
C	Remarks on the implementation	223
C.1	The experiments.m script	223
C.2	Implementation of the modified HHO method	225
C.2.1	Main method	225
C.2.2	Parameter study	229
C.3	Implementation of the extra-stabilized Crouzeix-Raviart method .	230
C.4	Implementation of the extra-stabilized Morley method	233
C.5	Notes on Crouzeix-Raviart and Morley method for comparison . .	237
D	Software	239
	Bibliography	241

1 Introduction

Motivation

The computation of eigenvalues for symmetric compact differential operators is a fundamental task in the numerical analysis with a well-understood a priori error analysis for conforming finite element methods (FEM) leading to optimal asymptotic convergence rates [BO91; Bof10]. The Rayleigh-Ritz (or) min-max principle shows that the discrete eigenvalues of conforming finite element methods are guaranteed upper bounds of the exact eigenvalues even in the pre-asymptotic range of coarse triangulations.

In practice, guaranteed lower bounds (GLB) can be more important in a safety analysis in computational mechanics, for the detection of spectral gaps, or for valid bounds of Sobolev embeddings. The eigenvalues of nonconforming finite element methods converge asymptotically from below [AD04; Yan+12; HHL14; YLB16]. In other words, they provide lower bounds under the assumption that the global mesh-size is sufficiently small, but it remains unclear whether a given triangulation belongs to the asymptotic regime. In very simple counter examples in [CG14a; CG14b] the eigenvalues computed with the Crouzeix-Raviart (resp. Morley) finite element method are indeed upper bounds of the continuous eigenvalues of the Laplacian (resp. bi-Laplacian). These papers introduce a simple post-processing that computes guaranteed lower eigenvalue bounds. However, the maximal mesh-size $h_{\max}$ enters as a global parameter in the bound and may cause a significant underestimation for adaptive meshes, when local mesh-refining leaves a few simplices coarse and $h_{\max}$ large. For example, in the convergence history plot for the principal Dirichlet eigenvalue of the Laplacian on the L-shaped domain in Fig. 1.1.d the involvement of $h_{\max}$ leads to steps in the convergence graph of the known post-processed guaranteed lower bound (blue dashed line with half filled markers) on adaptive meshes. This bound on the adaptive meshes is even worse than the uniform version, which coincides with the new extra-stabilized method (green line with filled markers). For the asymmetric dumbbell-slit domain in Fig. 1.2.d, where appropriate adaptive mesh-refinement refines mainly the square $(-1, 1)^2$ due to the form of the principal eigenfunction for the bi-Laplacian in Fig. 1.2.b, the post-processed bound fails completely. The post-processed bound (blue dashed line with half filled markers) on adaptive meshes does not converge at all. In both illustrated non-convex domains the principal eigenfunction is obviously not smooth, such that the empirical convergence rate for uniform refined meshes

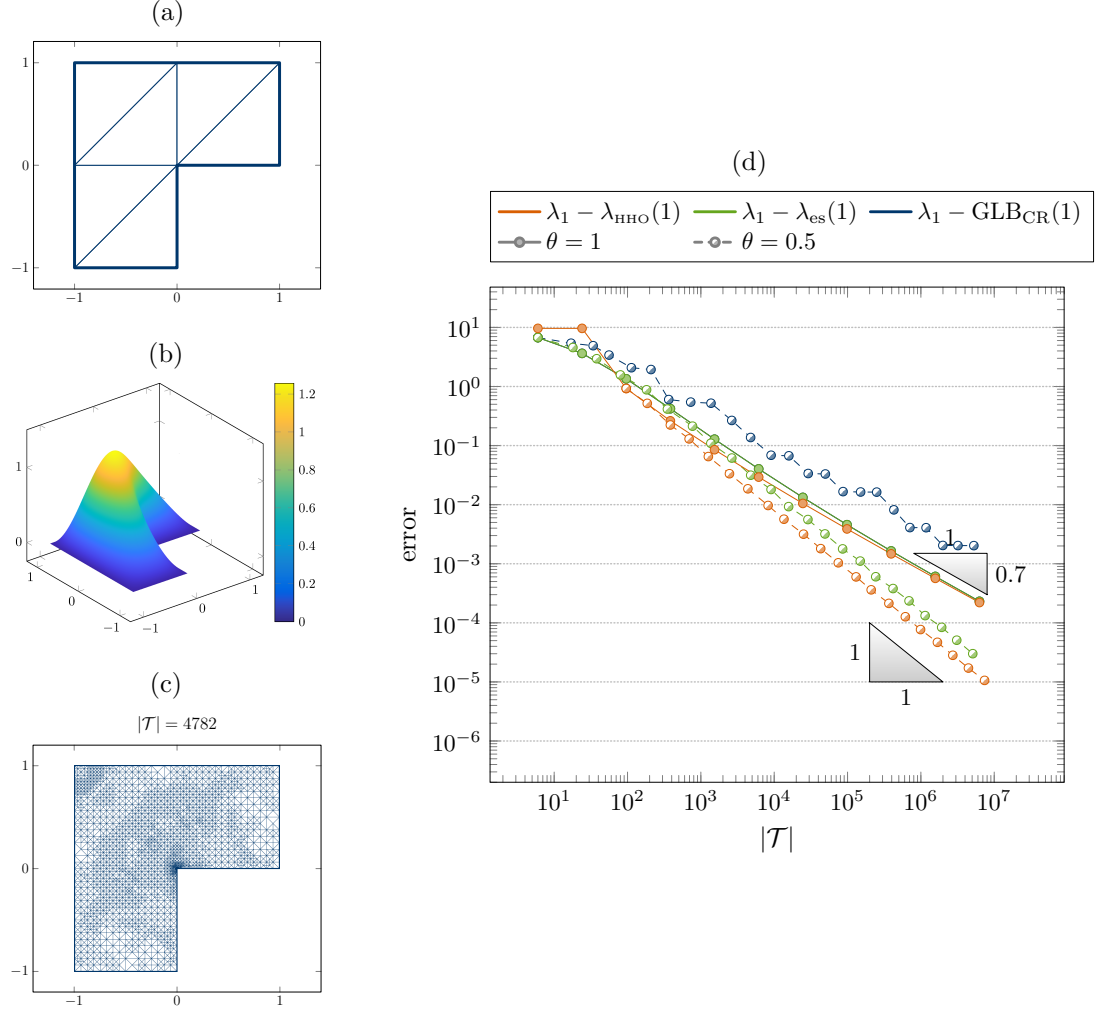


Figure 1.1: Experiments for the principal eigenvalue of the Laplacian on the L-shaped domain: (a) initial triangulation, (b) conforming P_1 approximation of the eigenfunction on a uniform red-refined mesh with $|\mathcal{T}| = 6291456$, (c) adaptive mesh with $|\mathcal{T}| = 4782$ computed with bulk parameter $\theta = 0.5$ and estimator (1.3), and (d) comparison of the distance of λ_1 and the new $\lambda_{\text{HHO}}(1)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(1)$ as well as the known $\text{GLB}_{\text{CR}}(1)$ (with $\kappa_1 = 0.2982$) computed on uniform ($\theta = 1$, solid line and filled markers) and adaptive ($\theta = 0.5$, dashed line and half filled markers) meshes

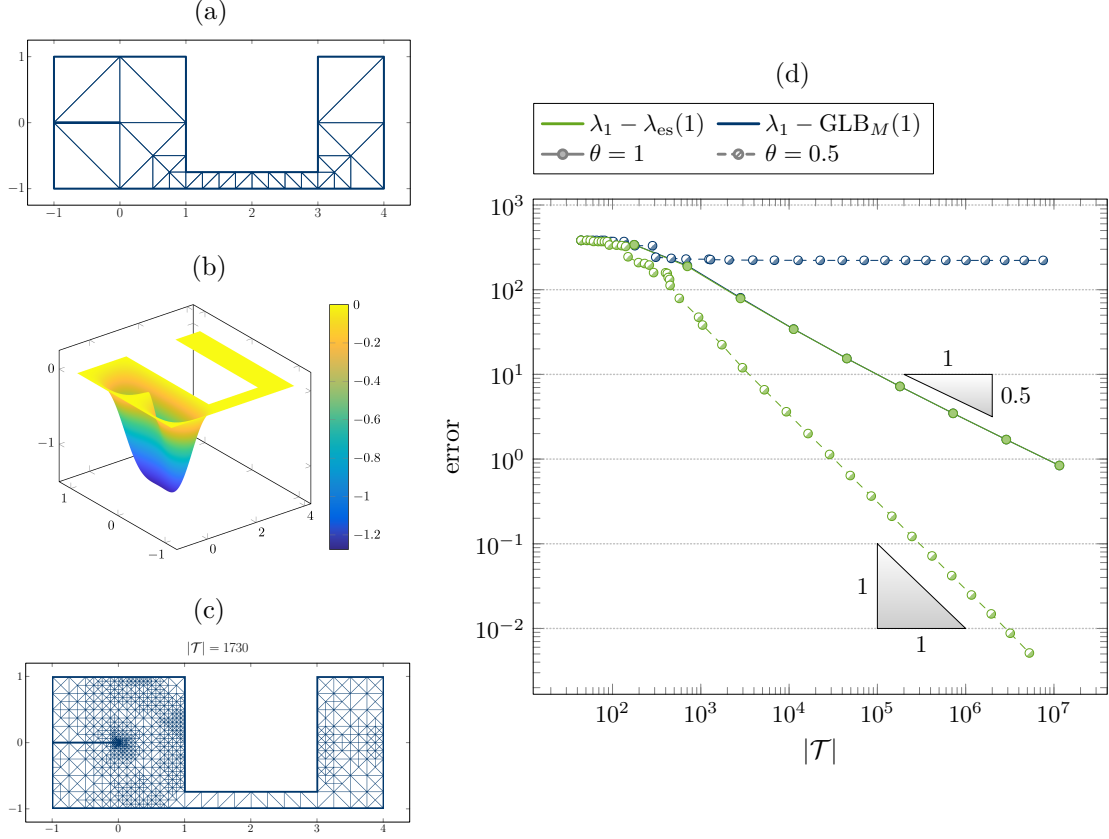


Figure 1.2: Experiments for the principal eigenvalue of the bi-Laplacian on the asymmetric dumbbell-slit domain: (a) initial triangulation, (b) conforming nodal interpolation of the Morley approximation of the eigenfunction on a uniform red-refined mesh with $|\mathcal{T}| = 11534336$, (c) adaptive mesh with $|\mathcal{T}| = 1730$ computed with bulk parameter $\theta = 0.5$ and estimator (1.3), and (d) comparison of the distance of λ_1 and the new $\lambda_{\text{es}}(1)$ as well as the known $\text{GLB}_M(1)$ (with $\kappa_2 = 0.2358$) computed on uniformly refined ($\theta = 1$, solid line and filled markers) and adaptive ($\theta = 0.5$, dashed line and half filled markers) meshes

is 0.7 on the L-shaped domain and 0.5 on the asymmetric dumbbell-slit, and adaptive mesh-refinement is necessary to recover the optimal convergence rate one for the lowest-order methods. This failure of the known bound motivates the design of novel schemes for the direct computation of guaranteed lower eigenvalue bounds without post-processing. The first example for the Laplace eigenvalue problem can be found in [CZZ20] for a hybridizable discontinuous Galerkin (HDG) method with Lehrenfeld-Schöberl stabilization, also studied under the label weak Galerkin scheme. The scheme proposed in [CZZ20] requires the fine-tuning of skeletal-based stabilization terms.

This thesis presents a modified hybrid high-order (HHO) method for the Laplace eigenvalue problem which leads to guaranteed lower bounds on the eigenvalues under a simplified tuning of the stabilization terms. In this HHO method the discrete unknowns are polynomials of degree $k \geq 0$ attached to the mesh faces and polynomials of degree $k + 1$ for the cell unknowns. For this method the optimal a priori convergence is established and the numerical studies show that with the right parameter choice even the lowest-order HHO method (line color orange) in Fig. 1.1.d improves the other guaranteed bounds.

For the extra-stabilized nonconforming method which computes guaranteed lower eigenvalue bounds for the Laplacian resp. bi-Laplacian, the a priori and a posteriori convergence analysis is presented. A slight generalization of the axioms of adaptivity [Car+14a; CR17] proves optimal convergence rates of an adaptive mesh-refining algorithm. The associated graph (line color green and half filled markers) in Fig. 1.1.d and Fig. 1.2.d shows the optimal empirical convergence rate one and the adaptive mesh in Fig. 1.1.c and Fig. 1.2.c depicts the expected mesh-refinement towards the reentrant corner resp. the end of the slit.

A historical overview and state of the art

A first overview of finite element methods for compact symmetric eigenvalue problems is presented in [SF73]. The finite element method is also one topic in the more abstract book [Cha83]. A standard reference is the review article [BO91] with estimates for general compact operators and several applications to second order problems as well as more recent overviews in [Bof10; SZ17]. The a priori error analysis for nonconforming discretization of eigenvalue problems dates back to [Ran79]. The first a posteriori error analysis in [Ver94] presents a general framework for nonlinear problems with application to second order eigenvalue problems, in [Lar00] duality techniques are applied, while [DPR03] uses the higher-order convergence of the L^2 error to prove equivalence of a standard estimator. Other a posteriori estimators are also available for eigenvalue problems, e.g., an averaging estimator in [MSZ06] and hierarchical estimators in [GO09; MM11; Ney02]. In [HR01] a posteriori error bounds for nonsymmetric eigenvalue problems are considered. The asymptotic convergence of adaptive finite element methods for

eigenvalue problems was established in [GG09; GMZ09; CG11]. The adaptive finite element method in [Car+14b] leads to an asymptotic error reduction and justifies the saturation for a class of estimators. For simple eigenvalues the proof of optimal convergence rates and quasi-optimal computational complexity of the adaptive method is given in [DXZ08; CG12] for conforming finite element methods, while [CGS15] establishes optimal convergence rates for the adaptive nonconforming Crouzeix-Raviart approximation of the principal eigenvalue. The revolutionary work [Car+14a] presents an axiomatic framework that unifies the optimality proofs that trace back to [Ste07; Cas+08]. As one example [Car+14a] covers the optimality of the adaptive finite element computation of simple eigenvalues of the Laplacian of [DXZ08; CG12]. The analysis of adaptive mesh-refinement for multiple eigenvalues and clusters of eigenvalues with optimal convergence rates can be found in [DHZ15; Gal15a]. In [Gal15b] the 2D Morley discretization of the biharmonic eigenvalue problem is analysed and optimal adaptive convergence rates are established even for eigenvalue clusters.

The main focus of this thesis is the computation of guaranteed lower eigenvalue bounds on adaptively refined meshes. Guaranteed upper and lower eigenvalue bounds for the symmetric algebraic eigenvalue problem have been an important topic in numerical linear algebra for decades and even software has been developed that takes round-off effects into account for a guaranteed outcome of the overall computation. This is part of the computational challenge in eigenvalue problems of partial differential equation (PDE). This thesis therefore restricts the attention to the discretization error. Throughout an approximation is called a guaranteed bound, if it is a valid bound under the idealized assumption of exact arithmetic and exact solve (even of the finite-dimensional discrete eigenvalue problems) and the many numerical linear algebra aspects are left aside. The word guaranteed eigenvalue bound is used to distinguish the bounds from the asymptotic lower bounds in (classical) nonconforming finite element schemes [AD04; LLX12; Yan+12; HHL14; YLB16]. It is so far not possible to decide for a given triangulation, if the nonconforming eigenvalue is a lower bound for the continuous one or not. The counter examples for coarse triangulation even of the unit square in [CG14a; CG14b] illustrate this problem. The last decade has seen a few guaranteed lower eigenvalue bounds on the basis of finite element methods.

One ansatz is to utilize some a posteriori post-processing that leads under additional a priori information on spectral gaps in [Can+17; Can+18; Can+20] to guaranteed lower eigenvalue bounds. A priori information on lower eigenvalue bounds and a combination of generalized Weinstein and Kato bounds with complementarity based estimators in [ŠV14] and flux reconstructions in the Lehrmann-Goerisch method in [Vej18a; Vej18b] as post-processing lead to guaranteed lower eigenvalue bounds as well.

An approach that works without any a priori information on the spectrum is the simple post-processing of the nonconforming FEM (Crouzeix-Raviart resp. Morley) for the m -Laplacian ($m = 1$ resp. $m = 2$) in [CG14a; CG14b; Liu15]. Given the

discrete nonconforming eigenvalue $\lambda_{\text{nc}}(j)$ of number $j \in \mathbb{N}$, the following

$$\text{GLB}_{\text{nc}}(j) := \frac{\lambda_{\text{nc}}(j)}{1 + \lambda_{\text{nc}}(j)\kappa_m^2 h_{\text{max}}^{2m}} \leq \lambda_j \quad (1.1)$$

is a guaranteed lower bound for the (unknown) continuous eigenvalue λ_j of the same number j . Therein the constants κ_m are well known and for any space dimension $n \geq 2$ analytically bounded by, e.g.,

$$\begin{aligned} \kappa_1^2 &:= \frac{1}{\pi^2} + \frac{1}{2n(n+1)(n+2)}, \\ \kappa_2^2 &:= \frac{\kappa_1^2}{\pi^2} + \left(\frac{\kappa_1}{\sqrt{2n(n+1)(n+2)}} + \sqrt{\frac{n\kappa_1^2 + 2\kappa_1}{2n^2(n-1)(n+1)}} \right)^2 \end{aligned} \quad (1.2)$$

and h_{max} denotes the maximal mesh-size of the triangulation. Unfortunately, the maximal mesh-size as global parameter can be very problematic as illustrated in Fig. 1.2.d. Given adaptive mesh-refinement with the standard nonconforming (Morley) estimator, the $\text{GLB}_{\text{nc}} = \text{GLB}_M$ fails in the example of the asymmetric dumbbell-slit domain.

A remedy are new methods, where the eigenfunction is approximated by two variables coupled with a fine-tuned stabilization and the discrete eigenvalue is a direct guaranteed lower eigenvalue bound. This is the ansatz in this work.

Main results

This thesis presents novel schemes for the direct computation of guaranteed lower eigenvalue bounds for the Dirichlet eigenvalues of the Laplace ($m = 1$) or bi-Laplace ($m = 2$) operator and extends the work in [CEP21a; CP21; CP22]. Given a polyhedral bounded Lipschitz domain $\Omega \subset \mathbb{R}^n$ with $n \geq 2$ and set of admissible triangulation $\mathbb{T}$, the weak form of the continuous problem seeks $(\lambda, u) \in \mathbb{R}^+ \times H_0^m(\Omega)$ with $\|u\|_{L^2(\Omega)} = 1$ and

$$a(u, v) = \lambda b(u, v) \quad \text{for all } v \in V := H_0^m(\Omega).$$

The energy scalar product in the Hilbert space $V := H_0^m(\Omega)$ reads $a(\bullet, \bullet) := (D^m \bullet, D^m \bullet)_{L^2(\Omega)}$ with the gradient $D^1 = \nabla$ ($m = 1$) or the Hessian D^2 ($m = 2$) while $b(u, v) := (u, v)_{L^2(\Omega)}$ denotes the L^2 scalar product. The infinite but countable many eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$ with $\lim_{j \rightarrow \infty} \lambda_j = \infty$ in (5.1) are enumerated in ascending order counting multiplicities. For varying discrete spaces V_h as well as bilinear forms a_h and b_h , the discrete problems seek $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h \setminus \{0\}$ with

$$a_h(u_h, v_h) = \lambda_h b_h(u_h, v_h) \quad \text{for all } v_h \in V_h.$$

The finite and positive algebraic eigenvalues $0 < \lambda_h(1) \leq \dots \leq \lambda_h(M) < \infty$ are enumerated in ascending order counting multiplicity as well. The three methods in this thesis, whose discrete eigenvalues $\lambda_h(j)$ are direct guaranteed lower bounds for the continuous one λ_j , if some easily verifiable condition on the mesh-size and parameter is satisfied, fit the displayed setting.

The first method for the Laplace eigenvalue problem ($m = 1$) is based on the framework of hybrid high-order (HHO) methods, which were introduced in [DPEL14; DPE15] for linear diffusion and locking-free linear elasticity and have been bridged to hybridizable discontinuous Galerkin (HDG) and nonconforming virtual element methods (ncVEM) in [CDPE16]. In [Cal+19] the standard HHO method was first used to discretize the Laplace eigenvalue problem. The discrete unknowns in the HHO methods are polynomials of degree $k \geq 0$ attached to the mesh faces and polynomials of degree $\ell \in \{k-1, k, k+1\}$, $\ell \geq 0$, attached to the mesh cells. The modified HHO method considers degree $\ell = k+1$ for the cell unknowns. Two key ingredients in HHO methods are a local gradient reconstruction operator and a local stabilization operator. The presented method differs from the literature in two points. The first novelty is that the gradient reconstruction combines an operator mapping to piecewise Raviart-Thomas functions of degree k and another reconstruction. Raviart-Thomas reconstructions were considered previously in [AEP18; DPDM18], but not yet combined, which turns out to be necessary in the present analysis. The second novelty is that the stabilization operator is not skeletal-based but cell-based.

The discrete space reads $V_{\text{HHO}} := P_{k+1}(\mathcal{T}) \times P_k(\mathcal{F}(\Omega))$ with the space of piecewise polynomials of degree at most $(k+1)$ on each simplex in the triangulation $P_{k+1}(\mathcal{T})$ and the space of piecewise polynomials of degree at most k on each interior face $P_k(\mathcal{F}(\Omega))$. The bilinear form $b_h := b_{\text{HHO}}$ represents the $L^2(\Omega)$ scalar product of the volume components, the gradient-like approximations in the bilinear form $a_h := a_{\text{HHO}}$ involve two reconstructions R and G_{RT} of the discrete unknowns in V_{HHO} . The precise definition of the linear maps $R : V_{\text{HHO}} \rightarrow L^2(\Omega; \mathbb{R})$ and $G_{RT} : V_{\text{HHO}} \rightarrow L^2(\Omega; \mathbb{R}^n)$ can be found in Subsection 4.2.1. Given two positive parameters $0 < \alpha < 1$ and $0 < \beta < \infty$, for any $v_{\text{HHO}} := (v_{\mathcal{T}}, v_{\mathcal{F}})$, $w_{\text{HHO}} := (w_{\mathcal{T}}, w_{\mathcal{F}}) \in V_{\text{HHO}}$, the bilinear forms read

$$\begin{aligned} a_{\text{HHO}}(v_{\text{HHO}}, w_{\text{HHO}}) &:= (G_{RT}(v_{\text{HHO}}), G_{RT}(w_{\text{HHO}}))_{L^2(\Omega)} \\ &\quad - \alpha \left((1 - \Pi_k) G_{RT}(v_{\text{HHO}}), (1 - \Pi_k) G_{RT}(w_{\text{HHO}}) \right)_{L^2(\Omega)} \\ &\quad + \beta (\nabla_{\text{pw}}(v_{\mathcal{T}} - R(v_{\text{HHO}})), \nabla_{\text{pw}}(w_{\mathcal{T}} - R(w_{\text{HHO}})))_{L^2(\Omega)}, \\ b_{\text{HHO}}(v_{\text{HHO}}, w_{\text{HHO}}) &:= b(v_{\mathcal{T}}, w_{\mathcal{T}}). \end{aligned}$$

In the source problem [AEP18; DPDM18] the Raviart-Thomas reconstruction G_{RT} of the gradient does not require a stabilization, but the second term in the definition of a_{HHO} has a negative sign and hence a stabilization with the reconstruction R in $P_{k+1}(\mathcal{T})$ is added. In this setting the algebraic eigenvalue

problem (4.2) has $M := \dim(P_{k+1}(\mathcal{T})) = \binom{k+1+n}{n} |\mathcal{T}|$ finite and positive algebraic eigenvalues. This thesis contains the a priori error analysis for discrete eigenvalues and eigenvectors and proves the guaranteed lower bound property

$$\tau^2 \beta + \delta^2 \min\{\lambda_j, \lambda_{\text{HHO}}(j)\} \leq \alpha < 1 \quad \text{implies} \quad \lambda_{\text{HHO}}(j) \leq \lambda_j,$$

with $\delta := \kappa h_{\max}$, where the constants κ and τ are related to the stability of L^2 -projections onto piecewise polynomial spaces, cf. Theorem 2.4 and Lemma 4.3. Notice that each of the conditions (i) $\tau^2 \beta + \delta^2 \lambda_j \leq \alpha < 1$ (a priori) and (ii) $\tau^2 \beta + \delta^2 \lambda_{\text{HHO}}(j) \leq \alpha < 1$ (a posteriori) implies $\lambda_{\text{HHO}}(j) \leq \lambda_j$. The a priori condition (i) can bound the maximal initial mesh-size by $h_{\max}^2 \leq (\alpha - \tau^2 \beta) / (\kappa^2 \mu)$ for any upper bound $\mu \geq \lambda_j$ computable by any conforming approximation or post-processing. The a posteriori condition can easily be verified as soon as $\lambda_{\text{HHO}}(j)$ is computed. This thesis contains a detailed analysis of the lowest-order version of this method with analytical bounds for the constants κ and τ . This version allows for comparison with the extra-stabilized method for $m = 1$.

The analysis for the extra-stabilized method is done in parallel for $m = 1, 2$ with the lowest-order Crouzeix-Raviart ($m = 1$) and Morley ($m = 2$) finite elements as nonconforming spaces $V(\mathcal{T})$. However, the framework with conditions for the interpolation operator and conforming companion of the nonconforming space is presented in a general form such that nonconforming finite elements of higher-order [WX13] could be adapted for the m -Laplace eigenvalue problems with $m \geq 3$. For $m = 1 = n - 1$ the extra-stabilized method coincides with the lowest-order version of the skeletal method in [CZZ20]. The new methods are parameter free and all constants are known for any space dimension and $m = 1, 2$. The discrete space reads $V_{\text{es}} := P_m(\mathcal{T}) \times V(\mathcal{T})$ and the discrete energy scalar product $a_h := a_{\text{es}}$ contains the L^2 scalar product of the piecewise derivatives of order m and some stabilization with the explicit constant $\kappa_m > 0$ in (1.2), while the bilinear form $b_h := b_{\text{es}}$ represents the L^2 scalar product of the piecewise polynomial components. For any $v_{\text{es}} = (v_{\text{pw}}, v_{\text{nc}})$, $w_{\text{es}} = (w_{\text{pw}}, w_{\text{nc}}) \in V_{\text{es}}$, the bilinear forms read

$$\begin{aligned} a_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= (D_{\text{pw}}^m v_{\text{nc}}, D_{\text{pw}}^m w_{\text{nc}})_{L^2(\Omega)} + \kappa_m^{-2} (h_{\mathcal{T}}^{-2m} (v_{\text{pw}} - v_{\text{nc}}), w_{\text{pw}} - w_{\text{nc}})_{L^2(\Omega)}, \\ b_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= b(v_{\text{pw}}, w_{\text{pw}}). \end{aligned}$$

The piecewise constant mesh-size function $h_{\mathcal{T}} \in P_0(\mathcal{T})$ has the value $h_{\mathcal{T}}|_T = h_T := \text{diam}(T)$ in each simplex $T \in \mathcal{T}$. Since $(V_{\text{es}}, a_{\text{es}})$ is a Hilbert space and b_{es} is a semi-scalar product with kernel $\{0\} \times V(\mathcal{T}) \subset V_{\text{es}}$, the algebraic eigenvalue problem has $M := \dim(P_m(\mathcal{T})) = \binom{m+n}{n} |\mathcal{T}|$ finite and positive algebraic eigenvalues. This thesis proves the following guaranteed lower bound property for any space dimension $n \geq 2$

$$\min\{\lambda_{\text{es}}(j), \lambda_j\} \kappa_m^2 h_{\max}^{2m} \leq 1 \quad \text{implies} \quad \lambda_{\text{es}}(j) \leq \lambda_j.$$

As above the a priori condition (i) $\lambda_j \kappa_m^2 h_{\max}^{2m} \leq 1$ could be used to bound the maximal initial mesh-size before the computation, while the a posteriori condition

(ii) $\lambda_{\text{es}}(j)\kappa_m^2 h_{\max}^{2m} \leq 1$ can easily be checked afterwards. In this work the extra-stabilized method is compared with the post-processing of the nonconforming method and for $m = 1$ with the lowest-order HHO method as well. A convergence analysis that utilizes the conforming companion operator proves optimal a priori convergence rates, while a medius analysis in the spirit of [CGS15] leads to best-approximation results. The latter in combination with the axioms of adaptivity framework [Car+14a; CR17] including a new generalized discrete reliability condition lead to optimal convergence rates in an adaptive mesh-refining algorithm. Therein the local estimator contribution reads

$$\eta^2(T) = (\eta(T))^2 = |T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 + |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2 \quad (1.3)$$

for any simplex $T \in \mathcal{T}$ with volume $|T|$ and the tangential components $[D^m u_{\text{nc}}]_F \times \nu_F$ of the jump of the (piecewise) gradient $D^1 = \nabla$ ($m = 1$) or Hessian D^2 ($m = 2$) along any side $F \in \mathcal{F}(T)$. A highlight of the convergence analysis is that it works for any space dimension $n \geq 2$ for the Laplacian and for $n = 2, 3$ for the bi-Laplacian. For the biharmonic eigenvalue problem the definition of the conforming companion operator utilized in the convergence analysis hinges on the choice of an appropriate C^1 -conforming space. In 2D the Hsieh-Clough-Tocher (HCT) macro finite element [Cia78] is convenient [Gal15b; VZ19]. For $n \geq 3$ [WF87] proposes a C^1 -conforming Clough-Tocher finite element, which misses additional constraints on the subtriangulation for $n \geq 4$ as [Sor09] proves a decade later. It appears still an open question if even for $n = 4$ all conditions can be satisfied simultaneously [Sor09, p.42, 1.40ff]. So this work includes an introduction to the Worsey-Farin (WF) finite element and the analysis of the associated nodal basis functions to generalize [Cia78, Thm. 6.1.3] for $n = 3$ in order to define a conforming companion for the Morley finite element in 3D. The generalized properties of the interpolation operator and conforming companion are useful beyond the scope of this thesis, cf. [CN21] for the usage of the conforming companion as a smoother for rough sources and in its role in the a posteriori error control.

Outline

This thesis is organized as follows. Chapter 2 introduces the necessary notations, preliminaries on triangulation in $\mathbb{R}^n$ and their adaptive mesh-refinement, summarizes some general properties of the model eigenvalue problem, and states frequently used inequalities. The aim of Chapter 3 is to define the interpolation operator and conforming companion for the Crouzeix-Raviart and Morley finite element. To this end a detailed and easy to read analysis of the Worsey-Farin finite element is included. The modified HHO method is analysed in Chapter 4, while Chapter 5 is concerned with the extra-stabilized method. The analysis of the optimal convergences of the adaptive mesh-refinement uses the axioms of

adaptivity framework reviewed and extended in Chapter 6. Chapter 7 presents two-dimensional numerical experiments that confirm the theoretical results for the lowest-order HHO and the extra-stabilized methods on non-convex domains. The results of the new methods are compared with the post-processed bounds from [CG14a; CG14b] and the treatment of multiple eigenvalues is tested empirically. After the conclusion and outlook in Chapter 8, Appendix A contains a list of common abbreviation, Appendix B some fundamental estimates used throughout the thesis to quantify as many constants as possible, and Appendix C some notes on the implementation of the provided software, see Appendix D.

Acknowledgement

Without questions there are many people who helped me with this thesis and I would like to use this opportunity to thank them.

First of all, I would like to thank my supervisor Prof. Carstensen for his constant support and invaluable guidance and advice in my studies. I am also very grateful to the numerical analysis working group for having a sympathetic ear for all my problems, I really enjoyed working with all of you. Special thanks to Philipp for your advice on this thesis and the many discussions, to Benedikt for the computations of conforming eigenvalues, as well as to Tien, Franz, and Hella for all the answers to my many questions. In addition, I thank Prof. Ern for the collaboration on HHO methods and appreciate the support of several institutions such as the Berlin Mathematical School (BMS), the European Network for Teaching and Research in Academic Liasion (CENTRAL), and the Oberwolfach-Leibniz-Graduate-Students program, allowing me to attend various workshops and conferences.

On the other hand, I cannot thank my family enough. I am very lucky to know, that they will always support and encourage me. If I manage to give the same feeling of safety to our little wonder, I am very glad indeed. I also want to thank my friends. Especially, Dji, Maren, and Linda, meeting you during my first week in Adlershof was definitely a piece of luck. Mathematics would have been a lot less fun without you. Finally, Lucian you make my life so much better and me so happy. Thank you for everything!

2 Preliminaries

This chapter shall clarify the notation on the continuous Lebesgue and Sobolev spaces in Section 2.1, give an overview on triangulation and adaptive mesh-refinement in Section 2.2, summarize properties of the continuous model partial differential equation (PDE) eigenvalue problem in Section 2.3, and conclude with some frequently used inequalities in Section 2.4.

Throughout this thesis, let $\Omega \subset \mathbb{R}^n$ denote a bounded open domain with polyhedral Lipschitz boundary $\partial\Omega$, e.g., [Gri85, Def. 1.2.1.1] or [Mon03, Def. 3.1]. Therein $n \geq 2$ denotes the space dimension. Furthermore $a \lesssim b$ abbreviates that there exists a generic constant C (independent of the mesh-size) with $a \leq Cb$, whereas $a \approx b$ abbreviates $a \lesssim b \lesssim a$.

2.1 Functional analysis background

Standard notation applies to Lebesgue and Sobolev spaces, e.g., [BS08; Eva10; EG15].

Notation in $\mathbb{R}^{\ell \times m}$

For $a, b \in \mathbb{R}^{\ell \times m}$, let $a \cdot b = a^\top b \in \mathbb{R}^{m \times m}$ and $a \otimes b = ab^\top \in \mathbb{R}^{\ell \times \ell}$. Let $e_k \in \mathbb{R}^m$ denote the canonical k -unit vector for $k = 1, \dots, m$ with $e_k(j) = \delta_{jk}$ for $1 \leq j, k \leq m$ and Kronecker delta δ_{jk} .

The notation $|\bullet|$ depends on the context and denotes the modulus, the Euclidean length of a vector, the cardinality of a finite set, the n , $(n-1)$, or $(n-2)$ -dimensional Lebesgue measure of a subset of $\mathbb{R}^n$, e.g., $|T|$ is the volume of a simplex $T \in \mathcal{T}$ and $|F|$ denotes the area of a face $F \in \mathcal{F}$ in 3D or the length of an edge in 2D.

Lebesgue spaces

Let $(\omega, \mathcal{A}, \mu)$ denote a measure space and $Y \subseteq \mathbb{R}^{\ell \times m}$ a finite-dimensional real Hilbert space with Euclidean scalar product $(\bullet, \bullet)$. The Lebesgue integral of any μ -measurable function $f : \omega \rightarrow Y$ reads $\int_\omega f \, d\mu$ with integral mean $\bar{f}_\omega f \, d\mu := \mu(\omega)^{-1} \int_\omega f \, d\mu$ for any ω with $0 < \mu(\omega) < \infty$. The target space Y and measure μ are not explicit in the notation of the L^2 seminorm

$$\|f\|_{L^2(\omega)}^2 := \int_\omega (f, f) \, d\mu,$$

but will always be clear from the context. The space of equivalence classes of square integrable function with respect to almost everywhere equality,

$$L^2(\omega, Y) := \{f : \omega \rightarrow Y : f \text{ is measurable with } \|f\|_{L^2(\omega)} < \infty\} \setminus \{\|\bullet\|_{L^2(\omega)} = 0\},$$

is a Hilbert space with the scalar product

$$(f, g)_{L^2(\omega)} := \int_{\omega} (f, g) \, d\mu \quad \text{for all } f, g \in L^2(\omega).$$

Abbreviate $L^2(\omega) := L^2(\omega; \mathbb{R})$.

Of interest is as well the space of (equivalence classes of) essentially bounded measurable functions $L^\infty(\omega)$ with norm

$$\|f\|_{L^\infty(\omega)} := \|f\|_\infty := \text{ess sup}\{|f(x)| : x \in \omega\}.$$

Given a Lebesgue-measurable set $\omega \subset \mathbb{R}^n$ and a Lebesgue-measurable function $f : \omega \rightarrow Y$, write $\int_{\omega} f \, dx$. The integral over a $(n-1)$ -dimensional hypersurface Γ with respect to the $(n-1)$ -dimensional Hausdorff measure reads $\int_{\Gamma} f \, ds$ and over $(n-2)$ -dimensional γ with respect to the $(n-2)$ -dimensional Hausdorff measure $\int_{\gamma} f \, d\sigma$.

Sobolev spaces

Let $C^\infty(\omega) := \bigcap_{k \in \mathbb{N}_0} C^k(\omega)$ denote the space of smooth functions, i.e., the intersection of the sets of k -times continuously differentiable function $C^k(\omega)$ for all $k \in \mathbb{N}_0$. The subspace of infinitely differentiable test function with compact support reads $C_c^\infty(\omega)$. For any $f \in C_c^\infty(\omega)$ and any multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ of length $|\alpha| := \sum_{j=1}^n \alpha_j$, define the partial derivative

$$D^\alpha f := \frac{\partial^{|\alpha|} f}{\partial x^\alpha} := \frac{\partial^{|\alpha|} f}{\partial x^{\alpha_1} \dots \partial x^{\alpha_n}}$$

with respect to α . For a bounded Lipschitz domain $\omega \subset \mathbb{R}^n$, the Sobolev space $H^k(\omega)$ consists of all square integrable function $f \in L^2(\omega)$ such that for any multi-index $\alpha \in \mathbb{N}_0^n$ with length $|\alpha| \leq k$, there exists some $g \in L^2(\omega)$ with

$$\int_{\omega} f D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_{\omega} g \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\omega).$$

These function are called k -times weakly differentiable with weak derivative $\partial^{|\alpha|} f / \partial x^\alpha := D^\alpha f := g$ with respect to α . In other words,

$$H^k(\omega) := \{f \in L^2(\omega) : D^\alpha f \in L^2(\omega) \text{ exists for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| \leq k\}.$$

For any $f \in H^k(\omega)$ define the Sobolev norm $\|f\|_{H^k(\omega)}$ via

$$\|f\|_{H^k(\omega)}^2 := \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^2(\omega)}^2 < \infty$$

and the Sobolev seminorm $|f|_{H^k(\omega)}$ via

$$|f|_{H^k(\omega)}^2 := \sum_{|\alpha|=k} \|D^\alpha f\|_{L^2(\omega)}^2 < \infty.$$

The closure of $C_c^\infty(\omega)$ with respect to the norm $\|\bullet\|_{H^k(\omega)}$ is denoted by $H_0^k(\omega)$. The finite-dimensional target space $Y \subseteq \mathbb{R}^{\ell \times m}$ maybe identified with $\mathbb{R}^L$ for some $L \in \mathbb{N}$ and analogous definition applied to vector-valued function. In other words, let $C_c^\infty(\omega; Y)$ denote the set of (component-wise) infinitely differentiable functions from ω to $Y \subseteq \mathbb{R}^L$ with compact support in ω and $H^k(\omega, Y)$ the set of Y -valued functions with components in $H^k(\omega)$. For any $f = (f_1, \dots, f_L) \in H^k(\omega; Y)$ define the Sobolev norm $\|f\|_{H^k(\omega)}$ via

$$\|f\|_{H^k(\omega)}^2 := \sum_{|\alpha| \leq k} \sum_{j=1}^L \|D^\alpha f_j\|_{L^2(\omega)}^2 < \infty$$

and the seminorm $|f|_{H^k(\omega)}$ via

$$|f|_{H^k(\omega)}^2 := \sum_{|\alpha|=k} \sum_{j=1}^L \|D^\alpha f_j\|_{L^2(\omega)}^2 < \infty.$$

The closure of $C_c^\infty(\omega; Y)$ with respect to $\|\bullet\|_{H^k(\omega)}$ is denoted by $H_0^k(\omega; Y)$ with $H_0^k(\omega; \mathbb{R}) := H_0^k(\omega)$.

For $k \in \mathbb{N}$ and $0 < s \leq 1$ the Sobolev-Slobodeckij norm $\|f\|_{H^{k+s}(\Omega)}$ is defined via

$$\|f\|_{H^{k+s}(\Omega)}^2 := \|f\|_{H^k(\omega)}^2 + \sum_{|\alpha|=k} \int_{\omega} \int_{\omega} \frac{|D^\alpha f(\xi) - D^\alpha f(\zeta)|^2}{|\xi - \zeta|^{n+2s}} dx(\xi) dx(\zeta)$$

and one way to define the broken Sobolev space

$$H^{k+s}(\omega) := \{f \in L^2(\omega) : \|f\|_{H^{k+s}(\omega)} < \infty\}.$$

Another ansatz is the definition as interpolation space between $H^k(\omega)$ and $H^{k+1}(\omega)$ [Tar07; BS08]. This thesis follows the second ansatz summarized in the following. For any Banach space $(X, \|\bullet\|)$, let $\mathcal{L}(X, X)$ denote the set of continuous and linear operators from X to X with norm $\|M\|_{\mathcal{L}(X, X)} := \sup_{x \in X \setminus \{0\}} \frac{\|Mx\|}{\|x\|} < \infty$. Consider two Banach spaces E_0, E_1 continuously embedded in some topological vector space $\mathcal{E}$. Then E is called an intermediate space, if $E_0 \cap E_1 \subset E \subset E_0 + E_1$ with continuous embeddings. If additionally for any linear function $A : E_0 + E_1 \rightarrow E_0 + E_1$ with $A \in \mathcal{L}(E_0, E_0) \cap \mathcal{L}(E_1, E_1)$ holds $A \in \mathcal{L}(E, E)$, E is called interpolation space between E_0 and E_1 . The interpolation space E is exact of exponent $0 < \theta < 1$, if E is an interpolation space and

$$\|A\|_{\mathcal{L}(E, E)} \leq \|A\|_{\mathcal{L}(E_0, E_0)}^{(1-\theta)} \|A\|_{\mathcal{L}(E_1, E_1)}^\theta.$$

A way to define an exact interpolation space of exponent $0 < \theta < 1$ is the complex method [Tar07, Def. 21.5], [BL76, Sec. 4.1]. Given two Banach spaces $(E_0, \|\bullet\|_0)$, $(E_1, \|\bullet\|_1)$, consider the space $\mathcal{F}(E_0, E_1)$ of functions with values in $E_0 + E_1$, which are bounded and continuous on the closed strip $\bar{S} := \{z \in \mathbb{C} : 0 \leq \operatorname{Re}(z) \leq 1\}$ and analytic on the open strip $S := \{z \in \mathbb{C} : 0 < \operatorname{Re}(z) < 1\}$. Moreover, suppose $f(iy)$ is bounded in E_0 and $f(1+iy)$ is bounded in E_1 for any $y \in \mathbb{R}$. The space $\mathcal{F}(E_0, E_1)$ is equipped with the norm $\|f\| := \max\{\sup_y \|f(iy)\|_0, \sup_y \|f(1+iy)\|_1\}$. For any $0 < \theta < 1$ the complex interpolation space reads

$$E := [E_0, E_1]_\theta := \{f(\theta) \in E_0 + E_1 : f \in \mathcal{F}(E_0, E_1)\}$$

with norm $\|a\|_{[E_0, E_1]_\theta} = \inf_{f(\theta)=a} \|f\|$ for any $a \in E$. In this thesis set

$$H^{k+s}(\omega) := [H^k(\omega), H^{k+1}(\omega)]_s.$$

For any bounded Lipschitz domain $\omega \subset \mathbb{R}^n$ the definitions are equivalent [BS08, Thm. 14.2.3]. The advantage of the interpolation space ansatz lies in the availability of interpolation constants, cf. Example 2.3 below.

For any domain $\omega \subset \mathbb{R}^n$ with Lipschitz boundary $\partial\omega$, this thesis also considers the trace space $H^s(\partial\omega)$ with $s \in \{\pm 1/2\}$. For any $\varphi \in C(\bar{\omega})$, i.e., smooth up to the boundary, the trace is easily defined as $\gamma(\varphi) := \varphi|_{\partial\omega} \in C(\partial\omega)$. The resulting linear operator γ has a surjective extension $\gamma : H^1(\omega) \rightarrow H^{1/2}(\partial\omega)$ [EG15, Thm. 4.6] with bounded L^2 norm, s.t. $\|\gamma(v)\|_{L^2(\partial\omega)} \lesssim \|v\|_{H^1(\omega)}$ for any $v \in H^1(\omega)$. The space $H^{1/2}(\partial\omega) := \gamma(H^1(\omega))$ is equipped with the minimal extension norm

$$\|g\|_{H^{1/2}(\partial\omega)} := \inf\{\|v\|_{H^1(\omega)} : v \in H^1(\omega) \text{ with } \gamma(v) = g\} \quad \text{for any } g \in H^{1/2}(\partial\omega).$$

Let $H^{-1/2}(\partial\omega)$ denote the dual space to $H^{1/2}(\partial\omega)$, such that $\langle \bullet, \bullet \rangle_{\partial\omega}$ denotes the duality brackets in $H^{1/2}(\partial\omega) \times H^{-1/2}(\partial\omega)$ that extend the scalar product in $L^2(\partial\omega)$ associated with the boundary $\partial\omega$.

Note, that the notion of traces allows another view on $H_0^k(\omega)$. In particular, $H_0^1(\omega) = \{f \in H^1(\omega) : f = 0 \text{ almost everywhere along } \partial\omega\}$ and $H_0^2(\omega) = \{f \in H^2(\omega) : f = \partial f / \partial \nu = 0 \text{ almost everywhere along } \partial\omega\}$ with normal derivative $\partial f / \partial \nu$ of f along $\partial\omega$.

Differential operators

For any sufficiently smooth function $f : \Omega \rightarrow \mathbb{R}^\ell$, let $Df = D^1 f$ denote the first (weak) derivative and $D^2 f$ the second. Hence, $Df \in \mathbb{R}^{\ell \times n}$ is the Jacobian matrix for sufficiently smooth $f : \Omega \rightarrow \mathbb{R}^\ell$. For $\ell = 1$, $D^1 := \nabla$ denotes the gradient and D^2 the Hessian. For any sufficiently smooth vector field $f : \Omega \rightarrow \mathbb{R}^n$, the divergence reads $\operatorname{div} f := \sum_{j=1}^n \partial f_j / \partial x_j = \operatorname{tr}(Df)$. For any sufficiently smooth function $f : \Omega \rightarrow \mathbb{R}$ the Laplacian reads $\Delta f := \operatorname{div} \nabla f = \sum_{j=1}^n \partial^2 f / \partial x_j^2$ and the

bi-Laplacian (also called biharmonic operator) reads $\Delta^2 f := \Delta \Delta f$. For vector resp. tensor fields the operators are applied row-wise.

Piecewise version with subscript ,pw' of all mentioned operators are used throughout; piecewise is understood with respect to the non-displayed triangulation $\mathcal{T}$.

2.2 Triangulation and piecewise spaces

Regular triangulation

Let $\mathcal{T}$ denote a shape-regular triangulation of the bounded polyhedral Lipschitz domain $\Omega \subset \mathbb{R}^n$ into n -simplices in the sense of Ciarlet [EG04; Bra07; BS08; BBF13]. Thereby an n -simplex T is the convex hull of $n + 1$ vertices $P_1, P_2, \dots, P_{n+1} \in \mathbb{R}^n$ with linearly independent differences $P_2 - P_1, \dots, P_{n+1} - P_1$. For any n -simplex $T \in \mathcal{T}$ (called simplex throughout) of diameter $h_T := \text{diam}(T)$ and volume $|T|$, let $\mathcal{V}(T)$ denote the set of its $(n + 1)$ vertices (0-subsimplices), let $\mathcal{F}(T)$ denote the set of its $(n + 1)$ sides ($(n - 1)$ -subsimplices), i.e., edges in 2D and faces in 3D, and let $\mathcal{E}(T)$ denote the set of its $n(n + 1)/2$ $(n - 2)$ -subsimplices, i.e., vertices in 2D and edges in 3D.

The triangulation $\mathcal{T}$ covers the domain Ω , in that $\bar{\Omega} = \bigcup_{T \in \mathcal{T}} T$. It is called regular, if the intersection $T_1 \cap T_2$ of any two distinct, non-disjoint simplices T_1 and T_2 in $\mathcal{T}$ is the shared sub-simplex $\text{conv}\{\mathcal{V}(T_1) \cap \mathcal{V}(T_2)\} = \partial T_1 \cap \partial T_2$ of their shared vertices. The shape-regularity of $\mathcal{T}$ means that there exists a constant $C_{\text{sr}} > 0$ such that $|T|^{1/n} \leq h_T \leq C_{\text{sr}} |T|^{1/n}$ holds for any $T \in \mathcal{T}$.

Associated with $\mathcal{T}$ are the following sets, $\mathcal{V} := \bigcup_{T \in \mathcal{T}} \mathcal{V}(T)$ (resp. $\mathcal{V}(\Omega)$ or $\mathcal{V}(\partial\Omega)$) the set of all (resp. interior or boundary) vertices, $\mathcal{F} := \bigcup_{T \in \mathcal{T}} \mathcal{F}(T)$ (resp. $\mathcal{F}(\Omega)$ or $\mathcal{F}(\partial\Omega)$) the set of all (resp. interior or boundary) sides, and $\mathcal{E} := \bigcup_{T \in \mathcal{T}} \mathcal{E}(T)$ (resp. $\mathcal{E}(\Omega)$ or $\mathcal{E}(\partial\Omega)$) the set of all (resp. interior or boundary) $(n - 2)$ -subsimplices. Moreover the following subsets are of interest. Given a vertex $z \in \mathcal{V}$, let $\mathcal{T}(z) := \{T \in \mathcal{T} : z \in \mathcal{V}(T)\}$ denote the set of all simplices with vertex z that cover the nodal patch $\omega_z := \text{int}(\bigcup \mathcal{T}(z))$. Analogously, given a side $F \in \mathcal{F}$, let $\mathcal{T}(F) := \{T \in \mathcal{T} : F \in \mathcal{F}(T)\}$ denote the simplices with side F that cover the side patch $\omega_F := \text{int}(\bigcup \mathcal{T}(F))$. Given a $(n - 2)$ -subsimplex $E \in \mathcal{E}$, let $\mathcal{T}(E) := \{T \in \mathcal{T} : E \in \mathcal{E}(T)\}$ denote the simplices with $(n - 2)$ -subsimplex E that cover the patch $\omega_E := \text{int}(\bigcup \mathcal{T}(E))$. For any simplex $T \in \mathcal{T}$ the set $\mathcal{T}(\Omega(T)) := \{K \in \mathcal{T} : \text{dist}(T, K) = 0\}$ of simplices in $\mathcal{T}$ near T has a bounded cardinality $|\mathcal{T}(\Omega(T))|$ and covers the closure of $\Omega(T) := \text{int}(\bigcup_{T \in \mathcal{T}(\Omega(T))} T) = \text{int}(\bigcup_{z \in \mathcal{V}(T)} \mathcal{T}(z))$.

Given any side $F \in \mathcal{F}$, assign its unit normal ν_F with a fixed orientation, while ν_T denotes the unit outward normal along the simplex boundary ∂T of $T \in \mathcal{T}$. For each boundary side $F \in \mathcal{F}(\partial\Omega) \cap \mathcal{F}(T)$ set $\nu_F := \nu_T|_F$. Given an interior side $F = \partial T_+ \cap \partial T_- \in \mathcal{F}(\Omega)$ shared by the simplices $T_+, T_- \in \mathcal{T}(F)$ with the unit normal ν_F , let T_+ denote the adjoint simplex with $\nu_{T_+}|_F = \nu_F$ and let T_- denote the simplex with $\nu_{T_-}|_F = -\nu_F$. The jump $[v]_F \in L^1(F)$ of a piecewise Lipschitz

continuous function v across $F \in \mathcal{F}$ reads

$$[v]_F(x) := \begin{cases} v|_{T_+}(x) - v|_{T_-}(x) & \text{at } x \in F = \partial T_+ \cap \partial T_- \in \mathcal{F}(\Omega), \\ v(x) & \text{at } x \in F \in \mathcal{F}(\partial\Omega). \end{cases}$$

For any $m \leq n + 1$ and $(m - 1)$ -simplex $K = \text{conv}\{P_1, \dots, P_m\} \subset \mathbb{R}^n$, let $\text{mid}(K) := \frac{1}{m} \sum_{j=1}^m P_j$ denote its centroid, $h_K := \text{diam}(K)$ its diameter, and $|K|$ its $(m - 1)$ -dimensional Lebesgue measure.

Piecewise polynomial spaces

For any $M \in \mathcal{T}$ or $M \in \mathcal{F}$, $k \in \mathbb{N}_0$, $\ell \in \mathbb{N}$, let $P_k(M; \mathbb{R}^\ell)$ denote the set of polynomials of total degree at most k in each component regarded as functions in $L^\infty(M; \mathbb{R}^\ell)$ and set

$$\begin{aligned} P_k(\mathcal{T}; \mathbb{R}^\ell) &:= \left\{ q \in L^\infty(\Omega; \mathbb{R}^\ell) : q|_T \in P_k(T; \mathbb{R}^\ell) \text{ for all } T \in \mathcal{T} \right\}, \\ P_k(\mathcal{F}; \mathbb{R}^\ell) &:= \left\{ q \in L^\infty(\mathcal{F}; \mathbb{R}^\ell) : q|_F \in P_k(F; \mathbb{R}^\ell) \text{ for all } F \in \mathcal{F} \right\}, \\ P_k(\mathcal{F}(\Omega); \mathbb{R}^\ell) &:= \left\{ q \in P_k(\mathcal{F}; \mathbb{R}^\ell) : q|_F = 0 \text{ for all } F \in \mathcal{F}(\partial\Omega) \right\} \end{aligned}$$

(omit $\mathbb{R}^\ell$ whenever $\ell = 1$). The vector space of piecewise polynomials $P_k(\mathcal{T})$ defines the spaces of piecewise polynomials that are globally continuous $S^k(\mathcal{T}) := P_k(\mathcal{T}) \cap C(\Omega) \subset H^1(\Omega)$ and $S_0^k(\mathcal{T}) := S^k(\mathcal{T}) \cap C_0(\Omega) \subset H_0^1(\Omega)$ with homogeneous boundary conditions. The nodal basis functions $\varphi_{P_j} = \varphi_j \in P_1(\mathcal{T})$ with $\varphi_{P_j}(P_k) = \delta_{jk}$ for all $P_j, P_k \in \mathcal{V}$ span $S^1(\mathcal{T})$, their restriction to on simplex $T \in \mathcal{T}$ is called barycentric coordinate.

The piecewise Raviart-Thomas space reads $RT_k^{\text{pw}}(\mathcal{T}) := P_k(\mathcal{T})x + P_k(\mathcal{T}; \mathbb{R}^n)$. The L^2 projection Π_0 onto $P_0(\mathcal{T})$ in $L^2(\Omega)$ is defined, for all $f \in L^2(\Omega)$, by $\Pi_0(f)|_T := \int_T f \, dx$ in $T \in \mathcal{T}$. The notation Π_0 omits the dependence on $\mathcal{T}$, if there is no risk of confusion; Π_0 acts componentwise on scalars and vectors. The piecewise constant function $h_{\mathcal{T}} \in P_0(\mathcal{T})$ takes the value $h_{\mathcal{T}}|_T = h_T := \text{diam}(T)$ in each simplex $T \in \mathcal{T}$ and $h_{\max} := \max_{T \in \mathcal{T}} h_T$ denotes the maximal mesh-size in $\mathcal{T}$. For any $k \in \mathbb{N}_0$ let Π_k denote the L^2 projection onto $P_k(\mathcal{T})$ omitting the dependence on $\mathcal{T}$, if there is no risk of confusion; Π_k acts component-wise on scalars and vectors. The k -th order oscillation of a given $f \in L^2(\Omega)$ with respect to $V := H_0^m(\Omega)$ reads $\text{osc}_k(f) := \|h_{\mathcal{T}}^m(1 - \Pi_k)f\|_{L^2(\Omega)}$. Moreover, consider the L^2 -projections $\Pi_{\mathcal{F},k} : L^2(\mathcal{F}) \rightarrow P_k(\mathcal{F})$, $\Pi_{F,k} : L^2(F) \rightarrow P_k(F)$ for all $F \in \mathcal{F}$, and $\Pi_{RT,k} : L^2(\Omega; \mathbb{R}^n) \rightarrow RT_k^{\text{pw}}(\mathcal{T})$. These projections act component-wise as well.

Let $H^\ell(T)$ abbreviate $H^\ell(\text{int}(T))$ for a compact $T \in \mathcal{T}$ with interior $\text{int}(T)$ and define the vector space $H^\ell(\mathcal{T}) := \{v \in L^2(\Omega) : v|_T \in H^\ell(T)\}$ of piecewise H^ℓ -functions. The same notation applies to norms of scalar- and vector-valued function. This leads to a piecewise energy scalar product a_{pw} , which reads

$$a_{\text{pw}}(v_{\text{pw}}, w_{\text{pw}}) := (D_{\text{pw}}^\ell v_{\text{pw}}, D_{\text{pw}}^\ell w_{\text{pw}})_{L^2(\Omega)} \quad \text{for any } v_{\text{pw}}, w_{\text{pw}} \in H^\ell(\mathcal{T})$$

and induces the semi-norm $\|\bullet\|_{\text{pw}} := a_{\text{pw}}(\bullet, \bullet)^{1/2}$. Moreover abbreviate

$$\langle \bullet, \bullet \rangle_{\partial \mathcal{T}} := \sum_{T \in \mathcal{T}} \langle \bullet, \bullet \rangle_{\partial T}$$

associated with the skeleton $\partial \mathcal{T}$.

The following two properties of the L^2 -projections are useful in the analysis. These properties are classical, see, e.g., [BS08, Lemma 4.3.8], [BBF13, Prop. 2.5.1], [DPE12, §1.4], and are stated without proof.

Lemma 2.1 (Commutation). $\Pi_k(\Pi_{RT,k}(f)) = \Pi_k(f) = \Pi_{RT,k}(\Pi_k(f))$ holds for all $f \in L^2(\Omega; \mathbb{R}^n)$.

Lemma 2.2 (Approximation). *The following holds for all $\ell = 1, \dots, k+1$, all $T \in \mathcal{T}$, and all $v \in H^\ell(T)$,*

$$\|(1 - \Pi_k)v\|_{L^2(T)} + h_T^{1/2} \|(1 - \Pi_k)v\|_{L^2(\partial T)} \leq C_{\text{apx}} h_T^\ell |v|_{H^\ell(T)}, \quad (2.1)$$

and for all $v \in H^\ell(T; \mathbb{R}^n)$,

$$\|(1 - \Pi_{RT,k})v\|_{L^2(T)} + h_T^{1/2} \|(1 - \Pi_{RT,k})v\|_{L^2(\partial T)} \leq C_{\text{apx}} h_T^\ell |v|_{H^\ell(T)}. \quad (2.2)$$

The constant C_{apx} depends on the shape-regularity of $\mathcal{T}$ and on the polynomial degree k , but is independent of the mesh-size h_T .

The following example comparable to the analysis in [BS08, Sec. 14.3] illustrates the approximation results for broken Sobolev spaces.

Example 2.3 (broken spaces). Suppose a function $v \in H^{\ell+s}(T)$ belongs to the broken Sobolev space $H^{\ell+s}(T) := [H^\ell(T); H^{\ell+1}(T)]_s$ for some $0 < s < 1$ and $\ell \in \mathbb{N}$ defined by interpolation. In this setting the ℓ -th weak derivative $D^\ell v \in H^s(T)$ belongs to $H^s(T) := [L^2(T); H^1(T)]_s$. The operator $A := 1 - \Pi_k$ for $k \in \mathbb{N}_0$, has as operator $A : L^2(T) \rightarrow L^2(T)$ the norm $\|A\|_{\mathcal{L}(L^2(T); L^2(T))} = 1$ and as operator $A : H^1(T) \rightarrow L^2(T)$ the norm $\|A\|_{\mathcal{L}(H^1(T); L^2(T))} \leq C_P h_T$ according to the Poincaré inequality from Lemma 2.11 below. Hence, the interpolation estimate [Tar07, Lem. 21.6] implies for $A : H^s(T) \rightarrow L^2(T)$ that $\|A\|_{\mathcal{L}(H^s(T); L^2(T))} \leq (C_P h_T)^s$, in other words,

$$\|(1 - \Pi_k)D^\ell v\|_{L^2(T)} \leq (C_P h_T)^s |v|_{H^{\ell+s}(T)}.$$

The following refined stability estimates for L^2 -projections play an important role in the analysis of the modified HHO method.

Theorem 2.4 (Refined stability estimates). *There exists $C_{st} \geq 1$ such that, for all $T \in \mathcal{T}$ and all $f \in H^1(T)$,*

$$\|\nabla(1 - \Pi_{k+1})(f)\|_{L^2(T)} \leq C_{st} \|(1 - \Pi_k)(\nabla f)\|_{L^2(T)}. \quad (2.3)$$

Moreover there is $\kappa > 0$ such that for all $T \in \mathcal{T}$ and all $f \in H^1(T)$,

$$\|(1 - \Pi_{k+1})(f)\|_{L^2(T)} \leq \kappa h_T \|(1 - \Pi_k)(\nabla f)\|_{L^2(T)}. \quad (2.4)$$

The constants C_{st} and κ depend on the shape-regularity of $\mathcal{T}$ and on the polynomial degree k , but are independent of the diameter h_T .

Remark 2.5 (Constants C_{st} and κ). The stability estimate (2.3) is established in [CZZ20] where it is shown that, for any $T \in \mathcal{T}$, the conditions

$$(H1) \ \Pi_{k+1}(1) = 1 \quad \text{and} \quad (H2) \ P_k(T; \mathbb{R}^n) \cap \nabla H^1(T) \subset \nabla P_{k+1}(T)$$

are equivalent to the existence of an h_T -independent constant $C_{st}(T) > 0$ such that (2.3) holds on $T \in \mathcal{T}$, and one then sets $C_{st} := \max\{C_{st}(T) : T \in \mathcal{T}\}$. The estimate $C_{st} \geq 1$ readily follows from the trivial bound $\|\nabla(1 - \Pi_{k+1})(f)\|_{L^2(T)} \geq \|(1 - \Pi_k)(\nabla f)\|_{L^2(T)}$ since $\nabla \Pi_{k+1}(f) \in P_k(T; \mathbb{R}^n)$. The Poincaré inequality in Lemma 2.11 below and (2.3) lead to (2.4) with $\kappa \leq C_P C_{st}$. For $k = 0$, an upper bound on the constant κ is introduced in Theorem 3.24.

Newest-vertex bisection

The adaptive mesh-refinement traces back to [Mau95; Tra97]. This section in any space dimension $n \geq 2$ is based on the notation in [Ste08; GSS14]. Any n -simplex $T = \text{conv}\{z_0, \dots, z_n\}$ is identified with the $(n+1)$ -tuple $(z_0, \dots, z_n)$ and refined by newest-vertex bisection (NVB), i.e., divided in two sub-simplices $T = T_1 \cup T_2$. For $n \geq 3$ the simplex needs to be tagged, which means that a type $\gamma \in \{0, \dots, n-1\}$ is added. The bisection of a tagged simplex $(z_0, \dots, z_n; \gamma)$ generates two tagged simplices, called children,

$$\begin{aligned} & \left(z_0, \frac{z_0 + z_n}{2}, z_1, \dots, z_\gamma, z_{\gamma+1}, \dots, z_{n-1}; (\gamma+1) \bmod n \right) \\ & \left(z_0, \frac{z_0 + z_n}{2}, z_1, \dots, z_\gamma, z_{n-1}, \dots, z_{\gamma+1}; (\gamma+1) \bmod n \right) \end{aligned}$$

with the convention that $(z_{\gamma+1}, \dots, z_{n-1})$ (resp. $(z_1, \dots, z_\gamma)$) are void for $\gamma = n-1$ (resp. $\gamma = 0$). So the edge $\text{conv}\{z_0, z_n\}$, called refinement edge, is cut into two. Note that for $n = 2$, the refinement edge in each child will be the edge opposite the new vertex $(z_0 + z_n)/2$.

An initial condition is necessary to guarantee that successive refinements of a regular triangulation $\mathcal{T}$ lead to a regular triangulation. Note that the tagged simplices $T = (z_0, \dots, z_n; \gamma)$ and $T_R := (z_n, z_1, \dots, z_\gamma, z_{n-1}, z_{n-2}, \dots, z_{\gamma+1}, z_0; \gamma)$ have the same children. Two tagged simplices T and K that share a common $(n-1)$ -dimensional side are called reflected neighbours, if the ordered sequences of vertices of either T or T_R coincide with that of K in all but one position. A regular triangulation $\mathcal{T}_0$ satisfies the initial condition, if all simplices are of the same type γ and any two neighbouring tagged simplices $T = (z_0, \dots, z_n; \gamma)$ and $K = (y_0, \dots, y_n; \gamma)$ satisfy

- (a) If $\text{conv}\{z_0, z_n\} \subset T \cap K$ or $\text{conv}\{y_0, y_n\} \subset T \cap K$, then T and K are reflected neighbours.
- (b) If $\text{conv}\{z_0, z_n\} \not\subset T \cap K \neq \emptyset$ or $\text{conv}\{y_0, y_n\} \not\subset T \cap K$, then any two neighbouring children of T and K are reflected neighbours.

For $n = 2$ this simplifies to the initial condition that the shared edge of T and K is either the refinement edge $\text{conv}\{z_0, z_2\} = \text{conv}\{y_0, y_2\}$ of both simplices or of none of them.

The initial condition guarantees that uniform refinements of any triangulation $\mathcal{T}_0$ are regular [Ste08, Thm. 4.3].

Admissible triangulations

Throughout this thesis, suppose that the initial triangulation $\mathcal{T}_0$ satisfies the initial condition. An admissible refinement $\mathcal{T}$ is a regular triangulation computed by successive admissible mesh-refinement of $\mathcal{T}_0$ with newest-vertex bisection, i.e., a sequence of newest-vertex bisection.

The set $\mathbb{T} := \mathbb{T}(\mathcal{T}_0)$ of all those triangulations is (uniformly) shape-regular [Tra97, p.128]. The shape regularity of $\mathbb{T}$ guarantees the existence of a constant $C_{\text{sr}} > 0$ such that $|T|^{1/n} \leq h_T \leq C_{\text{sr}}|T|^{1/n}$ holds for any $T \in \mathcal{T} \in \mathbb{T}$ with volume $|T|$. Moreover the shape regularity implies finite overlap of patches and their extensions in that $|\mathcal{T}(z)| \leq M_{\text{np}} < \infty$ for any $\mathcal{T} \in \mathbb{T}$ and $z \in \mathcal{V}$ and $M_{\text{ep}} := \sup_{T \in \mathcal{T} \in \mathbb{T}} |\mathcal{T}(\Omega(T))| \leq (n+1)M_{\text{np}} < \infty$.

For any $\mathcal{T} \in \mathbb{T}$, let $\mathbb{T}(\mathcal{T})$ abbreviate the set of all admissible refinements of $\mathcal{T}$. Note that for any simplex $T \in \hat{\mathcal{T}}$ in a refinement $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ of $\mathcal{T}$ there exists a simplex $K \in \mathcal{T}$ with $T \subseteq K$. For any $K \in \mathcal{T}$ set $\hat{\mathcal{T}}(K) := \{T \in \hat{\mathcal{T}} : T \subseteq K\}$. Let $\mathcal{T} \cap \hat{\mathcal{T}}$ denote the set of simplices that belong to $\mathcal{T}$ and $\hat{\mathcal{T}}$, i.e., are not refined (in $\hat{\mathcal{T}}$). Furthermore $\mathcal{T} \setminus \hat{\mathcal{T}}$ are the simplices in the coarse triangulation $\mathcal{T}$ that are refined (in $\hat{\mathcal{T}}$) and $\hat{\mathcal{T}} \setminus \mathcal{T}$ are the simplices in the fine triangulation $\hat{\mathcal{T}}$ that have been refined.

For any $N \in \mathbb{N}$ let $\mathbb{T}(N) := \{\mathcal{T} \in \mathbb{T} : |\mathcal{T}| - |\mathcal{T}_0| \leq N\}$ denote the set of admissible triangulation with at most N simplices more than $\mathcal{T}_0$.

For $0 < \varepsilon < 1$ let $\mathbb{T}(\varepsilon) := \{\mathcal{T} \in \mathbb{T} : h_{\text{max}} := \max_{T \in \mathcal{T}} \text{diam}(T) \leq \varepsilon\}$ denote the set of admissible triangulation with mesh-size $h_{\text{max}} \leq \varepsilon$. Note that the maximal mesh-size $h_{\text{max}} := \max_{T \in \mathcal{T}} h_T$ for any $T \in \mathcal{T}$ is bounded by $h_0 := \max_{T \in \mathcal{T}_0} h_T \leq \text{diam}(\Omega)$.

2.3 General eigenvalue problem

This section is based on [Bof10; BO91; SF73]. Given that the model problem in mind is the m -Laplacian (for $m = 1, 2$), the analysis is restricted to a compact and selfadjoint operator.

Introduction to spectral theory

Let H denote a Hilbert space with inner product $(\bullet, \bullet)_H$ and induced norm $\|\bullet\|_H$ and let $T \in \mathcal{L}(H; H)$ denote a bounded, linear, compact, and selfadjoint operator. Let $\rho(T) := \{z \in \mathbb{C} : (z - T)^{-1} \text{ is a bounded operator on } H\}$ denote the resolvent set of T and $\sigma(T) := \mathbb{C} \setminus \rho(T)$ the spectrum. In this setting $\sigma(T)$ is a countable or finite set of real numbers with only possible cluster point at zero. All positive elements of $\sigma(T)$ are eigenvalues with finite multiplicity.

The operator norm of T satisfies $\|T\| := \sup_{f \in H} \|Tf\|_H / \|f\|_H = \mu_{\max} < \infty$. Enumerate the eigenvalues $\mu \in \sigma(T) \setminus \{0\}$, such that $\mu_j \rightarrow 0$ for $j \rightarrow \infty$. The finite multiplicity implies that there exists $k \in \mathbb{N}$ such that $\mu_{k+1} < \mu_k$.

Let $C \subset \rho(T) \subset \mathbb{C}$ denote a circle such that there are no spectral values on C and C encloses $\mu \in \sigma(T)$ and no other element of $\sigma(T)$. Then the Riesz spectral projection associated with μ is defined via

$$E := E(\mu) := \frac{1}{2\pi i} \int_C (z - T)^{-1} dz.$$

Since $C \cap \sigma(T) = \emptyset$, the inverse $(z - T)^{-1}$ exists on C and $\|(z - T)^{-1}\|$ depends continuously on z , $(z - T)^{-1}$ is integrable. $E(\mu)$ is the projection onto the space of eigenfunctions associated with μ [Kat95]. In general, if C encloses more eigenvalues $\mu_1, \dots, \mu_k$, the image of $E(\mu_1, \dots, \mu_k)$ is $\bigoplus_{j=1}^k \ker(\mu_j - T)$.

On the other hand, define a discrete operator $T_h : H \rightarrow H$ (this allows different choices), which will be compact and continuous operator due to the regularity.

It remains to check that T_h is selfadjoint below. The first step in the convergence analysis is to prove that $T_h \rightarrow T$ in the strong norm for $h \rightarrow 0$ without rates, i.e., show that $T_h \in \mathcal{L}(H; H)$ is a member of a sequence of compact, selfadjoint operators that converge to T in operator norm,

$$\lim_{h \rightarrow 0} \|T - T_h\|_{\mathcal{L}(H; H)} = \lim_{h \rightarrow 0} \sup_{\substack{f \in H \\ \|f\|_H = 1}} \|Tf - T_h f\|_H = 0.$$

There exist $h_0 > 0$ (depending on C) such that for all $h \leq h_0$, there are no discrete eigenvalues of T_h on C (or even in an ε -environment of C), in other words $E_h := \frac{1}{2\pi i} \int_C (z - T_h)^{-1} dz$ is well defined. For discrete operators T_h on finite dimensional spaces the h corresponds to the mesh-size and this global smallness condition on the mesh-size is to be expected, such that enough discrete eigenvalues are computable etc. Given T_h compact and selfadjoint and E_h defined, the same argument as above shows $\text{im}(E_h) = \bigcup_{j=1}^k \ker(\mu_{h,j} - T_h)$. Since $T_h \rightarrow T$ in strong norm it follows $E_h \rightarrow E$ in strong norm. For the finite dimensional images this convergence leads to convergence of the images. In other words the discrete eigenvalues converge to the continuous eigenvalues and so do the eigenspaces. (In case of doubt another contour is possible.)

The second step concerns the multiplicity of the discrete eigenvalues. Since the number of discrete eigenvalues in C coincides with the number of continuous

eigenvalues counting multiplicities, there are no spurious modes inside C . There maybe spurious modes outside of C , but not beneath the first k eigenvalues. In other words, if $\mu_1 \leq \dots \leq \mu_k$ and $\mu_{1,h} \leq \dots \leq \mu_{k,h}$ are enumerated in ascending order counting multiplicities than holds $\mu_{h,j} \rightarrow \mu_j$ for $h \rightarrow 0$ and $1 \leq j \leq k$. The same holds for the reciprocal eigenvalues of the variationally posed eigenvalue problem below. Hence, choose k such that $\lambda_k < \lambda_{k+1}$ and h small enough (e.g. such that $k+1$ discrete eigenvalues are computable, then $\lambda_{h,j} \rightarrow \lambda_j$ for $h_{\max} \rightarrow 0$ and $1 \leq j \leq k$.

In this setting, with a bounded linear, compact, and selfadjoint operator $T \in \mathcal{L}(H; H)$ and a sequence of compact, selfadjoint operators $T_h \in \mathcal{L}(H; H)$ that converge to T in operator norm for $h \rightarrow 0$, the Babuška-Osborn theory [BO91] provides a priori convergence rates for the eigenvalues and eigenfunctions. Let $\mu \in \sigma(T) \setminus \{0\}$ be a non-zero eigenvalue of T with eigenspace $E_\mu = \ker(\mu - T)$ of dimension $J = \dim(E_\mu) \in \mathbb{N}$.

Theorem 2.6 (Convergence). *Suppose for any eigenvalue $\mu \in \sigma(T) \setminus \{0\}$ of multiplicity J , there exists J eigenvalues $\mu_{h,1}, \dots, \mu_{h,J}$ of T_h , that converge to μ as $h \rightarrow 0$. Let $E_\mu = \ker(\mu - T)$ denote the eigenspace of μ and let $E_h = \text{span}\{u_{h,j} \in E_h(\mu_{h,j}) := \ker(\mu_{h,j} - T_h) : j = 1, \dots, J\}$ abbreviate the span of the discrete eigenspaces $E_h(\mu_{h,j})$.*

Then there exist $h_0 > 0$ and constants $C_a, C_b \geq 0$ (that may depend on μ but are independent of h), such that for any $h \leq h_0$

$$\max_{1 \leq j \leq J} |\mu - \mu_{h,j}| \leq C_a \left(\sup_{\phi, \psi \in E_\mu \setminus \{0\}} \frac{|((T - T_h)\phi, \psi)_H|}{\|\phi\|_H \|\psi\|_H} + \|(T - T_h)|_{E_\mu}\|_{\mathcal{L}(E_\mu; H)}^2 \right)$$

and

$$\max_{\substack{u_h \in E_h \\ \|u_h\|_H=1}} \min_{\substack{u \in E_\mu \\ \|u\|_H=1}} \|u - u_h\|_H + \max_{\substack{u \in E_\mu \\ \|u\|_H=1}} \min_{\substack{u_h \in E_h \\ \|u_h\|_H=1}} \|u - u_h\|_H \leq C_b \|(T - T_h)|_{E_\mu}\|_{\mathcal{L}(E_\mu; H)}.$$

Proof. These are Theorems 7.2 and 7.4 in [BO91] for a selfadjoint operator T , see also [Bof10, Sec. 9] or [SZ17, Sec. 1.4.2]. $\square$

Variationally posed eigenvalue problems

This thesis considers in general variationally posed eigenvalue problems. Suppose $V \subset H$ a real Hilbert space continuously and dense embedded in the real Hilbert space H . Let $a : V \times V \rightarrow \mathbb{R}$ and $b : H \times H \rightarrow \mathbb{R}$ denote symmetric, continuous bilinear forms and seek $(\lambda, u) \in \mathbb{R}_+ \times V \setminus \{0\}$ such that

$$a(u, v) = \lambda b(u, v) \quad \text{for all } v \in V. \quad (2.5)$$

Assume that a is V -elliptic in that, there exists $\alpha > 0$ with $\|v\|_V^2 \leq \alpha a(v, v)$ for all $v \in V$, and that b is a scalar product in H . Let $T : H \rightarrow H$ denote the solution

operator, i.e., for any $f \in H$ let $Tf \in V$ denote the unique solution with

$$a(Tf, v) = b(f, v) \quad \text{for all } v \in V.$$

Suppose that T is compact and self-adjoint. In this setting, all positive elements in the spectrum $\sigma(T)$ of T are eigenvalues with finite multiplicity and their reciprocals are the eigenvalues in (2.5) with coinciding eigenspaces. The eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$ of (2.5) are enumerated in ascending order counting multiplicities and there exists some b -orthonormal (a -orthogonal) basis $(u_1, u_2, u_3, \dots)$ of corresponding eigenfunctions.

For any conforming discretization of (2.5) with a finite dimensional subspace $V_h \subset V$, that seeks $(\lambda_h, u_h) \in \mathbb{R}_+ \times V_h \setminus \{0\}$ with

$$a_h(u_h, v_h) = \lambda_h b_h(u_h, v_h) \quad \text{for all } v_h \in V_h, \quad (2.6)$$

the discrete solution operator $T_h : H \rightarrow H$ that maps $f \in H$ to the uniquely defined $T_h f$ with

$$a(T_h f, v_h) = b(f, v_h) \quad \text{for all } v_h \in V_h$$

is compact and the eigenpairs are in the same correspondence, i.e., the eigenvalues of T_h are reciprocals of λ_h and the eigenfunctions coincide.

The following min-max or Rayleigh-Ritz principle directly implies that conforming discretization lead to guaranteed upper eigenvalue bounds.

Lemma 2.7 (min-max-principle). *For any $j \in \mathbb{N}$, the j -th continuous eigenvalue λ_j of (2.5) satisfies*

$$\lambda_j = \min_{\substack{V_j \subset V \\ \dim V_j = j}} \max_{v \in V_j \setminus \{0\}} \frac{a(v, v)}{b(v, v)}.$$

Similarly, holds for any $j \in \mathbb{N}$ and the j -th discrete eigenvalue $\lambda_h(j)$ of (2.6) that

$$\lambda_h(j) = \min_{\substack{V_j \subset V_h \\ \dim V_j = j}} \max_{v_h \in V_j \setminus \{0\}} \frac{a_h(v_h, v_h)}{b_h(v_h, v_h)}.$$

Proof. The min-max principle is well-known, e.g., [WS72], [SF08], [Bof10, Prop. 7.2], or [SZ17, Sec. 1.4.3]. $\square$

The model eigenvalue problem

In this thesis, the Hilbert spaces of interest are $H := L^2(\Omega)$ and $V := H_0^m(\Omega)$. The energy scalar product $a : V \times V \rightarrow \mathbb{R}$ and the L^2 scalar product $b : H \times H \rightarrow \mathbb{R}$ are defined, for any $v, w \in V$ and for any $\tilde{v}, \tilde{w} \in H$, by

$$a(v, w) := (D^m v, D^m w)_{L^2(\Omega)} \quad \text{and} \quad b(\tilde{v}, \tilde{w}) := (\tilde{v}, \tilde{w})_{L^2(\Omega)}$$

and induce the energy norm $\|\bullet\| := a(\bullet, \bullet)^{1/2}$ in V and the L^2 norm $\|\bullet\|_{L^2(\Omega)} := b(\bullet, \bullet)^{1/2}$. Hence, V is in particular compactly embedded in H [Alt16, Thm. 10.9]. In other words the solution operator $T = \Delta^{-m}$ is compact and selfadjoint and Theorem 2.6 and Lemma 2.7 apply.

2.4 Inequalities

This subsection lists some well known inequalities, frequently used throughout this thesis.

Lemma 2.8 (Young's inequality, e.g., [Eva10, B.2.d]). *For any $a, b \geq 0$ and any $\varepsilon > 0$ holds*

$$ab \leq \frac{1}{2\varepsilon}a^2 + \frac{\varepsilon}{2}b^2.$$

Lemma 2.9 (Cauchy-Schwarz inequality, e.g., [Alt16, Lem. 2.2.2]). *Let $(X, \langle \bullet, \bullet \rangle)$ denote an inner product with induced norm $\|\bullet\| := \sqrt{\langle \bullet, \bullet \rangle}$, then any $x, y \in X$ satisfy*

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

In particular, any $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in \mathbb{R}^n$ satisfy $|x \cdot y| = |\sum_{j=1}^n x_j y_j| \leq |x| |y|$ with Euclidean norm $|\bullet|$ and any $f, g \in L^2(\Omega)$ with $\Omega \subset \mathbb{R}^n$ satisfy $|(f, g)_{L^2(\Omega)}| \leq \|f\|_{L^2(\Omega)} \|g\|_{L^2(\Omega)}$.

For any m -simplex T with $m \in \mathbb{N}$ the following trace identity and inequality hold.

Lemma 2.10 (trace identity and inequality). *Let $m \geq 2$. Suppose in the m -simplex $T = \text{conv}\{P, F\}$ the vertex $P \in \mathcal{V}(T)$ lies opposite to the side, $(m-1)$ -subsimplex, $F \in \mathcal{F}(T)$. Then any $v \in H^1(T)$ satisfies*

$$(a) \quad \int_F v \, ds = \int_T v \, dx + \frac{1}{m} \int_T (x - P) \cdot \nabla v \, dx,$$

$$(b) \quad \|v\|_{L^2(F)}^2 \leq \frac{|F|}{|T|} \|v\|_{L^2(T)} \left(\|v\|_{L^2(T)} + \frac{2h_T}{m} \|\nabla v\|_{L^2(T)} \right).$$

Proof. The m -simplex $T \subseteq \Omega$ is identified with $T \subset \mathbb{R}^m$. The trace identity (a) is a direct consequence from an integration by parts as in [CGR12, Lemma 2.1] for $m = 2$ and a special case of [CH18a, Lemma 2.6] for $m \geq 3$. Set $g(x) := (x - P)v(x)$ for all $x \in T$. Observe that $\text{div } g(x) = mv(x) + (x - P) \cdot \nabla v(x)$ and that along the boundary ∂T with unit outer normal field ν_T holds $(x - P) \cdot \nu_T = \text{dist}(P, \langle F \rangle) = m|T|/|F|$ for any $x \in F$ and $(x - P) \cdot \nu_T = 0$ for any $x \in G \in \mathcal{F}(T) \setminus \{F\}$. Hence, an integration by parts proves

$$\int_T \text{div } g \, dx = \int_T (mv + (x - P) \cdot \nabla v) \, dx = \int_{\partial T} g \cdot \nu_T \, ds = \frac{m|T|}{|F|} \int_F v \, ds.$$

This concludes the proof of (a). Let $v^2 \in H^1(T)$ in (a). Since $\nabla v^2 = 2v\nabla v$, the Cauchy-Schwarz inequality and $\max_{x \in T} |x - P| \leq h_T$ conclude the proof of the trace inequality (b). For reference see, e.g., [CF99; CB18] and [DPE12, Eq. (1.42)]. $\square$

Lemma 2.11 (Poincaré inequality). *Given any convex domain $\Omega \subset \mathbb{R}^n$ with diameter $\text{diam}(\Omega)$ and any function $f \in H^1(\Omega)$, it holds*

$$\left\| f - \int_{\Omega} f \, dx \right\|_{L^2(\Omega)} \leq C_P \text{diam}(\Omega) |f|_{H^1(\Omega)}$$

with optimal Payne-Weinberger constant $C_P := 1/\pi$ [PW60; Beb03]. In particular, for any simplex $T \subset \mathbb{R}^n$ and $f \in H^1(T)$ holds

$$\|(1 - \Pi_0)f\|_{L^2(T)} \leq C_P h_T \|\nabla f\|_{L^2(T)}$$

with optimal constant $C_P := 1/\pi$ [PW60; Beb03] for $n \geq 3$, $C_P := 1/j_{11}$ for $n = 2$ and the first positive root $j_{11} = 3.8317059702$ of the Bessel function of the first kind [LS10], and $C_P := 1/(\sqrt{2}\pi)$ for a right isosceles triangle [LS10, p.132].

Lemma 2.12 (Friedrichs inequality, e.g., [Bra07, Thm. 1.7]). *Given a bounded domain $\Omega \subset \mathbb{R}^n$ and any function $f \in H_0^m(\Omega)$, there exists a constant $C_F > 0$ that satisfies*

$$\|f\|_{L^2(\Omega)} \leq C_F |f|_{H^m(\Omega)}.$$

Lemma 2.13 (inverse estimate, e.g., [EG04, Sec. 1.7], [BS08, Sec. 4.5]). *Let $\Omega \subset \mathbb{R}^n$ denote a bounded Lipschitz domain and $\mathbb{T}$ a family of uniform shape-regular triangulation of Ω , suppose that $V(\mathcal{T})$ for $\mathcal{T} \in \mathbb{T}$ denotes a finite element space with $V(\mathcal{T})|_K \subset H^\ell(K) \cap H^m(K)$ for $0 \leq m \leq \ell$ and any $K \in \mathcal{T}$. Then there exists a constant $C_{\text{inv}} > 0$ such that for any triangulation $\mathcal{T} \in \mathbb{T}$, any $K \in \mathcal{T}$ of diameter $h_K := \text{diam}(K)$, and any $v \in V(\mathcal{T})$ holds*

$$|v|_{H^\ell(K)} \leq C_{\text{inv}} h_K^{m-\ell} |v|_{H^m(K)}.$$

There also exists a constant $c_{\text{inv}} > 0$ such that for any $\mathcal{T} \in \mathbb{T}$ and side $F \in \mathcal{F}$ of diameter $h_F := \text{diam}(F)$, and any $v \in V(\mathcal{T})$ holds

$$|v|_{H^\ell(F)} \leq c_{\text{inv}} h_F^{m-\ell} |v|_{H^m(F)}.$$

3 Discrete spaces

Throughout this thesis the nonconforming Crouzeix-Raviart and Morley finite element spaces and their respective interpolation and conforming companion operators are featured. This chapter summarizes all the required properties, which are of interest beyond this thesis, e.g., [CN21]. For the Crouzeix-Raviart space in Section 3.3 the interpolation and the conforming companion operator are known from the literature, e.g., [CGS15; CH18a; CZZ20], and necessary extension to any space dimension $n \geq 2$ are added. The Morley finite element space was extended to any space dimension in [MX06]. Section 3.4 analyses this finite element and the associated interpolation operator in any space dimension. The design of the conforming companion can follow the 2D discussions in [Gal15b; VZ19] in the spirit of [CGS15]: one subtle issue is the scaling of the nodal basis functions for the C^1 -conforming FEM as a generalization of [Cia78, Thm. 6.1.3] for the 2D Hsieh-Clough-Tocher (HCT) finite element utilized in [Gal15b; VZ19]. For any space dimension $n \geq 3$ the Clough-Tocher finite element has been proposed in [WF87]. More than two decades later, [Sor09] observed additional constraints on the choice of the subtriangulation of the macro element for $n \geq 4$ necessary for C^1 conformity, i.e., [WF87] is wrong for $n \geq 4$. It appears still an open problem, whether all conditions for $n = 4$ can be satisfied simultaneously [Sor09, p.42, 1.40ff]. Section 3.2 contains an elementary self-contained introduction to the Worsey-Farin (WF) FEM and the analysis of the associated nodal basis functions to generalize [Cia78, Thm. 6.1.3] for $n = 3$.

Throughout this chapter the following notation is used. Given $m \in \{1, \dots, n+1\}$ points $P_1, P_2, \dots, P_m \in \mathbb{R}^n$ with linearly independent differences $P_2 - P_1, \dots, P_m - P_1$, their convex hull $T = \text{conv}\{P_1, \dots, P_m\}$ is an $(m-1)$ -simplex (of positive $(m-1)$ -dimensional Hausdorff measure). The barycentric coordinates $(\lambda_1, \dots, \lambda_m)$ of a point $x \in \langle T \rangle := P_1 + \text{span}\{P_2 - P_1, \dots, P_m - P_1\}$ in the linear subspace solve the $(n+1) \times m$ linear system of equations

$$\begin{pmatrix} 1 & 1 & \dots & 1 \\ P_1 & P_2 & \dots & P_m \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} = \begin{pmatrix} 1 \\ x \end{pmatrix}. \quad (3.1)$$

It holds $x \in T$ if and only if $x \in \mathbb{R}^n$ satisfies (3.1) with $\lambda_1, \dots, \lambda_m \geq 0$. The relative interior of T contains all $x \in \mathbb{R}^n$ that satisfy (3.1) with $\lambda_1, \dots, \lambda_m > 0$. Hence, $T = \overline{\text{relint}(T)}$ and the relative boundary $\text{relbdy}(T) = \partial T := T \setminus \text{relint}(T)$ of T are all $x = \sum_{\mu=1}^m \lambda_\mu P_\mu \in \mathbb{R}^n$ with $\sum_{\mu=1}^m \lambda_\mu = 1$ such that $\lambda_1, \dots, \lambda_m \geq 0$ and $\lambda_j = 0$ for at least one $1 \leq j \leq m$.

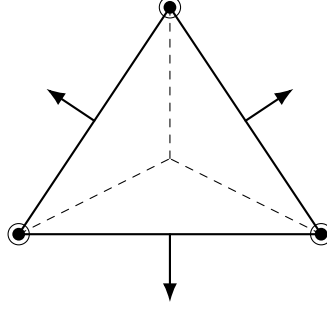


Figure 3.1: Mnemonic diagram of the Hsieh-Clough-Tocher finite element in 2D including the subtriangulation $HCT2D(K)$ with dashed lines

$L_j(f) = f(P_j)$	for $j = 1, 2, 3$
$L_{3\ell+j}(f) = \nabla f(P_j) \cdot e_\ell$	for $j = 1, 2, 3, \ell = 1, 2$
$L_{9+j}(f) = \nabla f(\text{mid}(F_j)) \cdot \nu_T _{F_j}$	for $j = 1, 2, 3$

Table 3.1: The 12 degrees of freedom $L_1, \dots, L_{12}$ of $HCT(T)$ for any $f \in C^1(T)$

The integration formula for barycentric coordinates on triangles can be extended to any space dimension $n \geq 2$ and $(m-1)$ -simplex $T \subset \mathbb{R}^n$ [EM73; VS18]. For all $\alpha_1, \dots, \alpha_m \in \mathbb{N}_0$ holds

$$\int_T \lambda_1^{\alpha_1} \dots \lambda_m^{\alpha_m} dx = \frac{|T|(\alpha_1)! \dots (\alpha_m)!(m-1)!}{(\alpha_1 + \dots + \alpha_m + (m-1))!}. \quad (3.2)$$

3.1 Hsieh-Clough-Tocher

In the two-dimensional setting, a conforming finite-dimensional subspace of $H_0^2(\Omega)$ is the Hsieh-Clough-Tocher (HCT) finite element [Cia78, Chap. 6]. This section defines the HCT space and summarizes some elementary properties.

The HCT finite element is piecewise cubic with respect to the following subtriangulation of a regular triangulation $\mathcal{T}$ of $\Omega \subset \mathbb{R}^2$. For any triangle $T \in \mathcal{T}$, let $HCT2D(T) := \{T_F : F \in \mathcal{F}(T)\}$ denote the triangulation of T into three sub-triangles $T_F := \text{conv}\{\text{mid}(T), F\}$ with edges $F \in \mathcal{F}(T)$ opposite the common vertex $\text{mid}(T)$. Define

$$HCT(\mathcal{T}) := \{v \in H_0^2(\Omega) : v|_T \in P_3(HCT2D(T)) \text{ for all } T \in \mathcal{T}\}. \quad (3.3)$$

The local degrees of freedom for $\psi \in HCT(\mathcal{T})$ on $T \in \mathcal{T}$ are the nodal values of the function $\psi(z)$, of the derivative $\nabla \psi(z)$ for $z \in \mathcal{V}(T) := \{P_1, P_2, P_3\}$ and the values of the normal derivatives $\partial \psi / \partial \nu_F(\text{mid}(F))$ at the midpoints of the edges $F \in \mathcal{F}(T) := \{F_1, F_2, F_3\}$, see Table 3.1.

Remark 3.1 (Properties of $\text{HCT}(\mathcal{T})$). (a) The Hsieh-Clough-Tocher element, i.e., $(T, P_3(\text{HCT}2D(T)) \cap C^1(T), (L_1, \dots, L_{12}))$, defines a finite element in the sense of Ciarlet according to [Cia78, Thm. 6.1.2].

(b) The notion of an almost affine family of finite element in [Cia78, Thm. 6.1.3, p.344] concerns the scaling of the basis functions dual to the degrees of freedom above. Suppose the degree of freedom L_ℓ for $\ell = 1, \dots, M := 4|\mathcal{V}(\Omega)| + |\mathcal{F}(\Omega)|$ is a directional derivative of order $\mu_\ell \in \{0, 1\}$ at the node $z_\ell \in \mathcal{N}(\Omega) := \mathcal{V}(\Omega) \cup \{\text{mid}(F) : F \in \mathcal{F}(\Omega)\}$ with nodal patch ω_ℓ of volume $h_\ell^2 := |\omega_\ell|$. Then the dual basis function ψ_ℓ with $L_j(\psi_\ell) = \delta_{j\ell}$ for all $j, \ell \in \{1, \dots, M\}$ satisfies

$$\|\psi_\ell\|_{H^s(\Omega)} \lesssim h_\ell^{1+\mu_\ell-s} \quad \text{for all } s = 0, 1, 2.$$

In particular, if a degree of freedom L_ℓ is associated with the derivative $\nabla\psi(z)$ in a vertex $z \in \mathcal{V}(K)$ for some $K \in \mathcal{T}$, then $\|\psi_\ell\|_{L^2(K)} \lesssim h_K^2$, cf. [Gal15b, Prop. 2.5] for the application of this scaling.

(c) The computation of the (complete and reduced) HCT basis functions is included in [BH81], while [Mey12] comments on a simplified calculation of reduced HCT basis functions and the implementation.

3.2 Worsey-Farin

This section aims at an elementary self-contained introduction to the Worsey-Farin (WF) FEM, C^1 -conforming functions piecewise cubic with respect to a particular subtriangulation, and the analysis of the associated nodal basis functions to generalize [Cia78, Thm. 6.1.3].

Subsection 3.2.1 contains an introduction to Bernstein polynomials [Boo87] in multivariate C^1 splines. This is used to discuss the smoothness of piecewise polynomials across an interface in a triangulation in Subsection 3.2.2. Unlike the HCT partition of each triangle in three subtriangles, the subdivision in the 3D WF finite element scheme [WF87; Sor09] in Definition 3.17 depends on the triangulation $\mathcal{T} \in \mathbb{T}$. Each tetrahedron $T \in \mathcal{T}$ is divided into 12 sub-tetrahedra with respect to a careful selection of center points c_F on each facet $F \in \mathcal{F}(T)$ of T and c_T inside the tetrahedron $T \in \mathcal{T}$: c_T is the midpoint of the incircle of T and c_F is the intersection of $F \in \mathcal{F}(\Omega)$ with the straight line through c_{T_+} and c_{T_-} for $T_\pm \in \mathcal{T}$ aligned to $F = \partial T_+ \cap \partial T_-$, while $c_F := \text{mid}(F)$ is simply the center of gravity for a triangle $F \in \mathcal{F}(\partial\Omega)$ on the boundary. Subsection 3.2.3 analyses the local subtriangulation and piecewise quadratic polynomials in this triangulation, while Theorem 3.18 in Subsection 3.2.6 guarantees the (uniform) shape-regularity of the resulting subtriangulation $\hat{\mathcal{T}}$ for $\mathcal{T} \in \mathbb{T}$. The 28 local degrees of freedom of WF for any $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\} \in \mathcal{T}$ are the evaluation of the function $f \in H^2(K)$ and its gradient ∇f at the vertices $Q_1, \dots, Q_4$ of

K and the evaluation of the gradient $\tau_E \times \nabla f(\text{mid}(E))$ at each edge midpoint $\text{mid}(E)$ for $E \in \mathcal{E}(T)$ with unit tangent vector τ_E . This determines $\nabla f(\text{mid}(E))$ in the non-tangential directions $\nu_{E,1}, \nu_{E,2}$ with $\text{span}\{\tau_E, \nu_{E,1}, \nu_{E,2}\} = \mathbb{R}^3$. Those 28 degrees of freedom define a finite element $(K, P_3(\hat{\mathcal{T}}(K)) \cap C^1(K), (L_1, \dots, L_{28}))$ in the sense of Ciarlet. Subsection 3.2.4 provides in Theorem 3.14 the explicit proof of this, since it is not included in [WF87]. That the facet center point c_F of an interior facet $F = \partial T_+ \cap \partial T_- \in \mathcal{F}(\Omega)$ shared by the two neighbouring tetrahedra $T_\pm \in \mathcal{T}$ belongs to the same straight line as their center points c_{T_+} and c_{T_-} implies C^1 conformity of $\text{WF}(\mathcal{T}) := P_3(\hat{\mathcal{T}}) \cap H_0^2(\Omega)$ in Theorem 3.16 in Subsection 3.2.5. Finally, Subsection 3.2.7 provides in Theorem 3.20 a scaling result analogue to [Cia78, Thm. 6.1.3] for WF and allows the design of the conforming companion for the Morley finite element in 3D in Section 3.4.

In other words, Subsection 3.2.1-3.2.3 prepare the proof that WF is a finite element in Subsection 3.2.4 and the global C^1 conformity proof in Subsection 3.2.5. In combination with the shape-regular global subtriangulation in Subsection 3.2.6 this allows for the scaling result Theorem 3.20 comparable to Remark 3.1.b which is necessary to analyse the conforming companion of the Morley finite element in 3D. The following analysis extends the supplement of [CP21].

3.2.1 Bernstein polynomials in a simplex

Given $m \in \{1, 2, 3, 4\}$ points $P_1, P_2, \dots, P_m \in \mathbb{R}^3$ with linearly independent differences $P_2 - P_1, \dots, P_m - P_1$, their convex hull $T = \text{conv}\{P_1, \dots, P_m\}$ is an $(m-1)$ -simplex (of positive $(m-1)$ -dimensional Hausdorff measure). The set of vertices $\mathcal{V}(T) = \{P_1, \dots, P_m\}$ of T is the set of extremal points and so uniquely defined. An ordering is fixed to identify the $(m-1)$ -simplex T with an ordered list $(P_1, \dots, P_m)$ of vertices. To keep track of the vertex set in a finite union of simplices below, $(P_1, \dots, P_m)$ will be replaced by $(P_{\sigma(1)}, \dots, P_{\sigma(m)})$ for global numbers $\sigma(1), \dots, \sigma(m)$ in a finite list of all vertices $\mathcal{V}$ in a regular triangulation $\mathcal{T}$ of $\Omega \subset \mathbb{R}^3$ into tetrahedra in the sense of Ciarlet. Let $\langle T \rangle := P_1 + \text{span}\{P_2 - P_1, \dots, P_m - P_1\}$ denote the unique $(m-1)$ -dimensional affine subspace $\langle T \rangle$ of $\mathbb{R}^3$ that contains $T \equiv (P_1, \dots, P_m)$. The barycentric coordinates $(\lambda_1, \dots, \lambda_m)$ of a point $x \in \langle T \rangle$ solve the $4 \times m$ linear system of equations (3.1). The dependence on x in $\lambda_j(x) = \lambda_j$ from (3.1) is typically not written out explicitly for brevity.

The space $P_k(T)$ of algebraic polynomials of total degree at most $k \in \mathbb{N}_0$ is seen as a subspace of $C^\infty(T)$ and allows many representations. The Bernstein polynomials form a natural basis of $P_k(T)$ with standard multi-index notation for $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}_0^m$ with length $|\alpha| = \alpha_1 + \dots + \alpha_m$ and

$$\frac{\lambda^\alpha}{\alpha!} := \frac{\lambda_1^{\alpha_1} \lambda_2^{\alpha_2} \dots \lambda_m^{\alpha_m}}{\alpha_1! \alpha_2! \dots \alpha_m!} \quad \text{at} \quad x = \sum_{\mu=1}^m \lambda_\mu P_\mu \in T. \quad (3.4)$$

(The convention is that $\lambda^\alpha/\alpha! \equiv 0$ if at least one of the indices $\alpha_1, \dots, \alpha_m$ of α is negative.) If the length $m \in \{1, 2, 3, 4\}$ is fixed, abbreviate $A_k := \{\alpha \in \mathbb{N}_0^m \mid |\alpha| =$

$k\} \subset \mathbb{N}_0^m$. The Bernstein polynomials of degree $k \in \mathbb{N}_0$ are all $\lambda^\alpha/\alpha!$ from (3.4) for $\alpha \in A_k$. In fact, any (real) polynomial $f \in P_k(T)$ has unique (real) coefficients $(c(\alpha)|\alpha \in A_k)$ called *ordinates* such that

$$f = k! \sum_{\alpha \in A_k} c(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at} \quad x = \sum_{\mu=1}^m \lambda_\mu P_\mu \in T. \quad (3.5)$$

The definition (3.5) makes sense also in the entire affine subspace $\langle T \rangle$ and immediately extends any $f \in P_k(T)$ to $f \in P_k(\langle T \rangle)$.

Remark 3.2 (gradient ∇f). The derivative of f from (3.5) at $x = \sum_{\mu=1}^m \lambda_\mu P_\mu \in T$ reads

$$\nabla f = k! \sum_{\alpha \in A_k} \sum_{\mu=1}^m c(\alpha) \frac{\lambda^{(\alpha - e_\mu)}}{(\alpha - e_\mu)!} \nabla \lambda_\mu = k! \sum_{\beta \in A_{k-1}} \frac{\lambda^\beta}{\beta!} \sum_{\mu=1}^m c(\beta + e_\mu) \nabla \lambda_\mu. \quad (3.6)$$

If the length of m is fixed (and there is no need to highlight the length m of the multi-indices from the context), then $e_j := (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{N}_0^m$ denotes the j -th canonical unit vector with the coefficients $e_j(\ell) = \delta_{j\ell}$ for $j, \ell \in \{1, \dots, m\}$.

It cannot be overemphasized that the ordinates $(c(\alpha)|\alpha \in A_k)$ of the polynomial f in (3.5) are neither coefficients of a monomic tensor basis $x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$ nor the values of the polynomial in general. However, note that $\lambda_j(x) = 0$ for some $x \in \partial T$ implies $\lambda^\alpha/\alpha! = 0$ vanishes for all $\alpha \in A_k$ with $\alpha_j \neq 0$. Hence, (3.5) for $x \in T$ with $\lambda_j(x) = 0$ reduces to $f(x) = k! \sum_{\substack{\alpha \in A_k \\ \alpha_j=0}} c(\alpha) \lambda^\alpha/\alpha!$. In particular for

any vertex P_j with barycentric coordinates $(\lambda_1, \dots, \lambda_m) = e_j$ of P_j (3.5) shows $c(ke_j) = f(P_j)$.

Some examples on the evaluation of f and ∇f on the boundary ∂T of a triangle ($m = 3$) and tetrahedron ($m = 4$) conclude this subsection for the sake of an illustration as well as future reference. The details are required in the analysis of Algorithm 1 later in Subsection 3.2.4.

Example 3.3 (tetrahedron, $k = 2, m = 4$). For $m = 4$ consider a tetrahedron T identified with its vertices (P_1, P_2, P_3, P_4) in $\mathbb{R}^3$ with edge midpoints $P_{k\ell} = (P_k + P_\ell)/2$ for $k, \ell = 1, \dots, 4$ and $k \neq \ell$. The ordinates $(c(\alpha)|\alpha \in A_2)$ define the quadratic polynomial $f = 2 \sum_{\alpha \in A_2} c(\alpha) \lambda^\alpha/\alpha! \in P_2(T)$, which assumes, for all $j, k, \ell = 1, \dots, 4$ and $k \neq \ell$, the following values

$$(a) \quad f(P_j) = c(2e_j),$$

$$(b) \quad 4f(P_{k\ell}) = c(2e_k) + 2c(e_k + e_\ell) + c(2e_\ell) = f(P_k) + 2c(e_k + e_\ell) + f(P_\ell).$$

Proof of (a). The first formula follows with $(\lambda_1, \dots, \lambda_4) = e_j$ at $x = P_j$ and the evaluation of the non-zero $\lambda^\alpha/\alpha! = 1^2/2! = 1/2$ for $\alpha = 2e_j$. $\square$

Proof of (b). At the edge midpoint $x = P_{k\ell}$ holds $(\lambda_1, \dots, \lambda_4) = (e_k + e_\ell)/2$ and $\lambda^\alpha/\alpha!$ is non-zero only for $\alpha = 2e_k, e_k + e_\ell$, and $2e_\ell$. The polynomial $\lambda^\alpha/\alpha!$ assumes the respective values $\lambda^\alpha/\alpha! = 1/8, 1/4, 1/8$. This leads to $f(P_{k\ell})/2 = c(2e_k)/8 + c(e_k + e_\ell)/4 + c(2e_\ell)/8$, i.e., with (a) to (b). $\square$

Example 3.4 (tetrahedron, $k = 3, m = 4$). In the notation of Example 3.4, the ordinates $(c(\alpha)|\alpha \in A_3)$ define the cubic polynomial $f = 6 \sum_{\alpha \in A_3} c(\alpha) \lambda^\alpha/\alpha! \in P_3(T)$ and its gradient $\nabla f \in P_2(T; \mathbb{R}^3)$. For all $\{j, k, \ell, m\} = \{1, 2, 3, 4\}$ it follows

- (a) $f(P_j) = c(3e_j)$,
- (b) $8f(P_{k\ell}) = f(P_k) + 3c(2e_k + e_\ell) + 3c(e_k + 2e_\ell) + f(P_\ell)$,
- (c) $(P_j - P_k) \cdot \nabla f(P_j) = 3(c(3e_j) - c(2e_j + e_k))$,
- (d) $\frac{4}{3}(P_j - P_{k\ell}) \cdot \nabla f(P_{k\ell}) = c(2e_k + e_j) + 2c(e_k + e_\ell + e_j) + c(2e_\ell + e_j) - 4f(P_{k\ell})$,
- (e) $\frac{8}{3}(P_k - P_{k\ell}) \cdot \nabla f(P_{k\ell}) = f(P_k) + c(2e_k + e_\ell) - c(e_k + 2e_\ell) - f(P_\ell)$.

Proof of (a)–(b). The formulas (a)–(b) follow with $(\lambda_1, \dots, \lambda_4) = e_j$ at $x = P_j$ and the only non-zero $\lambda^\alpha/\alpha! = 1/3!$ at $\alpha = 3e_j$ and $(\lambda_1, \dots, \lambda_4) = (e_k + e_\ell)/2$ at the midpoint $x = P_{k\ell}$ with possible $\alpha = 3e_k, 2e_k + e_\ell, e_k + 2e_\ell, 3e_\ell$ for respective non-zero values $\lambda^\alpha/\alpha! = 1/48, 1/16, 1/16, 1/48$. $\square$

Proof of (c). The polynomial $\lambda^\beta/\beta!$ vanishes at $x = P_j$ for all $\beta \in A_2 \setminus \{2e_j\}$, while $\lambda^{2e_j}/(2e_j)! = 1/2$ at $x = P_j$. Consequently,

$$\nabla f(P_j) = 3 \sum_{\mu=1}^4 c(2e_j + e_\mu) \nabla \lambda_\mu. \quad (3.7)$$

The barycentric coordinates satisfy $(P_j - P_k) \cdot \nabla \lambda_\mu = \lambda_\mu(P_j) - \lambda_\mu(P_k) = \delta_{j\mu} - \delta_{k\mu}$ and (c) follows from (3.7). $\square$

Proof of (d). At the midpoint $x = P_{k\ell}$, Example 3.3 shows $\lambda^\beta/\beta! = 1/8, 1/4, 1/8$ for $\beta = 2e_k, e_k + e_\ell, 2e_\ell \in A_2$ and $\lambda^\beta/\beta! = 0$ vanishes otherwise. Hence,

$$\nabla f(P_{k\ell}) = \frac{3}{4} \sum_{\mu=1}^4 \left(c(2e_\ell + e_\mu) + 2c(e_\ell + e_k + e_\mu) + c(2e_k + e_\mu) \right) \nabla \lambda_\mu. \quad (3.8)$$

Since $(P_j - P_{k\ell}) \cdot \nabla \lambda_\mu = \delta_{j\mu} - \frac{1}{2}(\delta_{k\mu} + \delta_{\ell\mu})$ is one for $\mu = j$, $-1/2$ for $\mu = k, \ell$, and vanishes otherwise, (d) follows from (3.8) and (b). $\square$

Proof of (e). Since $(P_k - P_{k\ell}) \cdot \nabla \lambda_\mu = \delta_{k\mu} - \frac{1}{2}(\delta_{k\mu} + \delta_{\ell\mu})$ is $1/2$ for $\mu = k$, $-1/2$ for $\mu = \ell$, and vanishes otherwise, (3.8) and (a) lead to (e). $\square$

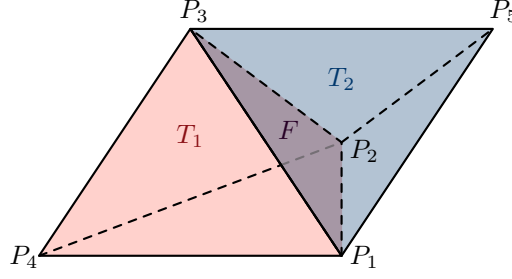


Figure 3.2: Illustration of two neighbouring tetrahedra $T_1 = \text{conv}\{P_1, P_2, P_3, P_4\}$ in red and $T_2 = \text{conv}\{P_1, P_2, P_3, P_5\}$ in blue with common face $F = \text{conv}\{P_1, P_2, P_3\}$ in violet

3.2.2 Smoothness across an interface

This subsection characterizes the function space $C^1(T_1 \cup T_2) \cap P_k(\{T_1, T_2\})$ for two tetrahedra T_1, T_2 with a common face $F = \partial T_1 \cap \partial T_2$. Fig. 3.2 illustrates the following situation. Suppose $T_1 = \text{conv}\{P_1, P_2, P_3, P_4\}$ and $T_2 = \text{conv}\{P_1, P_2, P_3, P_5\}$ share the face $F = \partial T_1 \cap \partial T_2 = \text{conv}\{P_1, P_2, P_3\}$ with unit normal ν_F . Recall $A_k := \{\alpha \in \mathbb{N}_0^4 \mid |\alpha| = k\} \subset \mathbb{N}_0^4$ and set $B_k := \{\alpha \in \mathbb{N}_0^3 \mid |\alpha| = k\} \subset \mathbb{N}_0^3$. In this context the notation $c_j(\beta; 0)$ below abbreviates $c_j(\beta; 0) := c_j((\beta_1, \beta_2, \beta_3, 0))$ for some $\beta \in B_k$ or $\beta \in B_{k-1}$ and $j = 1, 2$. Identify T_1 with (P_1, P_2, P_3, P_4) and T_2 with (P_1, P_2, P_3, P_5) . The function $g \in P_k(\{T_1, T_2\})$ is given in T_1 resp. T_2 as in (3.5) with ordinates $(c_1(\alpha) \mid \alpha \in A_k)$ resp. $(c_2(\alpha) \mid \alpha \in A_k)$,

$$g_j = g|_{T_j} = k! \sum_{\alpha \in A_k} c_j(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at} \quad x = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 + \lambda_4 P_\ell \in T_j \quad (3.9)$$

for $j = 1, 2$ with $\ell = \ell(1) = 4$ for $j = 1$ and $\ell = \ell(2) = 5$ for $j = 2$.

Lemma 3.5. *The polynomials $g_j \in P_k(T_j)$ with (3.9) for $j = 1, 2$ form a function $g \in P_k(\{T_1, T_2\})$ by $g|_{T_j} = g_j$ for $j = 1, 2$. Then*

(a) $g \in C^0(T_1 \cup T_2)$ if and only if for all $\beta \in B_k$

$$c_1(\beta; 0) = c_2(\beta; 0); \quad (3.10)$$

(b) $g \in C^1(T_1 \cup T_2)$ if and only if (3.10) and for all $\beta \in B_{k-1}$

$$\sum_{\mu=1}^4 c_1((\beta; 0) + e_\mu)(\nu_F \cdot \nabla \lambda_\mu|_{T_1}) = \sum_{\mu=1}^4 c_2((\beta; 0) + e_\mu)(\nu_F \cdot \nabla \lambda_\mu|_{T_2}). \quad (3.11)$$

Proof of (a). Recall that the common face $F = \partial T_1 \cap \partial T_2$ satisfies $F \subset \{\lambda_4 = 0\}$. Hence, (3.9) implies

$$g_j|_F = k! \sum_{\beta \in B_k} c_j(\beta; 0) \frac{\lambda^\beta}{\beta!} \quad \text{at} \quad x = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 \in F \quad \text{and} \quad j = 1, 2$$

with the abbreviation $c_j(\beta; 0) := c_j((\beta_1, \beta_2, \beta_3, 0))$. Therefore, $g_1|_F = g_2|_F$ for $g \in C^0(T_1 \cup T_2)$ is equivalent to

$$0 = \sum_{\beta \in B_k} (c_1(\beta; 0) - c_2(\beta; 0)) \lambda^\beta / \beta! \quad \text{at } x = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 \in F.$$

Since the Bernstein polynomials $(\lambda^\beta / \beta! | \beta \in B_k)$ form a basis of $P_k(F)$ the last statement is equivalent to (3.10). This proves (a). $\square$

Proof of (b). Since $g \in C^1(T_1 \cup T_2)$ includes $g \in C^0(T_1 \cup T_2)$, (3.10) has to hold. Additionally the continuity of the gradient at F is necessary. The gradient $\nabla g_j|_F$ from (3.6) reads at $x = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 \in F$

$$\nabla g_j|_F = k! \sum_{\beta \in B_{k-1}} \frac{\lambda^\beta}{\beta!} \Big|_F \sum_{\mu=1}^4 c_j(\beta + e_\mu) \nabla \lambda_\mu|_{T_j} = k! \sum_{\beta \in B_{k-1}} \frac{\lambda^\beta}{\beta!} \sum_{\mu=1}^4 c_j((\beta; 0) + e_\mu) \nabla \lambda_\mu|_{T_j}.$$

Hence, the continuity of the gradients $\nabla g_1|_F = \nabla g_2|_F$ is equivalent to

$$0 = \sum_{\beta \in B_{k-1}} \frac{\lambda^\beta}{\beta!} \sum_{\mu=1}^4 (c_1((\beta; 0) + e_\mu) \nabla \lambda_\mu|_{T_1} - c_2((\beta; 0) + e_\mu) \nabla \lambda_\mu|_{T_2})$$

at $x = \sum_{\mu=1}^3 \lambda_\mu P_\mu \in F$. Since the Bernstein polynomials $(\lambda^\beta / \beta! | \beta \in B_{k-1})$ form a basis of $P_{k-1}(F)$ the last statement is equivalent to

$$0 = \sum_{\mu=1}^4 (c_1((\beta; 0) + e_\mu) \nabla \lambda_\mu|_{T_1} - c_2((\beta; 0) + e_\mu) \nabla \lambda_\mu|_{T_2}) \in \mathbb{R}^3 \text{ for all } \beta \in B_{k-1}. \quad (3.12)$$

It remains to show, that (3.12) is equivalent to (3.11). In other words, to verify that solely the normal derivatives $\nu_F \cdot \nabla \lambda_\mu|_{T_j}$ for $j = 1, 2$ and $\mu = 1, 2, 3$ are of interest. Fix two tangential directions τ_1, τ_2 of unit length parallel to $\langle F \rangle$, such that τ_1, τ_2, ν_F form an orthonormal basis of $\mathbb{R}^3$. Then

$$\nabla \lambda_\mu|_{T_j} = (\nabla \lambda_\mu|_{T_j} \cdot \tau_1) \tau_1 + (\nabla \lambda_\mu|_{T_j} \cdot \tau_2) \tau_2 + (\nabla \lambda_\mu|_{T_j} \cdot \nu_F) \nu_F.$$

The components of the difference $\nabla \lambda_\mu|_{T_1} - \nabla \lambda_\mu|_{T_2}$ for $\mu = 1, 2, 3$ in $\mathbb{R}^3$ in the direction $\tau_1, \tau_2 \in \text{span}\{P_2 - P_1, P_3 - P_1\}$ parallel to $\langle F \rangle$ vanish owing to the Hadamard jump condition: The piecewise gradient ∇g of $g \in C^0(T_1 \cup T_2) \cap P_k(\{T_1, T_2\})$ jumps across the interface F and the jump $[\nabla g]_F \parallel \nu_F$ exclusively points in the normal direction. For $\mu = 1, 2, 3$ and the tangential direction $\tau_j \in \text{span}\{P_2 - P_1, P_3 - P_1\}$ for $j = 1, 2$, holds

$$\tau_j \cdot \nabla \lambda_\mu|_{T_1} = \frac{\partial}{\partial s} \lambda_\mu(x_0 + s\tau) \Big|_{s=0} = \tau_j \cdot \nabla \lambda_\mu|_{T_2} =: \tau_j \cdot \nabla \lambda_\mu \text{ at any } x_0 \in \text{relint}(F).$$

For $\mu = 4$, notice that $P_r - P_1 \perp \nu_F \parallel \nabla \lambda_4|_{T_j}$ for $r = 2, 3$ and $j = 1, 2$. In fact, suppose $\varrho_\ell > 0$ denotes the height of P_ℓ over the plane $\langle F \rangle$, then $\varrho_\ell \nabla \lambda_4|_{T_j} =$

$(-1)^j \nu_F$ for $\ell = 4, 5$, $j = 1, 2$ provided the orientation of ν_F is fixed from P_4 to P_5 in that ν_F is the outer unit normal in $F \subset \partial T_1$. In particular this means $\tau_j \cdot \nabla \lambda_4|_{T_1} = 0 = \tau_j \cdot \nabla \lambda_4|_{T_2}$ for $j = 1, 2$. Hence, (3.12) is recast for all $\beta \in B_{k-1}$ as the first identity in

$$\begin{aligned} & -\nu_F \sum_{\mu=1}^4 \left(c_1((\beta; 0) + e_\mu)(\nabla \lambda_\mu|_{T_1} \cdot \nu_F) - c_2((\beta; 0) + e_\mu)(\nabla \lambda_\mu|_{T_2} \cdot \nu_F) \right) \\ & = \sum_{j=1}^2 \tau_j \sum_{\mu=1}^3 \left(c_1((\beta; 0) + e_\mu) - c_2((\beta; 0) + e_\mu) \right) \nabla \lambda_\mu \cdot \tau_j = 0 \end{aligned}$$

with the equality of ordinates from (3.10) in the last step. The last identity is equivalent to (3.11). $\square$

The coefficient relations (3.10)–(3.11) concern the factors one in (3.10) or geometric factors $\nu_F \cdot \nabla \lambda|_{T_j}$ for $j = 1, 2$ in (3.11). However they do *not* explicitly display k and this gives rise to a factorization (cf. [Boo87, §14] in terms of B-forms). Suppose $(c_j(\alpha)|\alpha \in A_k)$ defines g_j in (3.9) for $F = \partial T_1 \cap \partial T_2 = \text{conv}\{P_1, P_2, P_3\}$ and $j = 1, 2$. Define for a fixed index $\xi \in \{1, 2, 3\}$ suppressed in the notation

$$g'|_{T_j} := g'_j := (k-1)! \sum_{\beta \in A_{k-1}} c_j(e_\xi + \beta) \lambda^\beta / \beta! \quad \text{for } j = 1, 2. \quad (3.13)$$

Then the composition $g' \in P_{k-1}(\{T_1, T_2\})$ from (3.13) is a piecewise polynomial of degree at most $k-1$.

Example 3.6 (g' for $k = 0, \dots, 3$). Abbreviate $\sum_{\mu \leq \ell} := \sum_{\substack{\mu, \ell=1 \\ \mu \leq \ell}}^4$.

- (a) If $k = 0$ and $g_j = c_j((0, 0, 0, 0))$, then $g'_j = 0$ by definition.
- (b) If $k = 1$ and $g_j = \sum_{\mu=1}^4 c_j(e_\mu) \lambda^{e_\mu}$, then $g'_j = c_j(e_\xi)$.
- (c) If $k = 2$ and $g_j = 2 \sum_{\mu \leq \ell} c_j(e_\mu + e_\ell) \frac{\lambda^{e_\mu + e_\ell}}{(e_\mu + e_\ell)!}$, then $g'_j = \sum_{\mu=1}^4 c_j(e_\xi + e_\mu) \lambda_\mu$; e.g., $g' = c_j((2, 0, 0, 0)) \lambda_1 + c_j((1, 1, 0, 0)) \lambda_2 + c_j((1, 0, 1, 0)) \lambda_3 + c_j((1, 0, 0, 1)) \lambda_4$ holds for $\xi = 1$.
- (d) If $k = 3$ and $g_j = 6 \sum_{\alpha \in A_3} c_j(\alpha) \frac{\lambda^\alpha}{\alpha!}$, then $g'_j = 2 \sum_{\mu \leq \ell} c_j(e_\xi + e_\mu + e_\ell) \frac{\lambda^{e_\mu + e_\ell}}{(e_\mu + e_\ell)!}$; e.g., for $\xi = 1$ holds

$$\begin{aligned} g'_j = & 2\lambda_1 \left(c_j((2, 1, 0, 0)) \lambda_2 + c_j((2, 0, 1, 0)) \lambda_3 + c_j((2, 0, 0, 1)) \lambda_4 \right) \\ & + 2 \left(c_j(1, 1, 1, 0) \lambda_2 \lambda_3 + c_j(1, 1, 0, 1) \lambda_2 \lambda_4 + c_j(1, 0, 1, 1) \lambda_3 \lambda_4 \right) \\ & + c_j((3, 0, 0, 0)) \lambda_1^2 + c_j((1, 2, 0, 0)) \lambda_2^2 + c_j((1, 0, 2, 0)) \lambda_3^2 + c_j((1, 0, 0, 2)) \lambda_4^2. \end{aligned}$$

Recall that the two tetrahedra T_1 and T_2 in Fig. 3.2 share the face $F = \partial T_1 \cap \partial T_2 = \text{conv}\{P_1, P_2, P_3\}$ with fixed unit normal ν_F . Suppose $g|_{T_j} := g_j$ in (3.9) for $j = 1, 2$ and define g' in (3.13) with respect to a fixed index $\xi \in \{1, 2, 3\}$. Let ℓ, m denote the two distinct indices with $\{\xi, \ell, m\} = \{1, 2, 3\}$ and $E := \text{conv}\{P_\ell, P_m\} \in \mathcal{E}(F)$ the edge opposite of P_ξ in F .

Lemma 3.7. *Under the present notation $g \in C^1(T_1 \cup T_2)$ is equivalent to the three conditions*

$$g' \in C^1(T_1 \cup T_2) \text{ as well as } g \text{ and } \frac{\partial g}{\partial \nu_F} \text{ are continuous along } E.$$

Proof. Without loss of generality fix $\xi = 1$ and $E = \text{conv}\{P_2, P_3\}$ throughout this proof. Lemma 3.5 characterizes $g \in C^1(T_1 \cup T_2)$ in (3.10)–(3.11); but it also applies to $g' \in P_{k-1}(\{T_1, T_2\})$ and shows that $g' \in C^1(T_1 \cup T_2)$ is equivalent to (3.14) for all $\beta \in B_{k-1}$ and (3.15) for all $\beta \in B_{k-2}$,

$$c_1(e_1 + (\beta; 0)) = c_2(e_1 + (\beta; 0)) \quad (3.14)$$

$$\sum_{\mu=1}^4 c_1(e_1 + (\beta; 0) + e_\mu) \nu_F \cdot \nabla \lambda_\mu|_{T_1} = \sum_{\mu=1}^4 c_2(e_1 + (\beta; 0) + e_\mu) \nu_F \cdot \nabla \lambda_\mu|_{T_2} \quad (3.15)$$

Observe that the conditions (3.14)–(3.15) concern $e_1 + (\beta; 0)$ for $\beta \in B_{k-1}$ and $\beta \in B_{k-2}$ and are included in (3.10)–(3.11). In other words (3.10)–(3.11) imply (3.14)–(3.15).

„ $\Rightarrow$ “ The first implication assumes $g \in C^1(T_1 \cup T_2)$ and Lemma 3.5 guarantees (3.10)–(3.11) and so (3.14)–(3.15). Recall that (3.14)–(3.15) imply $g' \in C^1(T_1 \cup T_2)$. Finally, $g \in C^1(T_1 \cup T_2)$ and so $\nabla g \in C^0(T_1 \cup T_2; \mathbb{R}^3)$ imply that g and $\frac{\partial g}{\partial \nu_F}$ are continuous along $E = \text{conv}\{P_2, P_3\}$.

„ $\Leftarrow$ “ The converse implication that assumes $g' \in C^1(T_1 \cup T_2)$, whence (3.14)–(3.15), and that g and $\frac{\partial g}{\partial \nu_F}$ are continuous along $E \subset \{\lambda_1 = 0 = \lambda_4\}$. First (3.14) implies some conditions of (3.10) and the remaining conditions read

$$c_1(0, \beta_2, \beta_3, 0) = c_2(0, \beta_2, \beta_3, 0) \text{ for all } \beta_2, \beta_3 \in \mathbb{N}_0 \text{ with } \beta_2 + \beta_3 = k,$$

in other words, concern $\beta \in B_k$ with $\beta_1 = 0$. On the other hand, the continuity of g along E implies $g_1|_E = g_2|_E$. In terms of (3.9), this reads

$$0 = g_1|_E - g_2|_E = k! \sum_{\substack{\beta \in B_k \\ \beta_1 = 0}} \left(c_1((\beta; 0)) - c_2((\beta; 0)) \right) \frac{\lambda^\beta}{\beta!} \quad (3.16)$$

at $x = \lambda_2 P_2 + \lambda_3 P_3 \in E$. Since the Bernstein polynomials $(\lambda^\beta / \beta! | \beta \in B_k, \beta_1 = 0)$ form a basis of $P_k(E)$, (3.16) implies the remaining condition in (3.10). Consequently, (3.14) and $g|_E \in C^0(E)$ prove $g \in C^0(T_1 \cup T_2)$. Second, (3.15) implies

the conditions of (3.11) except those for $\beta \in B_{k-1}$ with $\beta_1 = 0$. Recall that (3.9) and Remark 3.2 show

$$\nu_F \cdot \nabla g_j|_E = k! \sum_{\substack{\beta \in B_{k-1} \\ \beta_1=0}} \frac{\lambda^\beta}{\beta!} \sum_{\mu=1}^4 c_j((\beta; 0) + e_\mu)(\nu_F \cdot \nabla \lambda_\mu|_{T_j}) \text{ at } x = \lambda_2 P_2 + \lambda_3 P_3 \in E$$

for $j = 1, 2$. Deduce that $\frac{\partial(g_1 - g_2)}{\partial \nu_F} = \nu_F \cdot \nabla(g_1 - g_2) = 0$ in E implies $c_1((\beta; 0)) = c_2((\beta; 0))$ for all $\beta \in B_{k-1}$ with $\beta_1 = 0$. Consequently, (3.14)–(3.15) and $g|_E$, $\frac{\partial g}{\partial \nu_F}|_E \in C^0(E)$ imply (3.10)–(3.11). $\square$

3.2.3 Piecewise quadratic polynomials in a WF partition of a tetrahedron

Given a tetrahedron $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\} \subset \mathbb{R}^3$ of a positive volume, select five center points c_K, c_1, c_2, c_3, c_4 in K as follows

- (a) $c_K \in \text{int}(K)$, i.e., $\text{dist}(c_K, \partial K) > 0$,
- (b) $c_m \in \text{relint}(F_m)$, i.e., $c_m = \sum_{\mu=1}^4 \lambda_\mu Q_\mu$ for $\lambda_1 + \dots + \lambda_4 = 1$ and $\lambda_m = 0$ and all other $\lambda_j > 0$, for the face F_m of K that is opposite to the vertex Q_m for $m = 1, 2, 3, 4$.

Definition 3.8 (WF partition in 3D). Given c_K, c_1, c_2, c_3, c_4 with (a)–(b) in the tetrahedron $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\}$, define the regular triangulation $WF3D(K) := \hat{\mathcal{T}}$ to be the set of 12 subtetrahedra obtained as the convex hull $\text{conv}\{Q_j, Q_k, c_K, c_m\}$ of two distinct vertices Q_j and Q_k , $1 \leq j < k \leq 4$, the center point c_K of the tetrahedron, and the center $c_m \in \text{relint}(F_m)$ of the face F_m of K opposite to the vertex Q_m for $m \in \{1, 2, 3, 4\} \setminus \{j, k\}$.

Each face $F_m \in \mathcal{F}(K)$ is a triangle partitioned in $WF3D(K)$ by connecting the center $c_m \in \text{relint}(F_m)$ with its vertices $\mathcal{V}(F_m)$ as in the $HCT2D$ partition of the triangle F_m . In fact there are two steps in the partition illustrated in Fig. 3.3. First, the four tetrahedra $T(m) := \text{conv}\{c_K, F_m\}$ for $m = 1, 2, 3, 4$ partition K as displayed in Fig. 3.3.a. Second, $\hat{\mathcal{T}}(c_m) := \{T \in \hat{\mathcal{T}}(K) \mid c_m \text{ vertex of } T\}$ partitions each of the tetrahedra $T(1), T(2), T(3), T(4)$ in three subtetrahedra; Fig. 3.3.b illustrates the partition of $T(1)$. The following Lemma 3.9 adopts the notation of a WF triangulation $\hat{\mathcal{T}} = WF3D(K)$ and considers each $T(m) = \text{conv}\{c_K, F_m\}$ for $m \in \{1, \dots, 4\}$ with its decomposition $\hat{\mathcal{T}}(T(m)) = \{T \in \hat{\mathcal{T}} \mid T \subset T(m)\} =: \hat{\mathcal{T}}(c_m)$ into three subtetrahedra.

Lemma 3.9. *Any $f \in P_k(\hat{\mathcal{T}}(c_m))$ for $k = 0, 1, 2$ and $m = 1, 2, 3, 4$ satisfies $f \in C^1(T(m))$ if and only if $f \in P_k(T(m))$ is a global polynomial in $T(m)$.*

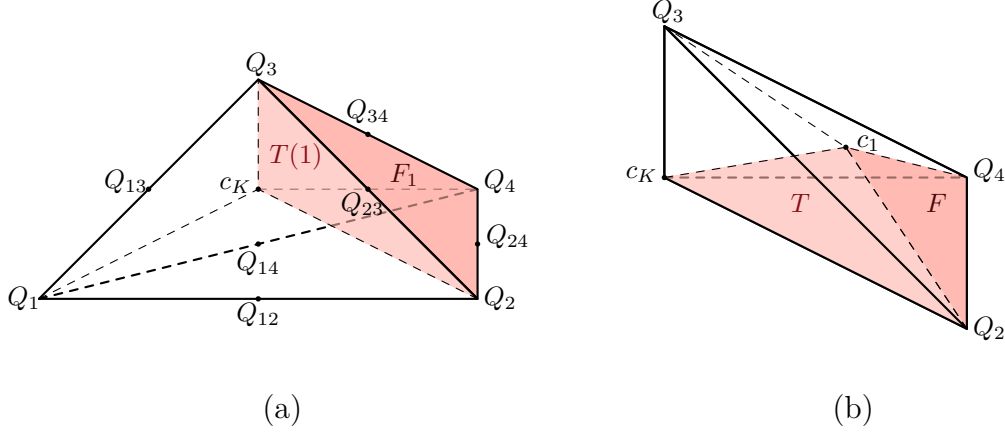


Figure 3.3: Illustration of the division of $K = \text{conv}\{Q_1, \dots, Q_4\}$ with center $c_K \in \text{int}(K)$ into $T(1), \dots, T(4)$ in (a) with $T(1) = \text{conv}\{c_K, F_1\}$ marked in red. Further subdivision of $T(1)$ with center $c_1 \in \text{int}(F_1)$ of face F_1 in the WF partition in (b) with one subsimplex $T = \text{conv}\{c_k, F\} \in \text{WF3D}(K)$ with face $F = \text{conv}\{c_1, Q_2, Q_4\} \subset F_1$ marked in red

Proof. Without loss of generality let $m = 1$ and $T(1) = \text{conv}\{c_K, Q_2, Q_3, Q_4\}$ as displayed in Fig. 3.3.b. The assertion is trivial for $k = 0$ and immediate for $k = 1$, because continuity means $\nabla f(c_1) = \nabla f|_T(c_1)$ for all $T \in \hat{\mathcal{T}}(c_1)$ and leads to a global constant vector ∇f . Suppose $k = 2$ and $f \in C^1(T(1)) \cap P_2(\hat{\mathcal{T}}(c_1))$. The function values $f(z)$ at the vertices $z \in \{c_K, Q_2, Q_3, Q_4\}$ of $T(1)$ and at the edge midpoints $z \in \{Q_{jk} := (Q_j + Q_k)/2 : 2 \leq j < k \leq 4\} \cup \{(c_K + Q_j)/2 : 2 \leq j \leq 4\}$ of $T(1)$ define the quadratic Lagrange interpolation $I_2 f \in P_2(T(1))$ in the tetrahedron $T(1)$. Subtract this (global) quadratic Lagrange interpolation $I_2 f \in P_2(T(1))$ from f . The difference $(f - I_2 f)(z) = 0$ vanishes for each vertex and each edge midpoint of $T(1)$. Since the assertion $f \in P_2(T(1))$ is equivalent to $f - I_2 f \equiv 0$, assume without loss of generality that f vanishes at these ten distinct points on the boundary $\partial T(1)$ and conclude $f \equiv 0$ below.

Identify any subtetrahedron $T = \text{conv}\{c_1, c_K, Q_j, Q_k\} \in \hat{\mathcal{T}}(c_1)$ for distinct $2 \leq j < k \leq 4$ with (c_1, c_K, Q_j, Q_k) . The Bernstein basis of $f|_T \in P_2(T)$ leads to ordinates $(c_T(\alpha)|\alpha \in A_2)$ in

$$f|_T = 2 \sum_{\alpha \in A_2} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} \text{ at } x = \lambda_1 c_1 + \lambda_2 c_K + \lambda_3 Q_j + \lambda_4 Q_k \in T \in \hat{\mathcal{T}}(c_1).$$

Since $f|_T$ is a quadratic function in each $T \in \hat{\mathcal{T}}(c_1)$, it is in particular a quadratic function on each face $F_{jk} := \text{conv}\{c_K, Q_j, Q_k\}$ for $2 \leq j < k \leq 4$ opposite to c_1 in T . The quadratic function $f|_{F_{jk}} \in P_2(F_{jk})$ vanishes at the vertices c_K, Q_j, Q_k and at the midpoints $Q_{jk}, (c_K + Q_j)/2$, and $(c_K + Q_k)/2$ of the edges of the triangle

F_{jk} . Hence,

$$(f|_T)\Big|_{F_{jk}} = 2 \sum_{\substack{\alpha \in A_2 \\ \alpha_1=0}} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} = 0 \quad \text{at } x = \lambda_2 c_K + \lambda_3 Q_j + \lambda_4 Q_k \in F_{jk}$$

vanishes and $c_T(0; \beta) = 0$ for all $\beta \in \mathbb{N}_0^3$ with $|\beta| = 2$ and $T \in \widehat{\mathcal{T}}(c_1)$. This leads to

$$f|_T = 2\lambda_1 \sum_{\beta \in A_1} \frac{c_T(e_1 + \beta)}{(1 + \beta_1)} \frac{\lambda^\beta}{\beta!} \quad \text{at } x = \lambda_1 c_1 + \lambda_2 c_K + \lambda_3 Q_j + \lambda_4 Q_k \in T \in \widehat{\mathcal{T}}(c_1). \quad (3.17)$$

The continuity conditions in Lemma 3.7 with $\xi = 1$ in (3.13) motivate the piecewise affine function

$$f'|_T := \sum_{\beta \in A_1} c_T(e_1 + \beta) \frac{\lambda^\beta}{\beta!} = \sum_{\mu=1}^4 c_T(e_1 + e_\mu) \lambda_\mu \quad \text{at } x \in T \in \widehat{\mathcal{T}}(c_1) \quad (3.18)$$

with ordinates $(c_T(e_1 + e_\mu) : \mu = 1, \dots, 4)$ of $f|_T$ for each $T \in \widehat{\mathcal{T}}(c_1)$. Since $f \in C^1(T(1))$ and any two tetrahedra $T_+, T_- \in \widehat{\mathcal{T}}(c_1)$ share a face $F = \partial T_+ \cap \partial T_- = \text{conv}\{c_1, c_K, Q_j\}$ for some $2 \leq j \leq 4$ with common vertex c_1 , Lemma 3.7 shows $f' \in C^1(T_+ \cup T_-)$. The repeated application of Lemma 3.7 leads to $f' \in C^1(T(1))$. Since $f' \in P_1(\widehat{\mathcal{T}}(c_1))$ is piecewise affine, the already proven assertion of this lemma for $k = 1$ implies that $f' \in C^1(T(1)) \cap P_1(\widehat{\mathcal{T}}(c_1)) = P_1(T(1))$ is one global affine function in $\mathbb{R}^3$. Let $\varphi_1 \in S^1(\widehat{\mathcal{T}}(c_1)) := P_1(\widehat{\mathcal{T}}(c_1)) \cap C(T(1))$ denote the first-order nodal basis function with $\varphi_1(c_1) = 1$ and $\varphi_1(c_K) = 0 = \varphi_1(Q_j)$ for any $2 \leq j \leq 4$. In other words $\varphi_1|_T = \lambda_1$ restricted to any $T \in \widehat{\mathcal{T}}(c_1)$ is the first barycentric coordinate λ_1 in T . A comparison of f in (3.17) with f' in (3.18) reveals

$$f = 2f'\varphi_1 - \gamma\varphi_1^2 \quad \text{in } C^1(T(1)),$$

where $\gamma|_T := c_T(2e_1)$ for any $T \in \widehat{\mathcal{T}}(c_1)$ defines the piecewise constant function $\gamma \in P_0(\widehat{\mathcal{T}}(c_1))$. The piecewise quadratic f in C^1 and the piecewise affine $\nabla f = 2(f' - \gamma\varphi_1)\nabla\varphi_1 + 2\varphi_1\nabla f'$ are continuous in $T(1)$. In particular ∇f is continuous at the boundary $\partial T(1) \setminus \partial K := \bigcup_{1 \leq j < k \leq 4} F_{jk} \subset \{\varphi_1 = 0\}$ for $F_{jk} := \text{conv}\{c_K, Q_j, Q_k\}$ without the face $F_1 \in \mathcal{F}(K)$, where $\varphi_1 = 0$ vanishes. Hence, $f'\nabla\varphi_1$ is continuous there as well. Since $\nabla\varphi_1$ jumps at c_K and $Q_j \in \partial T(1)$, this implies $f'(Q_j) = 0 = f'(c_K)$ for any $2 \leq j \leq 4$. Thus the affine function $f' \in P_1(T(1))$ vanishes and $f = \gamma\varphi_1^2 \in C^1(K)$ leads to $\gamma = 0$ as well. This shows that $f \in C^1(T(1)) \cap P_2(\widehat{\mathcal{T}}(c_1))$ with $f|_{F_{jk}} = 0$ for all $1 \leq j < k \leq 4$ implies $f \equiv 0$. $\square$

Lemma 3.10. *Let $\widehat{\mathcal{T}} = WF3D(K)$ be a WF partition of K and $k = 0, 1, 2$. Then $f \in P_k(\widehat{\mathcal{T}})$ satisfies $f \in C^1(K)$ if and only if $f \in P_k(K)$ is a global polynomial in K .*

Proof. The proof is similar to that of Lemma 3.9. The assertion is trivial for $k = 0$ and immediate for $k = 1$. Suppose $k = 2$ and $f \in C^1(K) \cap P_2(\hat{\mathcal{T}})$. In particular $f|_{T(m)} \in C^1(T(m)) \cap P_2(\hat{\mathcal{T}}(c_m))$ for any $m = 1, 2, 3, 4$ and Lemma 3.9 shows $f|_{T(m)} \in P_2(T(m))$. Hence, $f \in P_2(\tilde{\mathcal{T}})$ holds for $\tilde{\mathcal{T}} := \{T(1), \dots, T(4)\}$. The five vertices in the triangulation $\tilde{\mathcal{T}}$ are c_K, Q_1, Q_2, Q_3, Q_4 and each subtetrahedra $T(m) = \text{conv}\{c_K, Q_j, Q_k, Q_\ell\} \in \tilde{\mathcal{T}}$ for $1 \leq j < k < \ell \leq 4$ and $\{j, k, \ell, m\} = \{1, 2, 3, 4\}$ will be identified with (c_K, Q_j, Q_k, Q_ℓ) . The Bernstein basis of $f|_{T(m)} \in P_2(T(m))$ leads to ordinates $(c_m(\alpha)|_{\alpha \in A_2})$ in

$$f|_{T(m)} = 2 \sum_{\alpha \in A_2} c_m(\alpha) \frac{\lambda^\alpha}{\alpha!} \text{ at } x = \lambda_1 c_K + \lambda_2 Q_j + \lambda_3 Q_k + \lambda_4 Q_\ell \in T(m) \in \tilde{\mathcal{T}}$$

for $m = 1, \dots, 4$. The subtraction of the global quadratic Lagrange interpolation on K shows that, without loss of generality, f vanishes at the vertices $Q_1, \dots, Q_4$ and the edge midpoints $Q_{jk} := (Q_j + Q_k)/2$ for $1 \leq j < k \leq 4$. Since f is a quadratic function on $\partial K = F_1 \cup \dots \cup F_4 = \{\lambda_1 = 0\}$,

$$f|_{F_m} = 2 \sum_{\substack{\alpha \in A_2 \\ \alpha_1=0}} c_m(\alpha) \frac{\lambda^\alpha}{\alpha!} = 0 \text{ at } x = \lambda_2 Q_j + \lambda_3 Q_k + \lambda_4 Q_\ell \in F_m$$

vanishes. Hence, $c_m(0; \beta) = 0$ for all $\beta \in \mathbb{N}_0^3$ with $|\beta| = 2$ and $\ell = 1, \dots, 4$. Consequently,

$$f|_{T(m)} = 2\lambda_1 \sum_{\beta \in A_1} \frac{c_m(e_1 + \beta)}{(1 + \beta_1)} \frac{\lambda^\beta}{\beta!} \text{ at } x = \lambda_1 c_K + \lambda_2 Q_j + \lambda_3 Q_k + \lambda_4 Q_\ell \in T(m) \in \tilde{\mathcal{T}}.$$

Since all $T(m) \in \tilde{\mathcal{T}}$ are face connected with common vertex c_K , define f' for $\xi = 1$ as in (3.13). Repeated applications of Lemma 3.7 show $f' \in C^1(K) \cap P_1(\hat{\mathcal{T}})$ and therefore $f' \in P_1(K)$ by the proven assertion of this lemma. The comparison of f and f' concludes the proof as that of Lemma 3.9. Since the arguments apply verbatim, further details are omitted. $\square$

3.2.4 The WF finite element

Given a tetrahedron K and its WF partition $\hat{\mathcal{T}} = \text{WF3D}(K)$ from Definition 3.8, this section characterizes

$$\text{WF}(K) := P_3(\hat{\mathcal{T}}) \cap C^1(K)$$

as a 28-dimensional vector space such that the degrees of freedom (dof) $L_1, \dots, L_{28}$ in Table 3.2 are linear independent and the triple $(K, \text{WF}(K), (L_1, \dots, L_{28}))$ forms a finite element in the sense of Ciarlet. To describe the degrees of freedom from Table 3.2 for the tetrahedron $K = \text{conv}\{Q_1, \dots, Q_4\}$, determine, for each edge

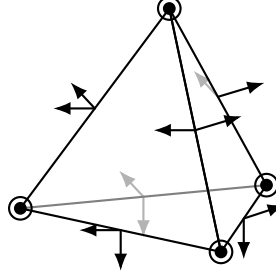


Figure 3.4: Mnemonic diagram of the Wosey-Farin finite element in 3D

$L_\mu(f) = f(Q_\mu)$	for $\mu = 1, \dots, 4$
$L_{4\ell+\mu}(f) = \nabla f(Q_\mu) \cdot e_\ell$	for $\mu = 1, \dots, 4, \ell = 1, 2, 3$
$L_{20-2^4-j+2k+m}(f) = \nabla f(Q_{jk}) \cdot \nu_{jk,m}$	for $j = 1, 2, 3, k = j + 1, \dots, 4, m = 1, 2$

Table 3.2: The 28 degrees of freedom $L_1, \dots, L_{28}$ of $\text{WF}(K)$ for any $f \in C^1(K)$

$E_{jk} = \text{conv}\{Q_j, Q_k\}$ of K with $1 \leq j < k \leq 4$ and midpoint $Q_{jk} = (Q_j + Q_k)/2$ a unit tangential vector $\tau_{jk} \parallel Q_j - Q_k$ and two additional unit vectors $\nu_{jk,1}, \nu_{jk,2} \in \mathbb{R}^3$ such that $\text{span}\{\tau_{jk}, \nu_{jk,1}, \nu_{jk,2}\} = \mathbb{R}^3$. Without loss of generality, assume that $(\tau_{jk}, \nu_{jk,1}, \nu_{jk,2})$ is an orthonormal basis of $\mathbb{R}^3$. The degrees of freedom with the enumeration in Table 3.2 are the evaluation of a function $f \in C^1(K)$ and its gradient ∇f at the vertices $Q_1, \dots, Q_4$ of K and the evaluation of the gradient $\nabla f(Q_{jk}) \cdot s$ in the direction $s = \nu_{jk,1}$ and $s = \nu_{jk,2}$ at each edge midpoint Q_{jk} . Throughout this section, the four vertices $Q_1, \dots, Q_4 \in \mathcal{V}$ of the tetrahedron K are opposite to the faces $F_1, \dots, F_4 \in \mathcal{F}$. The Algorithm 1 below with input $x_1, \dots, x_{28} \in \mathbb{R}$ for the 28 degrees of freedom computes $f \in P_3(\hat{\mathcal{T}})$ in terms of ordinates $(c_T(\alpha)|\alpha \in A_3, T \in \hat{\mathcal{T}})$ in

$$f|_T = 6 \sum_{\alpha \in A_3} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at } x = \sum_{\mu=1}^4 \lambda_\mu P_{\sigma(T,\mu)} \in T \in \hat{\mathcal{T}} \quad (3.19)$$

with the global enumeration $P_1, \dots, P_9$ of all the vertices in $\hat{\mathcal{T}}$ with $P_1 = c_K$. For each $T \in \hat{\mathcal{T}}$ let $P_{\sigma(T,1)} = P_1 = c_K$, $P_{\sigma(T,2)} = c_m$ for the unique $m \in \{1, 2, 3, 4\}$ with $\partial T \cap \partial K \subseteq F_m$, and the remaining vertices $P_{\sigma(T,3)}, P_{\sigma(T,4)} \in \{Q_1, \dots, Q_4\} \setminus \{Q_m\}$. The following algorithm is contained in [WF87].

Algorithm 1 Worsey-Farin-CT3D

Input: $x_1, \dots, x_{28} \in \mathbb{R}$, a tetrahedron $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\}$ with WF partition $\widehat{\mathcal{T}} = \text{WF3D}(K)$ and a local enumeration so that $T \equiv (P_{\sigma(T,1)}, \dots, P_{\sigma(T,4)})$ with $P_{\sigma(T,1)} = c_K$, $P_{\sigma(T,2)} = c_m$, and $P_{\sigma(T,3)}, P_{\sigma(T,4)} \in \{Q_1, \dots, Q_4\} \setminus \{Q_m\}$ for each $T \in \widehat{\mathcal{T}}(c_m)$ with $m \in \{1, \dots, 4\}$

- (a) A preprocessing computes $f(Q_{jk})$ and $\nabla f(Q_{jk})$ with $Q_{jk} := (Q_j + Q_k)/2$ for any $j, k = 1, \dots, 4$, note $Q_{jj} \equiv Q_j$, from the data $x_1, \dots, x_{28}$ representing $L_1(f), \dots, L_{28}(f)$, cf. Remark 3.11 below for further details.
- (b) For all $\ell = 1, \dots, 4$ and $1 \leq j < k \leq 4$,

$$g(Q_\ell) := f(Q_\ell) - \frac{1}{3}(Q_\ell - c_K) \cdot \nabla f(Q_\ell),$$

$$g(Q_{jk}) := \frac{1}{3}(c_K - Q_{jk}) \cdot \nabla f(Q_{jk}) + f(Q_{jk}).$$

Let $g \in P_2(K)$ denote the quadratic Lagrange interpolation of this data at the vertices $(Q_1, \dots, Q_4)$ and edge midpoints $(Q_{jk} : 1 \leq j < k \leq 4)$ of K and compute the ordinates $(c_T(\alpha) | \alpha \in A_3, \alpha_1 \geq 1, T \in \widehat{\mathcal{T}})$ of

$$g|_T = 2 \sum_{\beta \in A_2} c_T(e_1 + \beta) \frac{\lambda^\beta}{\beta!} \quad \text{at } x = \sum_{\mu=1}^4 \lambda_\mu P_{\sigma(T,\mu)} \in T \in \widehat{\mathcal{T}}.$$

- (c) For all $m = 1, \dots, 4$ let $f_m \in \text{HCT}(F_m) := P_3(\widehat{\mathcal{F}}(F_m)) \cap C^1(F_m)$ denote the 2D HCT finite element interpolation of the input data (resp. data from (a)) on the face F_m of K with unit normal ν_{F_m} . The degrees of freedom of $\text{HCT}(F_m)$ are the evaluation of f and the tangential derivatives $\nabla f \times \nu_{F_m}$ at the vertices of the triangle F_m as well as for each edge $E \in \mathcal{E}(F_m)$ with unit tangent vector τ_E of F_m the directional derivative in $\langle F_m \rangle$ perpendicular to the edge E , i.e., $\nabla f \cdot (\tau_E \times \nu_{F_m})$, evaluated in the edge midpoint. Compute coefficients $(c_T(\alpha) | \alpha \in A_3, \alpha_1 = 0, T \in \widehat{\mathcal{T}})$ such that

$$f_m|_{F_m \cap \partial T} = 6 \sum_{\beta \in A_3} c_T((0; \beta)) \frac{\lambda^\beta}{\beta!} \quad \text{at } x = \sum_{\mu=2}^4 \lambda_\mu P_{\sigma(T,\mu)} \in F_m \cap \partial T \text{ for } T \in \widehat{\mathcal{T}}.$$

Output: $(c_T(\alpha) | \alpha \in A_3, T \in \widehat{\mathcal{T}})$ and hence $f \in P_3(\widehat{\mathcal{T}})$ with $f|_T = 6 \sum_{\alpha \in A_3} c_T(\alpha) \lambda^\alpha / \alpha!$ at $x = \sum_{\mu=1}^4 \lambda_\mu P_{\sigma(T,\mu)} \in T \in \widehat{\mathcal{T}}$.

Remark 3.11 (preprocessing in Algorithm 1.a). Algorithm 1 aims at the computation of some global C^1 function f with prescribed values $L_\mu(f) = x_\mu$ for $\mu = 1, \dots, 28$. The step (a) provides $f(Q_{jk})$ and $\nabla f(Q_{jk})$ for any Q_{jk} as follows.

Step a.1. Suppose $j = k$, then $Q_{jj} = Q_j$ is a vertex and x_j directly represents $f(Q_j) = x_j$ and $x_{4\ell+j}$ represents $\frac{\partial f}{\partial x_\ell}(Q_j) = x_{4\ell+j}$ for $\ell = 1, 2, 3$.

Step a.2. Suppose $j < k$, then Q_{jk} is the midpoint of the edge $E = \text{conv}\{Q_j, Q_k\}$ from Q_j to Q_k with unit tangent vector τ_{jk} in the tetrahedron K . Consider $f|_E \in P_3(E)$ as a one-dimensional cubic polynomial that satisfies the following interpolation conditions. The function values $f(Q_\ell) = x_\ell$ and the one-dimensional derivative $f'(Q_\ell) = \nabla f(Q_\ell) \cdot \tau_{jk}$ at the vertices Q_j and Q_k (for $\ell = j, k$) are prescribed in Step a.1. The four one-dimensional Hermite interpolation conditions determine a unique cubic polynomial $f|_E \in P_3(E)$ along the edge E . This polynomial provides the values $f(Q_{jk})$ and $f'(Q_{jk}) = \nabla f(Q_{jk}) \cdot \tau_{jk}$. The final components of $\nabla f(Q_{jk})$ are prescribed by $x_{20-2^4-j+2k+m} = \nabla f(Q_{jk}) \cdot \nu_{jk,m}$ for $m = 1, 2$ and the fixed vectors $\nu_{jk,m}$ with $\text{span}\{\tau_{jk}, \nu_{jk,1}, \nu_{jk,2}\} = \mathbb{R}^3$.

Remark 3.12 (feasability of Algorithm 1). The data in Algorithm 1.b define a unique $g \in P_2(K)$, which has a unique representation in the Bernstein basis of each $T \in \hat{\mathcal{T}}$. Since the 2D HCT finite element is a finite element in the sense of Ciarlet, cf. Remark 3.1, $F_m \cap \partial T =: F \in \hat{\mathcal{F}}(F_m)$ and $f_m|_F \in P_3(F)$ allow for unique ordinates in Algorithm 1.c.

Remark 3.13 (parameter dependence of Algorithm 1). Suppose the input $x_1, \dots, x_{28} \in \mathbb{R}$ of Algorithm 1 is fixed, while the center points $c_K \in \text{int}(K)$ and $c_m \in \text{relint}(F_m)$ for $m = 1, 2, 3, 4$ may vary, i.e., the WF partition $\hat{\mathcal{T}} = \text{WF3D}(K)$ differs. The point is that the output $f \in P_3(\hat{\mathcal{T}}) \cap C^1(\Omega)$ depends continuously on these five parameters $c_K, c_1, \dots, c_4 \in \mathbb{R}^3$. The preprocessing in Algorithm 1.a depends only on the edges of K (see Remark 3.11) and is independent of the subtriangulation $\hat{\mathcal{T}}$. The values of $g \in P_2(K)$ in the vertices and edge midpoints in Algorithm 1.b depend directly on the choice of c_K . This dependence in g is affine, whence continuous. The 2D HCT function f_m on the face F_m of K for $m = 1, \dots, 4$ in Algorithm 1.c depends on the choice of the face center c_m . The coefficients of the function $f_m \in P_3(\hat{\mathcal{T}})$ for a given c_m are obtained through the solution of a linear system with an invertible matrix [Cia78, p.345, 1.26ff]; this is already exploited in [Cia78, p.346, 1.1ff]. Hence, the dependence of the ordinates $(c_T(\alpha) : \alpha \in A_3, T \in \hat{\mathcal{T}})$ on the points $c_K, c_1, \dots, c_4$ that define the subtriangulation $\hat{\mathcal{T}}$ is continuous. $\square$

The WF finite element is a finite element in the sense of Ciarlet as announced in [NW19]. Since [WF87; NW19] do not contain the proofs, this is carried out here for convenient reading.

Theorem 3.14 (WF finite element). (a) For any input $x_1, \dots, x_{28} \in \mathbb{R}$, the output $f \in C^1(K) \cap P_3(\hat{\mathcal{T}})$ of Algorithm 1 is continuously differentiable and

$$L_\mu f = x_\mu \quad \text{for } \mu = 1, \dots, 28. \quad (3.20)$$

(b) There exist at most one function $f \in C^1(K) \cap P_3(\widehat{\mathcal{T}})$ with (3.20).

(c) The triple $(K, C^1(K) \cap P_3(\widehat{\mathcal{T}}), (L_1, \dots, L_{28}))$ is a finite element in the sense of Ciarlet.

Proof of (a). This proof is divided in two steps.

Step 1 ($L_\mu f = x_\mu$ for $\mu = 1, \dots, 28$). The output $f \in P_3(\widehat{\mathcal{T}})$ is a piecewise cubic polynomial. To prove (3.20) fix one $T \in \mathcal{T}$, abbreviate $f_T := f|_T$, and check $L_\mu f_T = x_\mu$ for any degree of freedom L_μ , which can be evaluated for f_T .

Suppose $T := \text{conv}\{c_K, c_m, Q_j, Q_k\} \in \widehat{\mathcal{T}}$ for $1 \leq j < k \leq 4$ and $m \in \{1, \dots, 4\} \setminus \{j, k\}$ and identify T with (c_K, c_m, Q_j, Q_k) . The output f_T on $T \in \widehat{\mathcal{T}}$ reads

$$f_T = 6 \sum_{\alpha \in A_3} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at } x = \lambda_1 c_K + \lambda_2 c_m + \lambda_3 Q_j + \lambda_4 Q_k \in T.$$

There are ten degrees of freedom from Table 3.2 that can be evaluated in $T \equiv (c_K, c_m, Q_j, Q_k)$, namely $L_\ell, L_{4+\ell}, L_{8+\ell}, L_{12+\ell}$ for $\ell = j, k$, as well as $L_{20-2^4-j+2k+1}$ and $L_{20-2^4-j+2k+2}$.

Proof of (3.20) for $\mu = j, k$. Since the 2D HCT finite element in Algorithm 1.c interpolates the nodal values exactly, Example 3.4.a shows

$$L_j(f_T) = f_T(Q_j) = c_T(3e_3) = f_m|_T(Q_j) = x_j.$$

The proof of $L_k(f_T) = x_k$ follows by symmetry of j and k . Since $L_j(f_T) = x_j$ for any $T \in \widehat{\mathcal{T}}$, the output function $f \in P_3(\widehat{\mathcal{T}})$ is single valued and continuous at every vertex $Q_j \in \mathcal{V}$ of K for $j = 1, \dots, 4$ with $f(Q_j) = x_j$. $\square$

Proof of continuity at Q_{jk} . Remark 3.11 explains that the data $x_1, \dots, x_{28}$ allow the computation of a value x_{jk} in the edge midpoint in Algorithm 1.a. Algorithm 1.c interpolates x_{jk} such that $f_m|_{F_m}(Q_{jk}) = x_{jk}$. For any $T \in \widehat{\mathcal{T}}(T(m))$ identified with (c_K, c_m, Q_j, Q_k) as above, Example 3.4.b shows

$$\begin{aligned} f_T(Q_{jk}) &= \frac{1}{8} \left(f_T(Q_k) + 3c_T(2e_3 + e_4) + 3c_T(e_3 + 2e_4) + f_T(Q_j) \right) \\ &= \frac{1}{8} \left(x_k + 3c_T(2e_3 + e_4) + 3c_T(e_3 + 2e_4) + x_j \right) = f_m|_{F_m}(Q_{jk}) = x_{jk}. \end{aligned} \tag{3.21}$$

The result $f_T(Q_{jk}) = x_{jk}$ is independent of $m \in \{1, 2, 3, 4\} \setminus \{j, k\}$ and $T \in \widehat{\mathcal{T}}(T(m))$. Hence, the output function $f \in P_3(\widehat{\mathcal{T}})$ is single valued and continuous at every edge midpoint $Q_{jk} = (Q_j + Q_k)/2$ of K for $1 \leq j < k \leq 4$ with $f(Q_{jk}) = x_{jk}$. $\square$

Proof of (3.20) for $\mu = 4 + j, 8 + j, 12 + j, 4 + k, 8 + k, 12 + k$. Abbreviate $X_j := (x_{4+j}, x_{8+j}, x_{12+j})^\top$, recall $T \in \widehat{\mathcal{T}}(T(m))$ is identified with (c_K, c_m, Q_j, Q_k) . Observe that the gradient $\nabla f_T(Q_j) \in \mathbb{R}^3$ is uniquely determined by its scalar product with $Q_j - Q_k$, $Q_j - c_m$, and $Q_j - c_K$. The 2D HCT interpolation in Algorithm 1.c interpolates the tangential derivatives correctly. In other words $\nabla f_m|_{F_m}(Q_j) \times \nu_{F_m} = X_j \times \nu_{F_m}$ for the unit normal ν_{F_m} of F_m . The first two directions $Q_j - Q_k$ and $Q_j - c_m$ are parallel to the plane $\langle F_m \rangle$ and Example 3.4.c shows

$$\begin{aligned} (Q_j - Q_k) \cdot \nabla f_T(Q_j) &= 3(f_T(Q_j) - c_T(2e_3 + e_4)) = 3(x_j - c_T(2e_3 + e_4)) \\ &= (Q_j - Q_k) \cdot \nabla f_m|_{F_m}(Q_j) = (Q_j - Q_k) \cdot X_j, \\ (Q_j - c_m) \cdot \nabla f_T(Q_j) &= 3(f_T(Q_j) - c_T(e_2 + 2e_3)) = 3(x_j - c_T(e_2 + 2e_3)) \\ &= (Q_j - c_m) \cdot \nabla f_m|_{F_m}(Q_j) = (Q_j - c_m) \cdot X_j. \end{aligned}$$

The third direction $Q_j - c_K$ is not parallel to the plane $\langle F_m \rangle$. Example 3.4.c and 3.3.a imply

$$(Q_j - c_K) \cdot \nabla f_T(Q_j) = 3(f_T(Q_j) - c_T(e_1 + 2e_3)) = 3(x_j - g(Q_j)).$$

The choice of the nodal value $g(Q_j) \equiv x_j - \frac{1}{3}(Q_j - c_K) \cdot X_j$ in Algorithm 1.b implies

$$(Q_j - c_K) \cdot \nabla f_T(Q_j) = (Q_j - c_K) \cdot X_j.$$

This concludes the proof of $\nabla f_T(Q_j) = X_j = (x_{4+j}, x_{8+j}, x_{12+j})^\top$. The proof of $\nabla f_T(Q_k) = (x_{4+k}, x_{8+k}, x_{12+k})^\top$ follows by symmetry of j and k . Hence, the gradient of the output function $f \in P_3(\widehat{\mathcal{T}})$ is single valued and continuous at every vertex $Q_j \in \mathcal{V}$ of K for $j = 1, \dots, 4$ with $\nabla f(Q_j) = X_j$. $\square$

Proof of (3.20) for $\mu = 20 - 2^{4-j} + 2k + 1, 20 - 2^{4-j} + 2k + 2$. The remaining degrees of freedom $L_{20-2^{4-j}+2k+1}$ and $L_{20-2^{4-j}+2k+2}$ concern the gradient $\nabla f_T(Q_{jk})$ in the edge midpoint Q_{jk} . Remark 3.11 explains how $x_1, \dots, x_{28}$ determine a unique value X_{jk} in Algorithm 1.a that represents $\nabla f(Q_{jk})$ in Algorithm 1.b–c. Recall $f(Q_{jk}) = x_{jk}$ from (3.21) above. The next step is to verify that $\nabla f_T(Q_{jk}) = X_{jk}$ for $T \in \widehat{\mathcal{T}}(T(m))$ identified with (c_K, c_m, Q_j, Q_k) . The gradient $\nabla f_T(Q_{jk})$ of the output of Algorithm 1 is uniquely determined by its scalar product with $c_m - Q_{jk}$, $Q_j - Q_{jk}$, and $c_K - Q_{jk}$. In the directions $c_m - Q_{jk}$ and $Q_j - Q_{jk}$, parallel to the plane $\langle F_m \rangle$, the gradient is determined by f_m in Algorithm 1.c. Example 3.4.d–e lead to

$$\begin{aligned} (c_m - Q_{jk}) \cdot \nabla f_T(Q_{jk}) &= \frac{3}{4}(c_T(e_2 + 2e_3) + 2c_T(e_2 + e_3 + e_4) + c_T(e_2 + 2e_4)) - 3x_{jk} \\ &= (c_m - Q_{jk}) \cdot \nabla f_m|_{F_m}(Q_{jk}) = (c_m - Q_{jk}) \cdot X_{jk}, \\ (Q_j - Q_{jk}) \cdot \nabla f_T(Q_{jk}) &= \frac{3}{8}(x_j + c_T(2e_3 + e_4) - c_T(e_3 + 2e_4) - x_k) \\ &= (Q_j - Q_{jk}) \cdot \nabla f_m|_{F_m}(Q_{jk}) = (Q_j - Q_{jk}) \cdot X_{jk}. \end{aligned}$$

For the remaining direction $c_K - Q_{jk}$, Example 3.4.d and 3.3.b show

$$\begin{aligned}(c_K - Q_{jk}) \cdot \nabla f_T(Q_{jk}) &= \frac{3}{4} (c_T(e_1 + 2e_3) + 2c_T(e_1 + e_3 + e_4) + c_T(e_1 + 2e_4)) - 3x_{jk} \\ &= 3g(Q_{jk}) - 3x_{jk}.\end{aligned}$$

Since the quadratic polynomial $g \in P_2(K)$ in Algorithm 1.b assumes the value $g(Q_{jk}) \equiv \frac{1}{3}(c_K - Q_{jk}) \cdot X_{jk} + x_{jk}$ at Q_{jk} , it follows $(c_K - Q_{jk}) \cdot \nabla f_T(Q_{jk}) = (c_K - Q_{jk}) \cdot X_{jk}$. The last three displayed identities ensure $\nabla f_T(Q_{jk}) = X_{jk}$ for any $T \in \widehat{\mathcal{T}}(Q_{jk})$. Hence, the gradient of the output function $f \in P_3(\widehat{\mathcal{T}})$ is indeed single valued and continuous at every edge midpoint $Q_{jk} = (Q_j + Q_k)/2$ of K for $1 \leq j < k \leq 4$ with $\nabla f(Q_{jk}) = X_{jk}$. In particular, $L_{20-2^4-j+2k+1}(f_T) = \nabla f_T(Q_{jk}) \cdot \nu_{jk,1} = x_{20-2^4-j+2k+1}$ and $L_{20-2^4-j+2k+2}(f_T) = \nabla f_T(Q_{jk}) \cdot \nu_{jk,2} = x_{20-2^4-j+2k+2}$. This concludes the proof of (3.20). $\square$

Step 2 ($f \in C^1(K)$). It suffices to show $f|_{T_1 \cup T_2} \in C^1(T_1 \cup T_2)$ for any two tetrahedra $T_1, T_2 \in \widehat{\mathcal{T}}$ with common face $F := \partial T_1 \cap \partial T_2$. There are two types of common faces in $\widehat{\mathcal{T}}$.

Case 1. Assume $T_1, T_2 \in \widehat{\mathcal{T}}(c_m)$ for $m = 1, \dots, 4$. In other words, let $F = \text{conv}\{c_K, c_m, Q_j\}$ and identify T_1 with (c_K, c_m, Q_j, Q_k) and T_2 with (c_K, c_m, Q_j, Q_ℓ) for $\{j, k, \ell, m\} = \{1, 2, 3, 4\}$. The definition of $g \in P_2(K) \subset C^1(K)$ in Algorithm 1.b ensures $g|_{T_1 \cup T_2} = f'|_{T_1 \cup T_2}$ for $\xi = 1$ in (3.13) and $f' \in C^1(T_1 \cup T_2)$. Lemma 3.7 ensures that $f|_{T_1 \cup T_2} \in C^1(T_1 \cup T_2)$ is equivalent to $f' \in C^1(T_1 \cup T_2)$ and the continuity of f and $\nu_F \cdot \nabla f$ along $E := \text{conv}\{c_m, Q_j\}$. Hence, it remains to verify (i) continuity of f along E and (ii) continuity of $\nu_F \cdot \nabla f$ along E . Throughout the proof, abbreviate $c_\mu(\alpha) := c_{T_\mu}(\alpha)$ for $\mu = 1, 2$.

Proof of (i). For $\alpha \in A_3$ with $\alpha_1 = 0 = \alpha_4$, the continuity of f_m from Algorithm 1.c shows $c_1(\alpha) = c_2(\alpha)$. For $\alpha_1 \neq 0 = \alpha_4$, the continuity of g from Algorithm 1.b shows $c_1(\alpha) = c_2(\alpha)$. In particular, (3.10) implies that $f|_E$ is continuous along E . $\square$

Proof of (ii). Note, since the ordinates $c_1(e_2 + \alpha) = c_2(e_2 + \alpha)$ are equal for all $\alpha \in A_2$ with $\alpha_4 = 0$, Lemma 3.5 proves the continuity of the piecewise quadratic function $g'' \in P_2(\{T_1, T_2\})$ defined for $\mu = 1, 2$ by

$$g''|_{T_\mu} := 2 \sum_{\beta \in A_2} c_\mu(e_2 + \beta) \frac{\lambda^\beta}{\beta!} \quad \text{at } x = \lambda_1 c_K + \lambda_2 c_m + \lambda_3 Q_j + \lambda_4 P_{\sigma(T_\mu, 4)}, \quad (3.22)$$

i.e., $g'' \in C(T_1 \cup T_2)$. In order to prove the continuity of $\nu_F \cdot \nabla f$ along E , recall that $f|_{F_m} = f_m \in C^1(F_m)$ on the face $F_m := \text{conv}\{Q_\ell, Q_j, Q_k\} \supset E$ of K from Algorithm 1.c. Hence, the tangential derivative $\nu_{F_m} \times \nabla f$ is continuous along F_m and in particular along E . It remains to prove the continuity for one fixed direction $\tau := c_m - c_K \in \mathbb{R}^3 \setminus \{0\}$ oblique to $\langle F_m \rangle$. In $T \in \{T_1, T_2\}$, Remark 3.2

shows

$$\tau \cdot \nabla f|_T = 6 \sum_{\beta \in A_2} \sum_{\kappa=1}^4 c_T(\beta + e_\kappa) \frac{\lambda^\beta}{\beta!} (\tau \cdot \nabla \lambda_\kappa).$$

Since $\tau \cdot \nabla \lambda_\kappa = \lambda_\kappa(c_m) - \lambda_\kappa(c_K) = \delta_{2\kappa} - \delta_{1\kappa}$, this is

$$\tau \cdot \nabla f|_T = 6 \sum_{\beta \in A_2} \left(c_T(\beta + e_2) - c_T(\beta + e_1) \right) \frac{\lambda^\beta}{\beta!} = 3g''|_T - 3g|_T$$

with g'' from (3.22) and g from Algorithm 1.b. Since g'' and g are continuous in $T_1 \cup T_2$, $\nabla f|_{F_m}$ is continuous in particular along E . $\square$

Case 2. Assume $T_1, T_2 \in \widehat{\mathcal{T}}$ share a face $F = \text{conv}\{c_K, Q_j, Q_k\}$ and identify T_1 with (c_K, Q_j, Q_k, c_m) and T_2 with (c_K, Q_j, Q_k, c_ℓ) for $\{j, k, \ell, m\} = \{1, 2, 3, 4\}$. The proof is comparable to Case 1. The definition of $g \in P_2(K) \subset C^1(K)$ in Algorithm 1.b ensures $g|_{T_1 \cup T_2} = f'|_{T_1 \cup T_2}$ for $\xi = 1$ in (3.13) and $f' \in C^1(T_1 \cup T_2)$. According to Lemma 3.7 it remains to verify (i) continuity of f along E and (ii) continuity of $\nu_F \cdot \nabla f$ along E to guarantee $f|_{T_1 \cup T_2} \in C^1(T_1 \cup T_2)$. Abbreviate $c_\mu(\alpha) := c_{T_\mu}(\alpha)$ for $\mu = 1, 2$.

Proof of (i). Since $f|_E$ is uniquely determined in Algorithm 1.a, the equality of $f_m(x) = f_\ell(x)$ at $x = \lambda_2 Q_j + \lambda_3 Q_k \in E$ proves $c_1(\alpha) = c_2(\alpha)$ for $\alpha \in A_3$ with $\alpha_1 = 0 = \alpha_4$. The equality of the ordinates shows the continuity of f along E . $\square$

Proof of (ii). Since (each component of) $\nabla f|_{T_1}$ and $\nabla f|_{T_2}$ is a quadratic polynomial along E , which coincides at the three points Q_j, Q_{jk}, Q_k , it follows $\nabla f|_{T_1} = \nabla f|_{T_2}$ on E . In particular $\nu_F \cdot \nabla f$ is continuous along E . $\square$

This concludes the proof of (a). $\square$

Proof of (b). Suppose two functions $f_1, f_2 \in P_3(\widehat{\mathcal{T}}) \cap C^1(K)$ satisfy $L_\mu(f_j) = x_\mu$ for $\mu = 1, \dots, 28$ and $j = 1, 2$. Their difference $f := f_2 - f_1$ satisfies $L_\mu(f) = 0$ for all $\mu = 1, \dots, 28$. Identify $T := \text{conv}\{c_K, c_m, Q_j, Q_k\} \in \widehat{\mathcal{T}}$ with (c_K, c_m, Q_j, Q_k) for $1 \leq j < k \leq 4$ and $m \in \{1, 2, 3, 4\} \setminus \{j, k\}$. Let $(c_T(\alpha) | \alpha \in A_3, T \in \widehat{\mathcal{T}})$ denote the coefficients of $f \in C^1(K)$ in

$$f|_T = 6 \sum_{\alpha \in A_3} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at } x = \lambda_1 c_K + \lambda_2 c_m + \lambda_3 Q_j + \lambda_4 Q_k \in T.$$

Since $L_1(f) = \dots = L_{28}(f) = 0$, Remark 3.12 implies $f(Q_\ell) = f(Q_{jk}) = 0$ and $\nabla f(Q_\ell) = \nabla f(Q_{jk}) = 0$ for all $\ell = 1, \dots, 4$ and $1 \leq j < k \leq 4$. The restriction $f|_F$ of f to a face F of K is a 2D HCT finite element function with vanishing associated degrees of freedom. Hence, $f|_F \equiv 0$. Consequently,

$$0 = f|_{\partial K} = \sum_{\substack{\alpha \in A_3 \\ \alpha_1=0}} c_T(\alpha) \frac{\lambda^\alpha}{\alpha!} \quad \text{at } x = \lambda_2 c_m + \lambda_3 Q_j + \lambda_4 Q_k \in T \cap \partial K$$

vanishes. This implies $c_T(\alpha) = 0$ for all $\alpha \in A_3$ with $\alpha_1 = 0$. For $\alpha_1 \geq 0$, $c_T(\alpha)$ is related to $f' \in P_2(\widehat{\mathcal{T}})$ with

$$f'|_T := 2 \sum_{\beta \in A_2} c_T(e_1 + \beta) \frac{\lambda^\beta}{\beta!} \quad \text{at } x = \lambda_1 c_K + \lambda_2 c_m + \lambda_3 Q_j + \lambda_4 Q_k \in T. \quad (3.23)$$

Lemma 3.7 applies to any two $T_+, T_- \in \widehat{\mathcal{T}}$ that share a face $T_+ \cap T_- = F \in \widehat{\mathcal{F}}$. Hence, (3.23) defines $f' \in C^1(K) \cap P_2(\widehat{\mathcal{T}})$ and Lemma 3.10 shows $f' \in P_2(K)$. For any $T \in \widehat{\mathcal{T}}$ with vertex $Q_\ell \in \mathcal{V}(T) \cap \mathcal{V}(K)$ and $(\lambda_1, \dots, \lambda_4) = e_\kappa$ for $\kappa \in \{3, 4\}$, equation (3.7) in Example 3.4 reads

$$0 = \nabla f(Q_\ell) = 3 \sum_{\mu=1}^4 c_T(2e_\kappa + e_\mu) \nabla \lambda_\mu = 3c_T(e_1 + 2e_\kappa) \nabla \lambda_1$$

with $c_T(\alpha) = 0$ for $\alpha_1 = 0$ in the last equality. It follows with Example 3.3.a that $0 = c_T(e_1 + 2e_\kappa) = f'(Q_\ell)$ for any $1 \leq \ell \leq 4$. Since for any $T \in \widehat{\mathcal{T}}$ with vertices Q_j and Q_k the edge midpoint Q_{jk} has barycentric coordinates $(\lambda_1, \dots, \lambda_4) = e_3/2 + e_4/2$, equation (3.8) in Example 3.4 implies

$$\begin{aligned} 0 = \nabla f(Q_{jk}) &= \frac{3}{4} \sum_{\mu=1}^4 \left(c_T(2e_3 + e_\mu) + 2c_T(e_3 + e_4 + e_\mu) + c_T(2e_4 + e_\mu) \right) \nabla \lambda_\mu \\ &= \frac{3}{4} \left(c_T(e_1 + 2e_3) + 2c_T(e_1 + e_3 + e_4) + c_T(e_1 + 2e_4) \right) \nabla \lambda_1 \end{aligned}$$

with $c_T(\alpha) = 0$ for $\alpha_1 = 0$ in the last equality. Hence, from Example 3.3.b follows

$$0 = c_T(e_1 + 2e_3) + 2c_T(e_1 + e_3 + e_4) + c_T(e_1 + 2e_4) = 4f'(Q_{jk})|_T$$

for any $T \in \widehat{\mathcal{T}}$ with vertices Q_j and Q_k for $1 \leq j < k \leq 4$. This shows that $f' \in P_2(K)$ vanishes at all Lagrange points for a quadratic polynomial in K , whence $f' \equiv 0$. It follows $c_T(\alpha) = 0$ for all $T \in \widehat{\mathcal{T}}$ and $\alpha \in A_3$; in other words $f \equiv 0$. $\square$

Proof of (c). This follows directly from (a), which proves that Algorithm 1 allows the construction of $f \in P_3(\widehat{\mathcal{T}}) \cap C^1(K)$ for all input data $x_1, \dots, x_{28}$, and (b), which proves the uniqueness of the function with the values $x_1, \dots, x_{28}$ in the degrees of freedom $L_1, \dots, L_{28}$ in Table 3.2. $\square$

3.2.5 The WF on adjacent tetrahedra

This section is devoted to the compatibility of WF for two tetrahedra with a common face. The point of departure is a first Lemma on one tetrahedron. Let $\widehat{\mathcal{T}}$ denote the WF partition of a simplex K with vertices $Q_1, \dots, Q_4$ and opposite faces $F_1, \dots, F_4$. Recall, $c_m \in \text{relint}(F_m)$ and $c_K \in \text{relint}(K)$ from the WF partition in Definition 3.8, $T(m) = \text{conv}\{c_K, F_m\}$, and abbreviate $\tau := c_m - c_K \in \mathbb{R}^3 \setminus \{0\}$ as well as $\widehat{\mathcal{T}}(c_m) := \widehat{\mathcal{T}}(T(m))$ for one fixed $m \in \{1, \dots, 4\}$.

Lemma 3.15. *Let $f \in P_3(\widehat{\mathcal{T}}(c_m))$ and $g := \partial f / \partial \tau \in P_2(\widehat{\mathcal{T}}(c_m))$. Then $f \in C^1(T(m))$ if and only if $f|_{F_m} \in C^1(F_m)$ and $g \in P_2(T(m))$.*

Proof. Without loss of generality let $m = 4$, $\tau := c_4 - c_K$, and $T(4) = \text{conv}\{c_K, F_4\} = \text{conv}\{c_K, Q_1, Q_2, Q_3\}$.

„ $\Rightarrow$ “ The continuity of $f \in C^1(T(4))$ and its derivatives imply $f|_{F_4} \in C^1(F_4)$. Lemma 3.9 shows $P_2(\widehat{\mathcal{T}}(c_4)) \cap C^1(T(4)) = P_2(T(4))$. Since $g \in P_2(\widehat{\mathcal{T}}(c_4)) \cap C^0(T(4))$, it suffices to verify $g \in C^1(T(4))$. Fix the common side $F = \text{conv}\{c_K, c_4, Q_j\} = \partial T_1 \cap \partial T_2$ of two neighbouring tetrahedra $T_1, T_2 \in \widehat{\mathcal{T}}(c_4)$ identified with $T_1 = (c_K, c_4, Q_j, Q_k)$ and $T_2 = (c_K, c_4, Q_j, Q_\ell)$ for $\{j, k, \ell\} = \{1, 2, 3\}$. Since $f|_{T_j} = 6 \sum_{\alpha \in A_3} c_j(\alpha) \lambda^\alpha / \alpha!$ for $j = 1, 2$ defines $f \in C^1(T(m))$, Lemma 3.7 shows that $f'_\xi|_{T_j} := 2 \sum_{\beta \in A_2} c_j(e_\xi + \beta) \lambda^\beta / \beta!$ for $j = 1, 2$ in (3.13) forms $f'_\xi \in C^1(T_1 \cup T_2)$ for any $\xi \in \{1, 2, 3\}$. Notice that the formula for the derivative in (3.6) implies $\tau \cdot \nabla f|_{T_j} = 6 \sum_{\beta \in A_2} \sum_{\mu=1}^4 c_j(\beta + e_\mu) (\tau \cdot \nabla \lambda_\mu) \lambda^\beta / \beta!$ with $\tau \cdot \nabla \lambda_\mu = \lambda_\mu(c_4) - \lambda_\mu(c_K) = \delta_{2\mu} - \delta_{1\mu}$. Hence, it holds a.e. in T_j for $j = 1, 2$

$$g|_{T_j} = \tau \cdot \nabla f|_{T_j} = 6 \sum_{\beta \in A_2} \left(c_j(\beta + e_2) - c_j(\beta + e_1) \right) \frac{\lambda^\beta}{\beta!} = 3f'_2|_{T_j} - 3f'_1|_{T_j}.$$

This and $f'_1, f'_2 \in C^1(T_1 \cup T_2)$ show $g \in C^1(T_1 \cup T_2)$. Since $\widehat{\mathcal{T}}(c_4)$ is side-connected, it follows successively $g \in C^1(T(4)) \cap P_2(\widehat{\mathcal{T}}(c_4))$. Lemma 3.9 implies $g \in P_2(T(4))$.

„ $\Leftarrow$ “ Suppose $g \in P_2(T(4))$ and $f|_{F_4} \in C^1(F_4)$. A one-dimensional integration shows for all $x \in F_m$ and $t \in \mathbb{R}$ with $x + t\tau \in T(4)$ that

$$f(x + t\tau) = f(x) + \int_0^t g(x + s\tau) \, ds. \quad (3.24)$$

Notice that $0 < s < t$ and $x \in F_4$ with $x + t\tau \in T$ for some tetrahedron $T \in \widehat{\mathcal{T}}$ imply that $x + s\tau$ belongs to the same tetrahedron by definition of τ . The assumptions $g \in P_2(T(4))$ and $f|_{F_4} \in C^1(F_4)$ show in combination with (3.24) that $f \in C(T(4))$ and that $\sigma \cdot \nabla f$ is continuous for all $\sigma \in \text{span}\{\tau, Q_2 - Q_1, Q_3 - Q_1\} \in \mathbb{R}^3$ (in any direction parallel to the plane $\langle F_4 \rangle$ and in the oblique direction $\tau \notin \text{span}\{Q_2 - Q_1, Q_3 - Q_1\} \parallel \langle F_4 \rangle$). This implies ∇f in $C^0(T(4); \mathbb{R}^3)$ and concludes the proof. $\square$

The main result of this section requires some notation on two tetrahedra with a common side. Suppose that $K_+, K_- \in \mathcal{T}$ are two neighbouring tetrahedra in a regular triangulation $\mathcal{T}$ that share a common face $F = \partial K_+ \cap \partial K_-$. Suppose that the center points c_F, c_{K_+} , and c_{K_-} in the WF partition $\widehat{\mathcal{T}}(K_\pm)$ of Definition 3.8 belong to one straight line. In other words $c_{K_+} - c_F \parallel c_{K_-} - c_F$. Suppose the degrees of freedom for some piecewise WF (finite element) function $f \in \text{WF}(\{K_+, K_-\})$ with $f_\pm := f|_{K_\pm} \in \text{WF}(K_\pm) := C^1(K_\pm) \cap P_3(\widehat{\mathcal{T}}(K_\pm))$ coincide at $F = \text{conv}\{Q_1, Q_2, Q_3\}$. The latter means that for all $1 \leq \ell \leq 3$, $1 \leq j < k \leq 3$,

$$f_+(Q_\ell) = f_-(Q_\ell), \nabla f_+(Q_\ell) = \nabla f_-(Q_\ell), \nabla f_+(Q_{jk}) \times \tau_{jk} = \nabla f_-(Q_{jk}) \times \tau_{jk}. \quad (3.25)$$

(Recall $Q_{jk} := (Q_j + Q_k)/2$ is the midpoint of the edge $E_{jk} := \text{conv}\{Q_j, Q_k\}$ with unit tangent vector τ_{jk} of fixed orientation, e.g., $\tau_{jk} = (Q_k - Q_j)/|E_{jk}|$, in the triangle F .)

Theorem 3.16. *Under the present notation of 18 coinciding degrees of freedom (3.25), $f_{\pm}$ form a C^1 function $f \in C^1(K_+ \cup K_-)$ on the side patch $\omega(F) := \text{int}(K_+ \cup K_-)$ of F .*

Proof. Recall that the 2D HCT function $f_m|_{F_m} \in \text{HCT}(F_m)$ in Algorithm 1.c is computed from twelve degrees of freedom at F_m . For f_+ and f_- these twelve degrees of freedom coincide along $F_m \equiv F$ owing to the assumption (3.25). Hence, the interpolating 2D HCT function $f_{m\pm}$ in Algorithm 1.c is the same for each of the two tetrahedra $K_{\pm}$, i.e., $f_{m+}|_{K_+} = f_{m-}|_{K_-} =: f_m|_F \in \text{HCT}(F)$. Algorithm 1.c guarantees that $f_m|_F$ defines the WF function $f_{\pm}|_F$ at F . It follows $f_+|_F = f_-|_F = f_m|_F$ and so $f \in C^0(\omega(F))$. This and $f_{\pm} \in C^1(K_{\pm})$ also imply that the tangential derivatives $\nu_F \times \nabla f_+(x) = \nu_F \times \nabla f_-(x)$ coincide at any $x \in F$ owing to the Hadamard jump condition with unit normal vector ν_F of F . To conclude the proof of $f \in C^1(\omega(F))$, it remains to verify $\tau \cdot \nabla f_+(x) = \tau \cdot \nabla f_-(x)$ for the direction

$$\tau := \frac{c_{K_+} - c_F}{|c_{K_+} - c_F|} = -\frac{c_{K_-} - c_F}{|c_{K_-} - c_F|} \in \mathbb{R}^3 \setminus \{0\} \quad (3.26)$$

oblique to $\langle F \rangle$. The equality in (3.26) results from the assumption $c_{K_+} - c_F \parallel c_{K_-} - c_F$. Since $f_{\pm} \in C^1(K_{\pm})$, Lemma 3.15 implies that $g_+ := \tau \cdot \nabla f_+ \in P_2(K_+(m))$ and $g_- := \tau \cdot \nabla f_- \in P_2(K_-(m))$ are quadratic polynomials on $K_{\pm}(m) := \text{conv}\{c_{\pm}, F\}$, in particular $g_{\pm}|_F = \tau \cdot \nabla f_{\pm}|_F \in P_2(F)$. The continuity of $f_{\pm}$ at F and the equality (3.25) of the 18 degrees of freedom at F imply that the quadratic polynomials $g_+ \in P_2(K_+(m))$ and $g_- \in P_2(K_-(m))$ coincide at $Q_1, Q_2, Q_3, Q_{12}, Q_{13}$, and $Q_{23} \in F$ (see Remark 3.11 for the edge midpoints). The latter are the six Lagrange points of $P_2(F)$ in the triangle F and uniquely determine $g_{\pm}|_F \in P_2(F)$. Consequently, $g_+|_F = g_-|_F \in P_2(F)$. This shows that $\tau \cdot \nabla f$ is continuous at F and concludes the proof. $\square$

3.2.6 The shape-regularity of the WF partition

Let $\mathbb{T}$ denote the set of uniformly shape-regular triangulation $\mathcal{T}$ of a bounded polyhedral Lipschitz domain $\Omega \subset \mathbb{R}^3$ into tetrahedra: There exists a global constant $C_{\text{sr}} > 0$ with $|T|^{1/3} \leq h_T \leq C_{\text{sr}}|T|^{1/3}$ for any tetrahedra $T \in \mathcal{T} \in \mathbb{T}$ with diameter h_T and volume $|T|$. Throughout this section, let $\mathcal{V}$ denote set of vertices, $\mathcal{F}$ the set of faces, and $\mathcal{E}$ the set of edges in the triangulation $\mathcal{T} \in \mathbb{T}$. For any $E \in \mathcal{E}$ fix one unit tangent vector $\tau_E \parallel E$ and $\nu_{E,1}, \nu_{E,2} \in \mathbb{R}^3$ such that $(\tau_E, \nu_{E,1}, \nu_{E,2})$ form an orthonormal basis of $\mathbb{R}^3$.

Definition 3.17 (global WF partition). For any $T \in \mathcal{T}$, let c_T denote the midpoint of the incircle; for any interior face $F := \partial T_+ \cap \partial T_- \in \mathcal{F}(\Omega)$, let c_F be the

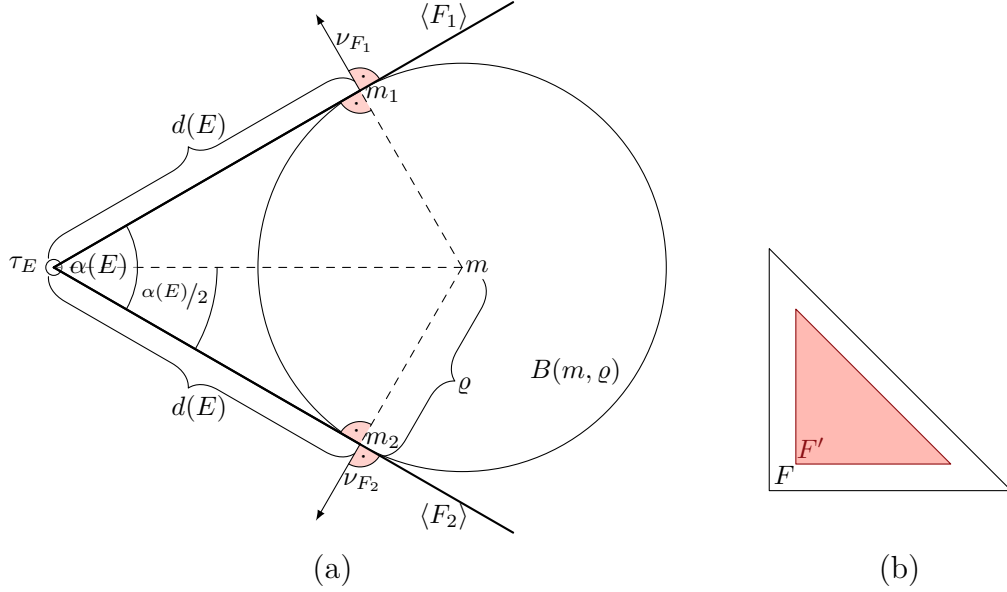


Figure 3.5: Notation in the profile of the intersection $\langle F_1 \rangle \cap \langle F_2 \rangle$ of two planes through the faces $F_1, F_2 \in \mathcal{F}(T)$ with common edge E indicated by τ_E as $\odot$ perpendicular to the plane depicted in (a) and of F' from (3.28) (in red) in (b)

intersection of F with the straight line through c_{T_+} and c_{T_-} of the neighbouring tetrahedra $T_{\pm} \in \mathcal{T}$; for any boundary face $F \in \mathcal{F}(\partial\Omega)$, let $c_F := \text{mid}(F)$ denote the center of gravity. Let $\hat{\mathcal{T}}$ denote the subtriangulation of $\mathcal{T}$, where any tetrahedron $T \in \mathcal{T}$ is partitioned in $\hat{\mathcal{T}}(T) = \text{WF3D}(T)$ according to Definition 3.8.

The center points c_T and c_F from Definition 3.17 satisfy the conditions (a)–(b) in Section 3.2.3 according to the following Theorem 3.18. In particular, the global WF partition in Definition 3.17 is well-defined.

Theorem 3.18 (shape regularity of $\hat{\mathcal{T}}$). *There exists a constant $\varepsilon > 0$ (that exclusively depends on $\mathbb{T}$), such that the points selected in Definition 3.17 satisfy $\text{dist}(c_T, \partial T) \geq \varepsilon h_T$ for all $T \in \mathcal{T} \in \mathbb{T}$ and $\text{dist}(c_F, \partial F) \geq \varepsilon h_F$ for all $F \in \mathcal{F}$. In particular, the global WF partition $\hat{\mathcal{T}}$ in Definition 3.17 is uniformly shape-regular with a shape-regularity constant that is bounded in terms of C_{sr} in the definition of $\mathbb{T}$.*

Proof. Step 1 – Preparation. First consider one tetrahedron $T \in \mathcal{T}$ with largest ball $B(m, \varrho) \subset T$ contained in T with midpoint m and radius ϱ and a common edge $E = \partial F_1 \cap \partial F_2 \in \mathcal{E}(T)$ of two of its faces $F_1, F_2 \in \mathcal{F}(T)$, $F_1 \neq F_2$.

Fig. 3.5.a illustrates the following notation. Let $\alpha(E)$ denote the dihedral angle of the distinct planes $\langle F_1 \rangle$ and $\langle F_2 \rangle$ through the faces $F_1, F_2 \in \mathcal{F}(T)$ with common

edge E . The two intersecting planes touch the ball $\bar{B}(m, \varrho)$ at the two points $m_j = \langle F_j \rangle \cap \partial B(m, \varrho) \in \text{int}(F_j)$ for $j = 1, 2$. The plane $\langle m, m_1, m_2 \rangle$ through the points m, m_1 , and m_2 is perpendicular to the unit tangent vector τ_E of the edge E . The distance $d(E)$ of m_j to the straight line $\langle E \rangle$ through E coincides for $j = 1$ and $j = 2$, $d(E) := \text{dist}(m_1, \langle E \rangle) = \text{dist}(m_2, \langle E \rangle)$, according to elementary geometry in a deltoid. The shape regularity of a tetrahedron allows many equivalent characterizations e.g. in terms of angles as discussed in [BKK08]. The lower and upper bounds of $\alpha(E)$ from [BKK08, Eq.(4)] lead in $\varrho = d(E) \tan(\alpha(E)/2)$ to the equivalence

$$d(E) \approx \varrho \approx h_T \approx h_F \quad \text{for all } E \in \mathcal{E}(T). \quad (3.27)$$

(Recall that the equivalence constants in the notation $\approx$ depend exclusively on C_{sr} .) This means that the point $m_F \in \partial B(m, \varrho) \cap F$ for the interior side F belongs to the subtriangle

$$F' := \{x \in F \mid \text{dist}(E, x) \geq d(E) \text{ for all } E \in \mathcal{E}(F)\} \ni m_F \quad (3.28)$$

depicted in Fig. 3.5.b. This and (3.27) will be employed in the sequel.

Step 2 – Distance conditions. Suppose that $F = \partial T_1 \cap \partial T_2$ is a common side of two distinct neighbouring tetrahedra $T_1, T_2 \in \mathcal{T}$. Then each T_j has a maximal incircle $B(m_j, \varrho_j) \subset T_j$ with midpoint m_j and radius ϱ_j and the touching point $m_{F,j} \in F'_j$ from (3.28). The shape regularity [BKK08] means $h_{T_1} \approx \varrho_1 \approx h_F \approx \varrho_2 \approx h_{T_2}$. Hence, the choice $m_j = c_{T_j}$ in Definition 3.17 guarantees $\varepsilon h_{T_j} \leq \text{dist}(c_{T_j}, \partial T) = \varrho_j \approx h_{T_j}$ for $j = 1, 2$ with a constant $\varepsilon \approx 1$.

Let $d_j(E)$ replaces the distance $d(E)$ for T replaced by T_j for $j = 1, 2$ in (3.28) to define F'_j . Then $F'_1 \subseteq F'_2$ or $F'_2 \subseteq F'_1$ holds. Let

$$F' := F'_1 \cup F'_2 = \left\{ x \in F \mid \text{dist}(E, x) \geq \min\{d_1(E), d_2(E)\} \text{ for all } E \in \mathcal{E}(F) \right\}$$

denote the bigger set. The face center c_F in Definition 3.17 lies in the convex hull of the two touching points $m_{F,1} = (\partial B(m_1, \varrho_1) \cap F) \in F'_1 \subseteq F'$ and $m_{F,2} = (\partial B(m_2, \varrho_2) \cap F) \in F'_2 \subseteq F'$. Since the triangle F' depicted in Fig. 3.5.b is convex, $c_F \in F'$. This and (3.27) (with $h_F \approx |F|^{1/2} \approx \varrho_1 \approx \varrho_2$ from shape regularity [BKK08]) prove that $\text{dist}(c_F, \partial F) \geq \min_{E \in \mathcal{E}(F)} \min_{j=1,2} d_j(E) \approx h_F$. This leads to $\varepsilon \approx 1$ and $\varepsilon h_F \leq \text{dist}(c_F, \partial F)$.

Step 3 – Shape regularity of $\hat{\mathcal{T}}$. Any tetrahedron $\hat{T} = \text{conv}\{c_T, c_F, Q_j, Q_k\} \in \hat{\mathcal{T}}$ is contained in one $T \in \mathcal{T}$ with center point c_T of the maximal incircle $B(c_T, \varrho) \subset T$ in T and face $F \in \mathcal{F}(T)$ such that $c_F \in \text{relint}(F)$ is the point $c_F \in F'$ from Definition 3.17 and $E = \text{conv}\{Q_j, Q_k\} \in \mathcal{E}(F)$ is an edge of F . Then

$$\begin{aligned} h_{\hat{T}} &:= \text{diam}(\hat{T}) \\ &= \max\{|c_T - c_F|, |c_T - Q_j|, |c_T - Q_k|, |c_F - Q_j|, |c_F - Q_k|, |Q_j - Q_k|\} \leq h_T. \end{aligned}$$

By construction and shape regularity of $\mathcal{T}$ with (3.27)–(3.28) it follows $h_T \approx \varrho \leq \min\{|c_T - c_F|, |c_T - Q_j|, |c_T - Q_k|\}$, $h_T \approx d(E) \leq \min\{|c_F - Q_j|, |c_F - Q_k|\}$, and $h_T \approx h_E = |Q_j - Q_k|$. In other words $h_{\hat{T}} \approx h_T$. The equivalence $h_{\hat{T}} \approx h_T \approx \varrho \approx |F|^{1/2}$ and the observation $\hat{T} = \text{conv}\{c_T, F\}$ imply

$$|\hat{T}| = \frac{|F|\varrho}{3} \approx h_T^3 \approx h_{\hat{T}}^3.$$

This guarantees the shape regularity of $\hat{\mathcal{T}}$ [BKK08, Eq.(1)] and the shape regularity constant solely depends on $\mathbb{T}$. □

3.2.7 The scaling of the WF basis functions

Throughout this section, $\hat{\mathcal{T}}$ is the WF partition of a triangulation $\mathcal{T} \in \mathbb{T}$ as in Definition 3.17 with set of vertices $\mathcal{V}$, set of faces $\mathcal{F}$, and set of edges $\mathcal{E}$. For any edge $E \in \mathcal{E}$ fix the orientation of a unit tangent vector τ_E along E as well as $\nu_{E,1}, \nu_{E,2} \in \mathbb{R}^3$ such that $(\tau_E, \nu_{E,1}, \nu_{E,2})$ form an orthonormal basis of $\mathbb{R}^3$. Set

$$\text{WF}(\mathcal{T}) := P_3(\hat{\mathcal{T}}) \cap H_0^2(\Omega). \quad (3.29)$$

Definition 3.19 (global degrees of freedom). The global $M = 4|\mathcal{V}(\Omega)| + 2|\mathcal{E}(\Omega)|$ degrees of freedom are $(L_{z,\mu}|z \in \mathcal{V}(\Omega), \mu = 1, \dots, 4)$ and $(L_{E,\mu}|E \in \mathcal{E}(\Omega), \mu = 1, 2)$ defined, for all $f \in C^1(\bar{\Omega})$, by

$$\begin{aligned} L_{z,1}f &:= f(z), \quad L_{z,\kappa+1} := \frac{\partial f}{\partial x_\kappa}(z) \quad \text{for } \kappa = 1, 2, 3; \\ L_{E,\mu}f &:= \frac{\partial f}{\partial \nu_{E,\mu}}(\text{mid}(E)) \quad \text{for } \mu = 1, 2. \end{aligned} \quad (3.30)$$

These M degrees of freedom are also enumerated as $L_1, \dots, L_M$. Note, that each degree of freedom L_ℓ for $\ell = 1, \dots, M$ is a directional derivative of order $\mu_\ell = 0$ or $\mu_\ell = 1$ at an interior node $z_\ell \in \mathcal{N}(\Omega) := \mathcal{V}(\Omega) \cup \{\text{mid}(E) : E \in \mathcal{E}(\Omega)\}$.

For each node $z_\ell \in \mathcal{N}(\Omega)$ let $\mathcal{T}(z_\ell) := \{T \in \mathcal{T} | z_\ell \in \partial T\}$ denote the set of adjacent simplices and let $\omega_\ell := \text{int}(\bigcup \mathcal{T}(z_\ell))$ denote the nodal patch with volume $h_\ell^3 := |\omega_\ell|$. Theorem 3.20 below extends the analysis for a quasi-affine family of finite elements in [Cia78, §6.1] to 3D and implies the correct scaling of the dual basis functions for $\text{WF}(\mathcal{T})$. The point is that Theorem 3.18 lead to uniform bounds in (b). Recall that $a \lesssim b$ abbreviates $a \leq Cb$ with a generic constant C that only depends the shape-regularity constant C_{sr} of $\mathcal{T} \in \mathbb{T}$. (The constants below depend on $\varepsilon > 0$ from Theorem 3.18 which depends solely on $\mathbb{T}$.)

Theorem 3.20 (scaling of WF basis functions). (a) *There exists a unique nodal basis $(\varphi_1, \dots, \varphi_M)$ of $\text{WF}(\mathcal{T})$ which satisfies $L_k(\varphi_\ell) = \delta_{k\ell}$ for $k, \ell = 1, \dots, M$. If $L_\ell = L_{z,\mu}$ for $z \in \mathcal{V}$, $\mu = 1, \dots, 4$, relabel $\varphi_{z,\mu} := \varphi_\ell$. If $L_\ell = L_{E,\mu}$ for $E \in \mathcal{E}$, $\mu = 1, 2$, relabel $\varphi_{E,\mu} := \varphi_\ell$.*

(b) Suppose the degree of freedom L_ℓ for $\ell = 1, \dots, M$ is a directional derivative of order $\mu_\ell \in \{0, 1\}$ at the node $z_\ell \in \mathcal{N}(\Omega)$ with nodal patch ω_ℓ of volume $h_\ell^3 := |\omega_\ell|$. Then

$$\|\varphi_\ell\|_{H^s(\Omega)} \lesssim h_\ell^{3/2+\mu_\ell-s} \quad \text{for all } s = 0, 1, 2.$$

Proof of (a). Consider one global degree of freedom L_ℓ with $\ell = 1, \dots, M$ from Definition 3.19 associated with the node $z_\ell \in \mathcal{N}(\Omega)$. For any $K \in \mathcal{T}(z_\ell)$ the duality translates in input data $(x_1, \dots, x_{28}) \in \mathbb{R}^{28}$ for Algorithm 1 as follows: Suppose $L_\ell = L_{E,\mu}$ for some edge $E = \text{conv}\{Q_j, Q_k\} \in \mathcal{E}(K)$ in K and $\mu \in \{1, 2\}$, i.e., $z_\ell = \text{mid}(E)$. Recall $(\tau_E, \nu_{E,1}, \nu_{E,2})$ is an orthonormal basis of $\mathbb{R}^3$ and $(\tau_{jk}, \nu_{jk,1}, \nu_{jk,2})$ for $E = E_{jk}$ denotes another basis of $\mathbb{R}^3$ in Subsection 3.2.4. Since $\tau_{jk} \parallel \tau_E$, it holds $\nu_{E,\mu} = \sum_{m=1}^2 \alpha_m \nu_{jk,m}$ for some $\alpha_1, \alpha_2 \in \mathbb{R}$. Let $x_{20-2^4-j+2k+m} = \alpha_m$ for $m = 1, 2$ (for the enumeration of Table 3.2). Algorithm 1 with input data $(0, \dots, 0, x_{20-2^4-j+2k+1}, x_{20-2^4-j+2k+2}, 0, \dots, 0) \in \mathbb{R}^{28}$ (and $\widehat{\mathcal{T}}(K)$) computes a function $\varphi_{\ell,K} \in \text{WF}(K) := P_3(\widehat{\mathcal{T}}(K)) \cap C^1(K)$ with

$$\begin{aligned} L_\ell(\varphi_{\ell,K}) &= \nu_{E,\mu} \cdot \nabla \varphi_{\ell,K}(\text{mid}(E)) = \alpha_1 \nu_{jk,1} \cdot \nabla \varphi_{\ell,K}(Q_{jk}) + \alpha_2 \nu_{jk,2} \cdot \nabla \varphi_{\ell,K}(Q_{jk}) \\ &= \alpha_1 x_{20-2^4-j+2k+1} + \alpha_2 x_{20-2^4-j+2k+2} = \alpha_1^2 + \alpha_2^2 = |\nu_{E,\mu}|^2 = 1 \end{aligned}$$

which vanishes $L_k(\varphi_{\ell,K}) = 0$ in all other degrees of freedom $k = 1, \dots, M$ with $k \neq \ell$. (For the second normal vector $\nu_{E,\kappa} = \sum_{m=1,2} \beta_m \nu_{jk,m}$ for some $\beta_1, \beta_2 \in \mathbb{R}$, $\kappa \in \{1, 2\} \setminus \{\mu\}$ holds $0 = \nu_{E,\kappa} \cdot \nu_{E,\mu} = \alpha_1 \beta_1 + \alpha_2 \beta_2$ and whence $L_{E,\kappa}(\varphi_{\ell,K}) = \sum_{m=1}^2 \beta_m x_{20-2^4-j+2k+m} = 0$.) Similarly suppose $L_\ell = L_{z,\mu}$ for some vertex $z = Q_j \in \mathcal{V}(K)$ in K and $\mu \in \{1, \dots, 4\}$. Set $x_{4(\mu-1)+j} = 1$. Algorithm 1 with input data $(0, \dots, 0, x_{4(\mu-1)+j}, 0, \dots, 0) \in \mathbb{R}^{28}$ (and $\widehat{\mathcal{T}}(K)$) computes a function $\varphi_{\ell,K} \in \text{WF}(K) := P_3(\widehat{\mathcal{T}}(K)) \cap C^1(K)$ with

$$\begin{aligned} L_\ell(\varphi_{\ell,K}) &= \varphi_{\ell,K}(Q_j) = x_j = 1 \quad \text{for } \mu = 1, \\ L_\ell(\varphi_{\ell,K}) &= e_{\mu-1} \cdot \varphi_{\ell,K}(Q_j) = x_{4(\mu-1)+j} = 1 \quad \text{for } \mu = 2, 3, 4, \end{aligned}$$

which vanishes $L_k(\varphi_{\ell,K}) = 0$ in all other degrees of freedom $k = 1, \dots, M$ with $k \neq \ell$. Let $\varphi_\ell \in P_3(\widehat{\mathcal{T}})$ denote the global function locally defined as $\varphi_\ell|_K := \varphi_{\ell,K}$ for any $K \in \mathcal{T}(z_\ell)$ (and $\varphi_\ell|_K = 0$ otherwise). In summary the functions $\varphi_1, \dots, \varphi_M \in P_3(\widehat{\mathcal{T}})$ satisfy the duality property in (a). For any two tetrahedra $K_\pm$ with common side $F = \partial K_+ \cap \partial K_- \in \mathcal{F}(\Omega)$ the respective center points $c_{K_+} \in \text{int}(K_+)$, $c_{K_-} \in \text{int}(K_-)$, and $c_F \in \text{relint}(F)$ in Definition 3.17 lie on a straight line and the values in the degrees of freedom in (3.25) along F coincide. Hence, Theorem 3.16 guarantees C^1 conformity of $\varphi_\ell|_{K_+ \cup K_-}$ on the side patch $\overline{\omega}(F) = K_+ \cup K_-$. Successive application of this argument leads to $\varphi_\ell \in C^1(\Omega)$ with support $\text{supp}(\varphi_\ell) = \overline{\omega}_\ell$ in the nodal patch of $z_\ell \in \mathcal{N}(\Omega)$. In particular this implies vanishing boundary conditions. Hence, $\varphi_\ell \in P_3(\widehat{\mathcal{T}}) \cap H_0^2(\Omega) = \text{WF}(\mathcal{T})$ is a WF function. For any $f \in \text{WF}(\mathcal{T})$ holds $f|_K \in \text{WF}(K)$ and along any interior face the values in the degrees of freedom (cf. (3.25)) coincide. Hence, the definition of $\varphi_\ell|_K = \varphi_{\ell,K}$ for any $K \in \mathcal{T}$ and Theorem 3.14 imply $f \in \text{span}\{\varphi_\ell : \ell = 1, \dots, M\}$. This concludes the proof of (a). $\square$

The proof of Theorem 3.20.b follows the methodology of [Cia78, Thm. 6.1.3] with an affine equivalent alternative finite element. Consider a tetrahedron $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\}$ with vertices $Q_1, \dots, Q_4 \in \mathcal{V}(K)$ and edges $E_{jk} := \text{conv}\{Q_j, Q_k\} \in \mathcal{E}(K)$ with midpoints $Q_{jk} := (Q_j + Q_k)/2$ for $1 \leq j < k \leq 4$.

Definition 3.21 (alternative set of linear functionals). Define $28 = 16 + 12$ linear functionals $\mathcal{L}(K) := (L_{jk}|j, k = 1, \dots, 4) \cup (L_{abc}|\{a, b, c, d\} = \{1, 2, 3, 4\}, a < b)$ on $K = \text{conv}\{Q_1, Q_2, Q_3, Q_4\}$, for $j, k = 1, \dots, 4$ and all pairwise distinct $a, b, c \in \{1, 2, 3, 4\}$ with $a < b$, by

$$L_{jk} := \begin{cases} \delta_{Q_j} & \text{if } j = k \\ (Q_k - Q_j) \cdot \delta_{Q_j} \nabla & \text{if } j \neq k \end{cases} \quad \text{and} \quad L_{abc} := (Q_c - Q_{ab}) \cdot \delta_{Q_{ab}} \nabla. \quad (3.31)$$

(Recall the point evaluation $\delta_P(f) = f(P)$ and $\delta_P \nabla(f) = \nabla f(P)$ for any $P \in \bar{\Omega}$ and $f \in C^1(\bar{\Omega})$.)

Lemma 3.22 (local basis). *There exists a constant $C > 0$ (that exclusively depends on $\mathbb{T}$), such that for any tetrahedron $K \in \mathcal{T} \in \mathbb{T}$ with WF partition $\hat{\mathcal{T}}(K)$, there exists a basis $(\varphi_{jk} : j, k = 1, \dots, 4) \cup (\varphi_{abc} : \{a, b, c, d\} = \{1, 2, 3, 4\}, a < b)$ of $\text{WF}(K)$ that satisfies the following. The 28 functions are dual to the linear functionals in $\mathcal{L}(K)$ from (3.31), in that*

$$L_{jk}(\varphi_{\ell m}) = \delta_{j\ell} \delta_{km}, \quad L_{abc}(\varphi_{def}) = \delta_{ad} \delta_{be} \delta_{cf}, \quad \text{and} \quad L_{jk}(\varphi_{abc}) = 0 = L_{abc}(\varphi_{jk}) \quad (3.32)$$

for all $j, k, \ell, m = 1, \dots, 4$, all pairwise distinct $a, b, c \in \{1, 2, 3, 4\}$ with $a < b$, and all pairwise distinct $d, e, f \in \{1, 2, 3, 4\}$ with $d < e$, and satisfy, for any $s = 0, 1, 2$, that

$$\sum_{j,k=1}^4 \|\varphi_{jk}\|_{H^s(K)} + \sum_{\substack{\{a,b,c,d\}=\{1,2,3,4\} \\ a < b}} \|\varphi_{abc}\|_{H^s(K)} \leq Ch_K^{3/2-s}. \quad (3.33)$$

Proof. Step 1 – linear independence. For each $K := \text{conv}\{Q_1, Q_2, Q_3, Q_4\} \in \mathcal{T} \in \mathbb{T}$ the linear functionals in $\mathcal{L}(K)$ from Definition 3.21 are linearly independent. The proof concerns $f \in \text{WF}(K) := P_3(\hat{\mathcal{T}}(K)) \cap C^1(K)$ with $L_{jk}f = 0 = L_{abc}f$ for all indices j, k and a, b, c . Hence, (3.31) implies that f vanishes at the vertices $Q_1, \dots, Q_4 \in \mathcal{V}(K)$ of K . Since for fixed $j \in \{1, 2, 3, 4\}$ the three vectors $Q_j - Q_k \in \mathbb{R}^3$ for $k \in \{1, 2, 3, 4\} \setminus \{j\}$ are linearly independent, (3.31) implies that ∇f vanishes at the vertices $Q_1, \dots, Q_4 \in \mathcal{V}(K)$ of K as well. Moreover the tangential derivative $\tau_{ab} \cdot \nabla f$ vanishes along each edge $E_{ab} := \text{conv}\{Q_a, Q_b\} \in \mathcal{E}(K)$ of K with unit tangent vector τ_{ab} for $1 \leq a < b \leq 4$. Since $\text{span}\{\tau_{ab}, Q_c - Q_{ab}, Q_d - Q_{ab}\} = \mathbb{R}^3$ for all $\{a, b, c, d\} = \{1, 2, 3, 4\}$, (3.31) implies that $\nabla f(Q_{ab})$ vanishes in all midpoints $Q_{ab} := (Q_a + Q_b)/2$ of edges $E_{ab} \in \mathcal{E}(K)$ of K as well. In conclusion $L_1 f = \dots = L_{28} f = 0$ for the WF degrees of freedom from Table 3.2 and so $f \equiv 0$ by Theorem 3.14. Hence, $(K, \text{WF}(K), \mathcal{L}(K))$ is a finite element in the sense of Ciarlet. Consequently, there exist nodal basis functions $\varphi_{jk} \in \text{WF}(K)$ and $\varphi_{abc} \in \text{WF}(K)$ with the required duality properties (3.33).

Step 2 – nodal basis in the reference tetrahedron $\hat{T}$. Let $\Phi : \hat{T} \rightarrow K$ denote an affine diffeomorphism from the reference tetrahedron $\hat{T} := \text{conv}\{0, e_1, e_2, e_3\}$ onto K . The center points in the WF partition $\hat{\mathcal{T}}$ in Definition 3.17 are input parameter in Algorithm 1. For each $K \in \mathcal{T}$ the center points $c_K \in \text{int}(K)$ and $c_m \in \text{relint}(F_m)$ for $m = 1, \dots, 4$ of the faces $F_1, \dots, F_4 \in \mathcal{F}(K)$ of K are mapped to five points in $\hat{T}$. Namely the point $\hat{c}_0 := \Phi^{-1}(c_K)$ and $\hat{c}_m := \Phi^{-1}(c_m) \in \hat{F}_m = \Phi^{-1}(F_m)$ for $m = 1, \dots, 4$. For each set of points $S := (\hat{c}_0, \hat{c}_1, \dots, \hat{c}_4) \in \mathbb{R}^{15}$ suppressed in the notation, there exists a set of nodal basis functions $\hat{\varphi}_{jk}$ and $\hat{\varphi}_{abc}$ for $j, k = 1, \dots, 4$ and for all $\{a, b, c, d\} = \{1, 2, 3, 4\}$ with $a < b$ associated with $\hat{\mathcal{L}} := \mathcal{L}(\hat{T})$ in (3.31). Theorem 3.18 shows that the parameters $S := (\hat{c}_0, \hat{c}_1, \dots, \hat{c}_4) \in \mathbb{R}^{15}$ belong to a compact set $\mathcal{C}$ of points in $\mathbb{R}^{15}$ such that the distance to the respective relative boundaries is positive. Note that for each fixed parameter $S \in \mathcal{C}$ the algorithm Algorithm 1 (with adapted preprocessing in step (a)) maps input data x_{jk} and x_{abc} onto a function $f \in \text{WF}(\mathcal{T})$ with $L_{jk}(f) = x_{jk}$ and $L_{abc}(f) = x_{abc}$ from $\hat{\mathcal{L}} := (L_{jk}|j, k = 1, \dots, 4) \cup (L_{abc}|\{a, b, c, d\} = \{1, 2, 3, 4\}, a < b)$. Remark 3.13 points out that the evaluation of this interpolation operator depends continuously on the parameters in the compact set $\mathcal{C}$. Hence, the norms of the nodal basis functions $\hat{\varphi}_{jk}$ and $\hat{\varphi}_{abc}$ for any $j, k = 1, \dots, 4$ and for all $\{a, b, c, d\} = \{1, 2, 3, 4\}$ with $a < b$ for the resulting WF partition of the reference tetrahedron $\hat{T}$ remain bounded by a universal constant $C_1 > 0$ as in [Cia78, Eq. (6.1.30)] for 2D,

$$\sum_{j,k=1}^4 \|\hat{\varphi}_{jk}\|_{H^s(\hat{T})} + \sum_{\substack{\{a,b,c,d\}=\{1,2,3,4\} \\ a < b}} \|\hat{\varphi}_{abc}\|_{H^s(\hat{T})} \leq C_1. \quad (3.34)$$

Note that the constant C_1 depends exclusively on $\mathcal{C}$ and so on $\varepsilon > 0$ from Theorem 3.18.

Step 3 – affine equivalence. Recall the affine diffeomorphism $\Phi : \hat{T} \rightarrow K$ from the reference tetrahedron $\hat{T} := \text{conv}\{0, e_1, e_2, e_3\}$ onto K from Step 2. The point is that the alternative finite element $(K, \text{WF}(K), \mathcal{L}(K))$ is affine equivalent to $(\hat{T}, \text{WF}(\mathcal{T}), \hat{\mathcal{L}})$ in that $\hat{\mathcal{L}} = \mathcal{L}(K) \circ \Phi$; cf. [Cia78; BS08] for the concept of affine equivalence. In particular the local nodal basis $\varphi_{jk} \in \text{WF}(K)$ and $\varphi_{abc} \in \text{WF}(K)$ associated with (3.31) and the local nodal basis functions $\hat{\varphi}_{jk} \in \text{WF}(\mathcal{T})$ and $\hat{\varphi}_{abc} \in \text{WF}(\mathcal{T})$ on the reference tetrahedron $\hat{T}$ satisfy

$$\varphi_{jk} = \hat{\varphi}_{jk} \circ \Phi^{-1} \quad \text{and} \quad \varphi_{abc} = \hat{\varphi}_{abc} \circ \Phi^{-1}$$

for any $j, k = 1, \dots, 4$ and all pairwise distinct $a, b, c \in \{1, 2, 3, 4\}$ with $a < b$. Standard scaling properties in [Cia78, Thm. 3.1.2–3.1.3] lead to a universal constant C_2 (that depends only on the reference tetrahedron $\hat{T}$) such that for any $s = 0, 1, 2$

$$\|\varphi_{abc}\|_{H^s(K)} \leq C_2 |K|^{1/2} \varrho_K^{-s} \|\hat{\varphi}_{abc}\|_{H^s(\hat{T})} \quad \text{and} \quad \|\varphi_{jk}\|_{H^s(K)} \leq C_2 |K|^{1/2} \varrho_K^{-s} \|\hat{\varphi}_{jk}\|_{H^s(\hat{T})}$$

with the radius ϱ_K of the biggest incircle in K . The shape regularity of $K \in \mathcal{T} \in \mathbb{T}$ guarantees $\varrho_K \approx h_K \approx |K|^{1/3}$ and so

$$|K|^{1/2} \varrho_K^{-s} \leq C_3 h_K^{3/2-s}.$$

The combination of the last two displayed estimates with (3.34) concludes the proof of (3.33) with $C = C_1 C_2 C_3$. □

The point in the proof of Theorem 3.20.b below is that each (global) nodal basis function φ_ℓ for $\ell = 1, \dots, M$ restricted to one tetrahedron $K \subset \text{supp}(\varphi_\ell)$ is identical with its interpolation in the local basis from Lemma 3.22, i.e.,

$$\varphi_\ell|_K = \sum_{j,k=1}^4 L_{jk}(\varphi_\ell) \varphi_{jk} + \sum_{\substack{\{a,b,c,d\}=\{1,2,3,4\} \\ a < b}} L_{abc}(\varphi_\ell) \varphi_{abc}. \quad (3.35)$$

The interpolation (3.35) and a triangle inequality show

$$\|\varphi_\ell\|_{L^2(K)} \leq \sum_{j,k=1}^4 |L_{jk}(\varphi_\ell)| \|\varphi_{jk}\|_{L^2(K)} + \sum_{\substack{\{a,b,c,d\}=\{1,2,3,4\} \\ a < b}} |L_{abc}(\varphi_\ell)| \|\varphi_{abc}\|_{L^2(K)}. \quad (3.36)$$

A careful analysis of the contributions $|L_{jk}(\varphi_\ell)|$ and $|L_{abc}(\varphi_\ell)|$ on the right-hand side of (3.36) for each of the three types of global nodal basis functions from Definition 3.19 and the application of the scaling in (3.33) shows

$$\|\varphi_\ell\|_{H^s(K)} \lesssim h_K^{3/2+\mu_\ell-s} \quad \text{for all } s = 0, 1, 2,$$

if the degree of freedom L_ℓ is a directional derivative of order $\mu_\ell \in \{0, 1\}$ at the node $z_\ell \in \mathcal{N}(K)$. Since the shape regularity of $\mathcal{T} \in \mathbb{T}$ implies $h_K^3 \approx |K|$ and the number of tetrahedra in the nodal patch ω_ℓ , the support of the basis function φ_ℓ , with volume $|\omega_\ell| = h_\ell^3$ is bounded, the last displayed estimate concludes the proof in each case.

Proof of Theorem 3.20.b for $\varphi_\ell = \varphi_{E,\mu}$ for $E \in \mathcal{E}(\Omega)$. First discuss the scaling for the basis function related to the global degree of freedom

$$L_{E,\mu} := \nu_{E,\mu} \cdot \delta_{\text{mid}(E)} \nabla \quad \text{for } \mu = 1, 2 \text{ and } E \in \mathcal{E}.$$

Consider one tetrahedron $K := \text{conv}\{Q_1, \dots, Q_4\} \in \mathcal{T}(E)$ with WF partition $\widehat{\mathcal{T}}(K)$. Suppose, without loss of generality, $E = \text{conv}\{Q_1, Q_2\} = E_{12}$. Note $\varphi_\ell|_K \in \text{WF}(K)$ satisfies $L_{E,\mu} \varphi_\ell = 1$ and all the other degrees of freedom of φ_ℓ in Definition 3.19 vanish. In particular, φ_ℓ and $\nabla \varphi_\ell$ vanish at $Q_1, \dots, Q_4$. This implies in (3.36), that

$$L_{jk}(\varphi_\ell) = 0 \quad \text{for all } j, k = 1, \dots, 4.$$

Along any edge $E_{jk} \in \mathcal{E}(K)$ with unit tangent vector τ_{jk} for $1 \leq j < k \leq 4$ consider $\varphi_\ell|_{E_{jk}} \in P_3(E_{jk})$ as a one-dimensional cubic polynomial which is uniquely determined by the values of the function and its first derivative in the endpoints of E_{jk} . In other words, the one-dimensional cubic polynomial $\varphi_\ell|_{E_{jk}} \in P_3(E_{jk})$ vanishes due to the four vanishing degrees of freedom. Hence, its one-dimensional derivative $\varphi'_\ell = \tau_{jk} \cdot \nabla \varphi_\ell = 0$ vanishes. Moreover for any edge $E_{jk} \in \mathcal{E}(K) \setminus \{E_{12}\}$ in K besides E_{12} the gradient $\nabla \varphi_\ell$ in the midpoint Q_{jk} vanishes in the direction $\nu_{jk,1}$ and $\nu_{jk,2}$. In other words $\nabla \varphi_\ell(Q_{jk}) = 0$ for all $(j, k) \neq (1, 2)$ with $1 \leq j < k \leq 4$. This implies

$$L_{abc}(\varphi_\ell) = 0 \quad \text{for all } (a, b, c) \neq (1, 2, 3) \text{ or } (a, b, c) \neq (1, 2, 4).$$

Consequently, (3.35) reads

$$\begin{aligned} \varphi_\ell &= L_{123}(\varphi_\ell)\varphi_{123} + L_{124}(\varphi_\ell)\varphi_{124} \\ &= (Q_3 - Q_{12}) \cdot \nabla \varphi_\ell(Q_{12})\varphi_{123} + (Q_4 - Q_{12}) \cdot \nabla \varphi_\ell(Q_{12})\varphi_{124}. \end{aligned}$$

Recall $\tau_{12} \cdot \nabla \varphi_\ell(Q_{12}) \equiv 0$ for the unit tangent vector $\tau_{12} \equiv \tau_E$ of the edge $E_{12} \equiv E$ and $L_{E,\kappa}(\varphi_\ell) = \nu_{E,\kappa} \cdot \nabla \varphi_\ell(Q_{12}) = 0$ for the normal $\nu_{E,\kappa}$ with $\kappa \in \{1, 2\} \setminus \{\mu\}$. Since $L_\ell(\varphi_\ell) = \nu_{E,\mu} \cdot \nabla \varphi_\ell(Q_{12}) = 1$, this implies for $k = 3, 4$

$$\begin{aligned} &|L_{12k}(\varphi_\ell)| \\ &= \left| (Q_k - Q_{12}) \cdot \left((\nu_{E,\mu} \cdot \nabla \varphi_\ell(Q_{12}))\nu_{E,\mu} + (\nu_{E,\kappa} \cdot \nabla \varphi_\ell(Q_{12}))\nu_{E,\kappa} + (\tau_E \cdot \nabla \varphi_\ell(Q_{12}))\tau_E \right) \right| \\ &= \left| (Q_k - Q_{12}) \cdot \nu_{E,\mu} \right| \leq h_K. \end{aligned}$$

For (3.36) this implies that

$$\|\varphi_\ell\|_{H^s(K)} = \|\varphi_{E,\kappa}\|_{H^s(K)} \leq h_K \sum_{k=3,4} \|\varphi_{12k}\|_{H^s(K)} \leq Ch_K^{5/2-s}$$

with the upper bound from Lemma 3.22 in the last step. As mentioned above, the last displayed estimate concludes the proof, since $\mathcal{T} \in \mathbb{T}$ is shape regular and the number of tetrahedra $|\mathcal{T}(E)|$ in the support of $\varphi_{E,\kappa}$ bounded. $\square$

Proof of Theorem 3.20.b for $\varphi_\ell = \varphi_{z,1}$ for $z \in \mathcal{V}(\Omega)$. Consider the basis function associated with

$$L_{z,1} := \delta_z \quad \text{for } z \in \mathcal{V}$$

and $K := \text{conv}\{Q_1, \dots, Q_4\} \in \mathcal{T}(z)$ with WF partition $\widehat{\mathcal{T}}(K)$. Suppose, without loss of generality, $z = Q_1$. Note $\varphi_\ell|_K \in \text{WF}(K)$ satisfies $L_{z,1}(\varphi_\ell) = \varphi_\ell(Q_1) = 1 = L_{11}(\varphi_\ell)$ and all the other degrees of freedom in Definition 3.19 of φ_ℓ vanish. Since φ_ℓ vanishes at Q_2, Q_3, Q_4 and $\nabla \varphi_\ell$ vanish at Q_1, Q_2, Q_3, Q_4 , (3.31) implies

$$L_{kk}(\varphi_\ell) = 0 \quad \text{and} \quad L_{jk}(\varphi_\ell) = 0 \quad \text{for } 1 \leq j < k \leq 4.$$

For any edge $E \in \mathcal{E}$ with midpoint $\text{mid}(E)$ holds $L_{E,1}(\varphi_\ell) = \nu_{E,1} \cdot \nabla f(\text{mid}(E)) = 0 = \nu_{E,2} \cdot \nabla \varphi_\ell(\text{mid}(E)) = L_{E,2}(\varphi_\ell)$. This implies for all pairwise distinct $a, b, c \in \{1, 2, 3, 4\}$ with $a < b$ and edges $E_{ab} \in \mathcal{E}(K)$ with unit tangent vector τ_{ab} and midpoint Q_{ab} , that

$$L_{abc}(\varphi_\ell) = (Q_c - Q_{ab}) \cdot \nabla \varphi_\ell(Q_{ab}) = (Q_c - Q_{ab}) \cdot (\tau_{ab} \cdot \nabla \varphi_\ell(Q_{ab}))\tau_{ab}.$$

Along any $E_{ab} \in \mathcal{E}(K)$ for $2 \leq a < b \leq 4$ in K without the vertex Q_1 with unit tangent vector τ_{ab} , consider $\varphi_\ell|_{E_{ab}} \in P_3(E_{ab})$ as a one-dimensional cubic polynomial. The degrees of freedom of this one-dimensional cubic polynomial vanish. Hence, the one-dimensional derivative $\varphi'_\ell = \tau_{ab} \cdot \nabla \varphi_\ell = 0$ vanishes and

$$L_{abc}(\varphi_\ell) = (Q_c - Q_{ab}) \cdot (\tau_{ab} \cdot \nabla \varphi_\ell(Q_{ab}))\tau_{ab} = 0 \quad \text{for } 2 \leq a < b \leq 4$$

and (3.35) reads

$$\varphi_\ell = L_{11}(f)\varphi_{11} + \sum_{\substack{2 \leq b, c \leq 4 \\ b \neq c}} L_{1bc}\varphi_{1bc} = \varphi_{11} + \sum_{\substack{2 \leq b, c \leq 4 \\ b \neq c}} (Q_c - Q_{1b}) \cdot ((\tau_{1b} \cdot \nabla \varphi_\ell(Q_{1b}))\tau_{1b})\varphi_{1bc}.$$

Along any edge $E = E_{1b} \in \mathcal{E}(K)$ for $b = 2, 3, 4$ with unit tangent vector $\tau_E \equiv \tau_{1b}$ consider $\varphi_\ell|_E \in P_3(E)$ as one-dimensional function uniquely determined by the given values of φ_ℓ and the tangential derivative $\tau_E \cdot \nabla \varphi_\ell$ in the endpoints Q_1 and Q_b of E . The parametrization $f_\tau(s)$ for $s \in (0, h)$ of $\varphi_\ell|_E$ with $f_\tau(0) = \varphi_\ell(Q_1) = 1$, $f_\tau(h) = \varphi_\ell(Q_b) = 0$, and derivative $f'_\tau = \tau_E \cdot \nabla \varphi_\ell$ vanishing in both endpoints, reads

$$f_\tau(s) = h^{-3}(s-h)^2(2s+h) \quad \text{with} \quad f'_\tau(s) = 6h^{-3}s(s-h).$$

Hence, $|\tau_E \cdot \nabla \varphi_\ell(Q_{1b})| = |f'_\tau(h/2)| = 1.5h^{-1} \leq C_4 h_K^{-1}$. In other words, for all $b = 2, 3, 4$,

$$|L_{1bc}(\varphi_\ell)| = |(Q_c - Q_{1b}) \cdot (\tau_{1b} \cdot \nabla \varphi_\ell(Q_{1b}))\tau_{1b}| \leq |(Q_c - Q_{1b}) \cdot \tau_{1b}| C_4 h_K^{-1} \leq C_4.$$

For (3.36) this implies that

$$\begin{aligned} \|\varphi_\ell\|_{H^s(K)} &= \|\varphi_{z,1}\|_{H^s(K)} = \|\varphi_{11}\|_{H^s(K)} + C_4 \sum_{\substack{2 \leq b, c \leq 4 \\ b \neq c}} \|\varphi_{1bc}\|_{H^s(K)} \\ &\leq \max\{1, C_4\} C h_K^{3/2-s} \end{aligned}$$

with the upper bound from Lemma 3.22 in the last step. As mentioned above this, the shape regularity of $\mathcal{T} \in \mathbb{T}$, and the boundedness of the number of tetrahedra $|\mathcal{T}(z)|$ in the support of $\varphi_{z,1}$ conclude the proof. $\square$

Proof of Theorem 3.20.b for $\varphi_\ell = \varphi_{z,\kappa+1}$ for $z \in \mathcal{V}(\Omega)$. Finally, regard the basis function associated with

$$L_{z,\kappa+1} := e_\kappa \cdot \delta_z \nabla \quad \text{for } \kappa = 1, 2, 3, \text{ and } z \in \mathcal{V}$$

and consider $K := \text{conv}\{Q_1, \dots, Q_4\} \in \mathcal{T}(z)$ with WF partition $\hat{\mathcal{T}}(K)$. Suppose, without loss of generality, $z = Q_1$. Note $\varphi_\ell|_K \in \text{WF}(K)$ satisfies $L_{z, \kappa+1}\varphi_\ell = \partial\varphi_\ell/\partial x_\kappa = 1$ and all the other degrees of freedom of φ_ℓ vanish. Since φ_ℓ vanishes at Q_1, Q_2, Q_3, Q_4 and $\nabla\varphi_\ell$ vanish at Q_2, Q_3, Q_4 , the definition in (3.31) shows

$$L_{jj}(\varphi_\ell) = 0 \quad \text{for } j = 1, \dots, 4 \quad \text{and} \quad L_{jk}(\varphi_\ell) = 0 \quad \text{for } 2 \leq j < k \leq 4.$$

For any edge $E \in \mathcal{E}$ with midpoint $\text{mid}(E)$ holds $L_{E,1}(\varphi_\ell) = \nu_{E,1} \cdot \nabla\varphi_\ell(\text{mid}(E)) = 0 = \nu_{E,2} \cdot \nabla\varphi_\ell(\text{mid}(E)) = L_{E,2}(\varphi_\ell)$. This implies as above for any edge $E_{ab} \in \mathcal{E}(K)$ with unit tangent vector τ_{ab} and midpoint Q_{ab} , that

$$L_{abc}(f) = (Q_c - Q_{ab}) \cdot \nabla\varphi_\ell(Q_{ab}) = (Q_c - Q_{ab}) \cdot (\tau_{ab} \cdot \nabla\varphi_\ell(Q_{ab}))\tau_{ab}.$$

Along any $E_{ab} \in \mathcal{E}(K)$ for $2 \leq a < b \leq 4$ in K without the vertex Q_1 , $\varphi_\ell|_{E_{ab}} \in P_3(E_{ab})$ considered as a one-dimensional cubic polynomial vanishes. Hence, the one-dimensional derivative $\varphi'_\ell = \tau_{ab} \cdot \nabla\varphi_\ell = 0$ vanishes and

$$L_{abc}(\varphi_\ell) = (Q_c - Q_{ab}) \cdot (\tau_{ab} \cdot \nabla\varphi_\ell(Q_{ab}))\tau_{ab} = 0 \quad \text{for } 2 \leq a < b \leq 4.$$

Therefore (3.35) reads

$$\begin{aligned} \varphi_\ell &= \sum_{k=2}^4 L_{1k}(\varphi_\ell)\varphi_{1k} + \sum_{\substack{2 \leq b, c \leq 4 \\ b \neq c}} L_{1bc}(\varphi_\ell)\varphi_{1bc} \\ &= \sum_{k=2}^4 (Q_k - Q_1) \cdot \nabla\varphi_\ell(Q_1)\varphi_{1k} + \sum_{\substack{2 \leq b, c \leq 4 \\ b \neq c}} (Q_c - Q_{1b}) \cdot ((\tau_{1b} \cdot \nabla\varphi_\ell(Q_{1b}))\tau_{1b})\varphi_{1bc}. \end{aligned}$$

Since the partial derivative $\partial\varphi_\ell/\partial x_m(Q_1) = 0$ for $m \in \{1, 2, 3\} \setminus \{\kappa\}$ vanishes and $\partial\varphi_\ell/\partial x_\kappa(Q_1) = L_{z, \kappa+1}(\varphi_\ell) = 1$, it holds for $k = 2, 3, 4$

$$|L_{1k}(\varphi_\ell)| = |(Q_k - Q_1) \cdot \nabla\varphi_\ell(Q_1)| = |(Q_k - Q_1) \cdot e_\kappa \partial\varphi_\ell/\partial x_\kappa(Q_1)| = |(Q_k - Q_1) \cdot e_\kappa| \leq h_K.$$

Along any edge $E = E_{1b} \in \mathcal{E}(K)$ for $b = 2, 3, 4$ with unit tangent vector $\tau_E = \tau_{1b}$ consider $\varphi_\ell|_E \in P_3(E)$ as one-dimensional function uniquely determined by the given values of φ_ℓ and the tangential derivative $\tau_E \cdot \nabla\varphi_\ell$ in the endpoints Q_1 and Q_b of E . Let $c := e_\kappa \cdot \tau_E \leq 1$. The parametrization $f_\tau(s)$ for $s \in (0, h)$ of $f|_E$ with $f_\tau(0) = \varphi_\ell(Q_1) = 0$, $f_\tau(h) = \varphi_\ell(Q_b) = 0$, and $f'_\tau = \tau \cdot \nabla\varphi_\ell$ with $f'_\tau(0) = c$ and $f'_\tau(h) = 0$, reads

$$f_\tau(s) = ch^{-2}(s-h)^2s \quad \text{with} \quad f'_\tau(s) = ch^{-2}(3s^2 - 4hs + h^2).$$

Hence, $|\tau_{1b} \cdot \nabla\varphi_\ell(Q_{1b})| = |f'_\tau(h/2)| = c/4 \leq 1$ and

$$|L_{1bc}(f)| = |(Q_c - Q_{1b}) \cdot (\tau_{1b} \cdot \nabla\varphi_\ell(Q_{1b}))\tau_{1b}| \leq |(Q_c - Q_{1b}) \cdot \tau_{1b}| \leq h_K.$$

For (3.36) this implies that

$$\|\varphi_\ell\|_{H^s(K)} = \|\varphi_{z,\kappa+1}\|_{H^s(K)} \leq h_K \sum_{k=2}^4 \|\varphi_{1k}\|_{H^s(K)} + h_K \sum_{\substack{2 \leq b,c \leq 4 \\ b \neq c}} \|\varphi_{1bc}\|_{H^s(K)} \leq Ch_K^{5/2-s}$$

with the upper bound from Lemma 3.22 in the last step. In combination with the shape regularity of $\mathcal{T} \in \mathbb{T}$ and the boundedness of the number of tetrahedra $|\mathcal{T}(z)|$ in the support of $\varphi_{z,\kappa+1}$, this concludes the proof. $\square$

3.3 Crouzeix-Raviart

The nonconforming P_1 finite element space named Crouzeix-Raviart finite element space [CR73; Bre15] is well known in the literature. This section summarizes the properties relevant throughout this thesis. After a short introduction to the space, Subsection 3.3.1 defines the interpolation operator and recalls its properties. Subsection 3.3.2 designs an appropriate conforming companion.

For any admissible triangulation $\mathcal{T} \in \mathbb{T}$ of $\Omega \subset \mathbb{R}^n$ with set $\mathcal{F}$ of sides ($(n-1)$ -subsimplices), the Crouzeix-Raviart finite element (with and without boundary conditions) reads

$$\begin{aligned} \text{CR}^1(\mathcal{T}) &:= \{v_{\text{CR}} \in P_1(\mathcal{T}) : v_{\text{CR}} \text{ is continuous at } \text{mid}(F) \text{ for all } F \in \mathcal{F}(\Omega)\}, \\ \text{CR}_0^1(\mathcal{T}) &:= \{v_{\text{CR}} \in \text{CR}^1(\mathcal{T}) : v_{\text{CR}}(\text{mid}(F)) = 0 \text{ for all } F \in \mathcal{F}(\partial\Omega)\}. \end{aligned}$$

The local degrees of freedom for any $T \in \mathcal{T}$ are the integral means of the function along the sides $F \in \mathcal{F}(T)$ in Table 3.3. The global degrees of freedom are labelled, for any $f \in H^1(\Omega)$, by

$$L_F(f) := \oint_F f \, ds \text{ for any } F \in \mathcal{F}.$$

Throughout this work, the dual side-oriented basis of $\text{CR}^1(\mathcal{T})$ is utilized. For any $T \in \mathcal{T}$, recall the barycentric coordinates $\lambda_P \in P_1(T)$ with $\lambda_P(Q) = \delta_{PQ}$

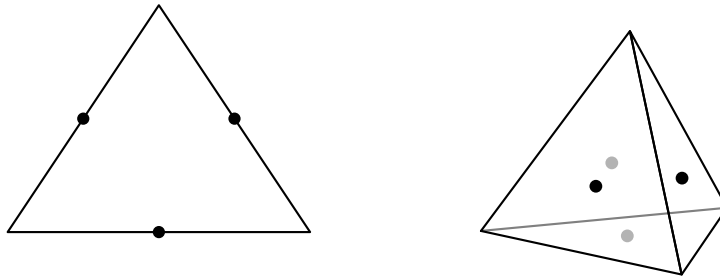


Figure 3.6: Mnemonic diagrams of the Crouzeix-Raviart finite element in 2D and 3D

$$L_j(f) := \int_{F_j} f \, ds \quad \text{for any } j = 1, \dots, |\mathcal{F}(T)|$$

Table 3.3: The local degrees of freedom for the Crouzeix-Raviart finite element for any $T \in \mathcal{T}$ and $f \in H^1(T)$

for all vertices $P, Q \in \mathcal{V}(T)$. The side-oriented Crouzeix-Raviart basis function associated with the side $F \in \mathcal{F}(T)$ opposite the vertex $P \in \mathcal{V}(T)$ in any simplex $T \in \mathcal{T}(F)$ reads

$$\psi_F|_T := 1 - n\lambda_P \quad (3.37)$$

and satisfies $L_E(\psi_F) = \int_E \psi_F \, ds = \psi_F(\text{mid}E) = \delta_{EF}$ for any $E \in \mathcal{F}$.

Remark 3.23 (Properties of $\text{CR}^1(\mathcal{T})$). (a) Given the local degrees of freedom in Table 3.3, the Crouzeix-Raviart element $(T, P_1(T), (L_1, \dots, L_{n+1}))$ is a finite element in the sense of Ciarlet, e.g., [EG04, Prop. 1.37].

(b) Let $T := \text{conv}\{P_1, \dots, P_{n+1}\} \in \mathcal{T}$, suppose that the side $F_j \in \mathcal{F}(T)$ lies opposite to the vertex P_j in T , and abbreviate $\psi_j := \psi_{F_j}$ and $\lambda_j := \lambda_{P_j}$ for any $j = 1, \dots, n+1$. The integration formula for barycentric coordinates (3.2) allows the straight forward computation of the local mass matrix. The mass matrix $M_{\text{CR}}(T) \in \mathbb{R}^{(n+1) \times (n+1)}$ has for any $j, k = 1, \dots, n+1$ the entries

$$\begin{aligned} (M_{\text{CR}}(T))_{jk} &:= \int_T \psi_j \psi_k \, dx = \int_T (1 - n\lambda_j - n\lambda_k + n^2\lambda_j\lambda_k) \, dx \\ &= |T| \left(1 - \frac{n!(2n)}{(n+1)!} + \frac{n!(1 + \delta_{jk})n^2}{(n+2)!} \right) = \frac{|T|(2 - n + n^2\delta_{jk})}{(n+1)(n+2)}. \end{aligned} \quad (3.38)$$

Since $\nabla \psi_j|_T = -n\nabla \lambda_j$, the definition of the barycentric coordinates (3.1) implies that the stiffness matrix $A_{\text{CR}}(T) \in \mathbb{R}^{(n+1) \times (n+1)}$ has the form

$$\begin{aligned} (A_{\text{CR}}(T))_{j,k=1,\dots,n+1} &:= \left(\int_T \nabla \psi_j \cdot \nabla \psi_k \, dx \right)_{j,k=1,\dots,n+1} = n^2 Q^\top Q \\ \text{with } Q &:= \begin{pmatrix} 1 & \cdots & 1 \\ P_1 & \cdots & P_{n+1} \end{pmatrix}^{-1} \begin{pmatrix} 0 & \cdots & 0 \\ I_{n \times n} \end{pmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}. \end{aligned} \quad (3.39)$$

3.3.1 Interpolation operator

Let $V := H_0^1(\Omega)$, where $\Omega \subset \mathbb{R}^n$ denotes a bounded Lipschitz domain with admissible triangulation $\mathcal{T} \in \mathbb{T}$ and piecewise (w.r.t. $\mathcal{T}$) energy scalar product $a_{\text{pw}}(\bullet, \bullet) := \int_\Omega (\nabla_{\text{pw}} \bullet \cdot \nabla_{\text{pw}} \bullet) \, dx$ inducing the energy norm $\|\bullet\|_{\text{pw}}^2 := a_{\text{pw}}(\bullet, \bullet)$.

Suppose $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ denotes an admissible refinement of $\mathcal{T} \in \mathbb{T}$ and recall the side-oriented nodal basis functions $\psi_F \in \text{CR}_0^1(\mathcal{T})$ from (3.37) for any $F \in \mathcal{F}(\Omega)$. Define the Crouzeix-Raviart interpolation operator $I_{\text{CR}} : V + \text{CR}_0^1(\hat{\mathcal{T}}) \rightarrow \text{CR}_0^1(\mathcal{T})$ as follows

$$I_{\text{CR}}(v) := \sum_{F \in \mathcal{F}(\Omega)} \int_F v \, ds \, \psi_F \quad \text{for any } v \in V + \text{CR}_0^1(\hat{\mathcal{T}}). \quad (3.40)$$

Hence, for all $F \in \mathcal{F}$ holds $L_F(I_{\text{CR}}v) = \int_F I_{\text{CR}}v \, ds = \int_F v \, ds = L_F(v)$. The following properties of the interpolation are known from the literature [CG14a; CG14b; CH18a; CZ20] with explicit constant κ_1 in any space dimension.

Theorem 3.24 (properties of I_{CR}). *Let $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ denote an admissible refinement of $\mathcal{T} \in \mathbb{T}$. There exist constants $\kappa_1, \hat{\kappa}_1 > 0$ such that the following statements hold.*

(a) *Any $v \in V + \text{CR}_0^1(\hat{\mathcal{T}})$ satisfies $\Pi_0 \nabla_{\text{pw}} v = \nabla_{\text{pw}} I_{\text{CR}} v$, in particular*

$$a_{\text{pw}}(v - I_{\text{CR}}v, w_{\text{CR}}) = 0 \quad \text{for all } w_{\text{CR}} \in \text{CR}_0^1(\mathcal{T}) \text{ and } v \in V + \text{CR}_0^1(\hat{\mathcal{T}}).$$

Hence, I_{CR} is the a_{pw} -orthogonal projection onto $\text{CR}_0^1(\mathcal{T})$ with

$$\|v - I_{\text{CR}}v\|_{\text{pw}} = \min_{v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})} \|v - v_{\text{CR}}\|_{\text{pw}} \quad \text{for any } v \in V + \text{CR}_0^1(\hat{\mathcal{T}}).$$

(b) *Any $v \in H^1(T)$ for $T \in \mathcal{T}$ satisfies*

$$\|v - I_{\text{CR}}v\|_{L^2(T)} \leq \kappa_1 h_T |v - I_{\text{CR}}v|_{H^1(T)}.$$

(c) *Any $\hat{v}_{\text{CR}} \in \text{CR}_0^1(\hat{\mathcal{T}})|_T$ in $T \in \mathcal{T}$ satisfies*

$$\|\hat{v}_{\text{CR}} - I_{\text{CR}}\hat{v}_{\text{CR}}\|_{L^2(T)} \leq \hat{\kappa}_1 h_T |\hat{v}_{\text{CR}} - I_{\text{CR}}\hat{v}_{\text{CR}}|_{H^1(T)}.$$

(d) *The operator I_{CR} acts as identity in non-refined simplices in that*

$$(1 - I_{\text{CR}})\hat{v}_{\text{CR}}|_T = 0 \quad \text{in } T \in \mathcal{T} \cap \hat{\mathcal{T}} \text{ for all } \hat{v}_{\text{CR}} \in \text{CR}_0^1(\hat{\mathcal{T}}).$$

The interpolation operator $\hat{I}_{\text{CR}}$ associated with $\text{CR}_0^1(\hat{\mathcal{T}})$ satisfies $I_{\text{CR}} \circ \hat{I}_{\text{CR}} = I_{\text{CR}}$ in $V + \text{CR}_0^1(\hat{\mathcal{T}})$.

Proof of (a). The first claim, $\Pi_0 \nabla_{\text{pw}} v = \nabla_{\text{pw}} I_{\text{CR}} v$, is a direct consequence from the definition (3.40) and a piecewise integration by parts. For any $T \in \mathcal{T}$ with admissible triangulation $\hat{\mathcal{T}}(T)$ and sides $\hat{\mathcal{F}}(T) := \{F \in \hat{\mathcal{F}} : F \subset \bar{T}\}$ holds

$$\begin{aligned} \int_T \nabla_{\text{pw}} v - \nabla I_{\text{CR}}v \, dx &= \sum_{\hat{T} \in \hat{\mathcal{T}}(T)} \int_{\hat{T}} \nabla v - \nabla I_{\text{CR}}v \, dx = \sum_{F \in \hat{\mathcal{F}}(T)} \nu_F \int_F [v]_F - [I_{\text{CR}}v]_F \, ds \\ &= \sum_{F \in \mathcal{F}(T)} \nu_T|_F \int_F v_F - I_{\text{CR}}v \, ds = 0, \end{aligned}$$

since the normal jump of $v \in V + \text{CR}_0^1(\hat{\mathcal{T}})$ and $I_{\text{CR}}v \in \text{CR}_0^1(\mathcal{T})$ vanishes for any side $F \in \hat{\mathcal{F}} \setminus \mathcal{F}$ that belongs only to the fine triangulation $\hat{\mathcal{T}}$. Finally, $w_{\text{CR}} \in \text{CR}_0^1(\mathcal{T}) \subset P_1(\mathcal{T})$ implies $\nabla_{\text{pw}} w_{\text{CR}} \in P_0(\mathcal{T}; \mathbb{R}^n)$ and concludes the proof. $\square$

Proof of (b). Theorem 2.1 in [CG14b] proves this estimate in 2D with the analytical constant $\kappa_1^2 \leq 1/8 + j_{1,1}^{-1} \leq 0.1932$ for the first positive root $j_{1,1} = 3.8317059702$ of the Besselfunction of the first kind [Tim85]. Theorem 3 in [CG14a] improves the analytical upper bound to $\kappa_1 \leq \sqrt{1/48 + j_{1,1}^{-1}} = 0.298234942888$. The numerical bound $\kappa_1 < 0.1893$ from [Liu15] is a slight improvement in 2D. In any space dimension $n > 2$ [CZZ20, Lem. A.1] computes the analytical bound $\kappa_1 \leq \sqrt{1/(2n(n+1)(n+2)) + \pi^{-2}}$. $\square$

Proof of (c). This is Theorem 3.5 in [CH18a] with explicit constant $\hat{\kappa}_1 = \sqrt{19/48} \leq 0.629153$ for $n = 2$ and $\hat{\kappa}_1 = \sqrt{101/180} \leq 0.749074$ for $n = 3$. $\square$

Proof of (d). The first part is, e.g., property (C3) in [CP20, Sec. 4.4]. The restriction of $\hat{v}_{\text{CR}} \in \text{CR}_0^1(\hat{\mathcal{T}})$ to $T \in \mathcal{T} \cap \hat{\mathcal{T}}$ is contained in $\hat{v}_{\text{CR}}|_T \in \text{span}\{\psi_F|_T : F \in \mathcal{F}(T)\}$ with the side-oriented shape functions $\psi_F \in \text{CR}^1(\mathcal{T})$ from (3.37). The duality property $\int_E \psi_F ds = \delta_{EF}$ for all $E, F \in \mathcal{F}$ implies the claim $I_{\text{CR}} \hat{v}_{\text{CR}}|_T = \hat{v}_{\text{CR}}|_T$. The second claim is a direct consequence from the duality property of the side-oriented basis functions on the fine level and the linearity of the integral. $\square$

3.3.2 Conforming companion

This section defines a conforming companion operator $J_{\text{CR}} : \text{CR}_0^1(\mathcal{T}) \rightarrow V := H_0^1(\Omega)$ for the Crouzeix-Raviart finite element that satisfies Theorem 3.25 below. The conforming companion operator J_{CR} is seen as a right-inverse of the Crouzeix-Raviart interpolation operator I_{CR} from (3.40) with an additional L^2 orthogonality.

Theorem 3.25 (properties of J_{CR}). *There exist constants $M_1, M_2, M_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) and a conforming companion $J_{\text{CR}} v_{\text{CR}} \in V$ for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$ with*

(a) $\|h_K^{-1}(1 - J_{\text{CR}})v_{\text{CR}}\|_{L^2(K)}^2 \leq M_1 \sum_{T \in \mathcal{T}(\Omega(K))} |K|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2$ for any $K \in \mathcal{T}$ with the set $\mathcal{T}(\Omega(K)) := \{T \in \mathcal{T} : \text{dist}(T, K) = 0\}$ of adjacent simplices, and the tangential jumps $[\nabla v_{\text{CR}}]_F \times \nu_F$ of the (piecewise) gradient along $F \in \mathcal{F}$,

(b) $\|h_{\mathcal{T}}^{-1}(1 - J_{\text{CR}})v_{\text{CR}}\|_{L^2(\Omega)} + \|(1 - J_{\text{CR}})v_{\text{CR}}\|_{\text{pw}} \leq \left(M_2 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \leq M_3 \min_{v \in V} \|v_{\text{CR}} - v\|_{\text{pw}},$

(c) J_{CR} is a right inverse to the interpolation I_{CR} in that $I_{\text{CR}} \circ J_{\text{CR}} = 1$ in $\text{CR}_0^1(\mathcal{T})$,

(d) the orthogonality $(1 - J_{\text{CR}})(\text{CR}_0^1(\mathcal{T})) \perp P_1(\mathcal{T})$ holds in $L^2(\Omega)$.

The design of the conforming companion is a generalization of [CGS15, Prop. 2.3] to any space dimension $n \geq 2$ and consists of three steps illustrated in Subsections 3.3.2.1–3.3.2.3. First an averaging operator also known as enrichment

operator is introduced, then appropriate corrections with side-bubble functions ensure the right-inverse property, and finally a correction with an explicit Riesz representation guarantees the L^2 -orthogonality.

3.3.2.1 Design of J_1

Definition 3.26 (enrichment J_1). First define $J_1 : \text{CR}_0^1(\mathcal{T}) \rightarrow S_0^1(\mathcal{T})$ by nodal averaging for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$,

$$(J_1 v_{\text{CR}})(z) := |\mathcal{T}(z)|^{-1} \sum_{T \in \mathcal{T}(z)} v_{\text{CR}}|_T(z) \quad \text{for all } z \in \mathcal{V}(\Omega),$$

followed by linear interpolation (plus homogeneous boundary conditions).

This is called an enrichment operator in [BS08] and also considered in the medius analysis in [Gud10; CPS12].

Lemma 3.27 (properties of J_1). *There exist constants $C_1, C_2, C_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) such that the enrichment operator $J_1 v_{\text{CR}} \in S_0^1(\mathcal{T})$ in Definition 3.26 satisfies for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$ and $\mathcal{T} \in \mathbb{T}$,*

(a) $\|h_K^{-1}(1 - J_1)v_{\text{CR}}\|_{L^2(K)}^2 \leq C_1 \sum_{T \in \mathcal{T}(\Omega(K))} |K|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2$
for any $K \in \mathcal{T}$ with the set $\mathcal{T}(\Omega(K))$ of adjacent simplices,

(b) $\|h_{\mathcal{T}}^{-1}(1 - J_1)v_{\text{CR}}\|_{L^2(\Omega)} + \|(1 - J_1)v_{\text{CR}}\|_{\text{pw}} \leq \left(C_2 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \leq C_3 \min_{v \in V} \|v_{\text{CR}} - v\|_{\text{pw}}.$

Proof of (b). The approximation property is well-known for the enrichment operator J_1 in 2D, e.g., [Car+12, Thm. 5.1] proves

$$\begin{aligned} \|h_{\mathcal{T}}^{-1}(1 - J_1)v_{\text{CR}}\|_{L^2(\Omega)}^2 &\lesssim \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 \\ &\lesssim \min_{v \in V} \|v_{\text{CR}} - v\|_{\text{pw}}^2. \end{aligned} \tag{3.41}$$

This estimate and an inverse estimate [BS08] conclude the proof. Similar arguments to [Car+12, Thm. 5.1] with an extended Curl definition show the property (3.41) for $n = 3$, cf. [CBJ02, Thm 3.2]. A generalization (of the local version of the first estimate) in (3.41) to higher dimensions is also outlined in the proof of [CGS13, Thm. 4.9].

Alternatively, the summation of the local approximation property (a) over all $K \in \mathcal{T}$ and a finite overlap argument for shape regular triangulation $\mathcal{T} \in \mathbb{T}$ lead to the first estimate, while the second claim follows with the arguments below

similar to [CP20, Lem. 3.1]. For each $K \in \mathcal{T}$ with adjacent simplices in $\Omega(K)$ and $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ define

$$\begin{aligned}\varrho_K(v_{\text{CR}}) &:= \left(|K|^{1/n} \sum_{F \in \mathcal{F}(\Omega(K))} \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \\ \mu_K(v_{\text{CR}}) &:= \min_{v \in V|_{\Omega(K)}} |v_{\text{CR}} - v|_{H^1(\Omega(K))}.\end{aligned}$$

Since the tangential derivative $[\nabla v]_F \times \nu_F = 0$ does not jump for any $v \in V$ and $F \in \mathcal{F}$, $\varrho_K(v_{\text{CR}}) = 0$ vanishes for any $v_{\text{CR}} \in (\text{CR}_0^1(\mathcal{T}) \cap V)|_{\Omega(K)}$. On the other hand $\mu_K(v_{\text{CR}}) = 0$ implies $v_{\text{CR}} \in (\text{CR}_0^1(\mathcal{T}) \cap V)|_{\Omega(K)}$. Since $\text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ is finite dimensional, this leads to some constant $C(K)$ depending on $\mathcal{T}(\Omega(K))$ and $\mathcal{F}(\Omega(K))$ such that

$$\varrho_K(v_{\text{CR}}) \leq C(K) \mu_K(v_{\text{CR}}) \quad \text{for all } v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}.$$

For any $\mathcal{T} \in \mathbb{T}$ there are only finitely many configurations of $\mathcal{T}(\Omega(K))$ possible [CP20, (A2)], hence $C(\mathbb{T}) := \sup_{K \in \mathcal{T} \in \mathbb{T}} C(K) < \infty$. This and a finite overlap argument conclude the proof. $\square$

Proof of (a). A closer investigation of the proofs in [Car+12; CBJ02; CGS13] reveals that the arguments are local and show (a). Alternatively, an inverse estimate argument comparable to the application of Lemma 3.1 in [CP20] is possible. For convenient reading, the arguments are summarized here. For each $K \in \mathcal{T}$ with adjacent simplices in $\Omega(K)$ and $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ define

$$\begin{aligned}\mu_K(v_{\text{CR}}) &:= \|h_K^{-1}(1 - J_1)v_{\text{CR}}\|_{L^2(K)} \\ \varrho_K(v_{\text{CR}}) &:= \left(\sum_{F \in \mathcal{F}(\Omega(K))} h_F \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2}.\end{aligned}$$

Definition 3.26 implies that $v_{\text{CR}} = J_1(v_{\text{CR}}|_{\Omega(K)})$ for all $v_{\text{CR}} \in (\text{CR}_0^1(\mathcal{T}) \cap V)|_{\Omega(K)}$. Hence, $(\text{CR}_0^1(\mathcal{T}) \cap V)|_{\Omega(K)}$ belongs to the kernel $\text{Ker}(\mu_K)$ of μ_K .

Since any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ is piecewise affine, continuous at the side midpoints, and vanishes at midpoints of boundary sides $F \subset \partial\Omega \cap \partial(\Omega(K))$, the jump $[v_{\text{CR}}]_F$ across each side $F \in \mathcal{F}$ is of the form $[v_{\text{CR}}]_F(x) = a \cdot (x - \text{mid}(F))$ for some $a \in \mathbb{R}^n$ and any $x \in F$. Since $a = [\nabla v_{\text{CR}}]_F$ and the normal ν_F is perpendicular to $(x - \text{mid}(F)) \perp \nu_F$ at any $x \in F$, the jumps $[v_{\text{CR}}]_F \equiv 0$ vanish if and only if the tangential jumps of the gradients $\|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 = 0$ vanish. Therefore, $\sum_{F \in \mathcal{F}(\Omega(K))} h_F \|[\nabla v_{\text{CR}}]_F \times \nu_F\|_{L^2(F)}^2 = 0$ implies that v_{CR} is continuous in $\Omega(K)$ and vanishes along each boundary side $F \in \mathcal{F}(\partial\Omega) \cap \mathcal{F}(\partial\Omega(K))$. This proves that $\varrho_K(v_{\text{CR}}) = 0$ implies for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ that $v_{\text{CR}} \in (\text{CR}_0^1(\mathcal{T}) \cap V)|_{\Omega(K)}$.

Since $\text{CR}_0^1(\mathcal{T})|_{\Omega(K)}$ is finite dimensional, this leads to some constant $C(K)$ depending on $\mathcal{T}(\Omega(K))$ and $\mathcal{F}(\Omega(K))$ such that

$$\mu_K(v_{\text{CR}}) \leq C(K) \varrho_K(v_{\text{CR}}) \quad \text{for all } v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})|_{\Omega(K)}.$$

For any $\mathcal{T} \in \mathbb{T}$ there are only finitely many configurations of $\mathcal{T}(\Omega(K))$ possible [CP20, (A2)], hence $C(\mathbb{T}) := \sup_{K \in \mathcal{T} \in \mathbb{T}} C(K) < \infty$. This and $h_F \approx |K|^{1/n}$ for any $F \in \mathcal{F}(\Omega(K))$ concludes the proof. $\square$

3.3.2.2 Design of J_2

The correction with the normalized side-bubble functions from Remark 3.28 guarantees that $J_2 : \text{CR}_0^1(\mathcal{T}) \rightarrow S_0^n(\mathcal{T})$ in Definition 3.29 satisfies the right-inverse property.

Remark 3.28 (correction functions I). Recall the barycentric coordinates $\lambda_P \in P_1(T)$ for any $T \in \mathcal{T}$ and $P \in \mathcal{V}(T)$. For any interior side $F \in \mathcal{F}(\Omega)$ and any adjacent simplex $T \in \mathcal{T}(F)$ in the side patch ω_F define the normalized side-bubble function

$$b_F|_T := \frac{\prod_{P \in \mathcal{V}(F)} \lambda_P}{\int_F \left(\prod_{P \in \mathcal{V}(F)} \lambda_P \right) ds} = \frac{(2n-1)!}{(n-1)!} \prod_{P \in \mathcal{V}(F)} \lambda_P \in P_n(T).$$

Then $b_F \in P_n(\mathcal{T}(F)) \cap H_0^1(\omega_F)$ extended by zero to $b_F \in S_0^n(\mathcal{T})$ satisfies $\int_E b_F ds = \delta_{EF}$ for any $E \in \mathcal{F}$.

Definition 3.29 (J_2). Given the normalized side-bubble function $b_F \in S_0^n(\mathcal{T})$ in Remark 3.28 for any $F \in \mathcal{F}(\Omega)$, define for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$

$$J_2 v_{\text{CR}} := J_1 v_{\text{CR}} + \sum_{F \in \mathcal{F}(\Omega)} \left(\int_F (v_{\text{CR}} - J_1 v_{\text{CR}}) ds \right) b_F \in S_0^n(\mathcal{T}).$$

Lemma 3.30 (properties of J_2). *There exists a constant $C_4 > 0$ such that the companion operator J_2 in Definition 3.29 satisfies for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$ and $\mathcal{T} \in \mathbb{T}$,*

- (a) $\|(1 - J_2)v_{\text{CR}}\|_{L^2(K)} \leq C_4 \|(1 - J_1)v_{\text{CR}}\|_{L^2(K)}$ for any $K \in \mathcal{T}$,
- (b) J_2 is a right inverse to the interpolation I_{CR} in that $I_{\text{CR}} \circ J_2 = 1$ in $\text{CR}_0^1(\mathcal{T})$.

Proof of (a). The first step in proof of [CP20, Thm. 6.1] shows for any $K \in \mathcal{T}$ and $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$, that $C_4 := 1 + (2n)! \sqrt{\frac{2^{n-3}(n+1)(n+2)}{(3n)!n!}}$ satisfies

$$\|v_{\text{CR}} - J_2 v_{\text{CR}}\|_{L^2(K)} \leq C_4 \|v_{\text{CR}} - J_1 v_{\text{CR}}\|_{L^2(K)}.$$

(It holds $C_4 = 1 + 2\sqrt{3/5} \leq 2.5492$ for $n = 2$ and $C_4 = 1 + 10/\sqrt{21} \leq 3.1822$ for $n = 3$.)

For convenient reading the proof is included here. Definition 3.29 and the triangle inequality show

$$\begin{aligned} \|v_{\text{CR}} - J_1 v_{\text{CR}}\|_{L^2(K)} &\leq \|v_{\text{CR}} - J_1 v_{\text{CR}}\|_{L^2(K)} \\ &\quad + \left\| \sum_{F \in \mathcal{F}(K)} \left| \int_F (v_{\text{CR}} - J_1 v_{\text{CR}})|_K \, ds \right| b_F \right\|_{L^2(K)}. \end{aligned}$$

The integration formula for barycentric coordinates (3.2) shows, that the local mass matrix for normalized side-bubble functions in $K \in \mathcal{T}$ is SPD and reads

$$B(K) := \left(\int_K b_E b_F \, dx \right)_{E, F \in \mathcal{F}(K)} = \left(\frac{2^{n-3}((2n)!)^2 |K|}{(3n)!n!} (1 + \delta_{EF}) \right)_{E, F \in \mathcal{F}(K)} \in \mathbb{R}^{(n+1) \times (n+1)}.$$

It has the multiple eigenvalue $\lambda_{\min} := 2^{n-3}((2n)!)^2 |K| / ((3n)!n!)$ and the simple eigenvalue $\lambda_{\max} := (n+2)\lambda_{\min}$. This proves

$$\left\| \sum_{F \in \mathcal{F}(K)} \left| \int_F (v_{\text{CR}} - J_1 v_{\text{CR}})|_K \, ds \right| b_F \right\|_{L^2(K)}^2 \leq \lambda_{\max} \sum_{F \in \mathcal{F}(K)} \left| \int_F (v_{\text{CR}} - J_1 v_{\text{CR}})|_K \, ds \right|^2.$$

Lemma B.4 of the appendix quantifies the constant in the discrete trace inequality and implies

$$\sum_{F \in \mathcal{F}(K)} \left| \int_F (v_{\text{CR}} - J_1 v_{\text{CR}})|_K \, ds \right|^2 \leq \frac{(n+1)}{|K|} \|v_{\text{CR}} - J_1 v_{\text{CR}}\|_{L^2(K)}^2.$$

The combination concludes the proof. $\square$

Proof of (b). The duality property $\int_E b_F \, ds = \delta_{EF}$ for any $E, F \in \mathcal{F}$ of the normalized side-bubble functions and Definition 3.29 implies $L_F(J_2 v_{\text{CR}}) = \int_F J_2 v_{\text{CR}} \, ds = \int_F v_{\text{CR}} \, ds = L_F(v_{\text{CR}})$ for any $F \in \mathcal{F}(\Omega)$. This and the definition of I_{CR} with the dual basis functions in (3.40) shows that J_2 satisfies (b). $\square$

3.3.2.3 Design of $J_3 =: J_{\text{CR}}$

An additional correction in Definition 3.32 guarantees the L^2 -orthogonality in Theorem 3.25.d. Let $b_T := (n+1)^{n+1} \prod_{P \in \mathcal{V}(T)} \lambda_P \in S_0^{n+1}(T)$ denote the normalized volume-bubble function with $\|b_T\|_{L^\infty(T)} = 1$ on each $T \in \mathcal{T}$ and (not renamed) zero extension $b_T \in S_0^{n+1}(\mathcal{T})$. In order to assure the L^2 orthogonality of $v_{\text{CR}} - J_{\text{CR}} v_{\text{CR}}$ onto $P_1(\mathcal{T})$, for each simplex $T \in \mathcal{T}$ a local correction with the product of b_T and a Riesz representation in $P_1(T)$ is utilized. For $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$, let $v_T \in P_1(T)$ denote the Riesz representation of the linear functional

$$\ell_T : P_1(T) \rightarrow \mathbb{R}, \quad w_T \mapsto \int_T (v_{\text{CR}} - J_2 v_{\text{CR}}) w_T \, dx$$

in the Hilbert space $P_1(T)$ endowed with the weighted L^2 scalar product $(b_T \bullet, \bullet)_{L^2(T)}$. This Riesz representation $v_T \in P_1(T)$ satisfies

$$(v_{\text{CR}} - J_2 v_{\text{CR}}, w_T)_{L^2(T)} = (b_T v_T, w_T)_{L^2(T)} \quad \text{for all } w_T \in P_1(T). \quad (3.42)$$

The property (3.42) suffices to prove that $J_{\text{CR}} v_{\text{CR}} := J_3 v_{\text{CR}} := J_2 v_{\text{CR}} + \sum_{T \in \mathcal{T}} v_T b_T$ satisfies Theorem 3.25. However, the Riesz representation can be computed directly with the following correction function.

Remark 3.31 (correction functions II). For any $T \in \mathcal{T}$ and $P \in \mathcal{V}(T)$, define the affine function

$$\Phi_{T,P} := \sqrt{\frac{(2n+1)!}{(n+1)^{n+1}(n+1)!|T|}} \left((\sqrt{2n+3} + 1) - (n+1)\sqrt{2n+3}\lambda_P \right) \in P_1(T).$$

This basis of $P_1(T)$ satisfies $\int_T \Phi_{T,P} \Phi_{T,Q} b_T dx = \delta_{PQ}$ for any $P, Q \in \mathcal{V}(T)$ as lengthy computation with (3.2) verify. Hence, the Riesz representation in (3.42) reads $v_T := \sum_{P \in \mathcal{V}(T)} \left(\int_T (v_{\text{CR}} - J_2 v_{\text{CR}}) \Phi_{T,P} dx \right) \Phi_{T,P}$.

In order to find the above basis, set $\Phi_{T,P} := a + b\lambda_P \in P_1(T)$ and compute $\int_T \Phi_{T,P} \Phi_{T,Q} b_T dx$ with (3.2) in terms of n and $|T|$. The orthonormality condition then yields a system of equations with the above solution for a and b .

In the spirit of [CGS15, Thm. 2.3] define the conforming companion $J_3 =: J_{\text{CR}}$ as follows.

Definition 3.32 ($J_3 =: J_{\text{CR}}$). Given the zero-extension of the normalized volume-bubble function $b_T := (n+1)^{n+1} \prod_{P \in \mathcal{V}(T)} \lambda_P \in S_0^{n+1}(T)$ and $\Phi_{T,P}$ in Remark 3.31 for any $T \in \mathcal{T}$ and $P \in \mathcal{V}(T)$, define for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$

$$J_3 v_{\text{CR}} := J_2 v_{\text{CR}} + \sum_{T \in \mathcal{T}} \sum_{P \in \mathcal{V}(T)} \left(\int_T (v_{\text{CR}} - J_2 v_{\text{CR}}) \Phi_{T,P} dx \right) \Phi_{T,P} b_T \in S_0^{n+2}(\mathcal{T}).$$

Lemma 3.33 (properties of $J_3 \equiv J_{\text{CR}}$). For any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$ the conforming companion $J_{\text{CR}} v_{\text{CR}} := J_3 v_{\text{CR}} \in S_0^{n+2}(\mathcal{T})$ satisfies Theorem 3.25.

Proof of Theorem 3.25.a. For any $K \in \mathcal{T}$, Definition 3.32, several triangle and Cauchy-Schwarz inequalities imply

$$\begin{aligned} \|v_{\text{CR}} - J_{\text{CR}} v_{\text{CR}}\|_{L^2(K)} &\leq \|v_{\text{CR}} - J_2 v_{\text{CR}}\|_{L^2(K)} \\ &\quad + \sum_{P \in \mathcal{V}(K)} \|v_{\text{CR}} - J_2 v_{\text{CR}}\|_{L^2(K)} \|\Phi_{K,P}\|_{L^2(K)}^2 \|b_K\|_{L^\infty(K)}. \end{aligned}$$

It holds $\|b_K\|_{L^\infty(K)} = 1$ and the integration formula (3.2) proves $\|\Phi_{K,P}\|_{L^2(K)}^2 = (2(2n+1)!)/((n+1)^n(n+2)n!)$. Hence, $C_5 := 1 + \frac{2(2n+1)!}{(n+1)^{n-1}(n+2)n!}$ satisfies

$$\|v_{\text{CR}} - J_3 v_{\text{CR}}\|_{L^2(K)} \leq C_5 \|v_{\text{CR}} - J_2 v_{\text{CR}}\|_{L^2(K)}.$$

The combination with Lemma 3.30.a implies

$$\|v_{\text{CR}} - J_{\text{CR}}v_{\text{CR}}\|_{L^2(K)} \leq C_4 C_5 \|v_{\text{CR}} - J_1 v_{\text{CR}}\|_{L^2(K)}. \quad (3.43)$$

Hence, Lemma 3.27.a concludes the proof of Theorem 3.25.a with $M_1 := C_1 C_4^2 C_5^2$. $\square$

Proof of Theorem 3.25.b. An inverse estimate as in Lemma 2.13, (3.43), and the summation over all $K \in \mathcal{T}$ imply with $C_6 := (1 + C_{\text{inv}})C_4 C_5$ that

$$\|h_{\mathcal{T}}^{-1}(1 - J_{\text{CR}})v_{\text{CR}}\|_{L^2(\Omega)} + \|(1 - J_{\text{CR}})v_{\text{CR}}\|_{\text{pw}} \leq C_6 \|h_{\mathcal{T}}^{-1}(1 - J_1)v_{\text{CR}}\|_{L^2(\Omega)}.$$

Hence, Lemma 3.27.b concludes the proof. $\square$

Proof of Theorem 3.25.c. Since $\Phi_{T,P}b_T \in P_{n+2}(\mathcal{T}) \cap H_0^1(T)$ vanishes along the boundary ∂T for any $T \in \mathcal{T}$ and $P \in \mathcal{V}(T)$, Lemma 3.30.b implies that

$$\oint_F J_{\text{CR}}v_{\text{CR}} \, ds = \oint_F J_2 v_{\text{CR}} \, ds = \oint_F v_{\text{CR}} \, ds \text{ for any } F \in \mathcal{F}(\Omega).$$

This and the definition of I_{CR} in (3.40) guarantee the right-inverse property. $\square$

Proof of Theorem 3.25.d. Since the functions $\Phi_{T,P}$ for any $T \in \mathcal{T}$ and $P \in \mathcal{V}(T)$ in Remark 3.31 extended by zero form a basis of $P_1(\mathcal{T})$, the duality property $\int_T \Phi_{T,P} \Phi_{T,Q} b_T \, dx = \delta_{PQ}$ for any $T \in \mathcal{T}$, $P, Q \in \mathcal{V}(T)$ guarantees the claimed orthogonality. (This is a direct consequence from (3.42).) $\square$

3.4 Morley

The Morley finite element [Mor68] is well established for two-dimensional plate problems and generalized in [MX06] to any space dimension. This section summarizes the properties relevant throughout this thesis. After a short introduction to the space, Subsection 3.4.1 analyzes the interpolation operator in any space

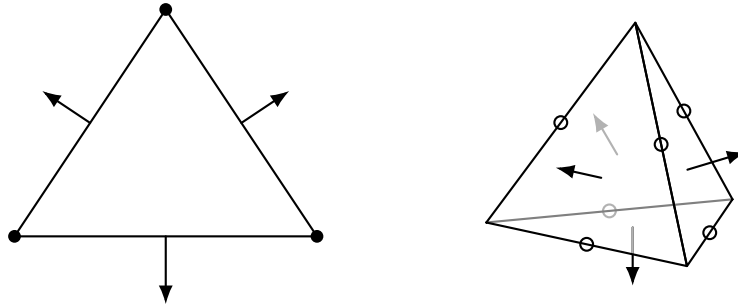


Figure 3.7: Mnemonic diagrams of the Morley finite element in 2D and 3D

$L_j(f) := \int_{F_j} \nabla f \cdot \nu_T _{F_j} \, ds \quad \text{for any } j = 1, \dots, \mathcal{F}(T) $ $L_{n+1+k}(f) := \int_{E_k} f \, d\sigma \quad \text{for any } k = 1, \dots, \mathcal{E}(T) $
--

Table 3.4: The local degrees of freedom for the Morley finite element for any $T \in \mathcal{T}$ and $f \in H^2(T)$

dimension and Subsection 3.4.2 focusses on the 3D case and designs an appropriate conforming companion.

Suppose the n -simplex $T \in \mathcal{T}$ has the set of vertices $\mathcal{V}(T) := \{P_1, \dots, P_{n+1}\}$. Recall the set $\mathcal{F}(T) := \{F_1, \dots, F_{|\mathcal{F}(T)|}\}$ of $(n-1)$ -subsimplices (faces in 3D) of cardinality $|\mathcal{F}(T)| = n+1$. Let $F_j := \text{conv}\{P_1, \dots, P_{j-1}, P_{j+1}, \dots, P_{n+1}\} \in \mathcal{F}(T)$ denote the side in T opposite to the vertex $P_j \in \mathcal{V}(T)$ in T . Recall the set $\mathcal{E}(T) := \{E_1, \dots, E_{|\mathcal{E}(T)|}\}$ of $(n-2)$ -subsimplices (edges in 3D) of cardinality $|\mathcal{E}(T)| = \binom{n+1}{2} = n(n+1)/2$. Let $E_{jk} := \text{conv}\{P_1, \dots, P_{j-1}, P_{j+1}, \dots, P_{k-1}, P_{k+1}, \dots, P_{n+1}\} \in \mathcal{E}(T)$ for any $1 \leq j < k \leq n+1$ denote the $(n-2)$ -subsimplex in the intersection $E \in \partial F_j \cap \partial F_k$ of the sides $F_j, F_k \in \mathcal{F}(T)$ in $T \in \mathcal{T}(E)$.

The local degrees of freedom for the Morley element [MX06, Def. 1] in an n -simplex $T \in \mathcal{T}$ are the integral means of the function $f \in H^2(T)$ along any $(n-2)$ -subsimplex $E \in \mathcal{E}(T)$ of T and of the normal derivative $\partial f / \partial \nu$ for each $(n-1)$ -subsimplex $F \in \mathcal{F}(T)$ of T in Table 3.4. Let the integral mean over a node $z \in \mathcal{V}$ be the point evaluation, to see that this reduces to the classical definition [Mor68] for $n = 2$.

Lemma 3.34 (Morley finite element). (a) *The degrees of freedom $L_1, \dots, L_{(n+1)(n+2)/2}$ uniquely determine $\int_F \nabla f \cdot \nu \, ds$ for any side $F \in \mathcal{F}(T)$ and $f \in H^2(T)$ in Table 3.4.*

(b) *The Morley element $(T, P_2(T), (L_1, \dots, L_{(n+1)(n+2)/2}))$ is a finite element in the sense of Ciarlet.*

Proof of (a). This is Lemma 1 in [MX06]. For any n -simplex $T \in \mathcal{T}$ and any $(n-1)$ -subsimplex $F = F_j \in \mathcal{F}(T)$, abbreviate $\nu = \nu_T|_F$. Then any constant vector $\alpha \in \mathbb{R}^n$ is split in its normal component $(\alpha \cdot \nu)\nu$ and its tangential component $\tau := \alpha - (\alpha \cdot \nu)\nu$. Hence, for any $f \in H^2(T)$

$$\int_F \nabla f \cdot \alpha \, ds = (\alpha \cdot \nu) \int_F \nabla f \cdot \nu \, ds + \int_F \nabla f \cdot \tau \, ds.$$

The first integral on the right-hand side is $|F_j|L_j(f)$. The second integral regarded as integration in $\mathbb{R}^{n-1}$ allows for an integration by parts. For any $E \in \mathcal{E}(F) :=$

$\{G \in \mathcal{E}(T) : G \subset F\}$ let ν_E denote the outer unit normal along the boundary of the $(n-1)$ -simplex F in $\mathbb{R}^{n-1}$, then

$$\int_F \nabla f \cdot \tau \, ds = \sum_{E \in \mathcal{E}(F)} \tau \cdot \nu_E \int_E f \, d\sigma.$$

Since each of these integrals along $E = E_k \in \mathcal{E}(F)$ on the right-hand side corresponds to $|E_k|L_{n+1+k}(f)$, this concludes the proof. $\square$

Proof of (b). This is Lemma 2 in [MX06]. Since for any $T \in \mathcal{T}$ the number of degrees of freedom and the dimension of $P_2(T)$ coincide, it suffices to show that any $f \in P_2(T)$ with vanishing degrees of freedom in Table 3.4 satisfies $f \equiv 0$. The essential step in the proof is the observation (a), that the degrees of freedom uniquely determine the integral mean of all first order derivatives $\int_F \nabla f \, ds$ for any $(n-1)$ -subsimplex $F \in \mathcal{F}(T)$. The claim follows as in [MX06, Lemma 2] with Greens formula or with the simple observation, that $\nabla f \in P_1(T; \mathbb{R}^n)$ is a Crouzeix-Raviart function in n -components with vanishing integral means (degrees of freedom for Crouzeix-Raviart) on all $F \in \mathcal{F}(T)$. $\square$

The $m := |\mathcal{F}| + |\mathcal{E}|$ global degrees of freedom are labelled, for any $f \in H^2(\Omega)$, by

$$L_F(f) := \int_F \nabla f \cdot \nu_F \, ds \text{ for any } F \in \mathcal{F} \quad \text{and} \quad L_E(f) := \int_E f \, d\sigma \text{ for any } E \in \mathcal{E},$$

where ν_F denotes the unit normal for any side $F \in \mathcal{F}$ with a fixed orientation. Section 6 in [CGH14] introduces the dual basis for the Morley finite element in 2D and specifies an implementation in 30 lines of Matlab. The nodal basis functions in [MX06, Thm. 1] read for $n \geq 3$ as follows.

Given any side $F \in \mathcal{F}$ with unit normal ν_F of fixed orientation, the support $\text{supp}(\phi_F) = \overline{\omega}_F$ of the basis function $\phi_F \in P_2(\mathcal{T})$ dual to L_F consists of all adjacent n -simplices. Suppose that $F = F_j$ is the side opposite the vertex P_j with barycentric coordinate λ_j in $T \in \mathcal{T}(F)$, then

$$\phi_F|_T := (\nu_T \cdot \nu_F) (\lambda_j(n\lambda_j - 2)) / (2|\nabla \lambda_j|). \quad (3.44)$$

Given a $(n-2)$ -subsimplex $E \in \mathcal{E}$, the support $\text{supp}(\phi_E) = \overline{\omega}_E$ of the basis function $\phi_E \in P_2(\mathcal{T})$ dual to L_E consists of all adjacent n -simplices in $\mathcal{T}(E)$. Suppose that $E = E_{jk}$ is the $(n-2)$ -subsimplex in the intersection $E \in \partial F_j \cap \partial F_k$ of the sides $F_j, F_k \in \mathcal{F}(T)$ in $T \in \mathcal{T}(E)$ with barycentric coordinates λ_j and λ_k in T associated with the opposite vertices P_j and P_k . Then

$$\begin{aligned} \phi_E|_T = & 1 - (n-1)(\lambda_j + \lambda_k) + n(n-1)\lambda_j\lambda_k \\ & - (n-1)\nabla \lambda_j \cdot \nabla \lambda_k \sum_{\ell \in \{j,k\}} \frac{\lambda_\ell(n\lambda_\ell - 2)}{2|\nabla \lambda_\ell|^2}. \end{aligned} \quad (3.45)$$

This defines the nodal basis functions: $L_G(\phi_F|_T) = \delta_{FG}$, $L_D(\phi_E|_T) = \delta_{DE}$, and $L_F(\phi_E|_T) = 0 = L_E(\phi_F|_T)$ follows for any n -simplex $T \in \mathcal{T}$, all $(n-1)$ -simplices $F, G \in \mathcal{F}(T)$, and all $(n-2)$ -simplices $D, E \in \mathcal{E}(T)$. The Morley finite element space without and with homogeneous boundary conditions read

$$\begin{aligned} M'(\mathcal{T}) &:= \text{span}\{\phi_F : F \in \mathcal{F}\} \oplus \text{span}\{\phi_E : E \in \mathcal{E}\} \subset P_2(\mathcal{T}), \\ M(\mathcal{T}) &:= \text{span}\{\phi_F : F \in \mathcal{F}(\Omega)\} \oplus \text{span}\{\phi_E : E \in \mathcal{E}(\Omega)\} \subset P_2(\mathcal{T}). \end{aligned}$$

Note that the Morley functions $v_M \in M(\mathcal{T})$ are single valued in the degrees of freedom in Table 3.4.

3.4.1 Interpolation operator

Let $V := H_0^2(\Omega)$ for some bounded Lipschitz domain $\Omega \subset \mathbb{R}^n$ with admissible triangulation $\mathcal{T} \in \mathbb{T}$ and piecewise (w.r.t. $\mathcal{T}$) energy scalar product $a_{\text{pw}}(\bullet, \bullet) := \int_{\Omega} (D_{\text{pw}}^2 \bullet : D_{\text{pw}}^2 \bullet) dx$ inducing the energy norm $\|\bullet\|_{\text{pw}}^2 := a_{\text{pw}}(\bullet, \bullet)$. Suppose $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ denotes an admissible refinement of $\mathcal{T} \in \mathbb{T}$ and recall the dual basis for the Morley FEM in (3.44)–(3.45). Define the Morley interpolation operator $I_M : V + M(\hat{\mathcal{T}}) \rightarrow M(\mathcal{T})$ as follows

$$I_M(v) := \sum_{F \in \mathcal{F}(\Omega)} \oint_F \nabla v \cdot \nu_F ds \phi_F + \sum_{E \in \mathcal{E}(\Omega)} \int_E v d\sigma \phi_E. \quad (3.46)$$

Hence, for all $E \in \mathcal{E}$ holds $L_E(I_M v) = \int_E I_M v d\sigma = \int_E v d\sigma = L_E(v)$ and for all $F \in \mathcal{F}$ holds $L_F(I_M v) = \oint_F \nu_F \cdot \nabla I_M v ds = \oint_F \nu_F \cdot \nabla v ds = L_F(v)$. The interpolation operator has the following important properties with explicit constants κ_1 and κ_2 in any space dimension $n \geq 2$ in (1.2).

Theorem 3.35 (properties of I_M). *Let $\hat{\mathcal{T}} \in \mathbb{T}$ denote an admissible refinement of $\mathcal{T} \in \mathbb{T}$. There exist constants $\kappa_1, \hat{\kappa}_1, \kappa_2, \hat{\kappa}_2 > 0$ such that the following statements hold.*

(a) *Any $v \in V + M(\hat{\mathcal{T}})$ satisfies $\Pi_0 D_{\text{pw}}^2 v = D_{\text{pw}}^2 I_M v$, in particular*

$$a_{\text{pw}}(v - I_M v, w_M) = 0 \text{ for all } w_M \in M(\mathcal{T}) \text{ and } v \in V + M(\hat{\mathcal{T}}).$$

Hence, I_M is the a_{pw} -orthogonal projection onto $M(\mathcal{T})$ with

$$\|v - I_M v\|_{\text{pw}} = \min_{v_M \in M(\mathcal{T})} \|v - v_M\|_{\text{pw}} \text{ for any } v \in V + M(\hat{\mathcal{T}}).$$

(b) *The piecewise gradient $\nabla_{\text{pw}} v_M \in \text{CR}_0^1(\mathcal{T})^n$ of any $v_M \in M(\mathcal{T})$ is a Crouzeix-Raviart function in n components and $\nabla_{\text{pw}} I_M v = I_{\text{CR}} \nabla_{\text{pw}} v$ for any $v \in V + M(\hat{\mathcal{T}})$ (with component-wise application of I_{CR}).*

(c) Any $v \in H^2(T)$ for $T \in \mathcal{T}$ satisfies

$$|v - I_M v|_{H^1(T)} \leq \kappa_1 h_T |v - I_M v|_{H^2(T)}$$

and any $\hat{v}_M \in M(\hat{\mathcal{T}})|_T$ satisfies

$$|\hat{v}_M - I_M \hat{v}_M|_{H^1(T)} \leq \hat{\kappa}_1 h_T |\hat{v}_M - I_M \hat{v}_M|_{H^2(T)}.$$

(d) Any $v \in H^2(T)$ for $T \in \mathcal{T}$ satisfies

$$\|v - I_M v\|_{L^2(T)} \leq \kappa_2 h_T^2 |v - I_M v|_{H^2(T)}.$$

(e) Any $\hat{v}_M \in M(\hat{\mathcal{T}})|_T$ for $T \in \mathcal{T}$ satisfies

$$\|\hat{v}_M - I_M \hat{v}_M\|_{L^2(T)} \leq \hat{\kappa}_2 h_T^2 |\hat{v}_M - I_M \hat{v}_M|_{H^2(T)}.$$

(f) The operator I_M acts as identity in non-refined simplices in that

$$(1 - I_M)\hat{v}_M|_T = 0 \quad \text{in } T \in \mathcal{T} \cap \hat{\mathcal{T}} \text{ for all } \hat{v}_M \in M(\hat{\mathcal{T}}).$$

The interpolation operator $\hat{I}_M$ associated with $M(\hat{\mathcal{T}})$ satisfies $I_M \circ \hat{I}_M = I_M$ in $V + M(\hat{\mathcal{T}})$.

Proof of (a). Lemma 3.34.a and the definition of I_M imply that $\oint_F \nabla I_M v \cdot \nu_F ds = \oint_F \nabla v \cdot \nu_F ds$ for any $F \in \mathcal{F}$. This and a piecewise integration by parts, where the normal jumps associated with $\hat{\mathcal{F}} \setminus \mathcal{F}$ vanish for $v \in V + M(\hat{\mathcal{T}})$, prove $\Pi_0 D_{\text{pw}}^2 v = D_{\text{pw}}^2 I_M v$ similar to the proof of Theorem 3.24.a. Since $w_M \in M(\mathcal{T}) \subset P_2(\mathcal{T})$, $D_{\text{pw}}^2 w_M \in P_0(\mathcal{T}; \mathbb{R}^{n \times n})$ concludes the proof. $\square$

Proof of (b). Since for any $v \in C^1(T)$ in $T \in \mathcal{T}$ the Morley degrees of freedom uniquely determine $\oint_F \nabla v ds$ for any $F \in \mathcal{F}(T)$ according to Lemma 3.34.a, $\oint_F \nabla I_M v ds = \oint_F \nabla v ds$. This implies that the integral mean of the piecewise gradient $\nabla_{\text{pw}} v_M \in P_1(\mathcal{T}; \mathbb{R}^n)$ of any $v_M \in M(\mathcal{T})$ is single valued for any side and hence $\nabla_{\text{pw}} v_M \in \text{CR}_0^1(\mathcal{T})^n$ is a Crouzeix-Raviart function in n components. The second claim, $\nabla_{\text{pw}} I_M v = I_{\text{CR}} \nabla v$, follows from the previous observation $\oint_F \nabla I_M v ds = \oint_F \nabla v ds$ and the definition of I_{CR} in (3.40). $\square$

Proof of (c). The combination of (b) and Theorem 3.24.b (resp. Theorem 3.24.c) for the choice $f = \partial v / \partial x_j \in H^1(T)$ (resp. $\hat{f} = \partial \hat{v}_M / \partial x_j \in \text{CR}_0^1(\mathcal{T})|_T$) proves the claim. $\square$

Proof of (d). This statement is known for $n = 2$ from [CG14a, Thm. 3] with $\kappa_2 \leq 0.257457844658$ and for $n > 2$ from [CP21, Thm. 2.1.b]. In 2D, [LSL19] provides the numerical bound $\kappa_2 < 0.07353$. The analytical constants [CG14a; CP21] can be slightly improved with the following arguments from Rui Ma [Ma21] which is presented here in adapted notation for completeness.

The repeated estimate for any m -simplex $K := \text{conv}\{P_1, \dots, P_{m+1}\}$ with midpoint $\text{mid}(K)$ is utilized throughout

$$\|\bullet - \text{mid}(K)\|_{L^2(K)}^2 \leq |K| h_K^2 m / (2(m+1)(m+2)). \quad (3.47)$$

The arguments follow the proof of Lemma A.1 in [CZZ20]. Suppose without loss of generality that $\text{mid}(K) = \frac{1}{m+1} \sum_{j=1}^{m+1} P_j = 0$ and recall that the local mass matrix of K has the entries $\int_K \lambda_{P_j} \lambda_{P_k} dx = \frac{|K|(1+\delta_{jk})}{(m+1)(m+2)}$. Hence,

$$\|\bullet - \text{mid}(K)\|_{L^2(K)}^2 = \int_K |x|^2 dx = \frac{|K|}{(m+1)(m+2)} \sum_{\ell=1}^{m+1} |P_\ell|^2$$

and elementary algebra with $\sum_{j=1}^{m+1} P_j = 0$ and $|P_j - P_k| \leq h_K$ for all $j, k = 1, \dots, m+1$ with $j \neq k$ leads to

$$\sum_{\ell=1}^{m+1} |P_\ell|^2 = \frac{1}{2(m+1)} \sum_{j,k=1}^{m+1} |P_j - P_k|^2 \leq \frac{h_K^2 m}{2}.$$

The combination of both displayed estimates proves (3.47).

Abbreviate $g := v - I_M v \in H^2(T)$. To improve the constant κ_2 below the orthogonal split $\|g\|_{L^2(T)}^2 = \|\Pi_0 g\|_{L^2(T)}^2 + \|(1 - \Pi_0)g\|_{L^2(T)}^2$ replaces $\|g\|_{L^2(T)} \leq \|(1 - I_{\text{CR}})g\|_{L^2(T)} + \|I_{\text{CR}}g\|_{L^2(T)}$ in [CG14a; CP21].

Note that $\|\Pi_0 g\|_{L^2(T)}^2 = |T|(\oint_T g dx)^2$. Consider the following integration by parts formula which improves the analytical bound on κ_2 (unlike a trace identity instead)

$$\int_T (x - \text{mid}(T)) \cdot \nabla g dx = \int_{\partial T} g (x - \text{mid}(T)) \cdot \nu_T ds - n \int_T g dx. \quad (3.48)$$

On the one hand, the Cauchy-Schwarz inequality, (3.47) for the n -simplex T , and (b) show

$$\begin{aligned} \left| \int_T (x - \text{mid}(T)) \cdot \nabla g dx \right| &\leq \|\bullet - \text{mid}(T)\|_{L^2(K)} \|\nabla g\|_{L^2(T)} \\ &\leq \frac{\kappa_1 h_T^2 \sqrt{|T|n}}{\sqrt{2(n+1)(n+2)}} \|D^2 g\|_{L^2(T)}. \end{aligned} \quad (3.49)$$

On the other hand, for any side $F \in \mathcal{F}(T)$ of T consider the sub-simplex $T_F = \text{conv}\{\text{mid}(T), F\}$. The volume of the sub-simplex $|T_F| = |T|/(n+1)$ due to the definition of $\text{mid}(T)$ is also computable as $|T_F| = |F| \text{dist}(\langle F \rangle, \text{mid}(T))/n$, where $\text{dist}(\langle F \rangle, \text{mid}(T))$ denotes the height of T_F over F or in other words the distance between $\text{mid}(T)$ and the plane $\langle F \rangle$ through the side F . Note that this distance $\text{dist}(\langle F \rangle, \text{mid}(T))$ is equal to $|(x - \text{mid}(T)) \cdot \nu_T|_F$. Hence the boundary integral in (3.48) is bounded by

$$\left| \int_{\partial T} g (x - \text{mid}(T)) \cdot \nu_T ds \right| \leq \frac{n|T|}{n+1} \sum_{F \in \mathcal{F}(T)} \left| \oint_F g ds \right|. \quad (3.50)$$

For $n = 2$ and any edge $F \in \mathcal{F}(T)$, Lemma 3 in [CG14a] bounds $\oint_F g \, ds$. For $n \geq 3$, each $(n-1)$ -subsimplex $F \in \mathcal{F}(T)$ with midpoint $\text{mid}(F)$ possesses a set of $(n-2)$ -subsimplces (edges in 3D) $\mathcal{E}(F) := \{E \in \mathcal{E}(T) : E \subset \partial F\} = \{E_1, \dots, E_n\}$. This set defines the sub-triangulation of F into $F_j := \text{conv}\{\text{mid}(F), E_j\} \subset F$ for $j = 1, \dots, n$. Since the function $g|_F := (v - I_M v)|_F \in H^1(F)$ satisfies $\oint_{E_j} g \, d\sigma = 0$ for any $j = 1, \dots, n$, the trace identity in Lemma 2.10.a for each $(n-1)$ -simplex $F_j \subset F$ with $2 \leq (n-1)$ and the Cauchy-Schwarz inequality prove

$$\begin{aligned} \oint_F g \, ds &= -\frac{1}{|F|(n-1)} \int_F (x - \text{mid}(F)) \cdot \nabla g \, ds \\ &\leq \frac{1}{|F|(n-1)} \|\bullet - \text{mid}(F)\|_{L^2(F)} \|\nabla g\|_{L^2(F)}. \end{aligned}$$

The combination with (3.47) for the $(n-1)$ -simplex F leads to

$$\left| \oint_F g \, ds \right| \leq \frac{h_F}{\sqrt{|F|}} \frac{1}{\sqrt{2(n-1)n(n+1)}} \|\nabla g\|_{L^2(F)}. \quad (3.51)$$

For $n = 2$, it holds that $g|_F \in H_0^1(F)$ and [CG14a, Lem. 3] implies that $(\oint_F g \, dx)^2 \leq h_F \|\partial g / \partial s\|_{L^2(F)} / 12$ for any edge $F \in \mathcal{F}(T)$ of length $h_F = |F|$. This is the estimate in (3.51) for $n = 2$. For any $n \geq 2$, the trace inequality Lemma 2.10.b for $\nabla g \in H^1(T)$ and Young's inequality show

$$\frac{|T|}{|F|} \|\nabla g\|_{L^2(F)}^2 \leq \min_{\alpha > 0} \left((1 + \alpha/2) \|\nabla g\|_{L^2(T)}^2 + 2h_T^2 / (\alpha n^2) \|D^2 g\|_{L^2(T)}^2 \right). \quad (3.52)$$

Since (c) implies $\|\nabla g\|_{L^2(T)} \leq \kappa_1 h_T \|D^2 g\|_{L^2(T)}$, the minimum in (3.52) is attained for $\alpha := 2/(\kappa_1 n)$ and recasts (3.52) as

$$\frac{|T|}{|F|} \|\nabla g\|_{L^2(F)}^2 \leq (\kappa_1^2 + 2\kappa_1/n) h_T^2 \|D^2 g\|_{L^2(T)}^2.$$

In combination with (3.48)–(3.51) and $h_F \leq h_T$ for any $F \in \mathcal{F}(T)$, this proves

$$\begin{aligned} \|\Pi_0 g\|_{L^2(T)} &= \frac{1}{\sqrt{|T|}} \left| \int_T g \, dx \right| \\ &\leq \frac{1}{n\sqrt{|T|}} \left| \int_T (x - \text{mid}(T)) \cdot \nabla g \, dx \right| + \frac{\sqrt{|T|}}{(n+1)} \sum_{F \in \mathcal{F}(T)} \left| \oint_F g \, ds \right| \\ &\leq h_T^2 \left(\frac{\kappa_1}{\sqrt{2n(n+1)(n+2)}} + \sqrt{\frac{n\kappa_1^2 + 2\kappa_1}{2n^2(n-1)(n+1)}} \right) \|D^2 g\|_{L^2(T)}. \end{aligned}$$

In addition, the Poincaré inequality in Lemma 2.11 followed by (c) bound the first term in the orthogonal split

$$\|(1 - \Pi_0)g\|_{L^2(T)} \leq \frac{h_T}{\pi} \|\nabla g\|_{L^2(T)} \leq h_T^2 \frac{\kappa_1}{\pi} \|D^2 g\|_{L^2(T)}.$$

The combination of the last two displayed estimates with the Pythagoras theorem conclude the proof with the analytical bound

$$\kappa_2^2 = \frac{\kappa_1^2}{\pi^2} + \left(\frac{\kappa_1}{\sqrt{2n(n+1)(n+2)}} + \sqrt{\frac{n\kappa_1^2 + 2\kappa_1}{2n^2(n-1)(n+1)}} \right)^2.$$

In the 2-dimensional case the Poincaré constant $1/\pi$ can be replaced by the optimal $1/j_{1,1}$ in 2D [LS10]. This leads to $\kappa_2 = 0.235882115439$ in 2D and $\kappa_2 = 0.154673265365$ in 3D. $\square$

Proof of (f). The first part is, e.g., the property (C3) in [CP20, Sec. 5.4] and follows from the fact that $\hat{v}_M|_T$ in $T \in \mathcal{T} \cap \hat{\mathcal{T}}$ is a function in $M(\mathcal{T})|_T$. Hence, the duality of the basis functions in (3.44)–(3.45) and the definition of the interpolation I_M in (3.46) implies $\hat{v}_M|_T = I_M \hat{v}_M|_T$ for $T \in \mathcal{T} \cap \hat{\mathcal{T}}$. The second claim is a direct consequence from the duality property of the basis functions on the fine level and the linearity of the integral. $\square$

Proof of (e). The arguments in Theorem 3.35.d and [CH18a] can be generalized to prove this with an explicit constant $\hat{\kappa}_2$. Alternatively the following soft analysis provides an extension of the 2D argumentation in [CP20, Sec. 5.2] to 3D and briefly accounts for the claim. Let $K \in \mathcal{T} \setminus \hat{\mathcal{T}}$ and set $\hat{w}_M := (\hat{v}_M - I_M \hat{v}_M)|_K$. It holds $\hat{w}_M \in M'(\hat{\mathcal{T}}(K))$ with the fine triangulation $\hat{\mathcal{T}}(K) := \{T \in \hat{\mathcal{T}} : T \subset K\}$ for the domain $\text{int}(K)$ replacing Ω , $\int_E \hat{w}_M \, d\sigma = 0$ for any $E \in \mathcal{E}(T)$ and $\int_F \nabla \hat{w}_M \cdot \nu_T \, ds = 0$ for any $F \in \mathcal{F}(T)$. Hence, Theorem 3.36 below resp. [Gal15b, Prop. 2.5–2.6] allow the definition of a conforming companion operator $\hat{J} : M'(\hat{\mathcal{T}}(K)) \rightarrow H^2(K)$ on the fine triangulation of a coarse simplex. In this case, as in Remark 3.51, there are non-homogeneous boundary condition to be considered. In the spirit of Definition 3.38 for any new boundary node $z \in \hat{\mathcal{V}}(\partial K) \setminus \mathcal{V}(\partial K)$ and any multi-index $\alpha \in \mathbb{N}_0^3$ with $0 \leq |\alpha| \leq 1$, set

$$D^\alpha(\hat{J} \hat{w}_M(z)) = |\hat{\mathcal{T}}(z) \cap \hat{\mathcal{T}}(K)|^{-1} \sum_{T \in \hat{\mathcal{T}}(z) \cap \hat{\mathcal{T}}(K)} (D^\alpha \hat{w}_M|_T)(z)$$

and for any new boundary edge $E \in \hat{\mathcal{E}}(\partial K) \setminus \mathcal{E}(\partial K)$ with unit tangential vector τ_E with a fixed orientation, set

$$\tau_E \times \nabla \hat{J} \hat{w}_M(\text{mid}(E)) := |\hat{\mathcal{T}}(E) \cap \hat{\mathcal{T}}(K)|^{-1} \sum_{T \in \hat{\mathcal{T}}(E) \cap \hat{\mathcal{T}}(K)} \tau_E \times (\nabla \hat{w}_M|_T)(\text{mid}(E)).$$

The correction in Subsections 3.4.2.2–3.4.2.3 lead to an companion with Theorem 3.36.a–c. The triangle inequality reads $\|\hat{w}_M\|_{L^2(K)} \leq \|\hat{w}_M - \hat{J} \hat{w}_M\|_{L^2(K)} + \|\hat{J} \hat{w}_M\|_{L^2(K)}$. The best-approximation property in Theorem 3.36.b proves $\|\hat{w}_M - \hat{J} \hat{w}_M\|_{L^2(K)} \lesssim h_K^2 \min_{w \in H^2(K)} \|D_{\text{pw}}^2(\hat{w}_M - w)\|_{L^2(K)} \leq h_K^2 \|D_{\text{pw}}^2 \hat{w}_M\|_{L^2(K)}$ for the

first term. Since $I_M \hat{J} \hat{w}_M = 0$ due to (f) and Theorem 3.36.c, the estimate (d) for the Morley interpolation of $\hat{J} \hat{w}_M \in H^2(K)$ followed by the stability property in Theorem 3.36.b of the companion operator proves

$$\begin{aligned} \|\hat{J} \hat{w}_M\|_{L^2(K)} &= \|\hat{J} \hat{w}_M - I_M \hat{J} \hat{w}_M\|_{L^2(K)} \leq \kappa_2 h_K^2 \|(1 - \Pi_0) D^2(\hat{J} \hat{w}_M)\|_{L^2(K)} \\ &\leq \kappa_2 h_K^2 \|D^2(\hat{J} \hat{w}_M)\|_{L^2(K)} \leq \kappa_2 h_K^2 (\|D_{\text{pw}}^2(\hat{J} \hat{w}_M - \hat{w}_M)\|_{L^2(K)} + \|D_{\text{pw}}^2 \hat{w}_M\|_{L^2(K)}) \\ &\lesssim h_K^2 \|D_{\text{pw}}^2 \hat{w}_M\|_{L^2(K)}. \end{aligned}$$

The combination of these estimates shows $\|\hat{w}_M\|_{L^2(K)} \lesssim h_K^2 \|D_{\text{pw}}^2 \hat{w}_M\|_{L^2(K)}$ for any $K \in \mathcal{T}$. This concludes the proof of (e). $\square$

3.4.2 Conforming companion

This section defines a conforming companion operator $J_M : M(\mathcal{T}) \rightarrow V := H_0^2(\Omega)$ for the Morley finite element that satisfies Theorem 3.36 below.

Theorem 3.36 (properties of J_M). *There exist constants $M_1, M_2, M_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) and a conforming companion $J_M v_M \in V := H_0^2(\Omega)$ for any $v_M \in M(\mathcal{T})$ with*

(a) $\|h_K^{-2}(1 - J_M)v_M\|_{L^2(K)}^2 \leq M_1 \sum_{T \in \mathcal{T}(\Omega(K))} |K|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2$ for any $K \in \mathcal{T}$ with the set $\mathcal{T}(\Omega(K)) := \{T \in \mathcal{T} : \text{dist}(T, K) = 0\}$ of adjacent simplices, and the tangential jumps $[D^2 v_M]_F \times \nu_F$ of the (piecewise) Hessian along $F \in \mathcal{F}$,

(b) $\|h_{\mathcal{T}}^{-2}(1 - J_M)v_M\|_{L^2(\Omega)} + \|(1 - J_M)v_M\|_{\text{pw}} \leq \left(M_2 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \leq M_3 \min_{v \in V} \|v_M - v\|_{\text{pw}},$

(c) J_M is a right inverse to the interpolation I_M in that $I_M \circ J_M = 1$ in $M(\mathcal{T})$,

(d) the orthogonality $(1 - J_M)(M(\mathcal{T})) \perp P_2(\mathcal{T})$ holds in $L^2(\Omega)$.

Since the design in 2D is up to the last step known, cf. [Gal15b; VZ19] for $n = 2$ and Theorem 3.36.a–c and Remark 3.37 below, this section concentrates on the design and proof of Theorem 3.36 in 3D. The conforming companion utilizes the Worsey-Farin finite element introduced in detail in Section 3.2. For the triangulation $\mathcal{T} \in \mathbb{T}$ of $\Omega \subset \mathbb{R}^3$ with set of edges $\mathcal{E}$ fix one unit tangent vector $\tau_E \parallel E$ and choose two unit vectors $\nu_{E,1}, \nu_{E,2} \in \mathbb{R}^3$ for any edge $E \in \mathcal{E}$ such that $(\tau_E, \nu_{E,1}, \nu_{E,2})$ form an orthonormal basis of $\mathbb{R}^3$, and define the WF partition $\hat{\mathcal{T}}$ as in Definition 3.17. Consider the conforming subspace with homogeneous boundary conditions as in (3.29)

$$\text{WF}(\mathcal{T}) := P_3(\hat{\mathcal{T}}) \cap H_0^2(\Omega).$$

The design in Subsection 3.4.2.1–3.4.2.4 below contains four steps. Given any Morley function $v_M \in M(\mathcal{T})$, first the enrichment $J_1 v_M \in \text{WF}(\mathcal{T})$ is computed via averaging of nodes. Two separate corrections are necessary to guarantee $L_E(v_M) = \int_E v_M d\sigma = L_E(J_M v_M)$ for any $E \in \mathcal{E}$ and $L_F(v_M) = \int_F \nabla v_M \cdot \nu_F ds = L_F(J_M v_M)$ for any $F \in \mathcal{F}$ in 3D. The integral means over the edges are corrected in J_2 . The integral means of the normal derivatives along the faces are corrected in J_3 . This guarantees that J_3 is a right inverse of I_M in that $I_M J_3 v_M = v_M$. Finally, the correction $J_4 v_M$ is designed such that its L^2 projection in $P_2(\mathcal{T})$ equals $v_M \in M(\mathcal{T})$, $\Pi_2 J_M v_M = v_M$. The corrections in the last two steps are mostly independent of the choice of the conforming space $\text{WF}(\mathcal{T})$ and possible in any space dimension.

Remark 3.37 (J_M in 2D). The conforming companion $J_G : M(\mathcal{T}) \rightarrow \text{HCT}(\mathcal{T}) + (P_5(\mathcal{T}) \cap H_0^2(\Omega))$ designed in [Gal15b, Prop. 2.5-2.6] satisfies Theorem 3.36.a–c. Note that [Gal15b] states only Theorem 3.36.b–c but a closer investigation reveals that the arguments in the proof (including the scaling in Remark 3.1.b) are local and verify Theorem 3.36.a as well. The strategy outlined above is similar to [Gal15b]. First an enrichment operator $J_{G1} : M(\mathcal{T}) \rightarrow \text{HCT}(\mathcal{T})$ is defined by averaging over the nodal (resp. edge patches) for the degrees of freedom of the HCT finite element in Table 3.1, while the degrees of freedom on the boundary $\partial\Omega$ are set to zero. Since the function evaluation in the vertices are degrees of freedom for the Morley and the HCT finite element in 2D only a correction of the integral means of the normal derivatives along the edges is necessary. This correction is as in Definition 3.45 with the function $\zeta_F|_T = 30(\nu_T \cdot \nu_E) \text{dist}(z_3, F) \lambda_1^2 \lambda_2^2 \lambda_3 \in P_5(T)$ (for $T = \text{conv}\{P_1, P_2, P_3\} = \text{conv}\{F, P_3\} \in \mathcal{T}$) as in Remark 3.44. Since the procedure proposed in Subsection 3.4.2.4 is possible in any space dimension $n \geq 2$ an operator $J_M : M(\mathcal{T}) \rightarrow \text{HCT}(\mathcal{T}) + (P_6(\mathcal{T}) \cap H_0^2(\Omega))$ satisfying Theorem 3.36 is available for $n = 2$.

3.4.2.1 Design of J_1

The definition of the enrichment operator J_1 depends on the choice of the conforming finite element space. In 2D the HCT space suffices and leads to Lemma 3.39 below, hence this section concentrates on the 3D case and the WF finite element. Determine the values for the degrees of freedom of $\text{WF}(\mathcal{T})$ by averaging the respective values of the Morley function on all adjacent simplices $T \in \mathcal{T}$ and assure homogeneous boundary conditions to define $J_1 : M(\mathcal{T}) \rightarrow \text{WF}(\mathcal{T}) \subset V$ as follows.

Definition 3.38 (enrichment J_1). For any vertex $z \in \mathcal{V}$ with nodal patch $\omega_z := \text{int}(\cup \mathcal{T}(z))$ for $\mathcal{T}(z) := \{T \in \mathcal{T} \mid z \in \mathcal{V}(T)\}$ of $|\mathcal{T}(z)|$ many tetrahedra with

the vertex z and for any multi-index $\alpha \in \mathbb{N}_0^3$ with $0 \leq |\alpha| \leq 1$, set

$$(D^\alpha J_1 v_M)(z) := \frac{1}{|\mathcal{T}(z)|} \sum_{T \in \mathcal{T}(z)} (D^\alpha v_M|_T)(z) \quad \text{for } z \in \mathcal{V}(\Omega)$$

(and $= 0$ for $z \in \mathcal{V}(\partial\Omega)$ owing to the homogeneous boundary conditions in V). For any edge $E \in \mathcal{E}$ with midpoint $\text{mid}(E)$, edge patch $\omega_E := \text{int}(\bigcup \mathcal{T}(E))$ for $\mathcal{T}(E) := \{T \in \mathcal{T} \mid E \in \mathcal{E}(T)\}$ of $|\mathcal{T}(E)|$ many tetrahedra with common edge E , and unit tangential vector τ_E with a fixed orientation, set

$$\tau_E \times \nabla J_1 v_M(\text{mid}(E)) := \frac{1}{|\mathcal{T}(E)|} \sum_{T \in \mathcal{T}(E)} \tau_E \times (\nabla v_M|_T)(\text{mid}(E)) \quad \text{for } E \in \mathcal{E}(\Omega)$$

(and $= 0$ for $E \in \mathcal{E}(\partial\Omega)$ owing to the homogeneous boundary conditions in V). The displayed definition means equality in the directions $\{\nu_{E,1}, \nu_{E,2}\}$ perpendicular to the tangential unit vector τ_E .

The enrichment operator J_1 satisfies the approximation condition. The analysis is comparable to the 2D case in [Gal15b, Prop. 2.5].

Lemma 3.39 (properties of J_1). *There exist constants $C_1, C_2, C_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) such that the enrichment operator $J_1 v_M \in \text{WF}(\mathcal{T})$ in Definition 3.38 satisfies for any $v_M \in M(\mathcal{T})$ and $\mathcal{T} \in \mathbb{T}$,*

(a) $\|h_K^{-2}(1 - J_1)v_M\|_{L^2(K)}^2 \leq C_1 \sum_{T \in \mathcal{T}(\Omega(K))} |K|^{1/3} \sum_{F \in \mathcal{F}(T)} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2$ for any $K \in \mathcal{T}$ with the set $\mathcal{T}(\Omega(K))$ of adjacent simplices,

(b) $\|h_{\mathcal{T}}^{-2}(1 - J_1)v_M\|_{L^2(\Omega)} + \|(1 - J_1)v_M\|_{\text{pw}} \leq \left(C_2 \sum_{T \in \mathcal{T}} |T|^{1/3} \sum_{F \in \mathcal{F}(T)} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \leq C_3 \min_{v \in V} \|v_M - v\|_{\text{pw}}.$

Proof of (a). Fix one tetrahedron $K \in \mathcal{T}$. The WF interpolation on K preserves the enrichment $J_1 v_M \in \text{WF}(K)$ and quadratic polynomials like the Morley function $v_M|_K \in P_2(K) \subset \text{WF}(K)$. Recall the nodal basis functions $\varphi_{z,1}|_K$, $\varphi_{z,\kappa+1}|_K$, and $\varphi_{E,\mu}|_K$ for any $z \in \mathcal{V}(K)$, $E \in \mathcal{E}(K)$, $\kappa = 1, 2, 3$, $\mu = 1, 2$, and the degrees of freedom of $\text{WF}(K)$ from Definition 3.19. Then the WF interpolation reads

$$\begin{aligned} (v_M - J_1 v_M)|_K &= \sum_{z \in \mathcal{V}(K)} \left((v_M|_K - J_1 v_M)(z) \varphi_{z,1} + \sum_{\kappa=1}^3 e_\kappa \cdot \nabla (v_M|_K - J_1 v_M)(z) \varphi_{z,\kappa+1} \right) \\ &\quad + \sum_{E \in \mathcal{E}(K)} \sum_{\mu=1}^2 \nu_{E,\mu} \cdot \nabla (v_M|_K - J_1 v_M)(\text{mid}(E)) \varphi_{E,\mu}. \end{aligned}$$

The triangle inequality and the scaling of the basis functions $\|\varphi_{z,1}\|_{L^2(K)} \lesssim h_K^{3/2}$, $\|\varphi_{z,\kappa+1}\|_{L^2(K)} \lesssim h_K^{5/2}$, and $\|\varphi_{E,\mu}\|_{L^2(K)} \lesssim h_K^{5/2}$ in Theorem 3.20 reveal for a constant

$c_1 \lesssim 1$ that

$$\begin{aligned}
& c_1^{-1} \|v_M - J_1 v_M\|_{L^2(K)}^2 \\
& \leq h_K^3 \sum_{z \in \mathcal{V}(K)} \left(|(v_M|_K - J_1 v_M)(z)|^2 + h_K^2 \sum_{\kappa=1}^3 |e_\kappa \cdot \nabla(v_M|_K - J_1 v_M)(z)|^2 \right) \\
& \quad + h_K^5 \sum_{E \in \mathcal{E}(K)} \sum_{\mu=1}^2 |\nu_{E,\mu} \cdot \nabla(v_M|_K - J_1 v_M)(\text{mid}(E))|^2. \tag{3.53}
\end{aligned}$$

In Definition 3.38 the given local values $L_j(v_M|_T)$ of v_M for every WF degree of freedom L_j for $j = 1, \dots, 28$ and every tetrahedron $T \in \mathcal{T}$ are averaged (and homogeneous boundary conditions ensured). This in mind, suppose $z \in \mathcal{V}(K)$ and analyse the term $|v_M|_K(z) - J_1 v_M(z)|$ first. There are three cases of interest that lead to the constant $C_z \lesssim 1$ in (3.57) below. The vertex $z \in \mathcal{V}(K)$ could be

Case 1. a boundary vertex $z \in \mathcal{V}(\partial\Omega)$, but K has no boundary side that contains z in that $\mathcal{F}(K) \cap \mathcal{F}(z) \cap \mathcal{F}(\partial\Omega) = \emptyset$,

Case 2. a boundary vertex $z \in \mathcal{V}(\partial\Omega) \cap F$ that belongs to a boundary side $F \in \mathcal{F}(\partial\Omega) \cap \mathcal{F}(K)$ of K , or

Case 3. an interior vertex $z \in \mathcal{V}(\Omega)$.

In Case 1 holds $J_1 v_M(z) = 0$ due to the homogeneous boundary conditions in V mirrored in Definition 3.38 since $z \in \mathcal{V}(\partial\Omega)$ is boundary vertex, but z does not belong to any boundary side of K . In this setting, Lemma B.3 in the appendix applies and contains the optimal constant. Choose a face-connected subset $\{T_1, \dots, T_k\} \subset \mathcal{T}(z)$ with $2 \leq k \leq |\mathcal{T}(z)| =: J_z \lesssim 1$, $T_k = K$, $z \in F_j := T_j \cap T_{j+1} \in \mathcal{F}$ for any $j = 1, \dots, k-1$, and boundary simplex T_1 such that $z \in F_0 \in \mathcal{F}(T_1) \cap \mathcal{F}(\partial\Omega)$. Set $x_j := v_M|_{T_j}(z) - J_1 v_M(z) = v_M|_{T_j}(z) \in \mathbb{R}$ for $j = 1, \dots, k$ and $x := (x_1, \dots, x_k) \in \mathbb{R}^k$. If $x \in \mathbb{R}^k \setminus \{0\}$, Lemma B.3 shows

$$\begin{aligned}
2(1 - \cos(\pi/(2k+1)))|x|^2 & \leq \sum_{j=1}^k (x_{j+1} - x_j)^2 + \min\{x_1^2, \dots, x_k^2\} \\
& \leq \sum_{j=1}^k (x_{j+1} - x_j)^2 + x_1^2.
\end{aligned}$$

The side-connectivity of $\{T_1, \dots, T_k\} \subset \mathcal{T}(z)$ and the definition of the jump imply, for any $j = 1, \dots, k-1$, that

$$|x_{j+1} - x_j| = |v_M|_{T_j}(z) - v_M|_{T_{j+1}}(z)| = |[v_M]_{F_j}(z)|$$

and $|x_1| = |v_M|_{T_1}(z)| = |[v_M]_{F_0}(z)|$. Hence, in Case 1 holds

$$|v_M|_K(z) - J_1 v_M(z)|^2 \leq |x|^2 \leq \frac{1}{2(1 - \cos(\pi/(2k+1)))} \sum_{j=0}^k |[v_M]_{F_j}(z)|^2. \tag{3.54}$$

In Case 2, for the boundary node $z \in \mathcal{V}(\partial\Omega) \cap F$ that belongs to a boundary side $F \in \mathcal{F}(\partial\Omega) \cap \mathcal{F}(K)$ of K , holds $J_1 v_M(z) = 0$ and the jump definition with $[v]_F := v$ along any boundary side, in particular along $F \in \mathcal{F}(\partial\Omega) \cap \mathcal{F}(K)$, shows

$$|v_M|_K(z) - J_1 v_M(z)| = |v_M|_K(z)| = |[v_M]_F(z)|. \quad (3.55)$$

In Case 3, for the interior vertex $z \in \mathcal{V}(\Omega)$, holds $J_1 v_M(z) = J_z^{-1} \sum_{T \in \mathcal{T}(z)} v_M|_T(z)$ with the abbreviation $J_z := |\mathcal{T}(z)| \lesssim 1$. This is the standard averaging and, e.g., contained in Step 3 in the proof of Theorem 6.1 in [CP20]. The arguments are repeated in the present notation for convenient reading. Choose an enumeration $\{T_1, \dots, T_{J_z}\}$ of $\mathcal{T}(z)$ such that the values $x_j := v_M|_{T_j}(z) - J_1 v_M(z) \in \mathbb{R}$ for $j = 1, \dots, J_z$ are ordered in the sense that $x_1 \leq x_2 \leq \dots \leq x_{J_z}$. The sum $\sum_{j=1}^{J_z} x_j = 0$ vanishes by definition of $J_1 v_M(z)$. Since $|v_M|_K(z) - J_1 v_M(z)| \leq \max_{T \in \mathcal{T}(z)} |v_M|_T - J_1 v_M(z)| = \max_{j=1, \dots, J_z} |x_j|$, Lemma B.2 in the appendix implies that

$$|v_M|_K(z) - J_1 v_M(z)|^2 \leq \frac{(J_z - 1)(2J_z - 1)}{6J_z} \sum_{j=1}^{J_z-1} |x_{j+1} - x_j|^2.$$

It remains a reordering to arrange for jump contributions on the right-hand side. To this end let $\mathcal{J} := \{\{\alpha, \beta\} : T_\alpha, T_\beta \in \mathcal{T}(z) \text{ and } \partial T_\alpha \cap \partial T_\beta \in \mathcal{F}(z)\}$ denote unordered index pairs of all tetrahedra in $\mathcal{T}(z)$ which share a side in $\mathcal{F}(z) := \{F \in \mathcal{F} | z \in \mathcal{V}(z)\}$. Since $\mathcal{T}(z)$ is side-connected, $\mathcal{J}$ is connected, in the sense that for all $\alpha, \beta \in \{1, \dots, J_z\}$ and $\alpha \neq \beta$ there are $k \in \mathbb{N}$ pairs $\{\alpha_1, \alpha_2\}, \{\alpha_2, \alpha_3\}, \dots, \{\alpha_k, \alpha_{k+1}\} \in \mathcal{J}$ with $\alpha_1 = \alpha$ and $\alpha_{k+1} = \beta$. Therefore some arguments in the language of graph theory in Lemma B.1 imply that

$$\sum_{j=1}^{J_z-1} |x_{j+1} - x_j|^2 \leq \sum_{\{\alpha, \beta\} \in \mathcal{J}} |x_\alpha - x_\beta|^2 = \sum_{\{\alpha, \beta\} \in \mathcal{J}} |(v_M|_{T_\alpha})(z) - (v_M|_{T_\beta})(z)|^2.$$

The combination of the last two displayed estimates with the jump definition shows in Case 3 that

$$|v_M|_K(z) - J_1 v_M(z)|^2 \leq \frac{(J_z - 1)(2J_z - 1)}{6J_z} \sum_{F \in \mathcal{F}(z)} |[v_M]_F(z)|^2. \quad (3.56)$$

Abbreviate

$$C_z := \max \left\{ 1, \frac{(J_z - 1)(2J_z - 1)}{6J_z}, \frac{1}{2(1 - \cos(\frac{\pi}{2J_z+1}))} \right\} \lesssim 1.$$

The combination of (3.54)–(3.56) implies for any vertex $z \in \mathcal{V}(K)$ of $K \in \mathcal{T}$ that

$$|v_M(z)|_K - J_1 v_M(z)|^2 \leq C_z \sum_{F \in \mathcal{F}(z)} |[v_M]_F(z)|^2. \quad (3.57)$$

Since J_1 is defined by nodal averaging for all degrees of freedom $L_1, \dots, L_{28}$ of WF (with homogeneous boundary conditions), the argumentation for $|v_M|_K(z) - J_1 v_M(z)|$ with Case 1–Case 3 applies verbatim for $|e_\kappa \cdot \nabla(v_M|_K - J_1 v_M)(z)|$ and $|\nu_{E,\mu} \cdot \nabla(v_M|_K - J_1 v_M)(\text{mid}(E))|$. Let $\mathcal{N} := \mathcal{V} \cup \{\text{mid}(E) | E \in \mathcal{E}\}$ denote the set of all vertices and edge midpoints. Let $\mathcal{T}(z) := \{T \in \mathcal{T} | z \in \partial T\}$ denote the set of all tetrahedra with node $z \in \mathcal{N}$ and $\mathcal{F}(z) := \{F \in \mathcal{F} | z \in \partial F\}$ set of all faces with node $z \in \mathcal{N}$. In abstract notation let J_A denote a general averaging operator; for any $k \in \mathbb{N}_0$, a piecewise polynomial $v \in P_k(\mathcal{T})$ of degree at most k , the tetrahedron $K \in \mathcal{T}$, and the node $z \in \mathcal{N}(K)$ set $J_A v(z) := |\mathcal{T}(z)|^{-1} \sum_{T \in \mathcal{T}(z)} v|_T(z)$ for each interior node $z \in \mathcal{N}(\Omega)$ and $J_A v(z) := 0$ else. Given this notation, $(D^\alpha J_1 v_M)(z) = J_A(D^\alpha v_M)(z)$ holds for $z \in \mathcal{V}$, $\alpha \in \mathbb{N}_0^3$ with $0 \leq |\alpha| \leq 1$, as well as, $\tau_E \times \nabla J_1 v_M(\text{mid}(E)) = J_A(\tau_E \times \nabla v_M)(\text{mid}(E))$ for $E \in \mathcal{E}$. In the above argumentation J_1 could be replaced J_A and $z \in \mathcal{V}$ by $z \in \mathcal{N}$ to obtain (3.57) in the general setting. Hence,

$$\begin{aligned} |e_\kappa \cdot \nabla(v_M|_K - J_1 v_M)(z)|^2 &\leq C_z \sum_{F \in \mathcal{F}(z)} |[e_\kappa \cdot \nabla v_M]_F(z)|^2, \\ |\nu_{E,\mu} \cdot \nabla(v_M|_K - J_1 v_M)(\text{mid}(E))|^2 &\leq C_E \sum_{F \in \mathcal{F}(E)} |[\nu_{E,\mu} \cdot \nabla v_M]_F(\text{mid}(E))|^2 \end{aligned}$$

with

$$C_E := \max \left\{ 1, \frac{(J_E - 1)(2J_E - 1)}{6J_E}, \frac{1}{2(1 - \cos(\frac{\pi}{2J_E + 1}))} \right\} \lesssim 1$$

for any edge $E \in \mathcal{E}$ with set of adjacent faces $\mathcal{F}(E) := \{F \in \mathcal{F} : E \in \mathcal{E}(F)\} = \mathcal{F}(\text{mid}(E))$ of cardinality $J_E := |\mathcal{F}(E)| \lesssim 1$.

Hence, (3.53) simplifies to

$$\begin{aligned} c_1^{-1} \|v_M - J_1 v_M\|_{L^2(K)}^2 &\leq h_K^3 \sum_{z \in \mathcal{V}(K)} C_z \sum_{F \in \mathcal{F}(z)} \left(|[v_M]_F(z)|^2 + h_K^2 |[\nabla v_M]_F(z)|^2 \right) \\ &\quad + h_K^5 \sum_{E \in \mathcal{E}(K)} C_E \sum_{F \in \mathcal{F}(E)} |[\nabla v_M]_F(\text{mid}(E))|^2 \end{aligned} \quad (3.58)$$

Note that the shape regularity of $\mathcal{T} \in \mathbb{T}$ guarantees that the number of tetrahedra J_z and J_E in the patches, i.e., C_z and C_E , are uniformly bounded for any $z \in \mathcal{V}$ and $E \in \mathcal{E}$.

For a fixed polynomial degree $k \in \mathbb{N}_0$ and any $z \in \mathcal{N} = \mathcal{V} \cup \{\text{mid}(E) | E \in \mathcal{E}\}$, there exists a constant $C > 0$, such that the following inverse estimate holds

$$|F| |[v]_F(z)|^2 \leq C \|[v]_F\|_{L^2(F)}^2 \quad \text{for any } v \in P_k(\mathcal{T}) \text{ and } F \in \mathcal{F}(z).$$

In the cases of interest, the optimal constant $C > 0$ is computed in Corollary B.5 in the appendix. It holds $|[v_M]_F(z)|^2 \leq 36 \|[v_M]_F\|_{L^2(F)}^2 / |F|$, $|[\nabla v_M]_F(z)|^2 \leq 9 \|[v_M]_F\|_{L^2(F)}^2 / |F|$, and $|[\nabla v_M]_F(\text{mid}(E))|^2 \leq 3 \|[v_M]_F\|_{L^2(F)}^2 / |F|$.

Since $f_E[v_M]_F ds = 0$ vanishes for all edges $E \in \mathcal{E}(F)$, the Poincaré-Friedrichs type estimate Lemma B.6 in the appendix implies with constant $c_F^2 := 1/j_{11}^2 + 1/4$

$$c_F^{-1} \|[v_M]_F\|_{L^2(F)} \leq h_F \|[\nabla v_M]_F\|_{L^2(F)} \leq h_K \|[\nabla v_M]_F\|_{L^2(F)}.$$

The combination of the last two displayed estimates recasts (3.58) as

$$\begin{aligned} c_1^{-1} \|v_M - J_1 v_M\|_{L^2(K)}^2 &\leq h_K^5 \sum_{z \in \mathcal{V}(K)} C_z \sum_{F \in \mathcal{F}(z)} \left(c_F^2 36 \|[\nabla v_M]_F\|_{L^2(F)}^2 / |F| + 9 \|[\nabla v_M]_F\|_{L^2(F)}^2 / |F| \right) \\ &\quad + h_K^5 \sum_{E \in \mathcal{E}(K)} C_E \sum_{F \in \mathcal{F}(E)} 3 \|[\nabla v_M]_F\|_{L^2(F)}^2 / |F|. \end{aligned}$$

Since for any edge $E = \text{conv}\{P_j, P_k\} \in \mathcal{E}$ with vertices $P_j, P_k \in \mathcal{V}(E)$, it holds $\mathcal{F}(E) \subset \mathcal{F}(P_j) \cup \mathcal{F}(P_k)$, the last displayed estimate simplifies for a constant $c_2 \lesssim 1$ to

$$c_2^{-1} h_K^{-5} \|v_M - J_1 v_M\|_{L^2(K)}^2 \leq \sum_{z \in \mathcal{V}(K)} \sum_{F \in \mathcal{F}(z)} \|[\nabla v_M]_F\|_{L^2(F)}^2 / |F|.$$

Recall that the integral mean of the gradient $f_F[\nabla v_M]_F ds$ vanishes for any side $F \in \mathcal{F}$ from Lemma 3.34.a. Hence, the Poincaré inequality in Lemma 2.11 shows

$$\|[\nabla v_M]_F\|_{L^2(F)} \leq h_F \pi^{-1} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}$$

with the tangential components $[D^2 v_M]_F \times \nu_F$ of the jump $[D^2 v_M]_F$ of the (piece-wise) Hessian D^2 along any side $F \in \mathcal{F}$. The combination of the last two displayed inequalities with the shape regularity $|F|^{1/2} \approx h_F \approx h_K \approx |K|^{1/3}$ for any $K \in \mathcal{T}$ and $F \in \mathcal{F}(K)$ concludes the proof. $\square$

Proof of (b). Summation of (a) over all $T \in \mathcal{T}$ and a finite overlap argument proves

$$\|h_{\mathcal{T}}^{-2} (1 - J_1) v_M\|_{L^2(\Omega)}^2 \lesssim \sum_{T \in \mathcal{T}} |T|^{1/3} \sum_{F \in \mathcal{F}(T)} \| [D^2 v_M]_F \times \nu_F \|_{L^2(F)}^2.$$

It remains to verify the best-approximation property. This step is comparable to the 2D analysis in [Gal15b, Prop. 2.3] but demands the observations on the Curl operator in 3D from [CBJ02, Thm 3.2]. For $\Psi = (\Psi_1, \Psi_2, \Psi_3)^\top \in H^1(\Omega)^{3 \times 3}$ the Curl is applied row-wise,

$$\text{Curl}(\Psi) := (\nabla \times \Psi_1, \nabla \times \Psi_2, \nabla \times \Psi_3)^\top.$$

First, suppose $F \in \mathcal{F}(\Omega) = \partial T_+ \cap \partial T_-$ is an interior side and recall the barycentric coordinates $\lambda_P \in P_1(T)$ for any vertex $P \in \mathcal{V}(T)$. Define the face-bubble function $b_F|_T := 60 \prod_{P \in \mathcal{V}(F)} \lambda_P \in P_3(T)$ for any $T \in \mathcal{T}(F)$. Then $b_F \in H_0^1(\omega_F)$ vanishes on

the boundary of the face patch $\omega_F := \text{int}(T_+ \cup T_-)$ and satisfies $\int_F b_F \, ds = |F|$ as well as $\|b_F\|_{L^\infty(\omega_F)} = 20/3$. Given $[D^2 v_M]_F \in \mathbb{R}^{3 \times 3}$, set $\Psi_F := b_F([D^2 v_M]_F \times \nu_F) \in H_0^1(\omega_F; \mathbb{R}^{3 \times 3})$. Since $[D^2 v_M]_F \in \mathbb{R}^{3 \times 3}$ is constant,

$$\|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2 = \|b_F^{1/2} [D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2.$$

For any $v \in H^2(\omega_E)$ the tangential jump $[D^2 v]_F \times \nu_F$ vanishes. Hence, an integration by parts shows

$$\begin{aligned} \|b_F^{1/2} [D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2 &= \int_F ([D^2 v_M]_F \times \nu_F) \Psi_F \, ds \\ &= \sum_{T \in \omega_F} \int_{\partial T} ((D^2 v_M - v) \times \nu_F) : \Psi_F \, ds = (\text{Curl} \Psi_F, D_{\text{pw}}^2(v_M - v))_{L^2(\omega_F)}. \end{aligned}$$

The Cauchy-Schwarz inequality, an inverse estimate, and the boundedness of $\|b_F\|_{L^\infty(\omega_E)}$ prove that the last term is bounded by

$$\begin{aligned} \|\text{Curl} \Psi_F\|_{L^2(\omega_F)} \|D_{\text{pw}}^2(v_M - v)\|_{L^2(\omega_F)} \\ \lesssim h_F^{-1/2} \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)} \|D_{\text{pw}}^2(v_M - v)\|_{L^2(\omega_F)}. \end{aligned}$$

This implies for any interior side $F \in \mathcal{F}(\Omega)$ that

$$h_F \|[D^2 v_M]_F \times \nu_F\|_{L^2(F)}^2 \lesssim \min_{v \in H^2(\omega_F)} \|D_{\text{pw}}^2(v_M - v)\|_{L^2(\omega_F)}^2. \quad (3.59)$$

Second, suppose $F \in \mathcal{F}(\partial\Omega)$ is a boundary face with face patch $\omega_F = T_+$. The face bubble $b_F := 60 \prod_{P \in \mathcal{V}(F)} \lambda_P \in H^1(\omega_F)$ vanishes on all faces $G \in \mathcal{F}(T_+) \setminus \{F\}$ of T_+ besides F . Hence, the argumentation above holds verbatim and shows (3.59) for $F \in \mathcal{F}(\partial\Omega)$.

This and (a) for any $K \in \mathcal{T}$ imply the local best-approximation property

$$h_K^{-4} \|v_M - J_1 v_M\|_{L^2(K)}^2 \lesssim \sum_{F \in \mathcal{F}(K)} \min_{v \in H^2(\omega_F)} \|D_{\text{pw}}^2(v_M - v)\|_{L^2(\omega_F)}^2.$$

The sum over all tetrahedra $K \in \mathcal{T}$ with the finite overlap of the face patches $(\omega_F : F \in \mathcal{F})$ yields the approximation property

$$\|h_{\mathcal{T}}^{-2} (1 - J_1) v_M\|_{L^2(\Omega)} \lesssim \min_{v \in H_0^2(\Omega)} \|v - v_M\|_{\text{pw}}.$$

The combination with the inverse estimate $\|(1 - J_1) v_M\|_{\text{pw}} \lesssim \|h_{\mathcal{T}}^{-2} (1 - J_1) v_M\|_{L^2(\Omega)}$ for the piecewise polynomials in $\text{WF}(\mathcal{T}) \subset P_3(\hat{\mathcal{T}})$ concludes the proof of (b). $\square$

3.4.2.2 Design of J_2

This step corrects the integral mean over each internal edge $E \in \mathcal{E}$ such that $\int_E J_2 v_M \, d\sigma = \int_E v_M \, d\sigma$. In 2D the equivalent degree of freedom is the evaluation at the vertices and the use of the conforming HCT finite element renders a correction needless, see Remark 3.37.

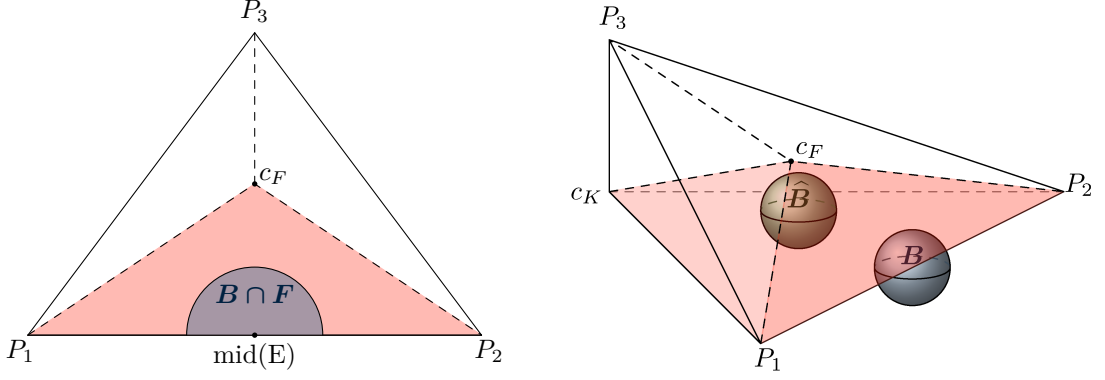


Figure 3.8: Subtetrahedron $T(4) = \text{conv}\{c_K, F_4\}$ of $K \in \mathcal{T}$ with face $F = F_4 = \text{conv}\{P_1, P_2, P_3\} \in \mathcal{F}(K)$ and edge $E = \text{conv}\{P_1, P_2\} \in \mathcal{E}(K)$ and dashed WF triangulation $\widehat{\mathcal{T}}(K)$ with $T = \text{conv}\{c_K, c_F, E\} \in \widehat{\mathcal{T}}$ marked in red, the support $\text{supp}(\xi_E) = B$ of ξ_E from Remark 3.40 in blue and the disjoint ball $\widehat{B} \subset T$ in green

Remark 3.40 (correction functions I). For any edge $E \in \mathcal{E}(\Omega)$ define the ball $B := B(\text{mid}(E), R_E) \subset \widehat{\omega}_E$ of radius $R_E > 0$ with midpoint $\text{mid}(E)$. The radius R_E is chosen maximal such that

- (i) B belongs to the edge patch $\widehat{\omega}_E := \text{int}\left(\bigcup_{T \in \widehat{\mathcal{T}}(E)} T\right)$ of E in the WF triangulation $\widehat{\mathcal{T}}$ and
- (ii) in any adjacent WF-subtetrahedra $T \in \widehat{\mathcal{T}}(E)$ exists a point $\widehat{c}_T \in T$ such that $\widehat{B} := B(\widehat{c}_T, R_E) \subset \text{int}(T)$ is a disjoint ball $B \cap \widehat{B} = \emptyset$ of the same size as B and lies in the interior of T .

Fig. 3.8.b illustrates the disjoint balls B and $\widehat{B}$. The shape regularity of $\mathcal{T} \in \mathbb{T}$ and $\widehat{\mathcal{T}}$ in Theorem 3.18 guarantees $h_T \approx R_E \approx h_E$. Set for any $E \in \mathcal{E}(\Omega)$

$$\xi_E(x) := \begin{cases} \frac{|E|}{R_E} \left(1 - 3\frac{|y|^2}{R_E^2} + 2\frac{|y|^3}{R_E^3}\right) & \text{for } x \in \overline{B} \text{ and } y := x - \text{mid}(E), \\ 0 & \text{for } x \notin \overline{B}. \end{cases}$$

This correction function satisfies $\xi_E \in C^1(\mathbb{R}^3) \cap H_0^2(B)$ with $\int_E \xi_D \, d\sigma = \delta_{ED}$ for any $E, D \in \mathcal{E}$ and the support $\text{supp}(\xi_E) = B \subset \widehat{\omega}_E$. To see this, note for any $x \in \partial B$ with $|x - \text{mid}(E)| = R_E$ holds $\xi_E(x) = 0$. Hence, $\xi_E \in C_c(\mathbb{R}^3)$. The gradient $\nabla \xi_E$ reads $\nabla \xi_E = 6|E|R_E^{-4}(|y| - R_E)y$ with $y = x - \text{mid}(E) \in \mathbb{R}^3$, vanishes along ∂B , and shows $\xi_E \in C^1(\mathbb{R}^3)$ with $\text{supp}(\xi_E) = \overline{B} \subset \omega_E$. Finally, integration along E verifies the scaling

$$\int_E \xi_E \, d\sigma = 2|E| \int_0^{R_E} (1/R_E - 3r^2/R_E^3 + 2r^3/R_E^4) \, dr = |E|.$$

The zero extension $\xi_E \in V$ is not renamed. $\square$

Definition 3.41 (J_2). Given $\xi_E \in V$ in Remark 3.40 for any $E \in \mathcal{E}$ define for any $v_M \in M(\mathcal{T})$

$$J_2(v_M) := J_1 v_M + \sum_{E \in \mathcal{E}(\Omega)} \left(\oint_E (v_M - J_1 v_M) d\sigma \right) \xi_E \in V.$$

Lemma 3.42 (properties of J_2). *There exists a constant $C_4 \approx 1$ (that exclusively depends on $\mathbb{T}$) such that the companion $J_2 v_M \in V$ satisfies for any $v_M \in M(\mathcal{T})$ and $\mathcal{T} \in \mathbb{T}$,*

$$(a) \quad \|(1 - J_2)v_M\|_{L^2(K)} \leq C_4 \|(1 - J_1)v_M\|_{L^2(K)} \text{ for any } K \in \mathcal{T},$$

$$(b) \quad \oint_E J_2 v_M d\sigma = \oint_E v_M d\sigma \text{ for any } E \in \mathcal{E}(\Omega).$$

Proof of (a). For any $K \in \mathcal{T}$, let $g := (v_M - J_1 v_M)|_K \in \text{WF}(K) \subset C^1(K)$. The Cauchy-Schwarz inequality and the discrete trace inequality Lemma 2.10.b for any face (2-simplex) $F \in \mathcal{F}(K)$ with edge (1-subsimplex) $E \in \mathcal{E}(F)$ show

$$\begin{aligned} \oint_E g d\sigma &\leq |E|^{-1/2} \|g\|_{L^2(E)} \leq |E|^{-1/2} \sqrt{\frac{|E|}{|F|}} \|g\|_{L^2(F)} (\|g\|_{L^2(F)} + h_F \|\nabla g\|_{L^2(F)}) \\ &\leq |F|^{-1/2} \sqrt{1 + c_{\text{inv}}} \|g\|_{L^2(F)} \end{aligned}$$

with an inverse estimate $\|\nabla g\|_{L^2(F)} \leq c_{\text{inv}} h_F^{-1} \|g\|_{L^2(F)}$ with a universal constant $c_{\text{inv}} \lesssim 1$ for any WF function $g|_F \in WF(\mathcal{T})|_F$ for any $F \in \mathcal{F}$ and $\mathcal{T} \in \mathbb{T}$ in the last step. The trace inequality Lemma 2.10.b for any face (2-subsimplex) $F \in \mathcal{F}$ of the tetrahedron (3-simplex) $K \in \mathcal{T}(F)$ reveals, that

$$\frac{|K|}{|F|} \|g\|_{L^2(F)}^2 \leq \|g\|_{L^2(K)} (\|g\|_{L^2(K)} + 2h_K/3 \|\nabla g\|_{L^2(K)}) \leq (1 + 2C_{\text{inv}}/3) \|g\|_{L^2(K)}^2$$

with an inverse estimate $\|\nabla g\|_{L^2(K)} \leq C_{\text{inv}} h_K^{-1} \|g\|_{L^2(K)}$ with a universal constant $C_{\text{inv}} \lesssim 1$ for any WF function $g|_K \in WF(\mathcal{T})|_K$ for any $K \in \mathcal{T}$ and $\mathcal{T} \in \mathbb{T}$ in the last step. The combination of the two displayed inequalities proves

$$\oint_E g d\sigma \leq \sqrt{1 + c_{\text{inv}}} \sqrt{1 + 2C_{\text{inv}}/3} |K|^{-1/2} \|g\|_{L^2(K)}.$$

Since straightforward computations reveal $\|\xi_E\|_{L^2(K)} = |E| \sqrt{38\pi R_E/315} \approx h_K^{3/2}$ for any $E \in \mathcal{E}(K)$ and ξ_E in Remark 3.40, the last displayed estimate and a triangle inequality prove

$$\begin{aligned} \|v_M - J_2 v_M\|_{L^2(K)} &\leq \|g\|_{L^2(K)} + \sum_{E \in \mathcal{E}(K) \cap \mathcal{E}(\Omega)} \left| \oint_E g d\sigma \right| \|\xi_E\|_{L^2(K)} \\ &\leq \left(1 + 6 \frac{|E| \sqrt{38\pi R_E}}{\sqrt{315}} \frac{\sqrt{1 + c_{\text{inv}}} \sqrt{1 + 2C_{\text{inv}}/3}}{|K|^{1/2}} \right) \|g\|_{L^2(K)}. \end{aligned}$$

The shape-regularity of $\mathcal{T} \in \mathbb{T}$ guarantees $h_K \approx |E| \approx R_E \approx |K|^{1/3}$ for any $K \in \mathcal{T}$ and $E \in \mathcal{E}(K)$ and so $|E|R_E^{1/2}|K|^{-1/2} \approx 1$. This concludes the proof of (a) with a constant $C_4 := 1 + 6 \frac{|E|\sqrt{38\pi R_E}}{\sqrt{315}} \frac{\sqrt{1+c_{\text{inv}}}\sqrt{1+2C_{\text{inv}}/3}}{|K|^{1/2}}$. $\square$

Proof of (b). This holds due to the duality condition of ξ_E in Remark 3.40, since for any $v_M \in M(\mathcal{T})$ and $E \in \mathcal{E}(\Omega)$

$$\begin{aligned} \oint_E J_2 v_M \, d\sigma &= \oint_E J_1 v_M \, d\sigma + \sum_{D \in \mathcal{E}(\Omega)} \left(\oint_D (v_M - J_1 v_M) \, d\sigma \right) \oint_D \xi_E \, d\sigma \\ &= \oint_E J_1 v_M \, d\sigma + \oint_E (v_M - J_1 v_M) \, d\sigma = \oint_E v_M \, d\sigma. \end{aligned}$$

$\square$

The following Lemma 3.43 provides more insight into the inverse estimates for functions of the type ξ_E in Remark 3.40. In particular standard inverse estimates hold for the operator $J_M \equiv J_4$ in Definition 3.49 (J_2 in Definition 3.41 or J_3 in Definition 3.45) below.

Lemma 3.43 (inverse estimates). *There exist constants $\tilde{c}_{\text{inv}}, \tilde{C}_{\text{inv}} \approx 1$ (that exclusively depends on $\mathbb{T}$), such that for any tetrahedron $T \in \mathcal{T} \in \mathbb{T}$, any face $F \in \mathcal{F}$, and $v_M \in M(\mathcal{T})$*

$$\begin{aligned} |v_M - J_M v_M|_{H^1(F)} &\leq \tilde{c}_{\text{inv}} h_F^{-1} \|v_M - J_M v_M\|_{L^2(F)} \\ |v_M - J_M v_M|_{H^1(T)} &\leq \tilde{C}_{\text{inv}} h_T^{-1} \|v_M - J_M v_M\|_{L^2(T)}. \end{aligned}$$

Notice that the H^1 -seminorm concerns the tangential gradient $\nu_F \times \nabla v_M$ for a face $F \in \mathcal{F}$ with unit normal ν_F of fixed orientation and the full gradient ∇v_M for a tetrahedra $T \in \mathcal{T}$.

Proof. The difference $v_M - J_M v_M$ can be rewritten for any $G \in \mathcal{T}$ or $G \in \mathcal{F}$ as

$$(v_M - J_M v_M)|_G = p_2 + g_{\text{WF}} + p_7 + p_{10} + \sum_{E \in \mathcal{E}(G)} \alpha_E \xi_E$$

with polynomials $p_2 \in P_2(G)$, $p_7 \in P_7(G)$, $p_{10} \in P_{10}(G)$, a WF function $g_{\text{WF}} \in P_3(\widehat{\mathcal{T}})$, $\alpha_E \in \mathbb{R}$, and ξ_E in Remark 3.40 with $\text{supp}(\xi_E) = B(\text{mid}(E), R_E) =: B$ for any edge $E \in \mathcal{E}(G)$.

Case 1. Suppose $G = F \in \mathcal{F}(T)$ is a face and without loss of generality regard $F \subset \mathbb{R}^2$. The key argument is that the support $\text{supp}(\xi_E) \subset \widehat{\omega}_E$ of ξ_E lies in the edge patch of $E \in \mathcal{E}$ in $\widehat{\mathcal{T}}$. In other words the WF function $g_{\text{WF}}|_{\overline{\omega}} \in P_3(\overline{\omega})$ is a cubic polynomial in $\overline{\omega} := \text{supp}(\xi_E) \cap F$ and the intersection $\omega = B \cap F \subset \mathbb{R}^2$ is

the half ball in $\mathbb{R}^2$ illustrated in Fig. 3.8.a. Hence, it remains to show, for all $p_{10} \in P_{10}(\omega)$ and $\alpha_E \in \mathbb{R}$, that

$$|p_{10} + \alpha_E \xi_E|_{H^1(\omega)} \lesssim h_E^{-1} \|p_{10} + \alpha_E \xi_E\|_{L^2(\omega)}.$$

Let $H_+ := \mathbb{R} \times (0, \infty)$ and consider the re-scaling $\varphi : B(0, R_E) \rightarrow B(0, 1)$, $x \mapsto \varphi(x) = x/R_E := \vartheta$ with Jacobean determinant $|\det D\varphi| = R_E^{-2}$ and abbreviate $\hat{f}(x) := f(xR_E)$ for $|x| < 1$, such that $f(x) = \hat{f}(x/R_E) = \hat{f}(\vartheta)$ and $\nabla_{\vartheta} \hat{f}(\vartheta) = R_E \nabla_x f(x)$. A substitution shows

$$\begin{aligned} \|\hat{f}\|_{L^2(B(0,1) \cap H_+)}^2 &= R_E^{-2} \|f\|_{L^2(B(0,R_E) \cap H_+)}^2, \\ |\hat{f}|_{H^1(B(0,1) \cap H_+)}^2 &= \int_{B(0,1) \cap H_+} |\nabla_y \hat{f}(y)|^2 dy = \int_{B(0,R_E) \cap H_+} |\nabla_{\vartheta} \hat{f}(\vartheta)|^2 |\det D\varphi| dx \\ &= \int_{B(0,R_E) \cap H_+} R_E^2 |\nabla_x f(x)|^2 |\det D\varphi| dx = |f|_{H^1(B(0,R_E) \cap H_+)}^2. \end{aligned}$$

For radius one there exists a constant $C_1 > 0$, such that for all $p_{10} \in P_{10}$ and $\alpha_E \in \mathbb{R}$,

$$|p_{10} + \alpha_E \xi_E|_{H^1(B(0,1) \cap H_+)} \leq \tilde{c}_{\text{inv}} \|p_{10} + \alpha_E \xi_E\|_{L^2(B(0,1) \cap H_+)}, \quad (3.60)$$

since $\|p_{10} + \alpha_E \xi_E\|_{L^2(B(0,1) \cap H_+)} = 0$ on the right-hand side of (3.60) implies on the left-hand side $|p_{10} + \alpha_E \xi_E|_{H^1(B(0,1) \cap H_+)} = 0$. The substitution shows for $\omega = B(0, R_E) \cap H^+$,

$$|p_{10} + \alpha_E \xi_E|_{H^1(\omega)} \leq \tilde{c}_{\text{inv}} R_E^{-1} \|p_{10} + \alpha_E \xi_E\|_{L^2(\omega)}$$

and $R_E \approx h_E \approx h_F \approx h_T$ concludes the proof since $\tilde{c}_{\text{inv}}$ is independent of any mesh-size factor.

Case 2. For a tetrahedron, the shape regularity of the WF triangulation $\hat{\mathcal{T}}$ as well as $\mathcal{T} \in \mathbb{T}$ allow the restriction to $T \in \hat{\mathcal{T}}$. Hence, it remains to show for all $p_{10} \in P_{10}(T)$ and $\alpha_E \in \mathbb{R}$,

$$|p_{10} + \alpha_E \xi_E|_{H^1(T)} \lesssim h_T^{-1} \|p_{10} + \alpha_E \xi_E\|_{L^2(T)}.$$

The key argument in this case is the existence of $\hat{B} = B(\hat{c}_T, R_E)$ with $\hat{B} \cap B = \emptyset$. A triangle inequality, the inverse estimate for $p_{10} \in P_{10}(T)$ with constant $c_{10} > 0$ and for ξ_E in Remark 3.40 with directly calculated constant show

$$|p_{10} + \alpha_E \xi_E|_{H^1(T)} \leq \frac{6\sqrt{6}}{\sqrt{19}R_E} \|\alpha_E \xi_E\|_{L^2(T)} + h_T^{-1} c_{10} \|p_{10}\|_{L^2(T)}.$$

Since the shape regularity ensures $R_E \approx h_T$, a triangle inequality leads to

$$\begin{aligned} |p_{10} + \alpha_E \xi_E|_{H^1(T)} &\lesssim h_T^{-1} \left(\|\alpha_E \xi_E\|_{L^2(T)} + \|p_{10}\|_{L^2(T)} \right) \\ &\leq h_T^{-1} \left(\|\alpha_E \xi_E + p_{10}\|_{L^2(T)} + 2\|p_{10}\|_{L^2(T)} \right). \end{aligned}$$

The shape regularity guarantees further that there exists a ball $\tilde{B} := B(\hat{c}_T, \tilde{R})$ with radius $\tilde{R} \approx h_T \approx R_E$ that contains T , $\hat{B} \subset T \subset \tilde{B}$. Standard scaling arguments for polynomials show for the polynomial extension $p_{10} \in P_{10}(\tilde{B})$ (not-relabelled), that

$$\|p_{10}\|_{L^2(T)} \leq \|p_{10}\|_{L^2(\tilde{B})} \lesssim \|p_{10}\|_{L^2(\hat{B})}.$$

The constant in the last step is bounded because the radius quotient $\tilde{R}/R_E \lesssim 1$ is bounded. Since the support of ξ_E satisfies $\text{supp}(\xi_E) \cap \hat{B} = \emptyset$, the estimate $\|p_{10}\|_{L^2(\hat{B})} = \|p_{10} + \alpha_E \xi_E\|_{L^2(\hat{B})} \leq \|p_{10} + \alpha_E \xi_E\|_{L^2(T)}$ concludes the proof. $\square$

3.4.2.3 Design of J_3

This step corrects the integral mean of the normal derivative over each interior side $F \in \mathcal{F}(\Omega)$ such that $\oint_F \nu_F \cdot \nabla J_3 v_M \, ds = \oint_F \nu_F \cdot \nabla v_M \, ds$. This correction is known in 2D from [Gal15b, Prop. 2.6] and (upto the necessity of an inverse estimate like Lemma 3.43 and the properties in Lemma 3.42 for the input) independent of the previous steps. Hence, the correction functions are defined for any $n \in \mathbb{N}$.

Remark 3.44 (correction functions II). The appropriate correction function is easily defined for any $n \geq 2$. Suppose $T = \text{conv}\{P_1, \dots, P_{n+1}\} \in \mathcal{T}$ denotes an n -simplex with face $F = \text{conv}\{P_1, P_2, \dots, P_n\} \in \mathcal{F}(T)$ opposite to the vertex $P_{n+1} \in \mathcal{V}(T)$. Recall the barycentric coordinate $\lambda_k \in P_1(T)$ in T associated with $P_k \in \mathcal{V}(T)$ for $k = 1, \dots, n+1$ and set

$$\zeta_F|_T := \frac{(3n-1)!}{2^n(n-1)!} (\nu_T \cdot \nu_F) \text{dist}(P_{n+1}, F) (\lambda_1 \dots \lambda_n)^2 \lambda_{n+1} \in P_{2n+1}(T)$$

for $T \in \mathcal{T}(F)$. (This reads $\zeta_F|_T = 30(\nu_T \cdot \nu_E) \text{dist}(z_3, F) \lambda_1^2 \lambda_2^2 \lambda_3 \in P_5(T)$ for $n = 2$ and $\zeta_F|_T = 7!/2(\nu_T \cdot \nu_E) \text{dist}(z_4, F) \lambda_1^2 \lambda_2^2 \lambda_3^2 \lambda_4 \in P_7(T)$ for $n = 3$.) The displayed formula defines $\zeta_F \in P_{2n+1}(\mathcal{T}(F)) \cap W_0^{2,\infty}(\omega_F)$ with $\oint_G \nabla \zeta_F \cdot \nu_G \, ds = \delta_{GF}$ for all sides $F, G \in \mathcal{F}$ in the face patch ω_F and allows for a (not relabelled) zero extension in V . To prove that ζ_F is continuously differentiable across interior edges $F \in \mathcal{F}(\Omega)$, note that for any $T = \text{conv}\{F, P_{n+1}\} \in \mathcal{T}(F)$ holds $\lambda_{n+1}|_F = 0$ and $\nabla \lambda_{n+1} = \text{dist}(P_{n+1}, F)^{-1} \nu_T$, hence $(\nabla \zeta_F|_T)|_F \nu_F = \frac{(3n-1)!}{2^n(n-1)!} (\nu_T \cdot \nu_F) \text{dist}(P_{n+1}, F) (\lambda_1 \dots \lambda_n)^2 (\nabla \lambda_{n+1} \nu_F) = \frac{(3n-1)!}{2^n(n-1)!} (\lambda_1 \dots \lambda_n)^2$. The careful scaling follows from the integration formula (3.2).

Definition 3.45 (J_3). Given $\zeta_F \in P_{2n+1}(\mathcal{T}) \cap V$ in Remark 3.44 define for any $v_M \in M(\mathcal{T})$

$$J_3(v_M) := J_2 v_M + \sum_{F \in \mathcal{F}(\Omega)} \left(\oint_F \nabla(v_M - J_2 v_M) \cdot \nu_F \, ds \right) \zeta_F \in V.$$

Lemma 3.46 (properties of J_3). *There exists a constant $C_5 \approx 1$ (that exclusively depends on $\mathbb{T}$) such that the companion $J_3 v_M \in V$ satisfies for any $v_M \in M(\mathcal{T})$ and $\mathcal{T} \in \mathbb{T}$,*

$$(a) \quad \|(1 - J_3)v_M\|_{L^2(K)} \leq C_5 \|(1 - J_2)v_M\|_{L^2(K)} \text{ for any } K \in \mathcal{T},$$

$$(b) \quad J_3 \text{ is a right inverse to the interpolation } I_M \text{ in that } I_M \circ J_3 = 1 \text{ in } M(\mathcal{T}).$$

Proof of (a). For any $K \in \mathcal{T}$, let $g := (v_M - J_2 v_M)|_K \in H^2(K)$. The Cauchy-Schwarz inequality, the trace inequality Lemma 2.10.b for any side $((n-1)$ -subsimplex) $F \in \mathcal{F}$ of the n -simplex $K \in \mathcal{T}(F)$, and an inverse estimate as in Lemma 3.43 reveal that

$$\begin{aligned} \int_F \nabla g \cdot \nu_F \, ds &\leq \frac{\tilde{c}_{\text{inv}}}{|F|^{1/2} h_F} \|g\|_{L^2(F)} \leq \frac{\tilde{c}_{\text{inv}}}{|K|^{1/2} h_F} \sqrt{\|g\|_{L^2(K)}^2 + \frac{2h_K}{n} \|g\|_{L^2(K)} \|\nabla g\|_{L^2(K)}} \\ &\leq |K|^{-1/2} h_F^{-1} \tilde{c}_{\text{inv}} \sqrt{1 + 2\tilde{C}_{\text{inv}}/n} \|g\|_{L^2(K)}. \end{aligned}$$

Straightforward computations reveal that $\|\zeta_F\|_{L^2(K)} \leq \frac{(3n-1)!}{(n-1)!} h_K \sqrt{\frac{2|K|6^n n!}{(5n+2)!}}$. In combination with the last displayed estimate and a triangle inequality, this proves

$$\begin{aligned} \|v_M - J_3 v_M\|_{L^2(K)} &\leq \|g\|_{L^2(K)} + \sum_{F \in \mathcal{F}(K) \cap \mathcal{F}(\Omega)} \left| \int_F g \, ds \right| \|\zeta_F\|_{L^2(K)} \\ &\leq \left(1 + (n+1) \frac{\tilde{c}_{\text{inv}} \sqrt{1 + 2\tilde{C}_{\text{inv}}/n}}{h_F} \frac{h_K (3n-1)!}{(n-1)!} \sqrt{\frac{2(6^n n!)}{(5n+2)!}} \right) \|g\|_{L^2(K)}. \end{aligned}$$

Since shape-regularity of $\mathcal{T} \in \mathbb{T}$ guarantees $h_K \approx h_F$ for any $K \in \mathcal{T}$ and $F \in \mathcal{F}(K)$, this concludes the proof of (a) with $C_5 := 1 + \frac{(n+1)(3n-1)! \tilde{c}_{\text{inv}} \sqrt{2(1+2\tilde{C}_{\text{inv}}/n)6^n n!} h_K}{(n-1)! \sqrt{(5n+2)!} h_F}$ for any $n \geq 2$ (where $\tilde{c}_{\text{inv}}, \tilde{C}_{\text{inv}} \lesssim 1$ depend on the space that contains $v_M - J_2 v_M$). $\square$

Proof of (b). Since $\zeta_F|_{\partial F} \equiv 0$ for any $F \in \mathcal{F}$, it holds $\int_E J_3 v_M \, ds = \int_E J_2 v_M \, ds$ for any $E \in \mathcal{E}$. Hence, Lemma 3.42.b implies $L_E(J_3 v_M) = L_E(v_M)$ for any $E \in \mathcal{E}(\Omega)$ and the duality property $\int_G \nabla \zeta_F \cdot \nu_G \, ds = \delta_{GF}$ for all sides $F, G \in \mathcal{F}$ in Remark 3.44 implies $L_F(J_3 v_M) = L_F(v_M)$ for any $F \in \mathcal{F}(\Omega)$. This and the definition of I_M in (3.46) proves that J_3 satisfies the right-inverse property (b). $\square$

3.4.2.4 Design of $J_4 =: J_M$

A local correction for each simplex $T \in \mathcal{T}$ with the product of a squared volume bubble-function and a Riesz representation in $P_2(T)$ assures the L^2 orthogonality of $v_M - J_M v_M$ onto $P_2(\mathcal{T})$. This section is written for any space dimension $n \geq 2$, such that the two-dimensional case in [Gal15b] is extended as promised in Remark 3.37.

Remark 3.47 (correction functions III). For any $T := \text{conv}\{P_1, \dots, P_{n+1}\} \in \mathcal{T}$, let $\lambda_k \in P_1(T)$ denote the barycentric coordinate associated with the vertex $P_k \in \mathcal{V}(T)$ in T for $k = 1, \dots, n+1$. Let

$$b_T := (n+1)^{2n+2} \prod_{k=1}^{n+1} \lambda_k^2 \in P_{2n+2}(T) \cap W_0^{2,\infty}(T) \subset V|_T$$

denote the squared volume bubble that satisfies $\|b_T\|_{L^\infty(T)} = 1$ with not renamed zero extension $b_T \in V$.

For any $v_M \in M(\mathcal{T})$, let $v_T \in P_2(T)$ denote the Riesz representation of the linear functional

$$\ell_T : P_2(T) \rightarrow \mathbb{R}, \quad w_T \mapsto \int_T (v_M - J_3 v_M) w_T \, dx$$

in the Hilbert space $P_2(T)$ endowed with the weighted L^2 scalar product $(b_T \bullet, \bullet)_{L^2(T)}$. The Riesz representation $v_T \in P_2(T)$ satisfies

$$(v_M - J_3 v_M, w_T)_{L^2(T)} = (b_T v_T, w_T)_{L^2(T)} \quad \text{for all } w_T \in P_2(T). \quad (3.61)$$

Given $b_T \in V$ in Remark 3.47 and v_T with (3.61), the conforming companion $J_4 v_M := J_3 v_M + \sum_{T \in \mathcal{T}} v_T b_T$ in Definition 3.49 satisfies Theorem 3.36. For completeness the Riesz representation v_T is computed below for $n = 2, 3$.

Remark 3.48 (correction functions IV). For any $T \in \mathcal{T}$ let $\Phi_{T,1}, \dots, \Phi_{T,L}$ for $L := (n+1)(n+2)/2$ denote a basis of $P_2(T)$ that satisfies $\int_T \Phi_{T,\ell} \Phi_{T,k} b_T \, dx = \delta_{\ell,k}$ for any $\ell, k = 1, \dots, L$, $b_T \in V$ in Remark 3.47, then the Riesz representation $v_T \in P_2(T)$ with (3.61) reads

$$v_T = \sum_{\ell=1}^L \left(\int_T (v_M - J_3 v_M) \Phi_{T,\ell} \, dx \right) \Phi_{T,\ell}.$$

These basis functions can be computed in terms of the barycentric coordinates (3.1) with (3.2). For any vertex $P \in \mathcal{V}(T)$ and any edge (1-subsimplex) $E = \text{conv}\{P, Q\}$ with vertices $P, Q \in \mathcal{V}(T)$ of T define

$$\Phi_{T,z} := \frac{1}{\sqrt{|T|}} (a + b\lambda_P + c\lambda_P^2) \quad \text{and} \quad \Phi_{T,E} := \frac{1}{\sqrt{|T|}} (d + e(\lambda_P + \lambda_Q) + f\lambda_P\lambda_Q).$$

For $n = 2$ the coefficients read

$$\begin{aligned} a &= -\frac{4(11\sqrt{21} + 27\sqrt{154} - 10\sqrt{462})}{297}, & b &= \frac{40(6\sqrt{154} - \sqrt{462})}{99}, \\ c &= -\frac{10\sqrt{154}}{3}, & d &= \frac{2(11\sqrt{14} - 20\sqrt{77} + 18\sqrt{231})}{99}, \\ e &= \frac{20(\sqrt{77} - 2\sqrt{231})}{33}, & f &= \frac{40\sqrt{231}}{9}, \end{aligned}$$

and for $n = 3$ the coefficients read

$$\begin{aligned} a &= \frac{540\sqrt{110/13} - 30\sqrt{462/13} - 30\sqrt{858}}{256}, & b &= \frac{120\sqrt{858} - 360\sqrt{1430}}{256}, \\ c &= \frac{315\sqrt{1430}}{128}, & d &= \frac{45(6\sqrt{15} - 13 - \sqrt{7})}{128} \sqrt{\frac{11}{13}}, \\ e &= \frac{45(\sqrt{143} - \sqrt{2145})}{64}, & f &= \frac{105\sqrt{2145}}{32}. \end{aligned}$$

Note that the L^2 norms of the basis functions read $\|\Phi_{T,z}\|_{L^2(T)}^2 = a^2 + \frac{2ab}{n+1} + \frac{2(2ac+b^2)}{(n+1)(n+2)} + \frac{12bc}{(n+1)(n+2)(n+3)}$ for any vertex $z \in \mathcal{V}(T)$ and $\|\Phi_{T,E}\|_{L^2(T)}^2 = d^2 + \frac{4ed}{n+1} + \frac{6e^2+2df}{(n+1)(n+2)} + \frac{8ef}{(n+1)(n+2)(n+3)} + \frac{4f^2}{(n+1)(n+2)(n+3)(n+4)}$ for any edge E of T . Renaming the basis functions allows the substitution in the formula for v_T .

Definition 3.49 ($J_4 \equiv J_M$). For any $v_M \in \mathcal{M}(T)$ define

$$J_M v_M := J_4 v_M := J_3 v_M + \sum_{T \in \mathcal{T}} v_T b_T.$$

with b_T in Remark 3.47 and the Riesz representation v_T with (3.61).

Lemma 3.50 (properties of $J_4 \equiv J_M$). *For any $v_M \in \mathcal{M}(T)$ the conforming companion $J_M v_M := J_4 v_M$ satisfies Theorem 3.36.*

Proof of Theorem 3.36.a. For any $K \in \mathcal{T}$, $w_K = v_K \in P_2(K)$, (3.61) and a Cauchy-Schwarz inequality imply

$$(b_K v_K, v_K)_{L^2(K)} = (v_M - J_3 v_M, v_K)_{L^2(K)} \leq \|v_M - J_3 v_M\|_{L^2(K)} \|v_K\|_{L^2(K)}. \quad (3.62)$$

The volume-bubble function b_K in Remark 3.47 generates an inverse estimate (cf. [Ver13, §3.6]) with a universal constant $C_{\text{inv}} \approx 1$. This and the bound $\|b_K\|_{L^\infty(K)} = 1$ show

$$C_{\text{inv}} \|w_K\|_{L^2(K)} \leq \|b_K^{1/2} w_K\|_{L^2(K)} \leq \|w_K\|_{L^2(K)} \quad \text{for all } w_K \in P_2(K). \quad (3.63)$$

The combination of (3.62)–(3.63) proves

$$C_{\text{inv}}^2 \|v_K\|_{L^2(K)}^2 \leq \|b_K^{1/2} v_K\|_{L^2(K)}^2 \leq \|v_M - J_3 v_M\|_{L^2(K)} \|v_K\|_{L^2(K)}.$$

Hence, $C_{\text{inv}}^2 \|v_K\|_{L^2(K)} \leq \|v_M - J_3 v_M\|_{L^2(K)}$. Definition 3.49 and $\|b_K\|_{L^\infty(K)} = 1$ imply

$$\|J_M v_M - J_3 v_M\|_{L^2(K)} = \|v_K b_K\|_{L^2(K)} \leq \|v_K\|_{L^2(K)} \leq C_{\text{inv}}^{-2} \|v_M - J_3 v_M\|_{L^2(K)}.$$

This holds for all $K \in \mathcal{T}$ and so a triangle inequality implies for $C_6 = (1 + C_{\text{inv}}^{-2})$

$$\begin{aligned} \|v_M - J_M v_M\|_{L^2(K)} &= \|v_M - J_3 v_M\|_{L^2(K)} + \|J_M v_M - J_3 v_M\|_{L^2(K)} \\ &\leq C_6 \|v_M - J_3 v_M\|_{L^2(K)}. \end{aligned}$$

Given the basis functions in Remark 3.48 a direct computation of the bound $C_6 \leq 1 + (n+1)\|\Phi_{K,z}\|_{L^2(K)}^2 + n(n+1)/2\|\Phi_{K,E}\|_{L^2(K)}^2$ as in the proof of Theorem 3.25.a is possible.

The combination with Lemma 3.46.a and Lemma 3.42.a proves

$$\|(v_M - J_M v_M)\|_{L^2(K)} \leq C_6 C_5 C_4 \|v_M - J_1 v_M\|_{L^2(K)}. \quad (3.64)$$

This and Lemma 3.39.a conclude the proof of Theorem 3.36.a with constant $M_1 := C_1 C_4^2 C_5^2 C_6^2$. $\square$

Proof of Theorem 3.36.b. An inverse estimate as in Lemma 3.43, (3.64), and a summation over all $K \in \mathcal{T}$ imply with $C_7 := (1 + \tilde{C}_{\text{inv}})C_4 C_5 C_6$ that

$$\|h_{\mathcal{T}}^{-2}(1 - J_M)v_M\|_{L^2(\Omega)} + \| (1 - J_M)v_M \|_{\text{pw}} \leq C_7 \|h_{\mathcal{T}}^{-2}(1 - J_1)v_M\|_{L^2(\Omega)}.$$

Hence, Lemma 3.39.b concludes the proof with $M_2 := C_7^2 C_2$ and $M_3 = C_7 C_3$. $\square$

Proof of Theorem 3.36.c. Since $(v_T b_T)|_{\partial T} \equiv 0 \equiv (\nabla(v_T b_T))|_{\partial T}$ vanishes along the boundary of any $T \in \mathcal{T}$, $L_E(J_M v_M) = L_E(J_3 v_M)$ and $L_F(J_M v_M) = L_F(J_3 v_M)$ hold for any $v_M \in M(\mathcal{T})$, $E \in \mathcal{E}$, and $F \in \mathcal{F}$. Hence, the definition of I_M in (3.46) implies $I_M(J_M v_M) = I_M(J_3 v_M)$ and $J_M \equiv J_4$ satisfies the right inverse property according to Lemma 3.46.b. $\square$

Proof of Theorem 3.36.d. Definition 3.49 and the identity (3.61) imply for all $T \in \mathcal{T}$ and $w_T \in P_2(T)$ that

$$(v_M - J_M v_M, w_T)_{L^2(T)} = (v_M - J_3 v_M, w_T)_{L^2(T)} - (b_T v_T, w_T)_{L^2(T)} = 0.$$

This proves the orthogonality. $\square$

Remark 3.51 (general boundary conditions). The construction in Sections 3.4.2.1–3.4.2.4 works for more general boundary conditions of $M(\mathcal{T})$. In Definition 3.38 the vanishing boundary values can be replaced by the same averaging formula as in the interior degrees of freedom. This simplifies the proof of Lemma 3.39 in that (3.56) holds directly for any $z \in \mathcal{V}$ etc. The corrections in Definition 3.41–3.45 are possible for boundary edges and sides as well and guarantee the right inverse property. Subsection 3.4.2.4 is independent of boundary conditions. $\square$

The outline of the design of the 3D conforming companion operator for the Morley finite element is also contained in the supplement of the preprint [CP21].

4 A modified HHO Method

This chapter introduces a scheme, which relies on the framework of hybrid high-order (HHO) methods and directly computes guaranteed lower bounds for the Dirichlet eigenvalues of the Laplacian under a simplified tuning of the stabilization terms. The discrete unknowns are polynomials of degree $k \geq 0$ attached to the mesh faces and polynomials of degree $\ell = k + 1$ attached to the mesh cells of the triangulation. The present method utilises a local gradient reconstruction operator mapping to piecewise Raviart-Thomas functions of degree k and a cell-based stabilization operator that contains another reconstruction.

The chapter is organized as follows. After a short introduction to HHO literature in Section 4.1, the modified method is presented in Section 4.2. Therefore Section 4.2 defines both reconstruction operators, the stability operator, and the discrete bilinear forms and introduces a few of their properties. Section 4.3 proves the guaranteed lower bound property (4.5) in Theorem 4.8. The a priori convergence analysis is contained in Section 4.4. Finally, Section 4.5 concentrates on the lowest-order version which is closely related to the Crouzeix-Raviart method and allows a comparison with the post-processed Crouzeix-Raviart bound (4.46).

4.1 Introduction to HHO methods

The hybrid high-order (HHO) methods have been introduced in [DPEL14] for linear diffusion and in [DPE15] for locking-free linear elasticity. The algebraic realization and implementation are discussed in [CDPE18]. The literature presents applications to various problems [CEP21b] such as, e.g., to the advection-diffusion equation [DPDE15], Stokes equation [DP+16], Leray-Lions equations [DPD17], hyperelasticity with finite deformation [AEP18], elliptic obstacle problems [CEG20], and recently to the biharmonic equations [DE22] as well as convex minimization problems [CT21; CT22]. A good introduction is contained in [EG21, Chapter 39], which presents the HHO method for the Poisson equation with homogeneous Dirichlet boundary conditions. In general polygonal meshes with polyhedral shaped cells are possible. For simplified notation consider a shape-regular triangulation $\mathcal{T}$ of $\Omega \subset \mathbb{R}^n$. The idea of the HHO methods is that the discrete solution consists of two variables, a face component $u_{\mathcal{F}} = (u_F)_{F \in \mathcal{F}} \in P_k(\mathcal{F})$ to approximate the traces and a cell component $u_{\mathcal{T}} = (u_T)_{T \in \mathcal{T}} \in P_{\ell}(\mathcal{T})$ to approximate the solution in the interior of the cells. The discrete function $v_h \in V_h := P_{\ell}(\mathcal{T}) \times P_k(\mathcal{F})$ restricted to $T \in \mathcal{T}$ reads $v_h|_T = (v_T, (v_F)_{F \in \mathcal{F}(T)}) \in P_{\ell}(T) \times P_k(\mathcal{F}(T)) = V_h|_T$.

In this nonconforming setting static condensation allows the (local) elimination of one of the components. In the standard HHO methods the cell variable is eliminated, whereas in the modified method this is the dominant variable. The main ingredients in HHO methods are a local reconstruction operator and a local stabilization operator. For any $T \in \mathcal{T}$, the standard local reconstruction R_T maps $v_h|_T \in V_h|_T$ onto $R_T(v_h|_T) \in P_m(T)$ with $\Pi_0 R_T(v_h|_T) = \Pi_0 v_T$ and

$$(\nabla R_T(v_h|_T), \nabla p)_{L^2(T)} = -(v_T, \Delta p)_{L^2(T)} + \sum_{F \in \mathcal{F}(T)} (v_F, \nabla p \cdot \nu_T)_{L^2(F)} \text{ for all } p \in P_m(T),$$

while the local stabilization $S_{\partial T}$ maps $v_h|_T \in V_h|_T$ for any $F \in \mathcal{F}(T)$ onto

$$S_{\partial T}(v_h|_T)|_F = \Pi_{F,k}(v_F - R_T(v_h|_T)|_F - \Pi_\ell(v_T - R_T(v_h|_T))|_F) \in P_k(F).$$

The operators are utilized, e.g., for the Poisson problem in [EG21] and for the approximation of the Laplace eigenvalue problem in [Cal+19] with $\ell = k$ and $m = k + 1$ and lead to $\mathcal{O}(h_{\max}^{k+1})$ convergence in H^1 for sufficiently smooth functions. The cell polynomial degree $\ell \geq 0$ can be chosen as $\ell \in \{k - 1, k, k + 1\}$ according to [CDPE16]. For $\ell = k - 1$ (and $k \geq 1$) the HHO method is comparable to the nonconforming virtual element method (ncVEM) from [DLM16], up to an equivalent form of the stabilization. Moreover, [CDPE16] bridges HHO to the hybridizable discontinuous Galerkin (HDG) methods introduced in [CGL09]. In the case $\ell = k + 1$, the simplified local stabilization reads $S_{\partial T}(v_h|_T)|_F = \Pi_{F,k}(v_F - v_T) \in P_k(F)$ for any $F \in \mathcal{F}(T)$. The latter is considered as Lehrenfeld-Schöberl discretization in the HDG context [LS16; Oik15]. All these methods are also connected to the weak Galerkin method [WY13]. The modified HHO method considered in this chapter uses cell variables of polynomial degree $\ell = k + 1 = m$. A novelty is, that the standard reconstruction above for $m = k + 1$ is combined with a Raviart-Thomas reconstruction, mapping $v_h|_T = (v_T, (v_F)_{F \in \mathcal{F}(T)}) \in V_h|_T$ for any $T \in \mathcal{T}$ onto the local gradient reconstruction $G_{RT,T}(v_h|_T) \in RT_k(T)$ with

$$(G_{RT,T}(v_h|_T), q)_{L^2(T)} = -(v_T, \operatorname{div} q)_{L^2(T)} + \sum_{F \in \mathcal{F}(T)} (v_F, q \cdot \nu_T)_{L^2(F)} \text{ for all } q \in RT_k(T)$$

as in, e.g., [AEP18; DPDM18]. The second novelty in the modified method is the cell-based stabilization $S_T(v_h|_T) = v_T - R_T(v_h|_T) \in P_{k+1}(T)$ for any $T \in \mathcal{T}$. The novel ansatz allows to prove the guaranteed lower bound property in Theorem 4.8 and still leads to the expected convergence rates in Theorem 4.13 below.

4.2 The modified HHO method

Let $\Omega \subset \mathbb{R}^n$ with $n \geq 2$ denote a polyhedral bounded Lipschitz domain with set of admissible triangulation $\mathbb{T}$. The weak formulation of the Laplace eigenvalue problem seeks $(\lambda, u) \in \mathbb{R}^+ \times H_0^1(\Omega)$ with

$$a(u, v) = \lambda b(u, v) \quad \text{for all } v \in V := H_0^1(\Omega) \quad \text{and} \quad \|u\|_{L^2(\Omega)} = 1, \quad (4.1)$$

where $a(u, v) := (\nabla u, \nabla v)_{L^2(\Omega)}$ denotes the energy scalar product and $b(u, v) := (u, v)_{L^2(\Omega)}$ the L^2 scalar product; the infinite but countable many eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$ with $\lim_{j \rightarrow \infty} \lambda_j = \infty$ in (4.1) are enumerated in ascending order counting multiplicities. For any regular triangulation $\mathcal{T} \in \mathbb{T}$, the discrete space reads $V_h := P_{k+1}(\mathcal{T}) \times P_k(\mathcal{F}(\Omega))$ with the space of piecewise polynomials of degree at most $(k+1)$ on each simplex in the triangulation $P_{k+1}(\mathcal{T})$ and the space of piecewise polynomials of degree at most k on each interior face $P_k(\mathcal{F}(\Omega))$. The discrete eigenvalue problem seeks $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ with

$$a_h(u_h, v_h) = \lambda_h b_h(u_h, v_h) \quad \text{for all } v_h \in V_h \quad \text{and} \quad b_h(u_h, u_h) = 1. \quad (4.2)$$

While the bilinear form b_h represents the $L^2(\Omega)$ scalar product b of the volume components, the gradient-like approximations in the bilinear form a_h involve two reconstructions R and G_{RT} of the discrete unknowns in V_h . The precise definition of the linear maps $R : V_h \rightarrow L^2(\Omega; \mathbb{R})$ and $G_{RT} : V_h \rightarrow L^2(\Omega; \mathbb{R}^n)$ can be found below in Subsection 4.2.1. Given two positive parameters $0 < \alpha < 1$ and $0 < \beta < \infty$, for any $u_h := (u_{\mathcal{T}}, u_{\mathcal{F}})$ and $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$, the bilinear forms read

$$a_h(u_h, v_h) := (G_{RT}(u_h), G_{RT}(v_h))_{L^2(\Omega)} \quad (4.3)$$

$$\begin{aligned} & - \alpha \left((1 - \Pi_k) G_{RT}(u_h), (1 - \Pi_k) G_{RT}(v_h) \right)_{L^2(\Omega)} \\ & + \beta (\nabla_{\text{pw}}(u_{\mathcal{T}} - R(u_h)), \nabla_{\text{pw}}(v_{\mathcal{T}} - R(v_h)))_{L^2(\Omega)}, \\ b_h(u_h, v_h) & := b(u_{\mathcal{T}}, v_{\mathcal{T}}). \end{aligned} \quad (4.4)$$

The Raviart-Thomas reconstruction G_{RT} of the gradient does not require a stabilization in the source problem [AEP18; DPDM18], but the second term on the right-hand side in (4.3) has a negative sign and another stabilization with the reconstruction R in $P_{k+1}(\mathcal{T})$ is added. The algebraic eigenvalue problem (4.2) has $M := \dim(P_{k+1}(\mathcal{T}))$ finite and positive algebraic eigenvalues $0 < \lambda_h(1) \leq \dots \leq \lambda_h(M) < \infty$ enumerated in ascending order counting multiplicities.

This chapter contains the a priori error analysis for discrete eigenvalues and eigenvectors and confirms optimal convergence rates. Moreover it proves, the following GLB property for the j -th discrete eigenvalue $\lambda_h(j)$ of (4.2) and the j -th eigenvalue λ_j of (4.1)

$$\tau^2 \beta + \delta^2 \min\{\lambda_j, \lambda_h(j)\} \leq \alpha < 1 \quad \text{implies} \quad \lambda_h(j) \leq \lambda_j, \quad (4.5)$$

with $\delta := \kappa h_{\max}$ and τ, κ related to the stability of L^2 -projections onto piecewise polynomial spaces, cf. Theorem 2.4 and Lemma 4.3 below. Notice that (4.5) means that each of the conditions (i) $\tau^2 \beta + \delta^2 \lambda_j \leq \alpha < 1$ (a priori) and (ii) $\tau^2 \beta + \delta^2 \lambda_h(j) \leq \alpha < 1$ (a posteriori) implies the GLB property. The following analysis extends [CEP21a].

4.2.1 Operators of interest

This section is devoted to the reconstruction and stability operators of the new method (4.2) and their properties. Set $V := H_0^1(\Omega)$. Let $k \geq 0$ be the polynomial degree and set

$$V_h := P_{k+1}(\mathcal{T}) \times P_k(\mathcal{F}(\Omega)).$$

The components of $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$ are $v_{\mathcal{T}} := (v_T)_{T \in \mathcal{T}}$, $v_{\mathcal{F}} := (v_F)_{F \in \mathcal{F}}$. Note that $v_F \equiv 0$ whenever $F \in \mathcal{F}(\partial\Omega)$ by definition of $P_k(\mathcal{F}(\Omega))$.

Let I denote the **interpolation operator** $I : V \rightarrow V_h$ such that $I(v) = (\Pi_{k+1}(v), \Pi_{\mathcal{F},k}(v)) \in V_h$ for all $v \in V$.

Define the **reconstruction operator** $R : V_h \rightarrow P_{k+1}(\mathcal{T})$ such that for any $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$, $R(v_h) \in P_{k+1}(\mathcal{T})$ is the unique function with $\Pi_0 R(v_h) = \Pi_0(v_{\mathcal{T}})$ and for all $p \in P_{k+1}(\mathcal{T})$

$$(\nabla_{\text{pw}} R(v_h), \nabla_{\text{pw}} p)_{L^2(\Omega)} = -(v_{\mathcal{T}}, \Delta_{\text{pw}} p)_{L^2(\Omega)} + \langle v_{\mathcal{F}}, \nabla_{\text{pw}} p \cdot \nu_{\mathcal{T}} \rangle_{\partial\mathcal{T}}. \quad (4.6)$$

Let G denote the **Galerkin projection** onto $P_{k+1}(\mathcal{T})$, i.e., for any $v \in H^1(\mathcal{T})$ holds $G(v) \in P_{k+1}(\mathcal{T})$ with $\Pi_0 G(v) = \Pi_0(v)$ and $a_{\text{pw}}((1-G)(v), p) = 0$ for all $p \in P_{k+1}(\mathcal{T})$.

Lemma 4.1. $R \circ I = G$ holds in $V := H_0^1(\Omega)$.

Proof. This identity is a direct consequence from an integration by parts and the definition of R and G [DPEL14; DPE15]. For any $v \in V$, $T \in \mathcal{T}$ and $p \in P_{k+1}(T)$ the definition of G and an integration by parts guarantee

$$\int_T \nabla Gv \cdot \nabla p \, dx = \int_T \nabla v \cdot \nabla p \, dx = - \int_T v \Delta p + \int_{\partial T} v \nabla p \cdot \nu_T \, ds.$$

Since $\Delta p \in P_{k-1}(T)$ and $\nabla p \cdot \nu_T \in P_k(\mathcal{F}(T))$ the definition of R implies

$$\int_T \nabla Gv \cdot \nabla p \, dx = - \int_T (\Pi_{k+1} v) \Delta p \, dx + \int_{\partial T} (\Pi_{\mathcal{F},k} v) \nabla p \cdot \nu_T \, ds = \int_T \nabla R I v \cdot \nabla p \, dx.$$

The observation, that the constant projections $\Pi_0 R I v = \Pi_0 \Pi_{k+1} v = \Pi_0 v = \Pi_0 Gv$ coincide, concludes the proof. $\square$

The **Raviart-Thomas reconstruction** $G_{RT} : V_h \rightarrow RT_k^{\text{pw}}(\mathcal{T})$ maps $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$ onto a unique $G_{RT}(v_h) \in RT_k^{\text{pw}}(\mathcal{T})$ such that for all $q_{RT} \in RT_k^{\text{pw}}(\mathcal{T})$

$$(G_{RT}(v_h), q_{RT})_{L^2(\Omega)} = -(v_{\mathcal{T}}, \text{div}_{\text{pw}} q_{RT})_{L^2(\Omega)} + \langle v_{\mathcal{F}}, q_{RT} \cdot \nu_{\mathcal{T}} \rangle_{\partial\mathcal{T}}. \quad (4.7)$$

Observe that G_{RT} directly reconstructs the gradient of the discrete function.

Lemma 4.2. $G_{RT} \circ I = \Pi_{RT,k} \circ \nabla$ holds in $V := H_0^1(\Omega)$.

Proof. This identity is a direct consequence of the operator definition and an integration by parts [AEP18; DPDM18]. For any $v \in V$, $T \in \mathcal{T}$ and $q_{RT} \in RT_k(T)$, the definition of $\Pi_{RT,k}$ and an integration by parts guarantee

$$\int_T q_{RT} \cdot \Pi_{RT,k} \nabla v \, dx = \int_T q_{RT} \cdot \nabla v \, dx = - \int_T v \operatorname{div} q_{RT} \, dx + \int_{\partial T} v q_{RT} \cdot \nu_T \, ds.$$

Since $\operatorname{div} q_{RT} \in P_k(T) \subset P_{k+1}(T)$ and $q_{RT} \cdot \nu_T \in P_k(\mathcal{F}(T))$ [BBF13, Prop.2.3.3], this is equal to

$$- \int_T (\Pi_{k+1} v) \operatorname{div} q_{RT} \, dx + \int_{\partial T} (\Pi_{\mathcal{F},k} v) q_{RT} \cdot \nu_T \, ds = \int_T q_{RT} \cdot G_{RT} I v \, dx.$$

Consequently, the difference $(\Pi_{RT,k} \nabla v - G_{RT} I v)$ is L^2 -orthogonal to $RT_k(T)$. Since both $\Pi_{RT,k} \nabla v \in RT_k(T)$ and $G_{RT} I v \in RT_k(T)$ are Raviart-Thomas function of order k , $\Pi_{RT,k} \nabla v - G_{RT} I v = 0$ follows. $\square$

The **stabilization operator** $S : V_h \rightarrow P_{k+1}(\mathcal{T})$ reads for any $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$

$$S(v_h) := v_{\mathcal{T}} - R(v_h). \quad (4.8)$$

Recall from Theorem 2.4, that C_{st} satisfies $\|\nabla(1 - \Pi_{k+1})(f)\|_{L^2(T)} \leq C_{st} \|(1 - \Pi_k)(\nabla f)\|_{L^2(T)}$ for all $T \in \mathcal{T}$ and $f \in H^1(T)$.

Lemma 4.3. *Given any $v \in V := H_0^1(\Omega)$, any simplex $T \in \mathcal{T}$, and $\tau^2 := C_{st}^2 - 1$, the stability term satisfies*

$$S(Iv) = \Pi_{k+1}(v) - G(v), \quad (4.9)$$

$$\|\nabla S(Iv)\|_{L^2(T)} \leq \tau \|(1 - \Pi_k)(\nabla v)\|_{L^2(T)}, \quad (4.10)$$

$$\|S(Iv)\|_{\text{pw}} \leq \tau \|(1 - \Pi_k)(\nabla v)\|_{L^2(\Omega)}. \quad (4.11)$$

Proof. The definition of $S(Iv)$ in (4.8) and Lemma 4.1 imply (4.9) directly. Since $\nabla_{\text{pw}}(1 - G)(v)$ is L^2 -orthogonal to $\nabla_{\text{pw}} P_{k+1}(\mathcal{T})$, the Pythagoras theorem proves for any $T \in \mathcal{T}$,

$$\begin{aligned} \|\nabla S(Iv)\|_{L^2(T)}^2 &= \|\nabla \Pi_{k+1}(v) - \nabla G(v)\|_{L^2(T)}^2 \\ &= \|\nabla(1 - \Pi_{k+1})(v)\|_{L^2(T)}^2 - \|\nabla(1 - G)(v)\|_{L^2(T)}^2. \end{aligned}$$

Theorem 2.4 bounds the term $\|\nabla(1 - \Pi_{k+1})(v)\|_{L^2(T)}^2$ by $C_{st}^2 \|(1 - \Pi_k)(\nabla v)\|_{L^2(T)}^2$, whereas the best approximation property of Π_k proves that

$$\|(1 - \Pi_k)(\nabla v)\|_{L^2(T)} \leq \|\nabla(1 - G)(v)\|_{L^2(T)}.$$

The combination finishes the proof of (4.10), and (4.11) follows by summation over all $T \in \mathcal{T}$. $\square$

Remark 4.4 (alternative stabilization). One could consider $b(h_{\mathcal{T}}^{-2}S(v_h), S(v_h))$ instead of the stabilization term $a_{\text{pw}}(S(v_h), S(v_h))$ in the bilinear form (4.3). In this case (4.9), an inverse estimate, and (4.11) imply for any $T \in \mathcal{T}$ and $v \in V$,

$$\|h_T^{-1}S(Iv)\|_{L^2(T)} \leq \tilde{\tau}\|(1 - \Pi_k)(\nabla v)\|_{L^2(T)}$$

to replace (4.10) and $\|h_{\mathcal{T}}^{-1}S(Iv)\|_{L^2(\Omega)} \leq \tilde{\tau}\|(1 - \Pi_k)(\nabla v)\|_{L^2(\Omega)}$ to replace (4.11). These replacements allow for the same results with the alternative stabilization in all the proofs below. The resulting method is analysed in the upcoming work [CGT22]. The lowest-order version of (4.2) allows for a direct estimate of τ in Lemma 4.21. However, for higher polynomial degrees the numerical approximation of τ is necessary. It turns out that the direct approximation of $\tilde{\tau}$ such that $\|\Pi_{k+1}(v) - G(v)\|_{L^2(T)} \leq \tilde{\tau}h_T\|(1 - \Pi_k)(\nabla v)\|_{L^2(T)}$ for any $v \in V$ is easier and robust with respect to the polynomial degree. In [CGT22] the constant $\tilde{C}_{st} \leq C_{st} \propto \sqrt{k+1}$ with

$$\|\nabla(1 - G)v\|_{L^2(T)} \leq \tilde{C}_{st}\|(1 - \Pi_k)\nabla v\|_{L^2(T)} \quad \text{for all } v \in H^1(T) \text{ and } T \in \mathcal{T}$$

allows to bound $\tilde{\tau} \leq C_P \tilde{C}_{st}$.

Remark 4.5 (piecewise evaluation). For all $T \in \mathcal{T}$, the reconstructions $R(v_h)|_T$ and $G_{RT}(v_h)|_T$ and the stabilization $S(v_h)|_T$ solely depend on the local data $v_T = v_{\mathcal{T}}|_T$ and $v_F = v_{\mathcal{F}}|_F$ for all $F \in \mathcal{F}(T)$ as illustrated in Section 4.1. Hence, all these quantities can be computed locally and in parallel.

4.2.2 Properties of the discrete bilinear forms

This section contains additional properties of the bilinear forms $a_h, b_h : V_h \times V_h \rightarrow \mathbb{R}$ from (4.3)–(4.4). Recall the global constants $0 < \alpha < 1$ and $0 < \beta < \infty$, the reconstructions in (4.6)–(4.7), and the stabilization in (4.8).

Remark 4.6 (alternative a_h). Since $(G_{RT}u_h, G_{RT}v_h)_{L^2(\Omega)} = ((1 - \Pi_k)G_{RT}u_h, (1 - \Pi_k)G_{RT}v_h)_{L^2(\Omega)} + (\Pi_k G_{RT}u_h, \Pi_k G_{RT}v_h)_{L^2(\Omega)}$ due to the L^2 -projection property of Π_k , the bilinear form a_h can be rewritten as follows for any $u_h, v_h \in V_h$,

$$\begin{aligned} a_h(u_h, v_h) &= (1 - \alpha) \left((1 - \Pi_k)G_{RT}(u_h), (1 - \Pi_k)G_{RT}(v_h) \right)_{L^2(\Omega)} \\ &\quad + (\Pi_k G_{RT}(u_h), \Pi_k G_{RT}(v_h))_{L^2(\Omega)} + \beta a_{\text{pw}}(S(u_h), S(v_h)). \end{aligned} \quad (4.12)$$

The vector space V_h is equipped with the norms $\|\bullet\|_h$ [DPEL14, Eq. (28)] and $\|\bullet\|_{a,h}$, defined for any $v_h = (v_{\mathcal{T}}, v_{\mathcal{F}}) = ((v_T)_{T \in \mathcal{T}}, (v_F)_{F \in \mathcal{F}}) \in V_h$ by

$$\|v_h\|_h^2 := \sum_{T \in \mathcal{T}} \left(\|\nabla v_T\|_{L^2(T)}^2 + \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_F - v_T\|_{L^2(F)}^2 \right), \quad (4.13)$$

$$\begin{aligned} \|v_h\|_{a,h}^2 &:= a_h(v_h, v_h) \\ &= \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 - \alpha \|(1 - \Pi_k)G_{RT}(v_h)\|_{L^2(\Omega)}^2 + \beta \|S(v_h)\|_{\text{pw}}^2 \\ &= (1 - \alpha) \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 + \alpha \|\Pi_k G_{RT}(v_h)\|_{L^2(\Omega)}^2 + \beta \|S(v_h)\|_{\text{pw}}^2. \end{aligned} \quad (4.14)$$

Since the homogeneous Dirichlet boundary conditions are embedded in the discrete space V_h , $\|\bullet\|_h$ defines a norm on V_h . The definiteness of $\|\bullet\|_{a,h}$ follows from Lemma 4.7 below. This guarantees that $a_h(\bullet, \bullet)$ is a scalar product on $V_h \times V_h$ which induces a norm on V_h . Consequently, the Lax-Milgram Lemma guarantees that the source problem (4.27) associated with a_h is well-posed.

Lemma 4.7. *There exist constants $\underline{\gamma}, \bar{\gamma} > 0$, independent of the mesh-size, such that*

$$\underline{\gamma}\|v_h\|_h^2 \leq a_h(v_h, v_h) \leq \bar{\gamma}\|v_h\|_h^2 \quad \text{for all } v_h \in V_h.$$

Proof. The Pythagoras theorem and the best approximation property of $\Pi_{F,k}$ show for any $T \in \mathcal{T}$, $F \in \mathcal{F}(T)$, and $v_h = (v_T, v_F) \in V_h$ with $v_T|_T = v_T$ and $v_F|_F = v_F$, that

$$\begin{aligned} \|v_T - v_F\|_{L^2(F)}^2 &= \|(1 - \Pi_{F,k})(v_T)\|_{L^2(F)}^2 + \|\Pi_{F,k}(v_T) - v_F\|_{L^2(F)}^2 \\ &\leq \|(1 - \Pi_0)(v_T)\|_{L^2(F)}^2 + \|\Pi_{F,k}(v_T) - v_F\|_{L^2(F)}^2. \end{aligned}$$

The trace inequality from Lemma 2.10 together with the Poincaré inequality from Lemma 2.11 and the shape-regularity of $\mathcal{T}$ lead to

$$\begin{aligned} &\|(1 - \Pi_0)(v_T)\|_{L^2(F)}^2 \\ &\leq \frac{|F|}{|T|} \|(1 - \Pi_0)(v_T)\|_{L^2(T)} \left(\|(1 - \Pi_0)(v_T)\|_{L^2(T)} + 2h_T/n \|\nabla v_T\|_{L^2(T)} \right) \\ &\leq C_P \frac{h_T^2 |F|}{|T|} (C_P + 2/n) \|\nabla v_T\|_{L^2(T)}^2 \leq c_2 h_F \|\nabla v_T\|_{L^2(T)}^2 \end{aligned}$$

with the abbreviation $c_2 := C_{\text{sr}} C_P (C_P + 2/n)$ throughout this proof. It is shown in [AEP18; DPDM18] (see, e.g., Lemma 1 in [AEP18]) that there exists a constant $c_1 > 0$ such that for any $T \in \mathcal{T}$,

$$\sum_{F \in \mathcal{F}(T)} h_F^{-1} \|\Pi_{F,k}(v_T) - v_F\|_{L^2(T)}^2 \leq c_1 \|G_{RT}(v_h)\|_{L^2(T)}^2.$$

The combination of the last three displayed inequalities leads to an upper bound of $h_F^{-1} \|v_T - v_F\|_{L^2(F)}^2$. The sum over all $T \in \mathcal{T}$ and $F \in \mathcal{F}(T)$ reads

$$\sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_T - v_F\|_{L^2(F)}^2 \leq c_1 \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 + c_2(n+1) \|v_{\mathcal{T}}\|_{\text{pw}}^2. \quad (4.15)$$

Let $\Pi_{\nabla P_{k+1}}$ denote the bounded L^2 -projection onto $\nabla_{\text{pw}} P_{k+1}(\mathcal{T})$ (the space of gradients of piecewise polynomials of degree $(k+1)$). The comparison of (4.6) with (4.7) proves that $\nabla_{\text{pw}} R(v_h) = \Pi_{\nabla P_{k+1}} G_{RT}(v_h)$ for all $v_h \in V_h$. Hence, it holds

$$\|R(v_h)\|_{\text{pw}} = \|\Pi_{\nabla P_{k+1}} G_{RT}(v_h)\|_{L^2(\Omega)} \leq \|G_{RT}(v_h)\|_{L^2(\Omega)}.$$

This, a triangle inequality, and the definition of $S(v_h)$ in (4.8) show that

$$\|v_{\mathcal{T}}\|_{\text{pw}} \leq \|v_{\mathcal{T}} - R(v_h)\|_{\text{pw}} + \|R(v_h)\|_{\text{pw}} \leq \|S(v_h)\|_{\text{pw}} + \|G_{RT}(v_h)\|_{L^2(\Omega)}. \quad (4.16)$$

The combination of (4.15)–(4.16) with the last identity from (4.14) shows that

$$\begin{aligned} \|v_h\|_h^2 &= \|v_{\mathcal{T}}\|_{\text{pw}}^2 + \sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_T - v_F\|_{L^2(F)}^2 \\ &\leq 2(1 + c_2(n+1)) \|S(v_h)\|_{\text{pw}}^2 + (2 + c_1 + 2c_2(n+1)) \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 \\ &\leq \underline{\gamma}^{-1} \left((1 - \alpha) \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 + \beta \|S(v_h)\|_{L^2(\Omega)}^2 \right) \leq \underline{\gamma}^{-1} \|v_h\|_{a,h}^2 \end{aligned}$$

with the constant $\underline{\gamma}^{-1} := \max\{2(1 + c_2(n+1))/\beta, (2 + c_1 + 2c_2(n+1))/(1 - \alpha)\}$. On the other hand, for any $T \in \mathcal{T}$, the definition of $G_{RT}(v_h)$ in (4.7) with the test function $q_{RT} = G_{RT}(v_h) \in RT_k^{\text{pw}}(\mathcal{T})$, an integration by parts, and the Cauchy-Schwarz inequality lead to

$$\begin{aligned} \|G_{RT}(v_h)\|_{L^2(T)}^2 &= (\nabla v_{\mathcal{T}}, G_{RT}(v_h))_{L^2(T)} + \langle v_{\mathcal{F}} - v_{\mathcal{T}}, G_{RT}(v_h) \cdot \nu_{\mathcal{T}} \rangle_{\partial T} \\ &\leq \|\nabla v_{\mathcal{T}}\|_{L^2(T)} \|G_{RT}(v_h)\|_{L^2(T)} \\ &\quad + \sum_{F \in \mathcal{F}(T)} h_F^{1/2} \|G_{RT}(v_h)\|_{L^2(F)} h_F^{-1/2} \|v_F - v_T\|_{L^2(F)}. \end{aligned}$$

The combination of the trace inequality from Lemma 2.10 and an inverse estimate for Raviart-Thomas functions with constant c_{inv} shows that $h_F^{1/2} \|G_{RT}(v_h)\|_{L^2(F)} \leq c_3 \|G_{RT}(v_h)\|_{L^2(T)}$ with $c_3 := \sqrt{C_{\text{sr}}(1 + 2c_{\text{inv}}/n)}$ for any $F \in \mathcal{F}(T)$. This and the Cauchy-Schwarz inequality in $\mathbb{R}^{n+1}$ lead to

$$\|G_{RT}(v_h)\|_{L^2(T)}^2 \leq 2\|\nabla v_{\mathcal{T}}\|_{L^2(T)}^2 + 2(n+1)c_3^2 \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_F - v_T\|_{L^2(F)}^2.$$

The sum over all $T \in \mathcal{T}$ reads

$$\|G_{RT}(v_h)\|_{L^2(\Omega)}^2 \leq 2\|v_{\mathcal{T}}\|_{\text{pw}}^2 + 2(n+1)c_3^2 \sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_F - v_T\|_{L^2(F)}^2 \leq c_4^2 \|v_h\|_h^2$$

with $c_4^2 := 2 \max\{1, (n+1)c_3^2\}$. The boundedness of the stability contribution follows from the triangle inequality, the above estimate $\|R(v_h)\|_{\text{pw}} \leq \|G_{RT}(v_h)\|_{L^2(\Omega)}$, and the last displayed estimate, in

$$\|S(v_h)\|_{\text{pw}} \leq \|v_{\mathcal{T}}\|_{\text{pw}} + \|G_{RT}(v_h)\|_{L^2(\Omega)} \leq (1 + c_4) \|v_h\|_h.$$

The second identity in (4.14) shows that $\|v_h\|_{a,h}^2 \leq \|G_{RT}(v_h)\|_{L^2(\Omega)}^2 + \beta \|S(v_h)\|_{\text{pw}}^2$. This concludes the proof with $\bar{\gamma} := \beta + (1 + \beta)c_4^2 + 2\beta c_4$. $\square$

4.2.3 Matrix eigenvalue problem

This section shall briefly illustrate the static condensation mentioned above. The algebraic realization of the eigenvalue problem (4.2) leads to a matrix eigenvalue problem with coefficient matrices and vector $(x_{\mathcal{T}}, x_{\mathcal{F}}) \in \mathbb{R}^{\dim P_{k+1}(\mathcal{T}) + \dim P_k(\mathcal{F}(\Omega))}$

$$\begin{pmatrix} A_{\mathcal{T}\mathcal{T}} & A_{\mathcal{T}\mathcal{F}} \\ A_{\mathcal{F}\mathcal{T}} & A_{\mathcal{F}\mathcal{F}} \end{pmatrix} \begin{pmatrix} x_{\mathcal{T}} \\ x_{\mathcal{F}} \end{pmatrix} = \lambda_h \begin{pmatrix} B_{\mathcal{T}\mathcal{T}} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_{\mathcal{T}} \\ x_{\mathcal{F}} \end{pmatrix} \quad \text{and} \quad x_{\mathcal{T}} \cdot B_{\mathcal{T}\mathcal{T}} x_{\mathcal{T}} = 1. \quad (4.17)$$

The bilinear form b_h solely depends on the volume components. The elimination of the face unknowns leads to the Schur complement

$$S_{\mathcal{T}\mathcal{T}} = A_{\mathcal{T}\mathcal{T}} - A_{\mathcal{T}\mathcal{F}} A_{\mathcal{F}\mathcal{F}}^{-1} A_{\mathcal{F}\mathcal{T}},$$

and the equivalent matrix eigenvalue problem

$$S_{\mathcal{T}\mathcal{T}} x_{\mathcal{T}} = \lambda_h B_{\mathcal{T}\mathcal{T}} x_{\mathcal{T}} \quad \text{and} \quad x_{\mathcal{T}} \cdot B_{\mathcal{T}\mathcal{T}} x_{\mathcal{T}} = 1. \quad (4.18)$$

The mass matrix $B_{\mathcal{T}\mathcal{T}} \in \mathbb{R}^{\dim P_{k+1}(\mathcal{T}) \times \dim P_{k+1}(\mathcal{T})}$ is positive definite and allows the approximation of $M := \dim(P_{k+1}(\mathcal{T}))$ eigenvalues and the application of the min-max principle in Lemma 2.7. The alternative formulation (4.18) indicates the relevance of the solution component $u_{\mathcal{T}} \in P_{k+1}(\mathcal{T})$ with coefficient vector $x_{\mathcal{T}} \in \mathbb{R}^{\dim P_{k+1}(\mathcal{T})}$ for the convergence analysis in Section 4.4.

4.3 Guaranteed lower eigenvalue bound property

This section proves property (4.5), in other words, that the modified HHO method (4.2) provides indeed guaranteed lower bounds for the continuous eigenvalues. Recall the global constants $0 < \alpha < 1$ and $0 < \beta < \infty$ as well as the constants C_{st} and κ from Theorem 2.4, set $\tau^2 := C_{st}^2 - 1$ and $\delta := \kappa h_{\max}$, such that for any $v \in V$ holds $\|(1 - \Pi_{k+1})(v)\|_{L^2(\Omega)} \leq \delta \|(1 - \Pi_k)(\nabla v)\|_{L^2(\Omega)}$ from (2.4) and $\|S(Iv)\|_{L^2(\Omega)} \leq \tau \|(1 - \Pi_k)(\nabla v)\|_{L^2(\Omega)}$ from (4.11).

Theorem 4.8. *For any $j = 1, \dots, M$, the j -th eigenvalue λ_j from (4.1) and the j -th eigenvalue $\lambda_h(j)$ from (4.2), $\tau^2 \beta + \delta^2 \min\{\lambda_j, \lambda_h(j)\} \leq \alpha < 1$ implies $\lambda_h(j) \leq \lambda_j$.*

Proof. Abbreviate $\lambda = \lambda_j$ from (4.1) and $\lambda_h = \lambda_h(j)$ from (4.2). Let $\phi_1, \dots, \phi_j$ denote the first j continuous b -orthonormal eigenfunctions of (4.1). Note that the min-max principle in Lemma 2.7 guarantees $\|\phi\|^2 \leq \lambda$ for any $\phi \in \text{span}\{\phi_1, \dots, \phi_j\}$ with $\|\phi\|_{L^2(\Omega)} = 1$.

Case 1. Assume $\Pi_{k+1}(\phi_1), \dots, \Pi_{k+1}(\phi_j) \in P_{k+1}(\mathcal{T})$ are linear dependent. Then there exists $\phi \in \text{span}\{\phi_1, \dots, \phi_j\}$ with $\|\phi\|_{L^2(\Omega)} = 1$ and $\Pi_{k+1}(\phi) = 0$. The estimate (2.4) in Theorem 2.4 implies for $\delta = \kappa h_{\max} > 0$ that

$$1 = \|\phi\|_{L^2(\Omega)} = \|(1 - \Pi_{k+1})(\phi)\|_{L^2(\Omega)} \leq \delta \|(1 - \Pi_k)(\nabla \phi)\|_{L^2(\Omega)}.$$

The Pythagoras theorem $\|\nabla\phi\|_{L^2(\Omega)}^2 = \|\Pi_k(\nabla\phi)\|_{L^2(\Omega)}^2 + \|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2$ and the min-max principle in Lemma 2.7 imply

$$\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2 \leq \|\nabla\phi\|_{L^2(\Omega)}^2 \leq \lambda.$$

The combination of the last two displayed inequalities reads $1 \leq \delta^2\lambda \leq \tau^2\beta + \delta^2\lambda$ with $\tau^2\beta \geq 0$. Hence, the condition $\tau^2\beta + \delta^2 \min\{\lambda, \lambda_h\} \leq \alpha < 1$ implies

$$\delta^2\lambda_h \leq \tau^2\beta + \delta^2\lambda_h \leq \alpha < 1 \leq \delta^2\lambda$$

and therefore $\lambda_h \leq \lambda$.

Case 2. Assume $\Pi_{k+1}(\phi_1), \dots, \Pi_{k+1}(\phi_j) \in P_{k+1}(\mathcal{T})$ are linear independent and $\tau^2\beta + \delta^2\lambda \leq \alpha < 1$. Since $\Pi_{k+1}(\phi_1), \dots, \Pi_{k+1}(\phi_j) \in P_{k+1}(\mathcal{T})$ are linear independent, the subspace $U_j := \text{span}\{I(\phi_1), \dots, I(\phi_j)\}$ of V_h has dimension j and b_h is positive definite on $U_j \times U_j$. Hence, the min-max principle from Lemma 2.7 for (4.2) (on the algebraic level) implies for the j -th discrete eigenvalue λ_h

$$\lambda_h \leq \max_{v_h \in U_j \setminus \{0\}} \frac{a_h(v_h, v_h)}{b_h(v_h, v_h)}. \quad (4.19)$$

Let $v_h = I\phi \in U_j \setminus \{0\}$ for $\phi \in \text{span}\{\phi_1, \dots, \phi_j\}$ be one maximizer in the upper bound of (4.19) and observe $\|\Pi_{k+1}\phi\|_{L^2(\Omega)} > 0$. Without loss of generality suppose $\|\phi\|_{L^2(\Omega)}^2 = 1$ and deduce from the min-max principle $\|\nabla\phi\|_{L^2(\Omega)}^2 \leq \lambda$. Then (4.19) ensures

$$\lambda_h b_h(I(\phi), I(\phi)) \leq a_h(I(\phi), I(\phi)). \quad (4.20)$$

First compute a lower bound for $b_h(I(\phi), I(\phi))$. The estimate (2.4) and the Pythagoras theorem with $\|\phi\|_{L^2(\Omega)}^2 = 1$ show that

$$\begin{aligned} 1 - \delta^2 \|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2 &\leq 1 - \|(1 - \Pi_{k+1})(\phi)\|_{L^2(\Omega)}^2 = \|\Pi_{k+1}(\phi)\|_{L^2(\Omega)}^2 \\ &= b_h(I(\phi), I(\phi)). \end{aligned} \quad (4.21)$$

Second compute an upper bound for $a_h(I(\phi), I(\phi))$. The alternative definition of a_h in (4.12) and the identity $G_{RT} \circ I = \Pi_{RT,k} \circ \nabla$ in V from Lemma 4.2 prove

$$\begin{aligned} a_h(I(\phi), I(\phi)) &= \|\Pi_k \Pi_{RT,k}(\nabla\phi)\|_{L^2(\Omega)}^2 + (1 - \alpha) \|(1 - \Pi_k) \Pi_{RT,k}(\nabla\phi)\|_{L^2(\Omega)}^2 \\ &\quad + \beta \|S(I\phi)\|_{\text{pw}}^2. \end{aligned} \quad (4.22)$$

The commuting property $\Pi_k \Pi_{RT,k} = \Pi_k$ from Lemma 2.1, the Pythagoras theorem, and $\|\nabla\phi\|_{L^2(\Omega)}^2 \leq \lambda$ lead to

$$\|\Pi_k \Pi_{RT,k} \nabla\phi\|_{L^2(\Omega)}^2 = \|\Pi_k \nabla\phi\|_{L^2(\Omega)}^2 \leq \lambda - \|(1 - \Pi_k) \nabla\phi\|_{L^2(\Omega)}^2. \quad (4.23)$$

Lemma 2.1 and the boundedness of $\Pi_{RT,k}$ with $\|\Pi_{RT,k}\| \leq 1$ show that

$$\|(1 - \Pi_k)\Pi_{RT,k}(\nabla\phi)\|_{L^2(\Omega)} = \|\Pi_{RT,k}(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)} \leq \|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}. \quad (4.24)$$

For $0 < \alpha < 1$, the combination of (4.22)–(4.24) with the bound for $\|S(I\phi)\|_{\text{pw}}$ in (4.11) proves that

$$a_h(I(\phi), I(\phi)) \leq \lambda + (\tau^2\beta - \alpha)\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2. \quad (4.25)$$

Together (4.20), (4.21), and (4.25) show that

$$\lambda_h(1 - \delta^2\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2) \leq \lambda - (\alpha - \tau^2\beta)\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2.$$

This and the assumption $\delta^2\lambda \leq \alpha - \tau^2\beta$ implies that

$$\lambda_h(1 - \delta^2\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2) \leq \lambda(1 - \delta^2\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2). \quad (4.26)$$

Observe that the boundedness of the projection $(1 - \Pi_k)$, the min-max principle for the continuous problem, and the assumption $\delta^2\lambda + \tau^2\beta \leq \alpha < 1$ imply

$$\delta^2\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2 \leq \delta^2\|\nabla\phi\|_{L^2(\Omega)}^2 \leq \delta^2\lambda \leq \delta^2\lambda + \tau^2\beta \leq \alpha < 1.$$

In other words, the factor $(1 - \delta^2\|(1 - \Pi_k)(\nabla\phi)\|_{L^2(\Omega)}^2)$ on both sides of (4.26) is positive and division concludes the proof of $\lambda_h \leq \lambda$.

Finish of the proof. In the remaining cases holds $\alpha < \tau^2\beta + \delta^2\lambda$, hence the a posteriori condition $\tau^2\beta + \delta^2\lambda_h \leq \alpha$ holds and implies $\lambda_h \leq \lambda$.

□

4.4 Convergence analysis

This section proves that the discrete eigenvalues and the volume components of the eigenfunction converge with the expected optimal rates. The analysis hinges on the Babuška-Osborn theory for the spectral approximation of compact selfadjoint operators and adapts the arguments devised to analyse the spectral approximation using the standard HHO method in [Cal+19]. The discrete problem in the Schur-complement notation (4.18) implies that the relevant solution component is the volume component $u_{\mathcal{T}} \in P_{k+1}(\mathcal{T})$.

4.4.1 Convergence analysis for the source problem

Given a right-hand side $f \in L^2(\Omega)$, the weak formulation of the Poisson model problem seeks $u \in V := H_0^1(\Omega)$

$$a(u, v) = b(f, v) \quad \text{for all } v \in V. \quad (4.27)$$

Let $T : L^2(\Omega) \rightarrow L^2(\Omega)$ denote the solution operator that maps the right-hand side $f \in L^2(\Omega)$ onto the solution $T(f) = u \in V$ of (4.27). Let $\sigma := \sigma_{\text{reg}}$ denote the index of elliptic regularity $\sigma > 0$ for the source problem of the Laplacian $-\Delta$ in V : Given any right-hand side $f \in L^2(\Omega)$, the weak solution $u \in V$ to $-\Delta u = f$ satisfies

$$u \in H^{1+\sigma}(\Omega) \text{ and } \|T(f)\|_{H^{1+\sigma}(\Omega)} = \|u\|_{H^{1+\sigma}(\Omega)} \leq C(\sigma)\|f\|_{L^2(\Omega)}. \quad (4.28)$$

This is well-established, cf. [Neč12; GT83; Dau88; Gri92; Agm10].

The source problem associated with the new method (4.2) seeks $u_h \in V_h$ such that

$$a_h(u_h, v_h) = b(f, v_{\mathcal{T}}) \quad \text{for all } v_h = (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h. \quad (4.29)$$

For simplified notation let $\hat{T}_h : L^2(\Omega) \rightarrow V_h$ with $\hat{T}_h(f) := u_h$ denote the associated solution operator. The operator $\hat{T}_h$ is uniformly bounded with respect to the mesh-size. This result is comparable to [Cal+19, Lemma 3.2].

Lemma 4.9 (boundedness of $\hat{T}_h$). *There exists a constant $C_0 > 0$ (that exclusively depends on $\mathbb{T}$) such that $\|\hat{T}_h(f)\|_{a,h} \leq C_0\|f\|_{L^2(\Omega)}$ for all $f \in L^2(\Omega)$.*

Proof. Let $\hat{T}_h(f) = u_h = (u_{\mathcal{T}}, u_{\mathcal{F}}) \in V_h$. The definition of $\hat{T}_h$ for (4.29) and the Cauchy-Schwarz inequality show

$$\|\hat{T}_h(f)\|_{a,h}^2 = b(f, u_{\mathcal{T}}) \leq \|f\|_{L^2(\Omega)} \|u_{\mathcal{T}}\|_{L^2(\Omega)}.$$

For the broken polynomial space $P_{k+1}(\mathcal{T})$ the discrete Poincaré inequality in the discontinuous Galerkin context, e.g., in [Arn82, Lem. 2.1] for $H^1(\mathcal{T})$ or [DPE10, Thm. 2.1], reads

$$C_{PdG} \|u_{\mathcal{T}}\|_{L^2(\Omega)} \leq \left(\sum_{T \in \mathcal{T}} \|\nabla u_{\mathcal{T}}|_T\|_{L^2(T)}^2 + \sum_{F \in \mathcal{F}} h_F^{-1} \|[u_{\mathcal{T}}]_F\|_{L^2(F)}^2 \right)^{1/2}.$$

Since $u_{\mathcal{F}} \in P_k(\mathcal{F}(\Omega))$ is single valued along the interior sides $F \in \mathcal{F}(\Omega)$ and vanishes along the boundary sides $F \in \mathcal{F}(\partial\Omega)$, the jump is bounded by

$$\|[u_{\mathcal{T}}]_F\|_{L^2(F)}^2 \leq 2 \left(\|u_{\mathcal{T}}|_{T_+} - u_{\mathcal{F}}|_F\|_{L^2(F)}^2 + \|u_{\mathcal{T}}|_{T_-} - u_{\mathcal{F}}|_F\|_{L^2(F)}^2 \right).$$

Hence, the definition (4.13) and Lemma 4.7 imply

$$2^{-1/2} C_{PdG} \|u_{\mathcal{T}}\|_{L^2(\Omega)} \leq \|u_h\|_h = \|\hat{T}_h(f)\|_h \leq \underline{\gamma}^{-1/2} \|\hat{T}_h(f)\|_{a,h}.$$

This concludes the proof with constant $C_0 := \sqrt{2\underline{\gamma}^{-1}} C_{PdG}^{-1}$. $\square$

The first component of $u_h = (u_{\mathcal{T}}, u_{\mathcal{F}}) \in V_h$ defines the operator

$$T_h : L^2(\Omega) \rightarrow P_{k+1}(\mathcal{T}) \subset L^2(\Omega) \quad \text{with} \quad T_h(f) := u_{\mathcal{T}}. \quad (4.30)$$

This operator T_h allows for the application of the Babuška-Osborn theory which needs a solution operator $T_h : L^2(\Omega) \rightarrow L^2(\Omega)$. The algebraic eigenvalue problem (4.18) proves that T_h is positive definite, selfadjoint, and determines the solution.

Remark 4.10 (solution operators). The Schur complement ansatz to rewrite (4.17) as reduced matrix eigenvalue problem (4.18) translates into the operator notation as follows. Let $Z_h : P_{k+1}(\mathcal{T}) \rightarrow P_k(\mathcal{F}(\Omega))$ denote the operator that maps $v_{\mathcal{T}} \in P_{k+1}(\mathcal{T})$ onto $Z_h(v_{\mathcal{T}}) \in P_k(\mathcal{F}(\Omega))$ with

$$a_h((v_{\mathcal{T}}, Z_h(v_{\mathcal{T}})), (0, w_{\mathcal{F}})) = 0 \quad \text{for all } w_{\mathcal{F}} \in P_k(\mathcal{F}(\Omega)).$$

Then $\hat{T}_h(f) = (T_h(f), (Z_h \circ T_h)(f))$ and $T_h : L^2(\Omega) \rightarrow L^2(\Omega)$ is solution operator to the source problem

$$a_h((u_{\mathcal{T}}, Z_h(u_{\mathcal{T}})), (v_{\mathcal{T}}, Z_h(v_{\mathcal{T}}))) = b(f, v_{\mathcal{T}}) \quad \text{for all } v_{\mathcal{T}} \in P_{k+1}(\mathcal{T}).$$

This is easily verified with test-functions of the form $v_h = (v_{\mathcal{T}}, 0) \in V_h$ and $v_h = (0, v_{\mathcal{F}}) \in V_h$, cf. [Cal+19, Lemma 3.2].

If $(\lambda_h, u_h) \in \mathbb{R}_+ \times V_h$ with $u_h = (u_{\mathcal{T}}, u_{\mathcal{F}})$ is an eigenpair of (4.2), then $(\lambda_h^{-1}, u_{\mathcal{T}}) \in \mathbb{R}_+ \times P_{k+1}(\mathcal{T})$ is an eigenpair of T_h . The analysis of the solution operator T_h , i.e., the error analysis for the solution component $u_{\mathcal{T}} \in P_{k+1}(\mathcal{T})$, based on the discrete error estimate in Theorem 4.11 below, follows the arguments of Section 4 in [Cal+19] (reduced to the present case of a symmetric bilinear form $a_h(\bullet, \bullet)$). The proofs are included below for convenient reading.

Theorem 4.11 (discrete error estimate). *There exist constants $C_1 \approx 1$ (that exclusively depends on $\mathbb{T}$) such that given any $f \in L^2(\Omega)$, the exact solution $u \in H^{1+\sigma}(\Omega) \cap V$ to (4.27) and the discrete solution $u_h \in V_h$ to (4.29) for any $\mathcal{T} \in \mathbb{T}$ with maximal mesh-size $h_{\max}$ satisfy the following estimate. If $u \in H^{1+s}(\Omega)$ for any $\sigma \leq s \leq k+1$, then*

$$\|u_h - Iu\|_{a,h} \leq C_1 h_{\max}^s |u|_{H^{1+s}(\Omega)}.$$

In other words, $T(f) \in H^{1+s}(\Omega)$ implies $\|\hat{T}_h(f) - IT(f)\|_{a,h} \leq C_1 h_{\max}^s |T(f)|_{H^{1+s}(\Omega)}$.

Proof. This proof adapts the arguments of [DPEL14, Thm. 8] and [Cal+19, Lemma 3.3] to the modified HHO method. Since $-\Delta u = f$ a.e. in Ω , a piecewise integration by parts shows, for any $v_h = (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$ with $v_{\mathcal{T}} = (v_T)_{T \in \mathcal{T}} \in P_{k+1}(\mathcal{T})$ and $v_{\mathcal{F}} = (v_F)_{F \in \mathcal{F}} \in P_k(\mathcal{F}(\Omega))$, that

$$\begin{aligned} a_h(u_h, v_h) &= b(f, v_{\mathcal{T}}) = (-\Delta u, v_{\mathcal{T}})_{L^2(\Omega)} \\ &= (\nabla u, \nabla_{\text{pw}} v_{\mathcal{T}})_{L^2(\Omega)} - \sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} \langle v_T|_T, \nabla u \cdot \nu_T \rangle_{L^2(F)} \\ &= (\nabla u, \nabla_{\text{pw}} v_{\mathcal{T}})_{L^2(\Omega)} + \sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} \langle v_F - v_T, \nabla u \cdot \nu_T \rangle_{L^2(F)}. \end{aligned} \quad (4.31)$$

The last equality holds since $v_{\mathcal{F}} = (v_F)_{F \in \mathcal{F}} \in P_k(\mathcal{F}(\Omega))$ is single-valued for $F \in \mathcal{F}(\Omega)$ and vanishes along $F \subset \partial\Omega$. The combination of the alternative form of a_h from (4.12) with Lemma 4.2 and Lemma 2.1 proves that

$$\begin{aligned} a_h(I(u), v_h) &= (\Pi_k G_{RT} I(u), G_{RT}(v_h))_{L^2(\Omega)} + (1-\alpha) \left((1-\Pi_k) G_{RT} I(u), G_{RT}(v_h) \right)_{L^2(\Omega)} \\ &\quad + \beta a_{\text{pw}}(S(Iu), S(v_h)) \\ &= \left(((1-\alpha)\Pi_{RT,k} + \alpha\Pi_k)(\nabla u), G_{RT}(v_h) \right)_{L^2(\Omega)} + \beta a_{\text{pw}}(S(Iu), S(v_h)), \end{aligned}$$

because $\Pi_k G_{RT} I(u) + (1-\alpha)(1-\Pi_k) G_{RT} I(u) = (1-\alpha) G_{RT} I(u) + \alpha \Pi_k G_{RT} I(u) = ((1-\alpha)\Pi_{RT,k} + \alpha\Pi_k)(\nabla u)$. This, the abbreviation $q_{RT} := ((1-\alpha)\Pi_{RT,k} + \alpha\Pi_k)(\nabla u) \in RT_k^{\text{pw}}(\mathcal{T})$, the definition of $G_{RT}(v_h)$ in (4.7), and an integration by parts show that

$$a_h(I(u), v_h) = (q_{RT}, \nabla_{\text{pw}} v_{\mathcal{T}})_{L^2(\Omega)} + \langle v_{\mathcal{F}} - v_{\mathcal{T}}, q_{RT} \cdot \nu_{\mathcal{T}} \rangle_{\partial\mathcal{T}} + \beta a_{\text{pw}}(S(Iu), S(v_h)).$$

In combination with (4.31) this leads to

$$\begin{aligned} a_h(I(u) - u_h, v_h) &= (q_{RT} - \nabla u, \nabla_{\text{pw}} v_{\mathcal{T}})_{L^2(\Omega)} + \langle v_{\mathcal{F}} - v_{\mathcal{T}}, (q_{RT} - \nabla u) \cdot \nu_{\mathcal{T}} \rangle_{\partial\mathcal{T}} \\ &\quad + \beta a_{\text{pw}}(S(Iu), S(v_h)) =: S_1 + S_2 + S_3. \end{aligned} \quad (4.32)$$

For any $T \in \mathcal{T}$, Lemma 2.2 and $0 < \alpha < 1$ proves that

$$\begin{aligned} \|q_{RT} - \nabla u\|_{L^2(T)} &\leq (1-\alpha) \|(1-\Pi_{RT,k})\nabla u\|_{L^2(T)} + \alpha \|(1-\Pi_k)\nabla u\|_{L^2(T)} \\ &\leq (1-\alpha) C_{\text{apx}} h_T^s |\nabla u|_{H^s(T)} + \alpha C_{\text{apx}} h_T^s |\nabla u|_{H^s(T)} \\ &\leq C_{\text{apx}} h_T^s |u|_{H^{1+s}(T)}. \end{aligned} \quad (4.33)$$

This observation and a Cauchy-Schwarz inequality bound S_1 via

$$|S_1| \leq \|q_{RT} - \nabla u\|_{L^2(\Omega)} \|\nabla_{\text{pw}} v_{\mathcal{T}}\|_{L^2(\Omega)} \leq C_{\text{apx}} h_{\max}^s |u|_{H^{1+s}(\Omega)} \|\nabla_{\text{pw}} v_{\mathcal{T}}\|_{L^2(\Omega)}. \quad (4.34)$$

The term S_2 is bounded similarly, the Cauchy-Schwarz inequality in $L^2(F)$, $\mathbb{R}^{|\mathcal{F}(T)|}$, and $\mathbb{R}^{|\mathcal{T}|}$ leads to

$$\begin{aligned} |S_2| &\leq \sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1/2} \|v_{\mathcal{F}}|_F - v_{\mathcal{T}}|_T\|_{L^2(F)} h_F^{1/2} \|q_{RT} - \nabla u\|_{L^2(F)} \\ &\leq \sum_{T \in \mathcal{T}} \sqrt{\sum_{F \in \mathcal{F}(T)} h_F \|q_{RT}|_T - \nabla u\|_{L^2(F)}^2} \sqrt{\sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_{\mathcal{F}}|_F - v_{\mathcal{T}}|_T\|_{L^2(F)}^2} \\ &\leq \sum_{T \in \mathcal{T}} \left(h_T^{1/2} \|q_{RT}|_T - \nabla u\|_{L^2(\partial T)} \sqrt{\sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_{\mathcal{F}}|_F - v_{\mathcal{T}}|_T\|_{L^2(F)}^2} \right) \\ &\leq \sqrt{\sum_{T \in \mathcal{T}} h_T \|q_{RT}|_T - \nabla u\|_{L^2(\partial T)}^2} \sqrt{\sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_{\mathcal{F}}|_F - v_{\mathcal{T}}|_T\|_{L^2(F)}^2}. \end{aligned}$$

For any $T \in \mathcal{T}$, Lemma 2.2 shows $h_T^{1/2} \|q_{RT} - \nabla u\|_{L^2(\partial T)} \leq C_{apx} h_T^s |u|_{H^{1+s}(T)}$ as in (4.33). This and the last displayed inequality prove

$$|S_2| \leq C_{apx} h_{\max}^s |u|_{H^{1+s}(\Omega)} \sqrt{\sum_{T \in \mathcal{T}} \sum_{F \in \mathcal{F}(T)} h_F^{-1} \|v_{\mathcal{F}}|_F - v_{\mathcal{T}}|_T\|_{L^2(F)}^2}.$$

In combination with (4.34) and Lemma 4.7, this shows

$$|S_1| + |S_2| \leq \sqrt{2/\underline{\gamma}} C_{apx} h_{\max}^s |u|_{H^{1+s}(T)} \|v_h\|_{a,h}. \quad (4.35)$$

The last term S_3 is controlled via Cauchy-Schwarz inequality, Lemma 4.3, and (2.1) in

$$\begin{aligned} |S_3| &\leq \beta \|SI(u)\|_{pw} \|S(v_h)\|_{pw} \leq \tau \sqrt{\beta} \|(1 - \Pi_k) \nabla u\|_{L^2(\Omega)} \sqrt{\beta} \|S(v_h)\|_{L^2(\Omega)} \\ &\leq C_{apx} \tau \sqrt{\beta} h_{\max}^s |u|_{H^{1+s}(\Omega)} \|v_h\|_{a,h}. \end{aligned}$$

This and (4.35) bound the sum in (4.32). That concludes the proof with $C_1 := C_{apx} (\sqrt{2/\underline{\gamma}} + \tau \sqrt{\beta})$, since $\|I(u) - u_h\|_{a,h} = \sup_{v_h \in V_h, \|v_h\|_{a,h}=1} a_h(I(u) - u_h, v_h)$. $\square$

Lemma 4.12. *There exist constants $C_2, C_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) such that given any $f \in L^2(\Omega)$, the exact solution $u \in H^{1+\sigma}(\Omega) \cap V$ to (4.27) and the component $u_{\mathcal{T}} \in P_{k+1}(\mathcal{T})$ in the solution $u_h = (u_{\mathcal{T}}, u_{\mathcal{F}}) \in P_{k+1}(\mathcal{T}) \times P_k(\mathcal{F}(\Omega)) = V_h$ to (4.29) for $\mathcal{T} \in \mathbb{T}$ with maximal mesh-size $h_{\max}$ satisfy (a)–(b).*

(a) *If $u \in H^{1+s}(\Omega)$ for some s with $\sigma \leq s \leq k+1$, then*

$$\|u - u_{\mathcal{T}}\|_{L^2(\Omega)} \leq C_2 h_{\max}^s \|u\|_{H^{1+s}(\Omega)}.$$

(b) *Given any eigenvalue λ of (4.1) with eigenspace $E(\lambda) \subset H^{1+t}(\Omega) \cap V$ for some t with $\sigma \leq t \leq k+1$, suppose $f, g \in E(\lambda)$. Then*

$$(u - u_{\mathcal{T}}, g)_{L^2(\Omega)} \leq h_{\max}^{2t} C_3 \lambda^{-1} (h_{\max}^2 + \lambda^{-1}) \|f\|_{H^{1+t}(\Omega)} \|g\|_{H^{1+t}(\Omega)}.$$

Proof of (a). Recall $u = T(f)$, $u_{\mathcal{T}} = T_h(f)$, and $u_h = (u_{\mathcal{T}}, u_{\mathcal{F}}) = \hat{T}_h(f)$. For any $g \in L^2(\Omega)$, the definition of the solution operators T_h , $\hat{T}_h$ and Remark 4.10 show

$$\begin{aligned} (u - u_{\mathcal{T}}, g)_{L^2(\Omega)} &= ((T - T_h)(f), g)_{L^2(\Omega)} = (T(f), g)_{L^2(\Omega)} - b(T_h(f), g) \\ &= (T(f), g)_{L^2(\Omega)} - a_h(\hat{T}_h(f), \hat{T}_h(g)). \end{aligned}$$

The definition of the interpolation operator I and the solution operator T_h imply

$$\begin{aligned} (u - u_{\mathcal{T}}, g)_{L^2(\Omega)} &= (T(f), g)_{L^2(\Omega)} - a_h(IT(f), \hat{T}_h(g)) + a_h(IT(f) - \hat{T}_h(f), \hat{T}_h(g)) \\ &= ((1 - \Pi_{k+1})T(f), g)_{L^2(\Omega)} + a_h(IT(f) - \hat{T}_h(f), \hat{T}_h(g)) \\ &=: S_4 + S_5. \end{aligned} \quad (4.36)$$

The Cauchy-Schwarz inequality and the projection estimate (2.1) show

$$S_4 \leq \|(1 - \Pi_{k+1})T(f)\|_{L^2(\Omega)} \|g\|_{L^2(\Omega)} \leq h_{\max}^{1+s} C_{\text{apx}} |T(f)|_{H^{1+s}(\Omega)} \|g\|_{L^2(\Omega)}.$$

The Cauchy-Schwarz inequality, Theorem 4.11, and Lemma 4.9 show

$$S_5 \leq \|IT(f) - \hat{T}_h(f)\|_{a,h} \|\hat{T}_h g\|_{a,h} \leq h_{\max}^s C_0 C_1 |T(f)|_{H^{1+s}(\Omega)} \|g\|_{L^2(\Omega)}.$$

These estimates for S_4 , S_5 in (4.36) conclude the proof with $C_2 := h_{\max} C_{\text{apx}} + C_0 C_1$. $\square$

Proof of (b). Note that for $f \in E(\lambda) \subset H^{1+t}(\Omega) \cap V$ holds $T(f) = f/\lambda \in E(\lambda)$. For any $f, g \in E(\lambda)$, (4.36) implies

$$(u - u_{\mathcal{T}}, g)_{L^2(\Omega)} = ((1 - \Pi_{k+1})T(f), g)_{L^2(\Omega)} + a_h(IT(f) - \hat{T}_h(f), \hat{T}_h(g)) =: S_4 + S_5.$$

To exploit the additional smoothness of $g \in H^{1+t}(\Omega)$ for S_4 , the projection property is combined with the Cauchy-Schwarz inequality and (2.1) to obtain

$$S_4 = ((1 - \Pi_{k+1})T(f), (1 - \Pi_{k+1})g)_{L^2(\Omega)} \leq \frac{C_{\text{apx}}^2}{\lambda} h_{\max}^{2(1+t)} |f|_{H^{1+t}(\Omega)} |g|_{H^{1+t}(\Omega)}. \quad (4.37)$$

The term S_5 needs a bit more algebraic manipulation,

$$S_5 = a_h(IT(f) - \hat{T}_h(f), \hat{T}_h(g) - IT(g)) + a_h(IT(f) - \hat{T}_h(f), IT(g)).$$

The Cauchy-Schwarz inequality for a_h and Theorem 4.11 prove for the first part

$$a_h(IT(f) - \hat{T}_h(f), \hat{T}_h(g) - IT(g)) \leq \lambda^{-2} h_{\max}^{2t} C_1^2 |f|_{H^{1+t}(\Omega)} |g|_{H^{1+t}(\Omega)}. \quad (4.38)$$

With the abbreviation $S_6 := a_h(IT(f), IT(g)) - a(T(f), T(g))$ the solution properties of T and $\hat{T}_h$ show for the second summand of S_5 that

$$\begin{aligned} a_h(IT(f) - \hat{T}_h(f), IT(g)) - S_6 &= a(T(f), T(g)) - a_h(\hat{T}_h(f), IT(g)) \\ &= (f, T(g))_{L^2(\Omega)} - (f, \Pi_{k+1}T(g))_{L^2(\Omega)} \\ &= ((1 - \Pi_{k+1})(f), (1 - \Pi_{k+1})T(g))_{L^2(\Omega)} \\ &\leq \lambda^{-1} C_{\text{apx}}^2 h_{\max}^{2(1+t)} |f|_{H^{1+t}(\Omega)} |g|_{H^{1+t}(\Omega)}, \end{aligned} \quad (4.39)$$

where the last inequality follows as in (4.37). The alternative form of a_h from (4.12) as well as a combination of Lemma 4.2 and Lemma 2.1 lead to

$$\begin{aligned} S_6 &= (\Pi_k \nabla T(f), \Pi_k \nabla T(g))_{L^2(\Omega)} - (\nabla T(f), \nabla T(g))_{L^2(\Omega)} + \beta a_{\text{pw}}(SIT(f), SIT(g)) \\ &\quad + (1 - \alpha) (\Pi_{RT,k}(1 - \Pi_k) \nabla T(f), \Pi_{RT,k}(1 - \Pi_k) \nabla T(g))_{L^2(\Omega)}. \end{aligned}$$

The Cauchy-Schwarz inequality, the boundedness of $\Pi_{RT,k}$, Lemma 4.3, and Lemma 2.2 prove

$$\begin{aligned}
S_6 &\leq \|(1 - \Pi_k)\nabla T(f)\|_{L^2(\Omega)}\|(1 - \Pi_k)\nabla T(g)\|_{L^2(\Omega)} + \beta\|SIT(f)\|_{pw}\|SIT(g)\|_{pw} \\
&\quad + (1 - \alpha)\|\Pi_{RT,k}(1 - \Pi_k)\nabla T(f)\|_{L^2(\Omega)}\|\Pi_{RT,k}(1 - \Pi_k)\nabla T(g)\|_{L^2(\Omega)} \\
&\leq (2 - \alpha + \beta\tau^2)\|(1 - \Pi_k)\nabla T(f)\|_{L^2(\Omega)}\|(1 - \Pi_k)\nabla T(g)\|_{L^2(\Omega)} \\
&\leq h_{\max}^{2t}(2 - \alpha + \beta\tau^2)\lambda^{-2}C_{\text{apx}}^2|f|_{H^{1+t}(\Omega)}|g|_{H^{1+t}(\Omega)}. \tag{4.40}
\end{aligned}$$

The combination of (4.38)–(4.40) proves $|S_5| \leq h_{\max}^{2t}c_5\|f\|_{L^2(\Omega)}\|g\|_{L^2(\Omega)}$ with $\lambda c_5 := \lambda^{-1}C_1^2 + h_{\max}^2C_{\text{apx}}^2 + \lambda^{-1}(2 - \alpha + \beta\tau^2)C_{\text{apx}}^2$. This and (4.37) conclude the proof with $C_3 := \max\{2C_{\text{apx}}^2, C_1^2 + (2 - \alpha + \beta\tau^2)C_{\text{apx}}^2\}$. $\square$

Note that Lemma 4.12.a and the elliptic regularity (4.28) show $\|T - T_h\|_{\mathcal{L}(L^2(\Omega))} \leq C_2C(\sigma)h_{\max}^\sigma$. In other words, Lemma 4.12.a implies the convergence $T_h \rightarrow T$ in $\mathcal{L}(L^2(\Omega))$ as $h_{\max} \rightarrow 0$.

4.4.2 Convergence rates for the eigenvalue problem

The preparations in Subsection 4.4.1 allow the proof of the optimal a priori convergence rates in Theorem 4.13 with fundamental arguments from [BO91].

Theorem 4.13 (a priori convergence). *Suppose λ is an eigenvalue of (4.1) of multiplicity μ with eigenspace $E(\lambda) \subset H^{1+t}(\Omega) \cap V$ for some t with $\sigma \leq t \leq k + 1$. Then there exist $\varepsilon, C_4 > 0$ such that any triangulation $\mathcal{T} \in \mathbb{T}(\varepsilon)$ with $h_{\max} \leq \varepsilon$ and the discrete space $V_h := P_{k+1}(\mathcal{T}) \times P_k(\mathcal{F}(\Omega))$ lead in (4.2) to exactly μ algebraic eigenvalues $\lambda_{h,1}, \dots, \lambda_{h,\mu}$ of (4.2) (counting multiplicities), that converge to λ as $h_{\max} \rightarrow 0$. Let $E_h := \text{span}\{u_h \in E_h(\lambda_{h,k}) : k = 1, \dots, \mu\}$ abbreviate the span of the discrete eigenspaces $E_h(\lambda_{h,k}) \subset V_h$ of $\lambda_{h,k}$ for $k = 1, \dots, \mu$. Then*

$$\begin{aligned}
&h_{\max}^{-t} \max_{k=1, \dots, \mu} |\lambda - \lambda_{h,k}| + \max_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \min_{\substack{u_h=(u_{\mathcal{T}}, u_{\mathcal{F}}) \in E_h \\ \|u_{\mathcal{T}}\|_{L^2(\Omega)}=1}} \|u - u_{\mathcal{T}}\|_{L^2(\Omega)} \\
&\quad + \max_{\substack{u_h=(u_{\mathcal{T}}, u_{\mathcal{F}}) \in E_h \\ \|u_{\mathcal{T}}\|_{L^2(\Omega)}=1}} \min_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \|u - u_{\mathcal{T}}\|_{L^2(\Omega)} \leq C_4 h_{\max}^t. \tag{4.41}
\end{aligned}$$

Given a discrete eigenfunction $u_h \in E_h$ with $\|u_{\mathcal{T}}\|_{L^2(\Omega)} = 1$, there exist a unit vector $u \in E(\lambda)$ such that

$$\|u_h - Iu\|_{a,h}^2 \leq C_5 h^{2t}.$$

Proof. Recall that Lemma 4.12.a implies the convergence $T_h \rightarrow T$ in $\mathcal{L}(L^2(\Omega))$ as $h_{\max} \rightarrow 0$. Suppose $(\lambda, u) \in \mathbb{R}^+ \times V$ denotes an eigenpair of (4.1) and $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ with $u_h = (u_{\mathcal{T}}, u_{\mathcal{F}})$ denotes an eigenpair of (4.2), then $(1/\lambda, u)$ is an eigenpair of T and $(1/\lambda_h, u_{\mathcal{T}})$ is an eigenpair of T_h . Hence, Theorem 2.6 implies for

any non-zero eigenvalue $1/\lambda$ of T with eigenspace $E(\lambda) = \ker(\lambda^{-1} - T)$ of dimension $\mu = \dim(E(\lambda)) \in \mathbb{N}$, that there exist exactly μ eigenvalues $1/\lambda_{h,1}, \dots, 1/\lambda_{h,\mu}$ of T_h , which converge to $1/\lambda$ as $h_{\max} \rightarrow 0$.

Note that for the finite-dimensional eigenspace $E(\lambda) \subset H^{1+t}(\Omega)$ there exists

$$C_t := \sup_{\phi \in E(\lambda)} \frac{\|\phi\|_{H^{1+t}(\Omega)}}{\|\phi\|_{L^2(\Omega)}} < \infty$$

such that $\|\phi\|_{H^{1+t}(\Omega)} \leq C_t \|\phi\|_{L^2(\Omega)}$ for all $\phi \in E(\lambda)$. Hence, Lemma 4.12.a and $T(\phi) = \phi/\lambda$ for all $\phi \in E(\lambda)$ shows

$$\|(T - T_h)|_{E(\lambda)}\|_{\mathcal{L}(E(\lambda); L^2(\Omega))} \leq \lambda^{-1} C_2 C_t h_{\max}^t$$

and Lemma 4.12.b bounds the term

$$\sup_{\phi, \psi \in E(\lambda) \setminus \{0\}} \frac{|((T - T_h)\phi, \psi)_{L^2(\Omega)}|}{\|\phi\|_{L^2(\Omega)} \|\psi\|_{L^2(\Omega)}} \leq h_{\max}^{2t} C_3 C_t^2 \lambda^{-1} (h_{\max}^2 + \lambda^{-1}).$$

Hence, Theorem 2.6 provides (4.41) with constant $C_4 := C_a C_t^2 (C_3 \lambda^{-1} (h_{\max}^2 + \lambda^{-1}) + \lambda^{-2} C_2^2) + C_b C_2 C_t \lambda^{-1}$.

The proof for the final claim is analogue to Corollary 4.6 in [Cal+19]. The properties of Π_{k+1} and the definition (4.30) show

$$\lambda_h(u_{\mathcal{T}}, u)_{L^2(\Omega)} = \lambda_h(u_{\mathcal{T}}, \Pi_{k+1}u)_{L^2(\Omega)} = a_h(u_h, Iu),$$

since Remark 4.10 implies $\lambda_h(u_{\mathcal{T}}, \Pi_{k+1}u)_{L^2(\Omega)} = a_h((u_{\mathcal{T}}, Z_h u_{\mathcal{T}}), (\Pi_{k+1}u, Z_h \Pi_{k+1}u)) + a_h((u_{\mathcal{T}}, Z_h u_{\mathcal{T}}), (0, \Pi_{\mathcal{F},k}u - Z_h \Pi_{k+1}u))$. Recall $S_6 = a_h(Iu, Iu) - a(u, u)$ for $g = f$ and $\|u\|_{L^2(\Omega)} = \|u_{\mathcal{T}}\|_{L^2(\Omega)} = 1$ by normalization. Then,

$$\begin{aligned} & a_h(u_h - Iu, u_h - Iu) \\ &= a_h(u_h, u_h) - 2a_h(u_h, Iu) + a_h(Iu, Iu) \\ &= \lambda_h \|u_{\mathcal{T}}\|_{L^2(\Omega)}^2 - 2\lambda_h(u_{\mathcal{T}}, u)_{L^2(\Omega)} + \lambda \|u\|_{L^2(\Omega)}^2 + S_6 \\ &= \lambda_h \|u_{\mathcal{T}}\|_{L^2(\Omega)}^2 - 2\lambda_h(u_{\mathcal{T}}, u)_{L^2(\Omega)} + \lambda_h \|u\|_{L^2(\Omega)}^2 - (\lambda_h - \lambda) \|u\|_{L^2(\Omega)}^2 + S_6 \\ &= \lambda_h \|u_{\mathcal{T}} - u\|_{L^2(\Omega)}^2 - \lambda_h + \lambda + S_6. \end{aligned}$$

This is a generalization of the Pythagorean eigenvalue error identity, cf. [SF08]. Since (4.41) bounds $\|u_{\mathcal{T}} - u\|_{L^2(\Omega)}^2$ and $\lambda - \lambda_h$ and (4.40) in the proof of Lemma 4.12.b controls S_6 , the last displayed estimate concludes the proof with $C_5 := (1 + \lambda_h C_4) C_4 + (2 - \alpha + \beta \tau^2) \lambda^{-2} C_t^2 C_{\text{apx}}^2$ (recall that $\lambda_h \leq \lambda$ for $h_{\max}$ small enough). $\square$

Remark 4.14. If the smoothness is optimal, i.e., $t = k+1$, the proven convergence is of order h^{2k+2} for the eigenvalues and h^{k+1} for the eigenvectors in the H^1 -seminorm.

Remark 4.15. Given the alternative stabilization in Remark 4.4 and a *simple* eigenpair $(\lambda, u) \in \mathbb{R}^+ \times (H^{1+t}(\Omega) \cap V)$ of (4.1), the L^2 error estimate is improved in the upcoming work [CGT22]. Utilizing an right-inverse enriching operator from [EZ20], Carstensen et. al. prove for the minimum of one and the index of elliptic regularity $0 \leq \sigma \leq 1$ and $\sigma \leq t \leq k+1$ that $\|u - u_{\mathcal{T}}\|_{L^2(\Omega)} \lesssim h_{\max}^{\sigma} \min_{v_{k+1} \in P_{k+1}(\mathcal{T})} \|\nabla_{\text{pw}}(u - v_{k+1})\|_{L^2(\Omega)} \lesssim h_{\max}^{\sigma+t}$. This best-approximation result allows for an efficient and reliable a posteriori error estimator.

4.5 Lowest-order case

This section is devoted to the analysis of the lowest-order case and provides a comparison to the Crouzeix-Raviart method. Throughout this section set $k \equiv 0$ so that $V_h := P_1(\mathcal{T}) \times P_0(\mathcal{F}(\Omega))$.

4.5.1 Comparison with the Crouzeix-Raviart method

Recall the nonconforming Crouzeix-Raviart space with side-oriented basis functions $\psi_F \in \text{CR}_0^1(\mathcal{T})$ with $\psi_F(\text{mid}(E)) = \delta_{EF}$ for all $F, E \in \mathcal{F}$ and the interpolation operator $I_{\text{CR}} : V \rightarrow \text{CR}_0^1(\mathcal{T})$ defined as $I_{\text{CR}}(f) := \sum_{F \in \mathcal{F}} f_F \int_F f \, ds \psi_F$ for any $f \in V$ from Section 3.3. The vector spaces $P_0(\mathcal{F}(\Omega))$ and $\text{CR}_0^1(\mathcal{T})$ of dimension $N := |\mathcal{F}(\Omega)|$ can be identified as follows. The extension operator $I_{\mathcal{F}} : P_0(\mathcal{F}(\Omega)) \rightarrow \text{CR}_0^1(\mathcal{T})$ maps bijectively $v_{\mathcal{F}} = (v_F)_{F \in \mathcal{F}} \in P_0(\mathcal{F}(\Omega))$ onto $I_{\mathcal{F}}(v_{\mathcal{F}}) := \sum_{F \in \mathcal{F}} v_F \psi_F$. This leads to $\Pi_{\mathcal{F},0} I_{\mathcal{F}}(v_{\mathcal{F}}) = v_{\mathcal{F}}$ for any $v_{\mathcal{F}} \in P_0(\mathcal{F}(\Omega))$ and $I_{\text{CR}}(f) = I_{\mathcal{F}} \Pi_{\mathcal{F},0}(f)$ for any $f \in V$. In other words, it holds $I_{\text{CR}} = I_{\mathcal{F}} \circ \Pi_{\mathcal{F},0}$.

Recall the operators R from (4.6), G_{RT} from (4.7), S from (4.8), and the barycenter $\text{mid}(K) := \frac{1}{m} \sum_{j=1}^m P_j \in K$ of $K := \text{conv}\{P_1, \dots, P_m\}$. The piecewise constant function $\text{mid}(\mathcal{T}) \in P_0(\mathcal{T}; \mathbb{R}^n)$ takes on each simplex $T \in \mathcal{T}$ the value $\text{mid}(\mathcal{T})|_T := \text{mid}(T)$. Abbreviate

$$\begin{aligned} s(\mathcal{T}) &:= 1/n^2 \Pi_0(|\bullet - \text{mid}(\mathcal{T})|^2) \in P_0(\mathcal{T}), \\ S(\mathcal{T}) &:= \Pi_0((\bullet - \text{mid}(\mathcal{T})) \otimes (\bullet - \text{mid}(\mathcal{T}))) \in P_0(\mathcal{T}; \mathbb{R}^{n \times n}). \end{aligned}$$

Lemma 4.16. *Any $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$ and $v_{\text{CR}} := I_{\mathcal{F}}(v_{\mathcal{F}}) \in \text{CR}_0^1(\mathcal{T})$ satisfy*

- (a) $R(v_h) = (1 - \Pi_0)(v_{\text{CR}}) + \Pi_0(v_{\mathcal{T}})$,
- (b) $G_{RT}(v_h) = \nabla_{\text{pw}} v_{\text{CR}} + \frac{1}{n} \frac{1}{s(\mathcal{T})} (\Pi_0(v_{\text{CR}} - v_{\mathcal{T}}))(\bullet - \text{mid}(\mathcal{T}))$,
- (c) $S(v_h) = (1 - \Pi_0)(v_{\mathcal{T}} - v_{\text{CR}})$.

Proof of (a). For any $p \in P_1(\mathcal{T})$, (4.6) and an integration by parts show that

$$\begin{aligned} a_{\text{pw}}(R(v_h), p) &= -(v_{\mathcal{T}}, \Delta_{\text{pw}} p)_{L^2(\Omega)} + \langle v_{\mathcal{F}}, \nabla_{\text{pw}} p \cdot \nu_{\mathcal{T}} \rangle_{\partial \mathcal{T}} = \langle v_{\text{CR}}, \nabla_{\text{pw}} p \cdot \nu_{\mathcal{T}} \rangle_{\partial \mathcal{T}} \\ &= a_{\text{pw}}(v_{\text{CR}}, p). \end{aligned}$$

Hence, $\nabla_{\text{pw}} R(v_h) = \nabla_{\text{pw}} v_{\text{CR}}$. The condition $\Pi_0 R(v_h) = \Pi_0(v_{\mathcal{T}})$ concludes the proof of (a). $\square$

Proof of (b). For any $q_{RT} \in RT_0(\mathcal{T})$, (4.7) and an integration by parts imply

$$\begin{aligned} (G_{RT}(v_h), q_{RT})_{L^2(\Omega)} &= -(v_{\mathcal{T}}, \text{div}_{\text{pw}} q_{RT})_{L^2(\Omega)} + \langle v_{\text{CR}}, q_{RT} \cdot \nu_{\mathcal{T}} \rangle_{\partial \mathcal{T}} \\ &= (v_{\text{CR}} - v_{\mathcal{T}}, \text{div}_{\text{pw}} q_{RT})_{L^2(\Omega)} + (\nabla_{\text{pw}} v_{\text{CR}}, q_{RT})_{L^2(\Omega)} \\ &= (\Pi_0(v_{\text{CR}} - v_{\mathcal{T}}), \text{div}_{\text{pw}} q_{RT})_{L^2(\Omega)} + (\nabla_{\text{pw}} v_{\text{CR}}, \Pi_0 q_{RT})_{L^2(\Omega)}. \end{aligned}$$

On the other hand, since any $q_{RT} \in RT_0^{\text{pw}}(\mathcal{T})$ can be written as

$$q_{RT} = \Pi_0 q_{RT} + n^{-1} \text{div}_{\text{pw}} q_{RT} (\bullet - \text{mid}(\mathcal{T})),$$

the Pythagoras theorem and the definition of $s(\mathcal{T})$ lead to

$$\begin{aligned} &(G_{RT}(v_h), q_{RT})_{L^2(\Omega)} \\ &= (\Pi_0 G_{RT}(v_h), \Pi_0 q_{RT})_{L^2(\Omega)} + ((1 - \Pi_0) G_{RT}(v_h), (1 - \Pi_0) q_{RT})_{L^2(\Omega)} \\ &= (\Pi_0 G_{RT}(v_h), \Pi_0 q_{RT})_{L^2(\Omega)} \\ &\quad + \frac{1}{n^2} (\text{div}_{\text{pw}} G_{RT}(v_h) (\bullet - \text{mid}(\mathcal{T})), \text{div}_{\text{pw}} q_{RT} (\bullet - \text{mid}(\mathcal{T})))_{L^2(\Omega)} \\ &= (\Pi_0 G_{RT}(v_h), \Pi_0 q_{RT})_{L^2(\Omega)} + (s(\mathcal{T}) \text{div}_{\text{pw}} G_{RT}(v_h), \text{div}_{\text{pw}} q_{RT})_{L^2(\Omega)}. \end{aligned}$$

The comparison of the last two displayed equalities shows that $s(\mathcal{T}) \text{div}_{\text{pw}} G_{RT}(v_h) = \Pi_0(v_{\text{CR}} - v_{\mathcal{T}})$ and $\Pi_0 G_{RT}(v_h) = \nabla_{\text{pw}} v_{\text{CR}}$. That completes the proof of (b). $\square$

Proof of (c). This follows directly from the definition (4.8) and (a). $\square$

Proposition 4.17. (a) *The bilinear forms in the lowest-order case read for all $u_h := (u_{\mathcal{T}}, u_{\mathcal{F}})$ and $v_h := (v_{\mathcal{T}}, v_{\mathcal{F}}) \in V_h$,*

$$\begin{aligned} a_h(u_h, v_h) &= a_{\text{pw}}(I_{\mathcal{F}}(u_{\mathcal{F}}), I_{\mathcal{F}}(v_{\mathcal{F}})) + \beta a_{\text{pw}}(u_{\mathcal{T}} - I_{\mathcal{F}}(u_{\mathcal{F}}), v_{\mathcal{T}} - I_{\mathcal{F}}(v_{\mathcal{F}})) \\ &\quad + (1 - \alpha) (s(\mathcal{T})^{-1} \Pi_0(I_{\mathcal{F}}(u_{\mathcal{F}}) - u_{\mathcal{T}}), \Pi_0(I_{\mathcal{F}}(v_{\mathcal{F}}) - v_{\mathcal{T}}))_{L^2(\Omega)}, \\ b_h(u_h, v_h) &= (u_{\mathcal{T}}, v_{\mathcal{T}})_{L^2(\Omega)}. \end{aligned}$$

(b) *Any lowest-order discrete eigenfunction $u_h := (u_{\mathcal{T}}, u_{\mathcal{F}}) \in V_h$ of (4.2) satisfies*

$$I_{\mathcal{F}}(u_{\mathcal{F}}) = \left(1 - \frac{\lambda_h s(\mathcal{T})}{1 - \alpha}\right) \Pi_0(u_{\mathcal{T}}) + \left(\left(1 - \frac{\lambda_h S(\mathcal{T})}{\beta}\right) \nabla_{\text{pw}} u_{\mathcal{T}}\right) \cdot (\bullet - \text{mid}(\mathcal{T})), \quad (4.42)$$

$$a_{\text{pw}}(I_{\mathcal{F}}(u_{\mathcal{F}}), v_{\text{CR}}) = \lambda_h (u_{\mathcal{T}}, v_{\text{CR}})_{L^2(\Omega)}, \quad \text{for all } v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T}). \quad (4.43)$$

Proof of (a). Lemma 4.16 allows the substitution of the operators in the lowest-order version of the alternative form of a_h from (4.12). This and the definition of $s(\mathcal{T})$ prove the claim. $\square$

Proof of (b). In the lowest-order version the bilinear forms in the discrete eigenvalue problem (4.2) can be replaced according to (a). For any test functions $v_h = (v_{\mathcal{T}}, 0) \in V_h$ with $v_{\mathcal{T}} \in P_1(\mathcal{T})$ this leads to

$$(1 - \alpha) \left(s(\mathcal{T})^{-1} \Pi_0(u_{\mathcal{T}} - I_{\mathcal{F}}(u_{\mathcal{F}})), v_{\mathcal{T}} \right)_{L^2(\Omega)} + \beta a_{\text{pw}}(u_{\mathcal{T}} - I_{\mathcal{F}}(u_{\mathcal{F}}), v_{\mathcal{T}}) = \lambda_h(u_{\mathcal{T}}, v_{\mathcal{T}})_{L^2(\Omega)}. \quad (4.44)$$

On the one hand side, the test function $\Pi_0(v_{\mathcal{T}}) \in P_0(\mathcal{T})$ in (4.44) shows that $\frac{1-\alpha}{s(\mathcal{T})} \Pi_0(u_{\mathcal{T}} - I_{\mathcal{F}}(u_{\mathcal{F}})) = \lambda_h \Pi_0(u_{\mathcal{T}})$ and equivalently

$$\Pi_0 I_{\mathcal{F}}(u_{\mathcal{F}}) = \left(1 - \frac{\lambda_h s(\mathcal{T})}{(1 - \alpha)} \right) \Pi_0(u_{\mathcal{T}}).$$

On the other hand side, the choice $(1 - \Pi_0)(v_{\mathcal{T}}) \in P_1(\mathcal{T})$ in (4.44) proves

$$\beta a_{\text{pw}}(u_{\mathcal{T}} - I_{\mathcal{F}}(u_{\mathcal{F}}), v_{\mathcal{T}}) = \lambda_h(u_{\mathcal{T}}, (1 - \Pi_0)(v_{\mathcal{T}}))_{L^2(\Omega)}.$$

Since $p_1 = \Pi_0(p_1) + \nabla_{\text{pw}} p_1 \cdot (\bullet - \text{mid}(\mathcal{T}))$ holds for any $p_1 \in P_1(\mathcal{T})$, the last displayed identity implies

$$\nabla_{\text{pw}} I_{\mathcal{F}}(u_{\mathcal{F}}) = \left(1 - \frac{\lambda_h S(\mathcal{T})}{\beta} \right) \nabla_{\text{pw}} u_{\mathcal{T}}$$

and concludes the proof of (4.42). Since the operator $I_{\mathcal{F}} : P_0(\mathcal{F}(\Omega)) \rightarrow \text{CR}_0^1(\mathcal{T})$ is surjective, test functions $v_h = (I_{\mathcal{F}}(v_{\mathcal{F}}), v_{\mathcal{F}}) \in V_h$ for any $v_{\mathcal{F}} \in P_0(\mathcal{F}(\Omega))$ in (4.2) prove (4.43). $\square$

For comparison, let $(\lambda_{\text{CR}}, \phi_{\text{CR}}) \in \mathbb{R}_+ \times \text{CR}_0^1(\mathcal{T})$ denote a Crouzeix-Raviart eigenpair with $\|\phi_{\text{CR}}\|_{L^2(\Omega)} = 1$ and

$$a_{\text{pw}}(\phi_{\text{CR}}, v_{\text{CR}}) = \lambda_{\text{CR}} b(\phi_{\text{CR}}, v_{\text{CR}}) \quad \text{for all } v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T}). \quad (4.45)$$

Assume the $N := |\mathcal{F}(\Omega)|$ algebraic eigenvalues $0 < \lambda_{\text{CR}}(1) \leq \lambda_{\text{CR}}(2) \leq \dots \leq \lambda_{\text{CR}}(N) < \infty$ of (4.45) are enumerated in ascending order counting multiplicities. Given the constant $\kappa_1 > 0$ from Theorem 3.24.b, [CG14b] proves as in Theorem 5.8 below that

$$\text{GLB}(j) := \frac{\lambda_{\text{CR}}(j)}{1 + \kappa_1^2 h_{\max}^2 \lambda_{\text{CR}}(j)} \leq \lambda_j. \quad (4.46)$$

Corollary 4.18 (comparison with Crouzeix-Raviart). *If $\lambda_{\text{CR}}(j)$ denotes the j -th eigenvalue of (4.45) and $\lambda_h(j)$ the j -th eigenvalue of (4.2) for $1 \leq j \leq N$, then $\lambda_h(j) \leq \lambda_{\text{CR}}(j)$.*

Proof. The proof is similar to [CZZ20, Thm.6.2] and added here for convenient reading. Let $\mathcal{U}(j) := \{U_h \subseteq V_h : \dim(U_h) = j\}$ denote the set of all j -dimensional subspaces of V_h . The discrete min-max principle in Lemma 2.7 characterizes the j -th discrete eigenvalue $\lambda_h(j)$ of (4.2) by

$$\lambda_h(j) = \min_{U_h \in \mathcal{U}(j)} \max_{v_h \in U_h} \frac{a_h(v_h, v_h)}{b_h(v_h, v_h)}. \quad (4.47)$$

Let $\phi_{\text{CR}}(1), \dots, \phi_{\text{CR}}(j)$ denote the b -orthonormal eigenfunctions of (4.45), then $U_{\text{CR}} := \text{span}\{(\phi_{\text{CR}}(1), \Pi_{\mathcal{F},0}\phi_{\text{CR}}(1)), \dots, (\phi_{\text{CR}}(j), \Pi_{\mathcal{F},0}\phi_{\text{CR}}(j))\} \in \mathcal{U}(j)$ has dimension $\dim(U_{\text{CR}}) = j$. Since $I_{\mathcal{F}}\Pi_{\mathcal{F},0}(v_{\text{CR}}) = v_{\text{CR}}$ for any $v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T})$, Proposition 4.17.a shows that any $v_{\text{CR},h} := (v_{\text{CR}}, \Pi_{\mathcal{F},0}v_{\text{CR}}) \in V_h$ satisfies

$$a_h(v_{\text{CR},h}, v_{\text{CR},h}) = a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) \quad \text{and} \quad b_h(v_{\text{CR},h}, v_{\text{CR},h}) = b(v_{\text{CR}}, v_{\text{CR}}).$$

This, the choice $U_{\text{CR}} \in \mathcal{U}(j)$ in (4.47), and the min-max principle for $\lambda_{\text{CR}}(j)$ in the algebraic eigenvalue problem (4.45) show

$$\lambda_h(j) \leq \max_{v_h \in U_{\text{CR}}} \frac{a_h(v_h, v_h)}{b_h(v_h, v_h)} = \max_{v_{\text{CR}} \in \text{span}\{\phi_{\text{CR}}(1), \dots, \phi_{\text{CR}}(j)\}} \frac{a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}})}{b(v_{\text{CR}}, v_{\text{CR}})} = \lambda_{\text{CR}}(j).$$

□

Theorem 4.19 (comparison with GLB). *Let $\lambda_h(j)$ denote the j -th discrete eigenvalue of (4.2) with parameters $0 < \alpha < 1$ and $\beta > 0$, $\lambda_{\text{CR}}(j)$ the j -th eigenvalue of (4.45), and $\text{GLB}(j)$ the associated bound computed in (4.46). If*

$$\theta := \frac{n^2}{(n+1)^2} \max\{1/\beta, 1/(n^2(1-\alpha))\} \leq \kappa_1^2$$

and $h_{\max}^2 \theta \lambda_h(j) \leq 1$, then $\text{GLB}(j) \leq \lambda_h(j)$.

Note, the condition in Theorem 4.19 is sufficient to guarantee $\text{GLB}(j) \leq \lambda_h(j)$ not necessary.

Proof. Let $(\lambda_h(\ell), \phi_h(\ell))$ for $1 \leq \ell \leq j$ denote the first j eigenpairs of (4.2), where $\phi_h(\ell) = (\phi_{\mathcal{T}}(\ell), \phi_{\mathcal{F}}(\ell)) \in V_h$ with $(\phi_{\mathcal{T}}(\ell), \phi_{\mathcal{T}}(k))_{L^2(\Omega)} = \delta_{\ell k}$ for $\ell, k = 1, \dots, j$. For any coefficient vector $\xi = (\xi_1, \dots, \xi_j) \in \mathbb{R}^j$ with Euclid norm $|\xi| = 1$ set

$$\begin{aligned} v_{\mathcal{F}} &:= \sum_{\ell=1}^j \xi_{\ell} \phi_{\mathcal{F}}(\ell), \quad v_{\text{CR}} := I_{\mathcal{F}} v_{\mathcal{F}}, \quad v_{\mathcal{T}} := \sum_{\ell=1}^j \xi_{\ell} \phi_{\mathcal{T}}(\ell), \\ \text{and } w_{\mathcal{T}} &:= \sum_{\ell=1}^j \xi_{\ell} \lambda_h(\ell) \phi_{\mathcal{T}}(\ell). \end{aligned} \quad (4.48)$$

Proposition 4.17.b and the orthogonal decomposition

$$p_1 = \Pi_0 p_1 + \nabla_{\text{pw}} p_1 \cdot (\bullet - \text{mid}(\mathcal{T})) \quad \text{for any } p_1 \in P_1(\mathcal{T}) \quad (4.49)$$

lead to

$$a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) = (w_{\mathcal{T}}, v_{\text{CR}})_{L^2(\Omega)}, \quad (4.50)$$

$$v_{\mathcal{T}} - v_{\text{CR}} = \frac{s(\mathcal{T})}{1-\alpha} \Pi_0 w_{\mathcal{T}} + \left(\frac{S(\mathcal{T})}{\beta} \nabla_{\text{pw}} w_{\mathcal{T}} \right) \cdot (\bullet - \text{mid}(\mathcal{T})). \quad (4.51)$$

Set $\mu := (v_{\mathcal{T}}, w_{\mathcal{T}})_{L^2(\Omega)} = \sum_{\ell=1}^j \xi_{\ell}^2 \lambda_h(\ell) \leq \lambda_h(j)$. The L^2 -orthogonal decomposition in (4.49) and (4.51) show

$$\rho := (w_{\mathcal{T}}, v_{\mathcal{T}} - v_{\text{CR}})_{L^2(\Omega)} = \frac{1}{(1-\alpha)} \|\Pi_0 w_{\mathcal{T}}\|_{s(\mathcal{T})}^2 + \frac{1}{\beta} \|\nabla_{\text{pw}} w_{\mathcal{T}}\|_{S(\mathcal{T})^2}^2 > 0. \quad (4.52)$$

On the one hand side, the abbreviation μ and ρ as well as (4.50) lead to

$$a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) = (w_{\mathcal{T}}, v_{\mathcal{T}})_{L^2(\Omega)} - (w_{\mathcal{T}}, v_{\mathcal{T}} - v_{\text{CR}})_{L^2(\Omega)} = \mu - \rho. \quad (4.53)$$

On the other hand, since $\|v_{\mathcal{T}}\|_{L^2(\Omega)} = 1$, the inverse triangle inequality proves

$$\|v_{\text{CR}}\|_{L^2(\Omega)} = \|v_{\mathcal{T}} - (v_{\mathcal{T}} - v_{\text{CR}})\|_{L^2(\Omega)} \geq 1 - \|v_{\mathcal{T}} - v_{\text{CR}}\|_{L^2(\Omega)}.$$

Since $|\bullet - \text{mid}(\mathcal{T})| \leq h_{\mathcal{T}} n / (n+1)$ and so $s(\mathcal{T}) \leq h_{\text{max}}^2 / (n+1)^2$, the orthogonal decomposition (4.51), the Pythagoras theorem, and the Cauchy-Schwarz inequality in $\mathbb{R}^n$ imply

$$\begin{aligned} & \|v_{\mathcal{T}} - v_{\text{CR}}\|_{L^2(\Omega)}^2 \\ &= \frac{1}{(1-\alpha)^2} \|s(\mathcal{T}) \Pi_0 w_{\mathcal{T}}\|_{L^2(\Omega)}^2 + \frac{1}{\beta^2} \|(S(\mathcal{T}) \nabla_{\text{pw}} w_{\mathcal{T}}) \cdot (\bullet - \text{mid}(\mathcal{T}))\|_{L^2(\Omega)}^2 \\ &\leq \frac{1}{(1-\alpha)^2 (n+1)^2} \|h_{\mathcal{T}} \Pi_0 w_{\mathcal{T}}\|_{s(\mathcal{T})}^2 + \frac{n^2}{\beta^2 (n+1)^2} \|h_{\mathcal{T}} \nabla_{\text{pw}} w_{\mathcal{T}}\|_{S(\mathcal{T})^2}^2. \end{aligned} \quad (4.54)$$

This, $h_{\mathcal{T}}|_T \leq h_{\text{max}}$ for any $T \in \mathcal{T}$, and the abbreviation θ and ρ show

$$\|v_{\mathcal{T}} - v_{\text{CR}}\|_{L^2(\Omega)}^2 \leq h_{\text{max}}^2 \theta \rho.$$

Hence, $\|v_{\text{CR}}\|_{L^2(\Omega)}$ is bounded from below in $\|v_{\text{CR}}\|_{L^2(\Omega)} \geq 1 - h_{\text{max}} \sqrt{\theta \rho}$. Given (4.52), the mentioned arguments and (4.49) allow to bound

$$\begin{aligned} \rho &= (1-\alpha)^{-1} \|\Pi_0 w_{\mathcal{T}}\|_{s(\mathcal{T})}^2 + \beta^{-1} \|\nabla_{\text{pw}} w_{\mathcal{T}}\|_{S(\mathcal{T})^2}^2 \\ &\leq \frac{h_{\text{max}}^2 n^2}{(n+1)^2} \max \left\{ 1/\beta, 1/(n^2(1-\alpha)) \right\} \left(\|\Pi_0 w_{\mathcal{T}}\|_{L^2(\Omega)}^2 + \|S(\mathcal{T})^{1/2} \nabla_{\text{pw}} w_{\mathcal{T}}\|_{L^2(\Omega)}^2 \right) \\ &= h_{\text{max}}^2 \theta \|w_{\mathcal{T}}\|_{L^2(\Omega)}^2 \leq h_{\text{max}}^2 \theta \lambda_h(j)^2. \end{aligned}$$

Thus, the condition $h_{\text{max}}^2 \theta \lambda_h(j) < 1$ guarantees $\|v_{\text{CR}}\|_{L^2(\Omega)} > 0$ cannot vanish. In particular, $\phi_{\text{CR}}(1), \dots, \phi_{\text{CR}}(j)$ are linear independent. In the following, suppose that the coefficients $\xi \in \mathbb{R}^j$ with $|\xi| = 1$ maximize the Rayleigh-Quotient

$$\|v_{\text{CR}}\|_{\text{pw}}^2 / \|v_{\text{CR}}\|_{L^2(\Omega)}^2 \leq (\mu - \rho) / (1 - h_{\text{max}} \sqrt{\theta \rho})^2.$$

The min-max-principle in Lemma 2.7 for (4.45) asserts $\lambda_{\text{CR}}(j) \leq \frac{\|v_{\text{CR}}\|_{\text{pw}}^2}{\|v_{\text{CR}}\|_{L^2(\Omega)}^2}$. Abbreviate $\varepsilon := h_{\max}^2 \kappa_1^2$. The function $f(x) := x/(1 + \varepsilon x)$ is monoton increasing for any $x > 0$, that implies

$$\begin{aligned} \text{GLB}(j) &= f(\lambda_{\text{CR}}(j)) \leq f\left(\frac{\mu - \rho}{(1 - h_{\max}\sqrt{\theta\rho})^2}\right) \\ &= \frac{\mu - \rho}{1 - 2h_{\max}\sqrt{\theta\rho} + (h_{\max}^2\theta - \varepsilon)\rho + \varepsilon\mu}. \end{aligned}$$

The new bound is guaranteed to improve (4.46), if the right-hand side is smaller or equal $\mu \leq \lambda_h(j)$. In other words, if $1 - \rho/\mu \leq 1 - 2h_{\max}\sqrt{\theta\rho} + (h_{\max}^2\theta - \varepsilon)\rho + \varepsilon\mu$. Since $2h_{\max}\sqrt{\theta\rho} \leq \rho/\mu + h_{\max}^2\theta\mu$, a sufficient condition reads

$$0 \leq (\varepsilon - h_{\max}^2\theta)(\mu - \rho) = h_{\max}^2(\kappa_1 - \theta)(\mu - \rho).$$

Given $\mu - \rho = a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) > 0$, this translates to the condition $\theta \leq \kappa_1^2$ as claimed. $\square$

Remark 4.20 (improvement of θ). In Theorem 4.19 (and Theorem 5.14 below) the constant θ depends on the bound of the integral mean $|\Pi_0| \bullet - \text{mid}(\mathcal{T})|$ by its maximum $h_{\mathcal{T}}n/(n+1)$. In 2D this estimate can easily be improved with the edge midpoint quadrature rule on $T = \text{conv}\{P_1, P_2, P_3\} \in \mathcal{T}$, which is exact for quadratic polynomials

$$\begin{aligned} |\Pi_0| \bullet - \text{mid}(\mathcal{T})|^2 &= \frac{1}{3} \sum_{E \in \mathcal{E}} (\text{mid}(T) - \text{mid}(E))^2 = \frac{1}{108} \sum_{\ell=1}^3 |2P_\ell - P_{\ell+1} - P_{\ell+2}|^2 \\ &\leq \frac{1}{108} \sum_{\substack{j,k=1 \\ j \neq k}}^3 (|E_j| + |E_k|)^2 \leq \frac{1}{27} \sum_{j=1}^3 |E_j|^2 \leq \frac{1}{9} h_T^2 < \frac{4}{9} h_T^2. \end{aligned}$$

In a triangulation into right-isosceles triangles assume without loss of generality $T = \text{conv}\{(0, 0), (0, h_T/\sqrt{2}), (h_T/\sqrt{2}, 0)\} \in \mathcal{T}$. Hence, the second equality above implies $|\Pi_0| \bullet - \text{mid}(\mathcal{T})|^2 = h_T^2/18$.

In the experiments in Chapter 7 with triangulation into right-isosceles triangles the condition from Theorem 4.19 (and Theorem 5.14 below) reads

$$\theta = \frac{1}{18} \max\{1/\beta, (1/(4 - 4\alpha))\} \leq \kappa_1^2$$

and is equivalent to $\alpha \leq 1 - 1/(72\kappa_1^2)$ and $1/(18\kappa_1^2) \leq \beta$.

4.5.2 Estimate on the constant τ

Recall the a posteriori condition $\tau^2\beta + \delta^2\lambda_h(j) \leq \alpha < 1$ in Theorem 4.8 sufficient for the j -th discrete eigenvalue $\lambda_h(j)$ in (4.2) to be a guaranteed lower bound for

the j -th exact eigenvalue λ_j in (4.1). Thus the constants τ and κ in $\delta := \kappa h_{\max}$ are essential for the choice of the parameters α and β . In the lowest-order case, the constant $\kappa = \kappa_1$ is analysed in Remark 2.5 and Theorem 3.24.b.

The constant $\tau := \sqrt{C_{st}^2 - 1}$ is more difficult to estimate, since the proof of the existence of C_{st} in [CZZ20, Thm 3.1] is non-constructive. Actually, the constant τ is only needed in Lemma 4.3 to bound $\|S(Iv)\|_{pw}$ for all $v \in V$. In the lowest-order case an inverse inequality allows to bound this quantity as follows.

Lemma 4.21 (bound on τ). *Let c_{inv} denote the constant associated with the inverse estimate in $P_1(\mathcal{T})$, i.e., $\|\nabla p_1\|_{L^2(T)} \leq c_{\text{inv}} h_T^{-1} \|p_1\|_{L^2(T)}$ for any $T \in \mathcal{T}$ and $p_1 \in P_1(T)$, and let C_P denote the Poincaré constant in Lemma 2.11. Then $\tau \leq C_P c_{\text{inv}}$ satisfies $\|S(Iv)\|_{pw} \leq \tau \|(1 - \Pi_0)(\nabla v)\|_{L^2(\Omega)}$ for all $v \in V := H_0^1(\Omega)$.*

Proof. Any $v \in V$ satisfies $\|S(Iv)\|_{pw} = \|\Pi_1(v) - I_{\mathcal{F}} \Pi_{\mathcal{F},0}(v)\|_{pw} = \|(\Pi_1 - I_{\text{CR}})(v)\|_{pw}$ according to Lemma 4.16.c. For any $T \in \mathcal{T}$, the projection properties of Π_0 and Π_1 and the inverse estimate for affine functions in $P_1(T)$ show that

$$\begin{aligned} h_T c_{\text{inv}}^{-1} \|\nabla(\Pi_1 - I_{\text{CR}})(v)\|_{L^2(T)} &= h_T c_{\text{inv}}^{-1} \|\nabla(1 - \Pi_0)\Pi_1(1 - I_{\text{CR}})(v)\|_{L^2(T)} \\ &\leq \|(1 - \Pi_0)\Pi_1(1 - I_{\text{CR}})(v)\|_{L^2(T)} = \|\Pi_1(1 - \Pi_0)(1 - I_{\text{CR}})(v)\|_{L^2(T)} \\ &\leq \|(1 - \Pi_0)(1 - I_{\text{CR}})(v)\|_{L^2(T)} \leq C_P h_T \|\nabla(1 - I_{\text{CR}})(v)\|_{L^2(T)}. \end{aligned}$$

This implies that $\|\nabla S(Iv)\|_{L^2(T)} \leq c_{\text{inv}} C_P \|\nabla(1 - I_{\text{CR}})(v)\|_{L^2(T)}$. The identity $\nabla I_{\text{CR}} = \Pi_0 \nabla$ from Theorem 3.24.a and summation over $\mathcal{T}$ conclude the proof. $\square$

For $n = 2$, the constant $c_{\text{inv}}(T)$ is computed in [CH18a, Lemma 4.10] in terms of the minimum angle ω_T^0 in $T \in \mathcal{T}$ as

$$c_{\text{inv}}(T)^2 = 24 \cot(\omega_T^0) \left(2 \cot(\omega_T^0) - \cot(2\omega_T^0) + ((2 \cot(\omega_T^0) - \cot(2\omega_T^0))^2 - 3)^{1/2} \right).$$

For a triangulation composed of right isosceles triangle holds $c_{\text{inv}} = \sqrt{72}$ and [LS10, p.132] improves the Poincaré constant $C_P = 1/j_{1,1}$ for general triangles to $C_P = 1/(\sqrt{2}\pi)$. Hence, Lemma 4.21 shows that $\tau \leq 6/\pi = 1.90986$. Lemma 4.21 gives a first analytical upper bound for the constant τ for the lowest-order method utilized in the numerical experiments in Chapter 7. For the alternative stabilization in Remark 4.4, the boundedness of Π_1 and Theorem 3.24.a–b show $\|h_{\mathcal{T}}^{-1} S(Iv)\|_{L^2(\Omega)} = \|h_{\mathcal{T}}^{-1} (1 - \Pi_0)(\Pi_1 - I_{\text{CR}})(v)\|_{L^2(\Omega)} \leq \|h_{\mathcal{T}}^{-1} (1 - I_{\text{CR}})(v)\|_{L^2(\Omega)} \leq \kappa_1 \|(1 - \Pi_0)(\nabla v)\|_{L^2(\Omega)}$ for any $v \in V$. In other words, the constant for this alternative stabilization satisfies $\tilde{\tau} \leq \kappa_1$ in the lowest-order case in any space dimension.

Sharper bounds for the parameter τ (in particular for higher polynomial degrees) may be computed by a related PDE eigenvalue problem on a reference domain; details and numerical examples are left for future work. In case of the alternative stabilization from Remark 4.4, [CGT22] provides the estimate $\tilde{\tau} \leq \sqrt{2}/\pi$ for triangles with maximal interior angle $\omega \leq \pi/2$ independent of the polynomial degree and utilizes it in numerical examples with higher polynomial degrees.

5 The extra-stabilized method

This section analyses an extra-stabilized method, which approximates the discrete solution with a piecewise polynomial of degree m and a nonconforming polynomial of the same degree. An additional stabilization coupling both components allows that the discrete eigenvalues are lower bounds of continuous eigenvalues of the m -Laplacian with homogeneous Dirichlet boundary provided a simple condition, which can be checked a posteriori, is satisfied. These direct guaranteed bounds that do not involve the maximal mesh-size $h_{\max}$ allow adaptive mesh-refinement. This chapter proves optimal convergence of the adaptive Algorithm 2 with help of the axioms of adaptivity [Car+14a; CR17] slightly adapted in Chapter 6. The analysis is done in parallel for $m = 1, 2$ with the lowest-order Crouzeix-Raviart ($m = 1$) and Morley ($m = 2$) finite elements as nonconforming spaces. However, the necessary conditions for the interpolation operator and conforming companion of the nonconforming space are generalized such that nonconforming finite elements of higher-order [WX13] could be adapted for the m -Laplace eigenvalue problems with $m \geq 3$. For $m = 1 = n - 1$ the method was introduced as lowest-order version of a skeletal method in [CZZ20].

This chapter is organized as follows. Section 5.1 introduces the discrete problem and contains the conditions for the interpolation operator I and the conforming companion J that allow the analysis in this chapter. These conditions are satisfied for the Crouzeix-Raviart and Morley finite element according to Sections 3.3–3.4. Afterwards Section 5.2 proves the guaranteed lower-bound property in Theorem 5.5. Section 5.3 compares the guaranteed lower bounds with the standard nonconforming results [CG14a; CG14b] and the modified HHO method from the previous chapter. Moreover, the section contains an equivalent reduced eigenvalue problem, which is useful in the convergence analysis for the solution components in Section 5.4. These preparations allow the proof of optimal adaptive convergence in Section 5.5.

5.1 Introduction to the extra-stabilized method

Let $\Omega \subset \mathbb{R}^n$ with $n \geq 2$ denote a polyhedral bounded Lipschitz domain with set of admissible triangulation $\mathbb{T}$. Throughout this chapter, the continuous eigenvalue problem concerns the Laplace ($m = 1$) or bi-Laplace ($m = 2$) operator. The weak

form seeks $(\lambda, u) \in \mathbb{R}^+ \times H_0^m(\Omega)$ with $\|u\|_{L^2(\Omega)} = 1$ and

$$a(u, v) = \lambda b(u, v) \quad \text{for all } v \in V := H_0^m(\Omega). \quad (5.1)$$

The energy scalar product in the Hilbert space $V := H_0^m(\Omega)$ reads $a(\bullet, \bullet) := (D^m \bullet, D^m \bullet)_{L^2(\Omega)}$ with the gradient $D^1 = \nabla$ ($m = 1$) or the Hessian D^2 ($m = 2$) and induces the energy norm $\|\bullet\| = \|D^m \bullet\|_{L^2(\Omega)}$ while $b(u, v) := (u, v)_{L^2(\Omega)}$ denotes the L^2 scalar product. The infinite but countable many eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$ with $\lim_{j \rightarrow \infty} \lambda_j = \infty$ in (5.1) are enumerated in ascending order counting multiplicities.

The discrete space $V_h = P_m(\mathcal{T}) \times V(\mathcal{T}) \subset P_m(\mathcal{T}) \times P_m(\mathcal{T})$ consists of piecewise polynomials of degree at most m on the admissible shape-regular triangulation $\mathcal{T} \in \mathbb{T}$ of $\Omega \subset \mathbb{R}^n$ into closed n -simplices (called simplex throughout). The nonconforming space $V(\mathcal{T})$ abbreviates the Crouzeix-Raviart finite element space $\text{CR}_0^1(\mathcal{T})$ [CR73] for $m = 1$ and the Morley finite element space $M(\mathcal{T})$ [Mor68; MX06] for $m = 2$. The algebraic eigenvalue problem seeks eigenpairs $(\lambda_h, u_h) \in \mathbb{R}^+ \times (V_h \setminus \{0\})$ with

$$a_h(u_h, v_h) = \lambda_h b_h(u_h, v_h) \quad \text{for all } v_h \in V_h. \quad (5.2)$$

The discrete scalar product a_h contains $a_{\text{pw}}(\bullet, \bullet) := (D_{\text{pw}}^m \bullet, D_{\text{pw}}^m \bullet)_{L^2(\Omega)}$ the scalar product of the piecewise derivatives of order m and some stabilization with explicit constant $\kappa_m > 0$ from Section 3.3–3.4, while the bilinear form b_h is the L^2 scalar product $b(\bullet, \bullet)$ of the piecewise polynomial components,

$$\begin{aligned} a_h(v_h, w_h) &:= a_{\text{pw}}(v_{\text{nc}}, w_{\text{nc}}) + \kappa_m^{-2} (h_{\mathcal{T}}^{-2m} (v_{\text{pw}} - v_{\text{nc}}), w_{\text{pw}} - w_{\text{nc}})_{L^2(\Omega)}, \\ b_h(v_h, w_h) &:= b(v_{\text{pw}}, w_{\text{pw}}) \quad \text{for all } v_h = (v_{\text{pw}}, v_{\text{nc}}), w_h = (w_{\text{pw}}, w_{\text{nc}}) \in V_h. \end{aligned}$$

The piecewise constant mesh-size function $h_{\mathcal{T}} \in P_0(\mathcal{T})$ has the value $h_{\mathcal{T}}|_T = h_T := \text{diam}(T)$ in each simplex $T \in \mathcal{T}$ and $h_{\max} := \max_{T \in \mathcal{T}} h_T$ denotes the maximal mesh-size. Notice that (V_h, a_h) is a Hilbert space and b_h is a semi-scalar product with kernel $\{0\} \times V(\mathcal{T}) \subset V_h$. Hence, the algebraic eigenvalue problem (5.2) has $M := \dim(P_m(\mathcal{T})) = \binom{m+n}{n} |\mathcal{T}|$ finite and positive algebraic eigenvalues $0 < \lambda_h(1) \leq \dots \leq \lambda_h(M) < \infty$ enumerated in ascending order counting multiplicity. Parts of the following analysis are contained in the recent preprints [CP21; CP22].

5.1.1 Properties of the interpolation operator

The operators I and J concern the (nonconforming) discrete space $V(\mathcal{T}) \subset P_m(\mathcal{T})$ and $V := H_0^m(\Omega)$ for an admissible triangulation $\mathcal{T} \in \mathbb{T}$ of $\Omega \subset \mathbb{R}^n$. An advantage of separate interest is that the analysis with I and J is performed simultaneous for $m \geq 1$. This section summarizes the properties the interpolation operator I has to satisfy in the analysis throughout this chapter. Suppose that for each

admissible triangulation $\mathcal{T} \in \mathbb{T}$ there exists a linear interpolation operator I onto $V(\mathcal{T})$ that is defined on $V + V(\widehat{\mathcal{T}})$ for any refinement $\widehat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ and that satisfies the following properties for universal positive constants κ_m and $\widehat{\kappa}_m$.

- (I1) Any $T \in \mathcal{T}$ and $v \in H^m(T)$ satisfy $\|v - Iv\|_{L^2(T)} \leq \kappa_m h_T^m |v - Iv|_{H^m(T)}$.
- (I2) The piecewise derivative D_{pw}^m of any $v \in V + V(\widehat{\mathcal{T}})$ satisfies $D_{\text{pw}}^m Iv = \Pi_0 D_{\text{pw}}^m v$.
- (I3) The operator I acts as identity in non-refined simplices in that $(1 - I)\widehat{v}_{\text{nc}}|_T = 0$ in $T \in \mathcal{T} \cap \widehat{\mathcal{T}}$ for all $\widehat{v}_{\text{nc}} \in V(\widehat{\mathcal{T}})$. The interpolation operator $\widehat{I}$ associated with $V(\widehat{\mathcal{T}})$ satisfies $I \circ \widehat{I} = I$ in $V + V(\widehat{\mathcal{T}})$.
- (I4) Any $T \in \mathcal{T}$ and $\widehat{v}_{\text{nc}} \in V(\widehat{\mathcal{T}})$ satisfy $\|\widehat{v}_{\text{nc}} - I\widehat{v}_{\text{nc}}\|_{L^2(T)} \leq \widehat{\kappa}_m h_T^m |\widehat{v}_{\text{nc}} - I\widehat{v}_{\text{nc}}|_{H^m(T)}$.

Note, that Theorem 3.24 proves (I1)–(I4) for $I = I_{\text{CR}}$ and $m = 1$ in Subsection 3.3.1 and Theorem 3.35 proves the same for $I = I_M$ and $m = 2$ in Subsection 3.4.1 in any space dimension. The following implications of (I1)–(I4) apply throughout this thesis and hold in particular for both examples.

Corollary 5.1 (properties of I). (a) Given $\widehat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$, any $v \in V + V(\widehat{\mathcal{T}})$ and $w_{\text{nc}} \in V(\mathcal{T})$ satisfy

$$a_{\text{pw}}(v - Iv, w_{\text{nc}}) = 0 \quad \text{and} \quad \|v - Iv\|_{\text{pw}} = \min_{v_{\text{nc}} \in V(\mathcal{T})} \|v - v_{\text{nc}}\|_{\text{pw}}.$$

(b) Any $v \in H^{m+s}(\Omega)$ with $0 \leq s \leq 1$ satisfies

$$\|(1 - I)v\|_{\text{pw}} \leq (h_{\max}/\pi)^s \|v\|_{H^{m+s}(\Omega)}.$$

(c) Any $v, w \in V$ and $v_{\text{nc}} \in V(\mathcal{T})$ satisfy $a_{\text{pw}}(v, v_{\text{nc}}) = a_{\text{pw}}(Iv, v_{\text{nc}})$ and

$$\begin{aligned} a_{\text{pw}}(v, (1 - I)w) &= a_{\text{pw}}((1 - I)v, (1 - I)w) \\ &\leq \min_{v_{\text{nc}} \in V(\mathcal{T})} \|v - v_{\text{nc}}\|_{\text{pw}} \min_{w_{\text{nc}} \in V(\mathcal{T})} \|w - w_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

(d) Any $w \in V$ and $v \in V + V(\mathcal{T})$ satisfy

$$b(v, (1 - I)w) \leq \|h_{\mathcal{T}}^m v\|_{L^2(\Omega)} \|h_{\mathcal{T}}^{-m} (1 - I)w\|_{L^2(\Omega)} \leq \kappa_m \|h_{\mathcal{T}}^m v\|_{L^2(\Omega)} \min_{w_{\text{nc}} \in V(\mathcal{T})} \|w - w_{\text{nc}}\|_{\text{pw}}.$$

Proof of (a). This is a direct consequence of (I2), since $D_{\text{pw}}^m w_{\text{nc}} \in P_0(\mathcal{T}; \mathbb{R}^{n^m})$. $\square$

Proof of (b). The piecewise Poincaré inequality in Lemma 2.11 and (I2) show for $s = 0$ and $s = 1$ that

$$\|(1 - I)v\|_{\text{pw}} = \|(1 - \Pi_0)D^m v\|_{L^2(\Omega)} \leq (h_{\max}/\pi)^s \|v\|_{H^{m+s}(\Omega)}. \quad (5.3)$$

Since $H^{m+s}(\Omega)$ is defined by complex interpolation of $H^m(\Omega)$ and $H^{m+1}(\Omega)$, each component of the derivative $D^m v$ belongs to $H^s(\Omega) \in [L^2(\Omega), H^1(\Omega)]_s$ in the complex interpolation space between $L^2(\Omega)$ and $H^1(\Omega)$ [Tar07]. The interpolation of (5.3) concludes the proof of (b). $\square$

Proof of (c). The first claim in (c) follows directly from the orthogonality in (a). The combination of (a) with the Cauchy-Schwarz inequality implies the second claim. $\square$

Proof of (d). The Cauchy-Schwarz inequality, the approximation property (II), and the best approximation property in (a) conclude the proof of (d). $\square$

5.1.2 Properties of the conforming companion

Given any simplex $T \in \mathcal{T} \in \mathbb{T}$ in a triangulation $\mathcal{T} \in \mathbb{T}$, recall that $\mathcal{V}(T)$ denotes the set of its vertices (0-subsimplices) and $\mathcal{F}(T)$ denotes the set of its sides $((n-1)$ -subsimplices). This section summarizes all properties of the conforming companion J that are needed throughout the analysis below. A linear operator $J : V(\mathcal{T}) \rightarrow V$ is called *conforming companion* if (J1)–(J4) hold with universal constants M_1, M_2, M_4 (that exclusively depend on $\mathbb{T}$).

(J1) J is a right inverse to the interpolation I in the sense that, $I \circ J = 1$ in $V(\mathcal{T})$.

(J2) $\|h_{\mathcal{T}}^{-m}(1 - J)v_{\text{nc}}\|_{L^2(\Omega)} + \|(1 - J)v_{\text{nc}}\|_{\text{pw}} \leq \left(M_1 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D_{\text{pw}}^m v_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2} \leq M_2 \min_{v \in V} \|v_{\text{nc}} - v\|_{\text{pw}}$
for any $v_{\text{nc}} \in V(\mathcal{T})$.

(J3) $(1 - J)(V(\mathcal{T})) \perp P_m(\mathcal{T})$ holds in $L^2(\Omega)$.

(J4) $|v_{\text{nc}} - Jv_{\text{nc}}|_{H^m(K)}^2 \leq M_4 \sum_{T \in \mathcal{T}(\Omega(K))} |K|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D_{\text{pw}}^m v_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2$ holds for any $v_{\text{nc}} \in V(\mathcal{T})$ and $K \in \mathcal{T}$ with the set $\mathcal{T}(\Omega(K)) := \{T \in \mathcal{T} : \text{dist}(T, K) = 0\}$ of adjacent simplices.

The properties are stated in this form for convenient quotation. Clearly (J4) implies parts of (J2) and the second inequality in (J2) is the efficiency of a posteriori error estimators and independent of the operator J . Note that Theorem 3.25 shows that $J = J_{\text{CR}}$ for $m = 1$ in Subsection 3.3.2 satisfies (J1)–(J4) for any space dimension $n \geq 2$, while Theorem 3.36 in Subsection 3.4.2 proves the same for $J = J_M$ for $m = 2$ in space dimension $n = 2, 3$.

The following implications of (J1)–(J4) apply throughout this work in particular for $m = 1, 2$.

Corollary 5.2 (properties of J). *Any $w \in V$ and $v_{\text{nc}} \in V(\mathcal{T})$ satisfy*

$$\begin{aligned} (a) \quad & b(v_{\text{nc}} - Jv_{\text{nc}}, w) = b(v_{\text{nc}} - Jv_{\text{nc}}, w - Iw) \leq \|v_{\text{nc}} - Jv_{\text{nc}}\|_{L^2(\Omega)} \|w - Iw\|_{L^2(\Omega)} \\ & \leq h_{\max}^{2m} \kappa_m M_2 \min_{v \in V} \|v_{\text{nc}} - v\|_{\text{pw}} \min_{w_{\text{nc}} \in V(\mathcal{T})} \|w - w_{\text{nc}}\|_{\text{pw}}; \\ (b) \quad & a_{\text{pw}}(v_{\text{nc}} - Jv_{\text{nc}}, w) = a_{\text{pw}}(v_{\text{nc}} - Jv_{\text{nc}}, Iw - w) \leq \|v_{\text{nc}} - Jv_{\text{nc}}\|_{\text{pw}} \|w - Iw\|_{\text{pw}} \\ & \leq M_2 \min_{v \in V} \|v - v_{\text{nc}}\|_{\text{pw}} \min_{w_{\text{nc}} \in V(\mathcal{T})} \|w - w_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

Proof of (a). The identity follows from the orthogonality in (J3) for $Iw \in P_m(\mathcal{T})$ and the Cauchy-Schwarz inequality implies the first inequality. Then the first term is controlled by (J2) and the second term by (I1) and Corollary 5.1.a. $\square$

Proof of (b). The identity follows from the right-inverse property (J1), which implies $v_{\text{nc}} - Jv_{\text{nc}} = (I - 1)Jv_{\text{nc}}$, and the orthogonality in Corollary 5.1.c. The latter also allows to bound the second term resulting from the Cauchy-Schwarz inequality. This and (J2) conclude the proof of (b). $\square$

Since the embedding $V \hookrightarrow L^2(\Omega)$ is bounded and the Friedrichs constant C_F satisfies the Friedrichs inequality, Lemma 2.12, the discrete Friedrichs inequality Corollary 5.3 below is a direct consequence of (J2).

Corollary 5.3 (discrete Friedrichs inequality). *The constant $C_{\text{dF}} = C_F(1 + M_2) + M_2 h_{\text{max}}^m$ satisfies $\|v_{\text{nc}}\|_{L^2(\Omega)} \leq C_{\text{dF}} \|v_{\text{nc}}\|_{\text{pw}}$ for all $v_{\text{nc}} \in V(\mathcal{T})$.*

Proof. The discrete Friedrichs inequality is a modification of the continuous Friedrichs inequality applied to $Jv_{\text{nc}} \in V$, where the difference $v_{\text{nc}} - Jv_{\text{nc}}$ is controlled by (J2).

On the one-hand side, (J2) with $h_{\mathcal{T}} \leq h_{\text{max}}$ controls

$$\|v_{\text{nc}} - Jv_{\text{nc}}\|_{L^2(\Omega)} \leq h_{\text{max}}^m M_2 \min_{v \in V} \|v_{\text{nc}} - v\|_{\text{pw}} \leq h_{\text{max}}^m M_2 \|v_{\text{nc}}\|_{\text{pw}}.$$

On the other hand, Lemma 2.12, a triangle inequality, and (J2) show

$$C_F^{-1} \|Jv_{\text{nc}}\|_{L^2(\Omega)} \leq \|Jv_{\text{nc}}\| \leq \|v_{\text{nc}} - Jv_{\text{nc}}\|_{\text{pw}} + \|v_{\text{nc}}\|_{\text{pw}} \leq (1 + M_2) \|v_{\text{nc}}\|_{\text{pw}}.$$

The displayed estimates and the triangle inequality $\|v_{\text{nc}}\|_{L^2(\Omega)} = \|v_{\text{nc}} - Jv_{\text{nc}}\|_{L^2(\Omega)} + \|Jv_{\text{nc}}\|_{L^2(\Omega)}$ conclude the proof. For Crouzeix-Raviart this connection is explored in [CH18a, Cor. 4.11]. $\square$

Remark 5.4 (consequences of (J4)). For any refinement $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ of a triangulation $\mathcal{T} \in \mathbb{T}$ let $\mathcal{R}_1 := \{K \in \mathcal{T} : \exists T \in \mathcal{T} \setminus \hat{\mathcal{T}} \text{ with } \text{dist}(K, T) = 0\} \subseteq \mathcal{T}$ denote the set of coarse but not fine simplices plus one layer of coarse simplices around. Then (J4) and a finite overlap argument for the shape-regular triangulation $\mathcal{T} \in \mathbb{T}$ imply that any $v_{\text{nc}} \in V(\mathcal{T})$ satisfies

$$\|D_{\text{pw}}^m(v_{\text{nc}} - Jv_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 \leq M_5 \sum_{T \in \mathcal{R}_1} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D_{\text{pw}}^m v_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2.$$

The superset $\mathcal{R}_1$ of $\mathcal{T} \setminus \hat{\mathcal{T}}$ serves as an simple example and could indeed be replaced by $\mathcal{T} \setminus \hat{\mathcal{T}}$ provided the conforming companion J is replaced by some $\tilde{J}$ that may depend on $\hat{\mathcal{T}}$ for some of the proofs; cf. [CP20, §6] for details in the two model problems with $m = 1$ or $m = 2$. $\square$

5.2 Guaranteed lower eigenvalue bound property

The proof of the guaranteed lower bound property in Theorem 5.5 requires the existence of an interpolation operator with the approximation property (I1) and the orthogonality $a_{\text{pw}}(v - Iv, w_{\text{nc}}) = 0$ for any $v \in V$ and $w_{\text{nc}} \in V(\mathcal{T})$ from Corollary 5.1.a, which is a direct consequence from (I2). Hence, this section proves that the discrete method (5.2) indeed provides GLBs for the continuous eigenvalues in *any* space dimension $n \geq 2$ for $m = 1, 2$.

Theorem 5.5 (GLB). *For any $k = 1, \dots, M$, the k -th eigenvalue λ_k from (5.1) and the k -th eigenvalue $\lambda_h(k)$ from (5.2) satisfy that $\min\{\lambda_h(k), \lambda_k\} \kappa_m^2 h_{\max}^{2m} \leq 1$ implies $\lambda_h(k) \leq \lambda_k$.*

The following Remark 5.6 introduces a constant κ'_m with (5.4) closely related to κ_m and comments on its value.

Remark 5.6. Let κ'_m denote the best possible constant in

$$\|(1 - \Pi_m)v\|_{L^2(\Omega)} \leq \kappa'_m h_{\max}^m \|(1 - I)v\|_{\text{pw}} \quad \text{for all } v \in V. \quad (5.4)$$

(a) The approximation property of Π_m and $V(\mathcal{T}) \subset P_m(\mathcal{T})$ imply (5.4) with $\kappa'_m \leq \kappa_m$.

(b) For $m = 1, 2$ successive (piecewise) Poincaré inequalities prove $\kappa'_m \leq 1/\pi^m$ and hence $\kappa'_m < \kappa_m$ for the analytical constants in (1.2). Given any $v \in V$ define the piecewise constant functions $a_0 \in P_0(\mathcal{T})$ with $a_0|_T := \int_T (1 - I)v \, dx \in \mathbb{R}$ and $a_1 \in P_0(\mathcal{T}; \mathbb{R}^n)$ with $a_1|_T := \int_T \nabla(1 - I)v \, dx \in \mathbb{R}^n$ for any $T \in \mathcal{T}$.

For $m = 1$, set $p_0(x) := a_0(x) \in P_0(\mathcal{T})$. Since $Iv - p_0 \in P_1(\mathcal{T})$, the (piecewise) Poincaré inequality in Lemma 2.11 shows

$$\|(1 - \Pi_1)v\|_{L^2(\Omega)} \leq \|(1 - I)v - p_0\|_{L^2(\Omega)} = \|(1 - \Pi_0)(1 - I)v\|_{L^2(\Omega)} \leq \frac{h_{\max}}{\pi} \|(1 - I)v\|_{\text{pw}}.$$

For $m = 2$ set $p_1(x) := a_0(x) + a_1 \cdot (x - \text{mid}(T)) \in P_1(\mathcal{T})$. Since $Iv - p_1 \in P_2(\mathcal{T})$, successive (piecewise) Poincaré inequalities prove

$$\begin{aligned} \|(1 - \Pi_2)v\|_{L^2(\Omega)} &\leq \|(1 - I)v - p_1\|_{L^2(\Omega)} = \|(1 - \Pi_0)((1 - I)v - a_1 \cdot (\bullet - \text{mid}(T)))\|_{L^2(\Omega)} \\ &\leq \frac{h_{\max}}{\pi} \|\nabla(1 - I)v - a_1\|_{L^2(\Omega)} = \frac{h_{\max}}{\pi} \|(1 - \Pi_0)\nabla(1 - I)v\|_{L^2(\Omega)} \\ &\leq \frac{h_{\max}^2}{\pi^2} \|(1 - I)v\|_{\text{pw}}. \end{aligned}$$

This concludes the proof of $\kappa'_m \leq 1/\pi^m$ for $m = 1, 2$.

(c) The computational bounds $\kappa_1 = 0.1893 > \kappa_1^*$ from [Liu15] and $\kappa_2 = 0.07353 > \kappa_2^*$ from [LSL19] in 2D are guaranteed upper bounds for some optimal κ_m^* with $\|v - Iv\|_{L^2(T)} \leq \kappa_m^* h_T^m |v - Iv|_{H^m(T)}$ for all $v \in H^m(T)$, $T \in \mathcal{T}$. Hence, for these bounds $\kappa'_m := \kappa_m^* < \kappa_m$.

Suppose (5.4) with $\kappa'_m < \kappa_m$ for Theorem 5.5, otherwise see Remark 5.7 below.

Proof of Theorem 5.5. Abbreviate $\lambda = \lambda_k$ from (5.1) and $\lambda_h = \lambda_h(k)$ from (5.2). Let $\phi_1, \dots, \phi_k \in V$ denote the first k continuous b -orthonormal eigenfunctions of (5.1); the min-max principle in Lemma 2.7 guarantees $|||\phi|||^2 \leq \lambda$ for any $\phi \in \text{span}\{\phi_1, \dots, \phi_k\}$ with $\|\phi\|_{L^2(\Omega)} = 1$. Recall that Π_m denotes L^2 projection onto $P_m(\mathcal{T})$.

Case 1. Assume the L^2 -projections $\Pi_m\phi_1, \dots, \Pi_m\phi_k$ are linear dependent. Then there exists some $\phi \in \text{span}\{\phi_1, \dots, \phi_k\}$ with $\|\phi\|_{L^2(\Omega)} = 1$ and $\Pi_m\phi = 0$, hence

$$1 = \|\phi\|_{L^2(\Omega)} = \|(1 - \Pi_m)\phi\|_{L^2(\Omega)} \leq \kappa'_m h_{\max}^m |||(1 - I)\phi|||_{\text{pw}} \quad (5.5)$$

with κ'_m from (5.4) in Remark 5.6 in the inequality. The Pythagoras theorem from Corollary 5.1.a reads

$$|||(1 - I)\phi|||_{\text{pw}}^2 + |||I\phi|||_{\text{pw}}^2 = |||\phi|||^2 \leq \lambda$$

with $|||\phi|||^2 \leq \lambda$ from the min-max principle in Lemma 2.7 for (5.1). The combination of the two displayed inequalities reads $1 \leq \lambda(\kappa'_m)^2 h_{\max}^{2m}$. Throughout this work, the values used for κ_m satisfy $\kappa'_m < \kappa_m$ according to Remark 5.6.b–c. In other words, it holds $1 \leq \lambda(\kappa'_m)^2 h_{\max}^{2m} < \lambda\kappa_m^2 h_{\max}^{2m}$. Hence, the a priori condition $\lambda\kappa_m^2 h_{\max}^{2m} \leq 1$ fails and (the remaining hypothesis of Theorem 5.5) $\lambda_h\kappa_m^2 h_{\max}^{2m} \leq 1$ holds. This proves $\lambda_h\kappa_m^2 h_{\max}^{2m} \leq 1 < \lambda\kappa_m^2 h_{\max}^{2m}$ and concludes the proof of $\lambda_h \leq \lambda$ in the first case.

Case 2. Assume that the projections $\Pi_m\phi_1, \dots, \Pi_m\phi_k$ are linear independent. Set $S_k := \text{span}\{(\Pi_m\phi_1, I\phi_1), \dots, (\Pi_m\phi_k, I\phi_k)\} \subset V_h$ with $\dim(S_k) = k$. Since b_h is positive definite on $S_k \times S_k$ (from the linear independence of $\Pi_m\phi_1, \dots, \Pi_m\phi_k$), the min-max principle in Lemma 2.7 for (5.2) shows

$$\lambda_h \leq \max_{v_h \in S_k \setminus \{0\}} \frac{a_h(v_h, v_h)}{b_h(v_h, v_h)}. \quad (5.6)$$

Let $v_h = (\Pi_m\phi, I\phi) \in S_k \setminus \{0\}$ be one maximizer in the upper bound of (5.6) and observe $\|\Pi_m\phi\|_{L^2(\Omega)} > 0$. Hence, suppose, without loss of generality, that $\|\phi\|_{L^2(\Omega)} = 1$ and deduce $|||\phi|||^2 \leq \lambda$ from the min-max principle in Lemma 2.7 for (5.1). The inequality (5.6) ensures for $v_h = (\Pi_m\phi, I\phi) \in S_k \setminus \{0\}$ that

$$\lambda_h \|\Pi_m\phi\|_{L^2(\Omega)}^2 = \lambda_h b_h(v_h, v_h) \leq a_h(v_h, v_h) = |||I\phi|||_{\text{pw}}^2 + \kappa_m^{-2} \|h_{\mathcal{T}}^{-m}(\Pi_m - I)\phi\|_{L^2(\Omega)}^2.$$

Since the piecewise constant mesh-size $h_{\mathcal{T}} \leq h_{\max}$ does not interact with the piecewise L^2 projections, the Pythagoras theorem shows

$$\begin{aligned} \|h_{\mathcal{T}}^{-m}(\Pi_m - I)\phi\|_{L^2(\Omega)}^2 &= \|h_{\mathcal{T}}^{-m}(1 - I)\phi\|_{L^2(\Omega)}^2 - \|h_{\mathcal{T}}^{-m}(1 - \Pi_m)\phi\|_{L^2(\Omega)}^2 \\ &\leq \|h_{\mathcal{T}}^{-m}(1 - I)\phi\|_{L^2(\Omega)}^2 - h_{\max}^{-2m} \|(1 - \Pi_m)\phi\|_{L^2(\Omega)}^2 \\ &\leq \kappa_m^2 |||(1 - I)\phi|||_{\text{pw}}^2 - h_{\max}^{-2m} + h_{\max}^{-2m} \|\Pi_m\phi\|_{L^2(\Omega)}^2 \end{aligned}$$

with (I1) and $\|(1 - \Pi_m)\phi\|_{L^2(\Omega)}^2 = 1 - \|\Pi_m\phi\|_{L^2(\Omega)}^2$ in the last step. The combination of the previous two displayed estimates reads

$$\lambda_h \|\Pi_m\phi\|_{L^2(\Omega)}^2 \leq \|I\phi\|_{\text{pw}}^2 + \|(1 - I)\phi\|_{\text{pw}}^2 - \kappa_m^{-2} h_{\max}^{-2m} + \kappa_m^{-2} h_{\max}^{-2m} \|\Pi_m\phi\|_{L^2(\Omega)}^2.$$

This, the Pythagoras theorem $\|I\phi\|_{\text{pw}}^2 + \|(1 - I)\phi\|_{\text{pw}}^2 = \|\phi\|^2$ from Corollary 5.1.a, and $\|\phi\|^2 \leq \lambda$ lead to

$$\lambda_h \|\Pi_m\phi\|_{L^2(\Omega)}^2 \leq \lambda - \kappa_m^{-2} h_{\max}^{-2m} + \kappa_m^{-2} h_{\max}^{-2m} \|\Pi_m\phi\|_{L^2(\Omega)}^2 \quad (5.7)$$

and imply

$$1 - \lambda \kappa_m^2 h_{\max}^{2m} \leq (1 - \lambda_h \kappa_m^2 h_{\max}^{2m}) \|\Pi_m\phi\|_{L^2(\Omega)}^2. \quad (5.8)$$

Without loss of generality assume $\lambda \kappa_m^2 h_{\max}^{2m} \leq 1$, otherwise the remaining hypothesis $\lambda_h \kappa_m^2 h_{\max}^{2m} \leq 1 < \lambda \kappa_m^2 h_{\max}^{2m}$ proves the claim. Since the linear independence of $\Pi_m\phi_1, \dots, \Pi_m\phi_k$ shows $\|\Pi_m\phi\|_{L^2(\Omega)} > 0$, (5.8) and $\lambda \kappa_m^2 h_{\max}^{2m} \leq 1$ imply $0 < 1 - \lambda_h \kappa_m^2 h_{\max}^{2m}$. This and $\|\Pi_m\phi\|_{L^2(\Omega)} \leq \|\phi\|_{L^2(\Omega)} = 1$ lead to $1 - \lambda \kappa_m^2 h_{\max}^{2m} \leq 1 - \lambda_h \kappa_m^2 h_{\max}^{2m}$ and conclude the proof of $\lambda_h \leq \lambda$ in the second case. $\square$

Remark 5.7 (relaxed conditions in Theorem 5.5). The proof of Theorem 5.5 is written for $\kappa'_m < \kappa_m$, which holds for the values used for κ_m for $m = 1, 2$ as in Remark 5.6.b–c. If in future computations $\kappa'_m < \kappa_m$ can not be verified, then $\kappa'_m \leq \kappa_m$ from Remark 5.6.a and the strict inequality $\min\{\lambda_h(k), \lambda_k\} \kappa_m^2 h_{\max}^{2m} < 1$ as condition imply $\lambda_h(k) \leq \lambda_k$ in Theorem 5.5. Since $1 \leq \lambda(\kappa'_m)^2 h_{\max}^{2m} \leq \lambda \kappa_m^2 h_{\max}^{2m}$ shows that the slightly stricter a priori condition $\lambda \kappa_m^2 h_{\max}^{2m} < 1$ fails in Case 1 and then the proof works verbatim.

Recall that the standard nonconforming eigenvalue problem seeks $(\lambda_{\text{nc}}, \phi_{\text{nc}}) \in \mathbb{R}^+ \times (V(\mathcal{T}) \setminus \{0\})$ with

$$a_{\text{pw}}(\phi_{\text{nc}}, v_{\text{nc}}) = \lambda_{\text{nc}}(\phi_{\text{nc}}, v_{\text{nc}})_{L^2(\Omega)} \quad \text{for all } v_{\text{nc}} \in V(\mathcal{T}). \quad (5.9)$$

Assume the $N := \dim(V(\mathcal{T}))$ algebraic eigenvalues $0 < \lambda_{\text{nc}}(1) \leq \lambda_{\text{nc}}(2) \leq \dots \leq \lambda_{\text{nc}}(N) < \infty$ of (5.9) are enumerated in ascending order counting multiplicities. The following Theorem 5.8 refines [CG14a; CG14b] for $n = 2$ and generalizes it for $n \geq 3$ stating for $k = 1, \dots, N$

$$\text{GLB}(k) := \frac{\lambda_{\text{nc}}(k)}{1 + \lambda_{\text{nc}}(k) \kappa_m^2 h_{\max}^{2m}} \leq \lambda_k \quad (5.10)$$

without the unnecessary separation condition $h_{\max}^{2m} \kappa_m^2 < (\sqrt{1 + k^{-1}} - 1)/\sqrt{\lambda_k}$.

Theorem 5.8 (GLB in (5.10)). *For any $k = 1, \dots, N$, the k -th discrete eigenvalue $\lambda_{\text{nc}}(k)$ of (5.1) leads in (5.10) to a guaranteed lower bound $\text{GLB}(k) \leq \lambda_k$ for the k -th eigenvalue λ_k of (5.1).*

Proof. Abbreviate $\lambda = \lambda_k$ from (5.1) and $\lambda_{\text{nc}} = \lambda_{\text{nc}}(k)$ from (5.9). Let $\phi_1, \dots, \phi_k$ denote the first k L^2 -orthonormal eigenfunctions in (5.1).

Case 1. Assume that the interpolations $I\phi_1, \dots, I\phi_k$ are linear dependent. Let $\phi \in \text{span}\{\phi_1, \dots, \phi_k\}$ with $\|\phi\|_{L^2(\Omega)} = 1$ and $I\phi = 0$. (I1) and the min-max principle for (5.1) show

$$1 = \|\phi\|_{L^2(\Omega)}^2 = \|(1 - I)\phi\|_{L^2(\Omega)}^2 \leq \kappa_m^2 h_{\max}^{2m} \|\phi\|_{\text{pw}}^2 = \kappa_m^2 h_{\max}^{2m} \|\phi\|^2 \leq \lambda \kappa_m^2 h_{\max}^{2m}.$$

It follows $\kappa_m^{-2} h_{\max}^{-2m} \leq \lambda$ and $\text{GLB}(k) = \lambda_{\text{nc}} / (1 + \lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m}) \leq \kappa_m^{-2} h_{\max}^{-2m} \leq \lambda$ concludes the proof.

Case 2. Assume $I\phi_1, \dots, I\phi_k$ are linearly independent. In this case the proofs in [CG14a; CG14b] do not need any separation condition to show (5.10) and (I1) allows for space dimension $n \geq 3$. The arguments are added here for convenient reading. If the interpolations $I\phi_1, \dots, I\phi_k$ are linear independent, then $S_k := \text{span}\{I\phi_1, \dots, I\phi_k\} \subset V(\mathcal{T})$ is a k -dimensional subspace of $V(\mathcal{T})$ and the min-max principle for the eigenvalue problem (5.9) assures

$$\lambda_{\text{nc}} \leq \max_{v_{\text{nc}} \in S_k \setminus \{0\}} \frac{\|v_{\text{nc}}\|_{\text{pw}}^2}{\|v_{\text{nc}}\|_{L^2(\Omega)}^2}. \quad (5.11)$$

Let $\phi \in \text{span}\{\phi_1, \dots, \phi_k\}$ with $\|\phi\|_{L^2(\Omega)} = 1$ such that $v_{\text{nc}} := I\phi \in S_k$ is the maximizer in (5.11). The orthogonality from Corollary 5.1.a in the Pythagoras theorem and the min-max principle on the continuous level with $\|\phi\|^2 \leq \lambda$ show

$$\|I\phi\|_{\text{pw}}^2 + \|\phi - I\phi\|_{\text{pw}}^2 = \|\phi\|^2 \leq \lambda. \quad (5.12)$$

To compute a lower bound for $\|I\phi\|_{L^2(\Omega)}^2$, recall $\|\phi\|_{L^2(\Omega)}^2 = 1$ to verify

$$\|I\phi\|_{L^2(\Omega)}^2 = \|\phi - I\phi - \phi\|_{L^2(\Omega)}^2 = 1 + \|\phi - I\phi\|_{L^2(\Omega)}^2 - 2 \int_{\Omega} (\phi - I\phi)\phi \, dx.$$

The Cauchy-Schwarz and Young's inequality show for $0 < t := \frac{\lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m}}{1 + \lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m}} < 1$

$$1 - t + (1 - t^{-1})\|\phi - I\phi\|_{L^2(\Omega)}^2 \leq \|I\phi\|_{L^2(\Omega)}^2.$$

The choice of t with $(1 - t^{-1}) = -(\lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m})^{-1} \leq 0$ and (I1) with $h_{\mathcal{T}} \leq h_{\max}$ lead to the following lower bound for $\|I\phi\|_{L^2(\Omega)}^2$

$$1 - t + (1 - t^{-1})\kappa_m^2 h_{\max}^{2m} \|\phi - I\phi\|_{\text{pw}}^2 = \frac{1}{1 + \lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m}} - \frac{1}{\lambda_{\text{nc}}} \|\phi - I\phi\|_{\text{pw}}^2. \quad (5.13)$$

The combination of (5.11)–(5.13) concludes the proof in that

$$\begin{aligned} \text{GLB}(k) &= \frac{\lambda_{\text{nc}}}{1 + \lambda_{\text{nc}} \kappa_m^2 h_{\max}^{2m}} \leq \lambda_{\text{nc}} \|I\phi\|_{L^2(\Omega)}^2 + \|\phi - I\phi\|_{\text{pw}}^2 \\ &\leq \|I\phi\|_{\text{pw}}^2 + \|\phi - I\phi\|_{\text{pw}}^2 \leq \lambda. \end{aligned}$$

□

Note that Theorem 5.5 and Theorem 5.8 do not require the condition (I2) directly, the orthogonality $a_{\text{pw}}(v - Iv, w_{\text{nc}}) = 0$ for any $v \in V$ and $w_{\text{nc}} \in V(\mathcal{T})$ from Corollary 5.1.a suffices.

5.3 Comparison and general remarks

A few general remarks on the extra-stabilized discrete eigenvalue problem are in order. Subsection 5.3.1 introduces an alternative reduced formulation, Subsection 5.3.2 compares the results with the nonconforming method, and Subsection 5.3.3 compares the method for $m = 1$ with the lowest-order modified HHO method. The following observation are useful throughout the remainder of this chapter.

Remark 5.9 (selected testfunctions). On the one-hand side, test functions of the form $v_h = (v_{\text{pw}}, 0) \in V_h$ with $v_{\text{pw}} \in P_m(\mathcal{T})$ in (5.2) imply for any $v_{\text{pw}} \in P_m(\mathcal{T})$

$$a_h(u_h, v_h) = \kappa_m^{-2} (h_{\mathcal{T}}^{-2m} (u_{\text{pw}} - u_{\text{nc}}), v_{\text{pw}})_{L^2(\Omega)} = \lambda_h b_h(u_h, v_h) = \lambda_h b(u_{\text{pw}}, v_{\text{pw}}).$$

In other words, the discrete solution $(\lambda_h, u_h) \in \mathbb{R}_+ \times V_h$ to (5.2) satisfies $(u_{\text{pw}} - u_{\text{nc}}) = \kappa_m^2 h_{\mathcal{T}}^{2m} \lambda_h u_{\text{pw}}$.

On the other hand, given any nonconforming function $v_{\text{nc}} \in V(\mathcal{T})$, $v_h = (v_{\text{nc}}, v_{\text{nc}}) \in V_h$ is an admissible test function in (5.2). Hence, the solution $(\lambda_h, u_h) \in \mathbb{R}_+ \times V_h$ satisfies

$$a_h(u_h, v_h) = a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = \lambda_h b_h(u_h, v_h) = \lambda_h b(u_{\text{pw}}, v_{\text{nc}}) \quad \text{for any } v_{\text{nc}} \in V(\mathcal{T}).$$

Remark 5.10 (fixed weight). Throughout this chapter, given a fixed $\lambda_h \in \mathbb{R}_+$ with $0 < \lambda_h < \kappa_m^{-2} h_{\text{max}}^{-2m}$, abbreviate

$$\delta := \frac{1}{1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}} - 1 = \frac{\lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}}{1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}} = \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m} (1 + \delta) \in P_0(\mathcal{T}). \quad (5.14)$$

The λ_h -dependence of δ in the notation is suppressed.

5.3.1 Reduced eigenvalue problem

Under the condition $\lambda_h \kappa_m^2 h_{\text{max}}^{2m} < 1$, the algebraic eigenvalue problem (5.2) is equivalent to a reduced form that seeks $(\lambda_h, u_{\text{nc}}) \in \mathbb{R}^+ \times (V(\mathcal{T}) \setminus \{0\})$ with

$$a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = \lambda_h b\left(\frac{u_{\text{nc}}}{1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}}, v_{\text{nc}}\right) \quad \text{for all } v_{\text{nc}} \in V(\mathcal{T}). \quad (5.15)$$

The formulation (5.15) is reduced in that the additional variables u_{pw} and v_{pw} in (5.2) are condensed out; (5.15) is a (simple) rational eigenvalue problem with

the same dimension and sparsity as the nonconforming eigenvalue problem (5.9). A solution $(\lambda_h, u_{\text{nc}})$ to (5.15) is also called an eigenpair and the (geometric) multiplicity ≥ 1 is the dimension of the eigenspace of all $u_{\text{nc}} \in V(\mathcal{T}) \setminus \{0\}$ so that $(\lambda_h, u_{\text{nc}})$ solves (5.15).

Proposition 5.11 (equivalence). (a) *If the eigenpair (λ_h, u_h) of (5.2) satisfies $\lambda_h < \kappa_m^{-2} h_{\max}^{-2m}$ with $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h \setminus \{0\}$, then $(\lambda_h, u_{\text{nc}})$ solves (5.15) and $u_{\text{pw}} = (1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m})^{-1} u_{\text{nc}}$.*

(b) *If $(\lambda_h, u_{\text{nc}})$ is a solution to (5.15) with $0 < \lambda_h < \kappa_m^{-2} h_{\max}^{-2m}$, then λ_h , u_{nc} , and $u_{\text{pw}} := (1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m})^{-1} u_{\text{nc}}$ in $u_h = (u_{\text{pw}}, u_{\text{nc}})$ form an eigenpair (λ_h, u_h) of (5.2).*

(c) *The set of eigenvalues of (5.2) in the open interval $(0, \kappa_m^{-2} h_{\max}^{-2m})$ is equal to the set of solutions λ_h to (5.15) in $(0, \kappa_m^{-2} h_{\max}^{-2m})$ counting (geometric) multiplicities.*

Proof of (a). Suppose that $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ is an eigenpair of (5.2). Given this λ_h with $0 < \lambda_h < \kappa_m^{-2} h_{\max}^{-2m}$ abbreviate δ as in (5.14) in Remark 5.10. Observe that test function $v_h = (v_{\text{pw}}, 0) \in V_h$ with $v_{\text{pw}} \in P_m(\mathcal{T})$ in Remark 5.9 imply $\kappa_m^{-2} h_{\mathcal{T}}^{-2m} (u_{\text{pw}} - u_{\text{nc}}) = \lambda_h u_{\text{pw}}$. This is equivalent to $u_{\text{pw}} = (1 + \delta) u_{\text{nc}}$. Moreover Remark 5.9 shows that test functions of the form $v_h = (v_{\text{nc}}, v_{\text{nc}}) \in V(\mathcal{T}) \times V(\mathcal{T}) \subset V_h$ in (5.2) lead to

$$a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = \lambda_h (u_{\text{pw}}, v_{\text{nc}})_{L^2(\Omega)} = \lambda_h ((1 + \delta) u_{\text{nc}}, v_{\text{nc}})_{L^2(\Omega)}.$$

Since this holds for all $v_{\text{nc}} \in V(\mathcal{T})$, $(\lambda_h, u_{\text{nc}})$ is a solution to (5.15). $\square$

Proof of (b). Suppose $(\lambda_h, u_{\text{nc}}) \in \mathbb{R}^+ \times V(\mathcal{T})$ is a solution to (5.15) with $0 < \lambda_h < \kappa_m^{-2} h_{\max}^{-2m}$, use the abbreviation δ in (5.14). Then $u_{\text{pw}} := (1 + \delta) u_{\text{nc}}$ and $u_h := (u_{\text{pw}}, u_{\text{nc}}) \in V_h$ in (5.15) lead to

$$a_h(u_h, v_h) = \lambda_h ((1 + \delta) u_{\text{nc}}, v_{\text{nc}})_{L^2(\Omega)} + (\kappa_m^{-2} h_{\mathcal{T}}^{-2m} \delta u_{\text{nc}}, v_{\text{pw}} - v_{\text{nc}})_{L^2(\Omega)}$$

for all $v_h = (v_{\text{pw}}, v_{\text{nc}}) \in V_h$. Recall $\kappa_m^{-2} h_{\mathcal{T}}^{-2m} \delta = \lambda_h (1 + \delta)$ from (5.14) to verify

$$a_h(u_h, v_h) = \lambda_h ((1 + \delta) u_{\text{nc}}, v_{\text{pw}})_{L^2(\Omega)} = \lambda_h b_h(u_h, v_h).$$

Hence, (λ_h, u_h) is an eigenpair of (5.2). $\square$

Proof of (c). The combination of (a)–(b) proves the equality of the solution sets for λ_h in (5.2) resp. (5.15) in the open interval $(0, \kappa_m^{-2} h_{\max}^{-2m})$. It remains to prove equality of the multiplicities. Let p resp. q denote the multiplicities of the eigenvalue λ_h in (5.2) resp. (5.15). The point is that $u_{\text{nc},1}, \dots, u_{\text{nc},\gamma} \in V(\mathcal{T})$ are linear independent if and only if $((1 + \delta) u_{\text{nc},1}, u_{\text{nc},1}), \dots, ((1 + \delta) u_{\text{nc},\gamma}, u_{\text{nc},\gamma})$ are linear independent in V_h for the fixed δ from (5.14) for λ_h with $0 < \lambda_h < \kappa_m^{-2} h_{\max}^{-2m}$. This and (a) resp. (b) imply $p \leq q$ resp. $q \leq p$. Consequently $p = q$ concludes the proof. $\square$

5.3.2 Comparison with the nonconforming method

Recall the standard nonconforming discretization of the eigenvalue problem in (5.9) and the post-processed guaranteed lower bound in (5.10). For $k \leq N := \dim(V(\mathcal{T}))$, the following Lemma 5.12 allows the placement of the discrete eigenvalue $\lambda_h(k)$ from (5.2) in $\text{GLB}(k) \leq \lambda_h(k) \leq \lambda_{\text{nc}}(k)$ between $\text{GLB}(k)$ from (5.10) and the nonconforming eigenvalue $\lambda_{\text{nc}}(k)$ from (5.9).

Lemma 5.12 (comparison I). *For any $k = 1, \dots, N$, the k -th algebraic eigenvalues $\lambda_h(k)$ of (5.15) and $\lambda_{\text{nc}}(k)$ of (5.9) satisfy $\lambda_h(k) \leq \lambda_{\text{nc}}(k)$. If $\lambda_h(k)$ satisfies $\lambda_h(k) \kappa_m^2 h_{\max}^{2m} < 1$, then $\text{GLB}(k) \leq \lambda_h(k)$ holds. For a uniform triangulation $\mathcal{T}$ with $h_{\max} = h_{\mathcal{T}}$ a.e. in Ω holds equality $\text{GLB}(k) = \lambda_h(k)$.*

Proof. The first result generalizes [CZZ20, Thm. 6.2] and is comparable to Corollary 4.18. As in Remark 5.9 the choice $v_h = (v_{\text{nc}}, v_{\text{nc}}) \in V_h$ in (5.2) implies

$$a_h(v_h, v_h) = a_{\text{pw}}(v_{\text{nc}}, v_{\text{nc}}) \quad \text{and} \quad b_h(v_h, v_h) = b(v_{\text{nc}}, v_{\text{nc}}) \quad \text{for any } v_{\text{nc}} \in V(\mathcal{T}).$$

This and the min-max principle for (5.2) and (5.9) conclude the proof of the first claim $\lambda_h(k) \leq \lambda_{\text{nc}}(k)$ as in Corollary 4.18.

The 2D version of the second claim for $m = 1$ is Theorem 6.4 in [CZZ20]. The proof therein can be adapted as follows. For $1 \leq j \leq k$ let $(\lambda_h(j), \phi_h(j)) \in \mathbb{R}^+ \times V_h$ denote the first k b_h -orthonormal eigenpairs of (5.2) with $\phi_h(j) = (\phi_{\text{pw}}(j), \phi_{\text{nc}}(j))$. Recall from Remark 5.9 that for any $v_{\text{nc}} \in V(\mathcal{T})$ and any $1 \leq j \leq k$

$$a_{\text{pw}}(\phi_{\text{nc}}(j), v_{\text{nc}}) = \lambda_h(j) b(\phi_{\text{pw}}(j), v_{\text{nc}}) \quad \text{and} \quad \phi_{\text{nc}}(j) - \phi_{\text{pw}}(j) = \lambda_h(j) \kappa_m^2 h_{\mathcal{T}}^{2m} \phi_{\text{nc}}(j). \quad (5.16)$$

For any $\xi = (\xi_1, \dots, \xi_k) \in \mathbb{R}^k$ with $|\xi|^2 = \sum_{j=1}^k \xi_j^2 = 1$ set

$$v_{\text{nc}} := \sum_{j=1}^k \xi_j \phi_{\text{nc}}(j), \quad v_{\text{pw}} := \sum_{j=1}^k \xi_j \phi_{\text{pw}}(j), \quad \text{and} \quad w_{\text{pw}} := \sum_{j=1}^k \xi_j \lambda_h(j) \phi_{\text{pw}}(j).$$

The b_h -orthonormality of the eigenfunctions in combination with (5.16) proves

$$a_{\text{pw}}(v_{\text{nc}}, v_{\text{nc}}) = b(w_{\text{pw}}, v_{\text{nc}}) \quad \text{and} \quad v_{\text{pw}} - v_{\text{nc}} = \kappa_m^2 h_{\mathcal{T}}^{2m} w_{\text{pw}}. \quad (5.17)$$

Abbreviate $\zeta := b(w_{\text{pw}}, v_{\text{pw}})$ and $\chi := \kappa_m^2 \|h_{\mathcal{T}}^m w_{\text{pw}}\|_{L^2(\Omega)}^2$. Then (5.17) leads to

$$\|v_{\text{nc}}\|_{\text{pw}}^2 = b(w_{\text{pw}}, v_{\text{nc}}) = b(w_{\text{pw}}, v_{\text{pw}}) - b(w_{\text{pw}}, v_{\text{pw}} - v_{\text{nc}}) = \zeta - \chi, \quad (5.18)$$

$$\begin{aligned} \|v_{\text{nc}}\|_{L^2(\Omega)} &= \|v_{\text{pw}} - \kappa_m^2 h_{\mathcal{T}}^{2m} w_{\text{pw}}\|_{L^2(\Omega)} \geq 1 - \kappa_m^2 h_{\max}^m \|h_{\mathcal{T}}^m w_{\text{pw}}\|_{L^2(\Omega)} \\ &= 1 - \kappa_m h_{\max}^m \sqrt{\chi} \end{aligned} \quad (5.19)$$

with $\|v_{\text{pw}}\|_{L^2(\Omega)} = 1$, $\|w_{\text{pw}}\|_{L^2(\Omega)}^2 = \sum_{j=1}^k \xi_j^2 \lambda_h(j)^2 \leq \lambda_h(k)^2$, and $h_{\mathcal{T}} \leq h_{\max}$ in (5.19). Since (5.19) and $\lambda_h(k) \kappa_m^2 h_{\max}^{2m} < 1$ imply $\|v_{\text{nc}}\|_{L^2(\Omega)} \geq 1 - \kappa_m^2 h_{\max}^{2m} \lambda_h(k) > 0$, the subspace $U_{\text{nc}} := \text{span}\{\phi_{\text{nc}}(1), \dots, \phi_{\text{nc}}(k)\} \subset V(\mathcal{T})$ has dimension k . In the

following, suppose that the coefficients $\xi \in \mathbb{R}^j$ with $|\xi| = 1$ maximize the Rayleigh-Quotient $a_{\text{pw}}(v_{\text{nc}}, v_{\text{nc}})/b(v_{\text{nc}}, v_{\text{nc}})$. Then the min-max principle for $\lambda_{\text{nc}}(k)$ of (5.9) and (5.18)–(5.19) imply that

$$\lambda_{\text{nc}}(k) \leq \frac{\|v_{\text{nc}}\|_{\text{pw}}^2}{\|v_{\text{nc}}\|_{L^2(\Omega)}^2} \leq \frac{\zeta - \chi}{(1 - \kappa_m h_{\text{max}}^m \sqrt{\chi})^2}.$$

The function $f(x) := x/(1 + \kappa_m^2 h_{\text{max}}^{2m} x)$ is monoton increasing for positive input $x > 0$. Therefore,

$$\begin{aligned} \text{GLB}(k) &= f(\lambda_{\text{nc}}(k)) \leq f\left(\frac{\zeta - \chi}{(1 - \kappa_m h_{\text{max}}^m \sqrt{\chi})^2}\right) = \frac{\zeta - \chi}{1 - 2\kappa_m h_{\text{max}}^m \sqrt{\chi} + \kappa_m^2 h_{\text{max}}^{2m} \zeta} \\ &\leq \frac{\zeta - \chi}{1 - \chi/\zeta} = \zeta. \end{aligned}$$

The second to last inequality follows with $2\kappa_m h_{\text{max}}^m \sqrt{\chi} \leq \chi/\zeta + \kappa_m^2 h_{\text{max}}^{2m} \zeta$ in the denominator. The estimate $\zeta = b(w_{\text{pw}}, v_{\text{pw}}) = \sum_{j=1}^k \xi_j^2 \lambda_h(j) \leq \lambda_h(k)$ concludes the proof of $\text{GLB}(k) \leq \lambda_h(k)$ for $\lambda_h(k) \kappa_m^2 h_{\text{max}}^{2m} < 1$.

On a uniform mesh with $h_{\text{max}} = h_{\mathcal{T}}$ a.e. in Ω the scaling on the right-hand side of the reduced problem (5.15) is constant, thus (5.15) and (5.9) are equivalent with $\lambda_h(k) = \lambda_{\text{nc}}(k)/(1 + \kappa_m^2 \lambda_{\text{nc}}(k) h_{\text{max}}^{2m}) = \text{GLB}(k)$. $\square$

Remark 5.13 (verification of the mesh-size condition). Lemma 5.12 and $\lambda_{\text{nc}}(k) < \kappa_m^{-2} h_{\text{max}}^{-2m}$ guarantee that the discrete eigenvalue $\lambda_h(k)$ satisfies $\lambda_h(k) < \kappa_m^{-2} h_{\text{max}}^{-2m}$. This is an a priori test sufficient for the applicability of Proposition 5.11 and Theorem 5.5. $\square$

5.3.3 Comparison with the lowest-order modified HHO method

The lowest-order version of the modified HHO method (4.2) in Section 4.5 features the discrete space $V_{\text{HHO}} := P_1(\mathcal{T}) \times P_0(\mathcal{F}(\Omega))$. Recall, the extension operator $I_{\mathcal{F}} : P_0(\mathcal{F}(\Omega)) \rightarrow \text{CR}_0^1(\mathcal{T})$ maps $v_{\mathcal{F}} = (v_F)_{F \in \mathcal{F}} \in P_0(\mathcal{F}(\Omega))$ bijectively onto $I_{\mathcal{F}}(v_{\mathcal{F}}) := \sum_{F \in \mathcal{F}} v_F \psi_F \in \text{CR}_0^1(\mathcal{T})$ with $I_{\text{CR}} = I_{\mathcal{F}} \circ \Pi_{\mathcal{F},0}$. In other words, the method features essentially the same space as the extra-stabilized Crouzeix-Raviart method (5.2). Hence, a comparison is in order. The notation in (4.2) and (5.2) is easily confused. Throughout this subsection the index ,es‘ replaces ,h‘ for the extra-stabilized method, e.g., λ_{es} denotes the discrete eigenvalue in (5.2), while the index ,HHO‘ is associated with the lowest-order HHO method, e.g., λ_{HHO} denotes the discrete eigenvalue in (4.2) with $k = 0$. The comparison proofs are rather similar and comparable to Theorem 4.19 and Lemma 5.12, they are included for completeness.

Theorem 5.14. *For $j = 1, \dots, M := \dim(P_1(\mathcal{T}))$, let $\lambda_{\text{es}}(j)$ denote the j -th discrete eigenvalue in the extra-stabilized Crouzeix-Raviart method (5.2) and $\lambda_{\text{HHO}}(j)$*

the j -th discrete eigenvalue of the lowest-order modified HHO method (4.2) with parameters $0 < \alpha < 1$ and $\beta > 0$.

(a) Given a regular triangulation $\mathcal{T}$ of $\Omega \subset \mathbb{R}^2$ into right-isosceles triangles, then $\tau := 72\kappa_1^2 \max\{(1-\alpha), \beta\} \leq 1$ implies $\lambda_{\text{HHO}}(j) \leq \lambda_{\text{es}}(j)$.

(b) Given any $\Omega \subset \mathbb{R}^n$, then $\theta := n^2/(n+1)^2 \max\{1/\beta, 1/(n^2(1-\alpha))\} \leq \kappa_1^2$ implies $\lambda_{\text{es}}(j) \leq \lambda_{\text{HHO}}(j)$.

The restriction to 2D and right-isosceles triangles in Theorem 5.14.a stems from Remark 4.20, which proves $|\Pi_0| \bullet -\text{mid}(\mathcal{T})|^2 = h_{\mathcal{T}}^2/18$ for these triangulations.

Proof of (a). The set up in this proof is comparable to Lemma 5.12. For $1 \leq j \leq k$ let $(\lambda_{\text{es}}(j), \phi_{\text{es}}(j)) \in \mathbb{R}^+ \times V_{\text{es}}$ denote the first k b_{es} -orthonormal eigenpairs of (5.2) with $\phi_{\text{es}}(j) = (\phi_{\text{pw}}(j), \phi_{\text{CR}}(j))$. For any $\xi = (\xi_1, \dots, \xi_j) \in \mathbb{R}^j$ with $|\xi|^2 = \sum_{\ell=1}^j \xi_{\ell}^2 = 1$ set

$$v_{\text{CR}} := \sum_{\ell=1}^j \xi_{\ell} \phi_{\text{CR}}(\ell), \quad v_{\text{pw}} := \sum_{\ell=1}^j \xi_{\ell} \phi_{\text{pw}}(\ell), \quad \text{and} \quad w_{\text{pw}} := \sum_{\ell=1}^j \xi_{\ell} \lambda_{\text{h}}(\ell) \phi_{\text{pw}}(\ell)$$

with $\|v_{\text{pw}}\|_{L^2(\Omega)} = 1$. Remark 5.9 and the b_{es} -orthonormal basis imply as in (5.17)

$$a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) = b(w_{\text{pw}}, v_{\text{CR}}) \quad \text{and} \quad v_{\text{pw}} - v_{\text{CR}} = \kappa_1^2 h_{\mathcal{T}}^2 w_{\text{pw}}. \quad (5.20)$$

Abbreviate $\zeta := b(w_{\text{pw}}, v_{\text{pw}}) \leq \|w_{\text{pw}}\|_{L^2(\Omega)} \leq \lambda_{\text{es}}(j)$ and $\chi := \kappa_1^2 \|h_{\mathcal{T}} w_{\text{pw}}\|_{L^2(\Omega)}^2 = b(v_{\text{pw}} - v_{\text{CR}}, w_{\text{pw}})$. The arguments that lead to (5.18) above, show

$$\|v_{\text{CR}}\|_{\text{pw}}^2 = b(w_{\text{pw}}, v_{\text{CR}}) = \zeta - \chi.$$

Since Remark 4.20 implies that $s(\mathcal{T}) = 1/4 \Pi_0(|\bullet - \text{mid}(\mathcal{T})|^2) = h_{\mathcal{T}}^2/72$ for the given triangulation $\mathcal{T}$, (5.20) proves

$$\|s(\mathcal{T})^{-1/2} \Pi_0(v_{\text{pw}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 = 72 \|h_{\mathcal{T}}^{-1} \Pi_0(h_{\mathcal{T}}^2 \kappa_1^2 w_{\text{pw}})\|_{L^2(\Omega)}^2 = 72 \kappa_1^4 \|h_{\mathcal{T}} \Pi_0 w_{\text{pw}}\|_{L^2(\Omega)}^2.$$

The projection properties of Π_0 , (5.20), and the inverse estimate in $P_1(\mathcal{T})$ with constant $c_{\text{inv}} = \sqrt{72}$ from [CH18a, Lemma 4.10] lead to

$$\kappa_1^{-4} \|\nabla_{\text{pw}}(v_{\text{pw}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 = \|h_{\mathcal{T}}^2 \nabla_{\text{pw}}(1 - \Pi_0) w_{\text{pw}}\|_{L^2(\Omega)}^2 \leq c_{\text{inv}}^2 \|h_{\mathcal{T}}(1 - \Pi_0) w_{\text{pw}}\|_{L^2(\Omega)}^2.$$

Set $v_{\text{HHO}} := (v_{\text{pw}}, \Pi_{0,\mathcal{F}} v_{\text{CR}}) \in V_{\text{HHO}}$. The last three displayed estimates bound the lowest-order energy scalar product from Proposition 4.17.a as follows

$$\begin{aligned} & a_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}}) \\ &= a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) + (1-\alpha) \|s(\mathcal{T})^{-1/2} \Pi_0(v_{\text{pw}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 + \beta \|\nabla_{\text{pw}}(v_{\text{pw}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 \\ &\leq \zeta - \chi + 72\kappa_1^4 \left((1-\alpha) \|h_{\mathcal{T}} \Pi_0 w_{\text{pw}}\|_{L^2(\Omega)}^2 + \beta \|h_{\mathcal{T}}(1 - \Pi_0) w_{\text{pw}}\|_{L^2(\Omega)}^2 \right) \\ &\leq \zeta - \chi + 72\kappa_1^4 \max\{(1-\alpha), \beta\} \|h_{\mathcal{T}} w_{\text{pw}}\|_{L^2(\Omega)}^2 \\ &= \zeta + (72\kappa_1^2 \max\{(1-\alpha), \beta\} - 1) \chi = \chi + (\tau - 1) \zeta. \end{aligned}$$

Moreover, $b_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}}) = \|v_{\text{pw}}\|_{L^2(\Omega)}^2 = 1$ holds, if $\xi \in \mathbb{R}^j$ has norm $|\xi| = 1$. Note that $V_j := \text{span}\{(\phi_{\text{pw}}(1), \Pi_{0,\mathcal{F}}\phi_{\text{CR}}(1)), \dots, (\phi_{\text{pw}}(j), \Pi_{0,\mathcal{F}}\phi_{\text{CR}}(j))\} \subset V_{\text{HHO}}$ is an j -dimensional subspace. In the following, suppose that the coefficients $\xi \in \mathbb{R}^j$ with $|\xi| = 1$ maximize the Rayleigh-Quotient $a_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}})/b_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}})$. Then the min-max principle for (4.2) and the condition $\tau \leq 1$ show

$$\lambda_{\text{HHO}}(j) \leq \frac{a_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}})}{b_{\text{HHO}}(v_{\text{HHO}}, v_{\text{HHO}})} \leq \chi + (\tau - 1)\zeta \leq \chi \leq \lambda_{\text{es}}(j).$$

This concludes the proof. $\square$

The condition in Theorem 5.14.a implies that the parameter choice $1 - 1/(72\kappa_1^2) \leq \alpha$ and $\beta \leq 1/(72\kappa_1^2)$ in the lowest-order modified HHO method does not lead to a better guaranteed lower bound than the extra-stabilized method (for a triangulation $\mathcal{T}$ of $\Omega \subset \mathbb{R}^2$ into right-isosceles triangles).

Proof of (b). The setting is similar to Theorem 4.19. Let $(\lambda_{\text{HHO}}(\ell), \phi_{\text{HHO}}(\ell))$ for $1 \leq \ell \leq j$ denote the first j eigenpairs of (4.2), where $\phi_{\text{HHO}}(\ell) = (\phi_{\mathcal{T}}(\ell), \phi_{\mathcal{F}}(\ell)) \in V_{\text{HHO}}$ with $(\phi_{\mathcal{T}}(\ell), \phi_{\mathcal{T}}(k))_{L^2(\Omega)} = \delta_{\ell k}$ for $\ell, k = 1, \dots, j$. For any coefficient vector $\xi = (\xi_1, \dots, \xi_j) \in \mathbb{R}^j$ with Euclid norm $|\xi| = 1$ set as in (4.48)

$$v_{\mathcal{F}} := \sum_{\ell=1}^j \xi_{\ell} \phi_{\mathcal{F}}(\ell), \quad v_{\text{CR}} := I_{\mathcal{F}} v_{\mathcal{F}}, \quad v_{\mathcal{T}} := \sum_{\ell=1}^j \xi_{\ell} \phi_{\mathcal{T}}(\ell), \quad \text{and} \quad w_{\mathcal{T}} := \sum_{\ell=1}^j \xi_{\ell} \lambda_h(\ell) \phi_{\mathcal{T}}(\ell).$$

Abbreviate $\mu := (v_{\mathcal{T}}, w_{\mathcal{T}})_{L^2(\Omega)} \leq \lambda_{\text{HHO}}(j)$ and conclude as in (4.52)

$$\rho := (v_{\mathcal{T}} - v_{\text{CR}}, w_{\mathcal{T}})_{L^2(\Omega)} = (1 - \alpha)^{-1} \|\Pi_0 w_{\mathcal{T}}\|_{s(\mathcal{T})}^2 + \beta^{-1} \|\nabla_{\text{pw}} w_{\mathcal{T}}\|_{S(\mathcal{T})^2}^2 > 0.$$

Hence, $a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) = \mu - \rho$ as in (4.53) and

$$\|h_{\mathcal{T}}^{-1}(v_{\mathcal{T}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 \leq \frac{1}{(1 - \alpha)^2(n + 1)^2} \|\Pi_0 w_{\mathcal{T}}\|_{s(\mathcal{T})}^2 + \frac{n^2}{\beta^2(n + 1)^2} \|\nabla_{\text{pw}} w_{\mathcal{T}}\|_{S(\mathcal{T})^2}^2$$

as in (4.54), i.e., $\|h_{\mathcal{T}}^{-1}(v_{\mathcal{T}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 \leq \theta\rho$. This implies for $v_{\text{es}} := (v_{\mathcal{T}}, v_{\text{CR}}) \in V_{\text{es}}$

$$a_{\text{es}}(v_{\text{es}}, v_{\text{es}}) = a_{\text{pw}}(v_{\text{CR}}, v_{\text{CR}}) + \kappa_1^{-2} \|h_{\mathcal{T}}^{-1}(v_{\mathcal{T}} - v_{\text{CR}})\|_{L^2(\Omega)}^2 \leq \mu + (\kappa_1^{-2}\theta - 1)\rho.$$

On the other hand, $b_{\text{es}}(v_{\text{es}}, v_{\text{es}}) = \|v_{\mathcal{T}}\|_{L^2(\Omega)}^2 = 1$ holds for $\xi \in \mathbb{R}^j$ with $|\xi| = 1$. Note that $V_j := \text{span}\{(\phi_{\mathcal{T}}(1), I_{\mathcal{F}}\phi_{\mathcal{F}}(1)), \dots, (\phi_{\mathcal{T}}(j), I_{\mathcal{F}}\phi_{\mathcal{F}}(j))\} \subset V_{\text{es}}$ is an j -dimensional subspace. In the following, suppose that the coefficients $\xi \in \mathbb{R}^j$ with $|\xi| = 1$ maximize the Rayleigh-Quotient $a_{\text{es}}(v_{\text{es}}, v_{\text{es}})/b_{\text{es}}(v_{\text{es}}, v_{\text{es}})$. Then the min-max principle for (5.2) and the condition $\theta \leq \kappa_1^2$ show

$$\lambda_{\text{es}}(j) \leq \frac{a_{\text{es}}(v_{\text{es}}, v_{\text{es}})}{b_{\text{es}}(v_{\text{es}}, v_{\text{es}})} \leq \mu + (\kappa_1^{-2}\theta - 1)\rho \leq \mu \leq \lambda_{\text{HHO}}(j).$$

This concludes the proof. $\square$

According to Remark 4.20, in the experiments in Chapter 7 with triangulation into right-isosceles triangles the condition reads $\theta = \frac{1}{18} \max\{1/\beta, (1/(4-4\alpha))\} \leq \kappa_1^2$. Hence, Theorem 5.14.b guarantees that for parameter $\alpha \leq 1 - 1/(72\kappa_1^2)$ and $1/(18\kappa_1^2) \leq \beta$ the lowest-order HHO approximation is not worse than the guaranteed lower bound from the extra-stabilized method. This condition is sufficient not necessary.

5.4 A priori convergence analysis

This section contains the a priori convergence analysis for the extra-stabilized method in the discrete eigenvalue problem (5.1). To utilize known results for the convergence analysis of the extra-stabilized eigenvalue problem (5.2), a reduction to only one solution component is helpful. First the Schur complement ansatz leads to an equivalent eigenvalue problem with eigenpair $(\lambda_h(k), u_{\text{pw}}) \in \mathbb{R}^+ \times P_m(\mathcal{T})$ as in Subsection 4.2.3 and allows the application of the Babuška-Osborn theory in Theorem 5.18. These results concede best approximation properties for the nonconforming solution component $u_{\text{nc}} \in V(\mathcal{T})$ in Theorem 5.20 based on the rational problem (5.15). The latter are fundamental for the a posteriori error analysis.

The first subsection applies the conforming companion J in a convergence analysis for the associated source problem. This and the Babuška-Osborn theory [BO91] prove a priori convergence for the discrete eigenvalue problem (5.2) and the standard nonconforming eigenvalue problem (5.9) in Subsection 5.4.2. Of course there are results available for the standard nonconforming problems. However, e.g., for the Morley example ($m = 2$) the paper [YLB16] for $n \geq 2$ follows [Ran79] for $n = 2$ and utilizes the trace inequality for second order derivatives $\partial^\alpha u / \partial x_\alpha$ for $|\alpha| = 2$ under the regularity assumption $u \in W^{3,p}(\Omega)$ for $4/3 < p \leq 2$. Those terms arise in an integration by parts in the classical a priori error analysis of the Morley FEM. This section circumvents this by using the companion operator J following [CGS13; Gal15b; CN21]. This works for any nonconforming method with (I1)–(I2) and an appropriate conforming companion that satisfies (J1)–(J2) and allows results for a general $u \in H^{m+\sigma}(\Omega)$ even for small σ with $0 < \sigma \leq 1$. The Babuška-Osborn theory leads to convergence results for the discrete eigenvalue λ_h and the eigenfunction component $u_{\text{pw}} \in P_m(\mathcal{T})$ in Theorem 5.18 independent of the multiplicity of the continuous eigenvalue λ in Subsection 5.4.2.

The second part, Subsection 5.4.3, concentrates on the nonconforming component $u_{\text{nc}} \in V(\mathcal{T})$ of the discrete eigenfunction to prepare the a posteriori error analysis in Section 5.5. The analysis in Subsection 5.4.3 adopts techniques from [CGS15] and utilizes the reduced problem (5.15) and an intermediate eigenvalue problem (5.33). This analysis is so far restricted to the case of a simple continuous eigenvalue.

Throughout the remainder of this chapter, assume without loss of generality that the maximal mesh-size $h_{\text{max}} \leq 1$ is smaller or equal one.

5.4.1 Convergence analysis for the source problem

Throughout this subsection, given $f \in L^2(\Omega)$, let $u \in V := H_0^m(\Omega)$ solve the source problem

$$a(u, v) = (f, v)_{L^2(\Omega)} \quad \text{for all } v \in V \quad (5.21)$$

and let $u_{\text{nc}} \in V(\mathcal{T})$ denote the discrete solution to the nonconforming source problem

$$a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = (f, v_{\text{nc}})_{L^2(\Omega)} \quad \text{for all } v_{\text{nc}} \in V(\mathcal{T}). \quad (5.22)$$

Let $\sigma := \min\{1, \sigma_{\text{reg}}\}$ denote the minimum of one for the approximation property and the index of elliptic regularity $\sigma_{\text{reg}} > 0$ for the source problem of the m -Laplacian $(-1)^m \Delta^m$ in $H_0^m(\Omega)$: Given any right-hand side $f \in L^2(\Omega)$, the weak solution $u \in V$ to $(-1)^m \Delta^m u = f$ satisfies

$$u \in H^{m+\sigma}(\Omega) \text{ and } \|u\|_{H^{m+\sigma}(\Omega)} \leq C(\sigma) \|f\|_{L^2(\Omega)}. \quad (5.23)$$

(This is well-established for $m = 1$ [Neč12; GT83; Dau88; Gri92; Agm10] and $m = 2$ in 2D [BR80] with $\sigma_{\text{reg}} > 1/2$ and otherwise a hypothesis throughout this work.)

Recall the m -th order oscillation $\text{osc}_m(f, \mathcal{T}) := \|h_{\mathcal{T}}^m(1 - \Pi_m)f\|_{L^2(\Omega)}$ of $f \in L^2(\Omega)$. The following volume efficiency estimate is applied throughout the analysis.

Lemma 5.15 (efficiency of the volume contribution). *There exists a constant $C_1 > 0$, such that the solution $u \in V$ to the continuous source problem (5.21) with right-hand side $f \in L^2(\Omega)$ satisfies*

$$C_1^{-1} \|h_{\mathcal{T}}^m \Pi_m f\|_{L^2(\Omega)} \leq \|u - Iu\|_{\text{pw}} + \text{osc}_m(f, \mathcal{T}). \quad (5.24)$$

Proof. The proof of the efficiency estimate of the volume residual is based on Verfürth's bubble-function methodology [Ver13], comparable to, e.g., [VNS07, Thm. 2], [Gal15a, Prop. 3.1], and given here for completeness.

Recall $\varphi_z \in S^1(\mathcal{T}) := P_1(\mathcal{T}) \cap C(\Omega)$ denotes the nodal basis function associated with the vertex $z \in \mathcal{V}$. For any $T \in \mathcal{T}$, let $b_T := (n+1)^{m(n+1)} \prod_{z \in \mathcal{V}(T)} \varphi_z^m \in P_{m(n+1)}(T) \cap W_0^{m, \infty}(T) \subset V$ denote the volume-bubble-function with $\text{supp}(b_T) = T$ and $\|b_T\|_{\infty} = 1$. An inverse estimate $\|p\|_{L^2(T)} \leq c_b \|p\|_{b_T}$ for any polynomial $p \in P_m(T)$ [Ver13, §3.6] leads to

$$c_b^{-2} \|\Pi_m f\|_{L^2(T)}^2 \leq \|\Pi_m f\|_{b_T}^2 = (\Pi_m f, f)_{b_T} - (\Pi_m f, f - \Pi_m f)_{b_T}. \quad (5.25)$$

The Cauchy-Schwarz inequality and $\|b_T\|_{\infty} = 1$ show $(\Pi_m f, f - \Pi_m f)_{b_T} \leq \|\Pi_m f\|_{L^2(T)} \|f - \Pi_m f\|_{L^2(T)}$. An integration by parts proves $\int_T D^m(b_T \Pi_m f) dx = 0$ since $b_T \Pi_m f \in H_0^m(T)$, i.e., $D^m(b_T \Pi_m f)$ is L^2 -orthogonal to $P_0(T)$. Recall that $u \in V$ solves the source problem (5.21) with right-hand side $f \in L^2(\Omega)$ and the

support of $b_T \Pi_m f \in V$ is T . This, $D_{\text{pw}}^m Iu \in P_0(\mathcal{T})$, and the Cauchy-Schwarz inequality result in

$$\begin{aligned} b(f, b_T \Pi_m f) &= a(u, b_T \Pi_m f) = (D^m u, D^m (b_T \Pi_m f))_{L^2(T)} \\ &\leq |u - Iu|_{H^m(T)} |b_T \Pi_m f|_{H^m(T)}. \end{aligned}$$

In combination with an inverse estimate for polynomials in $P_{m(n+2)}(T)$ with constant c_{inv} and $\|b_T\|_\infty = 1$, this bounds the remaining term in (5.25) via $b(f, b_T \Pi_m f) \leq c_{\text{inv}} h_T^{-m} |u - Iu|_{H^m(T)} \|\Pi_m f\|_{L^2(T)}$. It follows

$$c_b^{-2} h_T^m \|\Pi_m f\|_{L^2(T)} \leq h_T^m \|f - \Pi_m f\|_{L^2(T)} + c_{\text{inv}} |u - Iu|_{H^m(T)}.$$

The sum over all $T \in \mathcal{T}$ concludes the proof with constant $C_1 = c_b^2 \max\{1, c_{\text{inv}}\}$. $\square$

This allows the proof of Lemma 5.16 below.

Lemma 5.16 (error estimate in $V(\mathcal{T})$). *There exist constants $C_2, C_3 \approx 1$ (that exclusively depend on $\mathbb{T}$) such that given any $f \in L^2(\Omega)$, the exact solution $u \in H^{m+\sigma}(\Omega) \cap V$ to (5.21) and the discrete solution $u_{\text{nc}} \in V(\mathcal{T})$ to (5.22) for $\mathcal{T} \in \mathbb{T}$ with maximal mesh-size h_{max} satisfy (a)–(b).*

(a) *If $u \in H^{m+s}(\Omega)$ for some s with $\sigma \leq s \leq 1$, then*

$$\|u - u_{\text{nc}}\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_2 \left(h_{\text{max}}^s \|u\|_{H^{m+s}(\Omega)} + \text{osc}_m(f, \mathcal{T}) \right).$$

(b) *Given any eigenvalue λ of (5.1) with eigenspace $E(\lambda) \subset H^{m+t}(\Omega) \cap V$ for some t with $\sigma \leq t \leq 1$, suppose $f, g \in E(\lambda)$. Then*

$$|(u - u_{\text{nc}}, g)_{L^2(\Omega)}| \leq C_3 (\lambda^{-1} + h_{\text{max}}^{2m})^2 h_{\text{max}}^{2t} \|f\|_{H^{m+t}(\Omega)} \|g\|_{H^{m+t}(\Omega)}.$$

Proof of (a). Abbreviate $e := u - u_{\text{nc}}$ and $\hat{e} := J(Iu - u_{\text{nc}}) \in V$. Observe,

$$\|e\|_{\text{pw}}^2 = a_{\text{pw}}(u, e - \hat{e}) + a(u, \hat{e}) - a_{\text{pw}}(u_{\text{nc}}, e).$$

On the one hand, the split $e - \hat{e} = (1 - I)u + (J - 1)u_{\text{nc}} + (1 - J)Iu$, and Corollary 5.1.c, and Corollary 5.2.b show

$$\begin{aligned} a_{\text{pw}}(u, e - \hat{e}) &= a_{\text{pw}}(u, (1 - I)u) - a_{\text{pw}}(u, (1 - J)u_{\text{nc}}) + a_{\text{pw}}(u, (1 - J)Iu) \\ &\leq (1 + 2M_2) \|u - Iu\|_{\text{pw}} \|e\|_{\text{pw}}. \end{aligned}$$

On the other hand, Corollary 5.1.c implies $a_{\text{pw}}(u_{\text{nc}}, u) = a_{\text{pw}}(u_{\text{nc}}, Iu)$. Since $u \in V$ solves the source problem (5.21) and $u_{\text{nc}} \in V(\mathcal{T})$ solves the discrete source problem (5.22), this proves

$$\begin{aligned} a(u, \hat{e}) - a_{\text{pw}}(u_{\text{nc}}, e) &= a(u, \hat{e}) - a_{\text{pw}}(u_{\text{nc}}, Iu - u_{\text{nc}}) = b(f, \hat{e} - (Iu - u_{\text{nc}})) \\ &= b(f, (J - 1)(Iu - u_{\text{nc}})) \\ &= b((1 - \Pi_m)f, (J - 1)(Iu - u_{\text{nc}})) \end{aligned} \tag{5.26}$$

with the orthogonality (J3) in the last step. A weighted Cauchy-Schwarz inequality, the approximation property (J2), a triangle inequality, and Corollary 5.1.a lead to

$$\begin{aligned} a(u, \hat{e}) - a_{\text{pw}}(u_{\text{nc}}, e) &\leq \|h_{\mathcal{T}}^m(1 - \Pi_m)f\|_{L^2(\Omega)} \|h_{\mathcal{T}}^{-m}(J - 1)(Iu - u_{\text{nc}})\|_{L^2(\Omega)} \\ &\leq M_2 \text{osc}_m(f, \mathcal{T}) (\|u - Iu\|_{\text{pw}} + \|e\|_{\text{pw}}) \\ &\leq 2M_2 \text{osc}_m(f, \mathcal{T}) \|e\|_{\text{pw}}. \end{aligned}$$

The combination of the four previously displayed inequalities with Corollary 5.1.b implies that

$$\begin{aligned} \|e\|_{\text{pw}} &\leq (1 + 2M_2) \|u - Iu\|_{\text{pw}} + 2M_2 \text{osc}_m(f, \mathcal{T}) \\ &\leq c_1 \left(h_{\max}^s \|u\|_{H^{m+s}(\Omega)} + \text{osc}_m(f, \mathcal{T}) \right) \end{aligned} \quad (5.27)$$

with constant $c_1 := \max\{(1 + 2M_2)/\pi^s, 2M_2\}$. This concludes the proof for $\|e\|_{\text{pw}}$. For the estimate of the L^2 norm of $e := u - u_{\text{nc}}$, let $z \in V$ denote the solution to

$$a(z, v) = b(e, v) \quad \text{for all } v \in V. \quad (5.28)$$

This and the orthogonality in (J3) for $u_{\text{nc}} \in V(\mathcal{T}) \subset P_m(\mathcal{T})$ result in

$$\begin{aligned} \|e\|_{L^2(\Omega)}^2 &= b(Ju_{\text{nc}} - u_{\text{nc}}, e) + b(e, u - Ju_{\text{nc}}) = b(Ju_{\text{nc}} - u_{\text{nc}}, e) + a(z, u - Ju_{\text{nc}}) \\ &= b(Ju_{\text{nc}} - u_{\text{nc}}, u) + a_{\text{pw}}(z, u_{\text{nc}} - Ju_{\text{nc}}) + a_{\text{pw}}(z, e). \end{aligned} \quad (5.29)$$

Comparable to Corollary 5.2.a, the orthogonality property (J3) for $u_{\text{nc}} \in V(\mathcal{T}) \subset P_m(\mathcal{T})$, a Cauchy-Schwarz inequality, and (J2) show

$$\begin{aligned} b(Ju_{\text{nc}} - u_{\text{nc}}, u) &= b(Ju_{\text{nc}} - u_{\text{nc}}, u - u_{\text{nc}}) \leq \|Ju_{\text{nc}} - u_{\text{nc}}\|_{L^2(\Omega)} \|e\|_{L^2(\Omega)} \\ &\leq h_{\max}^m M_2 \|e\|_{\text{pw}} \|e\|_{L^2(\Omega)}. \end{aligned}$$

Corollary 5.2.b implies

$$a_{\text{pw}}(z, u_{\text{nc}} - Ju_{\text{nc}}) \leq M_2 \|z - Iz\|_{\text{pw}} \|e\|_{\text{pw}}.$$

Since u solves (5.21) and u_{nc} solves (5.22) as in (5.26), Corollary 5.1.c–d lead to

$$a_{\text{pw}}(z, e) = a(z, u) - a_{\text{pw}}(Iz, u_{\text{nc}}) = b(f, z - Iz) \leq \kappa_m \|h_{\mathcal{T}}^m f\|_{L^2(\Omega)} \|z - Iz\|_{\text{pw}}.$$

The (reduced) elliptic regularity (5.23) of Ω asserts $z \in H^{m+\sigma}(\Omega)$ and $\|z\|_{H^{m+\sigma}(\Omega)} \leq C(\sigma) \|e\|_{L^2(\Omega)}$. This and Corollary 5.1.b guarantee

$$\pi^\sigma \|z - Iz\|_{\text{pw}} \leq h_{\max}^\sigma \|z\|_{H^{m+\sigma}(\Omega)} \leq C(\sigma) h_{\max}^\sigma \|e\|_{L^2(\Omega)}.$$

A triangle inequality with $\text{osc}_m(f, \mathcal{T}) := \|h_{\mathcal{T}}^m(1 - \Pi_m)f\|_{L^2(\Omega)}$, Lemma 5.15, and Corollary 5.1.a imply

$$\|h_{\mathcal{T}}^m f\|_{L^2(\Omega)} \leq C_1 \|e\|_{\text{pw}} + (1 + C_1) \text{osc}_m(f, \mathcal{T}).$$

The combination of the last five displayed estimates bounds all summands on the right-hand side of (5.29). Since $\sigma \leq 1 \leq m$ and $h_{\max} \leq 1$, this implies

$$\begin{aligned} h_{\max}^{-\sigma} \|e\|_{L^2(\Omega)} &\leq M_2 \|e\|_{\text{pw}} + \frac{C(\sigma)}{\pi^\sigma} \left((M_2 + C_1 \kappa_m) \|e\|_{\text{pw}} + (1 + C_1) \kappa_m \text{osc}_m(f, \mathcal{T}) \right) \\ &\leq \left(\frac{C(\sigma)}{\pi^\sigma} (M_2 + C_1 \kappa_m) + M_2 \right) \|e\|_{\text{pw}} + \frac{C(\sigma)}{\pi^\sigma} (1 + C_1) \kappa_m \text{osc}_m(f, \mathcal{T}). \end{aligned}$$

Hence, (5.27) concludes the proof with constant $C_2 := c_1 \left(\frac{C(\sigma)}{\pi^\sigma} (M_2 + C_1 \kappa_m) + M_2 \right) + \frac{C(\sigma)}{\pi^\sigma} (1 + C_1) \kappa_m$. $\square$

Proof of (b). Since $u \in V$ solves (5.21) with the right-hand side $f \in E(\lambda) \subset V$, $u = f/\lambda \in H^{m+t}(\Omega) \cap V$. The definition of the oscillation, of $\kappa'_m \leq \kappa_m$ with (5.4) in Remark 5.6, and Corollary 5.1.b prove $\text{osc}_m(f, \mathcal{T}) \leq h_{\max}^{2m} \kappa'_m \|f - If\|_{\text{pw}} \leq h_{\max}^{2m+t} \kappa'_m / \pi^t \|f\|_{H^{m+t}(\Omega)}$. This and (a) show

$$\begin{aligned} \|u - u_{\text{nc}}\|_{\text{pw}} &\leq C_2 \left(h_{\max}^t \|u\|_{H^{m+t}(\Omega)} + \text{osc}_m(f, \mathcal{T}) \right) \\ &\leq C_2 h_{\max}^t \left(1/\lambda + h_{\max}^{2m} \kappa'_m / \pi^t \right) \|f\|_{H^{m+t}(\Omega)}. \end{aligned} \quad (5.30)$$

Since $g \in E(\lambda)$ is an eigenfunction in (5.1), $\lambda(u - Ju_{\text{nc}}, g)_{L^2(\Omega)} = a(u - Ju_{\text{nc}}, g)$. Corollary 5.1.c implies $a_{\text{pw}}(u_{\text{nc}}, g) = a_{\text{pw}}(u_{\text{nc}}, Ig)$. This shows the first equality in

$$\begin{aligned} \lambda(u - Ju_{\text{nc}}, g)_{L^2(\Omega)} &= a(u, g) - a_{\text{pw}}(u_{\text{nc}}, Ig) + a_{\text{pw}}(u_{\text{nc}} - Ju_{\text{nc}}, g) \\ &= (f, g - Ig)_{L^2(\Omega)} + a_{\text{pw}}(u_{\text{nc}} - Ju_{\text{nc}}, g). \end{aligned}$$

The second equality follows because u solves (5.21) and $u_{\text{nc}} \in V(\mathcal{T})$ solves (5.22) with right-hand side f . Corollary 5.2.b ensures

$$a_{\text{pw}}(u_{\text{nc}} - Ju_{\text{nc}}, g) \leq M_2 \|u - u_{\text{nc}}\|_{\text{pw}} \|g - Ig\|_{\text{pw}}.$$

The term $(f, g - Ig)_{L^2(\Omega)} = (f, JIg - Ig)_{L^2(\Omega)} + (f, g - JIg)_{L^2(\Omega)}$ is split into two. Corollary 5.2.a controls the first contribution

$$(f, JIg - Ig)_{L^2(\Omega)} \leq h_{\max}^{2m} \kappa_m M_2 \|f - If\|_{\text{pw}} \|g - Ig\|_{\text{pw}}.$$

Since $f \in E(\lambda)$ is an eigenfunction in (5.1), $\lambda(f, g - JIg)_{L^2(\Omega)} = a(f, g - JIg)$. The right-inverse property (J1) and Corollary 5.1.c show that $a(f, g - JIg) = a_{\text{pw}}(f, (1 - I)(g - JIg)) \leq \|f - If\|_{\text{pw}} \|g - JIg\|_{\text{pw}}$. A triangle inequality and (J2) ensure $\|g - JIg\|_{\text{pw}} \leq \|g - Ig\|_{\text{pw}} + \|Ig - JIg\|_{\text{pw}} \leq (1 + M_2) \|g - Ig\|_{\text{pw}}$. The combination proves

$$\lambda(f, g - JIg)_{L^2(\Omega)} \leq (1 + M_2) \|f - If\|_{\text{pw}} \|g - Ig\|_{\text{pw}}.$$

The combination of the last four displayed estimates reads

$$\frac{\lambda}{M_2} (u - Ju_{\text{nc}}, g)_{L^2(\Omega)} \leq \|g - Ig\|_{\text{pw}} \left(\left(\frac{1 + M_2^{-1}}{\lambda} + h_{\max}^{2m} \kappa_m \right) \|f - If\|_{\text{pw}} + \|u - u_{\text{nc}}\|_{\text{pw}} \right).$$

This, Corollary 5.1.b for $f, g \in H^{m+t}(\Omega) \cap V$, and (5.30) verify

$$(u - Ju_{\text{nc}}, g)_{L^2(\Omega)} \leq h_{\max}^{2t} \|f\|_{H^{m+t}(\Omega)} \|g\|_{H^{m+t}(\Omega)} \\ \times \left(\frac{1}{\lambda^2} \left(\frac{1 + M_2}{\pi^{2t}} + \frac{C_2 M_2}{\pi^t} \right) + \frac{h_{\max}^{2m}}{\lambda} \left(\frac{\kappa_m M_2}{\pi^{2t}} + \frac{\kappa'_m C_2 M_2}{\pi^{2t}} \right) \right).$$

For the term $(Ju_{\text{nc}} - u_{\text{nc}}, g)_{L^2(\Omega)}$, Corollary 5.2.a followed by Corollary 5.1.b and (5.30) show

$$(Ju_{\text{nc}} - u_{\text{nc}}, g)_{L^2(\Omega)} \leq h_{\max}^{2m} \kappa_m M_2 \|g - Ig\|_{\text{pw}} \|u - u_{\text{nc}}\|_{\text{pw}} \\ \leq h_{\max}^{2t} \|f\|_{H^{m+t}(\Omega)} \|g\|_{H^{m+t}(\Omega)} \left(\frac{h_{\max}^{2m}}{\lambda} + h_{\max}^{4m} \frac{\kappa'_m}{\pi^t} \right) \frac{\kappa_m C_2 M_2}{\pi^t}.$$

The combination of the last two displayed inequalities proves that $(u - u_{\text{nc}}, g)_{L^2(\Omega)} = (Ju_{\text{nc}} - u_{\text{nc}}, g)_{L^2(\Omega)} + (u - Ju_{\text{nc}}, g)_{L^2(\Omega)} \lesssim h_{\max}^{2t} (\lambda^{-1} + h_{\max}^{2m})^2 \|f\|_{H^{m+t}(\Omega)} \|g\|_{H^{m+t}(\Omega)}$. The bookkeeping of the multiplicative constants concludes the proof with $C_3 := \pi^{-2t} \max \left\{ 1 + M_2 + \pi^t C_2 M_2, (\kappa_m + \kappa'_m C_2 + \pi^t \kappa'_m \kappa_m C_2) M_2 / 2, \kappa'_m \kappa_m C_2 M_2 \right\}$. This constant can be simplified using the estimates $\kappa'_m \leq \kappa_m$ from Remark 5.6.a or $\kappa'_m \leq 1/\pi^m$ for $m = 1, 2$ from Remark 5.6.b. $\square$

In the remainder of this subsection, given any right-hand side $f \in L^2(\Omega)$, let $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$ denote the discrete solution to the extra-stabilized source problem

$$a_h(u_h, v_h) = (f, v_{\text{pw}})_{L^2(\Omega)} \quad \text{for all } v_h = (v_{\text{pw}}, v_{\text{nc}}) \in V_h. \quad (5.31)$$

As mentioned above, comparable to Section 4.4, the solution component of interest in (5.31) is $u_{\text{pw}} \in P_m(\mathcal{T})$, since (5.2) allows for the Schur complement method as in Subsection 4.2.3 for the HHO eigenvalue problem. The analysis of (5.31) reduces to that of Lemma 5.16 plus perturbation terms.

Lemma 5.17 (error estimate in V_h). *There exists a constant $C_{\text{pw}} > 0$ (that exclusively depends on $\mathbb{T}$), such that given any $f \in L^2(\Omega)$, the exact solution $u \in H^{m+\sigma}(\Omega) \cap V$ to (5.21) and the discrete solution $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$ to (5.31) for $\mathcal{T} \in \mathbb{T}$ with maximal mesh-size $h_{\max}$ satisfy (a)–(b).*

(a) *If $u \in H^{m+s}(\Omega)$ for some s with $\sigma \leq s \leq 1$, then*

$$\|u - u_{\text{pw}}\|_{\text{pw}} + h_{\max}^{-\sigma} \|u - u_{\text{pw}}\|_{L^2(\Omega)} \leq C_{\text{pw}} \left(h_{\max}^s \|u\|_{H^{m+s}(\Omega)} + \text{osc}_m(f, \mathcal{T}) \right).$$

(b) *Given any eigenvalue λ of (5.1) with eigenspace $E(\lambda) \subset H^{m+t}(\Omega) \cap V$ for some t with $\sigma \leq t \leq 1$, suppose $f, g \in E(\lambda)$ and C_3 from Lemma 5.16. Then*

$$|(u - u_{\text{pw}}, g)_{L^2(\Omega)}| \leq \left(C_3 (\lambda^{-1} + h_{\max}^{2m})^2 + \kappa_m^2 h_{\max}^{2(m-t)} \right) h_{\max}^{2t} \|f\|_{H^{m+t}(\Omega)} \|g\|_{H^{m+t}(\Omega)}.$$

Proof of (a). Given $v_{\text{pw}} \in P_m(\mathcal{T})$, the test-function $(v_{\text{pw}}, 0) \in V_h$ in (5.31) leads to $\kappa_m^{-2} h_{\mathcal{T}}^{-2m} (u_{\text{pw}} - u_{\text{nc}}) = \Pi_m f$, compare Remark 5.9. Thus $u_{\text{pw}} = u_{\text{nc}} + \kappa_m^2 h_{\mathcal{T}}^{2m} \Pi_m f$ and a triangle inequality shows

$$\begin{aligned} \|u - u_{\text{pw}}\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|u - u_{\text{pw}}\|_{L^2(\Omega)} &\leq \left(\|u - u_{\text{nc}}\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \right) \\ &\quad + \kappa_m^2 \left(\|h_{\mathcal{T}}^{2m} \Pi_m f\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|h_{\mathcal{T}}^{2m} \Pi_m f\|_{L^2(\Omega)} \right). \end{aligned} \quad (5.32)$$

For any $v_{\text{nc}} \in V(\mathcal{T})$, the test-function $(v_{\text{nc}}, v_{\text{nc}}) \in V(\mathcal{T}) \times V(\mathcal{T}) \subset V_h$ in (5.31) shows $a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = (f, v_{\text{nc}})_{L^2(\Omega)}$. In other words, the solution component $u_{\text{nc}} \in V(\mathcal{T})$ solves the nonconforming source problem (5.22). Lemma 5.16.a controls the first term in (5.32),

$$\|u - u_{\text{nc}}\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_2 \left(h_{\text{max}}^s \|u\|_{H^{m+s}(\Omega)} + \text{osc}_m(f, \mathcal{T}) \right).$$

An inverse estimate for $P_m(\mathcal{T})$ with constant $c_{\text{inv}} > 0$ and $0 < \sigma \leq 1 \leq m$ show

$$\|h_{\mathcal{T}}^{2m} \Pi_m f\|_{\text{pw}} + h_{\text{max}}^{-\sigma} \|h_{\mathcal{T}}^{2m} \Pi_m f\|_{L^2(\Omega)} \leq (c_{\text{inv}} + h_{\text{max}}^{m-\sigma}) \|h_{\mathcal{T}}^m \Pi_m f\|_{L^2(\Omega)}.$$

The efficiency estimate from Lemma 5.15 in combination with Corollary 5.1.b for $u \in H^{m+s}(\Omega)$ shows

$$C_1^{-1} \|h_{\mathcal{T}}^m \Pi_m f\|_{L^2(\Omega)} \leq h_{\text{max}}^s / \pi^s \|u\|_{H^{m+s}(\Omega)} + \text{osc}_m(f, \mathcal{T}).$$

The last two displayed inequalities bound the second term in (5.32) and that concludes the proof of (a) with $C_{\text{pw}} := C_2 + \kappa_m^2 C_1 (c_{\text{inv}} + h_{\text{max}}^{m-\sigma})$ with $h_{\text{max}}^{m-\sigma} \leq 1$. $\square$

Proof of (b). The substitution of $u_{\text{pw}} = u_{\text{nc}} + \kappa_m^2 h_{\mathcal{T}}^{2m} \Pi_m f$ from part (a) leads to

$$(u - u_{\text{pw}}, g)_{L^2(\Omega)} = (u - u_{\text{nc}}, g)_{L^2(\Omega)} - \kappa_m^2 (h_{\mathcal{T}}^{2m} \Pi_m f, g)_{L^2(\Omega)}.$$

Since u_{nc} solves the nonconforming source problem (5.22), Lemma 5.16.b controls $(u - u_{\text{nc}}, g)_{L^2(\Omega)}$. This and $(\Pi_m f, g)_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|g\|_{L^2(\Omega)}$ conclude the proof with $h_{\text{max}}^{2(m-t)} \leq 1$. $\square$

5.4.2 Convergence rates for the eigenvalue problem

The preparations in Subsection 5.4.1 allow the proof of the optimal a priori convergence rates in Theorem 5.18 with fundamental arguments from [BO91].

Theorem 5.18 (a priori convergence). *Suppose λ is an eigenvalue of (5.1) of multiplicity μ with eigenspace $E(\lambda) \subset H^{m+t}(\Omega) \cap V$ for some t with $\sigma \leq t \leq 1$. Then there exist $\varepsilon, C_4 > 0$ such that any triangulation $\mathcal{T} \in \mathbb{T}(\varepsilon)$ and the discrete space $V_h := P_m(\mathcal{T}) \times V(\mathcal{T})$ lead in (5.2) to exactly μ algebraic eigenvalues $\lambda_{h,1}, \dots, \lambda_{h,\mu}$ of (5.2) (counting multiplicities), that converge to λ as $h_{\text{max}} \rightarrow 0$.*

Let $E_h := \text{span}\{u_h \in E_h(\lambda_{h,k}) : k = 1, \dots, \mu\}$ abbreviate the span of the discrete eigenspaces $E_h(\lambda_{h,k}) \subset V_h$ of $\lambda_{h,k}$ for $k = 1, \dots, \mu$. Then

$$\begin{aligned} h_{\max}^{-t} \max_{k=1, \dots, \mu} |\lambda - \lambda_{h,k}| + h_{\max}^{-\sigma} \max_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \min_{\substack{u_h=(u_{\text{pw}}, u_{\text{nc}}) \in E_h \\ \|u_{\text{pw}}\|_{L^2(\Omega)}=1}} \|u - u_{\text{pw}}\|_{L^2(\Omega)} \\ + h_{\max}^{-\sigma} \max_{\substack{u_h=(u_{\text{pw}}, u_{\text{nc}}) \in E_h \\ \|u_{\text{pw}}\|_{L^2(\Omega)}=1}} \min_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \|u - u_{\text{pw}}\|_{L^2(\Omega)} \leq C_4 h_{\max}^t. \end{aligned}$$

Proof. Given any right-hand side $f \in L^2(\Omega)$ let $S(f) := u \in V$ denote the continuous solution to (5.21) and let $S_h(f) := u_{\text{pw}} \in P_m(\mathcal{T})$ denote the first component of the solution $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$ to (5.22). This defines solution operators $S : L^2(\Omega) \rightarrow L^2(\Omega)$ and $S_h : L^2(\Omega) \rightarrow L^2(\Omega)$. These are the appropriate solution operators for the Babuška-Osborn theory, cf. Section 4.4 for details on solution operators. Lemma 5.17.a implies the convergence $S_h \rightarrow S$ in the operator norm of $\mathcal{L}(L^2(\Omega))$ as $h_{\max} \rightarrow 0$. Suppose $(\lambda, \phi) \in \mathbb{R}^+ \times V$ denotes an eigenpair of (5.1) and $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ with $u_h = (u_{\text{pw}}, u_{\text{nc}})$ denotes an eigenpair of (5.2), then $(1/\lambda, \phi)$ is an eigenpair of S and $(1/\lambda_h, u_{\text{pw}})$ is an eigenpair of S_h . Hence, Theorem 2.6 implies for any non-zero eigenvalue $1/\lambda$ of S with eigenspace $E(\lambda) = \ker(\lambda^{-1} - S)$ of dimension $\mu = \dim(E(\lambda)) \in \mathbb{N}$, that there exist exactly μ eigenvalues $1/\lambda_{h,1}, \dots, 1/\lambda_{h,\mu}$ of S_h , which converge to $1/\lambda$ as $h_{\max} \rightarrow 0$. The error estimates for the selfadjoint operator S in Theorem 2.6 read (for any triangulation $T \in \mathbb{T}(\varepsilon)$ fine enough in the sense that $h_{\max} \leq \varepsilon$ for some $\varepsilon > 0$ and with generic constants which depend on λ)

$$\begin{aligned} \max_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \min_{\substack{u_h=(u_{\text{pw}}, u_{\text{nc}}) \in E_h \\ \|u_{\text{pw}}\|_{L^2(\Omega)}=1}} \|u - u_{\text{pw}}\|_{L^2(\Omega)} + \max_{\substack{u_h=(u_{\text{pw}}, u_{\text{nc}}) \in E_h \\ \|u_{\text{pw}}\|_{L^2(\Omega)}=1}} \min_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \|u - u_{\text{pw}}\|_{L^2(\Omega)} \\ \leq C_b \|(S - S_h)|_{E(\lambda)}\|_{\mathcal{L}(E(\lambda); L^2(\Omega))} \end{aligned}$$

and

$$\begin{aligned} \max_{1 \leq k \leq \mu} |\lambda - \lambda_{h,k}| &\lesssim \max_{1 \leq k \leq \mu} |\lambda^{-1} - \lambda_{h,k}^{-1}| \\ &\leq C_a \left(\sup_{\phi, \psi \in E(\lambda) \setminus \{0\}} \frac{|((S - S_h)\phi, \psi)_{L^2(\Omega)}|}{\|\phi\|_{L^2(\Omega)} \|\psi\|_{L^2(\Omega)}} + \|(S - S_h)|_{E(\lambda)}\|_{\mathcal{L}(E(\lambda); L^2(\Omega))}^2 \right). \end{aligned}$$

For the finite-dimensional eigenspace $E(\lambda) \subset H^{m+t}(\Omega)$ there exists

$$C_t := \sup_{\phi \in E(\lambda)} \frac{\|\phi\|_{H^{m+t}(\Omega)}}{\|\phi\|_{L^2(\Omega)}} < \infty$$

such that $\|\phi\|_{H^{m+t}(\Omega)} \leq C_t \|\phi\|_{L^2(\Omega)}$ for all $\phi \in E(\lambda)$. This and Lemma 5.17.a show

$$\|(S - S_h)|_{E(\lambda)}\|_{\mathcal{L}(E(\lambda); L^2(\Omega))} \leq h_{\max}^{t+\sigma} C_t C_{\text{pw}} \left(1/\lambda + h_{\max}^{2m} \kappa'_m / \pi^t \right),$$

since Remark 5.6 and Corollary 5.1.b lead (as an analog to (5.30)) to $\text{osc}(f, \mathcal{T}) \leq h_{\max}^{2m+t} \kappa'_m / \pi^t \|f\|_{H^{m+t}(\Omega)}$. Lemma 5.17.b bounds the remaining term

$$\sup_{\phi, \psi \in E(\lambda) \setminus \{0\}} \frac{|((S - S_h)\phi, \psi)_{L^2(\Omega)}|}{\|\phi\|_{L^2(\Omega)} \|\psi\|_{L^2(\Omega)}} \leq h_{\max}^{2t} C_t^2 \left(C_3(\lambda^{-1} + h_{\max}^{2m})^2 + \kappa_m^2 h_{\max}^{2(m-t)} \right).$$

This concludes the proof. $\square$

Lemma 5.16 allows the same convergence analysis for the standard nonconforming eigenvalue problem (5.9). Unlike, e.g., [Ran79; YLB16] for Morley ($m = 2$), the following Theorem 5.19 specifies the convergence rates for (5.9) directly in terms of $\sigma = \min\{1, \sigma_{\text{reg}}\}$ for the index of elliptic regularity σ_{reg} from (5.23) and the Sobolev regularity t of $E(\lambda)$ and is mentioned for convenience.

Theorem 5.19 (a priori convergence for (5.9)). *Given a non-zero eigenvalue λ of (5.1) of multiplicity μ , suppose that $E(\lambda) \subset H^{m+t}(\Omega) \cap V$ holds for some t with $\sigma \leq t \leq 1$. Then there exist $\varepsilon, C_5 > 0$ such that any $\mathcal{T} \in \mathbb{T}(\varepsilon)$ and the discrete space $V(\mathcal{T})$ lead in (5.9) to exactly μ algebraic eigenvalues $\lambda_{\text{nc},1}, \dots, \lambda_{\text{nc},\mu}$ that converge to λ as $h_{\max} \rightarrow 0$. Let $E_{\text{nc}} := \text{span}\{u_{\text{nc}} \in E_{\text{nc}}(\lambda_{\text{nc},k}) : k = 1, \dots, \mu\}$ denote the union of the discrete eigenspaces $E_{\text{nc}}(\lambda_{\text{nc},k}) \subset V(\mathcal{T})$ for $\lambda_{\text{nc},1}, \dots, \lambda_{\text{nc},\mu}$. Then*

$$\begin{aligned} h_{\max}^{-t} \max_{k=1, \dots, \mu} |\lambda - \lambda_{\text{nc},k}| + h_{\max}^{-\sigma} \max_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \min_{\substack{\phi_{\text{nc}} \in E_{\text{nc}} \\ \|\phi_{\text{nc}}\|_{L^2(\Omega)}=1}} \|u - \phi_{\text{nc}}\|_{L^2(\Omega)} \\ + h_{\max}^{-\sigma} \max_{\substack{\phi_{\text{nc}} \in E_{\text{nc}} \\ \|\phi_{\text{nc}}\|_{L^2(\Omega)}=1}} \min_{\substack{u \in E(\lambda) \\ \|u\|_{L^2(\Omega)}=1}} \|u - \phi_{\text{nc}}\|_{L^2(\Omega)} \lesssim h_{\max}^t. \end{aligned}$$

Proof. Define the solution operator $S_{\text{nc}} : L^2(\Omega) \rightarrow L^2(\Omega)$ which maps the right-hand side $f \in L^2(\Omega)$ onto $S_{\text{nc}}(f) := u_{\text{nc}} \in V(\mathcal{T})$ the nonconforming finite element solution of (5.22). The proof is verbatim to the proof of Theorem 5.18 with the results in Lemma 5.16 instead of Lemma 5.17 and so the details are omitted here. $\square$

5.4.3 Medius analysis

This section shows that (I1)–(I2) and (J1)–(J3) lead to best-approximation and error estimates in weaker Sobolev norms. The analysis assumes that $\lambda \equiv \lambda_k$ is a simple exact eigenvalue of (5.1) and applies the convergence results from Subsection 5.4.2.

5.4.3.1 Main result and layout of the proof

Throughout the remainder of this chapter, $k \in \mathbb{N}$ is the number of a *simple* exact eigenvalue $\lambda \equiv \lambda_k$. The aim of this subsection is the proof of Theorem 5.20 with the weight $\delta \in P_0(\mathcal{T})$ from (5.14) in $\|\bullet\|_{\delta}$, see Remark 5.22 below.

Theorem 5.20 (best-approximation). *Let $(\lambda, u) \in \mathbb{R}^+ \times V$ denote the k -th continuous eigenpair of (5.1) with a simple eigenvalue $\lambda \equiv \lambda_k$ and $\|u\|_{L^2(\Omega)} = 1$. There exist $\varepsilon_5 > 0$ and $C_0 > 0$ such that, for all $\mathcal{T} \in \mathbb{T}(\varepsilon_5) := \{\mathcal{T} \in \mathbb{T} : h_{\max} \leq \varepsilon_5\}$, there exists a discrete eigenpair $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ of number k of (5.2) with $\lambda_h \equiv \lambda_h(k)$, $u_h = (u_{\text{pw}}, u_{\text{nc}})$, $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$, and $b(u, u_{\text{nc}}) > 0$ such that*

(a) $\lambda_h(k)$ is a simple algebraic eigenvalue of (5.2) with $\lambda_k/2 \leq \lambda_h(k)$,

(b) $\lambda_h(j) \leq \lambda_j$ for all $j = 1, \dots, k+1$,

(c) $|\lambda - \lambda_h| + \|u - u_{\text{nc}}\|_{\text{pw}}^2 + h_{\max}^{-2\sigma} \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|u_{\text{nc}}\|_{\delta}^2 \leq C_0 \|u - Iu\|_{\text{pw}}^2$.

Some comments on related results and an outline of the proof of Theorem 5.20 are in order before Subsections 5.4.3.2–5.4.3.5 provide details. Throughout this sections the computation of constants values readability over optimality.

Remark 5.21 (consequences of Theorem 5.18). The analysis in Theorem 5.18 guarantees the convergence of the eigenvalues λ_h to λ and the component $u_{\text{pw}} \in P_m(\mathcal{T})$ to $u \in V$. The assumption that $\lambda = \lambda_k$ is a simple eigenvalue of (5.1) and the convergence $\lambda_h(k) \equiv \lambda_h \rightarrow \lambda$ as $h_{\max} \rightarrow 0$ lead to the existence of $\varepsilon_0 > 0$ such that the number $M := \dim(P_m(\mathcal{T}))$ of discrete eigenvalues of (5.2) is larger than $k+1$ and $\lambda_h(k-1) < \lambda_h(k) \equiv \lambda_h < \lambda_h(k+1)$ as well as $\lambda_k/2 \leq \lambda_h(k)$ for all $\mathcal{T} \in \mathbb{T}(\varepsilon_0)$. Then the eigenfunction $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h \setminus \{0\}$ is unique.

While the convergence analysis in Subsection 5.4.2 displays convergence of the component $u_{\text{pw}} \in P_m(\mathcal{T})$, this section focusses on the best-approximation results in the convergence analysis for $u_{\text{nc}} \in V(\mathcal{T})$, which is the solution component of interest in the estimator (5.54) below. Recall that $k \in \mathbb{N}$ is fixed and (λ, u) denotes the k -th eigenpair of (5.1) with a simple eigenvalue $\lambda \equiv \lambda_k > 0$ and $\|u\|_{L^2(\Omega)} = 1$. Set $\varepsilon_1 := \min\{\varepsilon_0, (2\lambda_{k+1}\kappa_m^2)^{-1/(2m)}\}$ and suppose $\mathcal{T} \in \mathbb{T}(\varepsilon_1)$. Let (λ_h, u_h) denote the k -th discrete eigenpair in (5.2) with $\lambda_h \equiv \lambda_h(k) > 0$, $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$, $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$, and $b(u, u_{\text{nc}}) \geq 0$.

Proof of Theorem 5.20.a. This follows directly from Remark 5.21 and the choice of $\varepsilon_1 := \min\{\varepsilon_0, (2\lambda_{k+1}\kappa_m^2)^{-1/(2m)}\}$. $\square$

Proof of Theorem 5.20.b. The choice $\varepsilon_1 := \min\{\varepsilon_0, (2\lambda_{k+1}\kappa_m^2)^{-1/(2m)}\}$ implies for all $j = 1, \dots, k$ that $\lambda_j \kappa_m^2 h_{\max}^{2m} \leq \lambda_{k+1} \kappa_m^2 \varepsilon_1^{2m} \leq 1/2$. Hence, Theorem 5.5 proves Theorem 5.20.b. $\square$

This section extends the analysis in [CGS15, Section 2–3] to a simultaneous analysis for Crouzeix-Raviart and Morley, and to the extra-stabilized discrete eigenvalue problem (5.2), and to higher space dimension. The following remarks recap the notation from Section 5.3.

Remark 5.22 (weight δ). Given the fixed $\lambda_h \equiv \lambda_h(k)$ above, recall

$$\delta := \frac{1}{1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}} - 1 = \frac{\lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}}{1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}} = \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m} (1 + \delta) \in P_0(\mathcal{T})$$

from (5.14) in Remark 5.10. This is the piecewise constant weight $\delta \in P_0(\mathcal{T})$ in the weighted L^2 norm $\|\bullet\|_\delta := \|\sqrt{\delta}\bullet\|_{L^2(\Omega)}$ on the left-hand side of Theorem 5.20.c. Notice that $h_{\max} \leq \varepsilon_1$ implies $\delta \leq \delta_{\max} := (1 - \lambda_h \kappa_m^2 h_{\max}^{2m})^{-1} - 1 \leq 1$. The constant $C_\delta := 2\lambda_h \kappa_m^2$ satisfies $\delta \leq C_\delta h_{\mathcal{T}}^{2m} \leq C_\delta h_{\max}^{2m}$ (because $\lambda_h \leq \lambda$ from Theorem 5.20.b) and δ converges to zero as the maximal mesh-size $h_{\max} \rightarrow 0$ approaches zero.

Remark 5.23 (equivalent problem). Since $\lambda_h \kappa_m^2 h_{\max}^{2m} \leq \lambda_{k+1} \kappa_m^2 \varepsilon_1^{2m} \leq 1/2$, (5.2) is equivalent to the reduced rational eigenvalue problem (5.15) that seeks $(\lambda_h, u_{\text{nc}}) \in \mathbb{R}^+ \times (V(\mathcal{T}) \setminus \{0\})$ with $a_{\text{pw}}(u_{\text{nc}}, v_{\text{nc}}) = \lambda_h (u_{\text{nc}} / (1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m}), v_{\text{nc}})_{L^2(\Omega)}$ for all $v_{\text{nc}} \in V(\mathcal{T})$ and $u_{\text{pw}} = (1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m})^{-1} u_{\text{nc}}$ according to Proposition 5.11.

Outline of the proof of Theorem 5.20.c. The remainder of this subsection provides an overview and clarifies the various steps for a reduction of ε_1 to ε_5 , before the technical details follow in the subsequent subsections.

The coefficient $(1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m})^{-1} = 1 + \delta \in P_0(\mathcal{T})$ with $\lambda_h \equiv \lambda_h(k)$ on the right-hand side of (5.15) is frozen in the intermediate eigenvalue problem.

Definition 5.24 (intermediate eigenvalue problem). Recall that $(\bullet, \bullet)_{1+\delta} := ((1 + \delta)\bullet, \bullet)_{L^2(\Omega)}$. Let $(\mu, \phi) \in \mathbb{R}^+ \times (V(\mathcal{T}) \setminus \{0\})$ solve the (algebraic) eigenvalue problem

$$a_{\text{pw}}(\phi, v_{\text{nc}}) = \mu(\phi, v_{\text{nc}})_{1+\delta} \quad \text{for all } v_{\text{nc}} \in V(\mathcal{T}). \quad (5.33)$$

The two coefficient matrices in (5.33) are symmetric positive definite and there exist $N := \dim V(\mathcal{T})$ (algebraic) eigenpairs $(\mu_1, \phi_1), \dots, (\mu_N, \phi_N)$ of (5.33). The eigenfunctions $\phi_1, \dots, \phi_N$ are $(\bullet, \bullet)_{1+\delta}$ -orthonormal and the eigenvalues $0 < \mu_1 \leq \dots \leq \mu_N$ are enumerated in ascending order counting multiplicities.

Since $\lambda_h \equiv \lambda_h(k)$ is an eigenvalue of the rational problem (5.15), $\lambda_h \in \{\mu_1, \dots, \mu_N\}$ belongs to the eigenvalues of (5.33). Lemma 5.27 below guarantees the convergence $|\mu_j - \lambda_h(j)| \rightarrow 0$ as $h_{\max} \rightarrow 0$ for $j = 1, \dots, k + 1$. Hence, there exist positive a $\varepsilon_2 \leq \min\{1/2, \varepsilon_1\}$ and a constant $M_6 > 0$ such that $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$ implies

(H1) $\mu_k = \lambda_h(k)$ is a simple algebraic eigenvalue of (5.33),

$$(H2) \quad \max_{\substack{j=1, \dots, N \\ j \neq k}} \frac{\lambda_k}{|\lambda_k - \mu_j|} \leq M_6,$$

see Corollary 5.29 below for details. The intermediate eigenvalue problem and the following associated source problem allow for the control of the extra-stabilization.

Definition 5.25 (auxiliary source problem). Suppose $z_{\text{nc}} \in V(\mathcal{T})$ solves

$$a_{\text{pw}}(z_{\text{nc}}, v_{\text{nc}}) = (\lambda u, v_{\text{nc}})_{1+\delta} \quad \text{for all } v_{\text{nc}} \in V(\mathcal{T}). \quad (5.34)$$

For any $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$, Lemma 5.30–5.31 below provide $C_6, C_7 > 0$ that satisfy

$$\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_6 \|u - z_{\text{nc}}\|_{L^2(\Omega)}, \quad (5.35)$$

$$C_7^{-1} \|u - z_{\text{nc}}\|_{L^2(\Omega)} \leq h_{\text{max}}^\sigma \|u - z_{\text{nc}}\|_{\text{pw}} + \|\delta\lambda u\|_{L^2(\Omega)}. \quad (5.36)$$

The proof of (5.35) in Subsection 5.4.3.3 extends [CGS15, Lem. 2.4]. The proof of (5.36) utilizes another continuous source problem with the right-hand side $u - Jz_{\text{nc}}$. For all $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$, Lemma 5.32 below provides a constant $C_8 > 0$ such that

$$C_8^{-1} \|u - z_{\text{nc}}\|_{\text{pw}} \leq \|u - Iu\|_{\text{pw}} + \|\delta\lambda u\|_{L^2(\Omega)}. \quad (5.37)$$

The proof of (5.37) below rests upon a decomposition of $\|u - z_{\text{nc}}\|_{\text{pw}}^2$ into terms controlled by the conditions (I1)–(I2) and (J1)–(J3). Since $h_{\text{max}} \leq 1$, the combination of (5.35)–(5.37) reads

$$\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_6 C_7 \left(C_8 h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + (1 + C_8) \|\delta\lambda u\|_{L^2(\Omega)} \right). \quad (5.38)$$

The control of $\|\delta\lambda u\|_{L^2(\Omega)}$ on the right-hand side of (5.38) consists of two steps and leads to $c_3 := 2\lambda^2 \kappa_m^2 C_6 C_7 (1 + C_8)$ and $\varepsilon_3 := \min\{\varepsilon_2, (2c_3)^{-1/2m}\}$. A triangle inequality $\|\delta\lambda u\|_{L^2(\Omega)} \leq \|\delta\lambda(u - u_{\text{nc}})\|_{L^2(\Omega)} + \|\delta\lambda u_{\text{nc}}\|_{L^2(\Omega)}$, the estimate $\delta \leq 2\lambda \kappa_m^2 h_{\text{max}}^{2m}$ in Remark 5.22, and (5.38) imply

$$\|\delta\lambda u\|_{L^2(\Omega)} \leq \frac{c_3 C_8 h_{\text{max}}^{2m}}{1 + C_8} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + c_3 h_{\text{max}}^{2m} \|\delta\lambda u\|_{L^2(\Omega)} + \|\delta\lambda u_{\text{nc}}\|_{L^2(\Omega)}.$$

The choice of ε_3 shows $c_3 h_{\text{max}}^{2m} \|\delta\lambda u\|_{L^2(\Omega)} \leq \|\delta\lambda u\|_{L^2(\Omega)}/2$ for any $\mathcal{T} \in \mathbb{T}(\varepsilon_3)$. Therefore

$$\|\delta\lambda u\|_{L^2(\Omega)} \leq C_8 / (1 + C_8) h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + 2 \|\delta\lambda u_{\text{nc}}\|_{L^2(\Omega)}. \quad (5.39)$$

Notice that $\|\delta u_{\text{nc}}\|_{L^2(\Omega)} \leq 2\lambda \kappa_m^2 h_{\text{max}}^m \|h_{\mathcal{T}}^m u_{\text{nc}}\|_{L^2(\Omega)}$ from Remark 5.22 allows for the application of the efficiency estimate

$$C_9^{-1} \|h_{\mathcal{T}}^m u_{\text{nc}}\|_{L^2(\Omega)} \leq h_{\text{max}}^m \|u - u_{\text{nc}}\|_{L^2(\Omega)} + \lambda^{-1} \|u - Iu\|_{\text{pw}}, \quad (5.40)$$

which is closely related to Lemma 5.15 and shown in Lemma 5.33. Abbreviate $c_4 := 4\lambda^2 \kappa_m^2 C_6 C_7 (1 + C_8) C_9$ and $C_{10} := 2C_6 C_7 (2C_8 + 4\lambda \kappa_m^2 (1 + C_8) C_9)$. The combination of (5.39)–(5.40) controls $\|\delta\lambda u\|_{L^2(\Omega)}$ in (5.38) and shows

$$\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq \frac{C_{10}}{2} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + c_4 h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)}. \quad (5.41)$$

The choice $\varepsilon_4 := \min\{\varepsilon_3, (2c_4)^{-1/2m}\} < 1$ shows $c_4 h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq \|u - u_{\text{nc}}\|_{L^2(\Omega)}/2$ for $\mathcal{T} \in \mathbb{T}(\varepsilon_4)$. This and (5.41) prove the central estimate in Theorem 5.20.c

$$\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_{10} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}}. \quad (5.42)$$

Note that (5.42) and Corollary 5.1.d from (I2) imply the convergence $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \rightarrow 0$ as $h_{\text{max}} \rightarrow 0$. This and some $\varepsilon_5 \leq \varepsilon_4$ ensures $b(u, u_{\text{nc}}) > 0$ for all $\mathcal{T} \in \mathbb{T}(\varepsilon_5)$. Based on this outline, it remains to prove (5.35)–(5.37), (5.40), and hence (5.42) and to identify $C_0, \dots, C_9$ below. The remaining estimates in Theorem 5.20.c follow in Subsection 5.4.3.5.

5.4.3.2 Intermediate eigenvalue problem

Recall $\varepsilon_1 := \min\{\varepsilon_0, (2\lambda_{k+1}\kappa_m^2)^{-1/(2m)}\}$ and that (λ_h, u_h) denotes the k -th eigenpair of (5.2) with $\lambda_h \equiv \lambda_h(k) > 0$, $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$, $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$, and $b(u, u_{\text{nc}}) \geq 0$. Recall the intermediate eigenvalue problem (5.33) and that $(\lambda_h, u_{\text{nc}}) \in \mathbb{R}^+ \times V(\mathcal{T})$ solves the rational eigenvalue problem (5.15).

Remark 5.26 ($\|\bullet\|_{1+\delta} \approx \|\bullet\|_{L^2(\Omega)}$). The weighted norm $\|\bullet\|_{1+\delta}$ is equivalent to the L^2 -norm. Since $\lambda_h \kappa_m^2 \varepsilon_1^{2m} < \lambda_{k+1} \kappa_m^2 \varepsilon_1^{2m} \leq 1/2$ and $1 \leq (1+\delta)|_T \leq 2$ for all $T \in \mathcal{T} \in \mathbb{T}(\varepsilon_1)$, $\|v_{\text{nc}}\|_{L^2(\Omega)} \leq \|v_{\text{nc}}\|_{1+\delta} \leq \sqrt{2}\|v_{\text{nc}}\|_{L^2(\Omega)}$ holds for any $v_{\text{nc}} \in V(\mathcal{T})$. $\square$

Lemma 5.27 (comparison of (5.2) with (5.33)). *Given $\mathcal{T} \in \mathbb{T}(\varepsilon_1)$, let $\lambda_h(j)$ denote the j -th eigenvalue of (5.2), and μ_j the j -th eigenvalue of (5.33) for any $j = 1, \dots, k+1$. Then*

$$(1 - \lambda_{k+1} \kappa_m^2 h_{\max}^{2m}) \mu_j \leq (1 - \lambda_h(j) \kappa_m^2 h_{\max}^{2m}) \mu_j \leq \lambda_h(j) \leq \mu_j + \lambda_h^2 \kappa_m^2 h_{\max}^{2m}. \quad (5.43)$$

The upper bound $\lambda_h(j) \leq \mu_j + 2\lambda_h^2 \kappa_m^2 h_{\max}^{2m}$ holds for all $j = 1, \dots, N := \dim V(\mathcal{T})$.

Proof of the upper bound. The eigenfunctions $\phi_1, \dots, \phi_N$ of (5.33) are $(\bullet, \bullet)_{1+\delta}$ -orthonormal, such that $a_{\text{pw}}(\phi_j, \phi_\ell) = \mu_j \delta_{j\ell}$ and $(\phi_j, \phi_\ell)_{1+\delta} = \delta_{j\ell}$ for all $j, \ell = 1, \dots, N$. Set $\psi_\ell := (1+\delta)\phi_\ell$ and $U_j := \text{span}\{(\psi_1, \phi_1), \dots, (\psi_j, \phi_j)\} \subset V_h$ for any $j, \ell = 1, \dots, N$. Since $\phi_1, \dots, \phi_N$ are linear independent, $\dim(U_j) = j$ and the discrete min-max principle in Lemma 2.7 for the algebraic eigenvalue problem (5.2) shows

$$\lambda_h(j) \leq \max_{v_h \in U_j \setminus \{0\}} a_h(v_h, v_h) / b_h(v_h, v_h). \quad (5.44)$$

The maximum in (5.44) is attained for some $v_h = (\psi, \phi) \in U_j \setminus \{0\}$ with $\phi = \sum_{\ell=1}^j \alpha_\ell \phi_\ell \in V(\mathcal{T})$, $\psi = \sum_{\ell=1}^j \alpha_\ell \psi_\ell = (1+\delta)\phi \in P_m(\mathcal{T})$, and $1 = \|\phi\|_{1+\delta}^2 = \sum_{\ell=1}^j \alpha_\ell^2$. Then $b_h(v_h, v_h) = \|(1+\delta)\phi\|_{L^2(\Omega)}^2 \geq 1$ and $a_h(v_h, v_h) = \|\phi\|_{\text{pw}}^2 + \|\kappa_m^{-1} h_{\mathcal{T}}^{-m}(\psi - \phi)\|_{L^2(\Omega)}^2$. Since $a_{\text{pw}}(\phi_j, \phi_\ell) = \mu_j \delta_{j\ell}$ for $\ell, j = 1, \dots, N$, $\sum_{\ell=1}^j \alpha_\ell^2 = 1$ implies $\|\phi\|_{\text{pw}}^2 = \sum_{\ell=1}^j \alpha_\ell^2 \mu_\ell \leq \mu_j$. Since $\delta = \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m} (1+\delta)$ a.e. in Ω , the stabilization term in a_h reads

$$\|\kappa_m^{-1} h_{\mathcal{T}}^{-m}(\psi - \phi)\|_{L^2(\Omega)}^2 = \|\kappa_m^{-1} h_{\mathcal{T}}^{-m} \delta \phi\|_{L^2(\Omega)}^2 \leq \lambda_h^2 \kappa_m^2 h_{\max}^{2m} \|(1+\delta)\phi\|_{L^2(\Omega)}^2.$$

The substitution in (5.44) concludes the proof of

$$\lambda_h(j) \leq \frac{\|\phi\|_{\text{pw}}^2 + \|\kappa_m^{-1} h_{\mathcal{T}}^{-m}(\psi - \phi)\|_{L^2(\Omega)}^2}{\|(1+\delta)\phi\|_{L^2(\Omega)}^2} \leq \mu_j + \lambda_h^2 \kappa_m^2 h_{\max}^{2m}$$

in (5.43) for $j = 1, \dots, N$. $\square$

Proof of the lower bound. Recall that $\lambda_h(j)\kappa_m^2 h_{\max}^{2m} \leq 1/2$ for $j = 1, \dots, k+1$, hence the arguments from Lemma 5.12 can be adapted as follows. Let $(\lambda_h(j), \phi_h(j)) \in \mathbb{R}^+ \times V_h$ denote the first b_h -orthonormal eigenpairs of (5.2) with $\phi_h(j) = (\phi_{\text{pw}}(j), \phi_{\text{nc}}(j))$ and for $\xi = (\xi_1, \dots, \xi_j) \in \mathbb{R}^j$ with $\sum_{\ell=1}^j \xi_\ell^2 = 1$, set

$$v_{\text{nc}} := \sum_{\ell=1}^j \xi_\ell \phi_{\text{nc}}(\ell), \quad v_{\text{pw}} := \sum_{\ell=1}^j \xi_\ell \phi_{\text{pw}}(\ell), \quad \text{and} \quad w_{\text{pw}} := \sum_{\ell=1}^j \xi_\ell \lambda_h(\ell) \phi_{\text{pw}}(\ell).$$

In this notation holds $\|v_{\text{pw}}\|_{L^2(\Omega)} = 1$ and $\|w_{\text{pw}}\|_{L^2(\Omega)} = \sqrt{\sum_{\ell=1}^j \xi_\ell^2 \lambda_h(\ell)^2} \leq \lambda_h(j)$. This and a Cauchy-Schwarz inequality in (5.18) imply

$$\|v_{\text{nc}}\|_{\text{pw}}^2 = b(w_{\text{pw}}, v_{\text{nc}}) \leq \lambda_h(j) \|v_{\text{nc}}\|_{L^2(\Omega)} \quad (5.45)$$

while the reverse triangle inequality in (5.19) and Remark 5.26 show

$$0 < 1 - \lambda_h(j)\kappa_m^2 h_{\max}^{2m} \leq \|v_{\text{pw}} - \kappa_m^2 h_{\mathcal{T}}^{2m} w_{\text{pw}}\|_{L^2(\Omega)} = \|v_{\text{nc}}\|_{L^2(\Omega)} \leq \|v_{\text{nc}}\|_{1+\delta}. \quad (5.46)$$

Let $v_{\text{nc}} = \sum_{\ell=1}^j \alpha_\ell \phi_{\text{nc}}(\ell) \in U_j$ denote a maximizer of $\|v_{\text{nc}}\|_{\text{pw}}^2 / \|v_{\text{nc}}\|_{1+\delta}^2$ with $\sum_{\ell=1}^j \alpha_\ell^2 = 1$. Since $\phi_{\text{nc}}(1), \dots, \phi_{\text{nc}}(j)$ are linear independent, the discrete min-max principle Lemma 2.7 for (5.33) and (5.45)–(5.46) show

$$\mu_j \leq \frac{\|v_{\text{nc}}\|_{\text{pw}}^2}{\|v_{\text{nc}}\|_{1+\delta}^2} \leq \frac{\|v_{\text{nc}}\|_{\text{pw}}^2}{\|v_{\text{nc}}\|_{L^2(\Omega)}^2} \leq \frac{\lambda_h(j)}{1 - \lambda_h(j)\kappa_m^2 h_{\max}^{2m}}.$$

Recall $\lambda_h(j) \leq \lambda_h(k+1) \leq \lambda_{k+1}$ from Theorem 5.5 to conclude the proof of the associated lower bound for all $j = 1, \dots, k+1$. $\square$

The subsequent corollaries adapt the notation $\mu_j, \lambda_h(j), \lambda_j$ from Lemma 5.27.

Corollary 5.28. *For any $j = 1, \dots, k+1$, it holds $|\mu_j - \lambda_h(j)| + |\mu_j - \lambda_j| \rightarrow 0$ as $h_{\max} \rightarrow 0$.*

Proof. The a priori convergence analysis Theorem 5.18 implies $\lim_{h_{\max} \rightarrow 0} \lambda_h(j) \rightarrow \lambda_j$. Observe $0 < \mu_j, \lambda_h(j)$ and $\lambda_{k+1}\kappa_m^2 h_{\max}^{2m} \leq 1/2$, hence the lower bound in (5.43) implies $\mu_j \leq 2\lambda_h(j) \leq 2\lambda_{k+1}$. This and Lemma 5.27 show $|\lambda_h(j) - \mu_j| \leq h_{\max}^{2m} \kappa_m^2 \max\{\lambda_h^2, \lambda_h(j)\mu_j\} \leq h_{\max}^{2m} \kappa_m^2 2\lambda_{k+1}^2 \rightarrow 0$ as $h_{\max} \rightarrow 0$. $\square$

Corollary 5.29. *There exists $0 < \varepsilon_2 \leq \min\{1/2, \varepsilon_1\}$ such that (H1)–(H2) hold for $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$.*

Proof. Corollary 5.28 and $\lambda_h = \lambda_h(k) \in \{\mu_1, \dots, \mu_N\}$ lead to $\varepsilon_a > 0$ such that $\lambda_h = \lambda_h(k) = \mu_k$ has the correct index k for all $\mathcal{T} \in \mathbb{T}(\varepsilon_a)$. It also leads to some $\varepsilon_b > 0$ such that $\mu_{k-1} < \mu_k < \mu_{k+1}$ for all $\mathcal{T} \in \mathbb{T}(\varepsilon_b)$. Then $\varepsilon_2 := \min\{1/2, \varepsilon_1, \varepsilon_a, \varepsilon_b\}$ and $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$ imply (H1)–(H2). $\square$

5.4.3.3 Proof of (5.35)–(5.36) for the L^2 error control

Recall $\lambda \equiv \lambda_k$, $\lambda_h \equiv \lambda_h(k)$, M_6 from (H2), δ from Remark 5.22, the norm equivalence from Remark 5.26, and the auxiliary source problem from Definition 5.25.

Lemma 5.30. *The constant $C_6 > 0$ satisfies (5.35) for all $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$, i.e., $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_6 \|u - z_{\text{nc}}\|_{L^2(\Omega)}$.*

Proof. For completeness first verify the following result from [CGS15, Eq. (2.8)]: Any $u, v \in L^2(\Omega)$ with $\|u\|_{L^2(\Omega)} = \|v\|_{L^2(\Omega)} = 1$ satisfy

$$(1 + b(u, v)) \|u - v\|_{L^2(\Omega)}^2 = 2 \min_{t \in \mathbb{R}} \|u - tv\|_{L^2(\Omega)}^2. \quad (5.47)$$

On one hand, the normalization implies

$$(1 + b(u, v)) \|u - v\|_{L^2(\Omega)}^2 = (1 + b(u, v)) (2 - 2b(u, v)) = 2(1 - b(u, v)^2).$$

On the other hand, the L^2 best-approximation and the normalization show

$$\min_{t \in \mathbb{R}} \|u - tv\|_{L^2(\Omega)}^2 = \|u - b(u, v)v\|_{L^2(\Omega)}^2 = 1 - b(u, v)^2.$$

The combination of the last two displayed identities proves (5.47).

Recall the eigenfunctions $\phi_1, \dots, \phi_N$ of (5.33), set $t := (z_{\text{nc}}, u_{\text{nc}})_{1+\delta} \|\phi_k\|_{L^2(\Omega)}^2$ and $v_{\text{nc}} := z_{\text{nc}} - tu_{\text{nc}}$. These abbreviations, the observation (5.47), and a triangle inequality, lead to

$$2^{-1/2} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq \|u - tu_{\text{nc}}\|_{L^2(\Omega)} \leq \|u - z_{\text{nc}}\|_{L^2(\Omega)} + \|v_{\text{nc}}\|_{L^2(\Omega)}. \quad (5.48)$$

Since $\phi_1, \dots, \phi_N$ are $(\bullet, \bullet)_{1+\delta}$ -orthonormal and form a basis of $V(\mathcal{T})$, there exist Fourier coefficients $\alpha_1, \dots, \alpha_N \in \mathbb{R}$ with $v_{\text{nc}} = \sum_{j=1}^N \alpha_j \phi_j$ and $\|v_{\text{nc}}\|_{1+\delta}^2 = \sum_{j=1}^N \alpha_j^2$. Since $(\lambda_h, u_{\text{nc}})$ with $\lambda_h \equiv \lambda_h(k)$ solves (5.15), (H1) implies $u_{\text{nc}} \in \text{span}\{\phi_k\}$ with $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$. Hence, $u_{\text{nc}} = \pm \phi_k / \|\phi_k\|_{L^2(\Omega)}$, $t = \pm (z_{\text{nc}}, \phi_k)_{1+\delta} \|\phi_k\|_{L^2(\Omega)}$, and $(u_{\text{nc}}, \phi_k)_{1+\delta} = \pm \|\phi_k\|_{L^2(\Omega)}^{-1}$. Consequently,

$$\alpha_k = (v_{\text{nc}}, \phi_k)_{1+\delta} = (z_{\text{nc}}, \phi_k)_{1+\delta} - t(u_{\text{nc}}, \phi_k)_{1+\delta} = 0.$$

Since $(u_{\text{nc}}, \phi_j)_{1+\delta} = 0$ for all $j = 1, \dots, N$ with $j \neq k$, $\alpha_j = (v_{\text{nc}}, \phi_j)_{1+\delta} = (z_{\text{nc}}, \phi_j)_{1+\delta}$. Since $\phi_j \in V(\mathcal{T})$ is an eigenfunction in (5.33) and $z_{\text{nc}} \in V(\mathcal{T})$ solves (5.34), it follows

$$\alpha_j = (z_{\text{nc}}, \phi_j)_{1+\delta} = \frac{1}{\mu_j} a_{\text{pw}}(z_{\text{nc}}, \phi_j) = \frac{\lambda}{\mu_j} (u, \phi_j)_{1+\delta}.$$

Hence, $(u - z_{\text{nc}}, \phi_j)_{1+\delta} = (\mu_j/\lambda - 1)\alpha_j$. These values for the coefficients α_j and the separation condition (H2) imply

$$\|v_{\text{nc}}\|_{1+\delta}^2 = \sum_{j \neq k} \alpha_j^2 = \sum_{j \neq k} \left| \frac{\lambda}{\mu_j - \lambda} \right| |\alpha_j| |(u - z_{\text{nc}}, \phi_j)_{1+\delta}| \leq M_6 \sum_{j \neq k} (u - z_{\text{nc}}, \alpha'_j \phi_j)_{1+\delta}$$

for a sign in $\alpha'_j \in \{\pm\alpha_j\}$ such that $|(u - z_{\text{nc}}, \alpha'_j \phi_j)_{1+\delta}| = (u - z_{\text{nc}}, \alpha'_j \phi_j)_{1+\delta}$ and with the abbreviation $\sum_{j \neq k} = \sum_{j=1, j \neq k}^N$. This and a Cauchy-Schwarz inequality show

$$M_6^{-1} \|v_{\text{nc}}\|_{1+\delta}^2 \leq \left(u - z_{\text{nc}}, \sum_{j \neq k} \alpha'_j \phi_j \right)_{1+\delta} \leq \|u - z_{\text{nc}}\|_{1+\delta} \|v_{\text{nc}}\|_{1+\delta}.$$

The norm equivalence in Remark 5.26 proves $\|v_{\text{nc}}\|_{L^2(\Omega)} \leq \|v_{\text{nc}}\|_{1+\delta} \leq \sqrt{2}M_6 \|u - z_{\text{nc}}\|_{L^2(\Omega)}$. This and (5.48) conclude the proof with $C_6 := \sqrt{2}(1 + \sqrt{2}M_6)$. $\square$

Lemma 5.31. *There exists a constant $C_7 > 0$ that satisfies (5.36) for any $T \in \mathbb{T}(\varepsilon_2)$, i.e., $C_7^{-1} \|u - z_{\text{nc}}\|_{L^2(\Omega)} \leq h_{\max}^\sigma \|u - z_{\text{nc}}\|_{\text{pw}} + \|\delta \lambda u\|_{L^2(\Omega)}$.*

Proof. Given the solution $z_{\text{nc}} \in V(\mathcal{T})$ to (5.34), let $w \in V := H_0^m(\Omega)$ solve

$$a(w, \varphi) = b(u - Jz_{\text{nc}}, \varphi) \quad \text{for all } \varphi \in V. \quad (5.49)$$

The test function $\varphi = u - Jz_{\text{nc}}$ in the auxiliary problem (5.49) leads to

$$\|u - Jz_{\text{nc}}\|_{L^2(\Omega)}^2 = a(w - JIw, u) + a_{\text{pw}}(w, z_{\text{nc}} - Jz_{\text{nc}}) + a(JIw, u) - a_{\text{pw}}(w, z_{\text{nc}}). \quad (5.50)$$

Since (J1) asserts $I(w - JIw) = 0$, Corollary 5.1.c and a triangle inequality show

$$\begin{aligned} a(w - JIw, u) &= a_{\text{pw}}((1 - I)(w - JIw), u) \\ &\leq (\|w - Iw\|_{\text{pw}} + \|Iw - JIw\|_{\text{pw}}) \|u - z_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

Then (J2) implies that $a(w - JIw, u) \leq (1 + M_2) \|w - Iw\|_{\text{pw}} \|u - z_{\text{nc}}\|_{\text{pw}}$. Corollary 5.2.b proves for the second term in the right-hand side of (5.50) that

$$a_{\text{pw}}(w, z_{\text{nc}} - Jz_{\text{nc}}) \leq M_2 \|w - Iw\|_{\text{pw}} \|u - z_{\text{nc}}\|_{\text{pw}}.$$

Corollary 5.1.c ensures $a_{\text{pw}}(w, z_{\text{nc}}) = a_{\text{pw}}(Iw, z_{\text{nc}})$. Since (λ, u) is an eigenpair of (5.1) and z_{nc} satisfies (5.34), this implies

$$a(u, JIw) - a_{\text{pw}}(z_{\text{nc}}, w) = b(\lambda u, JIw) - a_{\text{pw}}(z_{\text{nc}}, Iw) = \lambda b(u, JIw - Iw - \delta Iw).$$

In this subsection, the bound $\lambda \kappa_m^2 h_{\max}^{2m} \leq 1/2$ allows to simplify the constants. To this end modify Corollary 5.2.a as follows. Note, that for any $v_{\text{nc}} \in V(\mathcal{T})$, (J1), (I1), and Corollary 5.1.a show

$$\begin{aligned} \|(1 - J)v_{\text{nc}}\|_{L^2(\Omega)} &= \|(1 - I)(1 - J)v_{\text{nc}}\|_{L^2(\Omega)} \leq \kappa_m h_{\max}^m \|(1 - J)v_{\text{nc}}\|_{\text{pw}} \\ &\leq M_2 \kappa_m h_{\max}^m \min_{v \in V} \|v - v_{\text{nc}}\|_{\text{pw}} \end{aligned}$$

with (J2) in the last step. This and Corollary 5.1.a recast the estimate in Corollary 5.2.a for any $v_{\text{nc}} \in V(\mathcal{T})$ and $w \in V$ as

$$\begin{aligned} b(v_{\text{nc}} - Jv_{\text{nc}}, w) &\leq \|v_{\text{nc}} - Jv_{\text{nc}}\|_{L^2(\Omega)} \|w - Iw\|_{L^2(\Omega)} \\ &\leq h_{\text{max}}^{2m} \kappa_m^2 M_2 \min_{v \in V} \|v_{\text{nc}} - v\|_{\text{pw}} \min_{w_{\text{nc}} \in V(\mathcal{T})} \|w - w_{\text{nc}}\|_{\text{pw}}. \end{aligned} \quad (5.51)$$

The bound $\lambda \kappa_m^2 h_{\text{max}}^{2m} \leq 1/2$ and (5.51) imply $\lambda b(u, JIw - Iw) \leq M_2/2 \|u - z_{\text{nc}}\|_{\text{pw}} \|w - Iw\|_{\text{pw}}$. The discrete Friedrichs inequality in Corollary 5.3, (I2), and the boundedness of Π_0 imply $C_{\text{dF}}^{-1} \|Iw\|_{L^2(\Omega)} \leq \|Iw\|_{\text{pw}} = \|\Pi_0 D^m w\|_{L^2(\Omega)} \leq \|w\|_{H^m(\Omega)}$. The Cauchy-Schwarz inequality leads to

$$-b(\lambda u, \delta Iw) \leq \|\delta \lambda u\|_{L^2(\Omega)} \|Iw\|_{L^2(\Omega)} \leq C_{\text{dF}} \|\delta \lambda u\|_{L^2(\Omega)} \|w\|_{H^{m+\sigma}(\Omega)}.$$

This bounds the last term on the right-hand side of (5.50). The substitution in (5.50) result in

$$\|u - Jz_{\text{nc}}\|_{L^2(\Omega)}^2 \leq \left(1 + \frac{5M_2}{2}\right) \|u - z_{\text{nc}}\|_{\text{pw}} \|w - Iw\|_{\text{pw}} + C_{\text{dF}} \|\delta \lambda u\|_{L^2(\Omega)} \|w\|_{H^{m+\sigma}(\Omega)}.$$

Since $u - Jz_{\text{nc}} \in V \subset L^2(\Omega)$, the elliptic regularity (5.23) for (5.49) guarantees $w \in H^{m+\sigma}(\Omega)$ with $\|w\|_{H^{m+\sigma}(\Omega)} \leq C(\sigma) \|u - Jz_{\text{nc}}\|_{L^2(\Omega)}$ and Corollary 5.1.b shows $\|w - Iw\|_{\text{pw}} \leq (h_{\text{max}}/\pi)^\sigma C(\sigma) \|u - Jz_{\text{nc}}\|_{L^2(\Omega)}$. In combination with the last displayed estimate this implies

$$C(\sigma)^{-1} \|u - Jz_{\text{nc}}\|_{L^2(\Omega)} \leq (h_{\text{max}}/\pi)^\sigma (1 + 5M_2/2) \|u - z_{\text{nc}}\|_{\text{pw}} + C_{\text{dF}} \|\delta \lambda u\|_{L^2(\Omega)}.$$

The approximation property (J2) leads to $\|z_{\text{nc}} - Jz_{\text{nc}}\|_{L^2(\Omega)} \leq M_2 h_{\text{max}}^m \|u - z_{\text{nc}}\|_{\text{pw}}$. This, $0 < \sigma \leq 1 \leq m$, $h_{\text{max}} < 1$, and a triangle inequality show

$$\begin{aligned} \|u - z_{\text{nc}}\|_{L^2(\Omega)} &\leq \|Jz_{\text{nc}} - z_{\text{nc}}\|_{L^2(\Omega)} + \|u - Jz_{\text{nc}}\|_{L^2(\Omega)} \\ &\leq C_7 \left(h_{\text{max}}^\sigma \|u - z_{\text{nc}}\|_{\text{pw}} + \|\delta \lambda u\|_{L^2(\Omega)} \right) \end{aligned}$$

with the constant $C_7 := \max \{C(\sigma)(1 + 5M_2/2)/\pi^\sigma + M_2, C(\sigma)C_{\text{dF}}\}$. □

5.4.3.4 Proof of (5.37) and (5.40) for the energy error control

Recall δ from Remark 5.22 and that $z_{\text{nc}} \in V(\mathcal{T})$ is the solution to (5.34).

Lemma 5.32. *There exists a constant $C_8 > 0$ that satisfies (5.37) for all $\mathcal{T} \in \mathbb{T}(\varepsilon_2)$, i.e., $C_8^{-1} \|u - z_{\text{nc}}\|_{\text{pw}} \leq \|u - Iu\|_{\text{pw}} + \|\delta \lambda u\|_{L^2(\Omega)}$.*

Proof. Elementary algebra with $a_{\text{pw}}(z_{\text{nc}}, u) = a_{\text{pw}}(z_{\text{nc}}, Iu)$ from Corollary 5.1.c shows

$$\begin{aligned} \|u - z_{\text{nc}}\|_{\text{pw}}^2 &= a(u, u - JIu) + a_{\text{pw}}(u, Jz_{\text{nc}} - z_{\text{nc}}) + a(u, JIu - Jz_{\text{nc}}) \\ &\quad + a_{\text{pw}}(z_{\text{nc}}, z_{\text{nc}} - Iu). \end{aligned} \quad (5.52)$$

Corollary 5.1.c and Corollary 5.2.b control the first two terms in (5.52) via

$$\begin{aligned} & a(u, u - JIu) + a_{\text{pw}}(u, Jz_{\text{nc}} - z_{\text{nc}}) \\ &= a_{\text{pw}}(u, u - Iu) + a_{\text{pw}}(u, Iu - JIu) + a_{\text{pw}}(u, Jz_{\text{nc}} - z_{\text{nc}}) \\ &\leq (1 + M_2) \|u - Iu\|_{\text{pw}}^2 + M_2 \|u - Iu\|_{\text{pw}} \|u - z_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

Recall that (λ, u) is an eigenpair of (5.1) and z_{nc} satisfies (5.34). Consequently,

$$\begin{aligned} a(u, JIu - Jz_{\text{nc}}) + a_{\text{pw}}(z_{\text{nc}}, z_{\text{nc}} - Iu) &= b(\lambda u, JIu - Jz_{\text{nc}} + (1 + \delta)(z_{\text{nc}} - Iu)) \\ &= \lambda b(u, (J - 1)(Iu - z_{\text{nc}})) + b(\delta \lambda u, z_{\text{nc}} - Iu). \end{aligned}$$

The combination of $\lambda \kappa_m^2 h_{\text{max}}^{2m} \leq 1/2$ and (5.51) with a triangle inequality implies

$$\lambda b(u, (J - 1)(Iu - z_{\text{nc}})) \leq \frac{M_2}{2} \|u - Iu\|_{\text{pw}} (\|u - Iu\|_{\text{pw}} + \|u - z_{\text{nc}}\|_{\text{pw}}).$$

Since the Cauchy-Schwarz and triangle inequality show that $b(\delta \lambda u, z_{\text{nc}} - Iu) \leq \|\delta \lambda u\|_{L^2(\Omega)} (\|u - z_{\text{nc}}\|_{L^2(\Omega)} + \|u - Iu\|_{L^2(\Omega)})$, (I1) provides the first and (5.36) the second estimate in

$$\begin{aligned} b(\delta \lambda u, z_{\text{nc}} - Iu) &\leq \|\delta \lambda u\|_{L^2(\Omega)} (\|u - z_{\text{nc}}\|_{L^2(\Omega)} + \kappa_m h_{\text{max}}^m \|u - Iu\|_{\text{pw}}) \\ &\leq \|\delta \lambda u\|_{L^2(\Omega)} (C_7 h_{\text{max}}^\sigma \|u - z_{\text{nc}}\|_{\text{pw}} + C_7 \|\delta \lambda u\|_{L^2(\Omega)} + \kappa_m h_{\text{max}}^m \|u - Iu\|_{\text{pw}}). \end{aligned}$$

Since $h_{\text{max}}^m \|u - Iu\|_{\text{pw}} \leq h_{\text{max}}^\sigma \|u - z_{\text{nc}}\|_{\text{pw}}$ from Corollary 5.1.a, a weighted Young inequality shows $b(\delta \lambda u, z_{\text{nc}} - Iu) \leq ((C_7 + \kappa_m)^2 h_{\text{max}}^{2\sigma} + C_7) \|\delta \lambda u\|_{L^2(\Omega)}^2 + \|u - z_{\text{nc}}\|_{\text{pw}}^2/4$. The substitution of the displayed estimates in (5.52) shows

$$\begin{aligned} \|u - z_{\text{nc}}\|_{\text{pw}}^2 &\leq (1 + 3M_2/2) \|u - Iu\|_{\text{pw}}^2 + 3M_2/2 \|u - Iu\|_{\text{pw}} \|u - z_{\text{nc}}\|_{\text{pw}} \\ &\quad + ((C_7 + \kappa_m)^2 h_{\text{max}}^{2\sigma} + C_7) \|\delta \lambda u\|_{L^2(\Omega)}^2 + \|u - z_{\text{nc}}\|_{\text{pw}}^2/4. \end{aligned}$$

This and $3M_2/2 \|u - Iu\|_{\text{pw}} \|u - z_{\text{nc}}\|_{\text{pw}} \leq 9M_2^2/4 \|u - Iu\|_{\text{pw}}^2 + 1/4 \|u - z_{\text{nc}}\|_{\text{pw}}^2$ conclude the proof of (5.37) with $C_8^2 := 2 \max\{1 + 3M_2/2 + 9M_2^2/4, (C_7 + \kappa_m)^2 h_{\text{max}}^{2\sigma} + C_7\}$, note that $h_{\text{max}}^{2\sigma} \leq 1$. $\square$

The following Lemma 5.33 is closely related to the efficiency estimate of the volume contribution for the source problem in Lemma 5.15.

Lemma 5.33 (efficiency of the volume contribution). *There exists a constant $C_9 > 0$ that satisfies (5.40), i.e., $C_9^{-1} \|h_{\mathcal{T}}^m u_{\text{nc}}\|_{L^2(\Omega)} \leq h_{\text{max}}^m \|u - u_{\text{nc}}\|_{L^2(\Omega)} + \lambda^{-1} \|u - Iu\|_{\text{pw}}$.*

Proof. In the proof of Lemma 5.15 the interpolation $\Pi_m f \in P_m(\mathcal{T})$ can be replaced by $u_{\text{nc}} \in V(\mathcal{T}) \subset P_m(\mathcal{T})$. For any $T \in \mathcal{T}$, Verfürth's bubble-function methodology with $b_T := (n + 1)^{m(n+1)} \prod_{z \in \mathcal{V}(T)} \varphi_z^m \in P_{m(n+1)}(T) \cap W_0^{m,\infty}(T) \subset V$ and $\|p\|_{L^2(T)} \leq c_b \|p\|_{b_T}$ for all $p \in P_m(\mathcal{T})$ leads to

$$c_b^{-2} \|u_{\text{nc}}\|_{L^2(T)}^2 \leq \|u_{\text{nc}}\|_{b_T}^2 = (u_{\text{nc}}, u)_{b_T} - (u_{\text{nc}}, u - u_{\text{nc}})_{b_T}$$

instead of (5.25). A Cauchy-Schwarz inequality and $\|b_T\|_\infty = 1$ imply that $(u_{\text{nc}}, u - u_{\text{nc}})_{b_T} \leq \|u_{\text{nc}}\|_{L^2(T)} \|u - u_{\text{nc}}\|_{L^2(T)}$ and an integration by parts proves $D^m(b_T u_{\text{nc}})$ is L^2 -orthogonal to $P_0(T)$. Since (λ, u) is an eigenpair of (5.1) and $b_T u_{\text{nc}} \in W_0^{m,\infty}(T) \subset V$ an admissible test-function with support $T \in \mathcal{T}$, the orthogonality of $D^m(b_T u_{\text{nc}})$ to $D_{\text{pw}}^m Iu \in P_0(\mathcal{T})$ implies that

$$\lambda b(u, b_T u_{\text{nc}}) = a_{\text{pw}}(u, b_T u_{\text{nc}}) = (D^m u, D^m(b_T u_{\text{nc}}))_{L^2(T)} \leq |u - Iu|_{H^m(T)} |b_T u_{\text{nc}}|_{H^m(T)}.$$

An inverse estimate for polynomials in $P_{m(n+2)}(T)$ with the constant c_{inv} and $\|b_T\|_\infty = 1$ provide $c_b^{-2} h_T^m \|u_{\text{nc}}\|_{L^2(T)} \leq h_T^m \|u - u_{\text{nc}}\|_{L^2(T)} + c_{\text{inv}} \lambda^{-1} |u - Iu|_{H^m(T)}$ for all $T \in \mathcal{T}$. The sum over all $T \in \mathcal{T}$ concludes the proof $C_9 = c_b^2 \max\{1, c_{\text{inv}}\} = C_1$ as in Lemma 5.15. $\square$

5.4.3.5 Proof of Theorem 5.20.c

This section proves first the central estimate (5.42), i.e., there exist $\varepsilon_4, C_{10} > 0$, such that for any $\mathcal{T} \in \mathbb{T}(\varepsilon_4)$ it holds $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_{10} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}}$ as outlined in Subsection 5.4.3.1.

Proof of (5.42) for $\varepsilon_4 > 0$. The estimates (5.35)–(5.37) imply $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_6 C_7 (C_8 h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + (1 + C_8) \|\delta \lambda u\|_{L^2(\Omega)})$ in (5.38). Recall the abbreviation $c_3 := 2\lambda^2 \kappa_m^2 C_6 C_7 (1 + C_8)$. A triangle inequality, $C_\delta := 2\lambda \kappa_m^2$ in Remark 5.22, and (5.38) show

$$\begin{aligned} \|\delta \lambda u\|_{L^2(\Omega)} &\leq 2\lambda^2 \kappa_m^2 h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)} + \|\delta \lambda u_{\text{nc}}\|_{L^2(\Omega)} \\ &\leq \frac{c_3 C_8 h_{\text{max}}^{2m}}{1 + C_8} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + c_3 h_{\text{max}}^{2m} \|\delta \lambda u\|_{L^2(\Omega)} + \|\delta \lambda u_{\text{nc}}\|_{L^2(\Omega)}. \end{aligned}$$

Since $0 < \varepsilon_3 := \min\{\varepsilon_2, (2c_3)^{-1/2m}\}$ ensures $c_3 h_{\text{max}}^{2m} \leq 1/2$ for all $\mathcal{T} \in \mathbb{T}(\varepsilon_3)$, the absorption of $c_3 h_{\text{max}}^{2m} \|\delta \lambda u\|_{L^2(\Omega)}$ yields $\|\delta \lambda u\|_{L^2(\Omega)} \leq C_8 / (1 + C_8) h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + 2\|\delta \lambda u_{\text{nc}}\|_{L^2(\Omega)}$ in (5.39), which recasts (5.38) as

$$C_6^{-1} C_7^{-1} \|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq 2C_8 h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + 2(1 + C_8) \lambda \|\delta u_{\text{nc}}\|_{L^2(\Omega)}.$$

Remark 5.22 and the volume efficiency estimate in (5.40) control the last term in

$$\begin{aligned} (2\kappa_m^2)^{-1} \|\delta u_{\text{nc}}\|_{L^2(\Omega)} &\leq \lambda h_{\text{max}}^m \|h_{\mathcal{T}}^m u_{\text{nc}}\|_{L^2(\Omega)} \\ &\leq C_9 \lambda h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)} + C_9 h_{\text{max}}^m \|u - Iu\|_{\text{pw}}. \end{aligned}$$

Given $c_4 := 4\lambda^2 \kappa_m^2 C_6 C_7 (1 + C_8) C_9$ and $C_{10} := 2C_6 C_7 (2C_8 + 4\lambda \kappa_m^2 (1 + C_8) C_9)$, the last two displayed estimates lead to $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_{10} / 2 h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} + c_4 h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)}$ in (5.41). Since $\varepsilon_4 := \min\{\varepsilon_3, (2c_4)^{-1/2m}\} < 1$ ensures $c_4 h_{\text{max}}^{2m} \leq 1/2$, the absorption of $c_4 h_{\text{max}}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)}$ concludes the proof of the estimate (5.42). $\square$

Recall $0 < \varepsilon_5 \leq \varepsilon_4$ such that $b(u, u_{\text{nc}}) > 0$ for any $\mathcal{T} \in \mathbb{T}(\varepsilon_5)$.

Proof of Theorem 5.20.c for $\varepsilon_5 > 0$. Recall $\lambda_h \leq \lambda$ and $\|u\|_{L^2(\Omega)} = \|u_{\text{nc}}\|_{L^2(\Omega)} = 1$. Since (5.42) is shown above, the analysis of LHS $:= \lambda - \lambda_h + \|u - u_{\text{nc}}\|_{\text{pw}}^2 + \|u_{\text{nc}}\|_{\delta}^2$ on the left-hand side of Theorem 5.20.c remains.

The continuous eigenpair (λ, u) in (5.1) satisfies $\lambda = \|u\|^2$. The discrete eigenpair $(\lambda_h, u_{\text{nc}})$ solves (5.15) and so $\lambda_h = \|u_{\text{nc}}\|_{\text{pw}}^2 / \|u_{\text{nc}}\|_{1+\delta}^2$ with $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$. Hence,

$$\|u - u_{\text{nc}}\|_{\text{pw}}^2 = \lambda - 2a_{\text{pw}}(u, u_{\text{nc}}) + \lambda_h \|u_{\text{nc}}\|_{1+\delta}^2$$

and $\|u_{\text{nc}}\|_{1+\delta}^2 - 1 = b(\delta u_{\text{nc}}, u_{\text{nc}}) = \|u_{\text{nc}}\|_{\delta}^2$. These observations recast LHS in

$$\text{LHS} := \lambda - \lambda_h + \|u - u_{\text{nc}}\|_{\text{pw}}^2 + \|u_{\text{nc}}\|_{\delta}^2 = 2\lambda - 2a_{\text{pw}}(u, u_{\text{nc}}) + (1 + \lambda_h) \|u_{\text{nc}}\|_{\delta}^2.$$

Since $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1 = \|u\|_{L^2(\Omega)}$, $2b(u, u - u_{\text{nc}}) = \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2$ and $\lambda = \lambda b(u, u_{\text{nc}}) + \lambda b(u, u - u_{\text{nc}})$. This and that u is the eigenfunction in (5.1) show

$$\lambda = \lambda b(u, u_{\text{nc}} - Ju_{\text{nc}}) + a_{\text{pw}}(u, Ju_{\text{nc}}) + \lambda/2 \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2.$$

The combination of the last two displayed identities eventually leads to

$$\begin{aligned} \text{LHS} = & (1 + \lambda_h) \|u_{\text{nc}}\|_{\delta}^2 + \lambda \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2 + 2\lambda b(u, u_{\text{nc}} - Ju_{\text{nc}}) \\ & + 2a_{\text{pw}}(u, Ju_{\text{nc}} - u_{\text{nc}}). \end{aligned} \quad (5.53)$$

Recall $\sqrt{\delta} \leq \sqrt{2\lambda\kappa_m} h_{\mathcal{T}}^m \leq 1$ from Remark 5.22. This and the efficiency estimate in (5.40) imply that

$$\|u_{\text{nc}}\|_{\delta} \leq \sqrt{2\lambda\kappa_m} \|h_{\mathcal{T}}^m u_{\text{nc}}\|_{L^2(\Omega)} \leq C_9 \|u - u_{\text{nc}}\|_{L^2(\Omega)} + \sqrt{2/\lambda\kappa_m} C_9 \|u - Iu\|_{\text{pw}}$$

The estimate (5.42) proven above controls $\|u - u_{\text{nc}}\|_{L^2(\Omega)} \leq C_{10} h_{\text{max}}^{\sigma} \|u - Iu\|_{\text{pw}}$. The bound $2\lambda\kappa_m^2 h_{\text{max}}^{2m} \leq 1$ and the estimate (5.51) show that $2\lambda b(u, u_{\text{nc}} - Ju_{\text{nc}}) \leq M_2 \|u - Iu\|_{\text{pw}} \|u - u_{\text{nc}}\|_{\text{pw}}$. Corollary 5.2.b bounds the last term on the right hand-side of (5.53) in $2a_{\text{pw}}(u, Ju_{\text{nc}} - u_{\text{nc}}) \leq 2M_2 \|u - Iu\|_{\text{pw}} \|u - u_{\text{nc}}\|_{\text{pw}}$. Since $\lambda_h \leq \lambda$, the combination of these estimates lead in (5.53) to

$$\begin{aligned} \text{LHS} \leq & \left((1 + \lambda) C_9^2 (C_{10} h_{\text{max}}^{\sigma} + \sqrt{2/\lambda\kappa_m})^2 + \lambda C_{10}^2 h_{\text{max}}^{2\sigma} \right) \|u - Iu\|^2 \\ & + 3M_2 \|u - Iu\|_{\text{pw}} \|u - u_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

A Young inequality and the absorption of $\|u - u_{\text{nc}}\|_{\text{pw}}^2/2$ conclude the proof of Theorem 5.20.c with $C_0 := \max\{C_{10}^2, 2((1 + \lambda)C_9^2(C_{10} h_{\text{max}}^{\sigma} + \sqrt{2/\lambda\kappa_m})^2 + \lambda C_{10}^2 h_{\text{max}}^{2\sigma}) + 9M_2^2\}$. $\square$

5.5 Optimality of the adaptive algorithm

At first glance the discrete problem (5.2) involves a stabilization that is expected to generate the additional term $\kappa_m^{-2}|T|^{-2m/n}\|u_{\text{pw}} - u_{\text{nc}}\|_{L^2(T)}^2$ in the error estimator. The negative power of the mesh-size in the latter term prevents a reduction property [MNS02; Car+14a; CR17] and has to be circumvented. Since (5.2) is close to a nonconforming discrete eigenvalue problem without stabilization, see Corollary 5.28, the perturbation analysis eventually shows efficiency and reliability of the nonconforming error estimator (5.54) provided the mesh-size is sufficiently small. The careful verification relies on the medius analysis in the previous Subsection 5.4.3.

This section verifies the axioms of adaptivity from Chapter 6 sufficient for optimal convergence rates of the adaptive Algorithm 2 below to prove the following Theorem 5.34 in Subsection 5.5.5.

Theorem 5.34 (rate optimality of Algorithm 2). *Suppose that $\lambda_k = \lambda$ is a simple eigenvalue of (5.1), then there exist $\varepsilon > 0$ and $0 < \theta_0 \leq 1$ such that any initial triangulation $\mathcal{T}_0 \in \mathbb{T}(\varepsilon) := \{\mathcal{T} \in \mathbb{T} : h_{\max} := \max_{T \in \mathcal{T}} h_T \leq \varepsilon\}$ and bulk parameter θ with $0 < \theta \leq \theta_0$ in the adaptive Algorithm 2 imply, that the output $(\mathcal{T}_\ell)_{\ell \in \mathbb{N}_0}$ and $(\eta_\ell)_{\ell \in \mathbb{N}_0}$ of Algorithm 2 satisfies*

$$\sup_{\ell \in \mathbb{N}_0} (1 + |\mathcal{T}_\ell| - |\mathcal{T}_0|)^s \eta_\ell \approx \sup_{N \in \mathbb{N}_0} (1 + N)^s \min\{\eta(\mathcal{T}) : \mathcal{T} \in \mathbb{T} \text{ with } |\mathcal{T}| \leq |\mathcal{T}_0| + N\}$$

for any $s > 0$.

5.5.1 Adaptive algorithm

The adaptive algorithm [Dör96; MNS02; Car+14a; CR17] is based on the refinement indicator $\eta(T)$ defined in (5.54) below for any admissible triangulation $\mathcal{T} \in \mathbb{T}$ and any simplex $T \in \mathcal{T}$. Let $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ denote the k -th eigenpair of (5.2) with $\lambda_h := \lambda_h(k)$ and $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h$. For any simplex $T \in \mathcal{T}$ with volume $|T|$ and set of faces $\mathcal{F}(T)$, the local estimator contribution $\eta^2(T) = (\eta(T))^2$ reads

$$\eta^2(T) = |T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 + |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \| [D_{\text{pw}}^m u_{\text{nc}}]_F \times \nu_F \|_{L^2(F)}^2 \quad (5.54)$$

with the tangential components $[D_{\text{pw}}^m u_{\text{nc}}]_F \times \nu_F$ of the jump $[D_{\text{pw}}^m u_{\text{nc}}]_F$ along any face $F \in \mathcal{F}(T)$ and the (piecewise) gradient $D_{\text{pw}}^1 = \nabla_{\text{pw}}$ ($m = 1$) or Hessian D_{pw}^2 ($m = 2$). The global estimator for $\mathcal{T} \in \mathbb{T}$ reads $\eta^2(\mathcal{T}) = (\eta(\mathcal{T}))^2 := \sum_{T \in \mathcal{T}} \eta^2(T)$. Let $\mathbb{T} := \mathbb{T}(\mathcal{T}_0)$ denote the set of all admissible regular triangulations computed by successive newest-vertex bisection (NVB) [Ste08; GSS14] of a regular initial triangulation $\mathcal{T}_0$ of $\Omega \subset \mathbb{R}^3$. The AFEM algorithm with Dörfler marking and newest-vertex bisection abbreviates $\eta_\ell(T)$ for any $T \in \mathcal{T} := \mathcal{T}_\ell \in \mathbb{T}$ and $\eta_\ell^2 :=$

$\eta^2(\mathcal{T}_\ell) := \sum_{T \in \mathcal{T}_\ell} \eta_\ell^2(T)$. The selection of the set $\mathcal{M}_\ell$ in the step Mark of Algorithm 2 with *minimal cardinality* is possible at linear cost [PP20].

Algorithm 2 AFEM4EVP

Input: regular triangulation $\mathcal{T}_0$ and parameters $0 < \theta \leq 1$ and $k \in \mathbb{N}$
for $\ell = 0, 1, 2, \dots$ **do**
 Solve the discrete problem (5.2) exactly and compute the k -th algebraic eigenpair $(\lambda_\ell(k), u_\ell(k))$ with $\mathbf{u}_\ell(k) = (u_{\text{pw}}, u_{\text{nc}}) \in P_m(\mathcal{T}_\ell) \times V(\mathcal{T}_\ell)$ and $\mathcal{T}$ replaced by $\mathcal{T}_\ell$
 Compute $\eta_\ell(T)$ for any $T \in \mathcal{T}_\ell$ from (5.54) with $(\lambda_h, u_{\text{nc}}, \mathcal{T})$ replaced by $(\lambda_\ell(k), u_{\text{nc}}, \mathcal{T}_\ell)$
 Mark minimal subset $\mathcal{M}_\ell \subseteq \mathcal{T}_\ell$ with $\theta \eta_\ell^2 \leq \sum_{T \in \mathcal{M}_\ell} \eta_\ell^2(T)$
 Refine $\mathcal{T}_\ell$ with newest-vertex bisection to compute $\mathcal{T}_{\ell+1}$ with $\mathcal{M}_\ell \subseteq \mathcal{T}_\ell \setminus \mathcal{T}_{\ell+1}$
od
end for
Output: sequence of triangulations $(\mathcal{T}_\ell)_{\ell \in \mathbb{N}_0}$ with $(\lambda_\ell(k), \mathbf{u}_\ell(k))_{\ell \in \mathbb{N}_0}$ and $(\eta_\ell)_{\ell \in \mathbb{N}_0}$

The estimator η in (5.54) in Algorithm 2 is efficient and reliable, if the maximal mesh-size $h_{\max} \leq \varepsilon_6 := \min\{\varepsilon_5, (2\lambda(C_{10}^2 + \kappa_m M_2))^{-1/(2\sigma)}\}$.

Theorem 5.35 (reliability and efficiency). *There exist C_{rel} , C_{eff} , and $\varepsilon_6 > 0$ such that for any $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ holds*

$$C_{\text{eff}}^{-1} \eta(\mathcal{T}) \leq \|u - u_{\text{nc}}\|_{\text{pw}} \leq C_{\text{rel}} \eta(\mathcal{T}).$$

Proof of reliability. The reliability can be seen as consequence from Lemma 5.39 below for a sequence $\widehat{\mathcal{T}} = \widehat{\mathcal{T}}_\ell$ of uniform mesh-refinements of $\mathcal{T}$ in the limit as $\widehat{h}_{\max} \rightarrow 0$ for $\ell \rightarrow \infty$ and $\|u - \widehat{u}_{\text{nc}}\|_{\text{pw}} \rightarrow 0$ from Theorem 5.20.c. However, a direct proof is added here for convenient reading. Since (λ, u) is an eigenpair of (5.1) and $(\lambda_h, u_{\text{nc}})$ solves (5.15), Corollary 5.1.c and (J1) lead to

$$\begin{aligned} a_{\text{pw}}(u - u_{\text{nc}}, u - Ju_{\text{nc}}) &= a(u, u - Ju_{\text{nc}}) - a_{\text{pw}}(u_{\text{nc}}, I(u - Ju_{\text{nc}})) \\ &= \lambda b(u, u - Ju_{\text{nc}}) - \lambda_h(u_{\text{nc}}, Iu - u_{\text{nc}})_{(1+\delta)} \\ &= b(\lambda u - \lambda_h u_{\text{nc}}, u - u_{\text{nc}}) + b(\lambda u, u_{\text{nc}} - Ju_{\text{nc}}) \\ &\quad + b(\lambda_h u_{\text{nc}}, u - Iu) + b(\delta \lambda_h u_{\text{nc}}, u_{\text{nc}} - Iu). \end{aligned} \quad (5.55)$$

Elementary algebra with normalization of u and u_{nc} such that $b(u - u_{\text{nc}}, u + u_{\text{nc}}) = 0$, $\lambda_h \leq \lambda$, and the central estimate (5.42) in Theorem 5.20.c show

$$\begin{aligned} b(\lambda u - \lambda_h u_{\text{nc}}, u - u_{\text{nc}}) &= b\left(\frac{\lambda + \lambda_h}{2}(u - u_{\text{nc}}) + \frac{\lambda - \lambda_h}{2}(u + u_{\text{nc}}), u - u_{\text{nc}}\right) \\ &= \frac{\lambda + \lambda_h}{2} \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2 \leq C_{10}^2 \lambda h_{\max}^{2\sigma} \|u - Iu\|_{\text{pw}}^2. \end{aligned} \quad (5.56)$$

Corollary 5.2.a guarantees $b(\lambda u, u_{\text{nc}} - Ju_{\text{nc}}) \leq \lambda h_{\text{max}}^{2m} \kappa_m M_2 \|u - u_{\text{nc}}\|_{\text{pw}}^2$. A weighted Cauchy-Schwarz inequality and (I1) lead to

$$b(\lambda_h u_{\text{nc}}, u - Iu) \leq \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \|h_{\mathcal{T}}^{-m}(u - Iu)\|_{L^2(\Omega)} \leq \kappa_m \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \|u - Iu\|_{\text{pw}}.$$

Since $\delta \leq C_{\delta} h_{\mathcal{T}}^{2m}$ from Remark 5.22 and $h_{\text{max}}^m \leq h_{\text{max}}^{\sigma}$, the Cauchy-Schwarz inequality, (I1), a triangle inequality, and (5.42) yield

$$\begin{aligned} b(\delta \lambda_h u_{\text{nc}}, Iu - u_{\text{nc}}) &\leq C_{\delta} h_{\text{max}}^m \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} (\|u - Iu\|_{L^2(\Omega)} + \|u - u_{\text{nc}}\|_{L^2(\Omega)}) \\ &\leq C_{\delta} (\kappa_m + C_{10}) h_{\text{max}}^{m+\sigma} \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \|u - Iu\|_{\text{pw}}. \end{aligned}$$

The combination of the last four inequalities with the best approximation property in Corollary 5.1.a bounds the terms in (5.55) such that

$$\begin{aligned} a_{\text{pw}}(u - u_{\text{nc}}, u - Ju_{\text{nc}}) &\leq (C_{10}^2 + \kappa_m M_2) \lambda h_{\text{max}}^{2\sigma} \|u - u_{\text{nc}}\|_{\text{pw}}^2 \\ &\quad + (\kappa_m + C_{\delta}(\kappa_m + C_{10}) h_{\text{max}}^{m+\sigma}) \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \|u - u_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

On the other hand, the Cauchy-Schwarz inequality and (J2) show

$$a_{\text{pw}}(u - u_{\text{nc}}, Ju_{\text{nc}} - u_{\text{nc}}) \leq \|u - u_{\text{nc}}\|_{\text{pw}} \left(M_1 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2}.$$

The linearity of a_{pw} asserts $\|u - u_{\text{nc}}\|_{\text{pw}}^2 = a_{\text{pw}}(u - u_{\text{nc}}, Ju_{\text{nc}} - u_{\text{nc}}) + a_{\text{pw}}(u - u_{\text{nc}}, u - Ju_{\text{nc}})$. The combination of the last three estimates implies

$$\begin{aligned} \|u - u_{\text{nc}}\|_{\text{pw}} &\leq (C_{10}^2 + \kappa_m M_2) \lambda h_{\text{max}}^{2\sigma} \|u - u_{\text{nc}}\|_{\text{pw}} \\ &\quad + (\kappa_m + C_{\delta}(\kappa_m + C_{10}) h_{\text{max}}^{m+\sigma}) \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \\ &\quad + \left(M_1 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2 \right)^{1/2}. \end{aligned}$$

A reduction to $\varepsilon_6 := \min\{\varepsilon_5, (2\lambda(C_{10}^2 + \kappa_m M_2))^{-1/(2\sigma)}\}$ allows the absorption of $\lambda(C_{10}^2 + \kappa_m M_2) h_{\text{max}}^{2\sigma} \|u - u_{\text{nc}}\|_{\text{pw}} \leq \|u - u_{\text{nc}}\|_{\text{pw}}/2$. This leads to

$$\begin{aligned} \|u - u_{\text{nc}}\|_{\text{pw}}^2 &\leq 8(\kappa_m + C_{\delta}(\kappa_m + C_{10}) h_{\text{max}}^{m+\sigma})^2 \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)}^2 \\ &\quad + 8M_1 \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2. \end{aligned}$$

Since $\mathcal{T} \in \mathbb{T}$ is shape-regular with $h_T^{2m} \leq C_{\text{sr}}^{2m} |T|^{2m/n}$ for any $T \in \mathcal{T}$, this and the definition of η in (5.54) conclude the proof with $C_{\text{rel}}^2 := 8 \max\{M_1, C_{\text{sr}}^{2m}(\kappa_m + C_{\delta}(\kappa_m + C_{10}) h_{\text{max}}^{m+\sigma})^2\}$, note $h_{\text{max}}^{m+\sigma} \leq 1$. $\square$

Proof of efficiency. On the one hand side, the condition (J2) guarantees

$$\frac{M_1}{M_2^2} \sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D_{\text{pw}}^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2 \leq \min_{v \in V} \|v - u_{\text{nc}}\|_{\text{pw}}^2 \leq \|u - u_{\text{nc}}\|_{\text{pw}}^2.$$

On the other hand, the combination of $|T|^{1/n} \leq h_T$, $\lambda_h \leq \lambda$, and the efficiency of the volume term (5.40) with $\|u - Iu\|_{\text{pw}} \leq \|u - u_{\text{nc}}\|_{\text{pw}}$ from Corollary 5.1.a implies that

$$\begin{aligned} \sum_{T \in \mathcal{T}} |T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 &\leq \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\Omega)}^2 \\ &\leq 2C_9^2 \left(\lambda^2 h_{\max}^{2m} \|u - u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|u - u_{\text{nc}}\|_{\text{pw}}^2 \right). \end{aligned}$$

The central estimate (5.42) in Theorem 5.20.c concludes the proof with $C_{\text{eff}}^2 := M_2^2/M_1 + 2C_9^2 + 2C_9^2 C_{10}^2 \lambda^2 h_{\max}^{2m+2\sigma}$ and $h_{\max}^{2m+2\sigma} \leq 1$. $\square$

In the following, choose $\varepsilon_6 > 0$ from Theorem 5.35 and suppose $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$. The axioms of adaptivity depend on the following 2-level notation abbreviated in Table 5.1, which concerns one coarse triangulation $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ and one fine triangulation $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$. Let $(\lambda, u) \in \mathbb{R}^+ \times V$ denote the k -th continuous eigenpair of (5.1) with a *simple* eigenvalue $\lambda \equiv \lambda_k$ and the normalization $\|u\|_{L^2(\Omega)} = 1$. Note that both triangulation, $\mathcal{T}$ and $\hat{\mathcal{T}}$, belong to $\mathbb{T}(\varepsilon_6)$ and so the results from Theorem 5.20 and Theorem 5.35 hold for both discrete eigenpairs.

Definition 5.36 (2-level notation). Let $(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ (resp. $(\hat{\lambda}_h, \hat{u}_h) \in \mathbb{R}^+ \times \hat{V}_h$) with $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in V_h := P_m(\mathcal{T}) \times V(\mathcal{T})$ (resp. $\hat{u}_h = (\hat{u}_{\text{pw}}, \hat{u}_{\text{nc}}) \in \hat{V}_h := P_m(\hat{\mathcal{T}}) \times V(\hat{\mathcal{T}})$) denote the k -th discrete eigenpair of (5.2) with the simple algebraic eigenvalue $\lambda_h \equiv \lambda_h(k)$ (resp. $\hat{\lambda}_h \equiv \hat{\lambda}_h(k)$), the normalization $\|u_{\text{nc}}\|_{L^2(\Omega)} = 1$ (resp. $\|\hat{u}_{\text{nc}}\|_{L^2(\Omega)} = 1$), and the sign convention $b(u, u_{\text{nc}}) > 0$ (resp. $b(u, \hat{u}_{\text{nc}}) > 0$). Recall $\hat{h}_{\max} := \max_{T \in \hat{\mathcal{T}}} h_T \leq h_{\max} := \max_{T \in \mathcal{T}} h_T \leq \varepsilon_6$, $\lambda_h, \hat{\lambda}_h \leq \lambda \leq 2\lambda_h, 2\hat{\lambda}_h$ from Theorem 5.20.a–b, and δ from Remark 5.22 with its analogue $\hat{\delta} := (1 - \hat{\lambda}_h \kappa_m^2 h_{\hat{\mathcal{T}}}^{2m})^{-1} - 1 \in P_0(\hat{\mathcal{T}})$ on the fine level. The constant $C_{\delta} := 2\lambda \kappa_m^2$ satisfies $\delta \leq C_{\delta} h_{\mathcal{T}}^{2m}$ and $\hat{\delta} \leq C_{\delta} h_{\hat{\mathcal{T}}}^{2m}$. Recall the estimator $\eta^2(T)$ for any $T \in \mathcal{T}$ from (5.54) and define $\hat{\eta}^2(T)$, for any $T \in \hat{\mathcal{T}}$ with volume $|T|$ and the set of faces $\hat{\mathcal{F}}(T)$, by

$$\hat{\eta}^2(T) := |T|^{2m/n} \|\hat{\lambda}_h \hat{u}_{\text{nc}}\|_{L^2(T)}^2 + |T|^{1/n} \sum_{F \in \hat{\mathcal{F}}(T)} \|[D_{\text{pw}}^m \hat{u}_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2. \quad (5.57)$$

The sum conventions $\eta^2(\mathcal{M}) := \sum_{T \in \mathcal{M}} \eta^2(T)$ for $\mathcal{M} \subset \mathcal{T}$ and $\hat{\eta}^2(\hat{\mathcal{M}}) := \sum_{T \in \hat{\mathcal{M}}} \hat{\eta}^2(T)$ for $\hat{\mathcal{M}} \subset \hat{\mathcal{T}}$ from Table 5.1 apply throughout this section. Abbreviate the distance function

$$\delta^2(\mathcal{T}, \hat{\mathcal{T}}) := \|\lambda_h u_{\text{nc}} - \hat{\lambda}_h \hat{u}_{\text{nc}}\|_{L^2(\Omega)}^2 + \|u_{\text{nc}} - \hat{u}_{\text{nc}}\|_{\text{pw}}^2. \quad (5.58)$$

5.5.2 Stability and reduction

This section provides the proof of the axioms (A1)–(A2).

$(\lambda_h, u_h) \in \mathbb{R}^+ \times V_h$ k -th eigenpair in (5.2) with $u_h = (u_{\text{pw}}, u_{\text{nc}}) \in P_m(\mathcal{T}) \times V(\mathcal{T})$ $\ u_{\text{nc}}\ _{L^2(\Omega)} = 1, b(u, u_{\text{nc}}) > 0$ $\lambda_h \leq \lambda \leq 2\lambda_h$ $h_{\max} := \max_{T \in \mathcal{T}} h_T \leq \varepsilon_6$ $\delta := (1 - \lambda_h \kappa_m^2 h_{\mathcal{T}}^{2m})^{-1} - 1 \leq C_\delta h_{\mathcal{T}}^{2m} \leq 1$ $\eta^2(T)$ from (5.54) for $T \in \mathcal{T}$ $\eta^2(\mathcal{M}) := \sum_{T \in \mathcal{M}} \eta^2(T)$ for $\mathcal{M} \subseteq \mathcal{T}$	$(\hat{\lambda}_h, \hat{u}_h) \in \mathbb{R}^+ \times \hat{V}_h$ k -th eigenpair in (5.2) with $\hat{u}_h = (\hat{u}_{\text{pw}}, \hat{u}_{\text{nc}}) \in P_m(\hat{\mathcal{T}}) \times V(\hat{\mathcal{T}})$ $\ \hat{u}_{\text{nc}}\ _{L^2(\Omega)} = 1, b(u, \hat{u}_{\text{nc}}) > 0$ $\hat{\lambda}_h \leq \lambda \leq 2\hat{\lambda}_h$ $\hat{h}_{\max} := \max_{T \in \hat{\mathcal{T}}} h_T \leq \varepsilon_6$ $\hat{\delta} := (1 - \hat{\lambda}_h \kappa_m^2 h_{\hat{\mathcal{T}}}^{2m})^{-1} - 1 \leq C_\delta h_{\hat{\mathcal{T}}}^{2m} \leq 1$ $\hat{\eta}^2(T)$ from (5.57) for $T \in \hat{\mathcal{T}}$ $\hat{\eta}^2(\hat{\mathcal{M}}) := \sum_{T \in \hat{\mathcal{M}}} \hat{\eta}^2(T)$ for $\hat{\mathcal{M}} \subseteq \hat{\mathcal{T}}$
--	--

Table 5.1: 2-level notation with respect to $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ (left) and an admissible refinement $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ (right) including some consequences from Theorem 5.20

Theorem 5.37 (stability and reduction). *There exist $\Lambda_1, \Lambda_2 > 0$, and $0 < \rho_2 := 2^{-1/(4n)} < 1$ such that, for any $\mathcal{T}$ and $\hat{\mathcal{T}}$ from Definition 5.36, the following holds*

$$(A1) \text{ Stability. } |\eta(\mathcal{T} \cap \hat{\mathcal{T}}) - \hat{\eta}(\mathcal{T} \cap \hat{\mathcal{T}})| \leq \Lambda_1 \delta(\mathcal{T}, \hat{\mathcal{T}}),$$

$$(A2) \text{ Reduction. } \hat{\eta}(\hat{\mathcal{T}} \setminus \mathcal{T}) \leq \rho_2 \eta(\mathcal{T} \setminus \hat{\mathcal{T}}) + \Lambda_2 \delta(\mathcal{T}, \hat{\mathcal{T}}).$$

Proof. A reverse triangle inequality in $\mathbb{R}^L$ for the number $L := |\mathcal{T} \cap \hat{\mathcal{T}}|$ of simplices in $\mathcal{T} \cap \hat{\mathcal{T}}$ and one for each common simplex $T \in \mathcal{T} \cap \hat{\mathcal{T}}$ and each of its faces $F \in \mathcal{F}(T)$ lead to

$$\begin{aligned}
& |\eta(\mathcal{T} \cap \hat{\mathcal{T}}) - \hat{\eta}(\mathcal{T} \cap \hat{\mathcal{T}})|^2 \\
& \leq \sum_{T \in \mathcal{T} \cap \hat{\mathcal{T}}} \left(|T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 - |T|^{2m/n} \|\hat{\lambda}_h \hat{u}_{\text{nc}}\|_{L^2(T)}^2 \right. \\
& \quad \left. + |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \| [D_{\text{pw}}^m u_{\text{nc}}]_F \times \nu_F \|_{L^2(F)}^2 - \| [D_{\text{pw}}^m \hat{u}_{\text{nc}}]_F \times \nu_F \|_{L^2(F)}^2 \right) \\
& \leq \sum_{T \in \mathcal{T} \cap \hat{\mathcal{T}}} \left(|T|^{2m/n} \|\lambda_h u_{\text{nc}} - \hat{\lambda}_h \hat{u}_{\text{nc}}\|_{L^2(T)}^2 + |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \| [D_{\text{pw}}^m (u_{\text{nc}} - \hat{u}_{\text{nc}})]_F \times \nu_F \|_{L^2(F)}^2 \right).
\end{aligned}$$

The discrete jump control from [CR17, Lem. 5.2] with constant $C_{\text{jc}}(\ell)$ (that only depends on the shape-regularity of $\mathbb{T}$ and the polynomial degree $\ell \in \mathbb{N}_0$) reads

$$\sum_{T \in \mathcal{T}} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[g]_F\|_{L^2(F)}^2 \leq C_{\text{jc}}(\ell)^2 \|g\|_{L^2(\Omega)}^2 \quad \text{for any } g \in P_\ell(\mathcal{T}) \text{ and } \mathcal{T} \in \mathbb{T}.$$

The tangential part of the jump is bounded by the whole jump and the discrete jump control holds in particular for $g := D_{\text{pw}}^m(u_{\text{nc}} - \hat{u}_{\text{nc}}) \in P_0(\hat{\mathcal{T}})$. Hence, the

combination of the two displayed estimates concludes the proof of (A1) with $\Lambda_1^2 = \max \left\{ \max_{T \in \mathcal{T}_0} |T|^{2m/n}, C_{\text{jc}}(0)^2 \right\}$.

For any simplex $K \in \mathcal{T} \setminus \widehat{\mathcal{T}}$, let $\widehat{\mathcal{T}}(K) := \{T \in \widehat{\mathcal{T}} : T \subset K\}$ denote its fine triangulation. The newest-vertex bisection guarantees $|T| \leq |K|/2$ for the volume $|T|$ of any $T \in \widehat{\mathcal{T}}(K)$. This, triangle inequalities, and $(a+b)^2 \leq (1+\beta)a^2 + (1+1/\beta)b^2$ for $a, b, \beta \geq 0$ show

$$\begin{aligned} \widehat{\eta}^2(\widehat{\mathcal{T}}(K)) &\leq \sum_{T \in \widehat{\mathcal{T}}(K)} \left(|T|^{2m/n} \left(\|\widehat{\lambda}_h \widehat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(T)} + \|\lambda_h u_{\text{nc}}\|_{L^2(T)} \right)^2 \right. \\ &\quad \left. + |T|^{1/n} \sum_{F \in \widehat{\mathcal{F}}(T)} \left(\| [D^m(\widehat{u}_{\text{nc}} - u_{\text{nc}})]_F \times \nu_F \|_{L^2(F)} + \| [D^m u_{\text{nc}}]_F \times \nu_F \|_{L^2(F)} \right)^2 \right) \\ &\leq (1+\beta) \left(\frac{|K|^{2m/n}}{2^{2m/n}} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 + \frac{|K|^{1/n}}{2^{1/n}} \sum_{F \in \mathcal{F}(K)} \| [D^m u_{\text{nc}}]_F \times \nu_F \|_{L^2(F)}^2 \right) \\ &\quad + (1+1/\beta) \sum_{T \in \widehat{\mathcal{T}}(K)} \left(|T|^{2m/n} \|\widehat{\lambda}_h \widehat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(T)}^2 \right. \\ &\quad \left. + |T|^{1/n} \sum_{F \in \widehat{\mathcal{F}}(T)} \| [D^m(\widehat{u}_{\text{nc}} - u_{\text{nc}})]_F \times \nu_F \|_{L^2(F)}^2 \right), \end{aligned}$$

since $D_{\text{pw}}^m u_{\text{nc}} \in P_0(\mathcal{T})$ does not jump along $F \in \widehat{\mathcal{F}} \setminus \mathcal{F}$. Hence, the definition of η in (5.54) and the choice $\beta = 2^{1/(2n)} - 1 > 0$ with $2^{-1/n}(1+\beta) = 2^{-1/(2n)}$ imply

$$\begin{aligned} \widehat{\eta}^2(\widehat{\mathcal{T}}(K)) &\leq 2^{-1/(2n)} \eta^2(K) + (1+1/\beta) \sum_{T \in \widehat{\mathcal{T}}(K)} \left(|T|^{2m/n} \|\lambda_h u_{\text{nc}} - \widehat{\lambda}_h \widehat{u}_{\text{nc}}\|_{L^2(T)}^2 \right. \\ &\quad \left. + |T|^{1/n} \sum_{F \in \widehat{\mathcal{F}}(T)} \| [D_{\text{pw}}^m(\widehat{u}_{\text{nc}} - u_{\text{nc}})]_F \times \nu_F \|_{L^2(F)}^2 \right), \end{aligned}$$

The summation over all $K \in \mathcal{T} \setminus \widehat{\mathcal{T}}$ and the above jump control conclude the proof of (A2) with $\Lambda_2^2 = 2^{1/(2n)} / (2^{1/(2n)} - 1) \max \left\{ \max_{T \in \mathcal{T}_0} |T|^{2m/n}, C_{\text{jc}}(0)^2 \right\}$ and $\rho_2^2 = 2^{-1/(2n)}$. The arguments for (A1)–(A2) are similar for other problems; cf., e.g., [Cas+08; Car+14a; CR17; CH18b] for more details. $\square$

5.5.3 Discrete reliability

This section proves (A3_ε). Recall that for any $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ Theorem 5.35 proves reliability and efficiency of the estimator in (5.54). Given the 2-level notation of Definition 5.36 with respect to $\mathcal{T}$ and $\widehat{\mathcal{T}}$, let $\mathcal{R}_1 := \{K \in \mathcal{T} : \exists T \in \mathcal{T} \setminus \widehat{\mathcal{T}} \text{ with } \text{dist}(K, T) = 0\} \subseteq \mathcal{T}$ denote the set of coarse but not fine simplices plus one layer of coarse simplices around.

Theorem 5.38 (discrete reliability). *There exist constants $\Lambda_3, M_3 > 0$ such that $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ with maximal mesh-size $h_{\max} \leq \varepsilon_6$ and $\epsilon_3 := M_3 h_{\max}^{2\sigma}$ imply*

(A3_ε) *Discrete reliability.* $\delta^2(\mathcal{T}, \hat{\mathcal{T}}) \leq \Lambda_3 \eta^2(\mathcal{R}_1) + \epsilon_3 \eta^2(\mathcal{T})$.

Lemma 5.39–5.40 prepare the proof of the discrete reliability in Theorem 5.38 below. Let $\hat{I} : V + V(\hat{\mathcal{T}}) \rightarrow V(\hat{\mathcal{T}})$ denote the interpolation operator on the fine level $\hat{\mathcal{T}}$. According to (I3) the interpolation I acts on non-refined simplices as identity in $V(\hat{\mathcal{T}})$ and $I \circ \hat{I} = I$. In other words, $(I - \hat{I})v = (I - 1)\hat{I}v$ vanishes on $\mathcal{T} \cap \hat{\mathcal{T}}$ for any $v \in V + V(\hat{\mathcal{T}})$. This and a Cauchy-Schwarz inequality show, for any $v \in V + V(\hat{\mathcal{T}})$ and any $w \in V + V(\mathcal{T}) + V(\hat{\mathcal{T}})$, that

$$\begin{aligned} |b((I - \hat{I})v, w)| &\leq \|(I - \hat{I})v\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \|w\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}, \\ |a_{\text{pw}}((I - \hat{I})v, w)| &\leq \|D_{\text{pw}}^m(I - \hat{I})v\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \|D_{\text{pw}}^m w\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \end{aligned} \quad (5.59)$$

with the abbreviation $\|\bullet\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 := \sum_{T \in \mathcal{T} \setminus \hat{\mathcal{T}}} \|\bullet\|_{L^2(T)}^2$ for the L^2 -norm on the set of refined simplices.

Lemma 5.39 (distance control I). *There exists a constant $C_{11} > 0$ such that for any $\mathcal{T} \in \mathbb{T}(\varepsilon_5)$, the difference $e := \hat{u}_{\text{nc}} - u_{\text{nc}}$ of the two nonconforming solution components satisfies*

$$\begin{aligned} C_{11}^{-1} \|e\|_{\text{pw}}^2 &\leq \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + \|e\|_{L^2(\Omega)}^2 \\ &\quad + \|\delta u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)}^2. \end{aligned}$$

Proof. This is shown by arguments comparable to the reliability proof in Theorem 5.35. Corollary 5.1.c shows $a_{\text{pw}}(e, \hat{u}_{\text{nc}} - Ju_{\text{nc}}) = a_{\text{pw}}(\hat{u}_{\text{nc}}, \hat{u}_{\text{nc}} - \hat{I}Ju_{\text{nc}}) - a_{\text{pw}}(u_{\text{nc}}, I(\hat{u}_{\text{nc}} - Ju_{\text{nc}}))$. Since $(\lambda_h, u_{\text{nc}})$ and $(\hat{\lambda}_h, \hat{u}_{\text{nc}})$ solve (5.15), this and (J1) lead to

$$\begin{aligned} a_{\text{pw}}(e, \hat{u}_{\text{nc}} - Ju_{\text{nc}}) &= b(\hat{\lambda}_h \hat{u}_{\text{nc}}, (1 + \hat{\delta})(\hat{u}_{\text{nc}} - \hat{I}Ju_{\text{nc}})) - b(\lambda_h u_{\text{nc}}, (1 + \delta)(I\hat{u}_{\text{nc}} - u_{\text{nc}})) \\ &= b(\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}, e) + b(\hat{\lambda}_h \hat{u}_{\text{nc}}, \hat{\delta}e) - b(\lambda_h u_{\text{nc}}, \delta e) \\ &\quad + b(\hat{\lambda}_h \hat{u}_{\text{nc}}, (1 + \hat{\delta})(u_{\text{nc}} - \hat{I}Ju_{\text{nc}})) + b(\lambda_h u_{\text{nc}}, (1 + \delta)(\hat{u}_{\text{nc}} - I\hat{u}_{\text{nc}})). \end{aligned} \quad (5.60)$$

Elementary algebra with $\|u_{\text{nc}}\|_{L^2(\Omega)} = \|\hat{u}_{\text{nc}}\|_{L^2(\Omega)} = 1$ shows as in (5.56)

$$\begin{aligned} b(\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}, e) &= \frac{\hat{\lambda}_h + \lambda_h}{2} \|e\|_{L^2(\Omega)}^2 + \frac{\hat{\lambda}_h - \lambda_h}{2} b(\hat{u}_{\text{nc}} + u_{\text{nc}}, \hat{u}_{\text{nc}} - u_{\text{nc}}) \\ &= \frac{\hat{\lambda}_h + \lambda_h}{2} \|e\|_{L^2(\Omega)}^2. \end{aligned}$$

Cauchy-Schwarz inequalities verify

$$b(\hat{\lambda}_h \hat{u}_{\text{nc}}, \hat{\delta}e) - b(\lambda_h u_{\text{nc}}, \delta e) \leq \|e\|_{L^2(\Omega)} (\hat{\lambda}_h \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} + \lambda_h \|\delta u_{\text{nc}}\|_{L^2(\Omega)}).$$

Since $1 + \hat{\delta} \leq 2$ and $\hat{\lambda}_h \leq \lambda$ from Table 5.1, the right inverse property (J1) and (5.59) with a weighted Cauchy-Schwarz inequality result in

$$\begin{aligned} b((1 + \hat{\delta})\hat{\lambda}_h\hat{u}_{\text{nc}}, u_{\text{nc}} - \hat{I}Ju_{\text{nc}}) &= b((1 + \hat{\delta})\hat{\lambda}_h\hat{u}_{\text{nc}}, (I - \hat{I})Ju_{\text{nc}}) \\ &\leq 2\|h_{\mathcal{T}}^m\lambda\hat{u}_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}\|h_{\mathcal{T}}^{-m}(I - \hat{I})Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}. \end{aligned}$$

A triangle inequality and the bound $\lambda/\lambda_h \leq 2$ from Theorem 5.20.a imply $\|h_{\mathcal{T}}^m\lambda\hat{u}_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} - h_{\text{max}}^m\lambda\|e\|_{L^2(\Omega)} \leq \|h_{\mathcal{T}}^m\lambda u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \leq 2\|h_{\mathcal{T}}^m\lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}$. Since the interpolation operators I and $\hat{I}$ satisfy (I3)–(I4), it follows that

$$\|h_{\mathcal{T}}^{-m}(I - \hat{I})Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} = \|h_{\mathcal{T}}^{-m}(1 - I)\hat{I}Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \leq \hat{\kappa}_m\|D_{\text{pw}}^m(1 - I)\hat{I}Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}.$$

Recall that $D_{\text{pw}}^m u_{\text{nc}} \in P_0(\mathcal{T}; \mathbb{R}^{(n+1)^m})$ is piecewise constant. The condition (I2) and the L^2 -orthogonal projections Π_0 (resp. $\hat{\Pi}_0$) onto $P_0(\mathcal{T})$ (resp. $P_0(\hat{\mathcal{T}})$) lead to the estimate

$$\begin{aligned} \hat{\kappa}_m^{-1}\|h_{\mathcal{T}}^{-m}(I - \hat{I})Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} &\leq \|(\Pi_0 - \hat{\Pi}_0)D^m Ju_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \\ &= \|(\Pi_0 - \hat{\Pi}_0)D_{\text{pw}}^m(Ju_{\text{nc}} - u_{\text{nc}})\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \leq \|D_{\text{pw}}^m(Ju_{\text{nc}} - u_{\text{nc}})\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}. \end{aligned}$$

The estimate (5.59) with a weighted Cauchy-Schwarz inequality and $1 + \delta \leq 2$ from Table 5.1 imply the first inequality and (I4) and the best approximation property from Corollary 5.1.a the second in

$$\begin{aligned} b(\lambda_h u_{\text{nc}}, (1 + \delta)(\hat{u}_{\text{nc}} - I\hat{u}_{\text{nc}})) &= b(\lambda_h u_{\text{nc}}, (1 + \delta)(\hat{I} - I)\hat{u}_{\text{nc}}) \\ &\leq 2\|h_{\mathcal{T}}^m\lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}\|h_{\mathcal{T}}^{-m}(\hat{u}_{\text{nc}} - I\hat{u}_{\text{nc}})\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \leq 2\hat{\kappa}_m\|h_{\mathcal{T}}^m\lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}\|e\|_{\text{pw}}. \end{aligned}$$

The combination of the six previously displayed estimates and $\lambda_h, \hat{\lambda}_h \leq \lambda$ in (5.60) eventually lead to

$$\begin{aligned} a_{\text{pw}}(e, \hat{u}_{\text{nc}} - Ju_{\text{nc}}) &\leq \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \left(2\hat{\kappa}_m\lambda h_{\text{max}}^m\|e\|_{L^2(\Omega)} + 4\hat{\kappa}_m\|h_{\mathcal{T}}^m\lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \right) \\ &\quad + \lambda\|e\|_{L^2(\Omega)} \left(\|e\|_{L^2(\Omega)} + \|\delta u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{\delta}\hat{u}_{\text{nc}}\|_{L^2(\Omega)} \right) + 2\hat{\kappa}_m\|e\|_{\text{pw}}\|h_{\mathcal{T}}^m\lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}. \end{aligned}$$

Additionally, Corollary 5.2.b and (5.59) show

$$\begin{aligned} a_{\text{pw}}(e, Ju_{\text{nc}} - u_{\text{nc}}) &= a_{\text{pw}}((1 - I)e, Ju_{\text{nc}} - u_{\text{nc}}) = a_{\text{pw}}((\hat{I} - I)\hat{u}_{\text{nc}}, Ju_{\text{nc}} - u_{\text{nc}}) \\ &\leq \|D_{\text{pw}}^m(1 - I)e\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}\|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})}. \end{aligned}$$

Condition (I2) and the boundedness of Π_0 show $\|D_{\text{pw}}^m(1 - I)e\|_{L^2(\mathcal{T}\setminus\hat{\mathcal{T}})} \leq \|e\|_{\text{pw}}$. This and the combination of the two previously displayed estimates with a triangle

inequality eventually prove

$$\begin{aligned}
\|e\|_{\text{pw}}^2 &= a_{\text{pw}}(e, Ju_{\text{nc}} - u_{\text{nc}}) + a_{\text{pw}}(e, \hat{u}_{\text{nc}} - Ju_{\text{nc}}) \\
&\leq \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \left(\|e\|_{\text{pw}} + 2\hat{\kappa}_m \lambda h_{\text{max}}^m \|e\|_{L^2(\Omega)} + 4\hat{\kappa}_m \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \right) \\
&\quad + \lambda \|e\|_{L^2(\Omega)} \left(\|e\|_{L^2(\Omega)} + \|\delta u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} \right) + 2\hat{\kappa}_m \|e\|_{\text{pw}} \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \\
&\leq (1 + 4\hat{\kappa}_m^2 + \hat{\kappa}_m^2 \lambda^2 h_{\text{max}}^{2m}) \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + \|\delta u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)}^2 \\
&\quad + (1 + \lambda + \lambda^2/2) \|e\|_{L^2(\Omega)}^2 + (1 + 4\hat{\kappa}_m^2) \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + \|e\|_{\text{pw}}^2/2
\end{aligned}$$

with weighted Young inequalities in the last step. This concludes the proof with $C_{11} := 2 \max\{1 + 4\hat{\kappa}_m^2 + \hat{\kappa}_m^2 \lambda^2 h_{\text{max}}^{2m}, 1 + \lambda + \lambda^2/2\}$. $\square$

Lemma 5.40 (distance control II). *There exists a constant $C_{12} > 0$ such that $\|\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{u}_{\text{nc}} - u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} + \|\delta u_{\text{nc}}\|_{L^2(\Omega)} \leq C_{12} h_{\text{max}}^\sigma \|u - u_{\text{nc}}\|_{\text{pw}} \leq C_{12} C_{\text{rel}} h_{\text{max}}^\sigma \eta(\mathcal{T})$ holds for any $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$.*

Proof. Triangle inequalities and the normalization $\|u\|_{L^2(\Omega)} = 1$ show

$$\|\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} \leq \lambda_h \|u - u_{\text{nc}}\|_{L^2(\Omega)} + \hat{\lambda}_h \|u - \hat{u}_{\text{nc}}\|_{L^2(\Omega)} + |\hat{\lambda}_h - \lambda_h|.$$

Theorem 5.20.c and Corollary 5.1.b imply that $|\lambda - \lambda_h| \leq C_0 \|u - Iu\|_{\text{pw}}^2 \leq C_0 (h_{\text{max}}/\pi)^{2\sigma} \|u\|_{H^{m+\sigma}(\Omega)}^2$. Since the eigenfunction $u \in V$ in (5.1) solves the source problem with right-hand side $\lambda u \in L^2(\Omega)$, (5.23) implies $\|u\|_{H^{m+\sigma}(\Omega)} \leq C(\sigma) \|\lambda u\|_{L^2(\Omega)} = C(\sigma) \lambda$. The same arguments apply to $|\lambda - \hat{\lambda}_h|$. This and $\hat{h}_{\text{max}}^\sigma \|u - \hat{I}u\|_{\text{pw}} \leq h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}}$ result in

$$|\hat{\lambda}_h - \lambda_h| \leq |\lambda - \lambda_h| + |\lambda - \hat{\lambda}_h| \leq \frac{2C_0 C(\sigma) \lambda}{\pi^\sigma} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}}.$$

Recall $\lambda_h, \hat{\lambda}_h \leq \lambda$, $\|\delta u_{\text{nc}}\|_{L^2(\Omega)} \leq C_\delta^{1/2} h_{\text{max}}^m \|u_{\text{nc}}\|_\delta$, and $\|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} \leq C_\delta^{1/2} \hat{h}_{\text{max}}^m \|\hat{u}_{\text{nc}}\|_\delta$ from Table 5.1. The last two displayed estimates, a triangle inequality, and Theorem 5.20.c show

$$\begin{aligned}
&\|\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{u}_{\text{nc}} - u_{\text{nc}}\|_{L^2(\Omega)} + \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} + \|\delta u_{\text{nc}}\|_{L^2(\Omega)} \\
&\leq (\lambda + 1) (\|u - u_{\text{nc}}\|_{L^2(\Omega)} + \|u - \hat{u}_{\text{nc}}\|_{L^2(\Omega)}) + C_\delta^{1/2} (h_{\text{max}}^m \|u_{\text{nc}}\|_\delta + \hat{h}_{\text{max}}^m \|\hat{u}_{\text{nc}}\|_\delta) \\
&\quad + \frac{2C_0 C(\sigma) \lambda}{\pi^\sigma} h_{\text{max}}^\sigma \|u - Iu\|_{\text{pw}} \\
&\leq 2 \left((C_0 C(\sigma) \lambda / \pi^\sigma + C_0^{1/2} (1 + \lambda)) h_{\text{max}}^\sigma + (C_\delta C_0)^{1/2} h_{\text{max}}^m \right) \|u - Iu\|_{\text{pw}}.
\end{aligned}$$

with $\|u - \hat{I}u\|_{\text{pw}} \leq \|u - Iu\|_{\text{pw}}$ and $\hat{h}_{\text{max}} \leq h_{\text{max}}$. Since Corollary 5.1.a implies $\|u - Iu\|_{\text{pw}} \leq \|u - u_{\text{nc}}\|_{\text{pw}}$ and $h_{\text{max}}^m \leq h_{\text{max}}^\sigma$, this concludes the proof of the first bound in Lemma 5.40 with $C_{12} := 2C_0 C(\sigma) \lambda / \pi^\sigma + 2C_0^{1/2} (1 + \lambda + C_\delta^{1/2})$. The second claim follows from the reliability in Theorem 5.35. $\square$

Proof of Theorem 5.38. Recall that Lemma 5.39 shows

$$\begin{aligned} C_{11}^{-1} \|\hat{u}_{\text{nc}} - u_{\text{nc}}\|_{\text{pw}}^2 &\leq \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 \\ &\quad + \|\hat{u}_{\text{nc}} - u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|\hat{\delta} \hat{u}_h\|_{L^2(\Omega)}^2 + \|\delta u_{\text{nc}}\|_{L^2(\Omega)}^2. \end{aligned}$$

This and Lemma 5.40 lead with $M_3 := C_{12}^2 C_{\text{rel}}^2 \max\{1, C_{11}\}$ to

$$\begin{aligned} \delta^2(\mathcal{T}, \hat{\mathcal{T}}) &= \|\hat{\lambda}_h \hat{u}_{\text{nc}} - \lambda_h u_{\text{nc}}\|_{L^2(\Omega)}^2 + \|\hat{u}_{\text{nc}} - u_{\text{nc}}\|_{\text{pw}}^2 \\ &\leq C_{11} \|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + C_{11} \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 + M_3 h_{\text{max}}^{2\sigma} \eta^2(\mathcal{T}). \end{aligned}$$

The shape regularity $|T|^{1/n} \leq h_T \leq C_{\text{sr}} |T|^{1/n}$ for any $T \in \mathcal{T} \in \mathbb{T}$ guarantees

$$C_{\text{sr}}^{-2m} \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 \leq \sum_{T \in \mathcal{T} \setminus \hat{\mathcal{T}}} |T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2 \leq \sum_{T \in \mathcal{R}_1} |T|^{2m/n} \|\lambda_h u_{\text{nc}}\|_{L^2(T)}^2$$

with $\mathcal{T} \setminus \hat{\mathcal{T}} \subseteq \mathcal{R}_1$ in the last step. Remark 5.4 asserts

$$\|D_{\text{pw}}^m(u_{\text{nc}} - Ju_{\text{nc}})\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})}^2 \leq M_5 \sum_{T \in \mathcal{R}_1} |T|^{1/n} \sum_{F \in \mathcal{F}(T)} \|[D_{\text{pw}}^m u_{\text{nc}}]_F \times \nu_F\|_{L^2(F)}^2.$$

The combination with the local estimator definition in (5.54) and the sum convention in Table 5.1 concludes the proof of $(A3_\epsilon)$, $\delta^2(\mathcal{T}, \hat{\mathcal{T}}) \leq \Lambda_3 \eta^2(\mathcal{R}_1) + \epsilon_3 \eta^2(\mathcal{T})$ with $\Lambda_3 := C_{11} \max\{C_{\text{sr}}^{2m}, M_5\}$ and $M_3 := C_{12}^2 C_{\text{rel}}^2 \max\{1, C_{11}\}$ in $\epsilon_3 := M_3 h_{\text{max}}^{2\sigma}$. $\square$

5.5.4 Quasiorthogonality

The quasiorthogonality in Theorem 5.41 below concerns the outcome $(\mathcal{T}_j)_{j \in \mathbb{N}_0}$ of Algorithm 2. Let $u_j \in V(\mathcal{T}_j)$ abbreviate the nonconforming component of the discrete solution $\mathbf{u}_j = (u_{\text{pw}}, u_{\text{nc}}) =: (u_{\text{pw}}, u_j) \in P_m(\mathcal{T}_j) \times V(\mathcal{T}_j)$ with $b(u, u_j) > 0$, $\|u_j\|_{L^2(\Omega)} = 1$, and $\lambda_j(k) \leq \lambda$ the associated eigenvalue from Algorithm 2 on the level $j \in \mathbb{N}_0$. Recall the distance

$$\delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) = \|\lambda_j(k) u_j - \lambda_{j+1}(k) u_{j+1}\|_{L^2(\Omega)}^2 + \|u_j - u_{j+1}\|_{\text{pw}}^2 \quad (5.61)$$

for the triangulations $\mathcal{T}_j$ and $\mathcal{T}_{j+1}$. Set $h_0 := \max_{T \in \mathcal{T}_0} h_T$ and recall $\epsilon_6 > 0$ from Theorem 5.35.

Theorem 5.41 (quasiorthogonality). *For any $0 < \beta \leq C_{\text{eff}}^2 / C_{\text{rel}}^2$, there exist $\Lambda_4, \tilde{\Lambda}_4$, and $\epsilon_4 := \tilde{\Lambda}_4(\beta + h_0^{2\sigma}(1 + \beta^{-1})) > 0$, such that $\mathcal{T}_0 \in \mathbb{T}(\epsilon_6)$ implies that the output $(\eta_j)_{j \in \mathbb{N}_0}$ and $(\mathcal{T}_j)_{j \in \mathbb{N}_0}$ of Algorithm 2 satisfies for any $\ell, L \in \mathbb{N}_0$*

$$(A4_\epsilon) \text{ Quasiorthogonality. } \sum_{j=\ell}^{\ell+L} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) \leq \Lambda_4 (1 + \beta^{-1}) \eta_\ell^2 + \epsilon_4 \sum_{j=\ell}^{\ell+L} \eta_j^2.$$

The following Lemma 5.42, which is in the 2-level notation of Definition 5.36 for readability, prepares the proof of Theorem 5.41 below.

Lemma 5.42 (2-level quasiorthogonality). *There exists $C_{\text{qo}} > 0$ such that, for $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$,*

$$a_{\text{pw}}(u - \hat{u}_{\text{nc}}, u_{\text{nc}} - \hat{u}_{\text{nc}}) \leq C_{\text{qo}} \left(h_{\text{max}}^\sigma \|u - u_{\text{nc}}\|_{\text{pw}} + \|h_{\mathcal{T}}^m \lambda u\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \right) \|u - \hat{u}_{\text{nc}}\|_{\text{pw}}$$

holds.

Proof. Since $(\lambda_h, u_{\text{nc}})$ (resp. $(\hat{\lambda}_h, \hat{u}_{\text{nc}})$) solves (5.15) with respect to $\mathcal{T}$ and $\delta \in P_0(\mathcal{T})$ (resp. $\hat{\mathcal{T}}$ and $\hat{\delta} \in P_0(\hat{\mathcal{T}})$ from Table 5.1), Corollary 5.1.c and elementary algebra show that

$$\begin{aligned} a_{\text{pw}}(u_{\text{nc}} - \hat{u}_{\text{nc}}, u - \hat{u}_{\text{nc}}) &= a_{\text{pw}}(u_{\text{nc}}, I(u - \hat{u}_{\text{nc}})) - a_{\text{pw}}(\hat{u}_{\text{nc}}, \hat{I}u - \hat{u}_{\text{nc}}) \\ &= b(\lambda_h u_{\text{nc}}(1 + \delta), I(u - \hat{u}_{\text{nc}})) - b(\hat{\lambda}_h \hat{u}_{\text{nc}}(1 + \hat{\delta}), \hat{I}u - \hat{u}_{\text{nc}}) \\ &= b(\lambda_h u_{\text{nc}}(1 + \delta) - \hat{\lambda}_h \hat{u}_{\text{nc}}(1 + \hat{\delta}), \hat{I}u - \hat{u}_{\text{nc}}) + (\lambda_h u_{\text{nc}}, (I - \hat{I})(u - \hat{u}_{\text{nc}}))_{1+\delta}. \end{aligned} \quad (5.62)$$

The Cauchy-Schwarz inequality, $\lambda_h, \hat{\lambda}_h \leq \lambda$, and Lemma 5.40 in the last step prove

$$\begin{aligned} t_1 &:= b(\lambda_h u_{\text{nc}}(1 + \delta) - \hat{\lambda}_h \hat{u}_{\text{nc}}(1 + \hat{\delta}), \hat{I}u - \hat{u}_{\text{nc}}) \\ &\leq \left(\|\lambda_h u_{\text{nc}} - \hat{\lambda}_h \hat{u}_{\text{nc}}\|_{L^2(\Omega)} + \lambda_h \|\delta u_{\text{nc}}\|_{L^2(\Omega)} + \hat{\lambda}_h \|\hat{\delta} \hat{u}_{\text{nc}}\|_{L^2(\Omega)} \right) \|\hat{I}u - \hat{u}_{\text{nc}}\|_{L^2(\Omega)} \\ &\leq \max\{1, \lambda\} C_{12} h_{\text{max}}^\sigma \|u - u_{\text{nc}}\|_{\text{pw}} \|\hat{I}u - \hat{u}_{\text{nc}}\|_{L^2(\Omega)}. \end{aligned}$$

The discrete Friedrichs inequality from Lemma 2.12 with respect to $V(\hat{\mathcal{T}})$, (I2), and the boundedness of the L^2 -projection $\hat{\Pi}_0$ onto $P_0(\hat{\mathcal{T}})$ lead to

$$C_{\text{dF}}^{-1} \|\hat{I}u - \hat{u}_{\text{nc}}\|_{L^2(\Omega)} \leq \|\hat{I}u - \hat{u}_{\text{nc}}\|_{\text{pw}} = \|\hat{\Pi}_0 D_{\text{pw}}^m(u - \hat{u}_{\text{nc}})\|_{L^2(\Omega)} \leq \|u - \hat{u}_{\text{nc}}\|_{\text{pw}}.$$

Consequently, $t_1 \leq \max\{1, \lambda\} C_{12} C_{\text{dF}} h_{\text{max}}^\sigma \|u - u_{\text{nc}}\|_{\text{pw}} \|u - \hat{u}_{\text{nc}}\|_{\text{pw}}$. Since $1 + \delta \leq 2$ from Table 5.1, a weighted Cauchy-Schwarz inequality in (5.59) shows

$$t_2 := (\lambda_h u_{\text{nc}}, (I - \hat{I})(u - \hat{u}_{\text{nc}}))_{1+\delta} \leq 2 \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} \|h_{\mathcal{T}}^{-m} (I - \hat{I})(u - \hat{u}_{\text{nc}})\|_{L^2(\Omega)}.$$

Since (I3) implies $I(\hat{I}u) = Iu$, (I4) for I and (I2) for I and $\hat{I}$ with bounded L^2 projections show that $\|h_{\mathcal{T}}^{-m} (I - \hat{I})(u - \hat{u}_{\text{nc}})\|_{L^2(\Omega)} = \|h_{\mathcal{T}}^{-m} (1 - I)(\hat{I}u - \hat{u}_{\text{nc}})\|_{L^2(\Omega)} \leq \hat{\kappa}_m \| (1 - I)\hat{I}(u - \hat{u}_{\text{nc}}) \|_{\text{pw}} \leq \hat{\kappa}_m \|u - \hat{u}_{\text{nc}}\|_{\text{pw}}$. On the other hand, $\lambda_h \leq \lambda$, a triangle inequality, the central estimate (5.42) in Theorem 5.20.c, and Corollary 5.1.a imply

$$\begin{aligned} \|h_{\mathcal{T}}^m \lambda_h u_{\text{nc}}\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} &\leq \|h_{\mathcal{T}}^m \lambda u\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} + \lambda h_{\text{max}}^m \|u - u_{\text{nc}}\|_{L^2(\Omega)} \\ &\leq \|h_{\mathcal{T}}^m \lambda u\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} + C_{10} \lambda h_{\text{max}}^{m+\sigma} \|u - u_{\text{nc}}\|_{\text{pw}}. \end{aligned}$$

Therefore, $t_2 \leq 2\hat{\kappa}_m (\|h_{\mathcal{T}}^m \lambda u\|_{L^2(\mathcal{T} \setminus \hat{\mathcal{T}})} + C_{10} \lambda h_{\text{max}}^{m+\sigma} \|u - u_{\text{nc}}\|_{\text{pw}}) \|u - \hat{u}_{\text{nc}}\|_{\text{pw}}$. Hence, the upper bound $t_1 + t_2$ in (5.62) is controlled and the above estimates lead to the assertion with $C_{\text{qo}} := \max\{2\hat{\kappa}_m, \max\{1, \lambda\} C_{12} C_{\text{dF}} + 2C_{10} \lambda \hat{\kappa}_m h_{\text{max}}^m\}$ with $h_{\text{max}} \leq 1$. $\square$

Proof of Theorem 5.41. Recall $h_0 := \max_{T \in \mathcal{T}_0} h_T$, that $u_j \in V(\mathcal{T}_j)$ is the non-conforming component of the discrete solution $\mathbf{u}_j = (u_{\text{pw}}, u_{\text{nc}}) =: (u_{\text{pw}}, u_j) \in P_m(\mathcal{T}_j) \times V(\mathcal{T}_j)$, that $\lambda_j(k) \leq \lambda$ is the associated eigenvalue from Algorithm 2, and that $\eta_j := \eta(\mathcal{T}_j)$ denotes the estimator on the j -th level for $\ell \leq j \leq \ell + L$. Since $\mathcal{T}_j, \mathcal{T}_{j+1} \in \mathbb{T}(\mathcal{T}_0)$ for $\ell \leq j \leq \ell + L$ and $\mathcal{T}_0 \in \mathbb{T}(\varepsilon_6)$, the definition of the distance in (5.61) and Lemma 5.40 show

$$\delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) \leq \|u_j - u_{j+1}\|_{\text{pw}}^2 + C_{12}^2 C_{\text{rel}}^2 h_0^{2\sigma} \eta_j^2.$$

Elementary algebra for $\|u - u_j\|_{\text{pw}}^2 = \|u - u_{j+1} - (u_j - u_{j+1})\|_{\text{pw}}^2$, Lemma 5.42, and two weighted Young inequalities show

$$\begin{aligned} \|u_j - u_{j+1}\|_{\text{pw}}^2 - \|u - u_j\|_{\text{pw}}^2 + \|u - u_{j+1}\|_{\text{pw}}^2 &= 2a_{\text{pw}}(u - u_{j+1}, u_j - u_{j+1}) \\ &\leq 2C_{\text{qo}} \left(h_0^\sigma \|u - u_j\|_{\text{pw}} + \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})} \right) \|u - u_{j+1}\|_{\text{pw}} \\ &\leq \frac{2C_{\text{qo}}^2}{\beta} h_0^{2\sigma} C_{\text{rel}}^2 \eta_j^2 + \beta C_{\text{rel}}^2 \eta_{j+1}^2 + \frac{2C_{\text{qo}}^2}{\beta} \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2 \end{aligned}$$

with the reliability from Theorem 5.35 in the last step. Theorem 5.35 controls the telescoping sum

$$\sum_{j=\ell}^{\ell+L} (\|u - u_j\|_{\text{pw}}^2 - \|u - u_{j+1}\|_{\text{pw}}^2) = \|u - u_\ell\|_{\text{pw}}^2 - \|u - u_{\ell+L+1}\|_{\text{pw}}^2 \leq C_{\text{rel}}^2 \eta_\ell^2 - C_{\text{eff}}^2 \eta_{\ell+L+1}^2.$$

Since $\beta \leq C_{\text{eff}}^2 / C_{\text{rel}}^2$ implies $(\beta C_{\text{rel}}^2 - C_{\text{eff}}^2) \eta_{\ell+L+1}^2 \leq 0$, the last three displayed estimates allow for the following upper bound

$$\begin{aligned} \sum_{j=\ell}^{\ell+L} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) &= \sum_{j=\ell}^{\ell+L} \|u_j - u_{j+1}\|_{\text{pw}}^2 + C_{12}^2 C_{\text{rel}}^2 h_0^{2\sigma} \eta_j^2 \\ &\leq \sum_{j=\ell}^{\ell+L} \left(\|u - u_j\|_{\text{pw}}^2 - \|u - u_{j+1}\|_{\text{pw}}^2 + 2a_{\text{pw}}(u - u_{j+1}, u_j - u_{j+1}) + C_{12}^2 C_{\text{rel}}^2 h_0^{2\sigma} \eta_j^2 \right) \\ &\leq C_{\text{rel}}^2 \eta_\ell^2 - C_{\text{eff}}^2 \eta_{\ell+L+1}^2 + \left(\left(\frac{2C_{\text{qo}}^2}{\beta} + C_{12}^2 \right) h_0^{2\sigma} + \beta \right) C_{\text{rel}}^2 \sum_{k=\ell}^{\ell+L} \eta_j^2 + \beta C_{\text{rel}}^2 \eta_{\ell+L+1}^2 \\ &\quad + \frac{2C_{\text{qo}}^2}{\beta} \sum_{j=\ell}^{\ell+L} \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2 \\ &\leq C_{\text{rel}}^2 \eta_\ell^2 + \left(\left(\frac{2C_{\text{qo}}^2}{\beta} + C_{12}^2 \right) h_0^{2\sigma} + \beta \right) C_{\text{rel}}^2 \sum_{k=\ell}^{\ell+L} \eta_j^2 + \frac{2C_{\text{qo}}^2}{\beta} \sum_{j=\ell}^{\ell+L} \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2. \quad (5.63) \end{aligned}$$

Recall that $h_{\mathcal{T}_j}|_T := \text{diam}(T)$ for any $T \in \mathcal{T}_j$ and compare it with the piecewise constant function $\tilde{h}_j \in P_0(\mathcal{T}_j)$ defined by

$$\tilde{h}_j|_T := |T|^{1/n} \leq h_T \leq C_{\text{sr}} |T|^{1/n} \quad \text{for any } T \in \mathcal{T}_j \text{ and } j \in \mathbb{N}_0$$

with the upper bound from the shape-regularity. Then $\tilde{h}_j \approx h_{\mathcal{T}_j} \in P_0(\mathcal{T}_j)$ and $\tilde{h}_j \in P_0(\mathcal{T}_j)$ satisfies the reduction $\tilde{h}_{j+1} \leq \tilde{h}_j/2^{1/n}$ a.e. in the set of refined simplices $\bigcup (\mathcal{T}_j \setminus \mathcal{T}_{j+1})$. Since each refined simplex is at least halved,

$$\tilde{h}_j^m = \frac{2^{m/n}}{\sqrt{4^{m/n} - 1}} \sqrt{\tilde{h}_j^{2m} - \frac{\tilde{h}_j^{2m}}{4^{m/n}}} \leq \frac{2^{m/n}}{\sqrt{4^{m/n} - 1}} \sqrt{\tilde{h}_j^{2m} - \tilde{h}_{j+1}^{2m}} \text{ a.e. in } \bigcup (\mathcal{T}_j \setminus \mathcal{T}_{j+1}).$$

This allows for a telescoping sum in

$$\begin{aligned} C_{\text{sr}}^{-2m} \frac{4^{m/n} - 1}{4^{m/n}} \sum_{j=\ell}^{\ell+L} \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2 &\leq \frac{4^{m/n} - 1}{4^{m/n}} \sum_{j=\ell}^{\ell+L} \|\tilde{h}_j^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2 \\ &\leq \sum_{j=\ell}^{\ell+L} \left\| \sqrt{\tilde{h}_j^{2m} - \tilde{h}_{j+1}^{2m}} \lambda u \right\|_{L^2(\Omega)}^2 = \int_{\Omega} \sum_{j=\ell}^{\ell+L} (\tilde{h}_j^{2m} - \tilde{h}_{j+1}^{2m}) (\lambda u)^2 dx \\ &= \int_{\Omega} (\tilde{h}_{\ell}^{2m} - \tilde{h}_{\ell+L+1}^{2m}) (\lambda u)^2 dx \leq \|\tilde{h}_{\ell}^m \lambda u\|_{L^2(\Omega)}^2. \end{aligned}$$

Since $\tilde{h}_{\ell} \leq h_{\mathcal{T}_{\ell}} \leq h_0 := \max_{T \in \mathcal{T}_0} h_T \leq \varepsilon_6$, a triangle inequality implies

$$\|\tilde{h}_{\ell}^m \lambda u\|_{L^2(\Omega)}^2 \leq 2(\lambda/\lambda_{\ell}(k))^2 \|\tilde{h}_{\ell}^m \lambda_{\ell}(k) u_{\ell}\|_{L^2(\Omega)}^2 + 2\lambda^2 h_0^{2m} \|u - u_{\ell}\|_{L^2(\Omega)}^2.$$

Theorem 5.20.a and the estimator definition (5.54) with $\tilde{h}_{\ell}|_T := |T|^{1/n}$ for any $T \in \mathcal{T}_{\ell}$ show that $(\lambda/\lambda_{\ell}(k))^2 \|\tilde{h}_{\ell}^m \lambda_{\ell}(k) u_{\ell}\|_{L^2(\Omega)}^2 \leq 4\eta_{\ell}^2$. The combination of the central estimate (5.42) in Theorem 5.20, Corollary 5.1.a and the reliability in Theorem 5.35 imply $\|u - u_{\ell}\|_{L^2(\Omega)}^2 \leq h_0^{2\sigma} C_{10}^2 C_{\text{rel}}^2 \eta_{\ell}^2$. Hence, $\sum_{j=\ell}^{\ell+L} \|h_{\mathcal{T}_j}^m \lambda u\|_{L^2(\mathcal{T}_j \setminus \mathcal{T}_{j+1})}^2 \leq 2C_{\text{sr}}^{2m} \frac{4^{m/n}}{4^{m/n} - 1} (4 + h_0^{2m+2\sigma} C_{10}^2 C_{\text{rel}}^2 \lambda^2) \eta_{\ell}^2$. The substitution in (5.63) concludes the proof of the quasiorthogonality in

$$\begin{aligned} \sum_{j=\ell}^{\ell+L} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) &\leq \left(C_{\text{rel}}^2 + \frac{4}{\beta} C_{\text{qo}}^2 C_{\text{sr}}^{2m} \frac{4^{m/n}}{4^{m/n} - 1} (4 + h_0^{2m+2\sigma} C_{10}^2 C_{\text{rel}}^2 \lambda^2) \right) \eta_{\ell}^2 \\ &\quad + \left(\left(\frac{2C_{\text{qo}}^2}{\beta} + C_{12}^2 \right) h_0^{2\sigma} + \beta \right) C_{\text{rel}}^2 \sum_{k=\ell}^{\ell+L} \eta_j^2 \\ &= \Lambda_4 (1 + \beta^{-1}) \eta_{\ell}^2 + \epsilon_4 \sum_{j=\ell}^{\ell+L} \eta_j^2 \end{aligned}$$

with constants $\Lambda_4 := \max\{C_{\text{rel}}^2, C_{\text{qo}}^2 C_{\text{sr}}^{2m} \frac{4^{m/n+1}}{4^{m/n} - 1} (4 + h_0^{2m+2\sigma} C_{10}^2 C_{\text{rel}}^2 \lambda^2)\}$ and $\tilde{\Lambda}_4 := C_{\text{rel}}^2 \max\{1, 2C_{\text{qo}}^2, C_{12}^2\}$ in $\epsilon_4 := \tilde{\Lambda}_4 (\beta + h_0^{2\sigma} (1 + \beta^{-1})) > 0$. $\square$

5.5.5 Proof of optimality

The proven properties (A1)–(A4_ε) are the axioms of adaptivity in [Car+14a; CR17] and known to imply the equivalence of estimator values for optimal triangulation

and triangulation generated by the adaptive Algorithm 2 in Theorem 5.34. Compared to [Car+14a; CR17] the discrete reliability in Theorem 5.38 is extended in that (A3_ε) includes the additional term $M_3 h_{\max}^{2\sigma} \eta^2(\mathcal{T})$. Minor modifications of the arguments in [Car+14a; CR17] prove that (A1)–(A4_ε) imply (6.2). This is stated and proven as Theorem 6.1 in Chapter 6.

Proof of Theorem 5.34. The Algorithm 2 in Theorem 5.34 is a particular case of the general adaptive Algorithm 3. Theorem 5.37, 5.38, and 5.41 guarantee (A1)–(A4_ε) with $\tilde{\Lambda}_3 := 0$, $\epsilon_3 := M_3 h_{\max}^{2\sigma}$, and $\epsilon_4 := \tilde{\Lambda}_4(\beta + h_0^{2\sigma}(1 + 1/\beta)) > 0$, and $\mathcal{R}(\mathcal{T}, \hat{\mathcal{T}}) := \mathcal{R}_1 := \{K \in \mathcal{T} : \exists T \in \mathcal{T} \setminus \hat{\mathcal{T}} \text{ with } \text{dist}(K, T) = 0\}$. Let $\varepsilon_7 := \{\varepsilon_6, (2\Lambda_1^2 M_3)^{-1/(2\sigma)}\}$ such that $\epsilon_3 < \Lambda_1^{-2}$. Select ρ_{12} and Λ_{12} as described in Section 6.2 below and abbreviate $c_5 := (1 - \rho_{12})/(2\Lambda_{12}\tilde{\Lambda}_4)$, $\beta := \min\{C_{\text{eff}}^2/C_{\text{rel}}^2, c_5/2\}$, and

$$\varepsilon := \varepsilon_8 := \min\left\{\varepsilon_7, ((c_5 - \beta)/(1 + 1/\beta))^{1/(2\sigma)}\right\}. \quad (5.64)$$

Then the constants satisfy the condition in Theorem 6.1, $\hat{\Lambda}_3(\Lambda_1^2 + \Lambda_2^2) = 0 < 1$, $\epsilon_3 \Lambda_1^2 \leq 1/2 < 1$, and $\epsilon_4 \leq (1 - \rho_{12})/(2\Lambda_{12})$, and Theorem 6.1 implies Theorem 5.34. $\square$

Remark 5.43 (convergence rates of the error). The reliability and efficiency in Theorem 5.35 provide the equivalence $\|u - u_\ell\|_{\text{pw}} \approx \eta_\ell(\mathcal{T}_\ell)$. This and Theorem 5.34 lead to optimal convergence rates for the error as well.

Remark 5.44 (smallness assumptions on $\varepsilon_5, \varepsilon_6, \varepsilon_7, \varepsilon_8$). The reduction to ε_5 guarantees the best approximation result in Theorem 5.20, while $\varepsilon_6 := \min\{\varepsilon_5, (2\lambda(C_{10}^2 + \kappa_m M_2))^{-1/(2\sigma)}\}$ is sufficient for reliability in Theorem 5.35, and ε_7 allows the definition of Θ , ρ_{12} , and Λ_{12} . Optimal rates follow with $\varepsilon := \varepsilon_8$ from (5.64). Since C_{10} from (5.42), $c_5 := (1 - \rho_{12})/(2\Lambda_{12}\tilde{\Lambda}_4)$, and M_3 are ≈ 1 , independent of the mesh-size, $\varepsilon_6 = \min\{\varepsilon_5, \mathcal{O}(1)\}$, $\varepsilon_7 = \min\{\varepsilon_6, \mathcal{O}(1)\}$, and $\varepsilon_8 = \min\{\varepsilon_7, \mathcal{O}(1)\}$ are *not* expected to be dramatically smaller than ε_5 .

5.5.6 Global convergence

The results on the asymptotic convergence rates justify that a small initial mesh-size guarantees the asymptotic convergence from the beginning. Although the reasons are presented in several steps for $\varepsilon_0, \dots, \varepsilon_8$, the computation of ε_8 may be cumbersome and a huge overestimation in practice. To guarantee global convergence without a priori knowledge of ε_8 , we may modify the marking step in Algorithm 2 as follows: Enlarge the set $\mathcal{M}_\ell$ in Algorithm 2 by one simplex of maximal mesh-size in $\mathcal{T}_\ell$. This guarantees that the maximal mesh-size tends to zero as the level $\ell \rightarrow \infty$. Consequently there exists some $L \in \mathbb{N}$ such that $\mathcal{T}_\ell \in \mathbb{T}(\varepsilon_8)$ for all $\ell = L, L + 1, L + 2, \dots$. Relabel $\mathcal{T}_L$ by $\mathcal{T}_0$ so that Theorem 5.34 leads to optimal convergence rates for $\eta_L, \eta_{L+1}, \eta_{L+2}, \dots$, whence for the entire outcome of the adaptive algorithm. However, the constant in the overhead control [Ste08,

Thm. 6.1] depends on $\mathcal{T}_L$ and this possibly enlarges the equivalence constants in (6.2).

Remark 5.45 (modification with global convergence). The modified algorithm of Subsection 5.5.6, with $\mathcal{T}_L, \mathcal{T}_{L+1}, \dots$ has no influence on the constants $1/2 \leq \Theta(1 + \Lambda_1^2 \Lambda_3) \leq 1$, $\Lambda_4 \leq \Lambda_{\text{qo}} \leq 2\Lambda_4 + 1/\Lambda_{12}$, $1 + (\Lambda_1^2 + \Lambda_2^2)\Lambda_3 \leq \Lambda_{\text{mon}}^2 \leq \left(1 + \sqrt{(\Lambda_1^2 + \Lambda_2^2)(\Lambda_3 + \Lambda_1^{-2}/2)}\right)^2$ analyzed in Chapter 6. But Λ_{BDdV} in the overhead control (6.10) (e.g. [Ste08, Thm. 6.1]) depends on $\mathcal{T}_L$ and could become larger (when replacing $\mathcal{T}_0$ by $\mathcal{T}_L$) and leads to larger equivalence constants in Theorem 6.1. The asymptotic convergence rate remains optimal and the choice of θ is not affected.

Remark 5.46 (parameter choice in praxis). In a practical computation, a suggestion is uniform mesh-refinement until the eigenvalue λ_k of interest is resolved in that $5h_{\text{max}}$ is smaller or equal the estimated wavelength of λ_k . This triangulation serves as initial triangulation in $\mathcal{T}_0$ in the modified algorithm of Subsection 5.5.6 with some bulk parameter θ significantly smaller than $(1 - \Lambda_1^2 \Lambda_3)^{-1}$. In this way, the pre-asymptotic range is (hopefully) kept small while the asymptotic convergence rate remains optimal.

6 A review and extension of the axioms of adaptivity

The framework (A1)–(A4_ε) in Section 5.5 is a modification of [Car+14a; CR17] with a more general discrete reliability (A3_ε). Theorem 6.1 below proves that the modified axioms are sufficient for optimal convergence rates of the adaptive Algorithm 3 with Dörfler marking and newest-vertex bisection [Car+14a, Algorithm 2.2].

6.1 The general adaptive algorithm and axioms of adaptivity

On level $\ell \in \mathbb{N}_0$ of the general purpose adaptive Algorithm 3 there is given a regular triangulation $\mathcal{T}_\ell$ of $\Omega \subset \mathbb{R}^n$ into closed simplices and an undisplayed discrete problem with a discrete solution u_ℓ . These allow for the computation of the local estimator $\eta_\ell(T)$ for all $T \in \mathcal{T}_\ell$ in the step compute. The step mark uses the sum convention $\eta_\ell^2(\mathcal{M}) := \sum_{T \in \mathcal{M}} \eta_\ell^2(T)$ for any $\mathcal{M} \subseteq \mathcal{T}_\ell$ and $\eta_\ell^2 := \eta_\ell^2(\mathcal{T}_\ell)$ to describe the condition the set of marked simplices $\mathcal{M}_\ell$ has to satisfy. The selection of a set $\mathcal{M}_\ell$ with *almost minimal cardinality* $|\mathcal{M}_\ell|$ in this step means that there exists a constant $\Lambda_{\text{opt}} \geq 1$ such that the cardinality satisfies $|\mathcal{M}_\ell| \leq \Lambda_{\text{opt}} |\mathcal{M}_\ell^*|$, where $\mathcal{M}_\ell^* \subset \mathcal{T}_\ell$ denotes some set of minimal cardinality $|\mathcal{M}_\ell^*|$ with $\theta \eta_\ell^2 \leq \sum_{T \in \mathcal{M}_\ell^*} \eta_\ell^2(T)$; cf. [Ste07; Car+14a; CR17] for details; this is more general than in Algorithm 2, which utilizes a minimal set $\mathcal{M}_\ell$ with $\Lambda_{\text{opt}} = 1$ constructed at linear cost in [PP20]. The step refine uses newest-vertex bisection to obtain a regular triangulation $\mathcal{T}_{\ell+1}$, where at least the set $\mathcal{M}_\ell$ is refined.

Algorithm 3

Input: regular initial triangulation $\mathcal{T}_0$ of $\Omega \subset \mathbb{R}^n$ and bulk parameter $0 < \theta \leq 1$

for $\ell = 0, 1, 2, \dots$ **do**

Solve the discrete problem for the discrete solution u_ℓ based on $\mathcal{T}_\ell$

Compute $\eta_\ell(T)$ for any $T \in \mathcal{T}_\ell$ with respect to the discrete solution

Mark almost minimal subset $\mathcal{M}_\ell \subseteq \mathcal{T}_\ell$ with $\theta \eta_\ell^2 \leq \eta_\ell^2(\mathcal{M}_\ell)$

Refine $\mathcal{T}_\ell$ with NVB to compute $\mathcal{T}_{\ell+1}$ with $\mathcal{M}_\ell \subseteq \mathcal{T}_\ell \setminus \mathcal{T}_{\ell+1}$ **od**

end for

Output: sequence of triangulations $(\mathcal{T}_\ell)_{\ell \in \mathbb{N}_0}$ with $(u_\ell)_{\ell \in \mathbb{N}_0}$ and $(\eta_\ell)_{\ell \in \mathbb{N}_0}$

Recall the set $\mathbb{T} := \mathbb{T}(\mathcal{T}_0)$ of all admissible triangulation computed by successive newest-vertex bisection (NVB) of a regular initial triangulation $\mathcal{T}_0$ (plus some initialization of tagged n -simplices) of the bounded polyhedral Lipschitz domain $\Omega \subset \mathbb{R}^n$ into closed simplices and the subset $\mathbb{T}(\mathcal{T})$ of admissible refinements of $\mathcal{T} \in \mathbb{T}$. For $N \in \mathbb{N}_0$, set $\mathbb{T}(N) := \{\mathcal{T} \in \mathbb{T} : |\mathcal{T}| \leq |\mathcal{T}_0| + N\}$. To analyse the error estimates $\eta_\ell(\mathcal{T}_\ell)$ and their rates and in particular to compare with error estimators $\eta(\mathcal{T}, \bullet)$ for *any* admissible triangulation $\mathcal{T} \in \mathbb{T}$, we need to assume that the error estimators are computable for any $\mathcal{T} \in \mathbb{T}$. This leads to a family $\eta(\mathcal{T}, \bullet) \in \mathbb{R}^{\mathcal{T}}$ of error estimators parametrized by $\mathcal{T} \in \mathbb{T}$ with $\eta(\mathcal{T}, K) \geq 0$ for all $K \in \mathcal{T}$. For any subset $\mathcal{M} \subseteq \mathcal{T} \in \mathbb{T}$, the sum convention reads

$$\eta^2(\mathcal{T}, \mathcal{M}) := \left(\eta(\mathcal{T}, \mathcal{M})\right)^2 := \sum_{T \in \mathcal{M}} \eta^2(\mathcal{T}, T) \quad \text{and} \quad \eta^2(\mathcal{T}) := \eta^2(\mathcal{T}, \mathcal{T}). \quad (6.1)$$

For any triangulation $\mathcal{T}_\ell$ in the Algorithm 3, we abbreviate $\eta_\ell(\bullet) := \eta(\mathcal{T}_\ell, \bullet)$ and $\eta_\ell := \eta_\ell(\mathcal{T}_\ell) \equiv \eta(\mathcal{T}_\ell, \mathcal{T}_\ell)$. Recall the Axioms (A1)–(A4_ε) with constants $\Lambda_1, \Lambda_2, \Lambda_3, \Lambda_4 > 0$, $\hat{\Lambda}_3, \epsilon_3, \epsilon_4 \geq 0$, $\Lambda_{\text{ref}} \geq 1$, and $0 < \rho_2 < 1$. For any $\mathcal{T} \in \mathbb{T}$ and admissible refinement $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$, there exists a set $\mathcal{R}(\mathcal{T}, \hat{\mathcal{T}}) \subseteq \mathcal{T}$ with $\mathcal{T} \setminus \hat{\mathcal{T}} \subseteq \mathcal{R}(\mathcal{T}, \hat{\mathcal{T}})$ and $|\mathcal{R}(\mathcal{T}, \hat{\mathcal{T}})| \leq \Lambda_{\text{ref}} |\mathcal{T} \setminus \hat{\mathcal{T}}|$, such that $\mathcal{T} \in \mathbb{T}$, $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$, $\mathcal{R}(\mathcal{T}, \hat{\mathcal{T}})$, and the output $(\mathcal{T}_k)_{k \in \mathbb{N}_0}$ and $(\eta_k)_{k \in \mathbb{N}_0}$ of Algorithm 3 satisfy (A1)–(A4_ε).

$$(A1) \quad \text{Stability.} \quad \left| \eta(\mathcal{T}, \mathcal{T} \cap \hat{\mathcal{T}}) - \eta(\hat{\mathcal{T}}, \mathcal{T} \cap \hat{\mathcal{T}}) \right| \leq \Lambda_1 \delta(\mathcal{T}, \hat{\mathcal{T}}).$$

$$(A2) \quad \text{Reduction.} \quad \eta(\hat{\mathcal{T}}, \hat{\mathcal{T}} \setminus \mathcal{T}) \leq \rho_2 \eta(\mathcal{T}, \mathcal{T} \setminus \hat{\mathcal{T}}) + \Lambda_2 \delta(\mathcal{T}, \hat{\mathcal{T}}).$$

$$(A3_\epsilon) \quad \text{Discrete reliability.} \quad \delta^2(\mathcal{T}, \hat{\mathcal{T}}) \leq \Lambda_3 \eta^2(\mathcal{T}, \mathcal{R}(\mathcal{T}, \hat{\mathcal{T}})) + \hat{\Lambda}_3 \eta^2(\hat{\mathcal{T}}) + \epsilon_3 \eta^2(\mathcal{T}).$$

$$(A4_\epsilon) \quad \text{Quasiorthogonality.} \quad \sum_{j=\ell}^{\ell+m} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) \leq \Lambda_4 \eta_\ell^2 + \epsilon_4 \sum_{j=\ell}^{\ell+m} \eta_j^2 \text{ for any } \ell, m \in \mathbb{N}_0.$$

The Axiom (A3_ε) replaces (A3) from [CR17] adding the extra term $\epsilon_3 \eta^2(\mathcal{T})$ on the right hand side.

6.2 The general optimality result

Theorem 6.1 below contains smallness assumptions for the constants $\hat{\Lambda}_3, \epsilon_3$, and ϵ_4 . In a typical application such as Theorem 5.34 the quantities $\hat{\Lambda}_3, \epsilon_3, \epsilon_4$ contain a power of the initial mesh-size $h_0 := \max_{T \in \mathcal{T}_0} h_T$ such that the assumptions are satisfied for a sufficiently fine initial triangulation $\mathcal{T}_0$. Given $\epsilon_3 < \Lambda_1^{-2}$, set $\Theta := (1 - \Lambda_1^2 \epsilon_3) / (1 + \Lambda_1^2 \Lambda_3)$. Any choice of μ and ξ with $0 < \mu < \rho_2^{-2} - 1$ and $0 < \xi < (1 - (1 + \mu)\rho_2^2)\Theta / (1 - \Theta)$ implies

$$\rho_{12} := \Theta \rho_2^2 (1 + \mu) + (1 - \Theta)(1 + \xi) < 1 \quad \text{and} \quad \Lambda_{12} := (1 + 1/\xi) \Lambda_1^2 + (1 + 1/\mu) \Lambda_2^2 < \infty.$$

Theorem 6.1 (rate optimality of the adaptive algorithm). *Suppose the axioms (A1)–(A_{4 ϵ}) hold with*

$$\Lambda_1^2 \epsilon_3 < 1, \quad \hat{\Lambda}_3(\Lambda_1^2 + \Lambda_2^2) < 1, \quad \text{and} \quad \epsilon_4 < (1 - \rho_{12})/\Lambda_{12}.$$

Then the output $(\mathcal{T}_\ell)_{\ell \in \mathbb{N}_0}$ and $(\eta_\ell)_{\ell \in \mathbb{N}_0}$ of Algorithm 3 with bulk parameter $0 < \theta < \Theta := (1 - \Lambda_1^2 \epsilon_3)/(1 + \Lambda_1^2 \Lambda_3)$ satisfies, for any $s > 0$, the equivalence

$$\sup_{\ell \in \mathbb{N}_0} (1 + |\mathcal{T}_\ell| - |\mathcal{T}_0|)^s \eta_\ell \approx \sup_{N \in \mathbb{N}_0} (1 + N)^s \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T}). \quad (6.2)$$

In other words, if the estimator $\eta(\mathcal{T})$ converges with rate $s > 0$ for some optimal selection of triangulations $\mathcal{T} \in \mathbb{T}$, then the output η_ℓ of Algorithm 3 converges with the same rate s .

6.3 Preparation of the optimality proof

The proof of Theorem 6.1 below reviews parts of the analysis in [Car+14a; CR17] for convenient reading. The relevant extensions are contained in Theorem 6.5 and Lemma 6.6.b below.

First (A1)–(A2) and the Dörfler marking strategy with bulk parameter $\theta < \Theta < 1$ imply the following estimator reduction (A12).

Lemma 6.2 (estimator reduction (A12)). *Suppose (A1)–(A2) and any choice of μ and ξ with $0 < \mu < \rho_2^{-2} - 1$ and $0 < \xi < (1 - (1 + \mu)\rho_2^2)\Theta/(1 - \Theta)$. Set*

$$\rho_{12} := \Theta \rho_2^2 (1 + \mu) + (1 - \Theta)(1 + \xi) < 1 \quad \text{and} \quad \Lambda_{12} := (1 + 1/\xi)\Lambda_1^2 + (1 + 1/\mu)\Lambda_2^2 < \infty.$$

Given any $\mathcal{T} \in \mathbb{T}$ and any admissible refinement $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$, such that the Dörfler marking condition $\theta \eta^2(\mathcal{T}) \leq \eta^2(\mathcal{T}, \mathcal{T} \setminus \hat{\mathcal{T}})$ holds for $0 < \theta < \Theta$, then

$$\eta^2(\hat{\mathcal{T}}) \leq \rho_{12} \eta^2(\mathcal{T}) + \Lambda_{12} \delta^2(\mathcal{T}, \hat{\mathcal{T}}) \quad (\text{A12})$$

In particular, the output $\eta_\ell := \eta(\mathcal{T}_\ell) \equiv \eta(\mathcal{T}_\ell, \mathcal{T}_\ell)$ and $\eta_{\ell+1} := \eta(\mathcal{T}_{\ell+1}) \equiv \eta(\mathcal{T}_{\ell+1}, \mathcal{T}_{\ell+1})$ from Algorithm 3 with $0 < \theta < \Theta$ satisfies (A12) for any $\ell \in \mathbb{N}_0$.

Proof. This is [Car+14a, Lem. 4.7] in adapted notation and a reduced version of [CR17, Thm. 4.1]. Recall that $(a + b)^2 \leq (1 + \beta)a^2 + (1 + 1/\beta)b^2$ for any positive $a, b, \beta > 0$. The axioms (A1)–(A2) apply to the decomposition $\eta^2(\hat{\mathcal{T}}) = \eta^2(\hat{\mathcal{T}}, \mathcal{T} \cap \hat{\mathcal{T}}) + \eta^2(\hat{\mathcal{T}}, \hat{\mathcal{T}} \setminus \mathcal{T})$ and show

$$\begin{aligned} \eta^2(\hat{\mathcal{T}}) &\leq (1 + \xi) \eta^2(\mathcal{T}, \mathcal{T} \cap \hat{\mathcal{T}}) + (1 + 1/\xi) \Lambda_1^2 \delta^2(\mathcal{T}, \hat{\mathcal{T}}) \\ &\quad + (1 + \mu) \rho_2^2 \eta^2(\mathcal{T}, \mathcal{T} \setminus \hat{\mathcal{T}}) + (1 + 1/\mu) \Lambda_2^2 \delta^2(\mathcal{T}, \hat{\mathcal{T}}) \\ &\leq (1 + \xi) \eta^2(\mathcal{T}) + ((1 + \mu) \rho_2^2 - (1 + \xi)) \eta^2(\mathcal{T}, \mathcal{T} \setminus \hat{\mathcal{T}}) + \Lambda_{12} \delta^2(\mathcal{T}, \hat{\mathcal{T}}) \end{aligned}$$

with abbreviation $\Lambda_{12} := (1 + 1/\xi)\Lambda_1^2 + (1 + 1/\mu)\Lambda_2^2 > 0$ in the last step. Note that $0 < \mu < \rho_2^{-2} - 1$ implies $(1 + \mu)\rho_2^2 \leq 1 < 1 + \xi$ and so $\eta^2(\widehat{\mathcal{T}}) \leq (1 + \xi)\eta^2(\mathcal{T}) + \Lambda_{12}\delta^2(\mathcal{T}, \widehat{\mathcal{T}})$. Since the Dörfler marking ensures $\Theta\eta^2(\mathcal{T}) \leq \eta^2(\mathcal{T}, \mathcal{T} \setminus \widehat{\mathcal{T}})$ and $((1 + \mu)\rho_2^2 - (1 + \xi)) < 0$, the last displayed inequality leads to the improved estimate

$$\eta^2(\widehat{\mathcal{T}}) \leq ((1 + \xi)(1 - \Theta) + (1 + \mu)\rho_2^2\Theta)\eta^2(\mathcal{T}) + \Lambda_{12}\delta^2(\mathcal{T}, \widehat{\mathcal{T}}).$$

The definition $\rho_{12} := \Theta\rho_2^2(1 + \mu) + (1 - \Theta)(1 + \xi)$ concludes the proof and the conditions $\Theta \leq 1$, $0 < \mu$, and $0 < \xi < (1 - (1 + \mu)\rho_2^2)\Theta/(1 - \Theta)$ ensure $0 < \rho_{12} < 1$. $\square$

The axioms (A12) and (A4 $_{\epsilon}$) as well as smallness assumptions on ϵ_4 imply the stronger quasiorthogonality (A4) below. Recall $\eta_{\ell} := \eta(\mathcal{T}_{\ell}) \equiv \eta(\mathcal{T}_{\ell}, \mathcal{T}_{\ell})$ for any $\ell \in \mathbb{N}_0$.

Lemma 6.3 (quasiorthogonality (A4)). *Suppose (A12) holds with $0 < \rho_{12} < 1$ and $\Lambda_{12} > 0$ and (A4 $_{\epsilon}$) with $0 < \epsilon_4 < (1 - \rho_{12})/\Lambda_{12}$, then the constant $\Lambda_{\text{qo}} := \Lambda_4 + \epsilon_4(1 + \Lambda_{12}\Lambda_4)/(1 - \rho_{12} - \epsilon_4\Lambda_{12}) > 0$ satisfies*

$$\sum_{j=\ell}^{\ell+m} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) \leq \Lambda_{\text{qo}}\eta_{\ell}^2 \quad \text{for any } \ell, m \in \mathbb{N}_0. \quad (\text{A4})$$

Proof. This is known as general quasiorthogonality in [Car+14a] and adapts [CR17, Thm. 3.1] to the notation at hand. For any $\ell, m \in \mathbb{N}_0$, (A12) applied for $\mathcal{T} = \mathcal{T}_j$ and $\widehat{\mathcal{T}} = \mathcal{T}_{j+1}$ with $\ell \leq j \leq \ell + m - 1$ implies

$$\sum_{j=\ell}^{\ell+m} \eta_j^2 = \eta_{\ell}^2 + \sum_{j=\ell+1}^{\ell+m} \eta_j^2 \leq \eta_{\ell}^2 + \rho_{12} \sum_{j=\ell}^{\ell+m} \eta_j^2 + \Lambda_{12} \sum_{j=\ell}^{\ell+m} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}). \quad (6.3)$$

This and (A4 $_{\epsilon}$) implies

$$(1 - \rho_{12}) \sum_{j=\ell}^{\ell+m} \eta_j^2 \leq \eta_{\ell}^2 + \Lambda_{12} \left(\Lambda_4 \eta_{\ell}^2 + \epsilon_4 \sum_{j=\ell}^{\ell+m} \eta_j^2 \right).$$

Rearranging the terms leads to $\sum_{j=\ell}^{\ell+m} \eta_j^2 \leq \eta_{\ell}^2(1 + \Lambda_{12}\Lambda_4)/(1 - \rho_{12} - \Lambda_{12}\epsilon_4)$ with a positive constant on the right-hand side due to $\epsilon_4 < (1 - \rho_{12})/\Lambda_{12}$. The substitution of this bound for $\sum_{j=\ell}^{\ell+m} \eta_j^2$ in (A4 $_{\epsilon}$) concludes the proof with $\Lambda_{\text{qo}} := \Lambda_4 + \epsilon_4(1 + \Lambda_{12}\Lambda_4)/(1 - \rho_{12} - \epsilon_4\Lambda_{12})$. $\square$

From (A12) and (A4) follows the plain and R -linear convergence on each level.

Lemma 6.4 (convergence). *Suppose (A4) and (A12) with $0 < \rho_{12} < 1$ and $0 < \Lambda_{12} < \infty$ and bulk parameter $0 < \theta < \Theta \leq 1$ in Algorithm 3. Then the output*

$(\eta_\ell)_{\ell \in \mathbb{N}_0}$ of Algorithm 3 and the constants $\Lambda_c := (1 + \Lambda_{12}\Lambda_{\text{qo}})/(1 - \rho_{12}) > 0$ and $q_c := \Lambda_c/(1 + \Lambda_c) < 1$ satisfy

$$\sum_{j=\ell}^{\ell+m} \eta_j^2 \leq \Lambda_c \eta_\ell^2 \quad \text{and} \quad \eta_{\ell+m}^2 \leq \frac{q_c^m}{1 - q_c} \eta_\ell^2 \quad \text{for any } \ell, m \in \mathbb{N}_0. \quad (6.4)$$

Proof. This result is [Car+14a, Lemma 4.9] and [CR17, Thm. 4.2].

Let $\ell, m \in \mathbb{N}_0$. The consequence (6.3) of (A12) from the previous proof and (A4) directly imply

$$(1 - \rho_{12}) \sum_{j=\ell}^{\ell+m} \eta_j^2 \leq \eta_\ell^2 + \Lambda_{12} \sum_{j=\ell}^{\ell+m} \delta^2(\mathcal{T}_j, \mathcal{T}_{j+1}) \leq (1 + \Lambda_{12}\Lambda_{\text{qo}}) \eta_\ell^2.$$

So plain convergence $\sum_{j=\ell}^{\ell+m} \eta_j^2 \leq \Lambda_c \eta_\ell^2$ holds with $\Lambda_c := (1 + \Lambda_{12}\Lambda_{\text{qo}})/(1 - \rho_{12}) > 0$. To proof the R -linear convergence set $\sigma_\ell^2 := \sum_{j=\ell}^{\infty} \eta_j^2$ for any $\ell \in \mathbb{N}_0$. In the limit $m \rightarrow \infty$, the plain convergence bounds $\sigma_{\ell+1}^2 := \sum_{j=\ell+1}^{\infty} \eta_j^2 \leq \sum_{j=\ell}^{\infty} \eta_j^2 \leq \Lambda_c \eta_\ell^2$. Adding $\Lambda_c \sigma_{\ell+1}^2$ on both sides leads to

$$(1 + \Lambda_c) \sigma_{\ell+1}^2 \leq \Lambda_c \left(\eta_\ell^2 + \sum_{j=\ell+1}^{\infty} \eta_j^2 \right) = \Lambda_c \sigma_\ell^2.$$

In other words, $\sigma_{\ell+1}^2 \leq q_c \sigma_\ell^2$ with $q_c := \Lambda_c/(1 + \Lambda_c) < 1$ for any $\ell \in \mathbb{N}_0$. The successive application of this contraction for $m \in \mathbb{N}_0$ and the observation $\sigma_{\ell+1}^2 \leq \Lambda_c \eta_\ell^2$ above conclude the proof in

$$\eta_{\ell+m}^2 \leq \sum_{j=\ell+m}^{\infty} \eta_j^2 = \sigma_{\ell+m}^2 \leq q_c^m \sigma_\ell^2 = q_c^m (\eta_\ell^2 + \sigma_{\ell+1}^2) \leq q_c^m (1 + \Lambda_c) \eta_\ell^2 = \frac{q_c^m}{1 - q_c} \eta_\ell^2.$$

□

On the other hand, (A1)–(A3) are sufficient for the quasimonotonicity (QM) and the comparison lemma. But the discrete reliability is relaxed in (A3_ε), so the proofs of (QM) and the comparison lemma are revisited below.

Theorem 6.5 (QM). *The axioms (A1), (A2), (A3_ε), and $\widehat{\Lambda}_3(\Lambda_1^2 + \Lambda_2^2) < 1$ imply the existence of $\Lambda_{\text{mon}} > 0$ such that $\eta(\widehat{\mathcal{T}}) \leq \Lambda_{\text{mon}} \eta(\mathcal{T})$ holds for any $\mathcal{T} \in \mathbb{T}$ and $\widehat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$.*

Proof. This proof extends [Car+14a, Lem. 3.5] and [CR17, Thm. 3.2]. The axioms (A1)–(A2) apply to the decomposition $\eta^2(\widehat{\mathcal{T}}) = \eta^2(\widehat{\mathcal{T}}, \mathcal{T} \cap \widehat{\mathcal{T}}) + \eta^2(\widehat{\mathcal{T}}, \widehat{\mathcal{T}} \setminus \mathcal{T})$ of the estimator of the fine triangulation $\widehat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ and show

$$\begin{aligned} \eta^2(\widehat{\mathcal{T}}) &\leq \left(\eta(\mathcal{T}, \mathcal{T} \cap \widehat{\mathcal{T}}) + \Lambda_1 \delta(\mathcal{T}, \widehat{\mathcal{T}}) \right)^2 + \left(\rho_2 \eta(\mathcal{T}, \mathcal{T} \setminus \widehat{\mathcal{T}}) + \Lambda_2 \delta(\mathcal{T}, \widehat{\mathcal{T}}) \right)^2 \\ &\leq (1 + 1/\alpha) \eta^2(\mathcal{T}) + (1 + \alpha) (\Lambda_1^2 + \Lambda_2^2) \delta^2(\mathcal{T}, \widehat{\mathcal{T}}) \end{aligned}$$

with $(a + b)^2 \leq (1 + \alpha)a^2 + (1 + 1/\alpha)b^2$ for any positive a, b and $0 < \alpha < ((\Lambda_1^2 + \Lambda_2^2)\hat{\Lambda}_3)^{-1} - 1$ in the second step. (For $\hat{\Lambda}_3 = 0$, the upper bound for $0 < \alpha < \infty$ is understood as infinity.) The Axiom $(A3_\epsilon)$ controls the distance $\delta^2(\mathcal{T}, \hat{\mathcal{T}})$ and leads to

$$\eta^2(\hat{\mathcal{T}}) \leq \left(1 + 1/\alpha + (1 + \alpha)(\Lambda_1^2 + \Lambda_2^2)(\Lambda_3 + \epsilon_3)\right) \eta^2(\mathcal{T}) + (1 + \alpha)(\Lambda_1^2 + \Lambda_2^2)\hat{\Lambda}_3 \eta^2(\hat{\mathcal{T}}).$$

Since $(1 + \alpha)(\Lambda_1^2 + \Lambda_2^2)\hat{\Lambda}_3 < 1$ due to the choice of α , this proves $\eta^2(\hat{\mathcal{T}}) \leq \Lambda_{\text{mon}}^2 \eta^2(\mathcal{T})$ for

$$\Lambda_{\text{mon}}^2 := \frac{1 + 1/\alpha + (1 + \alpha)(\Lambda_1^2 + \Lambda_2^2)(\Lambda_3 + \epsilon_3)}{1 - (1 + \alpha)(\Lambda_1^2 + \Lambda_2^2)\hat{\Lambda}_3}.$$

□

The convergence is guaranteed by (6.4) and the optimality requires the sufficient smallness of the bulk parameter $\theta < \Theta$ in the adaptive algorithm. This enters with the help of the comparison lemma, where some $\theta_0(\varkappa, \alpha)$ depends on parameter $\varkappa, \alpha$ that allow for $\theta \leq \theta_0(\varkappa, \alpha) < \Theta$. The lemma dates back to [Ste07].

Lemma 6.6 (comparison). *Suppose (QM), i.e., the axioms (A1), (A2), $(A3_\epsilon)$, and $\hat{\Lambda}_3(\Lambda_1^2 + \Lambda_2^2) < 1$. Let $0 < \varkappa < 1$, $0 < \alpha < \infty$, and let $s > 0$ satisfy*

$$M := \sup_{N \in \mathbb{N}_0} (N + 1)^s \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T}) < \infty.$$

Then for any level $\ell \in \mathbb{N}_0$, there exist $\hat{\mathcal{T}}_\ell \in \mathbb{T}(\mathcal{T}_\ell)$ and

$$\theta_0(\alpha, \varkappa) := \frac{1 - \varkappa^2 \left((1 + \alpha) + (1 + 1/\alpha)\Lambda_1^2 \hat{\Lambda}_3 \right) - (1 + 1/\alpha)\Lambda_1^2 \epsilon_3}{1 + (1 + 1/\alpha)\Lambda_1^2 \Lambda_3} < 1$$

such that

- (a) $\eta(\hat{\mathcal{T}}_\ell) \leq \varkappa \eta(\mathcal{T}_\ell) \leq \Lambda_{\text{mon}} M |\mathcal{T}_\ell \setminus \hat{\mathcal{T}}_\ell|^{-s}$ and
- (b) $\theta_0(\alpha, \varkappa) \eta^2(\mathcal{T}_\ell) \leq \eta^2(\mathcal{T}_\ell, \mathcal{R}_\ell)$ with $\mathcal{T}_\ell \setminus \hat{\mathcal{T}}_\ell \subseteq \mathcal{R}_\ell := \mathcal{R}(\mathcal{T}_\ell, \hat{\mathcal{T}}_\ell)$, $|\mathcal{R}_\ell| \leq \Lambda_{\text{ref}} |\mathcal{T}_\ell \setminus \hat{\mathcal{T}}_\ell|$, and $\Lambda_{\text{ref}} \geq 1$.

Proof of (a). This proof is adapted from [Car+14a, Prop. 4.12] resp. [CR17, Lem. 4.3] based on the overlay control (6.6) below and Theorem 6.5.

Without loss of generality suppose $\eta(\mathcal{T}_\ell) > 0$, otherwise $\hat{\mathcal{T}}_\ell = \mathcal{T}_\ell$ satisfies (a)–(b). Choose a minimal $N_\ell \in \mathbb{N}_0$ such that

$$(N_\ell + 1)^{-s} \leq \frac{\varkappa \eta(\mathcal{T}_\ell)}{\Lambda_{\text{mon}} M} < 1.$$

The right-hand side is bounded by 1, since the quasi-monotonicity from Theorem 6.5 and $0 < \varkappa < 1$ imply $\varkappa\eta(\mathcal{T}_\ell) < \eta(\mathcal{T}_\ell) \leq \Lambda_{\text{mon}}\eta(\mathcal{T}_0) \leq \Lambda_{\text{mon}}M$. Hence, $N_\ell \geq 1$ and the choice of the minimizer $N_\ell \in \mathbb{N}_0$ shows

$$(N_\ell + 1)^{-s} \leq \frac{\varkappa\eta(\mathcal{T}_\ell)}{\Lambda_{\text{mon}}M} < N_\ell^{-s}. \quad (6.5)$$

The definition of $M < \infty$ leads to some $\mathcal{T}' \in \mathbb{T}(N_\ell)$ with $(N_\ell + 1)^s\eta(\mathcal{T}') \leq M$. Let $\widehat{\mathcal{T}}_\ell := \mathcal{T}_\ell \otimes \mathcal{T}' \in \mathbb{T}(\mathcal{T}_\ell) \cap \mathbb{T}(\mathcal{T}')$ denote the overlay, i.e., the smallest common refinement of $\mathcal{T}_\ell$ and $\mathcal{T}'$.

This choice, Theorem 6.5, and (6.5) prove the first claim

$$\eta(\widehat{\mathcal{T}}_\ell) \leq \Lambda_{\text{mon}}\eta(\mathcal{T}') \leq \Lambda_{\text{mon}}M(N_{\ell+1} + 1)^{-s} \leq \varkappa\eta(\mathcal{T}_\ell).$$

To prove the second claim, the following overlay control (6.6) is necessary. For the overlay $\widehat{\mathcal{T}}_\ell := \mathcal{T}_\ell \otimes \mathcal{T}'$, [Ste07, p.257], [Cas+08, Lemma 3.7] show

$$|\widehat{\mathcal{T}}_\ell| + |\mathcal{T}_0| \leq |\mathcal{T}_\ell| + |\mathcal{T}'|. \quad (6.6)$$

This, $\mathcal{T}' \in \mathbb{T}(N_\ell)$, and (6.5) imply

$$|\mathcal{T}_\ell \setminus \widehat{\mathcal{T}}_\ell| \leq |\widehat{\mathcal{T}}_\ell| - |\mathcal{T}_\ell| \leq |\mathcal{T}'| - |\mathcal{T}_0| \leq N_\ell \leq \left(\frac{\Lambda_{\text{mon}}M}{\varkappa\eta(\mathcal{T}_\ell)} \right)^{1/s}.$$

Rearranging the terms concludes the proof. $\square$

Proof of (b). A modification of the proofs in [Car+14a, Prop. 4.12] or [CR17, Lem. 4.3] verifies (b). Axiom (A1) and (a) imply that

$$\eta(\mathcal{T}_\ell, \mathcal{T}_\ell \cap \widehat{\mathcal{T}}_\ell) \leq \eta(\widehat{\mathcal{T}}_\ell, \mathcal{T}_\ell \cap \widehat{\mathcal{T}}_\ell) + \Lambda_1\delta(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell) \leq \varkappa\eta(\mathcal{T}_\ell) + \Lambda_1\delta(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell). \quad (6.7)$$

Recall $\eta_\ell^2(\mathcal{M}_\ell) := \eta^2(\mathcal{T}_\ell, \mathcal{M}_\ell) := \sum_{T \in \mathcal{M}_\ell} \eta^2(\mathcal{T}_\ell, T)$ for any $\mathcal{M}_\ell \subset \mathcal{T}_\ell$ and $\eta_\ell := \eta(\mathcal{T}_\ell) \equiv \eta(\mathcal{T}_\ell, \mathcal{T}_\ell)$ and abbreviate $\widehat{\eta}_\ell := \eta(\widehat{\mathcal{T}}_\ell) \equiv \eta(\widehat{\mathcal{T}}_\ell, \widehat{\mathcal{T}}_\ell)$. A weighted Young inequality with $\alpha > 0$, the Axiom (A3_e) with abbreviation $\mathcal{R}_\ell := \mathcal{R}(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell)$, and (a) show that

$$\begin{aligned} \left(\varkappa\eta_\ell + \Lambda_1\delta(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell) \right)^2 &\leq (1+\alpha)\varkappa^2\eta_\ell^2 + (1+1/\alpha)\Lambda_1^2 \left(\Lambda_3\eta_\ell^2(\mathcal{R}_\ell) + \widehat{\Lambda}_3\widehat{\eta}_\ell^2 + \epsilon_3\eta_\ell^2 \right) \\ &\leq (1+\alpha)\varkappa^2\eta_\ell^2 + (1+1/\alpha)\Lambda_1^2 \left(\Lambda_3\eta_\ell^2(\mathcal{R}_\ell) + \widehat{\Lambda}_3\varkappa^2\eta_\ell^2 + \epsilon_3\eta_\ell^2 \right). \end{aligned} \quad (6.8)$$

Recall $\varkappa < 1$, $\alpha > 0$, and set

$$C_a := (1+\alpha)\varkappa^2 + (1+1/\alpha)\Lambda_1^2(\epsilon_3 + \widehat{\Lambda}_3\varkappa^2) \quad \text{and} \quad C_b := (1+1/\alpha)\Lambda_1^2\Lambda_3.$$

Then the combination of (6.7)–(6.8) reads

$$\eta_\ell^2(\mathcal{T}_\ell \cap \widehat{\mathcal{T}}_\ell) \leq C_a\eta_\ell^2 + C_b\eta_\ell^2(\mathcal{R}_\ell). \quad (6.9)$$

Since $\mathcal{T}_\ell \setminus \widehat{\mathcal{T}}_\ell \subseteq \mathcal{R}_\ell$, the estimate (6.9) implies

$$\eta_\ell^2 \leq \eta_\ell^2(\mathcal{R}_\ell) + \eta_\ell^2(\mathcal{T}_\ell \cap \widehat{\mathcal{T}}_\ell) \leq C_a \eta_\ell^2 + (1 + C_b) \eta_\ell^2(\mathcal{R}_\ell).$$

This proves (b) with

$$\frac{1 - C_a}{1 + C_b} = \frac{1 - \left((1 + \alpha) \varkappa^2 + (1 + 1/\alpha) \Lambda_1^2 (\epsilon_3 + \widehat{\Lambda}_3 \varkappa^2) \right)}{1 + (1 + 1/\alpha) \Lambda_1^2 \Lambda_3} = \theta_0(\varkappa, \alpha) < 1.$$

□

6.4 Proof of optimality

Proof of Theorem 6.1. The conclusion is comparable to [Car+14a, Proof of Theorem 4.1 (ii)] or [CR17, Section 4.3]. The function $\theta_0(\alpha, \varkappa)$ in Lemma 6.6.b is bounded from above by $\lim_{\alpha \rightarrow \infty} \theta_0(\alpha, 0) = (1 - \Lambda_1^2 \epsilon_3) / (1 + \Lambda_1^2 \Lambda_3)$ and there exist a choice of $0 < \varkappa < 1$ and $0 < \alpha < \infty$ such that $0 < \theta < \theta_0(\alpha, \varkappa) < \Theta$. This is the first formula on page 2655 in [CR17] and the remaining parts of the proof are almost verbatim to Case A in [CR17]. The arguments are summarized here for convenient reading. The choice of θ and Lemma 6.6.b show

$$\theta \eta^2(\mathcal{T}_\ell) \leq \theta_0(\alpha, \varkappa) \eta^2(\mathcal{T}_\ell) \leq \eta^2(\mathcal{T}_\ell, \mathcal{R}_\ell),$$

i.e., $\mathcal{R}_\ell := \mathcal{R}(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell)$ satisfies the Dörfler marking condition. Recall that $\mathcal{M}_\ell$ denotes the set of marked elements on level ℓ in Algorithm 3, while $\mathcal{M}_\ell^*$ with cardinality $|\mathcal{M}_\ell^*|$ denotes a minimal set of elements satisfying the Dörfler marking condition. Then there exists $\Lambda_{\text{opt}} \geq 1$ with $|\mathcal{M}_\ell| \leq \Lambda_{\text{opt}} |\mathcal{M}_\ell^*| \leq \Lambda_{\text{opt}} |\mathcal{R}_\ell|$. The control over $\mathcal{R}_\ell := \mathcal{R}(\mathcal{T}_\ell, \widehat{\mathcal{T}}_\ell)$ and Lemma 6.6.a ensure

$$|\mathcal{R}_\ell| \leq \Lambda_{\text{ref}} |\mathcal{T}_\ell \setminus \widehat{\mathcal{T}}_\ell| \leq \Lambda_{\text{ref}} \left(\Lambda_{\text{mon}} M / (\varkappa \eta_\ell) \right)^{1/s}.$$

Hence $|\mathcal{M}_\ell| \leq C_c M^{1/s} \eta_\ell^{-1/s}$ with $C_c := \Lambda_{\text{opt}} \Lambda_{\text{ref}} \Lambda_{\text{mon}}^{1/s} \varkappa^{-1/s}$. One important property of the NVB is the overhead control [BDD04, Thm. 2.4], [Ste08, Thm. 6.1]

$$|\mathcal{T}_\ell| - |\mathcal{T}_0| \leq \Lambda_{\text{BDdV}} \sum_{k=0}^{\ell-1} |\mathcal{M}_k| \tag{6.10}$$

with a universal constant Λ_{BDdV} that exclusively depends on $\mathcal{T}_0$. The combination of the above with the overhead control leads to

$$|\mathcal{T}_\ell| - |\mathcal{T}_0| \leq \Lambda_{\text{BDdV}} C_c M^{1/s} \sum_{k=0}^{\ell-1} \eta_k^{-1/s}. \tag{6.11}$$

The R-linear convergence (6.4) bounds the sum $\sum_{k=0}^{\ell-1} \eta_k^{-1/s}$ as in [CR17, Thm. 4.2.c]. For all $0 \leq k < \ell$, the second identity in (6.4) implies

$$\eta_k^{-1/s} \leq \eta_\ell^{-1/s} q_c^{(\ell-k)/(2s)} (1 - q_c)^{-1/(2s)}.$$

Hence the formula for the partial sum of the geometric series shows

$$\sum_{k=0}^{\ell-1} \eta_k^{-1/s} \leq \frac{\eta_\ell^{-1/s}}{(1 - q_c)^{1/(2s)}} \sum_{k=0}^{\ell-1} \left(q_c^{1/(2s)} \right)^{\ell-k} \leq C_d \eta_\ell^{-1/s} \quad (6.12)$$

with constant $C_d := q_c^{1/(2s)} / \left((1 - q_c)^{1/(2s)} (1 - q_c^{1/(2s)}) \right)$. The combination of (6.11)–(6.12) reads

$$|\mathcal{T}_\ell| - |\mathcal{T}_0| \leq \Lambda_{\text{BDdV}} C_c C_d M^{1/s} \eta_\ell^{-1/s}.$$

For $1 \leq |\mathcal{T}_\ell| - |\mathcal{T}_0|$ the last displayed inequality implies $(1 + |\mathcal{T}_\ell| - |\mathcal{T}_0|) \leq 2(|\mathcal{T}_\ell| - |\mathcal{T}_0|) \leq 2\Lambda_{\text{BDdV}} C_c C_d M^{1/s} \eta_\ell^{-1/s}$, while for $|\mathcal{T}_\ell| = |\mathcal{T}_0|$ holds $\eta_\ell = \eta(\mathcal{T}_0) \leq M$, i.e., $1 \leq M^{1/s} \eta_\ell^{-1/s}$. This concludes the proof of

$$(1 + |\mathcal{T}_\ell| - |\mathcal{T}_0|)^s \eta_\ell \leq \max\{1, (2\Lambda_{\text{BDdV}} C_c C_d)^s\} M \text{ with } M := \sup_{N \in \mathbb{N}_0} (N+1)^s \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T})$$

and so of “ $\lesssim$ ” in Theorem 6.1.

For the proof of the converse implication, assume, without loss of generality, that $0 < \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T})$ and so $0 < \eta_\ell$ for any $\ell \in \mathbb{N}_0$ with $N_\ell := |\mathcal{T}_\ell| - |\mathcal{T}_0| \leq N$. Since no refinement only occurs for $\eta_\ell = 0$, Algorithm 3 leads to $N_\ell < N_{\ell+1}$. Hence there exists a level ℓ with $N_\ell < N \leq N_{\ell+1}$ and $(N+1)^s \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T}) \leq (N_{\ell+1} + 1)^s \eta_\ell$. On each refinement level ℓ each simplex creates at most a finite number $K(n)$ (depending only on the spatial dimension n) of children in the next level $\ell + 1$ [GSS14]. In other words

$$\begin{aligned} \frac{(N_{\ell+1} + 1)}{(N_\ell + 1)} &\leq \frac{K(n)|\mathcal{T}_\ell| - |\mathcal{T}_0| + 1}{|\mathcal{T}_\ell| - |\mathcal{T}_0| + 1} = \frac{K(n)(|\mathcal{T}_\ell| - |\mathcal{T}_0| + 1) + (K(n) - 1)(|\mathcal{T}_0| - 1)}{|\mathcal{T}_\ell| - |\mathcal{T}_0| + 1} \\ &\leq K(n) + (K(n) - 1)(|\mathcal{T}_0| - 1) \end{aligned}$$

This concludes the proof of

$$(N+1)^s \min_{\mathcal{T} \in \mathbb{T}(N)} \eta(\mathcal{T}) \leq (K(n) + (K(n) - 1)(|\mathcal{T}_0| - 1))^s (N_\ell + 1)^s \eta_\ell \text{ with } N_\ell := |\mathcal{T}_\ell| - |\mathcal{T}_0|,$$

and so the proof of “ $\gtrsim$ ” in Theorem 6.1. Hence the rate optimality of Algorithm 3 stated in Theorem 6.1 holds. $\square$

7 Numerical experiments

This chapter presents numerical experiments in 2D based on the data structure and assembling from [CB18, Sec. 7.8]. The software is an extension of the Matlab package AFEM [CN09] maintained at the Humboldt-Universität and utilizes routines for Crouzeix-Raviart and Morley eigenvalue problems adapted from the thesis of Dietmar Gallistl [Gal14]. Some details on the additional functions implemented for this thesis can be found in Appendix C. The complete software is attached in Appendix D. The experiments were conducted on an Intel Xeon 8280L server with Matlab version R2020b. The calculation of the local system matrices was done in parallel on 12 kernels. All matrix eigenvalue problems are solved with the Matlab routine `eigs` exactly; the termination and round-off errors are small and neglected for simplicity. An algebraic bound [Par98, p. 15.9.1] could partly circumvent the issue of inexact solve as in [CG14b]. However, the discussion of interval arithmetic and perturbation analysis for inexact solve in numerical linear algebra is beyond this thesis.

7.1 Numerical realization

7.1.1 Domains and refinement

Fig. 7.1 displays the initial triangulations $\mathcal{T}_0$ for the numerical experiments below. In the logarithmically scaled convergence history plots the respective error or estimator is displayed with respect to the number of triangles $|\mathcal{T}|$ in the triangulation. Of course the modified HHO (4.2) and the extra-stabilized (5.2) methods have more degrees of freedom than the lowest-order nonconforming methods. However, the implementation uses the same local mass and stiffness matrices, see Appendix C, and an especially adapted eigensolver could utilize the special problem structure. The goal of the convergence plots is to compare the quality of the GLB on a given mesh.

The meshes in Fig. 7.1 are refined with uniform red-refinement (solid line and filled markers), i.e., in each step each triangle is divided in four congruent triangles, if the parameter $\theta = 1$. If $\theta < 1$, adaptive mesh-refinement with newest-vertex bisection is utilized, see Section 2.2. In this case, θ denotes the bulk parameter in the Dörfler marking algorithm. The standard Dörfler marking strategy (dashed lines and half-filled markers) is compared with the slightly modified algorithm described in Subsection 5.5.6, that ensures after the Dörfler marking that at

least one triangle of maximal mesh-size is marked for refinement (dotted lines and half-filled markers). The problem specific estimators η are described below. For multiple eigenvalues or a cluster of eigenvalues the cluster marking strategy [Gal15a] is available. In this case, the sum of the squared local estimates $\eta_j^2(T)$ for all indices $j \in J$ belonging to the cluster J

$$\eta_J^2(T) := \sum_{j \in J} \eta_j^2(T) \quad \text{for any } T \in \mathcal{T} \quad (7.1)$$

is used as local estimator contribution $\eta_J(T)$ in the adaptive algorithm.

7.1.2 Computation of reference eigenvalues

The reference eigenvalues for the geometries in Fig. 7.1 were computed by Benedikt Gräßle on adaptively refined meshes. For the Laplacian a conforming P_5 finite element method was utilized. The eigenvalues of the bi-Laplacian were computed with the conforming piecewise P_5 Agyris finite element [Grä22]. The results clearly show the significant digits.

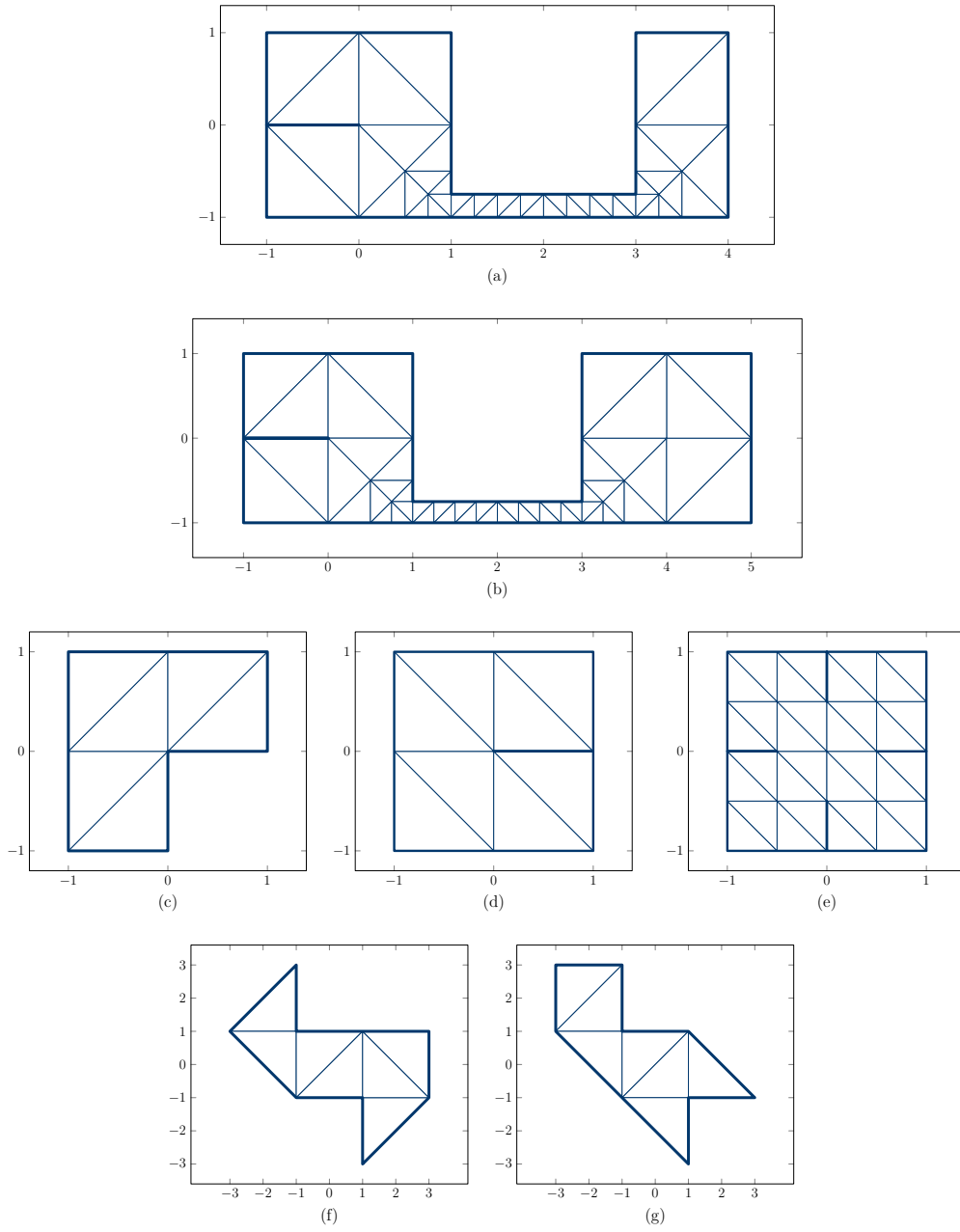


Figure 7.1: Initial triangulation $\mathcal{T}_0$ of (a) the asymmetric dumbbell-slit domain, (b) the dumbbell-slit domain, (c) the L-shaped domain, (d) the slit domain, and (e) the four-slit domain as well as the isospectral domains (f) and (g)

7.2 The Laplace eigenvalue problem

For the Laplace eigenvalue problem this section compares the guaranteed lower-bounds computed with the lowest-order HHO method (4.2), the extra-stabilized method (5.2) for $m = 1$, and the post-processing (4.46). Throughout this section, the energy scalar product reads $a(\bullet, \bullet) := (\nabla \bullet, \nabla \bullet)_{L^2(\Omega)}$ with piecewise version $a_{\text{pw}}(\bullet, \bullet) := (\nabla_{\text{pw}} \bullet, \nabla_{\text{pw}} \bullet)_{L^2(\Omega)}$ and the following notation applies. Let $(\lambda_j, u_j) \in \mathbb{R}_+ \times H_0^1(\Omega) \setminus \{0\}$ denote the j -th continuous Laplace eigenpair with

$$a(u_j, v) = \lambda_j b(u_j, v) \quad \text{for all } v \in H_0^1(\Omega).$$

Let $(\lambda_{\text{HHO}}(j), u_{\text{HHO}}(j)) \in \mathbb{R}_+ \times V_{\text{HHO}} \setminus \{0\}$ with $u_{\text{HHO}} := (u_{\mathcal{T}}, u_{\mathcal{F}}) \in V_{\text{HHO}} := P_1(\mathcal{T}) \times P_0(\mathcal{F}(\Omega))$ denote the j -th discrete eigenpair in the modified HHO method, that satisfies

$$a_{\text{HHO}}(u_{\text{HHO}}(j), v_{\text{HHO}}) = \lambda_{\text{HHO}}(j) b_{\text{HHO}}(u_{\text{HHO}}(j), v_{\text{HHO}}) \quad \text{for all } v_{\text{HHO}} \in V_{\text{HHO}}.$$

Given the bijective extension operator $I_{\mathcal{F}} : P_0(\mathcal{F}(\Omega)) \rightarrow \text{CR}_0^1(\mathcal{T})$ which maps $v_{\mathcal{F}} = (v_F)_{F \in \mathcal{F}} \in P_0(\mathcal{F}(\Omega))$ onto $I_{\mathcal{F}}(v_{\mathcal{F}}) := \sum_{F \in \mathcal{F}} v_F \psi_F \in \text{CR}_0^1(\mathcal{T})$ and $s(\mathcal{T}) := \Pi_0(|\bullet - \text{mid}(\mathcal{T})|^2)/4 \in P_0(\mathcal{T})$, the bilinear forms in Proposition 4.17.a read

$$\begin{aligned} a_{\text{HHO}}(v_{\text{HHO}}, w_{\text{HHO}}) &:= a_{\text{pw}}(I_{\mathcal{F}}(v_{\mathcal{F}}), w_{\mathcal{F}}) \\ &\quad + \frac{1-\alpha}{s(\mathcal{T})} \left(\Pi_0(I_{\mathcal{F}}(v_{\mathcal{F}}) - v_{\mathcal{T}}), \Pi_0(I_{\mathcal{F}}(w_{\mathcal{F}}) - w_{\mathcal{T}}) \right)_{L^2(\Omega)} \\ &\quad + \beta a_{\text{pw}}(v_{\mathcal{T}} - I_{\mathcal{F}}(v_{\mathcal{F}}), w_{\mathcal{T}} - I_{\mathcal{F}}(w_{\mathcal{F}})) \\ b_{\text{HHO}}(v_{\text{HHO}}, w_{\text{HHO}}) &:= (v_{\mathcal{T}}, w_{\mathcal{T}})_{L^2(\Omega)} \\ &\quad \text{for all } v_{\text{HHO}} = (v_{\mathcal{T}}, v_{\mathcal{F}}), w_{\text{HHO}} = (w_{\mathcal{T}}, w_{\mathcal{F}}) \in V_{\text{HHO}}. \end{aligned}$$

Let $(\lambda_{\text{es}}(j), u_{\text{es}}(j)) \in \mathbb{R}_+ \times V_{\text{es}} \setminus \{0\}$ with $u_{\text{es}}(j) := (u_{\text{pw}}(j), u_{\text{CR}}(j)) \in V_{\text{es}} := P_1(\mathcal{T}) \times \text{CR}_0^1(\mathcal{T})$ denote the j -th discrete eigenpair of the extra-stabilized Crouzeix-Raviart method, that satisfies

$$a_{\text{es}}(u_{\text{es}}(j), v_{\text{es}}) = \lambda_{\text{es}}(j) b_{\text{es}}(u_{\text{es}}(j), v_{\text{es}}) \quad \text{for all } v_{\text{es}} \in V_{\text{es}}.$$

The bilinear forms read

$$\begin{aligned} a_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= a_{\text{pw}}(v_{\text{CR}}, w_{\text{CR}}) + \kappa_1^{-2} (h_{\mathcal{T}}^{-2} (v_{\text{pw}} - v_{\text{CR}}), w_{\text{pw}} - w_{\text{CR}})_{L^2(\Omega)}, \\ b_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= b(v_{\text{pw}}, w_{\text{pw}}) \quad \text{for all } v_{\text{es}} = (v_{\text{pw}}, v_{\text{CR}}), w_{\text{es}} = (w_{\text{pw}}, w_{\text{CR}}) \in V_{\text{es}}. \end{aligned}$$

Recall from Subsection 5.3.3 that V_{HHO} can be identified with V_{es} due to the bijective extension operator $I_{\mathcal{F}} : P_0(\mathcal{F}(\Omega)) \rightarrow \text{CR}_0^1(\mathcal{T})$. Finally, let $(\lambda_{\text{CR}}(j), \phi_{\text{CR}}(j)) \in \mathbb{R}_+ \times \text{CR}_0^1(\mathcal{T}) \setminus \{0\}$ denote the j -th Crouzeix-Raviart eigenpair in (4.45) with

$$a_{\text{pw}}(\phi_{\text{CR}}(j), v_{\text{CR}}) = \lambda_{\text{CR}}(j) b(\phi_{\text{CR}}(j), v_{\text{CR}}) \quad \text{for all } v_{\text{CR}} \in \text{CR}_0^1(\mathcal{T}).$$

Recall the guaranteed lower eigenvalue bound from (4.46)

$$\text{GLB}_{\text{CR}}(j) := \frac{\lambda_{\text{CR}}(j)}{1 + \kappa_1^2 h_{\max}^2 \lambda_{\text{CR}}(j)} \leq \lambda_j.$$

7.2.1 Setting the parameter

In the two-dimensional case, the Crouzeix-Raviart interpolation constant $\kappa_1 = \sqrt{1/48 + j_{1,1}^{-1}} = 0.298234942888$ [CG14a] is slightly improved by the numerical bound $\kappa_1 = 0.1893$ from [Liu15] (see Theorem 3.24.b). In the figures below is clearly stated which constant κ_1 is selected. A darker version of the line color shall indicate the use of $\kappa_1 = 0.1893$ in comparison plots.

For the modified HHO method the choice of the parameter $0 < \alpha < 1$ and $0 < \beta < \infty$ is essential. The computable condition $\tau^2\beta + \delta^2\lambda_{\text{HHO}}(j) \leq \alpha < 1$ implies, see Theorem 4.8, that the discrete eigenvalue $\lambda_{\text{HHO}}(j)$ is a lower bound for the exact eigenvalue λ_j . This condition restricts the choice of the parameters $0 < \alpha < 1$ and $0 < \beta < \infty$. For right-isosceles triangles Lemma 4.21 shows that $\tau \leq 6/\pi \leq 1.90986$ and $\delta \leq \kappa_1 h_{\max}$. Corollary 4.18 proves that $\lambda_{\text{HHO}}(j) \leq \lambda_{\text{CR}}(j)$. This means that on a given triangulation $\mathcal{T}$ with known $\lambda_{\text{CR}}(j)$, the parameter choice $0 < \alpha < 1$ and $\beta = (\alpha - \delta^2\lambda_{\text{CR}}(j))/\tau^2 > 0$ leads to a guaranteed lower bound $\lambda_{\text{HHO}}(j)$ of the j -th exact eigenvalue λ_j . The choice of the parameter $\alpha = 0.5$ and $\beta = 0.1$ turns out to be sufficient in all experiments. If the parameters are chosen beforehand, the condition $\tau^2\beta + \delta^2\lambda_{\text{HHO}}(j) \leq \alpha$ may not be satisfied on a coarser mesh due to the impact of $h_{\max}$. In this case the computed value $\lambda_{\text{HHO}}(j)$ is replaced by zero (which is an obvious guaranteed lower bound).

For the extra-stabilized Crouzeix-Raviart method the computable condition $\lambda_{\text{es}}(j)\kappa_1^2 h_{\max}^2 \leq 1$ guarantees that $\lambda_{\text{es}}(j) \leq \lambda_j$, see Theorem 5.5. If this condition is not satisfied on a coarse mesh, the computed value $\lambda_{\text{es}}(j)$ is replaced by zero as well.

7.2.2 Adaptive mesh-refinement

The parameter $\theta < 1$ denotes the bulk parameter in the Dörfler marking algorithm for the adaptive mesh-refinement with newest-vertex bisection. The standard Dörfler marking strategy (dashed lines and half filled markers) is compared with the slightly modified algorithm from Subsection 5.5.6 which ensures that at least one triangle of maximal mesh-size is marked for refinement (dotted lines and half filled markers).

Theorem 5.34 states that the adaptive Algorithm 2 with local estimator contribution

$$\eta_{\text{es}}^2(T) := |T| \|\lambda_{\text{es}}(j)u_{\text{CR}}(j)\|_{L^2(T)}^2 + |T|^{1/2} \sum_{F \in \mathcal{F}(T)} \|[\nabla u_{\text{CR}}(j)]_F \times \nu_F\|_{L^2(F)}^2 \quad (7.2)$$

for any $T \in \mathcal{T}$ and tangential components $[\nabla u_{\text{CR}}]_F \times \nu_F$ of the jump of the (piece-wise) gradient $[\nabla u_{\text{CR}}]_F$ along any edge $F \in \mathcal{F}(T)$ recovers optimal convergence rates, if λ_j is a simple eigenvalue, the bulk parameter θ is small enough, and the initial triangulation is fine enough. For the extra-stabilized Crouzeix-Raviart method, this estimator drives the adaptive mesh-refinement. Theorem 5.35 proves

the reliability and efficiency with $\eta(\mathcal{T}) \approx \|u_j - u_{\text{CR}}(j)\|_{\text{pw}}$. Hence, the eigenvalue error $\lambda_j - \lambda_{\text{es}}(j)$ is expected to converge with the same rate as η^2 . Note that in all experiments below $\theta = 0.5$ is a sufficiently small bulk parameter and the initial triangulation is fine enough. A comparison of different values for θ is contained in Fig. 7.6.

For the analyzed HHO method, the a posteriori error analysis is open. For the method with alternative stabilization, see Remark 4.4, [CGT22] presents a reliable and efficient estimator. The local contribution for any $T \in \mathcal{T}$ in the lowest-order case read

$$\begin{aligned} \eta_{\text{HHO}}^2(T) := & |T| \|\lambda_{\text{es}}(j) u_{\mathcal{T}}\|_{L^2(T)}^2 + |T|^{1/2} \sum_{F \in \mathcal{F}(T) \cap \mathcal{F}(\Omega)} \|[\nabla I_{\mathcal{F}} u_{\mathcal{F}}(j)]_F \cdot \nu_F\|_{L^2(F)}^2 \\ & + |T|^{1/2} \sum_{F \in \mathcal{F}(T)} \|[\nabla I_{\mathcal{F}} u_{\mathcal{F}}(j)]_F \times \nu_F\|_{L^2(F)}^2 \end{aligned} \quad (7.3)$$

with normal $[\nabla I_{\mathcal{F}} u_{\mathcal{F}}(j)]_F \cdot \nu_F$ and tangential components $[\nabla I_{\mathcal{F}} u_{\mathcal{F}}(j)]_F \times \nu_F$ of the jump along any edge $F \in \mathcal{F}(T)$. For the modified HHO method, the estimator η_{HHO} drives the adaptive refinement on an experimental basis.

For the Crouzeix-Raviart method, the standard estimator with local contribution

$$\eta_{\text{CR}}^2(T) := |T| \|\lambda_{\text{CR}}(j) \phi_{\text{CR}}(j)\|_{L^2(T)}^2 + |T|^{1/2} \sum_{F \in \mathcal{F}(T)} \|[\nabla \phi_{\text{CR}}(j)]_F \times \nu_F\|_{L^2(F)}^2 \quad (7.4)$$

drives the adaptive mesh-refinement. The optimality of this estimator for the principal eigenvalue and even for inexact solve is proven in [CGS15].

Note that in (7.2)–(7.4) the number j of the eigenvalue in (7.1) is suppressed in the notation to improve the readability.

7.2.3 Displayed quantities

Fig. 7.1 displays the initial triangulations $\mathcal{T}_0$ for the numerical experiments below. The respective convergence history plots in Figures 7.3, 7.5, 7.8, 7.10, and 7.12 display the difference of the exact eigenvalue and the various guaranteed lower bounds for uniform mesh-refinement $\theta = 1$ (solid line and filled markers) and adaptive mesh-refinement plotted against the number of triangles $|\mathcal{T}|$. On uniform meshes the $\text{GLB}_{\text{CR}}(j)$ (line color blue) and $\lambda_{\text{es}}(j)$ (line color green) coincide, see Lemma 5.12, and only the green line is visible. Moreover, the error $\lambda_j - \lambda_{\text{HHO}}(j)$ (line color orange) is displayed. The number j of the eigenvalue is illustrated by different markers.

Fig. 7.13 exemplifies adaptive triangulations for the principal (or second) eigenvalue on the domains in Subsections 7.2.4–7.2.8.

Fig. 7.6 displays the differences $\lambda_1 - \lambda_{\text{es}}(1)$ (line color green) and $\lambda_1 - \lambda_{\text{HHO}}$ (line color orange) as well as the estimator values η_{es} (line color violet) and η_{HHO} (line color pink) for different bulk parameter $\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ (dashed line and

half filled markers) and uniform refinement $\theta = 1$ (solid line and filled markers). The size of the marker indicates the value of $\theta < 1$.

Figures 7.2, 7.4, 7.7, 7.9, and 7.11 display a parameter study for the different domains. The figures compare the guaranteed lower bound $\text{GLB}_{\text{CR}}(j)$ (dotted blue line) as well as $\lambda_{\text{es}}(j)$ (dotted green line) for $\lambda_{\text{es}}(j) \neq \text{GLB}_{\text{CR}}(j)$ with the guaranteed lower bound $\lambda_{\text{HHO}}(j)$ (dotted orange curve) computed with different choices of α (and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 > 10^{-3}$ from Subsection 7.2.1). In these graphs the dot-density of the plots indicates the triangulation $\mathcal{T}_\ell$ (the ℓ -th uniform refinement of the initial triangulation $\mathcal{T}_0$) in the computation. For comparison the eigenvalue approximation (assumed to be exact) is displayed as well (dark red line).

7.2.4 Dumbbell-slit domain

On the non-convex asymmetric dumbbell-slit domain $\Omega := ((-1, 4) \times (-1, 1)) \setminus ((-1, 0] \times \{0\} \cup [1, 3] \times [-0.75, 1))$ in Fig. 7.1.a, the principal eigenvalue approximated with the adaptive conforming P_5 finite elements with a three times uniformly refined initial mesh reads $\lambda_1 = 8.3677024309$. Comparable to the experiment in Fig. 1.2 for the bi-Laplacian, the uniform mesh-refinement leads to a reduced convergence rate of 0.5 in Fig. 7.3.a and the GLB_{CR} fails for adaptive mesh-refinement, while the new adaptive bounds recover the optimal rate. Fig. 7.2.a contains the parameter study for the HHO method and $\kappa_1 = 0.2982$ on this domain. Since the initial triangulation Fig. 7.1.a is not uniform, $\text{GLB}_{\text{CR}}(1) < \lambda_{\text{es}}(1)$ for the displayed red-refined triangulation. On at least twice uniform red-refined meshes $\alpha \in [0.28; 0.82]$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2$ implies $\lambda_{\text{es}} < \lambda_{\text{HHO}}$, in other words, the modified HHO method improves the other guaranteed lower bounds. For thrice uniform red-refined meshes, the α -interval in which this improvement occurs is larger, but the improvement is less significant. The choice $\kappa_1 = 0.1893$ has (up to the value of δ that ensures that λ_{HHO} is a lower bound) no influence on λ_{HHO} but improves GLB_{CR} and λ_{es} . Still the parameter choice $\alpha = 0.5$ and $\beta = 0.1$ leads to a minimal improvement on finer meshes. The convergence history plot Fig. 7.3.a compares the errors for $\kappa_1 = 0.2982$ with $\kappa_1 = 0.1893$. The smaller $\kappa_1 = 0.1893$ leads to a slight improvement of guaranteed lower bounds computed with the extra-stabilized and post-processed Crouzeix-Raviart method on coarse uniform meshes and a visible improvement on adaptive meshes. For the modified HHO method the only difference is that the first two uniform approximation λ_{HHO} satisfy the lower bound condition and are not replaced by zero. Otherwise the lines coincide, hence $\lambda_1 - \lambda_{\text{HHO}}(1)$ is displayed only for $\kappa_1 = 0.2982$ to avoid confusion. As mentioned above with the parameter choice $\alpha = 0.5$ and $\beta = 0.1$ the modified HHO method does not improve the extra-stabilized method with $\kappa_1 = 0.1893$ significantly. Moreover, Fig. 7.3.a compares the two described marking strategies. For the new methods the results do not differ and only a dashed line is visible.

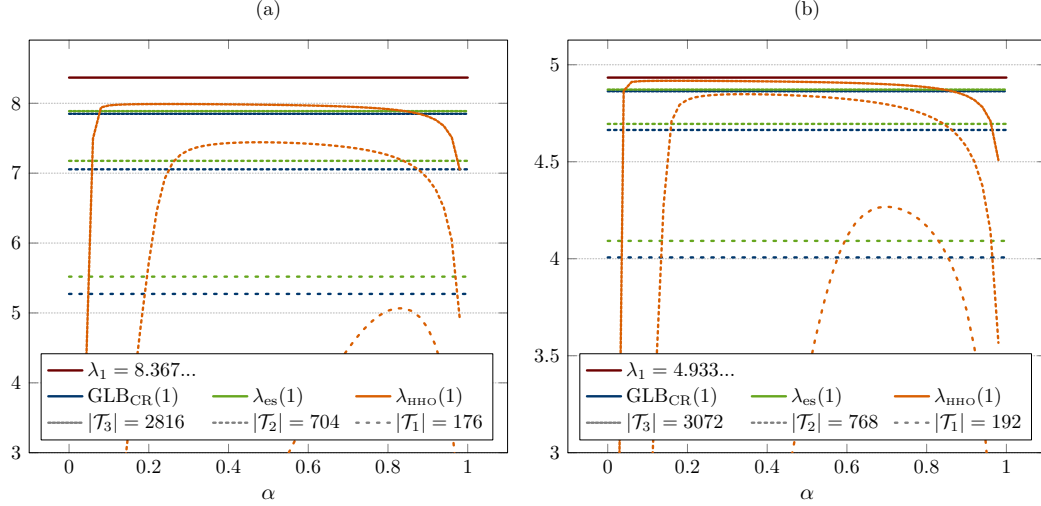


Figure 7.2: Comparison of $\text{GLB}_{\text{CR}}(1)$ and $\lambda_{\text{es}}(1)$ with $\lambda_{\text{HHO}}(1)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(1))/\tau^2 \in (0.003, 0.263)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of (a) the asymmetric dumbbell-slit domain and (b) the dumbbell-slit domain

The Crouzeix-Raviart post-processing benefits from the reduction of the maximal mesh-size in that there are flat steps instead of more or less constant errors. However, the known bound still fails for adaptive mesh-refinement.

Another related example is the dumbbell-slit domain $\Omega := ((-1, 5) \times (-1, 1)) \setminus ((-1, 0] \times \{0\} \cup [1, 3] \times [-0.75, 1))$ in Fig. 7.1.b. The adaptive conforming P_5 finite elements with a three times uniformly refined initial mesh approximated $\lambda_1 = 4.93378555456$ and $\lambda_2 = 8.36770243090$. On this domain the first eigenfunction could be described as bubble on $(-3, 5) \times (-1, 1)$ extended by zero and seems to belong to $H^2(\Omega)$. In other words, uniform mesh-refinement leads to the optimal convergence rate one in Fig. 7.3.b for all methods. Adaptive mesh-refinement improves the errors for the new methods, but the Crouzeix-Raviart post-processing fails again. In this case the HHO method with parameter $\alpha = 0.5$ and $\beta = 0.1$ improves the extra-stabilized method even with $\kappa_1 = 0.1893$ starting from the second uniform red-refinement. The parameter study in Fig. 7.2.b indicates that different parameters may lead to an improvement on the first uniform red-refined mesh for $\alpha \in [0.6, 0.82]$. The other observation are comparable to the asymmetric dumbbell-slit domain. The adaptive mesh-refinement for the dumbbell-slit domain is necessary for the second eigenvalue. For $\lambda_2 = 8.36770243090$ the results are similar to the principal eigenvalue of the asymmetric dumbbell-slit. The difference is that the new marking strategy improves the post-processed Crouzeix-Raviart bound less, since more refinements are necessary to reduce the maximal mesh-size.

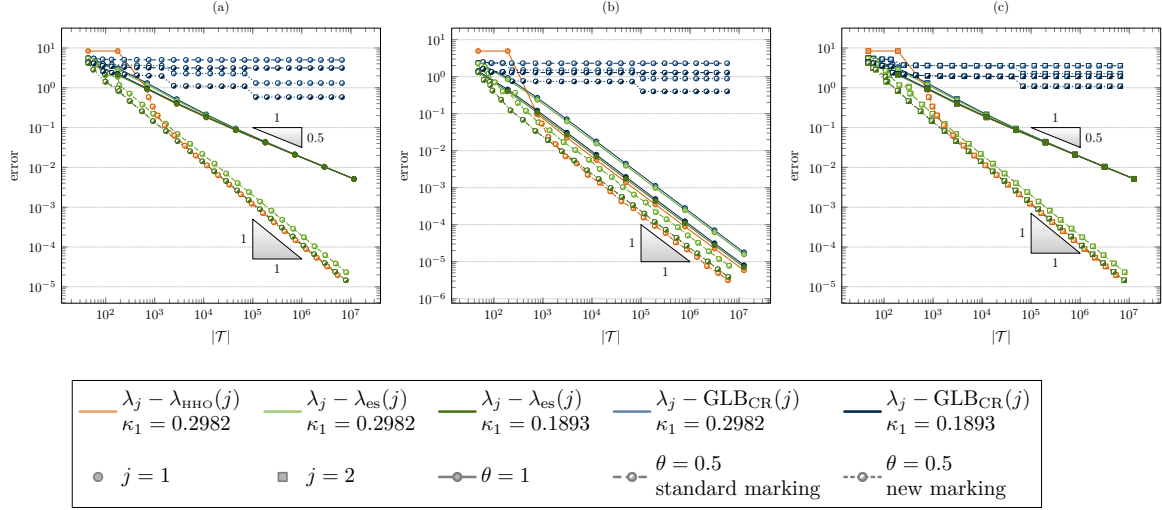


Figure 7.3: Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of (a) the asymmetric dumbbell-slit domain for $j = 1$, and on the dumbbell-slit domain for (b) $j = 1$ as well as (c) $j = 2$

7.2.5 L-shaped domain

On the non-convex L-shaped domain $\Omega := (-1, 1)^2 \setminus ([0, 1] \times (-1, 0])$, the principal eigenvalue $\lambda_1 = 9.639723844030$ is computed with an adaptive conforming P_5 finite element method. The associated eigenvector is apparently in $H_0^1(\Omega) \setminus H^2(\Omega)$ resulting in the reduced convergence rate 0.8 for uniform mesh-refinement in Fig. 7.5.a. As soon as the initial triangulation from Fig. 7.1.c is refined two times, $\lambda_{\text{HHO}}(1)$ improves the known bound $\lambda_{\text{es}}(1) = \text{GLB}_{\text{CR}}(1)$ with $\kappa_1 = 0.2982$ on the uniform meshes as illustrated in Fig. 7.4.a. For the choice $\kappa_1 = 0.1893$ a slight improvement is reached for a three times refined mesh. The adaptive mesh-refinement driven by the estimator (7.2) allows to recover the optimal convergence rate for $\lambda_1 - \lambda_{\text{es}}(1)$, while the estimator (7.3) allows the same for $\lambda_1 - \lambda_{\text{HHO}}(1)$ with a twice red-refined initial triangulation. Fig. 7.5.a compares the results using $\kappa_1 = 0.2982$, see Fig. 1.1, with the ones using $\kappa_1 = 0.1893$. The smaller κ_1 means that the HHO bound improves the extra-stabilized one only minimally. For the post-processed bound, the adaptive refinement leads to steps in the error plot and the results are worse than for comparable uniform meshes for both choices of κ_1 . In [TB06] the multiple eigenvalue $\lambda_{103} = \lambda_{104} = \lambda_{105} = 50\pi^2$ and the associated eigenfunction in $C^\infty(\Omega)$ are analyzed. Fig. 7.5.b shows for these eigenvalues the optimal convergence rate of one with uniform mesh-refinement. The similar plots obtained with adaptive mesh-refinement are omitted for brevity. The parameter choice $\alpha = 0.5$ and $\beta = 0.1$ guarantees the condition $\tau^2\beta + \delta^2\lambda_{\text{HHO}}(j) \leq \alpha$ for $\lambda_{103} = \lambda_{104} = \lambda_{105}$ from the sixth mesh. Fig. 7.4.b illustrates that the HHO

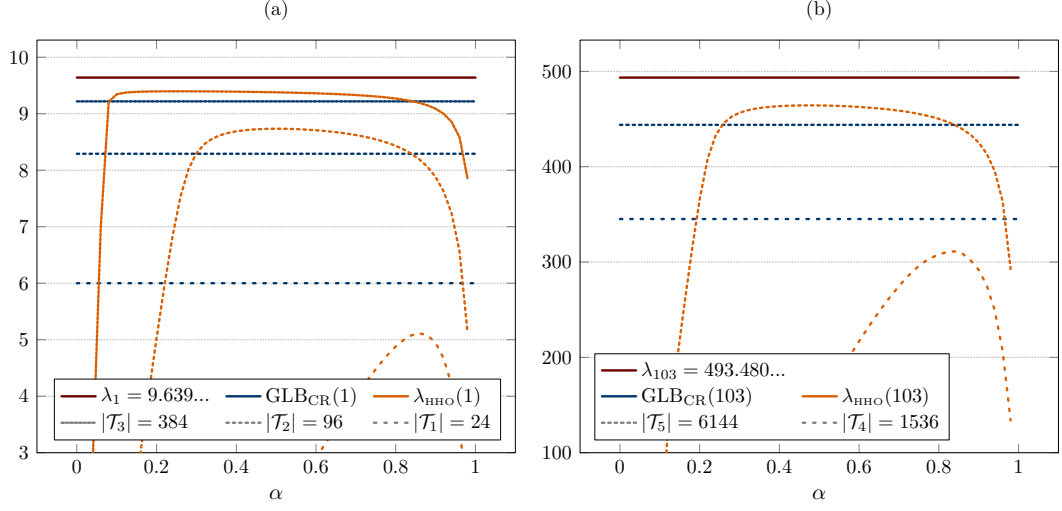


Figure 7.4: Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.262)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the L-shaped domain for (a) $j = 1$ and (b) $j = 103$

method improves the other bounds for the 103rd eigenvalue on the L-shaped domain on at least five times uniform refined meshes for an appropriate parameter choice.

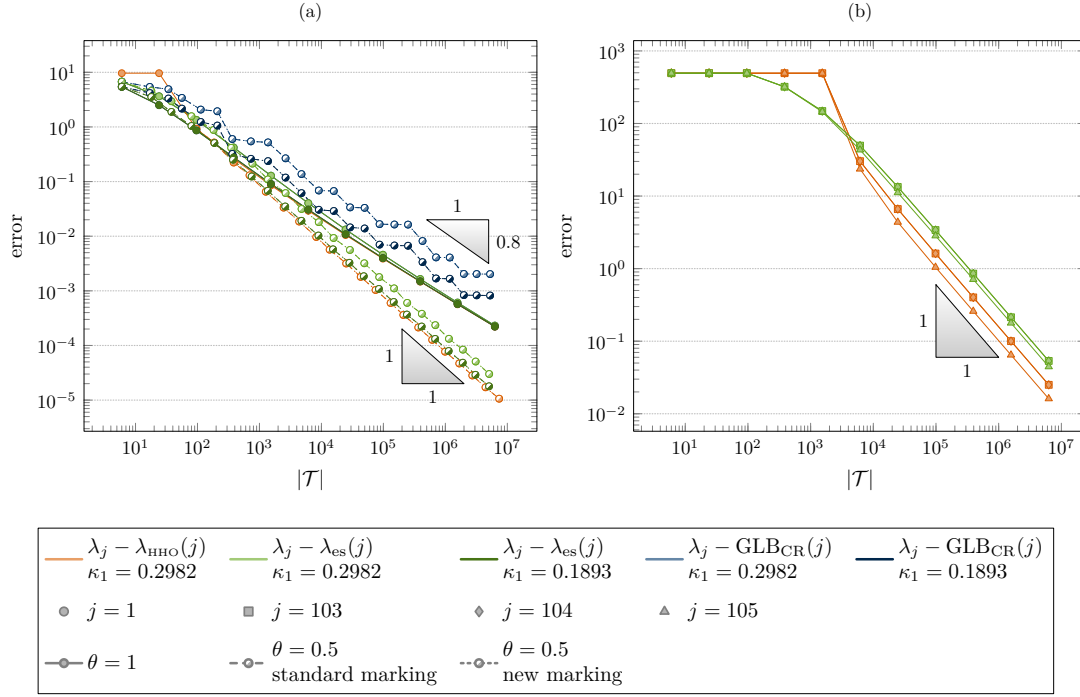


Figure 7.5: Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the L-shaped domain for (a) $j = 1$ and (b) $j \in \{103, 104, 105\}$

7.2.6 Slit domain

On the non-convex slit domain $\Omega := (-1, 1)^2 \setminus ([0, 1] \times \{0\})$, the principal eigenvalue $\lambda_1 = 8.37132971122$ and the sixth $\lambda_6 = 30.53599140845$ are computed with a conforming P_5 finite element method on adaptively refined meshes for comparison. Fig. 7.6.a displays the error $\lambda_1 - \lambda_{\text{es}}(1)$ and $\lambda_1 - \lambda_{\text{HHO}}(1)$ for adaptive refinement with different bulk parameters, while Fig. 7.6.b shows the associated estimators $\eta_{\text{es}}(\mathcal{T})$ and $\eta_{\text{HHO}}(\mathcal{T})$. The adaptive refinement for HHO starts from a twice uniform refined mesh with $|\mathcal{T}_2| = 128$ and the estimator values on the first two uniform meshes which are of another order are not displayed. The optimal empirical convergence rate of one for the error and 0.5 for the estimator are recovered for all bulk parameter $\theta < 1$. The error reduction justifies the choice $\theta = 0.5$ in the experiments, which balances the error, the size of the pre-asymptotic regime, and the number of necessary levels to reach an appropriate refinement of the domain. The first and sixth eigenfunction on the non-convex slit domain are seemingly in $H_0^1(\Omega) \setminus H^2(\Omega)$. For uniform mesh-refinement, this leads to the reduced convergence rates 0.5 for the first in Fig. 7.8.a and 0.6 for the sixth in Fig. 7.8.b. The AFEM algorithm with bulk parameter $\theta = 0.5$ driven by the estimator (7.2) and (7.3) allows to recover the optimal rates for both $\lambda_{\text{es}}(j)$ and $\lambda_{\text{HHO}}(j)$, $j \in \{1, 6\}$. The comparison of results with $\kappa_1 = 0.2982$ and $\kappa_1 = 0.1893$ shows a reduction of the error of the post-processed GLB and the extra-stabilized method. On uniform meshes this leads to $\lambda_{\text{es}}(j) = \text{GLB}_{\text{CR}}(j) \approx \lambda_{\text{HHO}}(j)$ only visible in orange. The parameter choice $\kappa_1 = 0.2982$, $\alpha = 0.5$, and $\beta = 0.1$ allows to compute sharper bounds with the HHO method on finer meshes. For the slit domain, Fig. 7.7 illustrates that the HHO method improves the other bounds on the moderately (two resp. three times for λ_1 resp. λ_6) uniformly refined triangulation for an appropriate parameter choice with a wider range of appropriate parameters on a finer mesh.

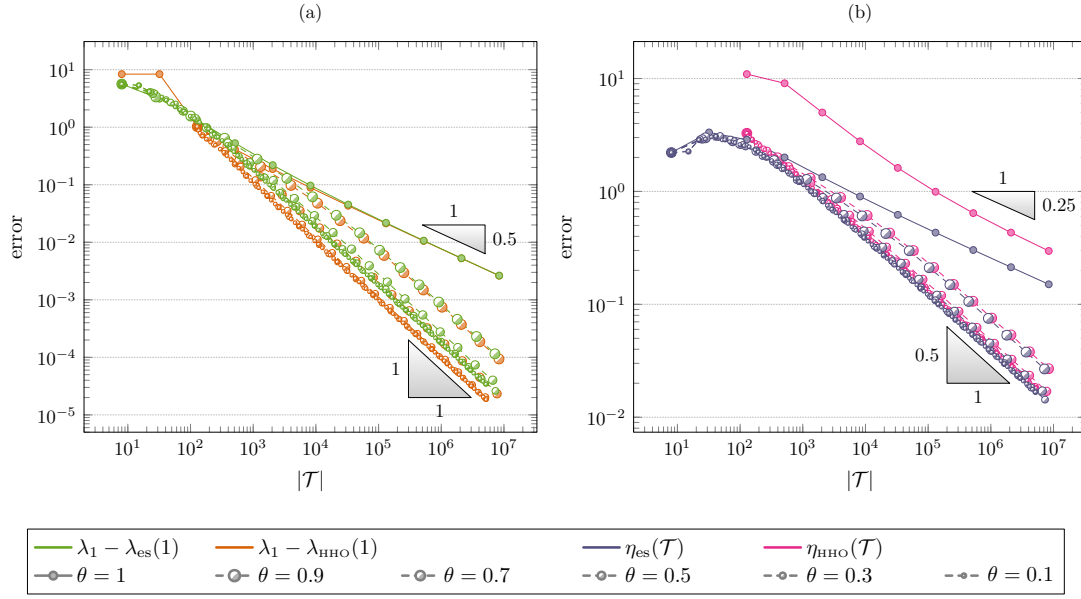


Figure 7.6: Comparison of (a) the distance of λ_1 and $\lambda_{\text{HHO}}(1)$ (with $\alpha = 0.5$ and $\beta = 0.1$) and $\lambda_{\text{es}}(1)$ (with $\kappa_1 = 0.2982$) and (b) the global estimator $\eta_{\text{HHO}}(\mathcal{T})$ and $\eta_{\text{es}}(\mathcal{T})$ computed on uniform ($\theta = 1$) and adaptive ($\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$) meshes of the slit domain

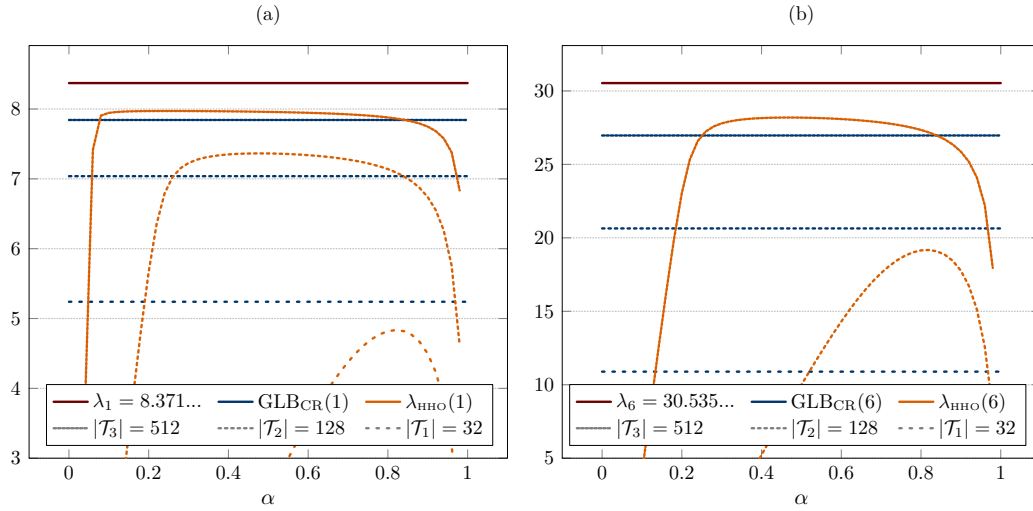


Figure 7.7: Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.002, 0.263)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the slit domain for (a) $j = 1$ and (b) $j = 6$

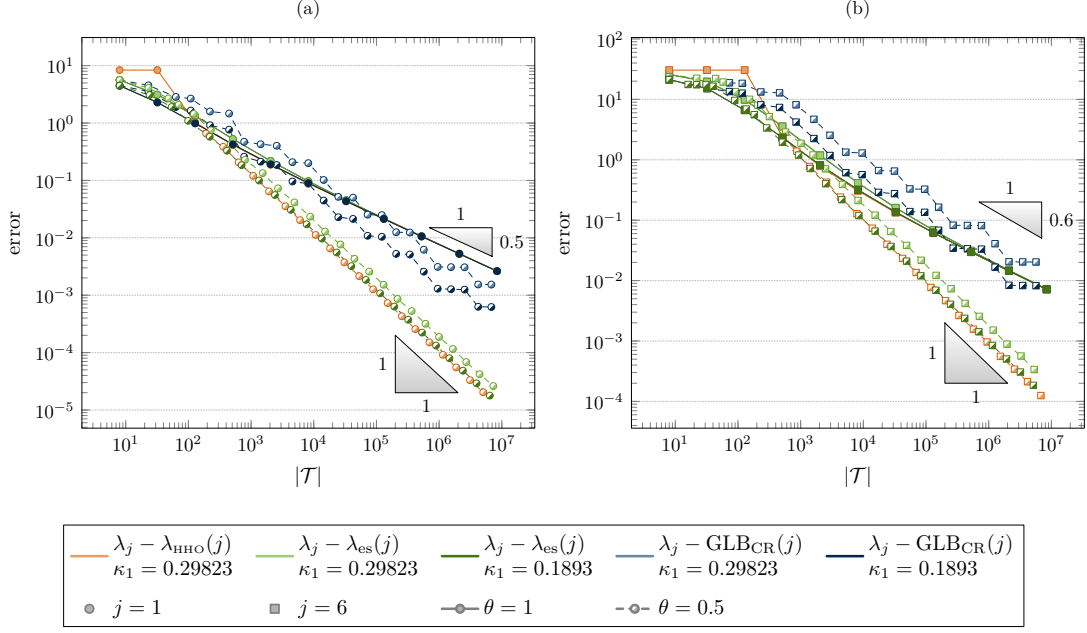


Figure 7.8: Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the slit domain for (a) $j = 1$ and (b) $j = 6$

7.2.7 Four-slit domain

The principal eigenvalue $\lambda_1 = 10.14739197017$ and the double eigenvalue $\lambda_2 = 17.66259617594 = \lambda_3$ on the four-slit domain $\Omega := (-1, 1)^2 \setminus ([0, 0.5) \times \{0\} \cup [0, -0.5) \times \{0\} \cup \{0\} \times [0, 0.5) \cup \{0\} \times [0, -0.5))$ of Fig. 7.1.e are approximated with a conforming P_5 finite element method on adaptively refined meshes. The three eigenfunction on the non-convex four slit domain are seemingly in $H_0^1(\Omega) \setminus H^2(\Omega)$. Uniform mesh-refinement leads to the reduced convergence rates 0.5 in Fig. 7.10. Adaptive mesh-refinement allows to recover the optimal convergence rate for the new methods, while the post-processed guaranteed lower bound suffers from the $h_{\max}$ involvement in form of steps. For the new methods cluster marking with an estimator of the form $\eta^2(T) = \sum_{j=2,3} \eta_j(T)^2$ does not improve the displayed results where the knowledge of the double eigenvalue is not used. For the post-processed $\text{GLB}_{\text{CR}}(3)$ the cluster marking (blue dashed line with diamonds filled at the top) improves the standard marking (blue dashed line with diamonds filled at the bottom). The HHO method with $\alpha = 0.5$ and $\beta = 0.1$ improves $\text{GLB}_{\text{CR}}(j) = \lambda_{\text{es}}(j)$ for $j \in \{1, 2\}$ and $\kappa_1 = 0.2982$ on at least once resp. twice uniform red-refined meshes as illustrated in Fig. 7.9. The improvement is more impressive on coarser uniform meshes.

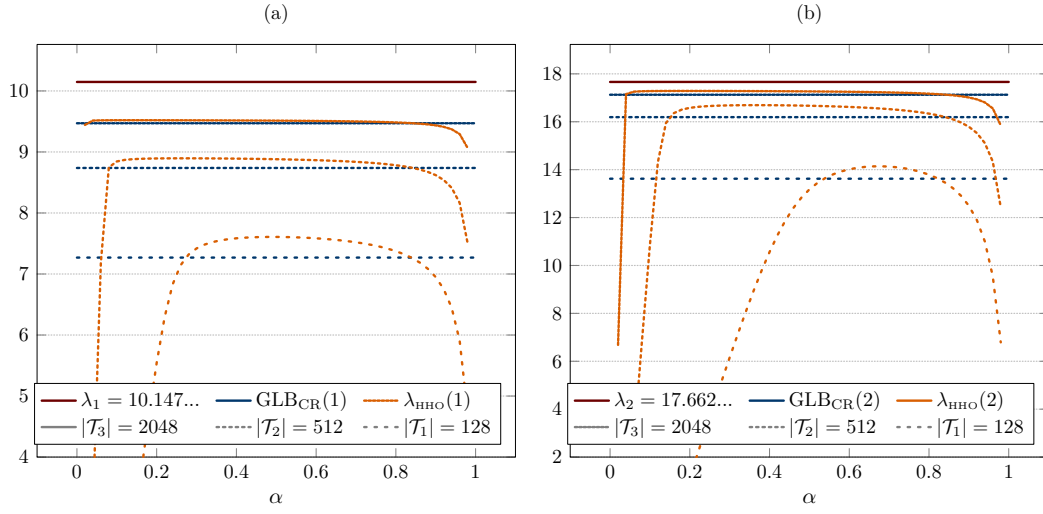


Figure 7.9: Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0,1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.267)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the four-slit domain for (a) $j = 1$ and (b) $j = 2$

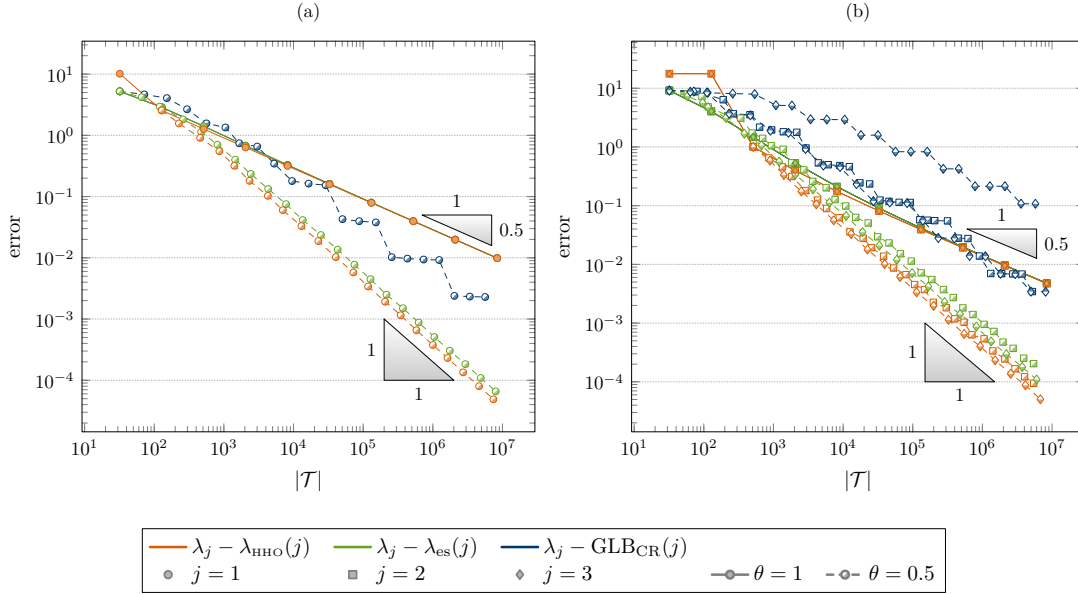


Figure 7.10: Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the four slit domain for (a) $j = 1$ and (b) $j \in \{2, 3\}$

7.2.8 Isospectral domains

The isospectral drums with the initial triangulation in Fig. 7.1.f and Fig. 7.1.g have the same eigenvalues. The paper [Dri97] displays approximations for the first 25 identical eigenvalues on these domains and [TB06] gives the approximation $\lambda_{50} = 54.187936$. Since the computed results are comparable, only the results for the domain in Fig. 7.1.f are displayed. Fig. 7.12 presents convergence plots for the error in the principal and the fiftieth eigenvalue. The first eigenfunction for $\lambda_1 = 2.53794399980$ seems to belong to $H_0^1(\Omega) \setminus H^2(\Omega)$ and leads to the reduced convergence rate 0.7 for uniform mesh-refinement. The AFEM algorithm driven by (7.3) (after two uniform refinements) and driven by (7.2) recovers the optimal convergence rate for the direct lower bounds in Fig. 7.12.a, while the involvement of the maximal mesh-size means that adaptively computed GLB_{CR} are worse than uniform ones. In [TB06] are no remarks on the smoothness of the fiftieth eigenfunction. The numerical results displayed in Fig. 7.12.b suggest that this eigenfunction is indeed in $H^2(\Omega)$. Uniform and adaptive mesh-refinement lead to optimal convergence rates. For the HHO method the condition in Theorem 4.8 is not satisfied on some of the adaptive meshes, hence, the computed eigenvalues are replaced by zero. For the extra-stabilized method this happens on only two relatively coarse triangulation. The parameter studies in Fig. 7.11.a indicate that a parameter $\alpha \in [0.34, 0.82]$ and an appropriate β in the HHO method improve the other bounds for the principal eigenvalue on at least twice uniform red-refined meshes. For the fiftieth eigenvalue four refinements are necessary for improvement with $\alpha \in [0.46, 0.82]$. Note that two refinements are needed to even compute λ_{50} with Crouzeix-Raviart method.

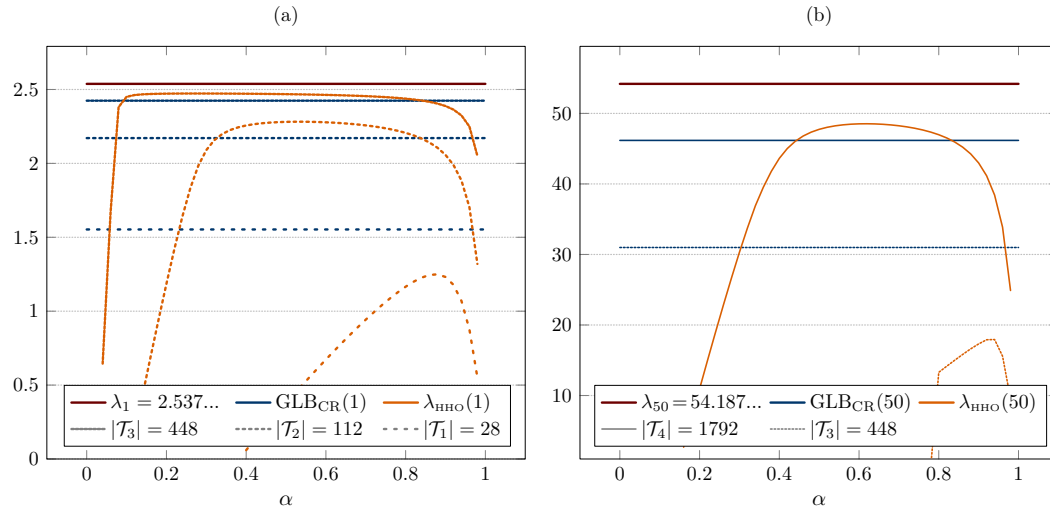


Figure 7.11: Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.262)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the first isospectral domain in 7.1.f for (a) $j = 1$ and (b) $j = 50$

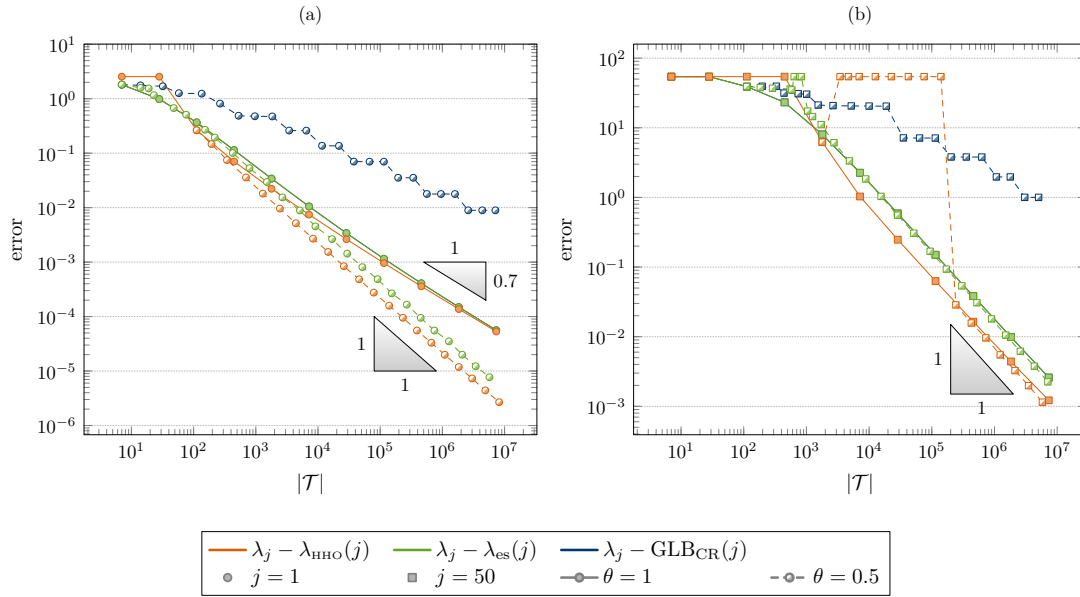


Figure 7.12: Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the first isospectral domain 7.1.f for (a) $j = 1$ and (b) $j = 50$

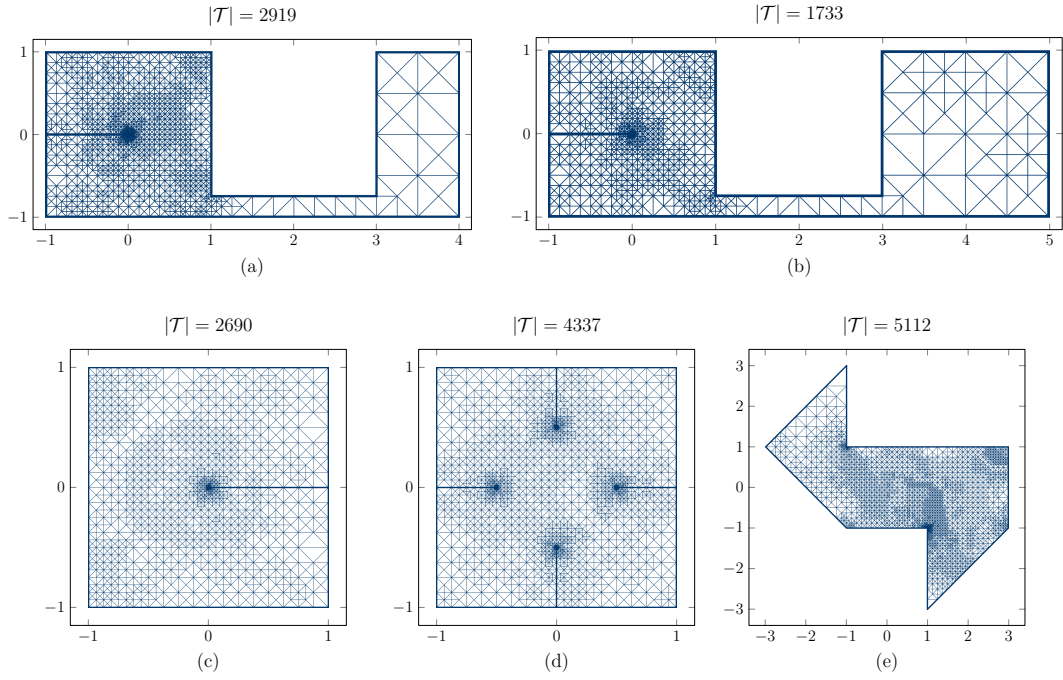


Figure 7.13: Adaptive triangulation $\mathcal{T}$ with $\theta = 0.5$ for the principal eigenvalue of (a) the asymmetric dumbbell-slit domain, (c) the slit domain, (d) the four-slit domain, and (e) the first isospectral domain as well as for (b) the second eigenvalue of the dumbbell-slit domain computed with (7.2) and $\kappa_1 = 0.2982$

7.2.9 Conclusions

This subsection summarizes the empirical observations of the numerical experiments in Subsections 7.2.4–7.2.8.

- (i) All experiments confirm the a priori convergence rates of Theorem 4.13 and Theorem 5.18. The convergence rate depends only on the smoothness of the approximated eigenfunction. For instance in Fig. 7.5.b and 7.12.b, the convergence rate is optimal for uniform refinement despite the reduced convergence rate in Fig. 7.5.a and 7.12.a for the principal eigenfunction.
- (ii) Theorem 4.13 and Theorem 5.18 predict convergence provided the initial mesh is sufficiently fine. In all examples the convergence rate is visible for moderate triangulations, so this restriction does not affect the numerical experiments too much.
- (iii) The new methods provide indeed guaranteed lower bounds, if the respective condition with the parameter in Subsection 7.2.1 is satisfied, fully confirming Theorem 4.8 and Theorem 5.5.
- (iv) For the eigenfunctions in $H_0^1(\Omega) \setminus H^2(\Omega)$, the AFEM algorithm recovers the optimal convergence rates with the experimental estimator (7.3) for the HHO method and with (7.2) for the extra-stabilized Crouzeix-Raviart method as proven in Theorem 5.34. In both cases the bulk parameter $\theta = 0.5$ and relatively coarse initial meshes suffice. The adaptive convergence rates illustrate the advantage of a direct lower bound compared to $\text{GLB}_{\text{CR}}(j)$, e.g., in Subsection 7.2.4, where the latter fails completely.
- (v) The alternative marking strategy seems not necessary for the optimal convergence rate. For the multiple eigenvalues a cluster algorithm was not needed for the new methods.
- (vi) The adaptive meshes for the extra-stabilized method with $\kappa_1 = 0.2982$ displayed in Fig. 7.13 show the expected refinements toward the reentrant corners resp. the endpoints of the slits. The other analysed methods lead to comparable and hence undisplayed adaptive meshes.
- (vii) The guaranteed lower bounds computed with the HHO method do not always improve $\lambda_{\text{es}}(j)$. The parameter studies do not contradict Theorem 4.19 (resp. Remark 4.20) and Theorem 5.14, but the sufficient conditions turn out to be rather vague. Remark 4.20 implies that $\alpha \leq 1 - 1/(72\kappa_1^2) = 0.8438$ (resp. $= 0.6124$) and $\beta \geq 1/(18\kappa_1^2) = 0.6246$ (resp. $= 1.5503$) for $\kappa_1 = 0.2982$ (resp. $\kappa_1 = 0.1893$) suffice for $\text{GLB}_{\text{CR}}(j) \leq \lambda_{\text{es}}(j) \leq \lambda_{\text{HHO}}(j)$. For the majority of the numerical experiments, the parameters $\alpha = 0.5$ and $\beta = 0.1$ lead to the best guaranteed lower bounds for the eigenvalues.

(viii) This first realization of the HHO method concerns the lowest-order case and illustrates that the scheme is competitive to other methodologies. For the appropriate parameter selection, the HHO scheme can provide sharper bounds than the extra-stabilized method. Numerical benchmarks with the higher-order versions of the HHO method are even more promising provided the mesh is adapted appropriately, cf. [CGT22] for a selection of experiments.

(ix) The extra-stabilized method is clearly competitive with the lowest-order HHO method. For the parameter choice $\kappa_1 = 0.1893$, the results are nearly indiscernible and the method is more robust with respect to the initial triangulation. In the displayed experiments the triangulation in Fig. 7.1 were sufficient and no additional red-refinement before the adaptive algorithm was necessary.

(x) The overall conclusion is that there exist examples, where the post-processed bound (4.46) may fail completely for localized triangulations while the new schemes are compatible with adaptive mesh-refinement.

7.3 The bi-Laplace eigenvalue problem

For the bi-Laplace eigenvalue problem this section compares the guaranteed lower-bounds computed with the extra-stabilized method (5.2) and the post-processed GLB_M (5.10) for $m = 2$. Throughout this section, the energy scalar product reads $a(\bullet, \bullet) := (D^2\bullet, D^2\bullet)_{L^2(\Omega)}$ with piecewise version $a_{\text{pw}}(\bullet, \bullet) := (D_{\text{pw}}^2\bullet, D_{\text{pw}}^2\bullet)_{L^2(\Omega)}$ and the following notation applies. Let $(\lambda_j, u_j) \in \mathbb{R}_+ \times H_0^2(\Omega) \setminus \{0\}$ denote the j -th continuous eigenpair with

$$a(u_j, v) = \lambda_j b(u_j, v) \quad \text{for all } v \in H_0^2(\Omega).$$

Let $(\lambda_{\text{es}}(j), u_{\text{es}}(j)) \in \mathbb{R}_+ \times V_{\text{es}} \setminus \{0\}$ with $u_{\text{es}}(j) := (u_{\text{pw}}(j), u_M(j)) \in V_{\text{es}} := P_2(\mathcal{T}) \times M(\mathcal{T})$ denote the j -th discrete eigenpair, that satisfies

$$a_{\text{es}}(u_{\text{es}}(j), v_{\text{es}}) = \lambda_{\text{es}}(j) b_{\text{es}}(u_{\text{es}}(j), v_{\text{es}}) \quad \text{for all } v_{\text{es}} \in V_{\text{es}}$$

with the bilinear forms

$$\begin{aligned} a_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= a_{\text{pw}}(v_M, w_M) + \kappa_2^{-2} (h_{\mathcal{T}}^{-4} (v_{\text{pw}} - v_M), w_{\text{pw}} - w_M)_{L^2(\Omega)}, \\ b_{\text{es}}(v_{\text{es}}, w_{\text{es}}) &:= b(v_{\text{pw}}, w_{\text{pw}}) \quad \text{for all } v_{\text{es}} = (v_{\text{pw}}, v_M), w_{\text{es}} = (w_{\text{pw}}, w_M) \in V_{\text{es}}. \end{aligned}$$

Finally, let $(\lambda_M(j), \phi_M(j)) \in \mathbb{R}_+ \times M(\mathcal{T})$ denote the j -th Morley eigenpair in (5.9) with

$$a_{\text{pw}}(\phi_M(j), v_{\text{CR}}) = \lambda_M(j) b(\phi_M(j), v_M) \quad \text{for all } v_M \in M(\mathcal{T})$$

and recall the guaranteed lower eigenvalue bound from (5.10)

$$\text{GLB}_M(j) := \frac{\lambda_M(j)}{1 + \kappa_2^2 h_{\max}^4 \lambda_M(j)} \leq \lambda_j.$$

7.3.1 Setting the parameter

In the present two-dimensional case the Morley interpolation constant $\kappa_2 = 0.257457844658$ [CG14a] is slightly improved by the analytical $\kappa_2 = 0.235882115439$ in Theorem 3.35.d and further improved by the numerical bound $\kappa_2 = 0.07353$ from [LSL19]. In the figures below is clearly marked which constant κ_2 is selected in the experiment. The difference between the results with analytical bounds $\kappa_2 = 0.23588$ and $\kappa_2 = 0.25745$ is negligible and not displayed. The use of $\kappa_2 = 0.07353$ is illustrated with a darker line color.

Under the condition $\kappa_2^2 h_{\max}^4 \lambda_{\text{es}}(k) \leq 1$, Theorem 5.5 guarantees $\lambda_{\text{es}}(j) \leq \lambda_j$. Otherwise (if $1 < \kappa_2^2 h_{\max}^4 \lambda_{\text{es}}(j)$ on a coarser mesh) the value $\lambda_{\text{es}}(j)$ is replaced by the guaranteed lower bound zero, but this never occurs in all the examples displayed below.

7.3.2 Adaptive mesh-refinement

If a parameter $\theta < 1$ is selected, adaptive mesh-refinement with newest-vertex bisection is utilized. In this case, θ denotes the bulk parameter in the Dörfler marking algorithm. The standard Dörfler marking strategy (dashed lines and half filled markers) is compared with the slightly modified algorithm described in Subsection 5.5.6, that ensures after the Dörfler marking that at least one triangle of maximal mesh-size is marked for refinement (dotted lines and half filled markers). Theorem 5.34 states that the adaptive Algorithm 2 with local estimator contribution for any $T \in \mathcal{T}$

$$\eta_{\text{es}}^2(T) = |T|^2 \|\lambda_{\text{es}} u_M\|_{L^2(T)}^2 + |T|^{1/2} \sum_{F \in \mathcal{F}(T)} \|[D^2 u_M]_F \times \nu_F\|_{L^2(F)}^2 \quad (7.5)$$

with the tangential components $[D^2 u_M]_F \times \nu_F$ of the jump $[D^2 u_M]_F$ of the (piecewise) Hessian along any edge $F \in \mathcal{F}$ recovers optimal convergence rates, if λ_j is a simple eigenvalue, the bulk parameter θ is small enough, and the initial triangulation is fine enough. Theorem 5.35 proves the reliability and efficiency with $\eta(\mathcal{T}) \approx \|u_j - u_M(j)\|_{\text{pw}}$. Hence, the eigenvalue error $\lambda_j - \lambda_{\text{es}}(j)$ is expected to converge with the same rate as $\eta_{\text{es}}^2(\mathcal{T})$.

For the Morley finite element method the standard estimator with local contribution

$$\eta_M^2(T) := |T|^2 \|\lambda_M(j) \phi_M(j)\|_{L^2(T)}^2 + |T|^{1/2} \sum_{F \in \mathcal{F}(T)} \|[D^2 \phi_M(j)]_F \times \nu_F\|_{L^2(F)}^2 \quad (7.6)$$

drives the adaptive mesh-refinement. According to [Gal15b, Sec. 4.3] this estimator with cluster marking leads to optimal convergence rate for any eigenvalue.

7.3.3 Displayed quantities

Fig. 7.1.a-e displays the initial triangulations $\mathcal{T}_0$ for the numerical experiments below. The respective convergence history plots in Figures 7.14, 7.16, and 7.18 display the difference of the exact eigenvalue and the two guaranteed lower bounds for uniform mesh-refinement $\theta = 1$ (solid line and filled markers) and adaptive mesh-refinement plotted against the number of triangles $|\mathcal{T}|$. On uniform meshes $\text{GLB}_M(j)$ (line color blue) and $\lambda_{\text{es}}(j)$ (line color green) coincide, see Lemma 5.12, and only the green line is visible. The number j of the eigenvalue is illustrated by different markers.

Fig. 7.17 displays the differences $\lambda_1 - \lambda_{\text{es}}(1)$ (line color green) as well as the estimator values η_{es} (line color violet) for different bulk parameter $\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ (dashed line and half filled markers) and uniform refinement $\theta = 1$ (solid line and filled markers). The size of the marker indicates the value of $\theta < 1$.

Fig. 7.13 shows adaptive triangulations computed with the extra-stabilized Morley method for the principal (resp. fourth) eigenvalue on the domains in Subsections 7.3.4–7.3.7.

7.3.4 Dumbbell-slit domain

On the non-convex asymmetric dumbbell-slit domain $\Omega := ((-1, 4) \times (-1, 1)) \setminus ((-1, 0] \times \{0\} \cup [1, 3] \times [-0.75, 1))$ in Fig. 7.1.a the principal eigenvalue $\lambda_1 = 386.78778163$ is approximated with the adaptive Agyris FEM with a three times red-refined initial triangulation. Fig. 1.2 with $\kappa_2 = 0.23588$ already indicates that the uniform mesh-refinement leads to a reduced convergence rate of 0.5 and that the $\text{GLB}_M(1)$ fails for adaptive mesh-refinement, while the new adaptive bounds recover the optimal rate. The convergence history plot Fig. 7.14.a compares the errors in Fig. 1.2 with the results for $\kappa_2 = 0.07353$. For uniform mesh-refinement holds $\lambda_{\text{es}}(1) \approx \text{GLB}_M(1)$ and only the green line for the new method is visible. The reduction of κ_2 only improves $\lambda_{\text{es}}(1)$ on coarse uniform and adaptive meshes. On adaptive meshes $\kappa_2 = 0.07353$ improves the value of GLB_M , but convergence is still not visible. Moreover, Fig. 7.14.a compares the two described marking strategies. For the new method the results do not differ and only a dashed line is visible. The Morley post-processing benefits from the reduction of the maximal mesh-size in that there are flat steps instead of the more or less constant error. However, the known bound still fails for adaptive mesh-refinement.

In the related example of the dumbbell-slit domain $\Omega := ((-1, 5) \times (-1, 1)) \setminus ((-1, 0] \times \{0\} \cup [1, 3] \times [-0.75, 1))$ in Fig. 7.1.b. The adaptive algorithm for the Agyris finite elements with a three times red-refined initial mesh approximated $\lambda_1 = 80.93254269$, $\lambda_2 = 336.64778361$, $\lambda_3 = 336.66603145$, and $\lambda_4 = 386.78778157$. On this domain the first three eigenfunctions seem to belong to $H^2(\Omega)$ with support in $(3, 5) \times (-1, 1)$, see Fig. 7.15. In other words uniform mesh-refinement leads to

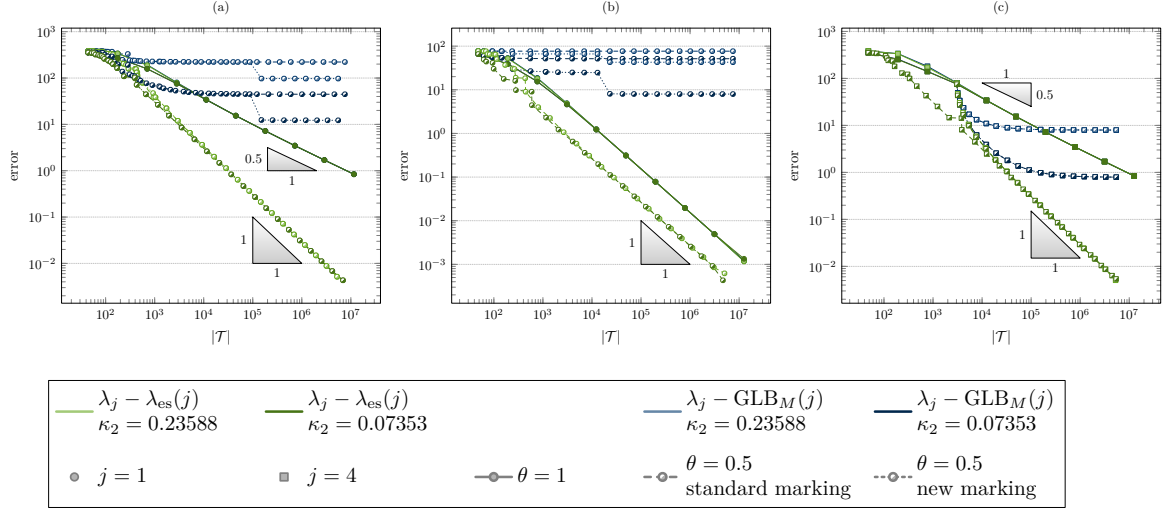


Figure 7.14: Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of (a) the asymmetric dumbbell-slit domain for $j = 1$, and of the dumbbell-slit domain for (b) $j = 1$ as well as (c) $j = 4$

the optimal convergence rate one for all methods in Fig. 7.14.b. Adaptive mesh-refinement improves the errors for the new method, but the Morley post-processing fails. The adaptive mesh-refinement is necessary for the fourth eigenvalue to recover the optimal convergence rate. The plot is very similar to the principal eigenvalue of the asymmetric dumbbell-slit. In this case the new marking strategy does not improve the results since more refinements are necessary to reduce the maximal mesh-size. An interesting observation is that the complex geometry and the closeness of the eigenvalues λ_2, λ_3 , and λ_4 lead to the necessity of four uniform red-refinements before the adaptive algorithm with (7.6) for the Morley method actually refines the left square that contains the support of u_4 , while the new method converges systematically even for coarse triangulations. The example provides striking numerical evidence for the superiority of the adaptive version of the extra-stabilized Morley eigensolver.

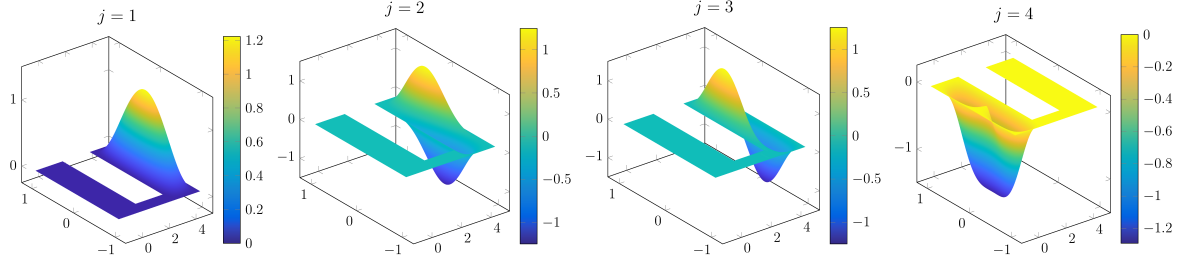


Figure 7.15: Nodal interpolation of the Morley approximation of the j -th eigenfunction of the dumbbell-slit domain computed on a uniformly red-refined mesh with $|\mathcal{T}| = 12582912$ for $j \in \{1, \dots, 4\}$

7.3.5 L-shaped domain

The principal eigenvalue $\lambda_1 = 418.97529285$ and the fourth $\lambda_4 = 1634.53378167$ on the non-convex L-shaped domain $\Omega := (-1, 1)^2 \setminus ([0, 1] \times (-1, 0])$ of Fig. 7.1.c are approximated with an adaptive Agyris finite element method. The associated first eigenfunction in $H_0^2(\Omega) \setminus H^3(\Omega)$ results in the reduced empirical convergence rate 0.66 for uniform mesh-refinement in Fig. 7.16.a. For the uniform mesh-refinement the choice of the parameter κ_2 does not influence the results and only $\lambda_1 - \lambda_{\text{es}}(1)$ computed with $\kappa_2 = 0.07353$ is visible. The adaptive mesh-refinement with (7.5) allows to recover the optimal convergence rate one for $\lambda_{\text{es}}(1)$. The choice of $\kappa_2 = 0.07353$ instead of $\kappa_2 = 0.23588$ improves $\text{GLB}_M(1)$ significantly on coarser meshes. The bound computed with $\kappa_2 = 0.23588$ suffers from the involvement of h_{max} visible in form of steps, while the choice $\kappa_2 = 0.07353$ leads to a nearly straight line in the convergence history plot. Undisplayed experiments on graded meshes [CB18] of the L-shaped domain, e.g., with grading parameter $\beta = 10/7$, recover the optimal convergence rates and confirm Lemma 5.12 as well. In contrast, the fourth eigenfunction seems to belong to $H^3(\Omega)$ and the optimal convergence rate one is reached with uniform mesh-refinement. Adaptive mesh-refinement and the two choices of κ_2 do not improve the results with the extra-stabilized method visibly. For the post-processed bound the adaptive mesh-refinement leads to worse results than the uniform refinement with clearly visible steps.

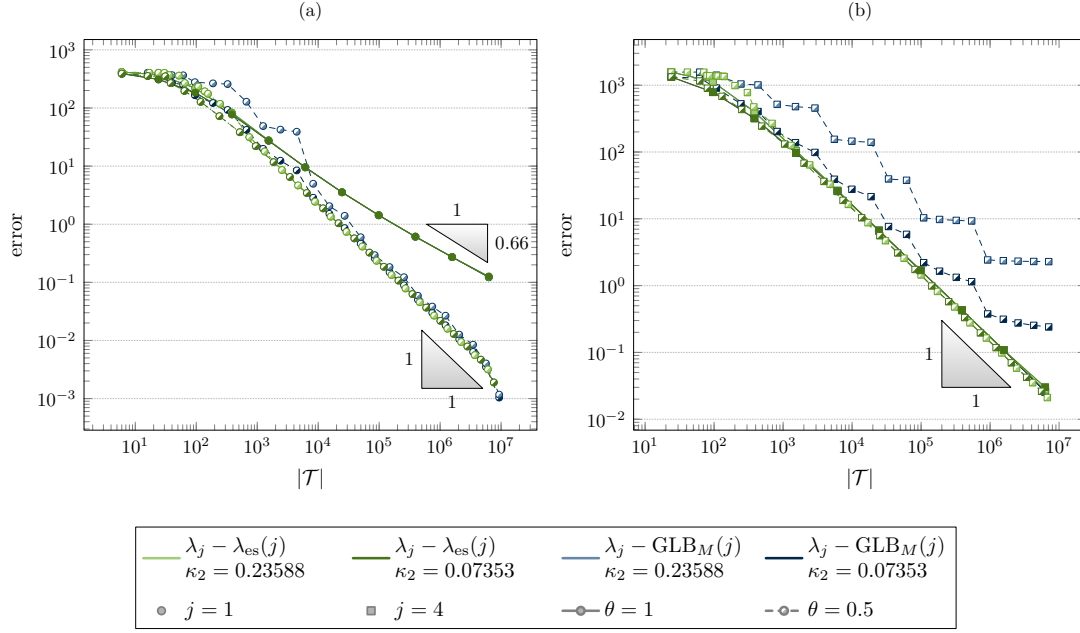


Figure 7.16: Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the L-shaped domain for (a) $j = 1$ and (b) $j = 4$

7.3.6 Slit domain

On the non-convex slit domain $\Omega := (-1, 1)^2 \setminus ([0, 1] \times \{0\})$, the principal eigenvalue approximated with the adaptive Agyris FEM reads $\lambda_1 = 418.97529285$. Fig. 7.17.a displays the error $\lambda_1 - \lambda_{\text{es}}(1)$ of the extra-stabilized method with $\kappa_2 = 0.23588$ and different bulk parameter $\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ in the adaptive algorithm. The first eigenfunction seems to belong to $H_0^2(\Omega) \setminus H^3(\Omega)$ because uniform mesh-refinement leads to the reduced convergence rate 0.5. Theorem 5.34 proves optimal convergence rates of the adaptive algorithm, if the initial mesh is fine enough and the bulk parameter small enough. Asymptotically Fig. 7.17.a shows the optimal convergence rate for all bulk parameter $\theta < 1$ without any initial mesh-refinement. The error reduction justifies the choice $\theta = 0.5$ in the experiments, which balances the error, the size of the pre-asymptotic regime, and the necessary number of levels to reach an appropriate refinement of the domain. Fig. 7.17.b displays the estimator $\eta_{\text{es}}(\mathcal{T})$ on different levels and confirms reliability and efficiency from Theorem 5.35. For the estimator values a preasymptotic range up to $|\mathcal{T}| \approx 10^3$ is clearly visible.

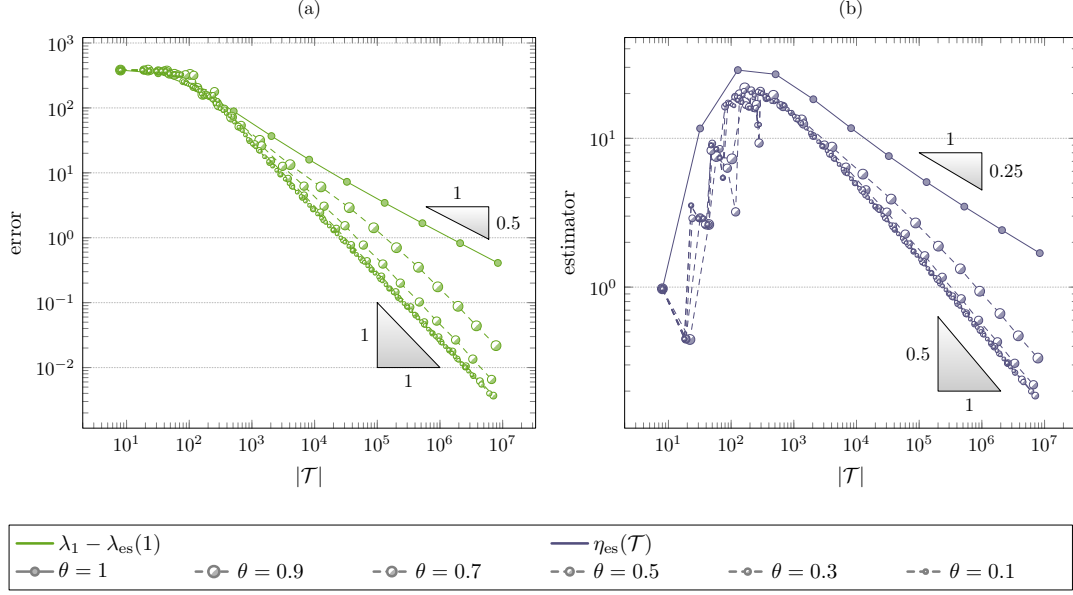


Figure 7.17: Comparison of (a) the distance of λ_1 and $\lambda_{\text{es}}(1)$ (with $\kappa_2 = 0.2358$) and (b) the global estimator $\eta_{\text{es}}(\mathcal{T})$ computed on uniform ($\theta = 1$) and adaptive ($\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$) meshes of the slit domain

7.3.7 Four-slit domain

On the four-slit domain $\Omega := (-1, 1)^2 \setminus ([0, 0.5] \times \{0\} \cup [0, -0.5] \times \{0\} \cup \{0\} \times [0, 0.5] \cup \{0\} \times [0, -0.5])$ of Fig. 7.1.e the principal eigenvalue $\lambda_1 = 830.21129191$ and the double eigenvalue $\lambda_3 \approx 1125.127623 \approx \lambda_4$ are approximated with the adaptive Agyris FEM. The associated eigenfunctions on the non-convex domain seem to belong to $H_0^2(\Omega) \setminus H^3(\Omega)$ because uniform mesh-refinement leads to the reduced convergence rates 0.5 for the principal eigenvalue in Fig. 7.18.a and 0.55 for the third and fourth in Fig. 7.18.b. For the uniform refinement $\text{GLB}_M(j) = \lambda_{\text{es}}(j)$ coincide in accordance with Lemma 5.12 and the difference between the choices $\kappa_2 = 0.23588$ and $\kappa_2 = 0.07353$ is indiscernible, hence only $\lambda_j - \lambda_{\text{es}}(j)$ for $\kappa_2 = 0.23588$ is visible in light green. The AFEM algorithm with bulk parameter $\theta = 0.5$ driven by the estimator (7.5) allows to recover the optimal convergence rate one for $\lambda_j - \lambda_{\text{es}}(j)$. The influence κ_2 is minimal for the extra-stabilized method and only $\kappa_2 = 0.23588$ is clearly visible. The adaptive GLB_M is clearly improved by $\kappa_2 = 0.07353$ instead of $\kappa_2 = 0.23588$, such that the step structure due to the involvement of $h_{\max}$ in Fig. 7.18.a becomes less prominent for $\kappa_2 = 0.07353$. In Fig. 7.18.b the improvement is obvious for the fourth eigenvalue without a cluster-marking algorithm. The clustering estimator (7.1) from [Gal15a] for the double eigenvalue $\lambda_3 = \lambda_4$ improves $\text{GLB}_M(4)$ (marked with squares with blue upper left half) such that it coincides with $\text{GLB}_M(3)$.

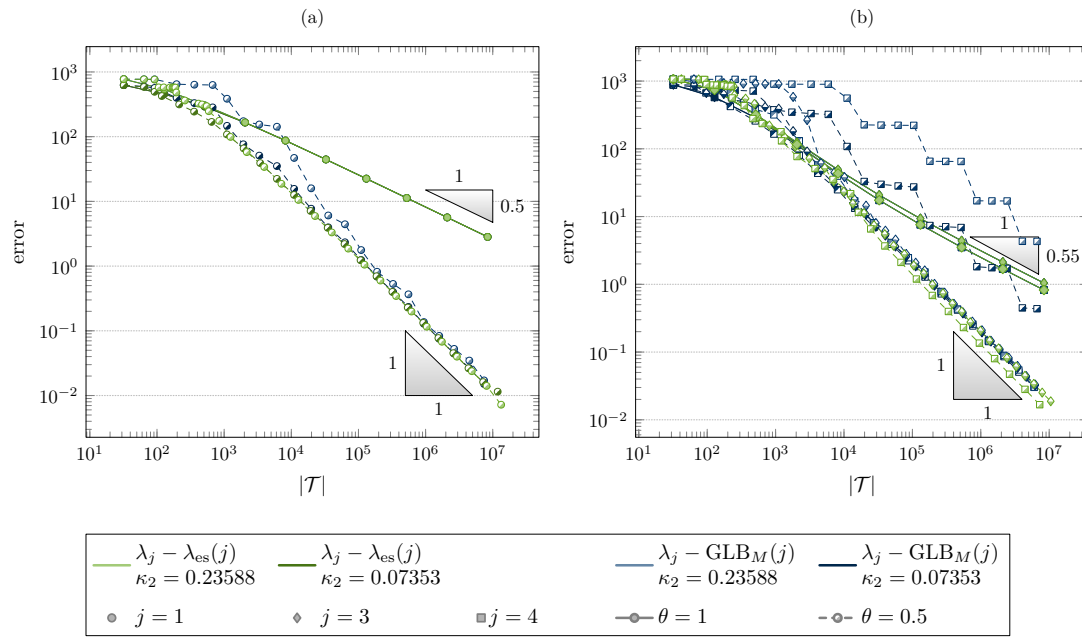


Figure 7.18: Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the four-slit domain for (a) $j = 1$ and (b) $j \in \{3, 4\}$

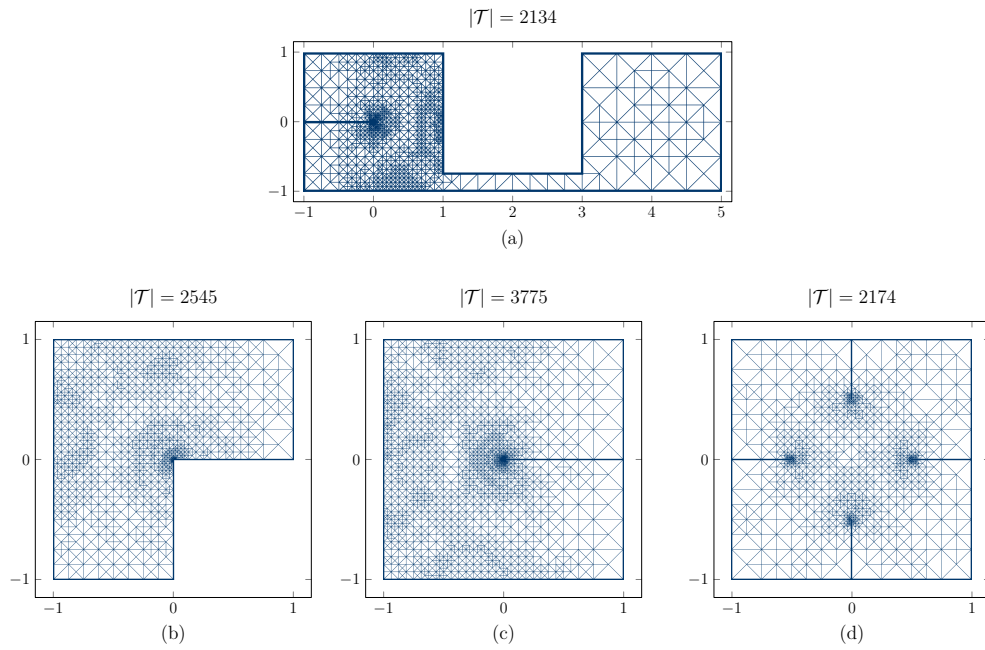


Figure 7.19: Adaptive triangulation $\mathcal{T}$ with $\theta = 0.5$ for (a) the fourth eigenvalue of the dumbbell-slit domain computed with (7.5) and $\kappa_2 = 0.07353$ as well as the principal eigenvalue of (b) the L-shaped domain, (c) the slit domain and (d) the four-slit domain computed with (7.5) and $\kappa_2 = 0.23588$

7.3.8 Conclusions

This subsection summarizes the empirical observations of the numerical experiments in Subsections 7.3.4–7.3.7. The results are comparable to the extra-stabilized Crouzeix-Raviart method.

(i) All experiments confirm the a priori convergence rates of Theorem 5.18. The empirical convergence rate depends only on the smoothness of the approximated eigenfunction. For instance in Fig. 7.14.b and Fig. 7.16.b the empirical convergence rate is optimal for uniform refinement despite the reduced convergence rate in Fig. 7.14.c and Fig. 7.16.a.

(ii) Theorem 5.18 predicts convergence for a sufficiently fine initial mesh. In all examples the convergence rate is visible even for moderately fine triangulations, so this restriction does not affect the numerical examples much.

(iii) If the condition on the mesh-size is satisfied, the extra-stabilized method provides indeed lower eigenvalue bounds in all numerical experiments and so confirms Theorem 5.5. All examples verify the comparison result $\text{GLB}_M(j) \leq \lambda_{\text{es}}(j) \leq \lambda_M(j)$ from Lemma 6.6.

(iv) The adaptive algorithm driven by the estimator (7.5) recovers the optimal empirical convergence rates in all examples for the extra-stabilized method and confirms Theorem 5.34 even for coarse initial triangulation and the moderate bulk parameter $\theta = 0.5$.

(v) The alternative marking strategy seems not necessary for the optimal convergence rate. For the double eigenvalue in Fig. 7.18.b a cluster algorithm was not needed for the extra-stabilized method.

(vi) The adaptive meshes for the extra-stabilized method with $\kappa_2 = 0.07353$ displayed in Fig. 7.19.a and $\kappa_2 = 0.23588$ displayed in Fig. 7.19.b-d show the expected refinements toward the reentrant corners resp. the endpoints of the slits. The choice $\kappa_2 = 0.07353$ leads to comparable meshes in fewer steps.

(vii) The constant $\kappa_2 = 0.07353$ from [LSL19] instead of $\kappa_2 = 0.23588$ leads to a significant improvement of the known bound (5.10) with adaptive mesh-refinement but only slightly improves the extra-stabilized method on coarse meshes.

(viii) The overall conclusion of the numerical experiments is that there exist examples, where the post-processed GLB (5.10) may fail completely for localized triangulations. In contrast, the new scheme is compatible with adaptive mesh-refinement.

8 Conclusion and outlook

Numerical experiments illustrate, that the post-processed guaranteed lower eigenvalue bounds [CG14a; CG14b] suffer from the involvement of the maximal mesh-size if the mesh is locally refined. This thesis presents three methods which are a remedy because the discrete eigenvalues are guaranteed lower bounds for the continuous eigenvalues provided a simply verifiable a priori or a posteriori condition holds. These novel methods allow for the adaptive mesh-refinement necessary for optimal convergence rates of non-smooth eigenfunction.

The modified HHO method computes guaranteed lower eigenvalue bounds for the Dirichlet eigenvalues of the Laplacian. In the lowest-order case all constants are bounded and for the appropriate parameter choice the sharpest bounds are computed. The method allows for higher polynomial degrees and therefore better convergence rates. For a scheme with a slightly modified stabilization the upcoming work [CGT22] provides numerical estimates for the constants for higher-polynomial degrees and an reliable and efficient estimator.

The extra-stabilized methods compute direct guaranteed lower eigenvalue bounds for the Dirichlet eigenvalues of the m -Laplacian without parameters. For the cases $m = 1$ and $m = 2$ the required interpolation constant κ_m is analytical bounded in any space dimension. The selected properties of the interpolation and conforming companion operator are of interest beyond the scope of this thesis as [CN21] illustrates. The interpolation operators and the appropriate conforming companion for Crouzeix-Raviart satisfy these properties in any space dimension $n \geq 2$. For the Morley finite element the definition of the companion operator extends the 2-dimensional work in [Gal15b] and provides the 3D version. To this end the detailed analysis of the Worsey-Farin finite element [WF87] in the spirit of Ciarlet [Cia78] is essential. In the general framework optimal a priori convergence rates and a detailed medius analysis for simple eigenvalues are established. The medius analysis and an extension of the axioms of adaptivity [Car+14a; CR17], which involves a generalized discrete reliability, allow to prove the convergence of an adaptive algorithm for the extra-stabilized bounds towards a simple eigenvalue with optimal rates.

The numerical experiments confirm the presented theory and show, e.g., the proven optimal convergence rates for coarse initial triangulation and moderate bulk parameter, both of which are assumed to be sufficiently small in the analysis. Future research could target the following open questions.

There seems to be a good chance to develop a general framework for guaranteed lower eigenvalue bounds for methods, like HHO, HDG, and VFEM, where the

solution is as well approximated by two variables, utilizing similarities in the arguments in this thesis and [CZZ20].

The presented framework for the extra-stabilized method could handle the case $m \geq 3$ as well. If the introduced conditions are established for appropriate finite elements, e.g., in the spirit of [WX13], the results would follow for the m -Laplacian of higher-order.

Of course higher accuracy needs higher-order schemes and the optimality for the adaptive modified HHO method is very interesting. The a posteriori error analysis for different HHO methods in the upcoming work [CGT22; Ber+23] in combination with the extended axioms of adaptivity in this thesis may allow results in this area.

Moreover future work should include – in contrast to the presented theory – (multi-level) inexact solver in the convergence analysis and provide guaranteed lower bounds for inexact solve. In this context the positive results in [CG11; CG14b; CGS15] for different methods may be of interest.

While the restriction to a simple eigenvalue is convenient in the presented optimality analysis, in the future the inclusion of eigenvalue clusters and multiple eigenvalue in the spirit of [DHZ15; Gal15a] should be considered. This could be a promising approach as the numerical experiments with multiple eigenvalues illustrate.

Finally, it would be interesting to compete the different guaranteed lower and upper eigenvalue bounds on the market.

A List of common notations

A.1 Elementary notation

$\mathbb{N}$	positive integers, i.e., $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	nonnegative integers, i.e., $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	integers, i.e., $\{0, \pm 1, \pm 2, \pm 3, \dots\}$
$\mathbb{R}$	field of real numbers
$\mathbb{R}^{j \times k}$	space of $j \times k$ matrices with real coefficients for $j, k \in \mathbb{N}$
$I_{m \times m}$	$m \times m$ unit matrix for $m \in \mathbb{N}$
$A^\top$	transpose of matrix $A \in \mathbb{R}^{j \times k}$
δ_{jk}	Kronecker δ , i.e., $\delta_{jk} = 1$ for $j = k$ and vanishes otherwise
$a \cdot b$	scalar product of $a, b \in \mathbb{R}^{\ell \times m}$, i.e., $a \cdot b = a^\top b \in \mathbb{R}^{m \times m}$
$a \otimes b$	tensor product of $a, b \in \mathbb{R}^{\ell \times m}$, i.e., $a \otimes b = ab^\top \in \mathbb{R}^{\ell \times \ell}$
$ x $	Euclidean length of a vector $x \in \mathbb{R}^m$, i.e., $ x = \sqrt{x \cdot x}$ or length of a multi-index $x \in \mathbb{N}_0^m$, i.e., $ x = \sum_{j=1}^m x_j$
$ A $	cardinality of a set A or n -, $(n-1)$ -, or $(n-2)$ -dimensional Lebesgue measure of $A \subset \mathbb{R}^n$
$(a, b), [a, b]$	open (resp. closed) real interval
$\emptyset$	empty set
$A \subseteq B$	A is a subset of B
$A \subset B, A \subsetneq B$	$A \subseteq B$ and $A \neq B$
$\bar{\omega}$	closure of $\omega \subset \mathbb{R}^n$ w.r.t. Euclidean topology
$\partial\omega$	boundary of $\omega \subset \mathbb{R}^n$ w.r.t. Euclidean topology
$\text{int}(\omega)$	interior of $\omega \subset \mathbb{R}^n$ w.r.t. Euclidean topology
$\text{conv}(\omega)$	convex hull of $\omega \subset \mathbb{R}^n$
$\text{span}(\omega)$	linear hull of $\omega \subset \mathbb{R}^n$
$\text{diam}(\omega)$	diameter of the set $\omega \subset \mathbb{R}^n$, i.e., $\text{diam}(\omega) := \max_{x, y \in \omega} x - y $

A.2 Measure theory notation and derivatives

$\int_\omega f \, dx$	n -dimensional Lebesgue integral of f over ω
$\int_\omega f \, ds$	$(n-1)$ -dimensional surface integral of f over ω
$\int_\omega f \, d\sigma$	$(n-2)$ -dimensional integral of f over ω
$f_\omega := \omega ^{-1} \int_\omega$	integral mean over ω with Lebesgue measure $ \omega < \infty$
$L^2(\omega)$	square integrable functions over ω

$\ f\ _{L^2(\omega)}$	L^2 norm of f over ω
$(f, g)_{L^2(\omega)}$	L^2 scalar product of f and g over ω
$L^\infty(\omega)$	essentially bounded measurable functions over ω
$\ f\ _{L^\infty(\omega)} := \ f\ _\infty$	essential supremum of f over ω
$H^m(\omega)$	Sobolev space of m -times weakly differentiable L^2 functions with weak derivative in L^2 over ω
$H^{m+s}(\omega)$	fractional order Sobolev space for $0 < s < 1$ over ω
$\ f\ _{H^m(\omega)},$	Sobolev norms of f over ω
$\ f\ _{H^{m+s}(\omega)}$	
$ f _{H^m(\omega)}$	Sobolev seminorm of f over ω , i.e., $\ D^m f\ _{L^2(\omega)}$
$C_c^\infty(\omega)$	space of smooth test function with compact support in ω
$H_0^m(\omega)$	closure of $C_c^\infty(\omega)$ w.r.t. $\ \bullet\ _{H^m(\omega)}$
$H^s(\partial\omega)$	(dual) trace spaces for $s \in \{\pm 1/2\}$
$\langle f, g \rangle_{\partial\omega}$	duality brackets in $H^{1/2}(\partial\omega) \times H^{-1/2}(\partial\omega)$ extending the scalar product in $L^2(\partial\omega)$
$D := \nabla, D^2$	derivative resp. gradient, Hessian
Δ, Δ^2	Laplacian, bi-Laplacian (biharmonic operator)
div	divergence
$\partial f / \partial v$	directional derivative w.r.t. $v \in \mathbb{R}^n$
$b(f, g)$	L^2 scalar product of f and g over Ω , i.e., $(f, g)_{L^2(\Omega)}$
$a(f, g)$	energy scalar product of f and g over Ω w.r.t. $V := H^m(\Omega)$, i.e., $(D^m f, D^m g)_{L^2(\Omega)}$

A.3 Triangulation notation

$\mathcal{T}_0$	initial triangulation of bounded Lipschitz domain $\Omega \subset \mathbb{R}^n$
$\mathbb{T}$	set of admissible triangulation by newest-vertex bisection of $\mathcal{T}_0$
$\mathcal{T} \in \mathbb{T}$	regular, admissible triangulation of $\Omega \subset \mathbb{R}^n$
$\mathbb{T}(\mathcal{T})$	set of admissible refinements of $\mathcal{T}$
$\mathbb{T}(\varepsilon)$	set of admissible triangulation with maximal mesh-size $\leq \varepsilon$, i.e., $\mathbb{T}(\varepsilon) := \{\mathcal{T} \in \mathbb{T} : h_{\max} := \max_{T \in \mathcal{T}} h_T \leq \varepsilon\}$
$\mathbb{T}(N)$	set of admissible triangulation with at most $N \in \mathbb{N}$ simplices more than $\mathcal{T}_0$, i.e., $\mathbb{T}(N) := \{\mathcal{T} \in \mathbb{T} : \mathcal{T} - \mathcal{T}_0 \leq N\}$
$\mathcal{V}, \hat{\mathcal{V}}$	set of all vertices in $\mathcal{T}$ (resp. $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$)
$\mathcal{V}, \hat{\mathcal{V}}$	set of all vertices in $\mathcal{T}$ (resp. $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$)
$\mathcal{V}(\Omega), \mathcal{V}(\partial\Omega)$	set of all interior (resp. boundary) vertices in $\mathcal{T}$
$\mathcal{E}, \hat{\mathcal{E}}$	set of all $(n-2)$ -subsimpllices in $\mathcal{T}$ (resp. $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$), e.g., vertices in 2D, edges in 3D
$\mathcal{E}(\Omega), \mathcal{E}(\partial\Omega)$	set of all interior (resp. boundary) $(n-2)$ -subsimpllices in $\mathcal{T}$
$\mathcal{F}, \hat{\mathcal{F}}$	set of all $(n-1)$ -subsimpllices called sides in $\mathcal{T}$ (resp. $\hat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$), e.g., edges in 2D, faces in 3D

$\mathcal{F}(\Omega), \mathcal{F}(\partial\Omega)$	set of all interior (resp. boundary) sides in $\mathcal{T}$
$\mathcal{V}(T), \mathcal{V}(F), \mathcal{V}(E)$	set of all $(n+1)$ (resp. n or $n-1$) vertices in $T \in \mathcal{T}$ (resp. $F \in \mathcal{F}$ or $E \in \mathcal{E}$)
$\mathcal{E}(T), \mathcal{E}(F)$	in 3D set of $\binom{n+1}{2}$ (resp. $\binom{n}{2}$) edges in $T \in \mathcal{T}$ (resp. $F \in \mathcal{F}$)
$\mathcal{F}(T)$	set of all $(n+1)$ sides in $T \in \mathcal{T}$
$\omega_F, \mathcal{T}(F)$	side patch of a all simplices in $\mathcal{T}$ with side $F \in \mathcal{F}$, i.e., $\mathcal{T}(F) := \{T \in \mathcal{T} : F \in \mathcal{F}(T)\}$ and $\omega_F := \text{int} \cup \mathcal{T}(F)$
$\omega_E, \mathcal{T}(E)$	(in 3D edge) patch of a all simplices in $\mathcal{T}$ with (edge) $E \in \mathcal{E}$, i.e., $\mathcal{T}(E) := \{T \in \mathcal{T} : E \in \mathcal{E}(T)\}$ and $\omega_E := \text{int} \cup \mathcal{T}(E)$
$\omega_z, \mathcal{T}(z)$	nodal patch of a all simplices in $\mathcal{T}$ with vertex $z \in \mathcal{V}$, i.e., $\mathcal{T}(z) := \{T \in \mathcal{T} : z \in \mathcal{V}(T)\}$ and $\omega_z := \text{int} \cup \mathcal{T}(z)$
$\Omega(T), \mathcal{T}(\Omega(T))$	set of all simplices $K \in \mathcal{T}$ with $\text{dist}(T, K) = 0$, i.e., $\mathcal{T}(\Omega(T)) := \{K \in \mathcal{T} : \text{dist}(T, K) = 0\}$ and $\Omega(T) := \text{int}(\cup_{T \in \mathcal{T}(\Omega(T))} T)$
$[v]_F$	jump of a function v across the side $F \in \mathcal{F}$
ν_F	unit normal of the side $F \in \mathcal{F}$
ν_T	unit outward normal along the boundary of the simplex $T \in \mathcal{T}$
$\text{mid}(T), \text{mid}(F), \text{mid}(E)$	centroid of a simplex $T \in \mathcal{T}$ (resp. a side $F \in \mathcal{F}$ or an $(n-2)$ -subsimplex $E \in \mathcal{E}$)
h_T, h_F, h_E	diameter of a simplex $T \in \mathcal{T}$ (resp. a side $F \in \mathcal{F}$ or an $(n-2)$ -subsimplex $E \in \mathcal{E}$)
$h_{\mathcal{T}}$	piecewise constant mesh-size function with $h_{\mathcal{T}} _T := h_T$ for any $T \in \mathcal{T}$
$P_k(\mathcal{T}; \mathbb{R}^\ell)$	piecewise w.r.t $\mathcal{T}$ polynomials of total degree $\leq k$ in ℓ components
$P_k(\mathcal{F}; \mathbb{R}^\ell),$ $P_k(\mathcal{F}(\Omega); \mathbb{R}^\ell)$	piecewise w.r.t $\mathcal{F}$ polynomials of total degree $\leq k$ in ℓ components (vanishing along the boundary)
$S^k(\mathcal{T}), S_0^k(\mathcal{T})$	piecewise polynomial global continuous function (vanishing along the boundary), i.e., $S^k(\mathcal{T}) := P_k(\mathcal{T}) \cap C(\Omega)$ and $S_0^k(\mathcal{T}) := S^k(\mathcal{T}) \cap C_0(\Omega)$
$RT_k^{\text{pw}}(\mathcal{T})$	piecewise Raviart-Thomas function of degree k , i.e., $RT_k^{\text{pw}}(\mathcal{T}) := P_k(\mathcal{T})x + P_k(\mathcal{T}; \mathbb{R}^n)$
Π_k	component-wise L^2 projection onto $P_k(\mathcal{T})$
$\Pi_{\mathcal{F},k}$	component-wise L^2 projection onto $P_k(\mathcal{F})$
$\Pi_{RT,k}$	component-wise L^2 projection onto $RT_k^{\text{pw}}(\mathcal{T})$
$\text{osc}_k(f)$	oscillation w.r.t. $V := H^m(\Omega)$, i.e., $\ h_{\mathcal{T}}^m(1 - \Pi_k)f\ _{L^2(\Omega)}$
$a \lesssim b$	$a \leq Cb$ for a constant independent of the mesh-size
$a \approx b$	$a \lesssim b \lesssim a$
pw	subscript index to indicate piecewise action of a differential operator or the energy scalar product w.r.t $\mathcal{T}$

B Some fundamental estimates

In order to provide explicit constants whenever possible the following observation are useful and maybe of independent interest.

The observations Lemma B.1, B.2, and B.4 can be found as Lemma B–D in [CP20], while Lemma B.3 is Lemma B.1 in the appendix of [CP21] with similar results in [CH18a, Lem 4.2]. The arguments in Lemma B.4 are adapted in Corollary B.5 which is of interest in the proof of Lemma 3.39. This proof utilizes the local Poincaré-Friedrichs inequality in Lemma B.6 as well.

Lemma B.1 utilizes some language of graph theory and concerns an undirected graph G as a pair $(\{1, \dots, n\}, \mathcal{E})$ of a set of vertices $\{1, \dots, n\}$ (fixed with n and so neglected in the notation) and a set $\mathcal{E}$ of edges $\{j, k\}$ with $j, k \in \{1, \dots, n\}$ and $j \neq k$. The graph G (identified with $\mathcal{E}$) is connected if for all $j, k \in \{1, \dots, n\}$ and $j \neq k$ there are $m \in \{1, 2, 3, \dots\}$ edges $\{\alpha_1, \alpha_2\}, \{\alpha_2, \alpha_3\}, \dots, \{\alpha_m, \alpha_{m+1}\} \in \mathcal{E}$ with $\alpha_1 = j$ and $\alpha_{m+1} = k$. The set of all connected graphs G over the set $\{1, \dots, n\}$ is identified with the set $\mathcal{C}(n)$ of all associated sets of edges $\mathcal{E}$; so $(\{1, \dots, n\}, \mathcal{E})$ is connected is abbreviated as $\mathcal{E} \in \mathcal{C}(n)$.

Given $x_1, \dots, x_n \in \mathbb{R}$, the goal is to minimize

$$f(\mathcal{E}) := \sum_{\{j,k\} \in \mathcal{E}} (x_j - x_k)^2 \text{ over all } \mathcal{E} \in \mathcal{C}(n).$$

Since permutations σ of the set $\{1, \dots, n\}$ transform $\mathcal{E} \in \mathcal{C}(n)$ into $\sigma(\mathcal{E}) := \{\{\sigma(j), \sigma(k)\} : \{j, k\} \in \mathcal{E}\} \in \mathcal{C}(n)$, without loss of generality, assume that the enumeration orders the real values $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$.

Lemma B.1. *Any vector $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ with $n \in \mathbb{N}$ and $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$ and the function $f(\mathcal{E}) := \sum_{\{j,k\} \in \mathcal{E}} (x_j - x_k)^2$ satisfy*

$$m(x) := \min_{\mathcal{E} \in \mathcal{C}(n)} f(\mathcal{E}) = \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2.$$

Proof. (1) For any $n \in \mathbb{N}$ and $x \in \mathbb{R}^n$ the minimum $m(x) := \min_{\mathcal{E} \in \mathcal{C}(n)} f(\mathcal{E})$ (where below f applies to any vector of any length) is attained for some $\mathcal{E} \in \mathcal{C}(n)$ with $|\mathcal{E}| = n - 1$.

It is known in graph theory that loops can be avoided by certain cuts and any cut means a reduction of the target functional. Therefore, we may and will assume without loss of generality, that $\mathcal{E} \in \mathcal{C}(n)$ is a tree. There is only a finite number of trees for a fixed number of vertices n and so the minimum $m(x)$ is attained for at least one of them.

(2) Given any $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$ it holds

$$m(x) := \min_{\mathcal{E} \in \mathcal{C}(n)} \sum_{\{j,k\} \in \mathcal{E}} (x_j - x_k)^2 = \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2.$$

The proof is by mathematical induction. The assertion holds for $n = 1$ (pathological) and $n = 2$ (trivial), so suppose it holds for some $n \geq 2$ and all $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$. Given $\hat{x} = (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1}$ with $x_1 \leq x_2 \leq \dots \leq x_n \leq x_{n+1}$, let $\hat{\mathcal{E}} \in \mathcal{C}(n+1)$ be a minimizer with $f(\hat{\mathcal{E}}) = m(\hat{x})$.

Step 1. Since $\hat{\mathcal{E}} \in \mathcal{C}(n+1)$ is connected, there exists a path $\{\alpha_1, \alpha_2\}, \{\alpha_2, \alpha_3\}, \dots, \{\alpha_m, \alpha_{m+1}\} \in \hat{\mathcal{E}}$ of length $m \in \mathbb{N}$ with $\alpha_1 = n+1$ and $\alpha_{m+1} = n$. The numbers $\alpha_1, \dots, \alpha_{m+1}$ can be chosen pairwise distinct (as loops may be excluded). Then $\mathcal{E}' := \hat{\mathcal{E}} \setminus \{\{n+1, \alpha_2\}\}$ and $\tilde{\mathcal{E}} := \mathcal{E}' \cup \{\{n, n+1\}\}$ lead to $\tilde{\mathcal{E}} \in \mathcal{C}(n+1)$ and $f(\hat{\mathcal{E}}) - f(\tilde{\mathcal{E}}) = (x_{n+1} - x_{\alpha_2})^2 - (x_n - x_{n+1})^2 \geq 0$ (since $x_{\alpha_2} \leq x_n \leq x_{n+1}$). Consequently, there exists a minimizer $\hat{\mathcal{E}} \in \mathcal{C}(n+1)$ with $\{n, n+1\} \in \hat{\mathcal{E}}$.

Step 2. For any $k \in \{1, 2, \dots, n-1\}$ with $\{k, n+1\} \in \hat{\mathcal{E}}$ consider $\mathcal{E}' := \hat{\mathcal{E}} \setminus \{\{k, n+1\}\}$ and $\tilde{\mathcal{E}} := \mathcal{E}' \cup \{\{k, n\}\}$. Then $\tilde{\mathcal{E}} \in \mathcal{C}(n+1)$ is connected (for $\{n, n+1\} \in \hat{\mathcal{E}} \in \mathcal{C}(n+1)$). Moreover, $f(\hat{\mathcal{E}}) - f(\tilde{\mathcal{E}}) = (x_{n+1} - x_k)^2 - (x_n - x_k)^2 \geq 0$ (since $x_k \leq x_n \leq x_{n+1}$).

Step 3. A finite number of changes as in *Step 2* leads to a minimizer $\hat{\mathcal{E}} \in \mathcal{C}(n+1)$ of f with $\{n, n+1\} \in \hat{\mathcal{E}}$ and $\{k, n+1\} \notin \hat{\mathcal{E}}$ for all $k = 1, \dots, n-1$.

Step 4. Given a minimizer $\hat{\mathcal{E}} \in \mathcal{C}(n+1)$ from *Step 3*, the set $\mathcal{E}' := \hat{\mathcal{E}} \setminus \{n+1, n\} \in \mathcal{C}(n)$ is connected and the induction hypothesis guarantees for $x := (x_1, \dots, x_n)$ that $m(x) = \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2 \leq f(\mathcal{E}')$ (where the abbreviation f applies to $\mathcal{E}'$ as well). Consequently,

$$m(\hat{x}) = f(\hat{\mathcal{E}}) = f(\mathcal{E}') + (x_{n+1} - x_n)^2 \geq \sum_{j=1}^n (x_{j+1} - x_j)^2.$$

Since $\{\{1, 2\}, \{2, 3\}, \dots, \{n, n+1\}\} \in \mathcal{C}(n+1)$ is in the competition with

$$f(\{\{1, 2\}, \{2, 3\}, \dots, \{n, n+1\}\}) = \sum_{j=1}^n (x_{j+1} - x_j)^2 \leq m(\hat{x}) = f(\hat{\mathcal{E}})$$

and $\hat{\mathcal{E}}$ is a minimizer, the claim $m(\hat{x}) = \sum_{j=1}^n (x_{j+1} - x_j)^2$ follows. □

The subsequent estimate holds with the optimal constant $(n-1)(2n-1)/(6n)$.

Lemma B.2. Any $x \in \mathbb{R}^n$, $n \in \mathbb{N}$, with vanishing sum $x \cdot (1, \dots, 1) = \sum_{j=1}^n x_j = 0$ satisfies

$$\max_{j=1}^n |x_j|^2 \leq \frac{(n-1)(2n-1)}{6n} \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2.$$

For any constant $C < (n-1)(2n-1)/(6n)$, there exists some $x \in \mathbb{R}^n$ with $x \cdot (1, \dots, 1) = 0$ and $C \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2 < \max_{j=1}^n |x_j|^2$.

Proof. The assertion holds for $n = 1$ (pathological) and $n = 2$ (trivial). A scaling argument for $n \geq 3$ proves that the multiplicative constant in the asserted inequality is the reciprocal of

$$\mu(n) := \min_{x \in \mathcal{A}(n)} f(x) \text{ for } f(x) := \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2$$

and $\mathcal{A}(n) := \{x \in \mathbb{R}^n : x \perp (1, \dots, 1) \text{ and } \|x\|_\infty = 1\}$. The arguments of Lemma B.1 (with a change of all signs if necessary) lead to the identity

$$\mu(n) = \min_{x \in \mathcal{B}(n)} f(x)$$

for $\mathcal{B}(n) = \{x \in \mathbb{R}^n : x \perp (1, \dots, 1) \text{ and } -1 \leq x_1 \leq x_2 \leq \dots \leq x_n = 1\}$. Any $x \in \mathcal{B}(n)$ is transformed into $y = (y_1, \dots, y_{n-1}) \in \mathbb{R}^{n-1}$ (recall $n \geq 3$) via

$$y_j := x_{j+1} - x_j \text{ for } j = 1, \dots, n-1, \text{ so that } f(x) = |y|^2 := \sum_{j=1}^{n-1} y_j^2, y \geq 0 \quad (\text{B.1})$$

(with $y \geq 0$ understood componentwise as $y_j \geq 0$ for $j = 1, \dots, n-1$). Since

$$x_k = 1 - \sum_{j=k}^{n-1} y_j \text{ for all } k = 1, \dots, n \quad (\text{B.2})$$

(the empty sum is zero), the condition $x \perp (1, \dots, 1)$ is equivalent to $n = (1, 2, \dots, n-1) \cdot y$ with the scalar product $\cdot$ in $\mathbb{R}^{n-1}$. The restriction $-1 \leq x_1 \leq x_2 \leq \dots \leq x_n = 1$ is equivalent to $y \geq 0$ and $(1, \dots, 1) \cdot y \leq 2$.

In conclusion, for $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^{n-1}$ with (B.1), $x \in \mathcal{B}(n)$ is equivalent to

$$y \in \mathcal{C}(n) := \{0 \leq y \in \mathbb{R}^{n-1} : (1, 2, \dots, n-1) \cdot y = n \text{ and } (1, \dots, 1) \cdot y \leq 2\}.$$

To determine $\mu(n) := \min_{x \in \mathcal{B}(n)} f(x) = \min_{y \in \mathcal{C}(n)} |y|^2$, suppose that $y \in \mathcal{C}(n)$ and utilize a Cauchy inequality for

$$n = (1, 2, \dots, n-1) \cdot y \leq |y| \sqrt{\sum_{j=1}^{n-1} j^2} = |y| \sqrt{\frac{(n-1)n(2n-1)}{6}}.$$

Consequently, $\frac{6n}{(n-1)(2n-1)} \leq |y|^2$. Since $y \in \mathcal{C}(n)$ is arbitrary, this proves one inequality in the claim

$$\mu(n) = \frac{6n}{(n-1)(2n-1)} \text{ for } n \in \mathbb{N}. \quad (\text{B.3})$$

To prove the reverse inequality, let $\lambda := n \left(\sum_{j=1}^{n-1} j^2 \right)^{-1} = \frac{6}{(n-1)(2n-1)} > 0$ and $y = \lambda(1, 2, \dots, n-1) \geq 0$ with $y \cdot (1, 2, \dots, n-1) = n$ and

$$(1, \dots, 1) \cdot y = \lambda \sum_{j=1}^{n-1} j = \frac{\lambda(n-1)n}{2} = \frac{3n}{2n-1} \leq 2 \quad \text{for } n \geq 2.$$

Consequently, $y \in \mathcal{C}(n)$ and $\mu(n) \leq |y|^2 = \frac{6n}{(n-1)(2n-1)}$. This concludes the proof of (B.3). It also proves the asserted optimality of the displayed constant. $\square$

A similar estimates with sharp constant reads.

Lemma B.3. *Any $x = (x_1, \dots, x_m) \in \mathbb{R}^m$, $m \in \mathbb{N}$ satisfies*

$$2 \left(1 - \cos \left(\frac{\pi}{2m+1} \right) \right) |x|^2 \leq \sum_{j=1}^{m-1} (x_{j+1} - x_j)^2 + \min\{x_1^2, \dots, x_m^2\}. \quad (\text{B.4})$$

The factor $2 \left(1 - \cos \left(\frac{\pi}{2m+1} \right) \right)$ in (B.4) is sharp in that there exists some $x \in \mathbb{R}^m \setminus \{0\}$ with equality in (B.4).

Proof. Abbreviate $\varrho(x)^2 := \sum_{j=1}^{m-1} (x_{j+1} - x_j)^2 + \min\{x_1^2, \dots, x_m^2\}$ for any $x = (x_1, \dots, x_m) \in \mathbb{R}^m \setminus \{0\}$, let $S := \{x \in \mathbb{R}^m \mid |x| = 1\}$ denote the unit sphere in $\mathbb{R}^m$, and $S_+ := \{x \in S \mid 0 \leq x_j \text{ for all } j = 1, \dots, m\}$ its positive half. Then $\varrho(|x_1|, \dots, |x_m|) \leq \varrho(x)$ follows from $(|x_{j+1}| - |x_j|)^2 \leq |x_{j+1} - x_j|^2$ for $j = 1, \dots, m-1$. This implies the equality after the definition of

$$\Lambda := \min_{x \in S} \varrho(x)^2 = \min_{x \in S_+} \varrho(x)^2.$$

Lemma B.1 shows that a minimum of the sum $\sum_{j=1}^{m-1} (x_{j+1} - x_j)^2$ is attained for ordered coefficients in the vector $x \in S_+$, hence

$$\Lambda = \min_{\substack{x \in S_+ \\ 0 \leq x_1 \leq x_2 \leq \dots \leq x_m}} \varrho(x)^2 = x \cdot Bx$$

with the tridiagonal matrix

$$B := \begin{pmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \dots & 0 & -1 & 1 \end{pmatrix}.$$

The positive eigenvalues $0 < \lambda_k = 2 \left(1 - \cos((2k-1)\pi/(2m+1)) \right)$ for $k = 1, \dots, m$ of the B are computed in [YC08, Thm.3.2.viii]. Hence, from the Rayleigh quotient

follows $\lambda_1|x|^2 \leq x \cdot Bx$ for any $x \in \mathbb{R}^m$ and $\Lambda = \lambda_1 = 2\left(1 - \cos\left(\pi/(2m+1)\right)\right)$. The eigenspace $E(\lambda_1)$ is spanned by $x = \left(\sin(\pi/(2m+1)), \sin(2\pi/(2m+1)), \dots, \sin(m\pi/(2m+1))\right) \in \mathbb{R}^m \setminus \{0\}$. Since $0 \leq \sin(k\pi/(2m+1)) \leq \sin((k+1)\pi/(2m+1))$ for any $k = 1, \dots, m-1$, the normed eigenvector $v := x/|x| \in E(\lambda_1) \cap S_+$ has ordered coefficients and leads to the equality $\Lambda = \varrho(v)^2$, in other words, $\Lambda = \lambda_1$ is the optimal constant. $\square$

Lemma B.4 is devoted to some discrete trace inequality for affine functions. The constants in Lemma B.4 are sharp. The first estimate in (B.5) is an equality for the constant function $f \equiv 1$ in any space dimension. The affine function f with $f|_{F_j} \equiv 1$ on $F_j \in \mathcal{F}(K)$ and $f(P_j) = -n/2$ at the vertex $P_j \in \mathcal{V}(K)$ opposite to F_j leads to an equality in the second estimate in (B.5). The third estimate in (B.5) is an equality for the affine function with $f|_{F_j} = 1$ and $f(P_j) = -(n+1)$.

Lemma B.4. *Let K be a n -simplex of positive volume $|K|$ with the set of sides $\mathcal{F}(K) = \{F_1, F_2, \dots, F_{n+1}\}$ and the set of vertices $\mathcal{V}(K) = \{P_1, P_2, \dots, P_{n+1}\}$ of its vertices. Then any $f \in P_1(K)$ satisfies*

$$\left\{ \sum_{k=1}^{n+1} \left| \int_{F_k} f \, ds \right|^2, \frac{n}{2} \max_{j=1, \dots, n+1} \left| \int_{F_j} f \, ds \right|^2, \max_{j=1, \dots, n+1} \frac{|f(P_j)|^2}{(n+1)} \right\} \leq \frac{n+1}{|K|} \|f\|_{L^2(K)}^2. \quad (\text{B.5})$$

Proof. Let P_k denote the vertex opposite to the side F_k in $K = \text{conv}\{F_k, P_k\}$ and set $x_k := f(P_k)$ for $k = 1, \dots, n+1$. Given $s := \sum_{k=1}^{n+1} x_k$ and $|x|^2 := \sum_{k=1}^{n+1} x_k^2$, for an affine $f \in P_1(K)$ and a side $F_k \in \mathcal{F}(K)$ the integral mean $\int_{F_k} f \, ds = f(\text{mid}(F_k)) = \sum_{j=1, j \neq k}^{n+1} x_j/n$ allows to rewrite

$$n^2 \sum_{k=1}^{n+1} \left| \int_{F_k} f \, ds \right|^2 = \sum_{k=1}^{n+1} \left(\sum_{j=1, j \neq k}^{n+1} x_j \right)^2 = \sum_{k=1}^{n+1} (s - x_k)^2 = |x|^2 + (n-1)s^2.$$

The Cauchy-Schwarz inequality implies $s^2 \leq (n+1)|x|^2$ and so $(n-2)s^2 \leq (n^2 - n - 2)|x|^2$, which is equivalent to

$$|x|^2 + (n-1)s^2 = \frac{1}{n+2}((n+2)|x|^2 + (n-2)s^2 + n^2s^2) \leq \frac{n^2}{n+2}(|x|^2 + s^2).$$

Hence, on the one hand side, $\sum_{k=1}^{n+1} |\int_{F_k} f \, ds|^2 \leq \frac{1}{n+2}(|x|^2 + s^2)$. On the other hand, the mass matrix for the barycentric coordinates reads

$$M := \left(\frac{|K|(1 + \delta_{jk})}{(n+1)(n+2)} \right)_{j,k=1, \dots, n+1} \in \mathbb{R}^{(n+1) \times (n+1)} \quad (\text{B.6})$$

and implies

$$\|f\|_{L^2(K)}^2 = \frac{|K|}{(n+1)(n+2)}(|x|^2 + s^2).$$

The combination concludes the proof of the first inequality in (B.5).

Without loss of generality assume $j = 1$ in the remaining estimates in (B.5). Let $x := (x_1, x_2, \dots, x_{n+1}) \in \mathbb{R}^{n+1}$, to deduce

$$n^2 \left| \int_{F_1} f \, ds \right|^2 = \left(\sum_{j=2}^{n+1} x_j \right)^2 = |x \cdot (0, 1, \dots, 1)|^2.$$

Recall, for any SPD matrix $A \in \mathbb{R}^{(n+1) \times (n+1)}$ and any vector $y \in \mathbb{R}^{n+1}$, $\|y\|_A^2 = y \cdot Ay$ denotes the associated weighted norm. Given the mass matrix M for the barycentric coordinates in (B.6), the Cauchy-Schwarz inequality implies

$$|x \cdot (0, 1, \dots, 1)| \leq \|(0, 1, \dots, 1)\|_{M^{-1}} \|x\|_M = \|(0, 1, \dots, 1)\|_{M^{-1}} \|f\|_{L^2(K)}.$$

An elementary calculation with the Sherman-Morrisson formula shows

$$M^{-1} = \left(\frac{(n+1)((n+2)\delta_{jk} - 1)}{|K|} \right)_{j,k=1,\dots,n+1} \in \mathbb{R}^{(n+1) \times (n+1)}. \quad (\text{B.7})$$

Hence, $\|(0, 1, \dots, 1)\|_{M^{-1}}^2 = 2n(n+1)/|K|$. The combination of this with the previous displayed formulas concludes the proof of the second inequality in (B.5).

In the above notation the Cauchy-Schwarz inequality implies

$$|f(P_1)| = |x \cdot (1, 0, \dots, 0)| \leq \|(1, 0, \dots, 0)\|_{M^{-1}} \|f\|_{L^2(K)}.$$

This and $\|(1, 0, \dots, 0)\|_{M^{-1}}^2 = (n+1)^2/|K|$ from (B.7) prove the third inequality in (B.5). $\square$

The arguments in Lemma B.4 can be adapted to different polynomial degrees and nodes. The following constants are computed for $n = 2$ with the application in Lemma 3.39 in mind.

Corollary B.5. *Let F denote an 2-simplex (a triangle) of positive volume $|F|$ with edges E_1, E_2, E_3 opposite the vertices P_1, P_2, P_3 , respectively.*

(a) *Given $f \in P_1(F)$, the following holds for $j = 1, 2, 3$*

$$|f(P_j)|^2 \leq \frac{9}{|F|} \|f\|_{L^2(F)}^2 \quad \text{and} \quad |f(\text{mid}(E_j))|^2 \leq \frac{3}{|F|} \|f\|_{L^2(F)}^2.$$

The first estimate is the last inequality in (B.5) for $n = 2$. The second estimate follows from the second in (B.5), since $\int_{E_j} f \, ds = f(\text{mid}(E_j))$.

(b) The Lagrange interpolation points for $P_2(F)$ are the vertices P_1, P_2, P_3 and the edge midpoints $\text{mid}(E_1), \text{mid}(E_2), \text{mid}(E_3)$. The dual basis functions in terms of barycentric coordinates read $2\lambda_1^2 - \lambda_1, 2\lambda_2^2 - \lambda_2, 2\lambda_3^2 - \lambda_3, 4\lambda_2\lambda_3, 4\lambda_1\lambda_3$, and $4\lambda_1\lambda_2$. The integration formula (3.2) implies that the associated mass matrix and its inverse read

$$M_2 = \frac{|F|}{180} \begin{pmatrix} 6 & -1 & -1 & -4 & 0 & 0 \\ -1 & 6 & -1 & 0 & -4 & 0 \\ -1 & -1 & 6 & 0 & 0 & -4 \\ -4 & 0 & 0 & 32 & 16 & 16 \\ 0 & -4 & 0 & 16 & 32 & 16 \\ 0 & 0 & -4 & 16 & 16 & 32 \end{pmatrix},$$

$$M_2^{-1} = \frac{1}{|F|} \begin{pmatrix} 36 & 6 & 6 & 6 & -3/2 & -3/2 \\ 6 & 36 & 6 & -3/2 & 6 & -3/2 \\ 6 & 6 & 36 & -3/2 & -3/2 & 6 \\ 6 & -3/2 & -3/2 & 39/4 & -27/8 & -27/8 \\ -3/2 & 6 & -3/2 & -27/8 & 39/4 & -27/8 \\ -3/2 & -3/2 & 6 & -27/8 & -27/8 & 39/4 \end{pmatrix}. \quad (\text{B.8})$$

Given $f \in P_2(F)$, the following holds for $j = 1, 2, 3$

$$|f(P_j)|^2 \leq \frac{36}{|F|} \|f\|_{L^2(F)}^2 \quad \text{and} \quad |f(\text{mid}(E_j))|^2 \leq \frac{39}{4|F|} \|f\|_{L^2(F)}^2.$$

Set $x = (f(P_1), f(P_2), f(P_3), f(\text{mid}(E_1)), f(\text{mid}(E_2)), f(\text{mid}(E_3))) \in \mathbb{R}^6$. The Cauchy-Schwarz inequality and the inverse mass matrix M_2^{-1} from (B.8) prove the claim

$$|f(P_j)| = |x \cdot e_j| \leq \|e_j\|_{M_2^{-1}} \|f\|_{L^2(K)} = \frac{36}{|F|} \|f\|_{L^2(K)},$$

$$|f(\text{mid}(E_j))| = |x \cdot e_{3+j}| \leq \|e_{3+j}\|_{M_2^{-1}} \|f\|_{L^2(K)} = \frac{39}{4|F|} \|f\|_{L^2(K)}.$$

Recall the Poincaré inequality with constant $C_P > 0$ from Lemma 2.11. The following local Poincaré-Friedrichs inequality is related to, e.g., [BS08, Thm. 10.6.12].

Lemma B.6. *Let $m \geq 2$ and T denote an m -simplex with diameter $h_T = \text{diam}(T)$ and side, $(m-1)$ -subsimplex, $F \in \mathcal{F}(T)$. Then any $v \in H^1(T)$ with vanishing integral mean $\oint_F v \, ds = 0$ satisfies*

$$\|v\|_{L^2(T)} \leq \sqrt{C_P^2 + 1/m^2} h_T \|\nabla v\|_{L^2(T)}.$$

Proof. Suppose that the vertex P lies opposite of the side F in $T = \text{conv}\{F, P\}$, then the trace identity from Lemma 2.10.a implies

$$0 = \oint_F v \, ds = \oint_T v \, dx + \frac{1}{m} \oint_T (x - P) \cdot \nabla v \, dx.$$

Since $|x - P| \leq h_T$ for any $x \in T$, it follows with Cauchy-Schwarz inequality

$$|T|^{1/2} \|\Pi_0 v\|_{L^2(T)} = \left| \int_T v \, dx \right| = \frac{1}{m} \left| \int_T (x - P) \cdot \nabla v \, dx \right| \leq \frac{1}{m} h_T |T|^{1/2} \|\nabla v\|_{L^2(T)}.$$

In combination with the Pythagoras theorem and the Poincaré inequality from Lemma 2.11, this concludes the proof of

$$\|v\|_{L^2(T)}^2 = \|(1 - \Pi_0)v\|_{L^2(T)}^2 + \|\Pi_0 v\|_{L^2(T)}^2 \leq (C_P^2 + 1/m^2) h_T^2 \|\nabla v\|_{L^2(T)}^2. \quad \square$$

C Remarks on the implementation

This chapter shall provide a brief overview of the software implemented for this thesis. A list of all files included in the submitted software is contained in Appendix D.

C.1 The `experiments.m` script

The script `experiments.m` allows to start any of the main functions and provides necessary information on the possible input. The input parameter are summarized in the following list for convenient reading.

- The input string `geom` contains the name of the domain Ω and determines the initial triangulation $\mathcal{T}_0$. The initial triangulations in Fig. 7.1.a-g illustrate the options `'DumbbellSlit_asymmetric'`, `'DumbbellSlit'`, `'Lshape'`, `'Slit'`, `'FourSlit'`, `'Drum1'`, `'Drum2'`, respectively. Available are as well `'DumbbellSlit_2'`, a dumbbell-slit domain with a shorter connecting tunnel $\Omega := ((-1, 4) \times (-1, 1)) \setminus ((-1, 0] \times \{0\} \cup [1, 2] \times [-0.75, 1))$, `'FourSlit_perturbed'`, a perturbed four-slit domain $\Omega := (-1, 1)^2 \setminus ([0, 0.501) \times \{0\} \cup [0, -0.5) \times \{0\} \cup \{0\} \times [0.5005, 0) \cup \{0\} \times [-0.499, 0))$, and the unit square $\Omega := (0, 1) \times (0, 1)$ with criss triangulation `'Square'` and with criss-cross triangulation `'Square_CrissCross'`.
- The integer `nrRedRefinements` fixes the number of initial red-refinements before the start of the experiment. In particular for the modified HHO method this should be chosen to be greater than zero.
- The integer `minNrElem` determines the minimal number of triangles in the final triangulation. Most experiments in Chapter 7 were computed with `minNrElem=5000000`.
- The double $0 < \text{theta} \leq 1$ determines the mesh-refinement. The choice `theta = 1` leads to uniform red-refinement of the mesh, while `theta < 1` denotes the bulk parameter in the Dörfler marking of the adaptive algorithm.
- The parameter `markPlus` allows option 0 for standard Dörfler marking and option 1 for the proposed marking strategy that ensures that at least one triangle of maximal mesh-size is marked for refinement.
- The integers `eValMin` $\leq$ `eValMax` determine the smallest and largest eigenvalue the algorithm will approximate, respectively.

The subfolder data contains comparison values (so called exact eigenvalues) for each possible domain. At least eigenvalue number one to fifteen are available. Most comparison values were computed by Benedikt Gräßle on adaptively refined meshes with a conforming P_5 finite element method for the Laplacian and a conforming piecewise P_5 Agyris finite element [Grä22] for the bi-Laplacian. For the Laplace eigenvalue problem on the L-shaped domain [TB06] provides additional approximation for $\lambda_{20}, \lambda_{50}$, and analytical values for $\lambda_{103} = \lambda_{104} = \lambda_{105}$. For the isospectral domains [Dri97] displays approximations for the first 25 identical Laplace eigenvalues and [TB06] approximates λ_{50} . If comparison values are not available, zero is used for the error computation instead.

Note, that $\text{eValMax} - \text{eValMin} > 1$ means that the estimator contributions for all eigenvalues are added as in (7.1) according to the cluster algorithm in [Gal15a].

- The parameter `savdata` allows the options 0 for no data storage and 1 for storage of the initial triangulation, computed values (e.g. number of degrees of freedom, number of triangles, eigenvalue approximation, GLBs), and errors (e.g. difference between compare values and eigenvalue approximation) after each solution step in a problem specific folder. The choice `savdata` = 2 means that the mesh (for $|\mathcal{T}| \leq 7000$) is also saved at each step in the subfolder triangulation.
- In the new methods the boolean `comparedata` decides if 0 only the chosen method is used for approximation or if 1 other GLB bounds are computed on the same mesh.
- The boolean `plotdata` in the new methods determines if triangulation and error plots are provided 1 or not 0 after the final computation step.
- The methods that approximate the Dirichlet eigenvalues of the Laplacian need the parameter `kappa1`, the constant from the Crouzeix-Raviart interpolation estimate Theorem 3.24.b. Proposed values are the analytical bound `kappa1` = 0.298234942888 from [CG14a] and the numerical bound `kappa1` = 0.1893 computed in [Liu15]. Moreover the modified HHO method needs the parameter $0 < \text{alpha} < 1$ and $0 < \text{beta}$ that appear in the bilinear form.
- The methods that approximate the Dirichlet eigenvalues of the bi-Laplacian need the parameter `kappa2`, the constant from the Morley interpolation estimate Theorem 3.35.d. Proposed values are the analytical bounds `kappa2` = 0.235882115439 from this thesis and `kappa2` = 0.25745784460 from [CG14a] as well as the numerical bound `kappa2` = 0.07353 computed in [LSL19]. Moreover, the boolean `numIntegration` determines if 1 the local matrices for the Morley method are computed with numerical integration as in [Gal14] or 0 the matrices are directly computed as explained below.

All functions use the data structure and assembling from [CB18, Sec. 7.8] and routines from the AFEM package [CN09]. Hence, on each level $\ell \in \mathbb{N}_0$, the

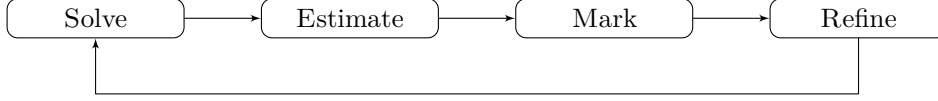


Figure C.1: AFEM cycle

triangulation $\mathcal{T}$ with set of vertices $\mathcal{V}$ and set of edges $\mathcal{F}$ is represented by $\mathbf{c4n} \in \mathbb{R}^{|\mathcal{V}| \times 2}$ (coordinates for nodes), $\mathbf{n4e} \in \mathbb{R}^{|\mathcal{T}| \times 3}$ (nodes for elements), $\mathbf{n4sDb} \in \mathbb{R}^{|\mathcal{F}(\partial\Omega)| \times 2}$ (nodes for sides on Dirichlet boundary), and the empty vector $\mathbf{n4sNb}$ (nodes for sides on Neumann boundary) for later inclusion of different types of boundary data.

The discrete problem is solved with the Matlab function `eigs`. The solution is combined with a `try-catch` instruction to avoid errors, if ,e.g., an eigenvalue is not computable on a coarse mesh. If a computation is not possible, a warning is printed, the output is set to zero, and a uniform red-refinement is forced. After the solution of the discrete problem, the (summed) local estimator contributions for `eValMin` to `eValMax` are provided as vector $\mathbf{eta4e} \in \mathbb{R}^{|\mathcal{T}|}$. If $\mathbf{theta} < 1$, the Dörfler marking is realized with the AFEM-function

```
n4sMarked = markBulk(n4e,eta4e,theta);
```

Given the maximal mesh-size $[\mathbf{hmax}, \mathbf{hmaxInd}] = \max(\text{computeLength4s}(\mathbf{c4n}, \mathbf{n4s}))$ computed with $\mathbf{n4s} = \text{computeN4s}(\mathbf{n4e}) \in \mathbb{R}^{|\mathcal{F}| \times 2}$ (nodes for sides), the new marking option is realized as

```
if markPlus == 1 && max(computeLength4s(c4n,n4sMarked)) < hmax
    n4sMarked = [n4sMarked; n4s(hmaxInd,:)];
end
```

The marked sides $\mathbf{n4sMarked}$ allow the adaptive refinement with the AFEM-function

```
[c4n,n4e,n4sDb,n4sNb] = refineBi3GB(c4n,n4e,n4sDb,n4sNb,n4sMarked);
```

to determine the triangulation on the next level. If $\mathbf{theta} = 1$, the uniform red-refined triangulation is computed with the AFEM-function

```
[c4n,n4e,n4sDb,n4sNb] = refineUniformRed(c4n,n4e,n4sDb,n4sNb);
```

C.2 Implementation of the modified HHO method

C.2.1 Main method

```
LaplaceEVP_HHOlowestorder(geom, nrRedRefinements, minNrElem, eValMin, ...
                           eValMax, theta, alpha, beta, kappa1, ...
                           savedata, plotdata, comparedata, markPlus)
```

This function runs the AFEM cycle in Fig. C.1 for the lowest-order HHO method (4.2) for $k = 0$ with the bilinear forms from Proposition 4.17.a. If `savedata > 0`, the data is saved in the folder `tmp/LaplaceEVP/HHO` with subfolder `markPlus` for `markPlus = 1`. The name of the files for the errors and values per level contains the geometry, the value for `kappa1`, `theta`, `alpha`, `beta`, and the number of the eigenvalue. If triangulation data is saved, the level and the numbers of all approximated eigenvalues are included as well. If `comparedata = 1`, on the same triangulation the Crouzeix-Raviart eigenvalues, the post-processed bound, and the extra-stabilized Crouzeix-Raviart eigenvalues are computed with parameter `kappa1` for comparison.

Given the number of eigenvalues $J = \text{eValMax} - \text{eValMin} + 1$ to be approximated, the solve function

```
solveLaplaceEVP_HHO(eValMin, eValMax, c4n, n4e, n4sDb, n4sNb, alpha, beta)
```

returns the matrices with the coefficient vectors $\mathbf{xT} \in \mathbb{R}^{3|\mathcal{T}| \times J}$ of the solution components $u_{\mathcal{T}}(j) \in P_1(\mathcal{T})$ and $\mathbf{xCR} \in \mathbb{R}^{|\mathcal{F}| \times J}$ of the solution component $u_{\mathcal{F}}(j) \in P_0(\mathcal{F}(\Omega))$ (with additional zero entries for the boundary sides) as well as the vector $\mathbf{eValHHO} \in \mathbb{R}^J$ of the discrete eigenvalues $\lambda_{\text{HHO}}(j)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ and the number of degrees of freedom $\text{nrDoFHHO} = 3|\mathcal{T}| + |\mathcal{F}(\Omega)|$.

Recall, that any $v_{\mathcal{F}}(j) \in P_0(\mathcal{F}(\Omega))$ has the same coefficient vector as its Crouzeix-Raviart extension $v_{\text{CR}} := I_{\mathcal{F}}(v_{\mathcal{F}}) \in \text{CR}_0^1(\mathcal{T})$. Hence, for the solution component $u_{\mathcal{F}}$ consider the edge-oriented Crouzeix-Raviart basis functions $\psi_E \in \text{CR}^1(\mathcal{T})$ with $\psi_E(\text{mid}(F)) = \delta_{EF}$ for any $E, F \in \mathcal{F}$. In order to save computational effort, for $P_1(\mathcal{T})$ choose the piecewise Crouzeix-Raviart function $\psi_E|_T$ for any $T \in \mathcal{T}$ and any edge $E \in \mathcal{F}(T)$ as basis.

In the lowest-order case the bilinear forms in the HHO method from Proposition 4.17.a read for all $u_{\text{HHO}} = (u_{\mathcal{T}}, u_{\text{CR}})$, $v_{\text{HHO}} = (v_{\mathcal{T}}, v_{\text{CR}}) \in V_{\text{HHO}} := P_1(\mathcal{T}) \times \text{CR}_0^1(\mathcal{T})$ with $\alpha = \text{alpha}$ and $\beta = \text{beta}$,

$$\begin{aligned} a_{\text{HHO}}(u_{\text{HHO}}, v_{\text{HHO}}) &= a_{\text{pw}}(u_{\text{CR}}, v_{\text{CR}}) + \frac{1-\alpha}{s(\mathcal{T})} \left(\Pi_0(u_{\text{CR}} - u_{\mathcal{T}}), \Pi_0(v_{\text{CR}} - v_{\mathcal{T}}) \right)_{L^2(\Omega)} \\ &\quad + \beta a_{\text{pw}}(u_{\mathcal{T}} - u_{\text{CR}}, v_{\mathcal{T}} - v_{\text{CR}}), \\ b_{\text{HHO}}(u_{\text{HHO}}, v_{\text{HHO}}) &= (u_{\mathcal{T}}, v_{\mathcal{T}})_{L^2(\Omega)}, \end{aligned}$$

with $s(\mathcal{T}) := \Pi_0(|\bullet - \text{mid}(\mathcal{T})|^2)/4 \in P_0(\mathcal{T})$. The discrete eigenvalue problem for each eigenvalue is rewritten as matrix eigenvalue problem, seeking $(\lambda_{\text{HHO}}, x) \in \mathbb{R}_+ \times \mathbb{R}^{3|\mathcal{T}| + |\mathcal{F}(\Omega)|}$ with $x = (x_{\mathcal{T}}, x_{\text{CR}}) \in \mathbb{R}^{3|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{F}(\Omega)|}$ such that

$$\left(\begin{pmatrix} \beta A_1 & -\beta A_3 \\ -\beta A_3^\top & (1+\beta)A_2 \end{pmatrix} + \begin{pmatrix} P_1^\top C P_1 & -P_1^\top C P_2 \\ -P_2^\top C^\top P_1 & P_2^\top C P_2 \end{pmatrix} \right) \begin{pmatrix} x_{\mathcal{T}} \\ x_{\text{CR}} \end{pmatrix} = \lambda_{\text{HHO}} \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_{\mathcal{T}} \\ x_{\text{CR}} \end{pmatrix},$$

where $A_1 \in \mathbb{R}^{3|\mathcal{T}| \times 3|\mathcal{T}|}$ denotes the stiffness matrix for $P_1(\mathcal{T})$, $A_2 \in \mathbb{R}^{|\mathcal{F}(\Omega)| \times |\mathcal{F}(\Omega)|}$ the stiffness matrix for $\text{CR}_0^1(\mathcal{T})$, $A_3 \in \mathbb{R}^{3|\mathcal{T}| \times |\mathcal{F}(\Omega)|}$ represents the L^2 scalar product of

the piecewise gradients of one function in $P_1(\mathcal{T})$ and one in $\text{CR}_0^1(\mathcal{T})$, $B \in \mathbb{R}^{3|\mathcal{T}| \times 3|\mathcal{T}|}$ is the mass matrix for $P_1(\mathcal{T})$, $P_1 \in \mathbb{R}^{3|\mathcal{T}| \times |\mathcal{T}|}$ represents the Π_0 projection from $P_1(\mathcal{T})$ to $P_0(\mathcal{T})$ and $P_2 \in \mathbb{R}^{|\mathcal{F}(\Omega)| \times |\mathcal{T}|}$ the Π_0 projection from $\text{CR}_0^1(\mathcal{T})$ to $P_0(\mathcal{T})$, and $C \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ the mass matrix for $P_0(\mathcal{T})$ weighted with the factor $(1 - \alpha)/s(\mathcal{T}) \in P_0(\mathcal{T})$. It suffices to assemble in a `parfor`-loop over all elements $T_k \in \mathcal{T}$ (numbering as in `n4e`) the local Crouzeix-Raviart mass matrices `Blocal` $\in \mathbb{R}^{3 \times 3 \times |\mathcal{T}|}$, stiffness matrices `Alocal` $\in \mathbb{R}^{3 \times 3 \times |\mathcal{T}|}$, and the local weight `Clocal` $\in \mathbb{R}^{|\mathcal{T}|}$ with `Clocal(k) = |Tk|(1 - α)/s(Tk)` to compute the matrices in the discrete eigenvalue problem. Remark 3.23.b contains `Blocal(:, :, k) = MCR(Tk)` and `Alocal(:, :, k) = ACR(Tk)` known from the standard Crouzeix-Raviart eigenvalue solver, e.g., [Gal14]. Since the edge-midpoint formula $\int_T f \, dx = |T|/3 \sum_{E \in \mathcal{E}(T)} f(\text{mid}(E))$ is exact for $f \in P_2(T)$, for the weight holds

$$s(T) = s(\mathcal{T})|_T = \frac{1}{12} \sum_{E \in \mathcal{F}} (\text{mid}(E) - \text{mid}(T)) \cdot (\text{mid}(E) - \text{mid}(T)).$$

The projection matrices P_1 and P_2 are assembled directly since $f_T \psi_E = 1/3$ for any $E \in \mathcal{F}$.

Given the matrix `s4e = computeS4e(n4e)  $\in \mathbb{R}^{3 \times |\mathcal{T}|}$` (sides for elements), the row indices vector `IT` (resp. `ICR`) and column indices vector `JT` (resp. `JCR`) for $P_1(\mathcal{T})$ (resp. $\text{CR}^1(\mathcal{T})$)

```
IT    = permute(reshape(repmat(1:3*nrElems,3,1),3,3, nrElems),[2,1,3]);
                                     % row indices P1 (local CR basis)
JT    = repmat(1:3*nrElems,3,1);    % column indices P1 (local CR basis)
s4eT  = s4e';
ICR   = [s4eT;s4eT;s4eT];          % row indices CR
JCR   = [s4eT(:),s4eT(:),s4eT(:)]'; % column indices CR
```

and the row indices vector `I0` for $P_0(\mathcal{T})$

```
I0    = repmat(1:nrElems,3,1);      % row indices for P0
```

allow the global assembly of the blocks in the matrix eigenvalue problem via `sparse` and `spdiags`

```
% blocks of stiffness matrix for a_pw(uCR,vCR)+beta a_pw(uT-uCR,vT-vCR)
A1    = sparse(IT(:), JT(:), Alocal(:));
A2    = sparse(ICR(:), JCR(:), Alocal(:));
A3    = sparse(IT(:), JCR(:), Alocal(:));

% blocks of stabilization (1-alpha)/s(T) (Pi0 (uCR-uT), Pi0 (vCR-vT))
C     = spdiags(Clocal, 0, nrElems, nrElems); % weighted mass matrix
P1    = sparse(I0(:), (1:3*nrElems)', ones(3*nrElems,1)/3);
                                     % projection Pi0uT
P2    = sparse(I0(:), s4eT(:), ones(3*nrElems,1)/3);
                                     % projection Pi0uCR

% block of the mass matrix (uT,vT)
B     = sparse(IT(:), JT(:), Blocal(:));
```

Before the solution of the matrix eigenvalue problem, the degrees of freedom for $\text{CR}_0^1(\mathcal{T})$ need to be considered. The number of sides `nrSides` and the vector `DbSides` $\in \mathbb{R}^{|\mathcal{F}(\partial\Omega)|}$ that contains the numbers of all Dirichlet boundary sides allow to compute

```
dof = setdiff(1:nrSides,DbSides);    % vector of degrees of freedom
                                     % (for the side component xCR, i.e.,
                                     % one per non-Dirichlet side)
```

and set

```
A2 = A2(dof, dof);
A3 = A3(:, dof);
P2 = P2(:,dof);
```

These blocks constitute the global mass matrix $\mathbf{AA} \in \mathbb{R}^{(3|\mathcal{T}|+|\mathcal{F}(\Omega)|) \times (3|\mathcal{T}|+|\mathcal{F}(\Omega)|)}$ and stiffness matrix $\mathbf{BB} \in \mathbb{R}^{(3|\mathcal{T}|+|\mathcal{F}(\Omega)|) \times (3|\mathcal{T}|+|\mathcal{F}(\Omega)|)}$. The matrix eigenvalue problem is directly solved with `eigs`, which utilizes some pseudo Schur complement and allows so for larger matrices

```
[V,D] = eigs(AA, BB, eValMax, 'sm');
[D,ind] = sort(diag(D));
x      = V(:,ind(eValMin:eValMax));
eVal   = D(eValMin:eValMax);
```

Finally, the solution components are recovered

```
xT      = x(1:3*nrElems,:);
xCR     = zeros(nrSides,nrEig);
xCR(dof,:) = x(3*nrElems+1:end,:);
```

The constants $\tau = 6/\pi = 1.90986$ for right-isosceles triangles from Lemma 4.21 and $\delta = \kappa_1 h_{\max}$ with $\kappa_1 = \text{kappa1}$ are used. If the condition $\tau\beta + \delta^2 \lambda_{\text{HHO}} \leq \alpha$ holds, the discrete eigenvalue $\lambda_{\text{HHO}}(j) = \text{eValHHO}(j - \text{eValMin} + 1)$ is a guaranteed lower bound for the exact eigenvalue λ_j and saved as $\text{GLBHHO}(j - \text{eValMin} + 1)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ in the main function (otherwise set $\text{GLBHHO}(j - \text{eValMin} + 1) = 0$). The estimator (7.3) introduced in [CGT22]

```
cur_eta4e = estimateLaplaceEVP_HHO_EtaElements(eValHHO(j), xCR(:,j), ...
                                                xT(:,j), c4n, n4e, n4sDb, n4sNb)
```

combines the AFEM functions `P0TangentJump` and `P0NormalJump` for the edge contributions and utilizes the choice of the piecewise Crouzeix-Raviart basis with local diagonal mass matrix. After splicing the solution components

```
xT4e      = reshape(xT,3,nrElems);    % coefficients for xT for elements
xCR4e     = reshape(xCR(s4e',:),'',3,nrElems);
                                     % coefficients for xCR for elements
```

a `parfor`-loop computes $|T|^2 \|u_{\mathcal{T}}\|_{L^2(T)}^2$ and $\nabla u_{\text{CR}}|_T$ for any $T \in \mathcal{T}$

```

parfor elem = 1 : nrElems
    coords    = coord4e(:, :, elem);      % coordinates for the nodes
    area      = area4e(elem);              % area of the element
    coeffxT    = xT4e(:, elem);            % coefficients xT
    coeffxCR   = xCR4e(:, elem);           % coefficients xCR

    grads     = [coords'; 1 1 1] \ [-2 0; 0 -2; 0 0]; % gradients for CR
    gradUCR(elem, :) = coeffxCR' * grads([3 1 2], :); % DxCR|_T
    volTermSq4e(elem) = area^2.*coeffxT' * coeffxT/3; % ||xT||^2
end

```

The mentioned AFEM-function compute the constant tangential jumps $\mathbf{TJump4sgrad}$ with entries $[\nabla u_{CR}]_F \times \nu_F$ and normal jumps $\mathbf{NJump4sgrad}$ with entries $[\nabla u_{CR}]_F \cdot \nu_F$ for any edge $F \in \mathcal{F}$. Given the vector $\mathbf{length4s} = \text{computeLength4s}(\mathbf{c4n}, \mathbf{n4s}) \in \mathbb{R}^{|\mathcal{F}|}$ (lengths for sides) and $\mathbf{intSides} \in \mathbb{R}^{|\mathcal{F}(\Omega)|}$ (numbers of all interior sides). The squared L^2 norm of the jumps are computed via

```

jump4s_L2normSq      = TJump4sgrad.^2 .* length4s;
                    % tangential jumps
jump4s_L2normSq(intSides) = jump4s_L2normSq(intSides) ...
                    + NJump4sgrad(intSides).^2.*length4s(intSides);
                    % tangential jumps and normal jumps for
                    % interior sides

```

Finally, the squared estimator contribution per element is computed

```

eta4eSq = eVal^2*volTermSq4e + sqrt(area4e).* sum(jump4s_L2normSq(s4e), 2);

```

C.2.2 Parameter study

```

LaplaceEVP_HHO_Parameter(geom, nrRedRefinements, minNrElem, theta, ...
                        eValMin, eValMax, kappa1, savedata, plotdata)

```

This method checks which parameter pairs $0 < \alpha < 1$ and $0 < \beta$ lead to guaranteed lower bounds with the modified HHO method. Set $J := \mathbf{eValMax} - \mathbf{eValMin} + 1$. If $\mathbf{savedata} = 1$, the computed guaranteed lower bounds and parameter combinations are saved for each triangulation in the folder `tmp/LaplaceEVP/HHO/Parameter` and the file name contains the geometry, the value for $\mathbf{kappa1}$, $\mathbf{theta}$, the number of the eigenvalue, and the number of refinements of the initial triangulation.

On each level the Laplace eigenvalue problem is once solved with the extra-stabilized and the standard Crouzeix-Raviart method to compute the eigenvalues $\mathbf{eValES} \in \mathbb{R}^J$ and $\mathbf{eValCR} \in \mathbb{R}^J$ as well as the eigenvector matrix $\mathbf{eVectCR} \in \mathbb{R}^{|\mathcal{F}| \times J}$. The values are used to compute the guaranteed lower bounds $\mathbf{GLBCR}$ in (4.46) and $\mathbf{GLBES} = \mathbf{eValES}$, if the a posteriori condition is satisfied. Given the constants $\tau = 6/\pi = 1.90986$ for right-isosceles triangles from Lemma 4.21 and $\delta = \kappa_1 h_{\max}$ with $\kappa_1 = \mathbf{kappa1}$, in a for-loop for $\alpha = 0.02 : 0.02 : 0.98$ an appropriate

$\text{beta} = (\text{alpha} - \delta^2 * \max(\text{eValCR}))/\tau^2$ is computed. If $\text{beta} > 10^{-3}$ to avoid badly conditioned matrix eigenvalue problems, the Laplace eigenvalue problem is solved with the parameter pair $(\text{alpha}, \text{beta})$ that guarantees that each discrete eigenvalue $\lambda_{\text{HHO}}(j)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ satisfies the a posteriori condition $\tau \text{beta} + \delta^2 \lambda_{\text{HHO}}(j) \leq \text{alpha}$ as a consequence of Corollary 4.18. The potential adaptive mesh-refinement is done with respect to the Crouzeix-Raviart eigenpair. Note that $\text{eValMax} > \text{eValMin}$ implies that beta is computed for eValMax and hence smaller than necessary for the other eigenvalues.

C.3 Implementation of the extra-stabilized Crouzeix-Raviart method

```
LaplaceEVP_extrastabilized(geom, nrRedRefinements, minNrElem, eValMin, ...
                           eValMax, theta, kappa1, savedata, plotdata, ...
                           comparedata, markPlus)
```

This function runs the AFEM cycle from Fig. C.1 for the extra-stabilized Crouzeix-Raviart method, (5.2) for $m = 1$. If $\text{savedata} > 0$, the data is saved in the folder `tmp/LaplaceEVP/extra_stabilized` with subfolder `markPlus` for $\text{markPlus} = 1$. The name of the files for the errors and computed values per level contains the geometry, the value for kappa1 , theta , and the number of the eigenvalue. If triangulation data is saved, the level and the numbers of all approximated eigenvalues are included as well. If $\text{comparedata} = 1$, the Crouzeix-Raviart eigenvalues and post-processed bound are computed on the same triangulation for comparison.

Given the number of eigenvalues $J = \text{eValMax} - \text{eValMin} + 1$ to be approximated, the solve function

```
solveLaplaceEVP_extrastabilized(eValMin, eValMax, c4n, n4e, n4sDb, ...
                                n4sNb, kappa1)
```

returns the matrices of the coefficient vectors $\mathbf{xPW} \in \mathbb{R}^{3|\mathcal{T}| \times J}$ of the solution components $u_{\text{pw}}(j) \in P_1(\mathcal{T})$ and $\mathbf{xCR} \in \mathbb{R}^{|\mathcal{F}| \times J}$ of the solution component $u_{\text{CR}}(j) \in \text{CR}^1(\mathcal{T})$ (with additional zero entries for the boundary sides) as well as the vector $\mathbf{eValES} \in \mathbb{R}^J$ of the discrete eigenvalues $\lambda_{\text{es}}(j)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ and the number of degrees of freedom $\text{nrDoFES} = 3|\mathcal{T}| + |\mathcal{F}(\Omega)|$.

Again the edge-oriented Crouzeix-Raviart basis functions $\psi_E \in \text{CR}^1(\mathcal{T})$ with $\psi_E(\text{mid}(F)) = \delta_{EF}$ for any $E, F \in \mathcal{F}$ are considered as basis for $\text{CR}_0^1(\mathcal{T})$ and their piecewise version $\psi_E|_T$ for any $T \in \mathcal{T}$ and any edge $E \in \mathcal{F}(T)$ for $P_1(\mathcal{T})$.

The discrete bilinear forms for any $u_{\text{es}} = (u_{\text{pw}}, u_{\text{CR}})$, $v_{\text{es}} = (v_{\text{pw}}, v_{\text{CR}})$, $\in V_{\text{es}} := P_1(\mathcal{T}) \times \text{CR}_0^1(\mathcal{T})$ read with $\kappa_1 = \text{kappa1}$

$$\begin{aligned} a_{\text{es}}(u_{\text{es}}, v_{\text{es}}) &:= a_{\text{pw}}(u_{\text{CR}}, v_{\text{CR}}) + \kappa_1^{-2} (h_{\mathcal{T}}^{-2} (u_{\text{pw}} - u_{\text{CR}}), v_{\text{pw}} - v_{\text{CR}})_{L^2(\Omega)}, \\ b_{\text{es}}(u_{\text{es}}, v_{\text{es}}) &:= b(u_{\text{pw}}, v_{\text{pw}}). \end{aligned}$$

This leads to a matrix eigenvalue problem, seeking $(\lambda_{\text{es}}, x) \in \mathbb{R}_+ \times \mathbb{R}^{3|\mathcal{T}|+|\mathcal{F}(\Omega)|}$ with $x = (x_{\text{pw}}, x_{\text{CR}}) \in \mathbb{R}^{3|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{F}(\Omega)|}$ such that

$$\begin{pmatrix} 0 & 0 \\ 0 & A \end{pmatrix} + \begin{pmatrix} C_1 & -C_3 \\ -C_3^\top & C_2 \end{pmatrix} \begin{pmatrix} x_{\text{pw}} \\ x_{\text{CR}} \end{pmatrix} = \lambda_{\text{es}} \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_{\text{pw}} \\ x_{\text{CR}} \end{pmatrix}. \quad (\text{C.1})$$

where $A \in \mathbb{R}^{|\mathcal{F}(\Omega)| \times |\mathcal{F}(\Omega)|}$ denotes the stiffness matrix for $\text{CR}_0^1(\mathcal{T})$, $B \in \mathbb{R}^{3|\mathcal{T}| \times 3|\mathcal{T}|}$ is the mass matrix for $P_1(\mathcal{T})$, $C_1 \in \mathbb{R}^{3|\mathcal{T}| \times 3|\mathcal{T}|}$ represents mass matrix for $P_1(\mathcal{T})$ weighted with $\kappa_1^{-2} h_{\mathcal{T}}^{-2} \in P_0(\mathcal{T})$, the same weight is contained in the mass matrix $C_2 \in \mathbb{R}^{|\mathcal{F}(\Omega)| \times |\mathcal{F}(\Omega)|}$ for $\text{CR}_0^1(\mathcal{T})$ and $C_3 \in \mathbb{R}^{3|\mathcal{T}| \times |\mathcal{F}(\Omega)|}$ which represents the L^2 scalar product of one function in $P_1(\mathcal{T})$ and one in $\text{CR}_0^1(\mathcal{T})$.

It suffices to assemble in a `parfor`-loop over all elements $T_k \in \mathcal{T}$ (numbering as in `n4e`) the local Crouzeix-Raviart mass matrices `Blocal` $\in \mathbb{R}^{3 \times 3 \times |\mathcal{T}|}$ and stiffness matrices `Alocal` $\in \mathbb{R}^{3 \times 3 \times |\mathcal{T}|}$. Remark 3.23.b contains the formula for `Blocal(:, :, k) = M_{\text{CR}}(T_k)` and `Alocal(:, :, k) = A_{\text{CR}}(T_k)`. Given the vectors `s4e = computeS4e(n4e)  $\in \mathbb{R}^{3 \times |\mathcal{T}|}$` (sides for elements) and `h4e = max(length4s(s4e), [], 2)` (diameter for element), the weight $\kappa_1^{-2} h_{\mathcal{T}}^{-2} \in P_0(\mathcal{T})$ is computed globally in

```
weight = repmat(h4e.^(-2), [1, 9])';
weight = weight(:) ./ kappa1^2;
```

The row indices vector `IPW` (resp. `ICR`) and column indices vector `JPW` (resp. `JCR`) for $P_1(\mathcal{T})$ (resp. $\text{CR}^1(\mathcal{T})$) read

```
IPW = permute(reshape(repmat(1:3*nrElems, 3, 1), 3, 3, nrElems), [2, 1, 3]);
                                % row indices piecewise P1
JPW = repmat(1:3*nrElems, 3, 1);
                                % column indices piecewise P1
s4eT = s4e';
ICR = [s4eT; s4eT; s4eT];
                                % row indices CR
JCR = [s4eT(:), s4eT(:), s4eT(:)]';
                                % column indices CR
```

and allow the global assembly of the blocks in the matrix eigenvalue problem via `sparse`

```
% local stiffness matrix a_pw(uCR, vCR)
A = sparse(ICR(:), JCR(:), Alocal(:));

% local mass matrices for 1/kappa1^2(h_T^{-2}) uPW-uCR, vPW-vCR)
C1 = sparse(IPW(:), JPW(:), weight.*Blocal(:));
C2 = sparse(ICR(:), JCR(:), weight.*Blocal(:));
C3 = sparse(IPW(:), JCR(:), weight.*Blocal(:));

% block of the mass matrix (uPW, vPW)
B = sparse(IPW(:), JPW(:), Blocal(:));
```

Before the solution of the matrix eigenvalue problem, the degrees of freedom for $\text{CR}_0^1(\mathcal{T})$ need to be considered. The number of sides `nrSides` and the vector `DbSides` $\in \mathbb{R}^{|\mathcal{F}(\partial\Omega)|}$ that contains the numbers of all Dirichlet boundary sides allow to compute

```
dof = setdiff(1:nrSides,DbSides); % vector of degrees of freedom
                                     % (for CR component xCR, i.e.,
                                     % one per non-Dirichlet side)
```

and set

```
A = A(dof, dof);
C2 = C2(dof, dof);
C3 = C3(:, dof);
```

These blocks constitute the global mass matrix $AA \in \mathbb{R}^{(3|\mathcal{T}|+|\mathcal{F}(\Omega)|) \times (3|\mathcal{T}|+|\mathcal{F}(\Omega)|)}$ and stiffness matrix $BB \in \mathbb{R}^{(3|\mathcal{T}|+|\mathcal{F}(\Omega)|) \times (3|\mathcal{T}|+|\mathcal{F}(\Omega)|)}$. The matrix eigenvalue problem is directly solved with `eigs`, which utilizes some pseudo Schur complement and allows for larger matrices

```
[V,D] = eigs(AA, BB, eValMax, 'sm');
[D,ind] = sort(diag(D));
x = V(:, ind(eValMin:eValMax));
eVal = D(eValMin:eValMax);
```

Finally, the solution components are recovered

```
xPW = x(1:3*nrElems,:);
xCR = zeros(nrSides, nrEig);
xCR(dof,:) = x(3*nrElems+1:end,:);
```

Since the mass and stiffness matrices for Crouzeix-Raviart are already computed, for `comparedata = 1` the function

```
solveLaplaceEVP_extrastabilized_compare(eValMin, eValMax, c4n, n4e, ...
                                         n4sDb, n4sNb, kappa1)
```

computes additionally the Crouzeix-Raviart eigenpair $eValCR \in \mathbb{R}^J$ and $eVectCR \in \mathbb{R}^{|\mathcal{F}(\Omega)| \times J}$ in

```
[VCR,DCR] = eigs(A, B, eValMax, 'sm');
DCR = sort(diag(DCR));
eValCR = DCR(eValMin:eValMax);
eVectCR(dof,:) = VCR(:,ind(eValMin:eValMax));
```

and $nrDoFCR = |\mathcal{F}(\Omega)|$ on the same mesh.

If the condition $\lambda_{es} h_{\max}^2 \leq 1/\kappa_1^2$ is satisfied, the discrete eigenvalue $\lambda_{es}(j) = eValES(j - eValMin + 1)$ is a guaranteed lower bound for the exact eigenvalue λ_j and saved as $GLBES(j - eValMin + 1)$ for $j = eValMin, \dots, eValMax$ in the main function (otherwise set $GLBES(j - eValMin + 1) = 0$).

The local estimator contributions for $j = 1, \dots, J$

```
cur_eta4e = estimateLaplaceEVP_CR_EtaElements(eValES(j), xCR(:,j), ...
                                              c4n, n4e, n4sDb, n4sNb)
```

are computed with the estimator for the Crouzeix-Raviart eigenvalue problem from [Gal14].

C.4 Implementation of the extra-stabilized Morley method

```
BiLaplaceEVP_extrastabilized(geom, nrRedRefinements, minNrElem, ...
                             eValMin, eValMax, theta, kappa2, ...
                             savedata, plotdata, comparedata, ...
                             numIntegration, markPlus)
```

This function runs the AFEM cycle from Fig. C.1 for the extra-stabilized Morley method, (5.2) for $m = 2$. For input `savedata` > 0 the data will be saved in the folder `tmp/BiLaplaceEVP/extra_stabilized` with subfolder `markPlus` for `markPlus = 1` and `withoutNumIntegration` for `numIntegration = 0`. The name of the files for the errors and computed values per level contains the geometry, the value for `kappa2`, `theta`, and the number of the eigenvalue. If triangulation data is saved, the level and the numbers of all approximated eigenvalues are included as well. If `comparedata = 1`, the Morley eigenvalues and post-processed bound are computed on the same triangulation for comparison.

Given the number of eigenvalues $J = \text{eValMax} - \text{eValMin} + 1$ to be approximated, the solve function

```
solveBiLaplaceEVP_extrastabilized(eValMin, eValMax, c4n, ...
                                   n4e, n4sDb, n4sNb, kappa2)
```

computes the matrices of the coefficient vectors $\mathbf{xPW} \in \mathbb{R}^{6|\mathcal{T}| \times J}$ of the solution components $u_{\text{pw}}(j) \in P_2(\mathcal{T})$ and $\mathbf{xM} \in \mathbb{R}^{(|\mathcal{V}|+|\mathcal{F}|) \times J}$ of the solution component $u_M(j) \in M(\mathcal{T})$ (with additional zero entries for the boundary vertices and sides) as well as the vector $\mathbf{eValES} \in \mathbb{R}^J$ of discrete eigenvalues $\lambda_{\text{es}}(j)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ and the number of degrees of freedom $\text{nrDoFES} = 6|\mathcal{T}| + |\mathcal{V}(\Omega)| + |\mathcal{F}(\Omega)|$. The local mass matrices $\mathbf{B}_{\text{local}} \in \mathbb{R}^{6 \times 6 \times |\mathcal{T}|}$ for the basis functions in (3.44)–(3.45) for each $T \in \mathcal{T}$ is returned to compute the volume contribution $\|\lambda_{\text{es}} u_M(j)\|_{L^2(T)}$ in the estimator for $j = \text{eValMin}, \dots, \text{eValMax}$. Additionally, the matrix $\mathbf{Hesse4e} \in \mathbb{R}^{2 \times 2 \times |\mathcal{T}| \times J}$ containing the piecewise constant Hessians of the solution component $u_M(j)$ for $j = \text{eValMin}, \dots, \text{eValMax}$, i.e., $\mathbf{Hesse4e}(:, :, k, j) = D^2 u_M(j)|_{T_k}$ for any $k = 1, \dots, |\mathcal{T}|$ is returned. The latter allows the computation of the jump terms $\|[D^2 u_M(j)]_F \times \nu_F\|_{L^2(F)}$ for any $F \in \mathcal{F}$ in the estimator. The assembly of the local matrices is more complex than for Crouzeix-Raviart, hence starting again in the estimator function would prolong the computation time unnecessarily.

The computation is comparable to Appendix C.3. The Morley basis functions (3.44)–(3.45) are used as basis for $M(\mathcal{T})$ and their piecewise version is used for $P_2(\mathcal{T})$. Of course, the matrix eigenvalue problem has the same form as in (C.1) with x_{CR} replaced by x_M , adapted dimension, i.e., $3|\mathcal{T}| = \dim(P_1(\mathcal{T}))$ replaced by $6|\mathcal{T}| = \dim(P_2(\mathcal{T}))$ and $|\mathcal{F}(\Omega)| = \dim(\text{CR}_0^1(\mathcal{T}))$ by $|\mathcal{V}(\Omega)| + |\mathcal{F}(\Omega)| = \dim(M(\mathcal{T}))$, and the weight $\kappa_2^{-2} h_{\mathcal{T}}^{-4} \in P_0(\mathcal{T})$ with $\kappa_2 = \text{kappa2}$.

In the function `solveBiLaplaceEVP_extrastabilized` the necessary local mass $\text{Blocal} \in \mathbb{R}^{6 \times 6 \times |\mathcal{T}|}$ and stiffness matrices $\text{Alocal} \in \mathbb{R}^{6 \times 6 \times |\mathcal{T}|}$ for Morley are assembled as in [Gal14] defining function handle for the Morley basis function as well as their Hessians and computing the local matrices with the AFEM-function `integrate` or by multiplying with the area (for constant integrands).

Let $\text{s4e} = \text{computeS4e}(\text{n4e}) \in \mathbb{R}^{3 \times |\mathcal{T}|}$ (sides for elements) and compute the vector $\text{h4e} = \max(\text{length4s}(\text{s4e}), [], 2) \in \mathbb{R}^{|\mathcal{T}|}$ (diameter for elements). The weight $\kappa_2^{-2} h_{\mathcal{T}}^{-4} \in P_0(\mathcal{T})$ is computed globally in

```
weight = repmat(h4e.^(-4),[1,9])';
weight = weight(:)./kappa2^2;
```

The row indices vector `IPW` (resp. `IM`) and column indices vector `JPW` (resp. `JM`) for $P_2(\mathcal{T})$ (resp. $M(\mathcal{T})$) read

```
IPW=permute(reshape(repmat(1:6*nrElem,6,1),6,6,nrElem),[2,1,3]);
                                % row indices piecewise P2
JPW=repmat(1:6*nrElem,6,1);
                                % column indices piecewise P2
dofM = [n4e,nrNodes+s4e]';
                                % Morley dof (with boundary)
IM = repmat(dofM',1,size(dofM,1))';
                                % row indices Morley
JM = repmat(dofM(:,1),size(dofM,1))';
                                % column indices Morley
```

The free components of the Morley functions are computed given the number of sides `nrSides`, the vector $\text{DbSides} \in \mathbb{R}^{|\mathcal{F}(\partial\Omega)|}$ with numbers of all Dirichlet boundary sides, as well as the vector $\text{DbNodes} \in \mathbb{R}^{|\mathcal{V}(\partial\Omega)|}$ with the numbers of all Dirichlet boundary nodes in

```
dof = setdiff(1:nrNodes+nrSides,[DbNodes,nrNodes+DbSides]);
                                % vector of degrees of freedom
                                % (for xM, i.e. one per non-Dirichlet
                                % node and side)
```

With this input, `IM` (resp. `JM`) replacing `ICR` (resp. `JCR`) the assembly is verbatim to Appendix C.3. The solution components are determined via

```
xPW = x(1:6*nrElem,:);
xM = zeros(nrNodes+nrSides,nrEig);
xM(dof,:) = x(6*nrElem+1:end,:);
```

The local Hessian `Hesse4e` is assembled utilizing the computed Hessians of the Morley basis functions and the coefficients `xM`.

Since the mass and stiffness matrices for Morley are already computed, for `comparedata = 1` the function

```
solveBiLaplaceEVP_extrastabilized_compare(eValMin,eValMax,...
                                           c4n,n4e,n4sDb,n4sNb,kappa2)
```

computes additionally the Morley eigenvalues $\text{eValM} \in \mathbb{R}^J$ and $\text{nrDoFM} = |\mathcal{V}(\Omega)| + |\mathcal{F}(\Omega)|$.

If the condition $\lambda_{\text{es}} h_{\max}^4 \leq 1/\kappa_2^2$ is satisfied, the discrete eigenvalue $\lambda_{\text{es}}(j) = \text{eValES}(j - \text{eValMin} + 1)$ is a guaranteed lower bound for the exact eigenvalue λ_j

and saved as $\text{GLBES}(j - \text{eValMin} + 1)$ for $j = \text{eValMin}, \dots, \text{eValMax}$ in the main function (otherwise set $\text{GLBES}(j - \text{eValMin} + 1) = 0$).

The local estimator contributions for $j = 1, \dots, J$

```
cur_eta4e = estimateBiLaplaceEVP_Morley_EtaElements(eValES(j), ...
    xM(:,j), c4n, n4e, n4sDb, n4sNb, Hesse4e(:,:,j), Blocal)
```

are computed with the estimator for Morley eigenvalue problems from [Gal14].

As an alternative to the computation of the local matrices in [Gal14], the functions

```
solveBiLaplaceEVP_extrastabilized_direct(eValMin, eValMax, c4n, n4e, ...
    n4sDb, n4sNb, kappa2)
```

and

```
solveBiLaplaceEVP_extrastabilized_compare_direct(eValMin, eValMax, ...
    c4n, n4e, n4sDb, n4sNb, kappa2)
```

compute the local matrices directly in a `parfor`-loop over all elements. The computation is inspired by the local stiffness matrix from [CGH14, Sec. 6.2, 6.5]. The stiffness matrix was adapted to the local enumeration of the sides in the AFEM-softwarepackage, where the first side is between the first and second vertex and the rest of the sides are enumerated counter-clockwise, while [CGH14] suppose the first side opposite the first vertex etc. The local mass matrix and the Hessians $\text{HesseB1local}, \dots, \text{HesseB6local} \in \mathbb{R}^{2 \times 2 \times |T|}$ of the basis functions are computed using the explicit form of the basis functions (3.44)–(3.45) and the integration formula (3.2). The resulting `parfor`-loop reads

```
I=[2 ,3 ,3 ,1 ,1 ,2];
parfor elem = 1 : nrElem
    area = area4e(elem);
    sg = [1;1;1] -2*(es4e(:,elem)== elem); % vector with sign for sides
    sg = sg([2;3;1]); % reodering for different side numbering

    G = [1 ,1 ,1; coord4e(:, :, elem)'] \ [0, 0; eye(2) ];
    % gradients for barycentric coordinates

    % auxillary matrices
    M = G*G';
    z = diag(M);
    zz = z.^(-2);
    R = repmat([M(2, 3); M(3, 1); M(1, 2)], 1, 2) ./ ...
        [z([2, 3, 1]), z([3, 1, 2])];
    v = [M(2,3);M(1,3);M(1,2)];

    H = [reshape(G([I, 3+I, 3+I]) * ...
        (kron(eye(9), [0, 2; 2, 0]) ...
        - (2 * diag( repmat( reshape(R', 1, 6) , 1, 3)))))* ...
        blkdiag(G(2:3), G([3, 1]), G(1:2), G(2:3), ...
        G([3 ,1]), G(1:2), G(5:6), ...
        G([6, 4]), G(4:5))', 3, 3 ); ...
        2 * diag(sg./ sqrt(z)) * G(:, [1, 2, 2]) .* G(:, [1, 1, 2])];
```

```

A = area * H * diag([1, 2, 1]) * H';
% stiffness matrix [CarstensenGallistlHu2014]
% (computation for E_j opposite P_j)
Alocal(:, :, elem) = A([1 2 3 6 4 5], [1 2 3 6 4 5]);
% reordering for different side numbering

% Blocal = area*[B1, B2; B2', B3];
B1 = ( (43/180 * ones(3,3) + 7/180 * eye(3)) ...
+ (1/36 * (sum(z,1)/prod(z,1))*(ones(3,3)-eye(3)) ...
+ 1/30*[0, zz(3), zz(2); zz(3), 0, zz(1); zz(2), zz(1), 0]) .* (v*v') ...
+ 1/12 * [0, R(1,2)+R(2,1), R(3,2)+R(1,1); ...
R(1,2)+R(2,1), 0, R(3,1)+R(2,2); ...
R(3,2)+R(1,1), R(3,1)+R(2,2), 0] ...
+ 4/45 * [0, R(1,1)+R(2,2), R(3,1)+R(1,2); ...
R(1,1)+R(2,2), 0, R(3,2)+R(2,1); ...
R(3,1)+R(1,2), R(3,2)+R(2,1), 0] ...
+ diag( sum(R,2)./6 + sum(R.^2,2)/30 + 2*prod(R,2)/36));
B2 = - ( (1/12 * ones(3,3) + 1/180 * eye(3)) ...
+ 1/36 * repmat( sum(R,2), 1, 3) ...
+ 1/180 * [0, R(1,1), R(1,2); R(2,2), 0, R(2,1); ...
R(3,1), R(3,2), 0]) .* (sg./sqrt(z))';
B3 = (1/36 * ones(3,3) + 1/180 * eye(3)) .* ((sg./sqrt(z)) ...
* (sg./sqrt(z))');
BM = area*[B1, B2; B2', B3];
% mass matrix
Blocal(:, :, elem) = BM([1 2 3 6 4 5], [1 2 3 6 4 5]);
% reordering for different side numbering

% Hessian for basis functions B1, ... , B6 (for E_j opposite P_j)
H1 = [-2*M(2,3)/z(2), 2; 2, -2*M(2,3)/z(3)];
H2 = [-2*M(1,3)/z(1), 2; 2, -2*M(1,3)/z(3)];
H3 = [-2*M(1,2)/z(1), 2; 2, -2*M(1,2)/z(2)];
HesseB1local(:, :, elem) = [G([2,3],1)'*H1*G([2,3],1), ...
G([2,3],1)'*H1*G([2,3],2); ...
G([2,3],1)'*H1*G([2,3],2), ...
G([2,3],2)'*H1*G([2,3],2)];
HesseB2local(:, :, elem) = [G([1,3],1)'*H2*G([1,3],1), ...
G([1,3],1)'*H2*G([1,3],2); ...
G([1,3],1)'*H2*G([1,3],2), ...
G([1,3],2)'*H2*G([1,3],2)];
HesseB3local(:, :, elem) = [G([1,2],1)'*H3*G([1,2],1), ...
G([1,2],1)'*H3*G([1,2],2); ...
G([1,2],1)'*H3*G([1,2],2), ...
G([1,2],2)'*H3*G([1,2],2)];
HesseB4local(:, :, elem) = 2*sg(1)/sqrt(z(1))*G(1,:)'*G(1,:);
HesseB5local(:, :, elem) = 2*sg(2)/sqrt(z(2))*G(2,:)'*G(2,:);
HesseB6local(:, :, elem) = 2*sg(3)/sqrt(z(3))*G(3,:)'*G(3,:);
end

```

The Hessians are reordered in the computation for Hesse4e by combining B6 with

the fourth entry of the coefficient vector, B4 with the fifth, and B5 with the sixth. For finer triangulations the direct assembly is a bit faster than [Gal14].

C.5 Notes on Crouzeix-Raviart and Morley method for comparison

For a straightforward comparison of the post-processed bounds from [CG14a; CG14b] with the new guaranteed lower bounds the optimal adaptive mesh-refinement for these methods was used. Therefore, the following two main function were implemented.

```
LaplaceEVP_CR(geom, nrRedRefinements, minNrElem, eValMin, eValMax, ...
               theta, kappa1, savedata, markPlus)
```

```
BiLaplaceEVP_Morley(geom, nrRedRefinements, minNrElem, eValMin, ...
                    eValMax, theta, kappa2, savedata, numIntegration, ...
                    markPlus)
```

In comparison to the functions in [Gal14], most for loops are parallelized and the functions compute and save the post-processed guaranteed lower bound with given parameter `kappa1` resp. `kappa2`. For the Morley method the newly implemented direct computation is an option for `numIntegration = 0`. The functions `LaplaceEVP_CR_all` and `BiLaplaceEVP_Morley_all` have no input parameter `kappa1` or `kappa2` and compute the post-processed bounds with all mentioned values for the constants κ_1 and κ_2 , respectively.

D Software

The software is published separately on the edoc-Server.

Content of the software archive

software_GLB	
afem_package	contains files taken from the AFEM package [CN09]
closure.m	
computeArea4e.m	
computeE4n.m	
computeE4s.m	
computeLength4s.m	
computeMid4e.m	
computeMid4s.m	
computeN4s.m	
computeNormal4e.m	
computeNormal4s.m	
computeS4e.m	
computeS4n.m	
computeTangent4e.m	
computeTangent4s.m	
cropFigure.m	
exportTriangulationPlot.m	
integrate.m	
loadGeometry.m	
markBulk.m	
P0NormalJump.m	
P0TangentJump.m	
plotTriangulation.m	
refineBi3GB.m	
refineUniformRed.m	
COPYING.txt	
data	contains *.dat-files with reference values for each geometry
estimate	
estimateBiLaplaceEVP_Morley_EtaElements.m	
estimateLaplaceEVP_CR_EtaElements.m	
estimateLaplaceEVP_HH0_EtaElements.m	
geometries	subfolders contain *.dat-files with c4n, n4e, n4sDb, n4sNb for each geometry
Drum_1	
Drum_2	
DumbbellSlit	
DumbbellSlit_2	
DumbbellSlit_asymmetric	
FourSlit	
FourSlit_perturbed	
Lshape	
Slit	
Square	
Square_CrissCross	
solve	
solveLaplaceEVP_HH0.m	
solveLaplaceEVP_extrastabilized_compare.m	
solveLaplaceEVP_extrastabilized.m	
solveLaplaceEVP_CR.m	
solveBiLaplaceEVP_Morley_direct.m	
solveBiLaplaceEVP_Morley.m	
solveBiLaplaceEVP_extrastabilized_direct.m	
solveBiLaplaceEVP_extrastabilized_compare_direct.m	
solveBiLaplaceEVP_extrastabilized_compare.m	
solveBiLaplaceEVP_extrastabilized.m	
BiLaplaceEVP_extrastabilized.m	
BiLaplaceEVP_Morley.m	
BiLaplaceEVP_Morley_all.m	
experiments.m	
LaplaceEVP_CR.m	
LaplaceEVP_CR_all.m	
LaplaceEVP_extrastabilized.m	
LaplaceEVP_HH0lowestorder.m	
LaplaceEVP_HH0_Parameter.m	
Licence.txt	
ReadMe.txt	

Bibliography

- [AEP18] M. Abbas, A. Ern, and N. Pignet. ‘Hybrid high-order methods for finite deformations of hyperelastic materials’. *Comput. Mech.* 62.4 (2018), pp. 909–928.
- [Agm10] S. Agmon. *Lectures on elliptic boundary value problems*. AMS Chelsea Publishing, Providence, RI, 2010, pp. x+216.
- [Alt16] H. W. Alt. *Linear functional analysis*. Universitext. Springer-Verlag London, Ltd., London, 2016, pp. xii+435.
- [AD04] M. G. Armentano and R. G. Durán. ‘Asymptotic lower bounds for eigenvalues by nonconforming finite element methods’. *Electron. Trans. Numer. Anal.* 17 (2004), pp. 93–101.
- [Arn82] D. N. Arnold. ‘An interior penalty finite element method with discontinuous elements’. *SIAM J. Numer. Anal.* 19.4 (1982), pp. 742–760.
- [BO91] I. Babuška and J. Osborn. ‘Eigenvalue problems’. In: *Handbook of numerical analysis, Vol. II*. North-Holland, Amsterdam, 1991, pp. 641–787.
- [Beb03] M. Bebendorf. ‘A note on the Poincaré inequality for convex domains’. *Z. Anal. Anwendungen* 22.4 (2003), pp. 751–756.
- [BL76] J. Bergh and J. Löfström. *Interpolation spaces. An introduction*. Grundlehren der Mathematischen Wissenschaften, No. 223. Springer-Verlag, Berlin-New York, 1976, pp. x+207.
- [BH81] M. Bernadou and K. Hassan. ‘Basis functions for general Hsieh-Clough-Tocher triangles, complete or reduced’. *Int. J. Numer. Methods Eng.* 17.5 (1981), pp. 784–789.
- [Ber+23] F. Bertrand, C. Carstensen, B. Gräßle, and N. T. Tran. ‘Stabilization-free HHO a posteriori error control’. *Numer. Math.* 154.3-4 (2023), pp. 369–408.
- [BDD04] P. Binev, W. Dahmen, and R. DeVore. ‘Adaptive finite element methods with convergence rates’. *Numer. Math.* 97.2 (2004), pp. 219–268.
- [BR80] H. Blum and R. Rannacher. ‘On the boundary value problem of the biharmonic operator on domains with angular corners’. *Math. Methods Appl. Sci.* 2.4 (1980), pp. 556–581.

- [Bof10] D. Boffi. ‘Finite element approximation of eigenvalue problems’. *Acta Numer.* 19 (2010), pp. 1–120.
- [BBF13] D. Boffi, F. Brezzi, and M. Fortin. *Mixed finite element methods and applications*. Vol. 44. Springer Series in Computational Mathematics. Springer, Heidelberg, 2013, pp. xiv+685.
- [Boo87] C. de Boor. ‘B-form basics’. In: *Geometric modeling*. SIAM, Philadelphia, PA, 1987, pp. 131–148.
- [Bra07] D. Braess. *Finite elements. Theory, fast solvers, and applications in elasticity theory*. Third edition. Cambridge University Press, Cambridge, 2007, pp. xviii+365.
- [BKK08] J. Brandts, S. Korotov, and M. Křížek. ‘On the equivalence of regularity criteria for triangular and tetrahedral finite element partitions’. *Comput. Math. Appl.* 55.10 (2008), pp. 2227–2233.
- [Bre15] S. C. Brenner. ‘Forty years of the Crouzeix-Raviart element’. *Numer. Methods Partial Differential Equations* 31.2 (2015), pp. 367–396.
- [BS08] S. C. Brenner and L. R. Scott. *The mathematical theory of finite element methods*. Third edition. Vol. 15. Texts in Applied Mathematics. Springer, New York, 2008, pp. xviii+397.
- [Cal+19] V. Calo, M. Cicuttin, Q. Deng, and A. Ern. ‘Spectral approximation of elliptic operators by the hybrid high-order method’. *Math. Comp.* 88.318 (2019), pp. 1559–1586.
- [Can+17] E. Cancès, G. Dusson, Y. Maday, B. Stamm, and M. Vohralík. ‘Guaranteed and robust a posteriori bounds for Laplace eigenvalues and eigenvectors: conforming approximations’. *SIAM J. Numer. Anal.* 55.5 (2017), pp. 2228–2254.
- [Can+18] E. Cancès, G. Dusson, Y. Maday, B. Stamm, and M. Vohralík. ‘Guaranteed and robust a posteriori bounds for Laplace eigenvalues and eigenvectors: a unified framework’. *Numer. Math.* 140.4 (2018), pp. 1033–1079.
- [Can+20] E. Cancès, G. Dusson, Y. Maday, B. Stamm, and M. Vohralík. ‘Guaranteed a posteriori bounds for eigenvalues and eigenvectors: multiplicities and clusters’. *Math. Comp.* 89.326 (2020), pp. 2563–2611.
- [CB18] C. Carstensen and S. C. Brenner. ‘Finite Element Methods’. In: *Encyclopedia of Computational Mechanics Second Edition*. Ed. by R. d. B. E. Stein and T. J. R. Hughes. John Wiley and Sons, 2018, pp. 1–47.
- [Car+12] C. Carstensen, M. Eigel, R. H. W. Hoppe, and C. Löbhard. ‘A review of unified a posteriori finite element error control’. *Numer. Math. Theory Methods Appl.* 5.4 (2012), pp. 509–558.

- [Car+14a] C. Carstensen, M. Feischl, M. Page, and D. Praetorius. ‘Axioms of adaptivity’. *Comput. Math. Appl.* 67.6 (2014), pp. 1195–1253.
- [Car+14b] C. Carstensen, J. Gedicke, V. Mehrmann, and A. Międlar. ‘An adaptive finite element method with asymptotic saturation for eigenvalue problems’. *Numer. Math.* 128.4 (2014), pp. 615–634.
- [CPS12] C. Carstensen, D. Peterseim, and M. Schedensack. ‘Comparison results of finite element methods for the Poisson model problem’. *SIAM J. Numer. Anal.* 50.6 (2012), pp. 2803–2823.
- [CR17] C. Carstensen and H. Rabus. ‘Axioms of adaptivity with separate marking for data resolution’. *SIAM J. Numer. Anal.* 55.6 (2017), pp. 2644–2665.
- [CBJ02] C. Carstensen, S. Bartels, and S. Jansche. ‘A posteriori error estimates for nonconforming finite element methods’. *Numer. Math.* 92.2 (2002), pp. 233–256.
- [CEP21a] C. Carstensen, A. Ern, and S. Puttkammer. ‘Guaranteed lower bounds on eigenvalues of elliptic operators with a hybrid high-order method’. *Numer. Math.* 149.2 (2021), pp. 273–304.
- [CF99] C. Carstensen and S. A. Funken. ‘Fully reliable localized error control in the FEM’. *SIAM J. Sci. Comput.* 21.4 (1999/00), pp. 1465–1484.
- [CG14a] C. Carstensen and D. Gallistl. ‘Guaranteed lower eigenvalue bounds for the biharmonic equation’. *Numer. Math.* 126.1 (2014), pp. 33–51.
- [CGH14] C. Carstensen, D. Gallistl, and J. Hu. ‘A discrete Helmholtz decomposition with Morley finite element functions and the optimality of adaptive finite element schemes’. *Comput. Math. Appl.* 68.12, part B (2014), pp. 2167–2181.
- [CGS13] C. Carstensen, D. Gallistl, and M. Schedensack. ‘Discrete reliability for Crouzeix-Raviart FEMs’. *SIAM J. Numer. Anal.* 51.5 (2013), pp. 2935–2955.
- [CGS15] C. Carstensen, D. Gallistl, and M. Schedensack. ‘Adaptive nonconforming Crouzeix-Raviart FEM for eigenvalue problems’. *Math. Comp.* 84.293 (2015), pp. 1061–1087.
- [CG11] C. Carstensen and J. Gedicke. ‘An oscillation-free adaptive FEM for symmetric eigenvalue problems’. *Numer. Math.* 118.3 (2011), pp. 401–427.
- [CG12] C. Carstensen and J. Gedicke. ‘An adaptive finite element eigenvalue solver of asymptotic quasi-optimal computational complexity’. *SIAM J. Numer. Anal.* 50.3 (2012), pp. 1029–1057.
- [CG14b] C. Carstensen and J. Gedicke. ‘Guaranteed lower bounds for eigenvalues’. *Math. Comp.* 83.290 (2014), pp. 2605–2629.

- [CGR12] C. Carstensen, J. Gedicke, and D. Rim. ‘Explicit error estimates for Courant, Crouzeix-Raviart and Raviart-Thomas finite element methods’. *J. Comput. Math.* 30.4 (2012), pp. 337–353.
- [CGT22] C. Carstensen, B. Gräßle, and N. T. Tran. ‘Adaptive hybrid high-order method for guaranteed lower eigenvalue bounds’. work in progress. 2022.
- [CH18a] C. Carstensen and F. Hellwig. ‘Constants in discrete Poincaré and Friedrichs inequalities and discrete quasi-interpolation’. *Comput. Methods Appl. Math.* 18.3 (2018), pp. 433–450.
- [CH18b] C. Carstensen and F. Hellwig. ‘Optimal convergence rates for adaptive lowest-order discontinuous Petrov-Galerkin schemes’. *SIAM J. Numer. Anal.* 56.2 (2018), pp. 1091–1111.
- [CN21] C. Carstensen and N. Nataraj. ‘A Priori and a Posteriori Error Analysis of the Crouzeix–Raviart and Morley FEM with Original and Modified Right-Hand Sides’. *Comput. Methods Appl. Math.* 21.2 (2021), pp. 289–315.
- [CN09] C. Carstensen and Numerische Analysis Group. ‘AFEM’. Unpublished MATLAB software package. 2009.
- [CP20] C. Carstensen and S. Puttkammer. ‘How to prove the discrete reliability for nonconforming finite element methods’. *J. Comput. Math.* 38.1 (2020), pp. 142–175.
- [CP21] C. Carstensen and S. Puttkammer. ‘Direct guaranteed lower eigenvalue bounds with optimal a priori convergence rates for the bi-Laplacian’ (2021). arXiv: 2105.01505.
- [CP22] C. Carstensen and S. Puttkammer. ‘Adaptive guaranteed lower eigenvalue bounds with optimal convergence rates’ (2022). arXiv: 2203.01028.
- [CT21] C. Carstensen and N. T. Tran. ‘Unstabilized hybrid high-order method for a class of degenerate convex minimization problems’. *SIAM J. Numer. Anal.* 59.3 (2021), pp. 1348–1373.
- [CT22] C. Carstensen and N. T. Tran. ‘Convergent adaptive hybrid higher-order schemes for convex minimization’. *Numer. Math.* 151.2 (2022), pp. 329–367.
- [CZZ20] C. Carstensen, Q. Zhai, and R. Zhang. ‘A skeletal finite element method can compute lower eigenvalue bounds’. *SIAM J. Numer. Anal.* 58.1 (2020), pp. 109–124.
- [Cas+08] J. M. Cascon, C. Kreuzer, R. H. Nochetto, and K. G. Siebert. ‘Quasi-optimal convergence rate for an adaptive finite element method’. *SIAM J. Numer. Anal.* 46.5 (2008), pp. 2524–2550.

- [Cha83] F. Chatelin. *Spectral approximation of linear operators*. Computer Science and Applied Mathematics. Academic Press, New York, 1983, pp. xix+458.
- [Cia78] P. G. Ciarlet. *The finite element method for elliptic problems*. Studies in Mathematics and its Applications, Vol. 4. North-Holland Publishing Co., Amsterdam-New York-Oxford, 1978, pp. xix+530.
- [CDPE18] M. Cicuttin, D. A. Di Pietro, and A. Ern. ‘Implementation of discontinuous skeletal methods on arbitrary-dimensional, polytopal meshes using generic programming’. *J. Comput. Appl. Math.* 344 (2018), pp. 852–874.
- [CEG20] M. Cicuttin, A. Ern, and T. Gudi. ‘Hybrid high-order methods for the elliptic obstacle problem’. *J. Sci. Comput.* 83.1 (2020), Paper No. 8, 18.
- [CEP21b] M. Cicuttin, A. Ern, and N. Pignet. *Hybrid high-order methods. A primer with applications to solid mechanics*. SpringerBriefs in Mathematics. Springer, Cham, 2021, pp. viii+136.
- [CDPE16] B. Cockburn, D. A. Di Pietro, and A. Ern. ‘Bridging the hybrid high-order and hybridizable discontinuous Galerkin methods’. *ESAIM Math. Model. Numer. Anal.* 50.3 (2016), pp. 635–650.
- [CGL09] B. Cockburn, J. Gopalakrishnan, and R. Lazarov. ‘Unified hybridization of discontinuous Galerkin, mixed, and continuous Galerkin methods for second order elliptic problems’. *SIAM J. Numer. Anal.* 47.2 (2009), pp. 1319–1365.
- [CR73] M. Crouzeix and P.-A. Raviart. ‘Conforming and nonconforming finite element methods for solving the stationary Stokes equations. I’. *Rev. Française Automat. Informat. Recherche Opérationnelle Sér. Rouge* 7.R-3 (1973), pp. 33–75.
- [DHZ15] X. Dai, L. He, and A. Zhou. ‘Convergence and quasi-optimal complexity of adaptive finite element computations for multiple eigenvalues’. *IMA J. Numer. Anal.* 35.4 (2015), pp. 1934–1977.
- [DXZ08] X. Dai, J. Xu, and A. Zhou. ‘Convergence and optimal complexity of adaptive finite element eigenvalue computations’. *Numer. Math.* 110.3 (2008), pp. 313–355.
- [Dau88] M. Dauge. *Elliptic boundary value problems on corner domains. Smoothness and asymptotics of solutions*. Vol. 1341. Lecture Notes in Mathematics. Springer-Verlag, Berlin, 1988, pp. viii+259.
- [DPD17] D. A. Di Pietro and J. Droniou. ‘A hybrid high-order method for Leray-Lions elliptic equations on general meshes’. *Math. Comp.* 86.307 (2017), pp. 2159–2191.

- [DPDE15] D. A. Di Pietro, J. Droniou, and A. Ern. ‘A discontinuous-skeletal method for advection-diffusion-reaction on general meshes’. *SIAM J. Numer. Anal.* 53.5 (2015), pp. 2135–2157.
- [DPDM18] D. A. Di Pietro, J. Droniou, and G. Manzini. ‘Discontinuous skeletal gradient discretisation methods on polytopal meshes’. *J. Comput. Phys.* 355 (2018), pp. 397–425.
- [DPE10] D. A. Di Pietro and A. Ern. ‘Discrete functional analysis tools for discontinuous Galerkin methods with application to the incompressible Navier-Stokes equations’. *Math. Comp.* 79.271 (2010), pp. 1303–1330.
- [DPE15] D. A. Di Pietro and A. Ern. ‘A hybrid high-order locking-free method for linear elasticity on general meshes’. *Comput. Methods Appl. Mech. Eng.* 283 (2015), pp. 1–21.
- [DPEL14] D. A. Di Pietro, A. Ern, and S. Lemaire. ‘An arbitrary-order and compact-stencil discretization of diffusion on general meshes based on local reconstruction operators’. *Comput. Methods Appl. Math.* 14.4 (2014), pp. 461–472.
- [DP+16] D. A. Di Pietro, A. Ern, A. Linke, and F. Schieweck. ‘A discontinuous skeletal method for the viscosity-dependent Stokes problem’. *Comput. Methods Appl. Mech. Eng.* 306 (2016), pp. 175–195.
- [DPE12] D. A. Di Pietro and A. Ern. *Mathematical aspects of discontinuous Galerkin methods*. Vol. 69. Mathématiques & Applications (Berlin). Springer, Heidelberg, 2012, pp. xviii+384.
- [DLM16] B. Ayuso de Dios, K. Lipnikov, and G. Manzini. ‘The nonconforming virtual element method’. *ESAIM Math. Model. Numer. Anal.* 50.3 (2016), pp. 879–904.
- [DE22] Z. Dong and A. Ern. ‘Hybrid High-Order and Weak Galerkin Methods for the Biharmonic Problem’. *SIAM J. Numer. Anal.* 60.5 (2022), pp. 2626–2656.
- [Dör96] W. Dörfler. ‘A convergent adaptive algorithm for Poisson’s equation’. *SIAM J. Numer. Anal.* 33.3 (1996), pp. 1106–1124.
- [Dri97] T. A. Driscoll. ‘Eigenmodes of isospectral drums’. *SIAM Review* 39.1 (1997), pp. 1–17.
- [DPR03] R. G. Durán, C. Padra, and R. Rodríguez. ‘A posteriori error estimates for the finite element approximation of eigenvalue problems’. *Math. Models Methods Appl. Sci* 13.8 (2003), pp. 1219–1229.
- [EM73] M. A. Eisenberg and L. E. Malvern. ‘On finite element integration in natural co-ordinates’. *Int. J. Numer. Methods Eng.* 7.4 (1973), pp. 574–575.

- [EG04] A. Ern and J.-L. Guermond. *Theory and practice of finite elements*. Vol. 159. Applied Mathematical Sciences. Springer-Verlag, New York, 2004, pp. xiv+524.
- [EG21] A. Ern and J.-L. Guermond. *Finite Elements II – Galerkin approximation, elliptic and mixed PDEs*. First edition. Vol. 72. Texts in Applied Mathematics. Springer, Cham, 2021, pp. ix+492.
- [EZ20] A. Ern and P. Zanotti. ‘A quasi-optimal variant of the hybrid high-order method for elliptic partial differential equations with H^{-1} loads’. *IMA J. Numer. Anal.* 40.4 (2020), pp. 2163–2188.
- [Eva10] L. C. Evans. *Partial differential equations*. Second edition. Vol. 19. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2010, pp. xxii+749.
- [EG15] L. C. Evans and R. F. Gariepy. *Measure theory and fine properties of functions*. Revised edition. Textbooks in Mathematics. CRC Press, Boca Raton, FL, 2015, pp. xiv+299.
- [Gal14] D. Gallistl. ‘Adaptive finite element computation of eigenvalues’. PhD thesis. Humboldt-Universität zu Berlin, Mathematisch-Naturwissenschaftliche Fakultät II, 2014.
- [Gal15a] D. Gallistl. ‘An optimal adaptive FEM for eigenvalue clusters’. *Numer. Math.* 130.3 (2015), pp. 467–496.
- [Gal15b] D. Gallistl. ‘Morley finite element method for the eigenvalues of the biharmonic operator’. *IMA J. Numer. Anal.* 35.4 (2015), pp. 1779–1811.
- [GSS14] D. Gallistl, M. Schedensack, and R. P. Stevenson. ‘A remark on newest vertex bisection in any space dimension’. *Comput. Methods Appl. Math.* 14.3 (2014), pp. 317–320.
- [GMZ09] E. M. Garau, P. Morin, and C. Zuppa. ‘Convergence of adaptive finite element methods for eigenvalue problems’. *Math. Models Methods Appl. Sci* 19.5 (2009), pp. 721–747.
- [GG09] S. Giani and I. G. Graham. ‘A convergent adaptive method for elliptic eigenvalue problems’. *SIAM J. Numer. Anal.* 47.2 (2009), pp. 1067–1091.
- [GT83] D. Gilbarg and N. S. Trudinger. *Elliptic partial differential equations of second order*. Second edition. Vol. 224. Grundlehren der Mathematischen Wissenschaften. Springer-Verlag, Berlin, 1983, pp. xiii+513.
- [Grä22] B. Gräßle. ‘Optimal multilevel adaptive FEM for the Argyris element’. *Comput. Methods Appl. Mech. Eng.* 399 (2022), Paper No. 115352.

- [Gri85] P. Grisvard. *Elliptic problems in nonsmooth domains*. Vol. 24. Monographs and Studies in Mathematics. Pitman (Advanced Publishing Program), Boston, MA, 1985, pp. xiv+410.
- [Gri92] P. Grisvard. *Singularities in boundary value problems*. Vol. 22. Recherches en Mathématiques Appliquées. Masson, Paris; Springer-Verlag, Berlin, 1992, pp. xiv+199.
- [GO09] L. Grubišić and J. S. Owall. ‘On estimators for eigenvalue/eigenvector approximations’. *Math. Comp.* 78.266 (2009), pp. 739–770.
- [Gud10] T. Gudi. ‘A new error analysis for discontinuous finite element methods for linear elliptic problems’. *Math. Comp.* 79.272 (2010), pp. 2169–2189.
- [HR01] V. Heuveline and R. Rannacher. ‘A posteriori error control for finite approximations of elliptic eigenvalue problems’. *Adv. Comput. Math.* 15.1-4 (2001), pp. 107–138.
- [HHL14] J. Hu, Y. Huang, and Q. Lin. ‘Lower bounds for eigenvalues of elliptic operators: by nonconforming finite element methods’. *J. Sci. Comput.* 61.1 (2014), pp. 196–221.
- [Kat95] T. Kato. *Perturbation theory for linear operators*. Classics in Mathematics. Springer-Verlag, Berlin, 1995, pp. xxii+619.
- [Lar00] M. G. Larson. ‘A posteriori and a priori error analysis for finite element approximations of self-adjoint elliptic eigenvalue problems’. *SIAM J. Numer. Anal.* 38.2 (2000), pp. 608–625.
- [LS10] R. S. Laugesen and B. A. Siudeja. ‘Minimizing Neumann fundamental tones of triangles: an optimal Poincaré inequality’. *J. Differential Equations* 249.1 (2010), pp. 118–135.
- [LS16] C. Lehrenfeld and J. Schöberl. ‘High order exactly divergence-free hybrid discontinuous Galerkin methods for unsteady incompressible flows’. *Comput. Methods Appl. Mech. Eng.* 307 (2016), pp. 339–361.
- [LSL19] S.-K. Liao, Y.-C. Shu, and X. Liu. ‘Optimal estimation for the Fujino-Morley interpolation error constants’. *Jpn. J. Ind. Appl. Math.* 36.2 (2019), pp. 521–542.
- [Liu15] X. Liu. ‘A framework of verified eigenvalue bounds for self-adjoint differential operators’. *Appl. Math. Comput.* 267 (2015), pp. 341–355.
- [LLX12] F. Luo, Q. Lin, and H. Xie. ‘Computing the lower and upper bounds of Laplace eigenvalue problem: by combining conforming and non-conforming finite element methods’. *Sci. China Math.* 55.5 (2012), pp. 1069–1082.

- [Ma21] R. Ma. *The constants for the L^2 error estimates of the interpolation of Morley-Wang-Xu elements*. Research seminar talk at Humboldt-Universität zu Berlin. 2021.
- [MSZ06] D. Mao, L. Shen, and A. Zhou. ‘Adaptive finite element algorithms for eigenvalue problems based on local averaging type a posteriori error estimates’. *Adv. Comput. Math.* 25.1-3 (2006), pp. 135–160.
- [Mau95] J. M. Maubach. ‘Local bisection refinement for n -simplicial grids generated by reflection’. *SIAM J. Sci. Comput.* 16.1 (1995), pp. 210–227.
- [MM11] V. Mehrmann and A. Międlar. ‘Adaptive computation of smallest eigenvalues of self-adjoint elliptic partial differential equations’. *Numer. Linear Algebra Appl.* 18.3 (2011), pp. 387–409.
- [Mey12] A. Meyer. ‘A simplified calculation of reduced HCT-basis functions in a finite element context’. *Comput. Methods Appl. Math.* 12.4 (2012), pp. 486–499.
- [MX06] W. Ming and J. Xu. ‘The Morley element for fourth order elliptic equations in any dimensions’. *Numer. Math.* 103.1 (2006), pp. 155–169.
- [Mon03] P. Monk. *Finite element methods for Maxwell’s equations*. Numerical Mathematics and Scientific Computation. Oxford University Press, New York, 2003, pp. xiv+450.
- [MNS02] P. Morin, R. H. Nochetto, and K. G. Siebert. ‘Convergence of adaptive finite element methods’. *SIAM Review* 44.4 (2002), pp. 631–658.
- [Mor68] L. S. D. Morley. ‘The Triangular Equilibrium Element in the Solution of Plate Bending Problems’. *Aeronaut. Quart.* 19.2 (1968), 149–169.
- [Neč12] J. Nečas. *Direct methods in the theory of elliptic equations*. Springer Monographs in Mathematics. Springer, Heidelberg, 2012, pp. xvi+372.
- [NW19] M. Neilan and M. Wu. ‘Discrete Miranda-Talenti estimates and applications to linear and nonlinear PDEs’. *J. Comput. Appl. Math.* 356 (2019), pp. 358–376.
- [Ney02] K. Neymeyr. ‘A posteriori error estimation for elliptic eigenproblems’. *Numer. Linear Algebra Appl.* 9.4 (2002), pp. 263–279.
- [Oik15] I. Oikawa. ‘A hybridized discontinuous Galerkin method with reduced stabilization’. *J. Sci. Comput.* 65.1 (2015), pp. 327–340.
- [Par98] B. N. Parlett. *The symmetric eigenvalue problem*. Vol. 20. Classics in Applied Mathematics. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 1998, pp. xxiv+398.

- [PW60] L. E. Payne and H. F. Weinberger. ‘An optimal Poincaré inequality for convex domains’. *Arch. Rational Mech. Anal.* 5 (1960), pp. 286–292.
- [PP20] C.-M. Pfeiler and D. Praetorius. ‘Dörfler marking with minimal cardinality is a linear complexity problem’. *Math. Comp.* 89.326 (2020), pp. 2735–2752.
- [Ran79] R. Rammacher. ‘Nonconforming finite element methods for eigenvalue problems in linear plate theory’. *Numer. Math.* 33.1 (1979), pp. 23–42.
- [ŠV14] I. Šebestová and T. Vejchodský. ‘Two-sided bounds for eigenvalues of differential operators with applications to Friedrichs, Poincaré, trace, and similar constants’. *SIAM J. Numer. Anal.* 52.1 (2014), pp. 308–329.
- [Sor09] T. Sorokina. ‘A C^1 multivariate Clough-Tocher interpolant’. *Constr. Approx.* 29.1 (2009), pp. 41–59.
- [Ste07] R. Stevenson. ‘Optimality of a standard adaptive finite element method’. *Found. Comput. Math.* 7.2 (2007), pp. 245–269.
- [Ste08] R. Stevenson. ‘The completion of locally refined simplicial partitions created by bisection’. *Math. Comp.* 77.261 (2008), pp. 227–241.
- [SF08] G. Strang and G. Fix. *An analysis of the finite element method*. Second edition. Wellesley-Cambridge Press, Wellesley, MA, 2008, pp. x+402.
- [SF73] G. Strang and G. J. Fix. *An analysis of the finite element method*. Prentice-Hall, Englewood Cliffs, 1973, pp. xiv+306.
- [SZ17] J. Sun and A. Zhou. *Finite element methods for eigenvalue problems*. Monographs and Research Notes in Mathematics. CRC Press, Boca Raton, FL, 2017, pp. xxii+343.
- [Tar07] L. Tartar. *An introduction to Sobolev spaces and interpolation spaces*. Vol. 3. Lecture Notes of the Unione Matematica Italiana. Springer, Berlin; UMI, Bologna, 2007, pp. xxvi+218.
- [Tim85] S. P. Timoshenko. *Theory of elastic stability*. Second edition. McGraw-Hill Book Co., Inc., New York-Toronto-London, 1985, pp. xvi+541.
- [Tra97] C. T. Traxler. ‘An algorithm for adaptive mesh refinement in n dimensions’. *Computing* 59.2 (1997), pp. 115–137.
- [TB06] L. N. Trefethen and T. Betcke. ‘Computed eigenmodes of planar regions’. In: *Recent advances in differential equations and mathematical physics*. Vol. 412. Contemp. Math. Amer. Math. Soc., Providence, RI, 2006, pp. 297–314.

- [VZ19] A. Veeseer and P. Zanotti. ‘Quasi-optimal nonconforming methods for second-order problems on domains with non-Lipschitz boundary’. In: *Numerical mathematics and advanced applications*. Vol. 126. Lect. Notes Comput. Sci. Eng. Springer, Cham, 2019, pp. 461–469.
- [VNS07] L. Beirão da Veiga, J. Niiranen, and R. Stenberg. ‘A posteriori error estimates for the Morley plate bending element’. *Numer. Math.* 106.2 (2007), pp. 165–179.
- [Vej18a] T. Vejchodský. ‘Flux reconstructions in the Lehmann-Goerisch method for lower bounds on eigenvalues’. *J. Comput. Appl. Math.* 340 (2018), pp. 676–690.
- [Vej18b] T. Vejchodský. ‘Three methods for two-sided bounds of eigenvalues—a comparison’. *Numer. Methods Partial Differential Equations* 34.4 (2018), pp. 1188–1208.
- [Ver94] R. Verfürth. ‘A posteriori error estimates for nonlinear problems. Finite element discretizations of elliptic equations’. *Math. Comp.* 62.206 (1994), pp. 445–475.
- [Ver13] R. Verfürth. *A posteriori error estimation techniques for finite element methods*. Numerical Mathematics and Scientific Computation. Oxford University Press, Oxford, 2013, pp. xx+393.
- [VS18] F. J. Vermolen and A. Segal. ‘On an integration rule for products of barycentric coordinates over simplexes in $\mathbb{R}^n$ ’. *J. Comput. Appl. Math.* 330 (2018), pp. 289–294.
- [WY13] J. Wang and X. Ye. ‘A weak Galerkin finite element method for second-order elliptic problems’. *J. Comput. Appl. Math.* 241 (2013), pp. 103–115.
- [WX13] M. Wang and J. Xu. ‘Minimal finite element spaces for $2m$ -th-order partial differential equations in R^n ’. *Math. Comp.* 82.281 (2013), pp. 25–43.
- [WS72] A. Weinstein and W. Stenger. *Methods of intermediate problems for eigenvalues. Theory and ramifications*. Mathematics in Science and Engineering, Vol. 89. Academic Press, New York-London, 1972, pp. xi+235.
- [WF87] A. J. Worsey and G. Farin. ‘An n -dimensional Clough-Tocher interpolant’. *Constr. Approx.* 3.2 (1987), pp. 99–110.
- [YLB16] Y. Yang, H. Li, and H. Bi. ‘The lower bound property of the Morley element eigenvalues’. *Comput. Math. Appl.* 72.4 (2016), pp. 904–920.
- [Yan+12] Y. Yang, Q. Lin, H. Bi, and Q. Li. ‘Eigenvalue approximations from below using Morley elements’. *Adv. Comput. Math.* 36.3 (2012), pp. 443–450.

- [YC08] W.-C. Yueh and S. S. Cheng. ‘Explicit eigenvalues and inverses of tridiagonal Toeplitz matrices with four perturbed corners’. *The ANZIAM J.* 49.3 (2008), pp. 361–387.

List of Figures

1.1	Experiments for the principal eigenvalue of the Laplacian on the L-shaped domain: (a) initial triangulation, (b) conforming P_1 approximation of the eigenfunction on a uniform red-refined mesh with $ \mathcal{T} = 6291456$, (c) adaptive mesh with $ \mathcal{T} = 4782$ computed with bulk parameter $\theta = 0.5$ and estimator (1.3), and (d) comparison of the distance of λ_1 and the new $\lambda_{\text{HHO}}(1)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(1)$ as well as the known $\text{GLB}_{\text{CR}}(1)$ (with $\kappa_1 = 0.2982$) computed on uniform ($\theta = 1$, solid line and filled markers) and adaptive ($\theta = 0.5$, dashed line and half filled markers) meshes . . .	2
1.2	Experiments for the principal eigenvalue of the bi-Laplacian on the asymmetric dumbbell-slit domain: (a) initial triangulation, (b) conforming nodal interpolation of the Morley approximation of the eigenfunction on a uniform red-refined mesh with $ \mathcal{T} = 11534336$, (c) adaptive mesh with $ \mathcal{T} = 1730$ computed with bulk parameter $\theta = 0.5$ and estimator (1.3), and (d) comparison of the distance of λ_1 and the new $\lambda_{\text{es}}(1)$ as well as the known $\text{GLB}_M(1)$ (with $\kappa_2 = 0.2358$) computed on uniformly refined ($\theta = 1$, solid line and filled markers) and adaptive ($\theta = 0.5$, dashed line and half filled markers) meshes	3
3.1	Mnemonic diagram of the Hsieh-Clough-Tocher finite element in 2D including the subtriangulation $\text{HCT}2D(K)$ with dashed lines	26
3.2	Illustration of two neighbouring tetrahedra $T_1 = \text{conv}\{P_1, P_2, P_3, P_4\}$ in red and $T_2 = \text{conv}\{P_1, P_2, P_3, P_5\}$ in blue with common face $F = \text{conv}\{P_1, P_2, P_3\}$ in violet	31
3.3	Illustration of the division of $K = \text{conv}\{Q_1, \dots, Q_4\}$ with center $c_K \in \text{int}(K)$ into $T(1), \dots, T(4)$ in (a) with $T(1) = \text{conv}\{c_K, F_1\}$ marked in red. Further subdivision of $T(1)$ with center $c_1 \in \text{int}(F_1)$ of face F_1 in the WF partition in (b) with one subsimplex $T = \text{conv}\{c_k, F\} \in \text{WF}3D(K)$ with face $F = \text{conv}\{c_1, Q_2, Q_4\} \subset F_1$ marked in red	36
3.4	Mnemonic diagram of the Wosey-Farin finite element in 3D . . .	39
3.5	Notation in the profile of the intersection $\langle F_1 \rangle \cap \langle F_2 \rangle$ of two planes through the faces $F_1, F_2 \in \mathcal{F}(T)$ with common edge E indicated by τ_E as $\odot$ perpendicular to the plane depicted in (a) and of F' from (3.28) (in red) in (b)	49

3.6	Mnemonic diagrams of the Crouzeix-Raviart finite element in 2D and 3D	59
3.7	Mnemonic diagrams of the Morley finite element in 2D and 3D . .	68
3.8	Subtetrahedron $T(4) = \text{conv}\{c_K, F_4\}$ of $K \in \mathcal{T}$ with face $F = F_4 = \text{conv}\{P_1, P_2, P_3\} \in \mathcal{F}(K)$ and edge $E = \text{conv}\{P_1, P_2\} \in \mathcal{E}(K)$ and dashed WF triangulation $\hat{\mathcal{T}}(K)$ with $T = \text{conv}\{c_K, c_F, E\} \in \hat{\mathcal{T}}$ marked in red, the support $\text{supp}(\xi_E) = B$ of ξ_E from Remark 3.40 in blue and the disjoint ball $\hat{B} \subset T$ in green	84
7.1	Initial triangulation $\mathcal{T}_0$ of (a) the asymmetric dumbbell-slit domain, (b) the dumbbell-slit domain, (c) the L-shaped domain, (d) the slit domain, and (e) the four-slit domain as well as the isospectral domains (f) and (g)	181
7.2	Comparison of $\text{GLB}_{\text{CR}}(1)$ and $\lambda_{\text{es}}(1)$ with $\lambda_{\text{HHO}}(1)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(1))/\tau^2 \in (0.003, 0.263)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of (a) the asymmetric dumbbell-slit domain and (b) the dumbbell-slit domain	186
7.3	Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of (a) the asymmetric dumbbell-slit domain for $j = 1$, and on the dumbbell-slit domain for (b) $j = 1$ as well as (c) $j = 2$	187
7.4	Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.262)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the L-shaped domain for (a) $j = 1$ and (b) $j = 103$	188
7.5	Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the L-shaped domain for (a) $j = 1$ and (b) $j \in \{103, 104, 105\}$	189
7.6	Comparison of (a) the distance of λ_1 and $\lambda_{\text{HHO}}(1)$ (with $\alpha = 0.5$ and $\beta = 0.1$) and $\lambda_{\text{es}}(1)$ (with $\kappa_1 = 0.2982$) and (b) the global estimator $\eta_{\text{HHO}}(\mathcal{T})$ and $\eta_{\text{es}}(\mathcal{T})$ computed on uniform ($\theta = 1$) and adaptive ($\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$) meshes of the slit domain	191
7.7	Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.002, 0.263)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the slit domain for (a) $j = 1$ and (b) $j = 6$	191
7.8	Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the slit domain for (a) $j = 1$ and (b) $j = 6$	192

7.9	Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.267)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the four-slit domain for (a) $j = 1$ and (b) $j = 2$	193
7.10	Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the four slit domain for (a) $j = 1$ and (b) $j \in \{2, 3\}$	193
7.11	Comparison of $\text{GLB}_{\text{CR}}(j)$ with $\lambda_{\text{HHO}}(j)$ for $\alpha \in (0, 1)$ and $\beta = (\alpha - \delta^2 \lambda_{\text{CR}}(j))/\tau^2 \in (0.003, 0.262)$ computed with $\kappa_1 = 0.2982$ on coarse uniform refinements of the first isospectral domain in 7.1.f for (a) $j = 1$ and (b) $j = 50$	195
7.12	Comparison of the distance of λ_j and $\lambda_{\text{HHO}}(j)$ (with $\alpha = 0.5$ and $\beta = 0.1$), $\lambda_{\text{es}}(j)$, and $\text{GLB}_{\text{CR}}(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the first isospectral domain 7.1.f for (a) $j = 1$ and (b) $j = 50$	195
7.13	Adaptive triangulation $\mathcal{T}$ with $\theta = 0.5$ for the principal eigenvalue of (a) the asymmetric dumbbell-slit domain, (c) the slit domain, (d) the four-slit domain, and (e) the fist isospectral domain as well as for (b) the second eigenvalue of the dumbbell-slit domain computed with (7.2) and $\kappa_1 = 0.2982$	196
7.14	Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of (a) the asymmetric dumbbell-slit domain for $j = 1$, and of the dumbbell-slit domain for (b) $j = 1$ as well as (c) $j = 4$	201
7.15	Nodal interpolation of the Morley approximation of the j -th eigenfunction of the dumbbell-slit domain computed on a uniformly red-refined mesh with $ \mathcal{T} = 12582912$ for $j \in \{1, \dots, 4\}$	202
7.16	Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the L-shaped domain for (a) $j = 1$ and (b) $j = 4$	203
7.17	Comparison of (a) the distance of λ_1 and $\lambda_{\text{es}}(1)$ (with $\kappa_2 = 0.2358$) and (b) the global estimator $\eta_{\text{es}}(\mathcal{T})$ computed on uniform ($\theta = 1$) and adaptive ($\theta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$) meshes of the slit domain	204
7.18	Comparison of the distance of λ_j and $\lambda_{\text{es}}(j)$ as well as $\text{GLB}_M(j)$ computed on uniform ($\theta = 1$) and adaptive ($\theta = 0.5$) meshes of the four-slit domain for (a) $j = 1$ and (b) $j \in \{3, 4\}$	205
7.19	Adaptive triangulation $\mathcal{T}$ with $\theta = 0.5$ for (a) the fourth eigenvalue of the dumbbell-slit domain computed with (7.5) and $\kappa_2 = 0.07353$ as well as the principal eigenvalue of (b) the L-shaped domain, (c) the slit domain and (d) the four-slit domain computed with (7.5) and $\kappa_2 = 0.23588$	206

C.1 AFEM cycle	225
--------------------------	-----

List of Tables

3.1	The 12 degrees of freedom $L_1, \dots, L_{12}$ of $\text{HCT}(T)$ for any $f \in C^1(T)$	26
3.2	The 28 degrees of freedom $L_1, \dots, L_{28}$ of $\text{WF}(K)$ for any $f \in C^1(K)$	39
3.3	The local degrees of freedom for the Crouzeix-Raviart finite element for any $T \in \mathcal{T}$ and $f \in H^1(T)$	60
3.4	The local degrees of freedom for the Morley finite element for any $T \in \mathcal{T}$ and $f \in H^2(T)$	69
5.1	2-level notation with respect to $\mathcal{T} \in \mathbb{T}(\varepsilon_6)$ (left) and an admissible refinement $\widehat{\mathcal{T}} \in \mathbb{T}(\mathcal{T})$ (right) including some consequences from Theorem 5.20	158

Selbständigkeitserklärung

Ich erkläre, dass ich die Dissertation selbstständig und nur unter Verwendung der von mir gemäß §7 Abs. 3 der Promotionsordnung der Mathematisch-Naturwissenschaftlichen Fakultät, veröffentlicht im Amtlichen Mitteilungsblatt der Humboldt-Universität zu Berlin Nr. 42/2018 am 11.07.2018 angegebenen Hilfsmittel angefertigt habe.

Berlin, den 06.10.2022, Sophie Louise Puttkammer